FED. TEST METHOD STD. NO. 601

April 12, 1955

SUPERSEDING
Fed. Spec. ZZ-R-601a
June 25, 1940

FEDERAL TEST METHOD STANDARD

RUBBER: SAMPLING AND TESTING

This standard was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies

Additional copies of this standard may be purchased from the General Services Administration, Business Service Center, Region 3, Seventh and D Streets SW, Washington 25, D. C., for $2 each

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INFORMATION SHEET

ON

FEDERAL TEST METHOD STANDARDS

This Federal Test Method Standard is issued in loose leaf form to permit the insertion or removal of new or revised sections and test methods.

All users of Federal Test Method Standards should keep them up to date by inserting revised or new sections and test methods as issued and removing superseded and canceled pages.

New and revised material and cancellations will be issued under Change Notices which will be numbered consecutively and will bear the date of issuance. Change Notices should be retained and filed in front of the Alphabetical Index of the Standard until such time as they are superseded by a reissue of the entire Standard.

(This Information Sheet should be filed immediately behind the front cover page of the Federal Test Method Standard.)

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FEDERAL TEST METHOD STANDARD
RUBBER: SAMPLING AND TESTING

Authority.—*This standard is issued pursuant to the Federal Property and Administration Services Act of 1949, as amended, and its application to the purchase of commodities referred to herein is mandatory on all Federal agencies*

SECTION 1
SCOPE AND CONTENTS

1. SCOPE

1.1 This standard describes the general physical and chemical methods for testing natural and synthetic rubber products for conformance with the requirements of Federal specifications It was prepared in order to eliminate unnecessary or undesirable variations in the general sampling and testing procedures This standard does not include special test methods applicable to certain natural and synthetic rubber products that are described in the appropriate detail specification, nor does it include all the test methods for natural and synthetic rubber materials used in industry In the case of conflict between the provisions of these methods and those of the individual test procedures in specifications for particular materials, the latter shall take precedence

1.2 General methods.—In this standard similar tests have been grouped together and given a group number, for example, all tension tests in the 4000 group, all geometrical measurements in the 2000 group, etc (See contents) Several of the groups of methods include a general method of test which contains general information, procedures, and instructions which are applicable to the methods of test described in the particular group It usually contains information and descriptions pertaining to generally applicable apparatus, test media, test conditions, conditioning of materials, precautions to be observed in conducting the tests, definitions of terms, etc , which are common to all or a majority of the methods in the group Insofar as applicable, the general method shall be considered a part of each method in the particular group, for example, method 4001 shall be considered a part of all of the methods in the 4000 group wherever applicable In case of conflict between the provisions of the general method and those of individual test procedures, the latter shall take precedence

1.3 New methods and revisions.—The methods in this standard are prepared in loose-leaf form for convenience in adding new methods and revising methods When it is necessary to correct or change an existing method, the method is rewritten and given a decimal number and the date the revised method is effective For example, Hardness Durometer, 3021, dated April 12, 1955, if and when rewritten, will be designated method 3021 1 indicating the first revision, and dated

When the revision of a method is promulgated, it automatically supersedes all previous editions of the method and shall be used whenever the method is referenced in a detail specification For example, when the method for determining tensile strength is referred to in a detail specification, it shall be listed as method 4111 and the revision in effect on the date of invitation for bids will form a part of the specification

New and revised test methods may be obtained from the same source as Federal Standards, as listed in section 8 notes

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2. CONTENTS

2.1 The contents of this standard are as follows

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2 Alphabetical Index of Test Methods
3 Numerical Index of Test Methods
4 Subject Index
5 Definitions of Terms
6 Sampling and Acceptance Procedures
7 Temperature and Humidity of Conditioning Room
8 Notes
9 Test Methods

Group

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2000 Geometrical Measurements
3000. Rheological Tests
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5000 Thermal Tests.
6000 Liquid Treatment Tests.
7000 Accelerated Aging Tests
8000. Adhesion Tests
9000 Electrical Tests
10000 Hydrostatic Tests.
11000 Hard Rubber
12000 Cellular Rubber.
13000 Tape
14000 Miscellaneous Physical Tests
15000 Chemical Analysis of Synthetic Rubber Compounds
16000 Chemical Analysis of Rubber Compounds and Packings.

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SECTION 2

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SECTION 5

DEFINITIONS OF TERMS

1. SCOPE

1.1 The following definitions are divided into two groups The first group covers general terms and the second group covers definitions applicable to sampling, inspection, and testing

1.2 The definitions of general terms in the first group include some of those encountered by personnel concerned with specifications and procurement of rubber and rubber products by the Federal Government Terms adequately defined by unabridged dictionaries are not generally included

1.3 The definitions applicable to sampling, inspection, and testing in the second group are mainly to be used in connection with the sampling for, and testing of, physical and chemical properties

2. GENERAL TERMS

2.1 Acetone extract. The material removed from the rubber compound by extraction with acetone The acetone removes resins, free sulfur; softeners, cellulose ethers and esters; naphthalene, part of any bituminous materials, mineral oil, and waxes, factice, and part or all of organic dyes, accelerators, and antioxidants or their decomposition products

2.2 Alcoholic potassium hydroxide extract.—The material removed from the rubber compound by extraction with alcoholic potassium hydroxide after the specimen has been extracted with acetone and chloroform The extract consists of a portion of the factice and any saponifiable material not removed by previous extraction (acetone-chloroform)

2.3 Cellular rubber.—A material containing cells or small hollow receptacles The cells may be either open and interconnecting or closed and not interconnecting

2.4 Chloroform extract.—The material removed from a vulcanized rubber compound by extraction with chloroform after the specimen has been extracted with acetone The chloroform removes a portion of the bituminous materials and may include a small amount of rubber for which no correction is made

2.5 Compression set.—The residual decrease in thickness of a specimen measured at a specified time after having been compressed for a definite time under specified conditions of load and temperature

2.6 Elongation.—The extension between bench marks produced by a tension force applied to the specimen It is expressed as a percentage of the original distance between the marks on the unstretched specimen Ultimate elongation is the elongation at the moment of rupture

2.7 Expanded rubber.—A closed-cell material usually made by subjecting a rubber compound to gas, such as nitrogen, under high pressure It may also be made by incorporating gas-forming ingredients in the compound

2.8 Flexural strength.—The transverse force expressed as fiber stress required to rupture a simple beam of specified dimensions when subjected to a transverse force

2.9 Free sulfur.—The sulfur that is removed by extraction with a 5-percent solution of sodium sulfite It includes only the elemental sulfur and does not include the sulfur that may be present in accelerators added to rubber compounds

2.10 Friction.—The term "friction" is used to indicate the adhesion between plies of fabric bonded with rubber or the adhesion between rubber and any other material to which it is bonded

2.11 Hard rubber.—Vulcanized rubber compounds having a ratio of combined sulfur to rubber hydrocarbon in excess of 15 percent

2.12 Inorganic sulfur.—Inorganic sulfur in a rubber compound includes (1) a portion of the sulfur originally added to the compound but which has since combined with a portion of the fillers and (2) sulfur determined as inorganic sulfides or sulfates

2.13 Latex foam rubber.—A rubber material with open or interconnecting cells made from rubber latices or liquid rubber

2.14 Median.—If numerical values are arranged in ascending or descending order, the median is (1) the middle value of the series if

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the number of values is odd and (2) the mean of the two middle values if the number of values is even

2.15 Mineral oil.—The portion of the unsaponifiable acetone extract that is soluble in absolute alcohol at −5° C (23° F) and that is soluble in carbon tetrachloride and is not decomposed by concentrated sulfuric acid

2.16 Natural rubber as compounded.—Rubber as compounded is approximately equivalent to the crude rubber used in the manufacture of a rubber compound Its computation is based on the arbitrary assumption that crude rubber consists of 94 percent rubber hydrocarbon and 6 percent resins, proteins, and other substances

2.17 Natural rubber by volume.—The percentage of the volume of a natural rubber compound occupied by the natural rubber as compounded, assuming the latter to have a specific gravity of 0 911 at 25° C (77° F)

2.18 Natural rubber hydrocarbon.—The characteristic and major component of crude or raw natural rubber It is considered to have the chemical formula $(C_5H_8)_x$

2.19 Natural rubber hydrocarbon by volume.—The percentage of the volume of a natural rubber compound occupied by the natural rubber hydrocarbon, assuming the latter to have a specific gravity of 0 911 at 25° C (77° F)

2.20 Organic extract.—Organic extract is calculated by subtracting the sulfur in extract from the total extract

2.21 Organic sulfur.—The sulfur in a rubber compound that is not removed during the extraction with acetone or with acetone-chloroform mixture and that is present in the rubber compound exclusive of that contained in inorganic sulfates or sulfides that may be present The organic sulfur is not intended to represent the sulfur of vulcanization The organic sulfur as determined includes (1) sulfur combined with rubber, (2) sulfur present in accelerators which is not removed by extraction with acetone or acetone-chloroform mixture, and (3) part of the sulfur present in factice and mineral rubber

2 22 **Skin or rind.**—The smooth outer surface of cellular rubber material It may be formed by contact with the surface of the mold or cover plates The skin or rind may also be formed by applying a solid rubber coating to the uncured rubber and vulcanizing them together

2.23 Sponge rubber.—An open-cell material made by incorporating into the rubber compound an inflating agent such as sodium bicarbonate that gives off gas which expands the rubber during vulcanization

2.24 Strain.—The elongation produced during a given period of time by a specified force per unit of cross-sectional area of the unstretched specimen

2.25 Strength of splice.—The force per unit of width necessary to separate the union of two pieces of soft vulcanized rubber

2.26 Sulfur in extract.—The sulfur that is removed from a rubber compound by extraction with acetone or acetone-chloroform mixture It includes not only elemental sulfur but sulfur present in soluble compounds

2.27 Tear resistance.—The ratio of the maximum force applied during tear of the specimen to the thickness of the unstretched specimen

2.28 Tensile strength.—The force per unit of the original cross-sectional area of the unstretched specimen which is applied at the time of rupture of the specimen It is calculated by dividing the breaking force in pounds by the cross section of the unstretched specimen in square inches

2.29 Tensile stress.—The force per unit of the original cross-sectional area of the unstretched specimen required to stretch the specimen to a stated elongation It is expressed in pounds of tension force per square inch at the stated elongation

2.30 Tension set.—The elongation remaining after the specimen has been stretched and held at a stated elongation for a definite period of time, then relieved of the force, and allowed to rest for a definite period of time. It is expressed as a percentage of the original distance between bench marks on the unstretched specimen

2.31 Total extract.—The material removed from vulcanized rubber compounds by extracting with a mixture of 32 percent acetone and 68 percent chloroform by volume It is ap-

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proximately equal to the sum of the acetone and chloroform extracts

2.32 Total sulfur.—All the sulfur present in the rubber compound exclusive of that contained in the inorganic sulfides and sulfates that may be present in fillers

2.33 Unsaponifiable acetone extract.—The portion of the acetone extract that is not saponified by a 1 N alcoholic potassium hydroxide solution

2.34 Voltage withstand.—The maximum potential gradient that a material can stand without rupture

2.35 Volume resistivity.—The resistance between two opposite faces of a centimeter cube of the material in absence of surface conduction

2.36 Waxy hydrocarbons.—The portion of the unsaponifiable acetone extract that is removed by extraction with absolute alcohol and that separates from the alcoholic solution on cooling to −5° C (23° F)

3. TERMS APPLICABLE TO SAMPLING, INSPECTION, AND TESTING

3.1 Acceptance number.—A number used in connection with a sampling plan such that if the number of defective test units in the sample for test is less than or equal to this number, the lot shall be accepted, provided that it is acceptable on the basis of visual and dimensional inspection and that the chemical composition as determined on the composite sample meets the specification requirements

3.2 Defective test unit.—A test unit that fails to conform to one or more of the requirements for which it has been tested

3.3 Defective unit.—A unit that fails to conform to one or more of the requirements for which it has been inspected

3.4 Delivery.—The quantity of rubber articles purchased under a single contract that is to be delivered at one specified time

3.5 Inspection.—Inspection refers to examination of the material for visual characteristics and the determination of any dimensional characteristic that does not necessitate destruction of the material

3.6 Lot.—A delivery or a portion of a delivery composed of material of any one type, grade, class, size, or composition A lot is either accepted or rejected as a whole on the basis of inspection and/or test carried out on a portion of the lot

3.7 Lot size.—The number of units of which the lot is composed

3.8 Rejection number.—A number used in connection with a sampling plan such that if the number of defective units or defective test units in the sample taken from the lot is equal to or greater than this number, the lot should be rejected

3.9 Sample.

3.9.1 *Composite sample*—A sample prepared from portions taken from each of the test units constituting the sample for test

3.9.2 *Sample for inspection*—A specified number of units taken from a lot for the purpose of inspection

3.9.3 *Sample for test*—A specified number of test units taken from a lot for the purpose of determining the physical and chemical properties for which requirements are specified

3.10 Sample size.

3.10.1 *Sample size for inspection*—The number of units of which the sample for inspection is composed

3.10.2 *Sample size for test*—The number of test units of which the sample for test is composed

3.11 Specimen.

3.11.1 *Specimen for chemical test*—The portion of a composite sample which is designated for a single chemical determination

3.11.2 *Specimen for physical test*—A test unit or a portion of a test unit designated for a particular physical test or for a dimensional measurement requiring destruction of the material In some cases, a specimen can be used for the determination of two or more physical properties The specimens taken from a single test unit for a given physical test shall so far as practicable be taken from different locations selected at random on the test unit Thus, if the test unit consists of several units, not all specimens shall be taken from the same unit

3.12 Test.—Test refers to the measurement of the physical and chemical properties of the material and to the determination of dimen-

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sional characteristics necessitating destruction of the material

3.13 Test result.

3.13.1 *Test result for chemical tests* —The arithmetic average of the values obtained for a particular chemical test from the required number of specimens taken from a single composite sample

3.13.2 *Test result for physical tests* —A single numerical value calculated by a prescribed procedure from all the measurements for a given physical property carried out on a single test unit In most cases, the test result is an average of the measurements, but there are instances where another function, such as the median, is used

3.14 Test unit.—The total quantity of material necessary to obtain test results for all the physical properties for which requirements are specified, including the portion of material to be used in the preparation of the composite sample for chemical analysis A test unit may consist of a unit, a portion of a unit, or two or more units In no case shall more than one test unit be taken from a single unit

3.15 Unit.—A rubber article in the form in which it is purchased, such as a pneumatic tire, a specified length of hose, a pair of surgical gloves, etc

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SECTION 6

SAMPLING AND ACCEPTANCE PROCEDURES

1. SCOPE

1.1 This section describes the sampling procedures for inspection and testing of rubber products and the procedures for acceptance of lots of rubber products

2. DEFINITION OF TERMS

2.1 Inspection and test.

2.1.1 *Inspection* —Inspection, as used in this standard, refers to examination of the material for visual characteristics and for the determination of any dimensional characteristic that does not necessitate destruction of the material

2.1.2 *Test* —Test, as used in this standard, refers to the measurement of the physical and chemical properties of the material and to the determination of dimensional characteristics necessitating destruction of the material

2.3 Delivery.—The quantity of rubber articles purchased under a single contract that is to be delivered at one specified time.

2.4 Lot.—A delivery or a portion of a delivery composed of material of any one type, grade, class, size, or composition (see 3 2) A lot is either accepted or rejected as a whole on the basis of inspection and/or test carried out on a portion of the lot

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2.5 Unit.—A rubber article in the form in which it is purchased, such as a pneumatic tire, a specified length of hose, a pair of surgical gloves, etc

2.6 Test unit.—The total quantity of material necessary to obtain test results, 2 11 1, for all the physical properties for which requirements are specified, including the portion of material to be used in the preparation of the composite sample for chemical analysis, 2 8 3 A test unit may consist of a unit, a portion of a unit, or two or more units. In no case shall more than one test unit be taken from a single unit

2.7 Specimen.

2.7.1 *Specimen for physical tests* —A test unit or a portion of a test unit designated for a particular physical test, or for a dimensional measurement requiring destruction of a unit

2.7.1.1 In some cases, a specimen can be used for the determination of two or more physical properties

2.7.1.2 The specimens taken from a single test unit for a given physical test shall so far as practicable be taken from different locations, selected at random on the test unit Thus, if the test unit consists of several units, not all specimens shall be taken from the same unit

2.7.2 *Specimen for chemical tests* —That portion of a composite sample, 2 8 3, which is designated for a single chemical determination

2.8 Sample.

2.8.1 *Sample for inspection* —A specified number of units taken from a lot for the purpose of inspection, 2 1 1

2.8.2 *Sample for test* —A specified number of test units taken from a lot for the purpose of determining the physical and chemical properties for which requirements are specified

2.8.3 *Composite sample* —For the purpose of determining the chemical properties, a portion of rubber shall be taken from each of the test units constituting the final sample for test The portions shall be composited as specified in method 1211 The total portion of rubber thus obtained is a *composite sample* Specimens for chemical tests, 2 7 2, are portions taken from the composite sample in accordance with the instructions given in the relevant test methods

2.9 Lot size.—The number of units, 2 5, of which the lot is composed

2.10 Sample size.

2.10.1 *Sample size for inspection* —The number of units, 2 5, of which the sample for inspection is composed

2.10.2 *Sample size for test* —The number of test units, 2 6, of which the sample for test is composed

2.11 Test result.

2.11.1 *Test result for physical tests* —A single numerical value calculated by a prescribed procedure from all the measurements for a given physical property carried out on a single test unit In most cases the test result is an average of the measurements, but there

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are instances where another function such as the median is used.

2.11.2 *Test result for chemical tests* —The arithmetic average of the values, obtained for a particular chemical test, of the required number of specimens, 2 7 2, taken from a single composite sample

2.11.3 Whereas a test result for a physical test, 2 11 1, is representative of a single test unit, a test result for a chemical test, 2 11 2, is representative of an entire sample for test

2.12 Defective unit.—A unit that fails to conform to one or more of the requirements for which it has been inspected, 2 1 1

2.13 Defective test unit.—A test unit that fails to conform to one or more of the requirements for which it has been tested, 2 1 2

2.14 Acceptance of lot.—The approval of a lot as conforming to contract and/or specification

2.15 Rejection of lot.—The disapproval of a lot as not conforming to contract and/or specification

2.16 Acceptance number.—A number used in connection with a sampling plan such that if the number of defective test units in the sample for test is less than or equal to this number, the lot should be accepted, provided that it is acceptable on the basis of inspection, 2 1 1, and that the chemical composition as determined on the composite sample, 2 8 3, meets the specification requirements

2.17 Rejection number.—A number used in connection with a sampling plan such that if the number of defective units or defective test units in the sample taken from the lot is equal to or greater than this number, the lot should be rejected

3. GENERAL SAMPLING PROCEDURES

3.1 Scope.—The term "sampling" is used throughout this standard to signify lot-by-lot acceptance sampling, the essential feature of which is that the contractor submits his product grouped into lots each of which is accepted or rejected in its entirety on the basis of the performance of one or more samples taken from it

3.2 Formation of lots.—For purposes of sampling, the manufacturer shall submit the delivery in lots, each of which is composed of material produced under essentially the same conditions, 2 4 Whereas no exact instructions for the formation of lots can be given that will cover all cases, the following rules shall be adhered to as far as is practicable

3.2.1 The material in any one lot shall be produced—

a From a single batch of raw materials or from component parts obtained from a single source

b By a single production method

c From a single production line

d During a single production shift

e From a single cure

3.2.2 The quantity of material included in a lot shall be as large as possible, subject to—

a The considerations immediately preceding,

b Limitation, if any, mentioned in the relevant detail specification

c The condition that all the material in the lot shall be reasonably accessible to the inspector for sampling purposes

3.2.3 No material shall be added to or removed from a lot by the contractor once the inspector has begun to draw samples from the lot However, if the inspector notices some damaged units (by accident or other circumstances not related to the manufacturing process) before beginning to sample, and if the damage is of such character that it is apparent to him that no other similarly damaged units are distributed throughout the lot, then the inspector shall have this damaged material removed from the lot before drawing a sample

3.3 Place of sampling.—Materials may be sampled, inspected, and/or tested either at the place of manufacture or at the point of delivery If the inspection and tests are to be made at the point of manufacture, the manufacturer shall provide a place for conducting the test, the necessary labor, equipment, and other facilities, and the manufacturer and/or the contractor shall notify the purchasing officer sufficiently in advance of the completion of manufacture of material to permit arrangement for inspection Unless otherwise specified in the invitation for bids, purchases by the Departments of the Navy, Army and Air Force (Department of Defense) shall be inspected and tested at the place of manufacture, whereas purchases by other departments

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and independent establishments of the Government shall be inspected and samples for test taken at the point of delivery

3.4 Method of sampling.—Units for inspection or for test shall be taken from the lot in accordance with a predetermined procedure designed first, to eliminate deliberate selection or omission of any particular unit by the inspector and second, to give each unit of the lot an equal or nearly equal chance of being included in the sample To this end, the method of sampling should conform as far as practicable to the following rules

Rule 1 Samples shall be taken proportionally to sublot sizes —When a lot is, for practical reasons, not formed according to all of the principles enumerated in 3 2, but is made up of sublots each of which is formed according to these principles, then the units of a sample from the lot shall be taken from each sublot in number proportional to the size of the sublot

Rule 2 Samples shall be taken from locations scattered throughout the lot —The object of this rule is to insure that the units taken for a sample shall not all be taken from the same portion of the lot, such as a single carton, layer, tier, etc

Rule 3 Samples shall be taken "blind"— Units shall be taken without regard to quality, that is, no conscious effort shall be made to include or exclude a particular unit by virtue of its apparent quality, except in the case of damaged units, under the conditions stated in 3 2 3

The rules for size of samples are given in 4 2 3 1 2 and 4 2 3 2 2

3.5 Substitute sample.—Specimens for test shall be obtained if possible from the articles to be tested In case it is not practicable to obtain suitable specimens from the test units of a lot, the manufacturer shall, if required to do so by the detail specification, or upon written request, furnish a substitute sample in the form of a piece of rubber having dimensions appropriate to the tests required, and an affidavit to the effect that this substitute sample is of the same material and equivalent cure as that used in the manufacture of the material in the lot

3.6 Unfrictioned fabric.—Whenever tests of the fabric from rubber articles are required in the detail specification, the manufacturer shall submit at least one square foot of unfrictioned fabric from each contract or order, and shall furnish an affidavit to the effect that the fabric is the same as the material used in the manufacture of the finished article However, if breaking strength tests of the fabric are required, the piece shall be one yard long and the full width of the fabric

3.7 Care of samples.—Between the time of taking a sample and its inspection and/or testing, the sample shall not be subjected to any conditions which may alter the quality of the material Thus, samples or specimens for tests shall not be heated, immersed in water, or subjected to any mechanical or chemical treatment prior to test except as specifically described herein or in the specification covering the article being tested Particular attention is necessary to avoid previous stretching Samples of materials with a high rubber content, such as surgeons' rubber gloves, shall not be exposed to direct sunlight or strong artificial light before or during accelerated aging in the oven or bomb

4. SAMPLING PLANS AND THEIR GENERAL CHARACTERISTICS

4.1 Scope.—In this section, consideration is given to criteria for acceptance or rejection of a lot on the basis of the performance of one or more samples taken from the lot

4.2 Sampling plans.

4.2.1 *Purpose* —A sampling plan aims to accept frequently lots that contain only a small proportion of nonconforming units and to reject frequently lots that contain a large proportion of nonconforming units

4.2.2 *Protection features*

4.2.2.1 *Protection to the contractor* —One essential feature of a sampling plan is the protection that it affords to the contractor, in the sense that, in the long run, lots which contain a certain small proportion of nonconforming units shall not be rejected in more than a specified percentage of cases in which they are submitted

4.2.2.2 *Protection to the consumer* —Another essential feature of a sampling plan is the protection that it affords to the consumer, in that

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it limits his risk of accepting a lot containing more than a certain proportion of nonconforming units

4.2.3 *Sampling for inspection and sampling for test* —The inspector shall first ascertain that the conditions of 3 2 1 and 3 2 2 are satisfied and shall act in accordance with 3 2 3 The sampling procedure adopted in this standard consists of first obtaining a sample for inspection, 2 1 1, and then in using all or a portion of that sample for tests, 2 1 2

4.2.3.1 *Sampling for inspection*

4.2.3.1.1 *Sampling procedure* —The sample required for inspection shall be taken from the lot and stored in accordance with 3 4, 3 5, 3 6, and 3 7

4.2.3.1.2 *Sample size* —Unless otherwise specified in the detail specification, the number of units to be taken from a lot for the purpose of inspection shall depend on the size of the lot, in accordance with table I

TABLE I —*Sampling for inspection*

Lot size	Sample size	Rejection number R
Units	*Units*	*Units*
14 or under	All	1
15 to 110	15	2
111 to 300	25	2
301 to 500	35	3
501 to 800	50	4
801 to 1,300	75	5
1,301 and over	110	7

4.2.3.1.3 *Instructions for use of table I* —The inspector shall locate in the first column of the table the size of the lot to be sampled The sampling plan to be used is the one given on the line corresponding to this lot size All units of the sample shall be inspected The number of nonconforming units shall be recorded The lot should be rejected and should not be tested for physical or chemical characteristics if the number of defective units in the sample equals or exceeds the "Rejection number R" If the number of defective units in the sample is less than the "Rejection number R," the lot should be sampled for physical and chemical tests, to determine its acceptability

4.2.3.1.3.1 *Dimensional measurements* —Inspection includes both visual examination and all specified dimensional measurements that do not require destruction of the material Therefore, in general, all units in the sample for inspection should be subjected to these measurements However, in certain cases, the importance of certain dimensional requirements may not be sufficient to warrant the time that would be spent by the inspector in measuring all units in the sample In such cases, it will be permissible, at the discretion of the inspector, to carry out the dimensional measurements on a portion of the sample for inspection The size of this portion shall not be less than 20 percent of the sample for inspection, and the units on which the measurements are made shall be taken at random from the inspection sample If one or more defective units are found in this portion, the complete inspection of the entire sample for inspection is mandatory

4.2.3.2 *Sampling for physical and chemical tests.*

4.2.3.2.1 *Sampling procedure* —The sample required for testing the lot for physical and chemical characteristics shall be taken from the *sample for inspection* in accordance with a random process. For this purpose, *each unit* of the inspection sample, including defective units, 2 12, shall be given an equal chance of being included in the sample for tests Should the sample for inspection be too small to provide the required number of test units (table II), additional test units shall be taken from the lot, using the same precautions as observed for obtaining the sample for inspection

4.2.3.2.2 *Groups of sampling plans for test* — In this standard several sampling plans for test are presented. They are classified into three groups according to the severity of requirements they impose on the over-all quality of the lot, and within each group according to the size of the lot submitted. Table II lists these plans in tabular form The selection of a sampling plan in any given case depends largely on the over-all quality required For guidance in such a selection, the protection features afforded by the plans in each of the three groups are tabulated in table III

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TABLE II —*Sampling for test*

Group	Lot size (number of units)	First sample			Second sample	Combined samples		
		Number of test units	Acceptance number	Rejection number	Number of test units	Number of test units	Acceptance number	Rejection number
A	100 to 1,000*.....	10	0	1	.....	.....	.....	.....
	1,001 to 5,000.....	20	0	2	40	60	1	2
	5,001 to 20,000.....	30	0	2	30	60	1	2
	Over 20,000.....	40	0	2	40	80	1	2
B	50 to 1,000*.....	5	0	1	.....	.....	.....	.....
	1,001 to 5,000.....	10	0	2	20	30	1	2
	5,001 to 20,000.....	15	0	2	15	30	1	2
	Over 20,000.....	20	0	2	20	40	1	2
C	50 to 200*.....	2	0	2	4	6	1	2
	201 to 500.....	3	0	2	3	6	1	2
	501 to 1,000.....	4	0	2	4	8	1	2
	1,001 to 5,000.....	5	0	2	5	10	1	2
	5,001 to 10,000.....	8	0	3	8	16	2	3
	Over 10,000.....	15	1	3	15	30	3	4

*For smaller lots, see 4 2 3 2 2 2

4.2.3.2.2.1 *Instructions for use of table II* — The inspector shall locate in the second column of the table under the group of plans specified in the detail specification, the size of the lot to be sampled The sampling plan to be used is the one given on the line corresponding to this lot size The inspector shall then proceed as follows

a. A first sample shall be taken consisting of a number of test units equal to the number in the column headed "First sample, number of test units."

b. Each of the test units of this sample shall be subjected to all of the required physical tests

c The number of defective test units thus found shall be recorded

d. The lot is considered to meet the requirements relating to physical tests if this number does not exceed the number given under the heading "First sample, acceptance number"

The lot shall be considered as having failed to meet the requirements of the detail specification if the number of defective test units in the first sample equals or exceeds the number given under the heading "First sample, rejection number"

If the number of defective test units in the first sample exceeds the acceptance number for the first sample, but is less than the rejection number for the first sample, the inspector shall proceed to *e*

e A second sample shall be taken, consisting of a number of test units equal to the number given under the heading. "Second sample, number of test units"

f Each of the test units of this sample shall be subjected to all of the required physical tests

g. The number of defective test units found in the second sample shall be recorded, and this number shall be added to the number of defective test units which were found in the first sample

h The lot is considered to meet the requirements relating to physical tests if the total number of defective test units found in the

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two samples does not exceed the number given under the heading "Combined samples, acceptance number"

c The lot shall be considered as having failed to meet the requirements of the detail specification and should be rejected, if the total number of defective test units found in the two samples is equal to or greater than the number given under the heading. "Combined samples, rejection number"

TABLE III *Protection afforded by sampling plans*

Protection features	Group of plans		
	A	B	C
Protection to producer			
The lot will be accepted at least 95 percent of the time if the proportion of the defective items contained in it does not exceed	*Percent* 0 5	*Percent* 1	*Percent* 5
Protection to consumer			
The lot will be rejected at least 95 percent of the time if the proportion of defective items contained in it reaches [1]	8 to 26	15 to 46	28 to 60.

[1] The larger of the 2 values applies to lots of less than 1,000 units in the case of groups A and B and to lots of less than 500 units in the case of group C As the lot size increases, the protection to the consumer increases For the larger lots listed in table II, in all 3 groups, the proportion of defective items that will be accepted 5 percent of the time is given by the smaller of the 2 values tabulated

4.2.3.2.2.2 *Small lots*—For lots containing less than 100 units, in cases in which group A plan is called for in the detail specification, or less than 50 units, in the case of group B or C plans, the amount of testing required in order to obtain degrees of protection comparable to those obtained by application of these sampling plans, in general, is excessive from an economical point of view On the other hand, there exists, in general, reasons for special vigilance in the case of small lots When originating from large producers, small lots may often be made up of residual units from several batches, while manufacturers with a rather small output may have more difficulty than larger firms in keeping their production under control Moreover, experience indicates that cases are not uncommon in which each unit of a small lot shows considerable departure from the requirements of the detail specification In deciding with respect to the desirability of sampling a small lot and with respect to the amount of testing to be performed on such lot, one should therefore be guided by—

a Economical consideration, as already stated

b Whatever knowledge the purchaser may have from past experience or from reliable information regarding the quality of the goods furnished by the manufacturer or the contractor in question If such experience or information regarding small lots is very favorable to the manufacturer or the contractor, then a small lot from the same origin may, in some cases, be considered as not necessitating the destruction of any of its units for test purposes On the other hand, if, on account of its origin the quality of a lot is subject to doubt, one or a few of its units should be tested Guiding consideration for the decision regarding acceptance or rejection of the lot will then be obtained by focusing one's attention not only on the number of departures from the specification observed in the sample but also on the degree of these departures, as evidenced by comparison of the value obtained from a given test and the corresponding specified value

5. REFERENCING

5.1 The sampling plans provided in this standard are given for convenient reference in preparing detail specifications and contracts The three groups of plans for test differ

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with respect to the protection they afford to the contractor and the consumer, and with respect to the amount of testing required for lots of a given size It is essential that every detail specification and/or contract that references these plans indicate specifically the particular group of plans that is to be used This information, together with the size of the lot, enables the inspector to determine, by reference to table II, the specific sampling plan to be used for testing each lot Detailed instructions for using this table are given immediately following the table

6. ACCEPTANCE AND REJECTION PROCEDURES

6.1 A lot is considered to meet the requirement of the detail specification, and should be accepted, if *all* of the following conditions are met

a The lot is acceptable as a result of inspection

b The lot is found to meet the requirements for physical tests under application of the specified sampling plan in table II

c The results of the chemical determinations made on the composite sample satisfy the requirements for chemical composition specified in the detail specification

6.2 A lot is considered as having failed to meet the requirements of the detail specification if one or more of the conditions in *a*, *b*, or *c* listed above are not fulfilled.

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SECTION 7

TEMPERATURE AND HUMIDITY OF CONDITIONING ROOM

1. CONDITIONING

1.1 **Standard atmosphere.**— Unless otherwise specified in the detail specification, all physical tests of rubber shall be made at a room temperature of 27°±5° C (80°±9° F), except in the case of dispute or disagreement between laboratories

1.2 **Dispute testing.**— In the case of dispute or disagreement between laboratories, unless otherwise specified, the specimen shall be conditioned in an atmosphere of 50±4 percent relative humidity and at a temperature of 23°±1 1° C (73.5°±2° F.). It is intended that the laboratory be maintained as closely as possible to 23° C and a relative humidity of 50 percent The tolerances indicated are meant to cover variations from the specified conditions which may occur at different locations in the room

1.3 **Time of conditioning.**— Unless otherwise specified in the detail specification, the specimens shall be conditioned for not less than 3 hours before being tested, and shall be tested in the same atmosphere

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SECTION 8

NOTES

1. DETAIL SPECIFICATIONS

1.1 Detail specifications for rubber and rubber products should specify which method shall be used for a particular test, such as method 8011 or 8021 for friction, method 3021, 3031, 3041, or 3051 for hardness, etc

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SEPARATION OF RUBBER FROM OTHER MATERIALS

1. SCOPE

1.1 This method is intended for use in removing the rubber from fabric or other material when necessary before conducting tests

2. MATERIAL

2.1 The piece for removal of fabric or other material shall be a portion of the test unit of sufficient size to permit the preparation of a specimen of the required size for test

3. APPARATUS

3.1 The apparatus shall be as follows·

3.1.1 Knife or other instrument for cutting the material

3.1.2 Iso-octane, as described in method 6001, medium No 4

4. PROCEDURE

4.1 If it is practicable, the separation shall be performed without the use of solvents The rubber shall be gripped as near the point of separation from the other material as possible and shall be removed slowly with a minimum amount of stretching If it is necessary to use a solvent in the separating operation, iso-octane shall be used If iso-octane is used, the rubber after removal of the fabric shall be placed so as to permit free evaporation of the solvent from all parts of its surface and shall be permitted to rest at least one hour before being tested

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

BUFFING

1. SCOPE

1.1 This method is intended for use in removing unevenness of surface from rubber which would interfere with measurement of thickness or subsequent tests, such as fabric impressions or corrugations caused by contact with fabric components or cloth wraps used for curing It is also applicable to reducing the thickness of materials where necessary before testing, the removal of glaze or skin coats from rubber where it may interfere with further examination or tests, and the removal of fabric backings from materials such as rubber-lined fire-hose

2. PIECE

2.1 Unless otherwise specified in the detail specification or applicable method of test, the piece for buffing shall consist of either the specimen for physical test as described in the applicable test method or a portion of the test unit of sufficient size and shape to permit the preparation of the specimen for physical test

3. APPARATUS

3.1 The apparatus shall consist of a grinder equipped with a motor-driven abrasive wheel of approximately 30 grit, 5 inches in diameter, and revolving at a speed of 2,500 to 3,500 revolutions per minute Larger diameter wheels may be used provided the speed is adjusted to give a surface speed equivalent to a 5-inch wheel operating at 2,500 to 3,500 revolutions per minute The grinder shall also be equipped with a slow feed in order that very little of the material will be removed at one cut

4. PROCEDURE

4.1 The face of the abrasive wheel shall be sharp during the buffing operation The abrasive wheel shall be adjusted to a speed of from 2,500 to 3,500 revolutions per minute when a 5-inch wheel is used. A very small amount of rubber or fabric shall be removed from the piece at one time in order to avoid damage to the rubber by overheating Buffing shall be carried just to the point where unevenness of gage disappears, except in the case of thick materials where it is necessary to reduce the thickness before testing When it is necessary to reduce the thickness before testing, it may be desirable to slice the material to the approximate required thickness with a sharp instrument before buffing If the material has a fabric backing, the piece shall be buffed until the fabric just disappears The final buffed surface shall be finished uniformly and smoothly The piece shall be permitted to rest at least 30 minutes between buffing and testing the specimen

4.2 Pieces from which specimens are to be taken for tension tests shall be buffed in the strip form before specimens are cut with a die

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COMPOSITE SAMPLE FOR CHEMICAL ANALYSIS

1. SCOPE

1.1 This method is intended for use in preparing a representative composite sample of the rubber compound from the test units of the sample for test in a form suitable for chemical analysis

2. COMPOSITE SAMPLE

2.1 The composite sample shall consist of equal portions taken from each test unit of the sample for test The composite sample shall weigh not less than 10 grams

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An airtight container

3.1.2 Holland cloth or equivalent

3.1.3 A laboratory mill

3.1.4 A pair of scissors or other cutting instrument

3.1.5 Sieves, U S Standard Sieve Series No 14 and 30, conforming to Federal Specification RR–S–366

3.1.6 A balance that will weigh to an accuracy of 0 5 gram

4. PROCEDURE

4.1 Each portion from each test unit shall be cleaned of any foreign material If it is required to determine specific gravity, a subsample shall be selected from the cleaned portions in accordance with method 14011 or 14021 The remaining portions designated for the composite sample and the remainder of each portion from which a specimen was taken for specific gravity test shall be weighed on a balance and adjusted to within 0 5 gram of the same weight All portions taken shall be ground and thoroughly blended by passing through a clean, cold, laboratory mill with the rollers set close Usually passing the material through the mill 4 or 5 times will be sufficient Vulcanized rubber will come from the mill in the form of a coarse powder that will form a porous sheet if milling is continued Crude, reclaimed, or unvulcanized rubber shall be sheeted to a thickness of no greater than 0 5 mm and then rolled in Holland or other cloth to prevent sticking

4.2 In the absence of a laboratory mill, the material may be cut with scissors to pass through a No 14 sieve

4.3 Pieces of hard rubber shall be reduced to powder by filing The powder shall be cleaned with a magnet and passed through a No 30 sieve The powder remaining on the sieve shall be reduced in size until all the material passes through the sieve

4.4 Each test unit of rubber cement shall be thoroughly mixed and portions of each test unit combined and thoroughly mixed The mixture shall be evaporated to dryness in a vacuum at a temperature not to exceed 30° C (86° F)

4.5 The composite sample shall be stored in a suitable airtight container in a dark place

4.6 Specimens for various analyses in groups 15000 and 16000 shall be taken from this composite sample

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GEOMETRICAL MEASUREMENTS, GENERAL

1. SCOPE

1.1 This group of methods is intended for use in determining the geometrical properties of rubber, rubber products, and specimens for physical tests Geometrical measurements are made for one of the following three purposes

1.1.1 To determine conformance with specification requirements regarding geometrical properties of whole units, such as the size of surgeons' gloves In such a case, determination of the geometrical property is considered as an inspection, and measurements are made on one or more locations of each unit in the sample for inspection

1.1.2 To determine conformance with specification requirements regarding particular functional parts of units, such as the thickness of the soles of rubber boots In such a case, determination of the geometrical property is considered as a test since it necessitates destruction of the material, and measurements are made on one or more specimens taken from each test unit

1.1.3 To determine geometrical values that are necessary in the calculation of test results for a physical property for which requirements are specified, such as the thickness of a specimen for tensile strength In such a case, measurements are made on the specimens on which the physical property in question is measured and the geometrical values found are only an adjunct to the determination of the physical property

1.2 Methods are described for measuring thickness, width, diameter, circumference, and length Some methods are of general application and some are applicable only to limited types of units, test units, or specimens for physical tests The method to be used for the measurement shall be as specified in the detail specification or appropriate test method Geometrical measurements shall not be made at places where any damage is apparent that may interfere with the geometrical property

2. SPECIMEN

2.1 Units or test units.—The specimen shall consist of a unit or test unit, or a portion of a unit or test unit, 1 1 1 and 1 1 2

2.1.1 It is not practicable to specify the size and shape of specimens for geometrical properties in all cases In general, the specimen shall consist of the largest single piece of material in the unit or in the specified part of the test unit which it is practicable to measure For example, in determining the height of rubber boots, the unit consists of a pair of boots and the specimen of a single boot On the other hand, in determining the thickness of fingertips of surgical gloves, the test unit may consist of two pairs of gloves, but the specimen is a single fingertip

2.2 Physical test.—The specimen shall be as described in the applicable test method, 1 1 3

3. THICKNESS AND WIDTH

3.1 Methods are described for determining the average, minimum, and maximum thickness and width of the unit, test unit, and specimen for physical test Minimum thickness and minimum width shall be interpreted as meaning thickness at the thinnest place and width at the narrowest place, respectively Maximum thickness and maximum width shall be interpreted as meaning thickness at the thickest place and width at the widest place, respectively

4. DIAMETER

4.1 In some of the methods the diameter is measured directly while in other methods the diameter is calculated from other measurements

5. CIRCUMFERENCE

5.1 In some of the methods the circumference is measured directly while in other methods the circumference is calculated from other measurements

6. THICKNESS OF RUBBER ARTICLES

6.1 When the thickness of a rubber part which has a rough surface is specified in the detail specification, as for example the tube or cover of hose, cover of belting, etc , measurement shall be made of a specimen that has been buffed just to the point where the fabric or other impressions disappear

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THICKNESS, MICROMETER, FLAT FOOT

1. SCOPE

1.1 This method is intended for use in determining the thickness of materials with flat surfaces such as sheet materials, coated fabrics, and fabricated rubber articles It is applicable to the determination of thickness of units, test units, or specimens for physical tests

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable

3. APPARATUS

3.1 The apparatus shall consist of a dial micrometer having a flat anvil not less than 0 25 inch in diameter and a flat presser foot 0 25±0 01 inch in diameter which exerts a total force of 3 0±0 1 ounce on the specimen, the force being applied by means of a weight The surfaces of the anvil and presser foot shall be parallel to within 0 0001 inch The dial shall be graduated to read to 0 001 inch

4. PROCEDURE

4.1 **Buffing.**—If the material has as uneven surface, such as fabric impressions or corrugations that may interfere with the test, for example, the tube or cover of hose, the material shall be buffed as described in method 1111

4.2 **Thickness measurement.**—The specimen shall be placed on the anvil of the micrometer and the presser foot lowered gently until it contacts the surface of the specimen The thickness shall be read from the dial to the nearest 0 001 inch If more than one measurement is made on the specimen, the measurements shall be equally spaced over the specimen

4.3 **Average thickness.**

4.3.1 *Unit or test unit* —Unless otherwise specified in the detail specification, a total of 15 measurements shall be made on the unit or test unit The measurements shall be made as described in 4 2, and if made on more than one specimen they shall be equally distributed as nearly as practicable among all the specimens from the unit or test unit

4.3.2 *Specimen for physical test* —Unless otherwise specified in the applicable test method, five measurements shall be made on the specimen The measurements shall be made as described in 4 2

4.4 **Minimum thickness.**

4.4.1 *Unit or test unit* —The unit or test unit shall be examined for the thinnest place If the thinnest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 2 If no thinnest place is found, measurements shall be made as described in 4 3 1 and the smallest value obtained shall be the minimum thickness of the unit or test unit

4.4.2 *Specimen for physical test* —The specimen shall be examined for the thinnest place If the thinnest place is found, a measurement shall be made at that place as described in 4 2 If no thinnest place is found, measurements shall be made as described in 4 3 2 and the smallest value obtained shall be the minimum thickness of the specimen

4.5 **Maximum thickness.**

4.5.1 *Unit or test unit* —The unit or test unit shall be examined for the thickest place If the thickest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 2 If no thickest place is found, measurements shall be made as described in 4 3 1 and the largest value obtained shall be the maximum thickness of the unit or test unit

4.5.2 *Specimen for physical test* —The specimen shall be examined for the thickest place If the thickest place is found, a measurement shall be made at that place as described in 4 2 If no thickest place is found, measurements shall be made as described in 4 3 2, and the largest value obtained shall be the maximum thickness of the specimen

5. RESULTS

5.1 **Unit or test unit.**

5.1.1 *Average thickness* —The average thickness of the unit or test unit shall be the average of all the values obtained, 4 3 1

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5.1.2 *Minimum thickness*

5.1.2.1 If a thinnest place is found in the unit or test unit, the minimum thickness of the unit or test unit shall be the thickness obtained at that place

5.1.2.2 If no thinnest place is found in the unit or test unit, the minimum thickness of the unit or test unit shall be the smallest of all of the values obtained when measurements are made as described in 4 3 1

5.1.3 *Maximum thickness*

5.1.3.1 If a thickest place is found in the unit or test unit, the maximum thickness of the unit or test unit shall be the thickness obtained at that place

5.1.3.2 If no thickest place is found in the unit or test unit, the maximum thickness of the unit or test unit shall be the largest of all of the values obtained when measurements are made as described in 4 3.1

5.1.4 The average, minimum, and maximum thicknesses of the unit or test unit shall be recorded to the nearest 0 001 inch.

5.2 **Specimen for physical test.**—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method

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THICKNESS, MICROMETER, SPHERICAL FOOT

1. SCOPE

1.1 This method is intended for use in determining the thickness of units, test units, or specimens for physical tests, with irregular or curved surfaces It is particularly applicable to compression set specimens and specimens, units, or test units with concave surfaces

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable

3. APPARATUS

3.1 The apparatus shall consist of a dial micrometer having a raised anvil ⅜ inch in diameter and a spherical presser foot 0 25±0 01 inch in diameter which exerts a total force of 3 0±0 1 ounce on the specimen, the force being applied by means of a weight The dial shall be graduated to read to 0 001 inch

4. PROCEDURE

4.1 The procedure shall be as described in method 2011

5. RESULTS

5.1 The results shall be as described in method 2011

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THICKNESS, OPTICAL

1. SCOPE

1.1 This method is intended for use in determining the thickness of units, test units, or specimens for physical tests, especially when minimum or maximum thickness is required. This method is recommended for determining the thickness of thin materials and walls of cylindrical layers of rubber such as tubing It is also applicable to materials referred to in methods 2011 and 2021 if the thickness is not too great This method is more accurate but more time consuming than methods 2011 and 2021

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable For units or test units with flat surfaces, the specimen shall be narrow and short, with full cross section cut perpendicular to the principal surfaces of the material For units or test units with cylindrical layers, the specimen shall have thin walls and shall be short, with full cross section cut perpendicular to the length axis of the cylinder

3. APPARATUS

3.1 The apparatus shall consist of a microscope equipped with devices capable of making measurements accurate to approximately 0 0002 inch

3.1.1 A suitable instrument is a microscope equipped with a 6- or 8-power eyepiece, an objective of 32-mm focal length, a mechanical stage, and a ruled glass disk or ocular micrometer

4. PROCEDURE

4.1 **Buffing.**—If the material has an uneven surface, such as fabric impressions or corrugations, that may interfere with the test, for example the tube or cover of hose, the material shall be buffed as described in method 1111

4.2 **Thickness measurement.**

4.2.1 *Flat surfaces* —If the specimen is so thin that it will not support itself edgewise, it may be mounted in a slit cut in a cork or other mounting device The ruled glass disk shall be placed on the cut surface of the cross section with the ruled surface in contact with the material The thickness, if in the field of view, shall be read directly from the rulings on the disk to the nearest 0 0002 inch If the thickness is such that it does not all lie in the field of view, the specimen shall be moved by means of the mechanical stage until the complete section thickness passes through the field The rulings on the disk passed over during the movement shall be counted and the distance recorded to the nearest 0 0002 inch When the scale is used in the eyepiece, the ruled glass disk or ocular micrometer shall be placed in the ocular and calibrated The specimen shall be placed on the stage with the surface to be measured perpendicular to the optical axis of the microscope so as to expose the full thickness The microscope shall be focused on the specimen and the thickness determined by counting the divisions of the ruled disk in the eyepiece that cover the distance from one edge of the specimen to the other and the distance recorded to the nearest 0 0002 inch If more than one measurement is made on a specimen, the measurements shall be equally spaced over the specimen

4.2.2 *Cylindrical layers* —Thickness measurements of cylindrical layers shall be made as described in 4 2 1, except that measurements shall be made of the wall thickness

4.3 **Average thickness.**

4.3.1 *Unit or test unit* —Unless otherwise specified in the detail specification, a total of six measurements shall be made on the unit or test unit The measurements shall be made as described in 4 2, and if made on more than one specimen they shall be equally distributed as nearly as practicable among all of the specimens from the unit or test unit

4.3.2 *Specimen for physical test* —Unless otherwise specified in the applicable test method, three measurements shall be made on the specimen The measurements shall be made as described in 4 2

4.4 **Minimum thickness.**

4.4.1 *Unit or test unit* —The unit or test unit shall be examined for the thinnest place

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If the thinnest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 2 If no thinnest place is found, measurements shall be made as described in 4 3 1, and the smallest value obtained shall be the minimum thickness of the unit or test unit

4.4.2 *Specimen for physical test* —The specimen shall be examined for the thinnest place If the thinnest place is found, a measurement shall be made at that place as described in 4 2 If no thinnest place is found, measurements shall be made as described in 4 3 2, and the smallest value obtained shall be the minimum thickness of the specimen

4.5 Maximum thickness.

4.5.1 *Unit or test unit* —The unit or test unit shall be examined for the thickest place If the thickest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 2 If no thickest place is found, measurements shall be made as described in 4.3 1, and the largest value obtained shall be the maximum thickness of the unit or test unit.

4.5.2 *Specimen for physical test* —The specimen shall be examined for the thickest place If the thickest place is found, a measurement shall be made at that place as described in 4 2 If no thickest place is found, measurements shall be made as described in 4 3.2 and the largest value obtained shall be the maximum thickness of the specimen

5. RESULTS

5.1 The results shall be as described in method 2011, except that the average, minimum, and maximum thickness of the unit or test unit shall be recorded to the nearest 0 0002 inch

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THICKNESS, MAGNETIC GAGE

1. SCOPE

1.1 This method is intended for use in determining the thickness of units, test units, or specimens for physical test with rubber coatings which adhere to the surface of mild steel or other magnetic material If the rubber coating does not adhere to the metal backing, the thickness of the intervening air gap will also be measured as rubber coating, thus giving a high value The indication of the gage will depend on the thickness of the rubber upon which the gage head is placed and to some extent upon the thickness and the magnetic qualities of the metal backing

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A magnetic thickness gage for measuring thicknesses up to 0 1 inch A suitable instrument is shown in figure 2041

3.1.2 *Thickness* —Standards 0 001, 0 003, 0 030, and 0 060 inch in thickness

3.1.3 Strips of uncoated metal with the same thickness and magnetic properties as those of the backing on the material to be tested

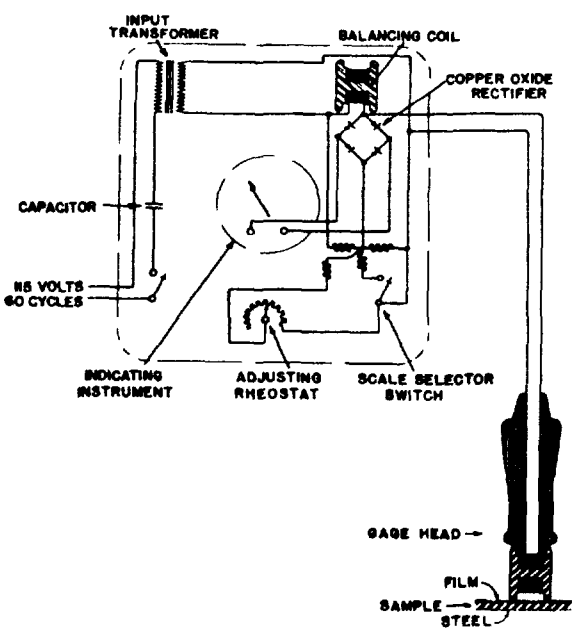

FIGURE 2041 —Thickness gage

4. PROCEDURE

4.1 Adjustment of instrument.—The instrument shall be connected to a source of current and allowed to warm up for at least 15 minutes The proper scale range for the estimated thickness shall be selected by means of the scale-selector switch The standard, 3 1 2, with thickness nearest to that of the coating to be tested, shall be placed over a strip of uncoated metal of the same thickness and magnetic properties as those of the backing on the specimen to be tested The gage head of the instrument shall be placed on the thickness standard so that both disks rest lightly on the standard The rheostat shall be adjusted until the pointer on the scale of the instrument indicates a thickness corresponding to that of the thickness standard

4.1.1 If a metal strip with properties the same as those of the backing on the specimen to be tested is not available, the instrument shall be adjusted as follows The proper scale range shall be selected The gage head shall be placed on the coating of the specimen to be tested and the reading on the scale of the instrument noted A thickness standard, 3 1 2, shall be inserted between the gage head and the specimen The rheostat shall be adjusted until the scale reading is equal to the original reading plus the reading of the thickness standard The operation shall be repeated two or three times to insure that the instrument is adjusted properly

4.2 Thickness determination.—The gage head of the instrument shall be placed on the specimen so that both disks rest lightly on the rubber coating The thickness of the coating shall be read directly from the scale All specimens shall be measured at a distance of at least ½ inch from the edge of the specimen In other respects, the procedure shall be as described in method 2011

5. RESULTS

5.1 The results shall be as described in method 2011

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WIDTH, NARROW UNITS

1. SCOPE

1.1 This method is intended for use in determining the width of narrow units, test units, or specimens for physical tests Two types of apparatus are permitted Use of the apparatus described in 3 1 1 is recommended in case of dispute as to which type of apparatus to use and where high accuracy is desired This method is more accurate and more time consuming than method 2121

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable For units or test units, the specimen shall be narrow with full width exposed

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in method 2031

3.1.2 Any apparatus which will measure the width to within ±0 001 inch and which can be used in such a manner that compression of the specimen is negligible

4. PROCEDURE

4.1 Apparatus in 3.1.1.

4.1.1 Width measurement.—When the apparatus described in method 2031 is used, the ruled glass disk shall be placed on the flat surface of the specimen with the ruled surface in contact with the material The width, if in the field of view, shall be read directly from the rulings on the disk to the nearest 0 0002 inch If the width is such that it does not all lie in the field of view, the specimen shall be moved by means of the mechanical stage until the complete width passes through the field The rulings on the disk passed over during the movement shall be counted and the distance recorded to the nearest 0 0002 inch When the scale is used in the eyepiece, the ruled glass disk or ocular micrometer shall be placed in the ocular and calibrated The specimen shall be placed on the stage with the surface to be measured perpendicular to the optical axis of the microscope so as to expose the full width The microscope shall be focused on the specimen and the width determined by counting the divisions of the ruled disk in the eyepiece that cover the distance from one edge of the specimen to the other and the distance recorded to the nearest 0 0002 inch If more than one measurement is made on the specimen the measurements shall be equally spaced over the specimen

4.1.2 Average width.

4.1.2.1 *Unit or test unit*—Unless otherwise specified in the detail specification, a total of six measurements shall be made on the unit or test unit The measurements shall be made as described in 4.1.1, and if made on more than one specimen they shall be equally distributed as nearly as practicable among all of the specimens from the unit or test unit

4.1.2.2 *Specimen for physical test*—Unless otherwise specified in the applicable test method, three measurements shall be made on the specimen The measurements shall be made as described in 4 1 1

4.1.3 Minimum width.

4.1.3.1 *Unit or test unit*—The unit or test unit shall be examined for the narrowest place If the narrowest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 1 1 If no narrowest place is found, measurements shall be made as described in 4 1 2 1 and the smallest value obtained shall be the minimum width of the unit or test unit

4.1.3.2 *Specimen for physical test*—The specimen shall be examined for the narrowest place If the narrowest place is found, a measurement shall be made at that place as described in 4 1 1 If no narrowest place is found, measurements shall be made as described in 4 1 2 2 and the smallest value obtained shall be the minimum width of the specimen

4.1.4 Maximum width.

4.1.4.1 *Unit or test unit*—The unit or test unit shall be examined for the widest place If the widest place is found, the specimen shall be taken so as to include it, and a measurement

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shall be made at that place as described in 4 1 1 If no widest place is found, measurements shall be made as described in 4 1 2 1, and the largest value obtained shall be the maximum width of the unit or test unit

4.1.4.2 *Specimen for physical test* —The specimen shall be examined for the widest place If the widest place is found, a measurement shall be made at that place as described in 4 1 1 If no widest place is found, measurements shall be made as described in 4 1 2 2, and the largest value obtained shall be the maximum width of the specimen

4.2 Apparatus in 3.1.2.

4.2.1 Average, minimum, or maximum width shall be determined as described in 4 1, except that (1) the apparatus in 3 1 2 shall be used, (2) the number of measurements shall be 9 instead of 6 in 4 1 2 1, and (3) the measurements shall be made to within 0 001 inch in any manner that does not distort the specimen

5. RESULTS

5.1 Unit or test unit.

5.1.1 *Average width* —The average width of the unit or test unit shall be the average of all the values obtained, 4 1 2 1

5.1.2 *Minimum width*

5.1.2.1 If a narrowest place is found in the unit or test unit, the minimum width of the unit or test unit shall be the width obtained at that place

5.1.2.2 If no narrowest place is found in the unit or test unit, the minimum width of the unit or test unit shall be the smallest of all of the values obtained when measurements are made as described in 4 1 2 1

5.1.3 *Maximum width*

5.1.3.1 If a widest place is found in the unit or test unit, the maximum width of the unit or test unit shall be the width obtained at that place

5.1.3.2 If no widest place is found in the unit or test unit, the maximum width of the unit or test unit shall be the largest of all the values obtained when measurements are made as described in 4 1 2 1

5.1.4 The average, minimum, and maximum width of the unit or test unit shall be recorded to the nearest 0 0002 inch when the apparatus in 3 1 1 is used, and 0.001 inch when the apparatus in 3 1 2 is used

5.2 Specimens for physical test.—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method.

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WIDTH, SCALE OR TAPE

1. SCOPE

1.1 This method is intended for use in determining the width of units, test units, or specimens for physical tests In certain cases, this method is also applicable to determining the length of short units, test units, or specimens for physical tests

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable

3. APPARATUS

3.1 The apparatus shall consist of a scale or tape graduated to 0 01 or 1/64 inch for narrow specimens and 1/32 inch for wider specimens

4. PROCEDURE

4.1 Width measurement.—The specimen shall be placed on a flat surface so as to expose the full width The graduated scale or tape shall be applied over the specimen in such a manner that the width is not distorted The width shall be read from the scale or tape to the nearest 0 01, 1/64, or 1/32 inch, as applicable If more than one measurement is made on the specimen, the measurements shall be equally spaced over the specimen

4.2 Average width.

4.2.1 *Unit or test unit* —Unless otherwise specified in the detail specification, a total of nine measurements shall be made on the unit or test unit The measurements shall be made as described in 4 1, and if made on more than one specimen they shall be equally distributed as nearly as practicable among all the specimens from the unit or test unit

4.2.2 *Specimen for physical test* —Unless otherwise specified in the applicable test method, three measurements shall be made on the specimen The measurements shall be made as described in 4 1

4.3 Minimum width.

4.3.1 *Unit or test unit* —The unit or test unit shall be examined for the narrowest place If the narrowest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 1 If no narrowest place is found, measurements shall be made as described in 4 2 1, and the smallest value obtained shall be the minimum width of the unit or test unit

4.3.2 *Specimen for physical test* —The specimen shall be examined for the narrowest place If the narrowest place is found, a measurement shall be made at that place as described in 4 1 If no narrowest place is found, measurements shall be made as described in 4 2 2, and the smallest value obtained shall be the minimum width of the specimen

4.4 Maximum width.

4.4.1 *Unit or test unit* —The unit or test unit shall be examined for the widest place If the widest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 1 If no widest place is found, measurements shall be made as described in 4 2 1, and the largest value obtained shall be the maximum width of the unit or test unit

4.4.2 *Specimen for physical test* —The specimen shall be examined for the widest place If the widest place is found, a measurement shall be made at that place as described in 4 1 If no widest place is found, measurements shall be made as described in 4 2 2, and the largest value obtained shall be the maximum width of the specimen

5. RESULTS

5.1 The results shall be as described in method 2111, except that the average, minimum, and maximum width of the unit or test unit shall be recorded to the nearest 0 01, 1/64, or 1/32 inch as applicable

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

DIAMETER, OPTICAL

1. SCOPE

1.1 This method is intended for use in determining the outer and inner diameter of relatively rigid units, test units, or specimens for physical tests, with circular cross section It is particularly applicable to small hose and rubber tubing with a diameter of less than ½ inch This method is more accurate but more time consuming than method 2221

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable For units or test units, the specimen shall be short with full cross section cut perpendicular to the length axis

3. APPARATUS

3.1 The apparatus shall be as described in method 2031

4. PROCEDURE

4.1 The ruled glass disk shall be placed on the cut surface of the cross section with the ruled surface in contact with the material The diameter, if in the field of view, shall be read directly from the rulings on the disk to the nearest 0 0002 inch If the diameter is such that it does not all lie in the field of view, the specimen shall be moved by means of the mechanical stage until the complete diameter passes through the field The rulings on the disk passed over during the movement shall be counted and the distance recorded to the nearest 0 0002 inch One set of measurements shall be made on each cross section measured A set of measurements is two measurements made on the same cross section at right angles to each other When a scale is used in the eyepiece, the ruled glass disk or ocular micrometer shall be placed in the ocular and calibrated The specimen shall be placed on the stage with the surface to be measured perpendicular to the optical axis of the microscope so as to expose the full cross section The microscope shall be focused on the specimen and the diameter determined by counting the divisions of the ruled disk in the eyepiece that cover the distance from one edge of the specimen to the other and the distance recorded to the nearest 0 0002 inch One set of measurements shall be made on each cross section measured A set of measurements is two measurements made on the same cross section at right angles to each other

4.2 Unit or test unit.—Unless otherwise specified in the detail specification, a total of 6 measurements (3 sets) shall be made on the unit or test unit The measurements shall be made as described in 4.1

4.3 Specimen for physical test.—Unless otherwise specified in the applicable test method, 4 measurements (2 sets) shall be made on the specimen The measurements shall be made as described in 4 1.

5. RESULTS

5.1 Unit or test unit.

5.1.1 The diameter of the unit or test unit shall be the average of all the values obtained, 4 2

5.1.2 The diameter of the unit or test unit shall be recorded to the nearest 0 0002 inch

5.2 Specimen for physical test.—The number of specimens to be tested and the treatment of data (average, etc.) shall be as required in the applicable test method

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DIAMETER, SCALE OR TAPE

1. SCOPE

1.1 This method is intended for use in determining the outer and inner diameter of relatively rigid units, test units, or specimens for physical tests, with circular cross section It is particularly applicable to hose and rubber tubing with a diameter greater than ½ inch

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable For units or test units, the cross section of the specimen shall be the full cross section cut perpendicular to the length axis

3. APPARATUS

3.1 The apparatus shall consist of a scale or tape graduated to 1/64 inch for specimens one inch or less in diameter, or 1/32 inch for specimens greater than one inch in diameter

4. PROCEDURE

4.1 The specimen shall be placed on a flat surface so as to expose the full circular cross section The graduated scale or tape shall be applied over the specimen in such a manner that the diameter is not distorted The diameter of the specimen shall be the distance between the two graduations of the scale which are tangent to the circumference of the specimen The diameter shall be recorded to the nearest 1/64 or 1/32 inch, as applicable One set of measurements shall be made on each cross section measured A set of measurements is two measurements made on the same cross section at right angles to each other

4.2 Unit or test unit.—Unless otherwise specified in the detail specification, a total of 4 measurements (2 sets) shall be made on the unit or test unit The measurements shall be made as described in 4 1

4.3 Specimen for physical test.—Unless otherwise specified in the applicable test method, 4 measurements (2 sets) shall be made on each specimen The measurements shall be made as described in 4 1

5. RESULTS

5.1 Unit or test unit.

5.1.1 The diameter of the unit or test unit shall be the average of all the values obtained, 4 2

5.1.2 The diameter of the unit or test unit shall be recorded to the nearest 1/64 inch if the diameter is one inch or less, or 1/32 inch if the diameter is greater than one inch

5.2 Specimen for physical test.—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

DIAMETER, CIRCUMFERENCE METHOD

1. SCOPE

1.1 This method is intended for use in determining the diameter of units, test units, or specimens for physical tests, where it is more practicable to measure the circumference and calculate the diameter The method for determining the circumference shall be as specified in the detail specification or as required in the appropriate method of test Methods 2311, 2341, and 2351 describe procedures for the direct determination of the circumference

2. SPECIMEN

2.1 The specimen shall be as described in the method of test specified in the detail specification for determining the circumference

3. APPARATUS

3.1 The apparatus shall be as described in methods 2311, 2341, or 2351 as specified in the detail specification for determining the circumference

4. PROCEDURE

4.1 The circumference shall be determined as described in method 2311, 2341, or 2351 as specified in the detail specification

5. RESULTS

5.1 Calculation.—The diameter shall be calculated for each place measured, 4.1, as follows

$$\text{Diameter, inches,} = \frac{C}{\pi}$$

Where
C is the circumference at the place measured, 4 1, inches

5.2 Unit or test unit.

5.2.1 The diameter of the unit or test unit shall be the average of all the values, 5 1, obtained on the unit or test unit

5.2.2 When the circumference is determined as described in method 2311, the diameter of the unit or test unit shall be recorded to the nearest 0 001 inch.

5.2.3 When the circumference is determined as described in method 2341, the diameter of the unit or test unit shall be recorded to the nearest 0 015 inch

5.2.4 When the circumference is determined as described in method 2351, the diameter of the unit or test unit shall be recorded to the nearest 0 01 inch

5.3 Specimen for physical test.—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method

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CIRCUMFERENCE, OUTER, WIRE OR THREAD

1. SCOPE

1.1 This method is intended for use in determining the outer circumference of relatively rigid units, test units, or specimens for physical tests, with circular cross section, such as small hose, thick-walled tubing, etc It is particularly applicable to units, test units, or specimens for physical test with an outer circumference of 3 inches or less This method is less time consuming and less accurate than method 2321 for measuring materials less than 1½ inches in outer circumference

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable For units or test units, the specimen shall have the full circumference of the unit

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Fine wire or thread with a length at least 10 times the circumference of the specimen.

3.1.2 Convenient equipment for wrapping the wire or thread around the specimen under a given tension

3.1.3 Convenient equipment for measuring within ±0 02 inch the length of wire or thread which is wrapped around the specimen

3.1.4 Satisfactory equipment for controlling the tension of the wire or thread during wrapping and for measuring the length of thread or wire in a given number of turns Suitable equipment is shown in figure 2311

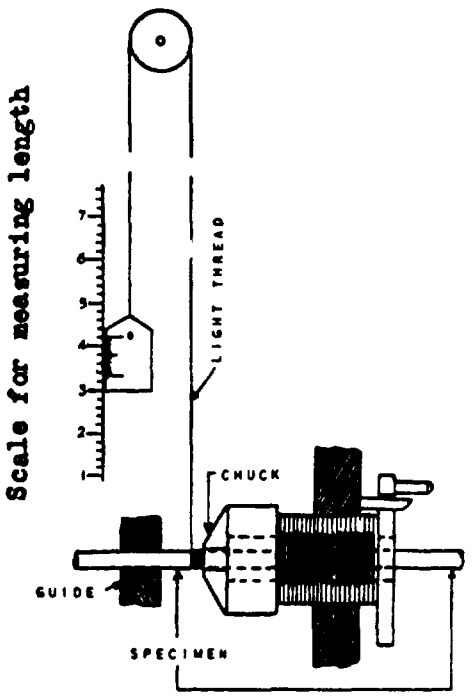

FIGURE 2311 —Apparatus for measuring circumference

4. PROCEDURE

4.1 The specimen shall be placed in the testing apparatus and one end of the wire or thread attached to it by means of a chuck The other end of the wire or thread shall be placed over the pulley and attached to the 10-gram weight The weight moves along a scale which indicates the length of the wire or thread wound around the specimen At least 10 complete turns of the wire or thread shall be wrapped around the specimen under the load of 10 grams, and the number of turns recorded as N The wraps of wire or thread shall be just in contact with each other, without any overlapping The length of N turns of wire or thread shall be read from the scale and the value recorded to the nearest 0 02 inch as L If more than one measurement is made on a specimen, the measurements shall be equally spaced over the specimen

4.1.1 The diameter of the wire or thread shall be measured with a microscope as described in method 2211

4.2 Unit or test unit.—Unless otherwise specified in the detail specification, a total of three measurements shall be made on the unit or test unit The measurements shall be made as described in 4 1, and if made on more than one specimen they shall be equally distributed as nearly as practicable among all the specimens from the unit or test unit

4.3 Specimen for physical test.—Unless otherwise specified in the applicable test method, two measurements shall be made on

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the specimen The measurements shall be made as described in 4 1

5. RESULTS

5.1 Calculation.—The circumference shall be calculated for each place measured, as follows

$$\text{Circumference, inches,} = \frac{L}{N} - \pi D$$

Where

L is the length of the wire or thread for N turns at the place measured, inches

N is the corresponding number of turns of the wire or thread

D is the diameter of the wire or thread, 4 1 1, inch

5.2 Unit or test unit.

5.2.1 The circumference of the unit or test unit shall be the average of all the values, 5 1, obtained on the unit or test unit

5.2.2 The circumference of the unit or test unit shall be recorded to the nearest 0.002 inch.

5.3 Specimen for physical test.—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method

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CIRCUMFERENCE, DIAMETER METHOD, OPTICAL

1. SCOPE

1.1 This method is intended for use in determining the outer or inner circumference of relatively rigid units, test units, or specimens for physical tests, with circular cross section It is particularly applicable to small hose and rubber tubing with a circumference of less than 1½ inches This method is more accurate but more time consuming than method 2311

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable For units or test units, the specimen shall be short with full cross section cut perpendicular to the length axis

3. APPARATUS

3.1 The apparatus shall be as described in method 2031.

4. PROCEDURE

4.1 The diameter shall be determined as described in method 2211

5. RESULTS

5.1 Calculation.—The outer or inner circumference shall be calculated for each place measured, 4 1, as follows.

Circumference, inches, $= \pi D$

Where

D is the outer or inner diameter at the place measured inches, 4 1

5.2 Unit or test unit.

5.2.1 The circumference of the unit or test unit shall be the average of all values, 5 1, obtained on the unit or test unit.

5.2.2 The circumference of the unit or test unit shall be recorded to the nearest 0 001 inch

5.3 Specimens for physical test.—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method

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Method 2331
April 12, 1955

This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

CIRCUMFERENCE, SCALE

1. SCOPE

1.1 This method is intended for use in determining the outer and inner circumference of units, test units, or specimens for physical tests, with irregular cross section, such as gloves, rubber bands, etc

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable For units or test units, the specimen shall have the full circumference of the unit

3. APPARATUS

3.1 The apparatus shall consist of a scale graduated to 0 01, 1/64, 1/32, or 1/8 inch, as required in the detail specification

4. PROCEDURE

4.1 Outer circumference measurement.—The specimen shall be pressed flat on a smooth surface and the graduated scale applied over the specimen in such a manner that it is not appreciably distorted The distance between the two outer edges of the specimen shall be read from the scale to the nearest 0 01, 1/64, 1/32, or 1/8 inch, as required, and recorded as D

4.1.1 *Unit or test unit*—Unless otherwise specified in the detail specification, a total of three measurements shall be made on the unit or test unit The measurements shall be made as described in 4 1, and if made on more than one specimen they shall be equally distributed as nearly as practicable among all of the specimens from the unit or test unit

4.1.2 *Specimen for physical test*—Unless otherwise specified in the applicable test method, two measurements shall be made on each specimen The measurement shall be made as described in 4 1

4.2 Inner circumference measurement.—The specimen shall be placed on edge on a flat surface and pressed flat between two straight edges and the graduated scale applied over the inner edges of the specimen in such a manner that it is not appreciably distorted The distance between the two inner edges of the specimen shall be read from the scale to the nearest 0 01, 1/64, 1/32, or 1/8 inch, as required, and recorded as d

4.2.1 *Unit or test unit*—Unless otherwise specified in the detail specification, a total of three measurements shall be made on the unit or test unit The measurements shall be made as described in 4 2, and if made on more than one specimen they shall be equally distributed as nearly as practicable among all the specimens from the unit or test unit

4.2.2 *Specimen for physical test*—Unless otherwise specified in the applicable test method, two measurements shall be made on the specimen The measurement shall be made as described in 4 2

5. RESULTS

5.1 Calculation.—The circumference shall be calculated for each place measured, 4 1 1 or 4 1 2 or 4 2 1 or 4 2 2, as follows

Outer circumference Circumference, inches, $=2D$
Inner circumference Circumference, inches, $=2d$

Where

D is the distance between the two outer edges at the place measured, 4 1, inches
d is the distance between the two inner edges at the place measured, 4 2, inches

5.2 Unit or test unit.

5.2.1 The circumference of the unit or test unit shall be the average of all the values, 5 1, obtained on the unit or test unit

5.2.2 The circumference of the unit or test unit shall be recorded to the nearest 0 01, 1/64, 1/32, or 1/8 inch, as required

5.3 Specimens for physical test.—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method

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CIRCUMFERENCE, INNER, MANDREL

1. SCOPE

1.1 This method is intended for use in determining the inner circumference of units, test units, or specimens for physical tests, with circular cross section It is particularly applicable to units, test units, or specimens for physical tests, with a diameter of less than 6 inches, such as rubber hose, inner tubes, etc.

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable For units or test units, the specimen shall have the full circumference of the unit

3. APPARATUS

3.1 The apparatus shall consist of a tapered mandrel graduated to 0 05 inch or finer

4. PROCEDURE

4.1 The specimen shall be carefully fitted over the mandrel in such a manner that it is not distorted The graduation on the mandrel that just comes in contact with the inner circumference of the specimen shall be read directly from the mandrel and the value recorded to the nearest 0 05 inch as the inner circumference of the specimen

4.2 Unit or test unit.—Unless otherwise specified in the detail specification, a total of three measurements shall be made on the unit or test unit The measurements shall be made as described in 4 1, and if made on more than one specimen they shall be equally distributed as nearly as practicable among all the specimens from the unit or test unit

4.3 Specimen for physical tests.—Unless otherwise specified in the detail specification, two measurements shall be made on the specimen The measurements shall be made as described in 4 1

5. RESULTS

5.1 Unit or test unit.

5.1.1 The circumference of the unit or test unit shall be the average of all the values obtained in 4 2

5.1.2 The circumference of the unit or test unit shall be recorded to the nearest 0 05 inch

5.2 Specimen for physical test.—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method

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(formerly Fed Spec ZZ-R-601a)

CIRCUMFERENCE, OUTER, TAPE

1. SCOPE

1.1 This method is intended for use in determining the outer circumference of relatively rigid units, test units, or specimens for physical tests, with circular cross section It is particularly applicable to articles with circumference greater than 3 inches, such as hose

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable For units or test units, the specimen shall have the full circumference of the unit.

3. APPARATUS

3.1 The apparatus shall consist of a flexible steel measuring tape graduated to 1/32 inch or finer on both edges

4. PROCEDURE

4.1 The tape shall be applied around the specimen in such a manner that it is in contact with the specimen at all points without distortion of the specimen The circumference shall be read directly from the tape and the value recorded to the nearest 1/32 inch If more than one measurement is made on a specimen, the measurements shall be equally spaced over the specimen

4.2 **Unit or test unit.**—Unless otherwise specified in the detail specification, a total of six measurements shall be made on the unit or test unit The measurements shall be made as described in 4 1, and if made on more than one specimen they shall be equally distributed as nearly as practicable among all the specimens from the unit or test unit

4.3 **Specimen for physical tests.**—Unless otherwise specified in the applicable test method, three measurements shall be made on the specimen The measurements shall be made as described in 4 1

5. RESULTS

5.1 **Unit or test unit.**

5.1.1. The circumference of the unit or test unit shall be the average of all the values obtained, 4 2

5.1.2 The circumference of the unit or test unit shall be recorded to the nearest 1/32 inch

5.2 **Specimen for physical test.**—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method.

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CIRCUMFERENCE, DIAMETER, SCALE OR TAPE

1. SCOPE

1.1 This method is intended for use in determining the inner and outer circumference of units, test units, or specimens for physical tests, with circular cross section, by measuring the diameter and calculating the circumference It is particularly applicable to articles with diameters greater than 6 inches, such as dredging sleeves, hose, etc

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable For units or test units, the specimen shall have the full circumference of the unit

3. APPARATUS

3.1 The apparatus shall consist of a scale or tape graduated to ¼ or ½ inch as required in the detail specification

4. PROCEDURE

4.1 The specimen shall be placed on a flat surface so as to expose the full circular cross section The graduated scale or tape shall be applied over the specimen in such a manner that the diameter is not distorted The diameter of the specimen shall be the distance between the two graduations of the scale or tape which are tangent to the circumference of the specimen and shall be recorded to the nearest ¼ or ½ inch, as required One set of measurements shall be made on each cross section measured A set of measurements is two measurements made on the same cross section at right angles to each other

4.2 Unit or test unit.—Unless otherwise specified in the detail specification, a total of measurements (3 sets) shall be made on the unit or test unit The measurements shall be made as described in 4 1

4.3 Specimen for physical test.—Unless otherwise specified in the applicable test method, 4 measurements (2 sets) shall be made on the specimen The measurements shall be made as described in 4 1

5. RESULTS

5.1 Calculation.—The outer or inner circumference shall be calculated for each place measured, 4 1, as follows

$$\text{Circumference, inches, } = \pi D$$

Where
D is the outer or inner diameter at the place measured, inches

5.2 Unit or test unit.

5.2.1 The circumference of the unit or test unit shall be the average of all the values, 5 1, obtained on the unit or test unit

5.2.2 The circumference of the unit or test unit shall be recorded to the nearest ¼ or ½ inch, as required

5.3 Specimen for physical test.—The number of specimens to be tested and the treatment of the data (average, etc) shall be as required in the applicable test method

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

LENGTH

1. SCOPE

1.1 This method is intended for use in determining the length of units, test units, or specimens for physical tests Method 2121 is also applicable to the determination of length of short articles

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 2001, as applicable

3. APPARATUS

3.1 The apparatus shall consist of a scale or tape graduated to 1/32, 1/16, or 1/8 inch as required in the detail specification

4. PROCEDURE

4.1 The specimen shall be placed on a flat surface so as to expose the full length The graduated scale or tape shall be applied over the specimen in such a manner that the length is not appreciably distorted The length shall be read from the scale to the nearest 1/32, 1/16, or 1/8 inch, as required

4.2 **Unit or test unit.**—Unless otherwise specified in the detail specification, 2 measurements shall be made on the unit or test unit. The measurement shall be made as described in 4 1

4.3 **Specimen for physical test.**—Unless otherwise specified in the applicable test method, 2 measurements shall be made on the specimen The measurements shall be made as described in 4 1

5. RESULTS

5.1 **Unit or test unit.**

5.1.1 The length of the unit or test unit shall be the average of all the values obtained, 4 2

5.1.2 The length of the unit or test unit shall be recorded to the nearest 1/32, 1/16, or 1/8 inch, as required

5.3 **Specimen for physical test.**—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method

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HARDNESS, DUROMETER

1. SCOPE

1.1 This method is intended for use in determining the hardness of soft vulcanized rubber compounds by means of a durometer. For the purpose of this method, a durometer is any instrument which conforms to the requirements in 3. Typical instruments of this type are the Shore durometer and the Rex hardness gage shown in figures 3021A and 3021C, respectively

1.2 Hardness determinations using the durometer should not be made on single or plied-up specimens with curved surfaces.

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit at least ¾ by ¾ inch, or ¾ inch in diameter if circular in shape.

2.1.1 Unless otherwise specified in the detail specification, the specimen shall have a thickness of at least ¼ inch, method 2011 The principal surfaces of the specimen shall be as nearly parallel as practicable.

2.2 Unless otherwise specified in the detail specification, materials thinner than ¼ inch shall be plied-up to obtain the required thickness

3. APPARATUS

3.1 The apparatus shall be as follows·

3.1.1 A durometer consisting of the following parts:

3.1.1.1 *Presser foot*—A presser foot which is pierced by a hole not greater than ⅛ inch in diameter and the center of which is not closer than ¼ inch to the edge of the presser foot in any direction.

3.1.1.2 *Indenter point.*—An indenter point made of hardened steel conforming to the shape and size shown in figure 3021B. The extension of the indenter beyond the surface of the presser foot of the instrument at zero reading is 0.100+0.000 or −0.010 inch. The indenter point operates through the hole in the presser foot.

3.1.1.3 *Spring.*—A spring within the instrument that will force the indenter point to extend from the face of the presser foot as far as permitted by the material being tested. When the spring is calibrated according to method 3025, the calibration curve produced shall not deviate from the standard curve, figure 3025C, more than ±4 grams at any point. The

FIGURE 3021A —Shore durometer

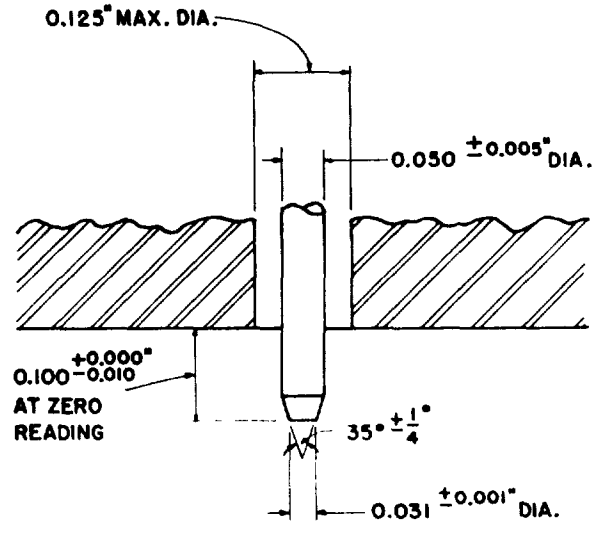

FIGURE 3021B —Indenter point for durometer

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FIGURE 3021C —Rex hardness gage.

straight line is determined by the coordinates zero dial reading and 56-gram load at one end and 100 dial readings and 822-gram load at the other, as shown in figure 3025C

3.1.1.4 *Indicator gage*—A dial gage on which the amount of extension of the indenter point beyond the face of the presser foot is read in terms of graduations ranging from zero for full extension to 100 for zero extension. The reading shall be 100 when the indenter and presser foot are pressed down firmly on a flat piece of plate glass. At zero reading, the indenter shall extend 0.100 inch +0.000 or −0.010 inch beyond the presser foot.

3.1.2 A stop watch or other timing device that will indicate the time in seconds

4. PROCEDURE

4.1 If the material has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111.

4.2 The specimen shall be free from mechanical damage The specimen shall be placed in a horizontal position on a firm, smooth surface The durometer shall be held in a vertical position and applied to the specimen while keeping the presser foot parallel to the surface of the specimen The instrument shall be lowered as rapidly as possible without shock. The pressure applied shall be just sufficient to insure firm contact of the presser foot with the specimen. No hardness determination shall be made with the center of the indenter point less than ⅜ inch from the edge of the specimen and no determination shall be made unless all of that area of the presser foot comprised by a radius of ¼ inch in all directions from the center of the indenter is resting on the specimen.

4.3 Unless otherwise specified in the detail specification, the indentation reading shall be taken immediately after firm contact has been established between the presser foot and the specimen, except in the case of instruments employing a maximum indicator, in which case the reading can be taken at the convenience of the operator. The indentation shall be read from the dial of the gage and the value recorded to the nearest whole scale unit.

4.4 When a time interval for reading the instrument, as for instance 15 seconds, is specified in the detail specification, the instrument shall be held with the presser foot in contact with the specimen without change of position

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or pressure for the required time, after which the reading shall be taken and the value recorded to the nearest whole scale unit This method of reading shall not be used with instruments equipped with a maximum indicating device

5. RESULTS

5.1 Unless otherwise specified in the detail specification, five specimens from each test unit shall be tested

5.2 The hardness of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see section 5

5.3 The hardness of the test unit shall be recorded to the nearest whole scale reading

5.4 The time interval, if any, allowed before reading the instrument shall be recorded

5.5 If a plied-up specimen is tested, it shall be recorded

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CALIBRATION OF DUROMETER

1. SCOPE

1.1 This method is intended for use in calibrating the durometer described in method 3021. The method applies to the checking of instruments that should be returned to the factory for adjustment and calibration and to instruments which may be adjusted and calibrated in the laboratory.

2. APPARATUS

2.1 The apparatus shall be as follows:

2.1.1 *Balance*.—A balance of the Harvard trip type with two pans, one of which is equipped with a calibrating foot about 0.1 inch in height and 0 050 inch in diameter, slightly cupped to accommodate the indenter point to prevent interference between the presser foot and pan when the durometer is placed in position on the pan as shown in figure 3025A. A weight equal to that of the calibrating foot is added to the opposite pan

2.1.2 A stand or other equipment for supporting the durometer in a vertical position on the calibrating foot

2.1.3 Weights of such denomination as to provide varying loads from 50 to 850 grams

2.1.4 A piece of plate glass.

2.1.5 A very small end wrench that will fit linkage 4 in figure 3025B

2.1.6 A very small screwdriver

2.1.7 A special wrench that will fit over the indenter and inside of the hole in the presser foot

3. PROCEDURE

3.1 The instrument shown in figure 3021A may be adjusted and calibrated in the laboratory The instrument shown in figure 3021C is not adapted for adjustment in the laboratory and should be returned to the manufacturer for adjustment; however, it may be calibrated in the laboratory Calibration of both instruments at least once a month is recommended.

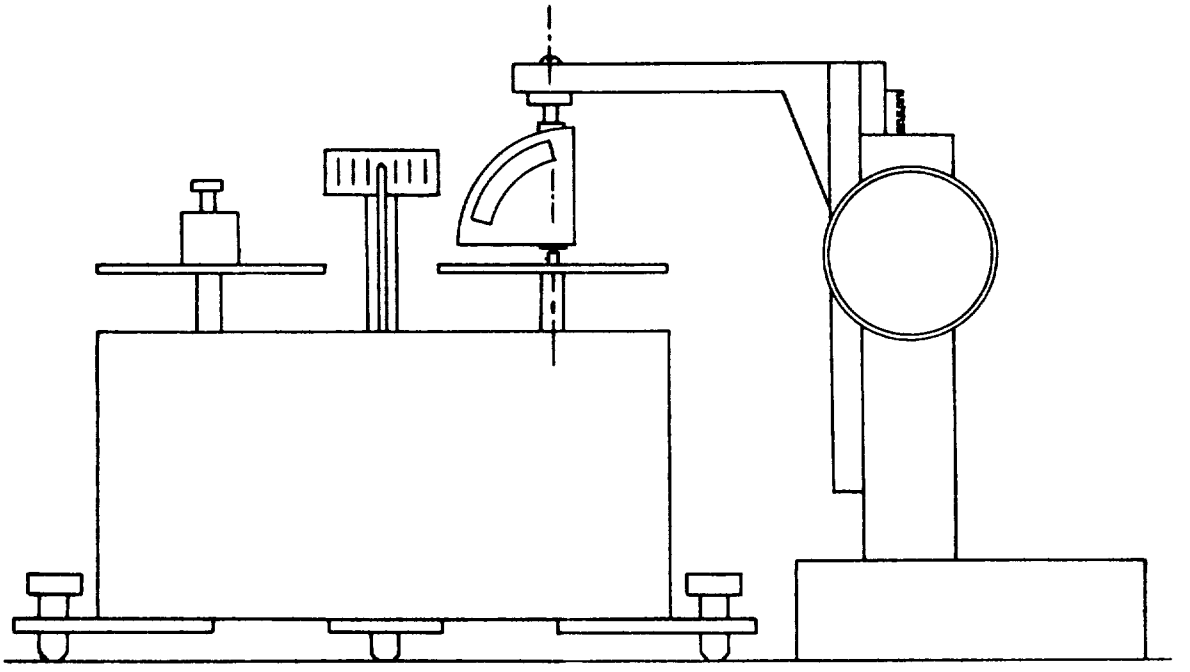

FIGURE 3025A.—Balance and apparatus for calibration of spring

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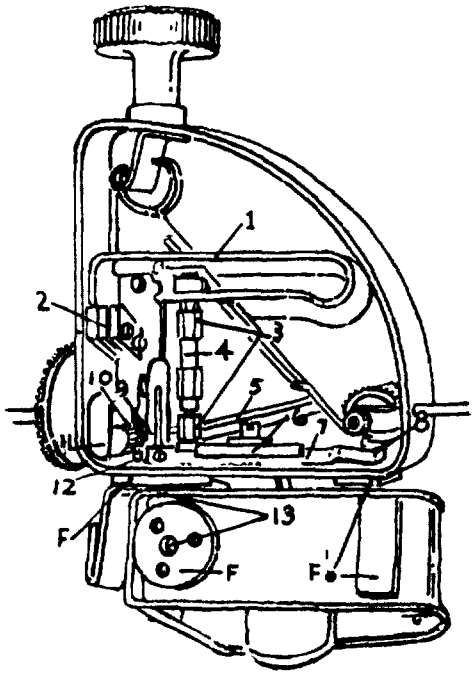

FIGURE 3025B —Shore durometer

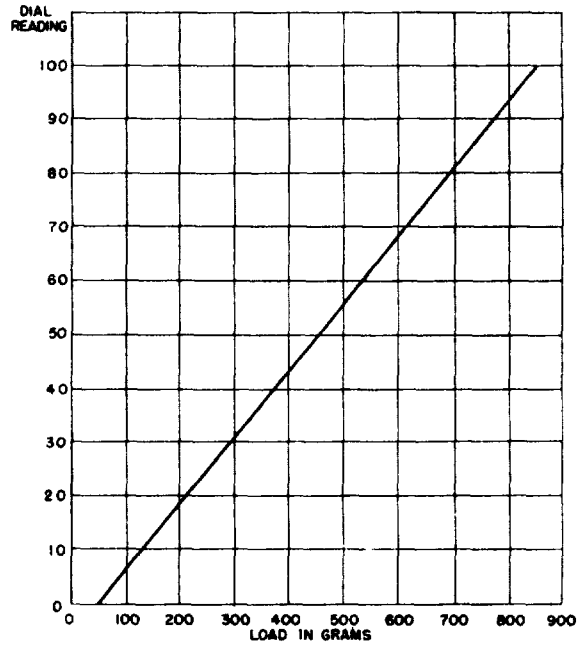

FIGURE 3025C.—Calibration curve

3.2 Calibration of spring.—The durometer shall be aligned on the stand with the indenter point resting on the calibrating foot, as shown in figure 3025A Various weights shall be added to the other pan of the balance and the corresponding reading recorded from the dial of the durometer Eight to ten readings equally spaced over the range of the instrument using weight increments of 100 or 75 grams are recommended A calibration curve shall be prepared by plotting the dial readings against the load in grams from the readings obtained If the curve deviates more than ±4 grams at any point from the standard curve shown in figure 3025C, the instrument shown in figure 3021A shall be adjusted according to 3.3 and the instrument shown in 3021C shall be sent to the manufacturer for adjustment

3.3 Adjustment of the instrument, fig. 3021A.

3.3.1 Three adjustments may be made which will change the hardness reading of the instrument, namely, the position of slide 6, length of linkage 4, and length of pin 13 shown in figure 3025B.

3.3.2 The face of the pin shall be examined under a microscope If the shape of the indenter point is not in conformity with figure 3021B, it shall be replaced or reground

3.3.3 The instrument shall be opened by removing the small screws in the back and cleaned carefully The bearings shall be set to minimize friction and the pointer adjusted, parallax eliminated, and backlash in the rack and pinion reduced

3.3.4 The back shall be replaced The screws are not necessary if the back fits snugly in place The back shall be in place while readings are being taken, as it is part of the bearing for the beam The instrument shall be tried on a piece of plate glass If it does not read 100, the back shall be removed, the setscrew loosened, and the indenter turned in or out on its thread until 100 can be read on the dial The setscrew shall be tightened, the durometer placed on the calibration stand, and the spring calibrated as described in 3 2

3.3.5 If the curve lies consistently above or below the standard curve, figure 3025C, linkage 4 shall be slightly lengthened or shortened. A

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new curve shall be constructed and by trial and error the spring tension adjusted to obtain a curve which is within ±4 grams of the standard curve at all points. If the curve crosses the standard, correction shall be made by adjusting slide 6 on the beam Sliding it toward the indenter swings the curve clockwise, thus reducing the slope of the curve It may be necessary to make both adjustments or to regulate the spring tension by adjustment where the spring is attached to the case

3.3.6 When a curve is obtained that does not deviate more than ±4 grams at any point from the standard curve shown in figure 3025C, the case shall be closed and the screws tightened

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HARDNESS, PLASTOMETER

1. SCOPE

1.1 This method is intended for use in determining the hardness of rubber by measuring the depth of indentation produced by a spherical-ended plunger of definite size under a stated load exerted for a specified time It may be used for determining the hardness of rubber of all types except hard rubber

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit at least 1¼ by 1¼ inches, or 1¼ inches in diameter if circular in shape.

2.1.1 Unless otherwise specified in the detail specification, the specimen shall have a thickness of at least ¼ inch, method 2011

2.2 For flat specimens, the principal surfaces shall be as nearly parallel as practicable

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An apparatus consisting essentially of an indenter acting under a dead-weight load and a suitable gage for indicating the depth of indentation The parts are so mounted in a supporting frame that the indenter and weight may be independently raised or lowered vertically, permitting the indenter to rest on the surface of the specimen and the weight to be subsequently applied to the indenter A vibrator is attached to the frame to overcome frictional forces in the tester. A suitable apparatus is shown in figure 3031

3.1.1.1 *Supporting frame* —The supporting frame consists of a parallel base and top plates held apart by bolts that are threaded to carry two intermediate movable plates The weight is suspended from the upper movable plate by means of a hollow tube having a shoulder which rests on the plate The indicator gage is attached to a pedestal that is supported by the lower movable plate The indenter is attached to the spindle of the gage by means of a shaft that passes freely through all of the plates and the weight tube The movable plates are equipped with a screw, sprocket, and chain arrangements for raising and lowering them The base plate has 4 feet attached by ball joints which permit use of the instrument either on a flat horizontal surface or on a curved surface such as rubber-covered rolls The base plate is also provided with an opening by which the entire instrument may be mounted on a rod and secondary base that allows clearance for specimens having dimensions larger than can be accommodated when using the regular feet The instrument shall be equipped with levels and a leveling stand for adjusting the indenter shaft to a vertical position

3.1.1.2 *Indenter* —The indenter consists of a vertical steel shaft attached at the upper end to the spindle of the gage and having at the lower end a spherical steel point 0 1250±0 0005 inch or 0 2500±0 0005 inch in diameter, made of highly polished, noncorrodible hard metal properly treated to resist wear The indenter shaft passes through the center of the loading weight and is provided with a suitable shoulder so that the dead-weight load may be fully applied to the spherical end by proper screw adjustment of the positions of the movable plates

3.1.1.3 *Weight* —The weight consists of a metal cylinder and a tube weighing 1,000±0 01 gram, placed concentric with the indenter shaft but not touching it except when resting on the shoulder during application

3.1.1.4 *Indicator gage* —The dial gage for measuring the depth of indentation is graduated to 0 01 mm and is actuated by vertical movement of the indenter

3.1.1.5 *Vibrator* —A small vibrator which operates continuously during a test to overcome any friction in the apparatus An instrument which vibrates approximately 120 times per second, with an amplitude of vibration of the fixed platform of the tester varying from approximately 0 00008 to 0 00019, has been found satisfactory

3.1.1.5.1 Such a vibrator may consist of a simple cored solenoid fastened to the top of a C-shaped piece of strap steel The lower part of the solenoid core is a loose cylindrical piece of steel with a shoulder at its lower end

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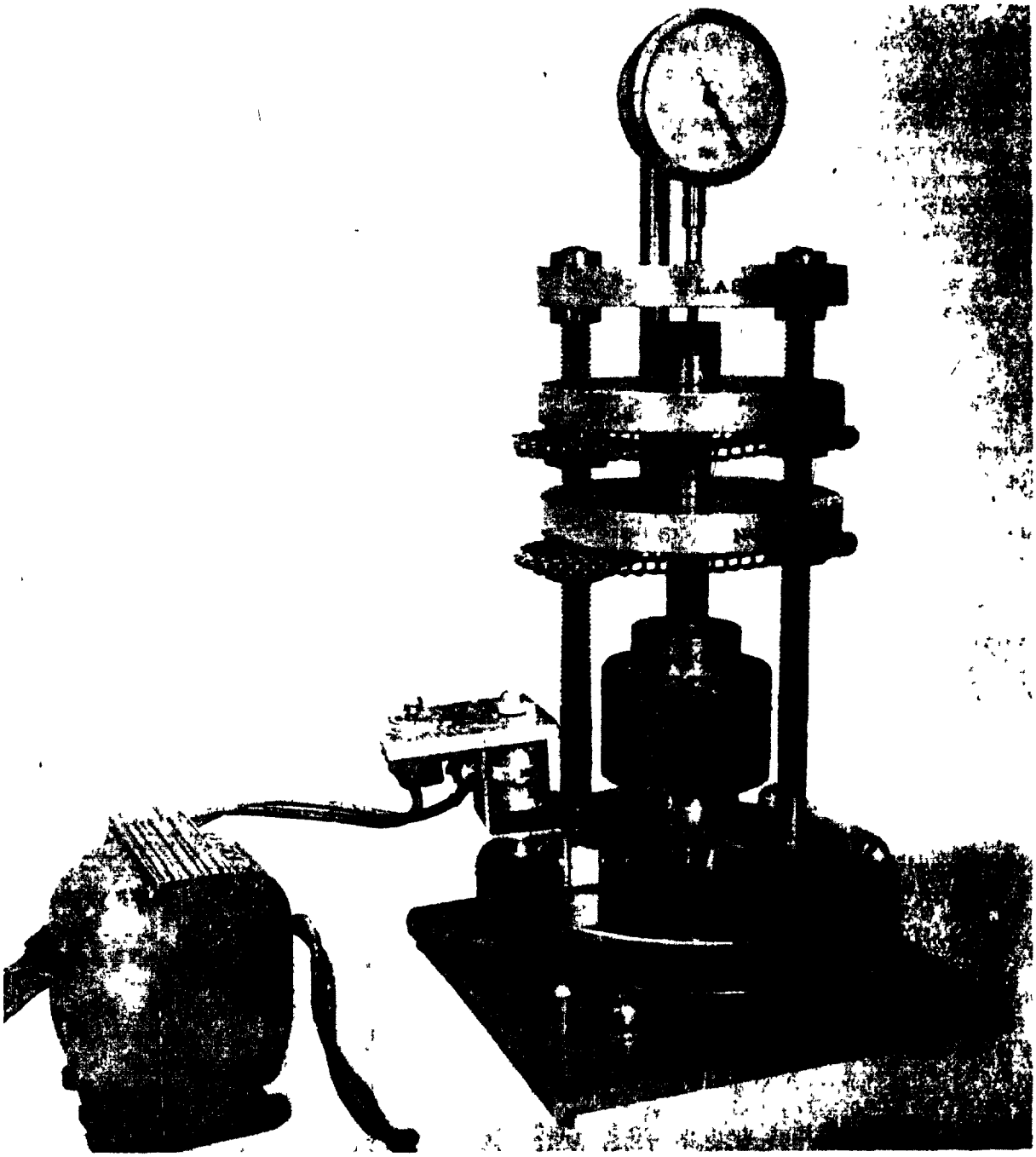

FIGURE 3031 —Plastometer

When the current is off, the loose-cored section rests on a screw projecting from the bottom of the C-shaped support When the current is on, the loose-cored section is attached to the solenoid but is restrained by a flat split ring of spring brass through which the cored section passes, and which bears on the shoulder of the cored section The cored section is thus

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caused to vibrate by the alternating forces of attraction by the solenoid, and repulsion by the flat ring acting as a spring A small plate bearing a toggle switch for closing the circuit to the solenoid is fastened to the top of the C-shaped piece of strap steel by means of the same bolt that holds the solenoid The bottom of the C-shaped piece is fastened by screws to a portion of a brass nut that serves to clamp the vibrator to one upright post of the tester

3.1.2 Specimen holder.—A holder consisting of a piece of metal approximately ¼-inch thick with small knobs or projections at three points on the bottom The upper surface of the metal should have a circular depression not less than 1½ inches in diameter and approximately 1/32 inch in depth The specimen may be secured to the holder by means of melted paraffin

3.1.3 A stopwatch or other timing device that will indicate the time in seconds

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the indenter point 0 1250 inch in diameter shall be used when a reading of less than 200 is obtained on the specimen by means of the 0 1250-inch point

4.2 Unless otherwise specified in the detail specification, the indenter point 0 2500 inch in diameter shall be used when a reading of 200 or more is obtained on the specimen by means of the 0 1250-inch point

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4.3 When the indenter with the 0 1250-inch point is used, the indentation shall be read from the dial gage 60 ± 1 second after application of the load

4.4 When the indenter with 0 2500-inch point is used, the indentation shall be read from the dial gage 15 ± 1 second after application of the load

4.5 If the material has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111

4.6 Flat specimen.—The specimen shall be free from mechanical damage The specimen shall be dusted lightly with talc so that the flow of rubber beneath the indenter point is unrestricted The specimen shall be placed on a firm base with the surface to be tested in a horizontal position and the instrument adjusted so that the indenter is vertical and will remain vertical during the test

4.6.1 It may be advantageous to mount the specimen in the holder described in 3 1 2 The vertical shaft and the indenter point shall be lowered until the center of the spherical end rests on the specimen at least ½ inch from any edge of the specimen and the indicator hand of the gage has made three revolutions The hand on the indicator gage shall then be adjusted to the zero position The vibrator shall be started, the one-kilogram weight lowered without shock onto the indenter shaft, and the supporting plate lowered still further so that the weight rests fully on the indenter as shown by a space of approximately 1/16 inch between the supporting plate and the shoulder of the weight tube The weight shall be allowed to remain on the indenter shaft for the required time, 4 3 or 4 4, the indentation read from the dial of the gage, and the value recorded to the nearest 0 01 mm

4.7 Curved specimen.—The specimen shall be placed so that the center line of the indenter shaft is perpendicular to the tangent at the point of contact of the indenter. The remainder of the procedure for determining the hardness shall be as described in 4 6

5. RESULTS

5.1 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.2 The hardness of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see section 5

5.3 The hardness of the test unit shall be recorded to the nearest 0.01 mm

5.4 The diameter of the indenter point shall be recorded

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HARDNESS, ASTM HARDNESS NUMBER

1. SCOPE

1.1 This method is intended for use in determining the hardness of rubber by measuring the depth of indentation by a spherical-ended plunger of definite size under a stated load exerted for a specified time It is applicable to the more common grades of rubber but not to extremely hard or extremely soft rubbers The indentation is expressed in ASTM hardness numbers (one hardness number is equal to 0 001-inch indentation)

1.2 For the purpose of this method, the hardness of rubber is that property by virtue of which the surface and adjoining layers resist indentation when subjected to a definite pressure by an indentation element of standard size, the contact surface of which conforms in shape to a portion of a true spherical surface

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit at least 2 by 2 inches, or 2 inches in diameter if circular in shape

2.1.1 Unless otherwise specified in the detail specification, the specimen shall have a thickness of at least ¼ inch, method 2011 The principal surfaces of the specimen shall be as nearly parallel as practicable

2.2 Unless otherwise specified in the detail specification, materials thinner than ¼ inch shall be plied-up to obtain the required thickness

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An apparatus consisting essentially of an indenter acting under a dead-weight load, an annular presser foot exerting a definite force on the surface of the specimen, a suitable gage for indicating the depth of indentation, and a specimen support

3.1.1.1 *Indenter* —The indenter consists of a rigid vertical shaft with the lower end made of highly polished, noncorrodible hard metal properly treated to resist wear The lower end is finished to a true hemisphere 0 0938 ± 0 0005 inch in diameter The maximum diameter of that part of the shaft that passes through the presser foot shall not exceed that of the indenter point The indenter is actuated by a freely acting deadweight of 3 pounds ±0 25 ounce When the penetrator is counterbalanced, less than 0 25 ounce is required to overcome friction and to produce perceptible motion of the penetrator The indenter acts through the opening in the center of the presser foot

3.1.1.2 *Presser foot* —The presser foot is 0 625 ± 0 0005 inch in diameter and has a centrally located hole 0 109 ± 0 005 inch in diameter through which the indenter passes The presser foot is sufficiently thick to insure rigidity, and the edges are rounded to a radius of 1/64 inch The inner surface of the presser foot is parallel to and concentric with the shaft of the indenter The contact surface of the presser foot is at right angles to the inner surface and parallel to the surface that supports the specimen The presser foot shall be subjected to a freely acting load of 5 pounds ±0 75 ounce

3.1.1.3 *Indicator gage* —The dial gage for measuring the depth of indentation is graduated to 0 001 inch and is actuated by vertical movement of the indenter The gage is mounted so as to be rigid with the presser foot and shall indicate the indentation over the whole range within ±0 0005 inch

3.1.1.4 The specimen support consists of a table or other smooth, flat horizontal surface with minimum linear dimensions of 2 by 2 inches

3.1.2 A stopwatch or other timing device that will indicate the time in seconds

4. PROCEDURE

4.1 If the material has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111

4.2 The specimen shall be free from mechanical damage The specimen shall be supported on a smooth, flat, horizontal surface The presser foot shall be lowered until in contact with the surface of the specimen, and the indenter lowered without shock until the full

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load is applied The center of the spherical indenter shall be at least ½ inch from any edge of the specimen Thirty seconds ±1 second after application of the full load the indentation shall be read from the dial of the gage and the value recorded to the nearest 0 001 inch

5. RESULTS

5.1 Unless otherwise specified in the detail specification, five specimens from each test unit shall be tested

5.2 The hardness of the test unit shall be median of the values obtained from the specimens tested For definition of median, see section 5

5.3 The hardness of the test unit shall be recorded to the nearest hardness number One hardness number is equal to 0 001 inch indentation

5.4 If a plied-up specimen is tested, it shall be recorded

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HARDNESS, INDENTOMETER

1. SCOPE

1.1 This method is intended for use in determining the hardness of rubber by measuring the depth of indentation produced by a spherical-ended plunger of definite size under a stated load exerted for a specified time. It may be used for determining the hardness of rubber of all types except hard rubber

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit at least 2 by 2 inches, or 2 inches in diameter if circular in shape.

2.1.1 Unless otherwise specified in the detail specification, the specimen shall have a thickness of at least ¼ inch, method 2011 The principal surfaces of the specimen shall be as nearly parallel as practicable

2.2 Unless otherwise specified in the detail specification, materials thinner than ¼ inch shall be plied-up to obtain the required thickness

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An apparatus consisting essentially of a loading and indentation measuring assembly suspended from a rigid cast-iron framework and an adjustable horizontal platform mounted directly below the indenter. A vibrator is attached to the frame to overcome frictional forces in the tester Suitable apparatus is shown in figures 3051A and 3051B

3.1.1.1 *Indenter* —The spherical indenter, 0 1250±0 0005 inch in diameter, is attached to the lower end of an indicator gage, *C*, and contacts the upper surface of the specimen

3.1.1.2 *Indicator gage* —The dial gage for measuring the amount of indentation when the major load is applied to the indenter is graduated to 0 01 mm One revolution of the main pointer corresponds to one-mm indenter travel The dial gage is equipped with a revolution counter and has a range of 10 mm

3.1.1.3 *Minor load* —The minor load, 90 grams, consists of the load produced by means of a spring tension plus the weight of the indenter

3.1.1.4 *Major load*.—The major load, 1,000 grams, consists of *I* and *K* plus the minor load The major load is applied to the indenter shaft with relatively little friction by lowering the weight onto a concentric tapered shaft which is an extension of the indenter stem Since the center of gravity of the major load is lower than the pointed end of the tapered shaft, as shown in figure 3051C, the load is balanced on the shaft during the test The major load is lowered and balanced on the pointed end of the shaft, without undue friction or impact, by helical movement of sleeve support, *J*, which is moved by rotation of a pin guided in the helical slot of the cylinder, *H*, mounted above and concentric with the indicator shaft

3.1.1.5 *Leveling apparatus* —The instrument is equipped with a fixed screw at the front center of the base and two knurled leveling screws, *B*, located under the back of the base for accurate vertical alignment A spirit level, *A*, is attached for leveling the instrument

3.1.1.6 The position of the specimen platform is controlled by a lock screw, *F*, and an elevating and lowering pinion, *G*

3.1.1.7 *Weight* —The 454-gram weight, shown at left in figure 3051A, with specimen contact surface ⅝ inch outside diameter and ⅜ inch inside diameter may be applied to the specimen concentrically with the indenter, if necessary to prevent the specimen from bowing away from the platform at the start of the test

3.1.1.8 The small vibrator, *E*, attached to the bracket, *D*, operates continuously during a test to overcome any friction in the apparatus

3.1.2 A stopwatch or other timing device that will indicate the time in seconds

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, indentation readings shall be made 60±1 second after application of the major load

4.2 If the material has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111

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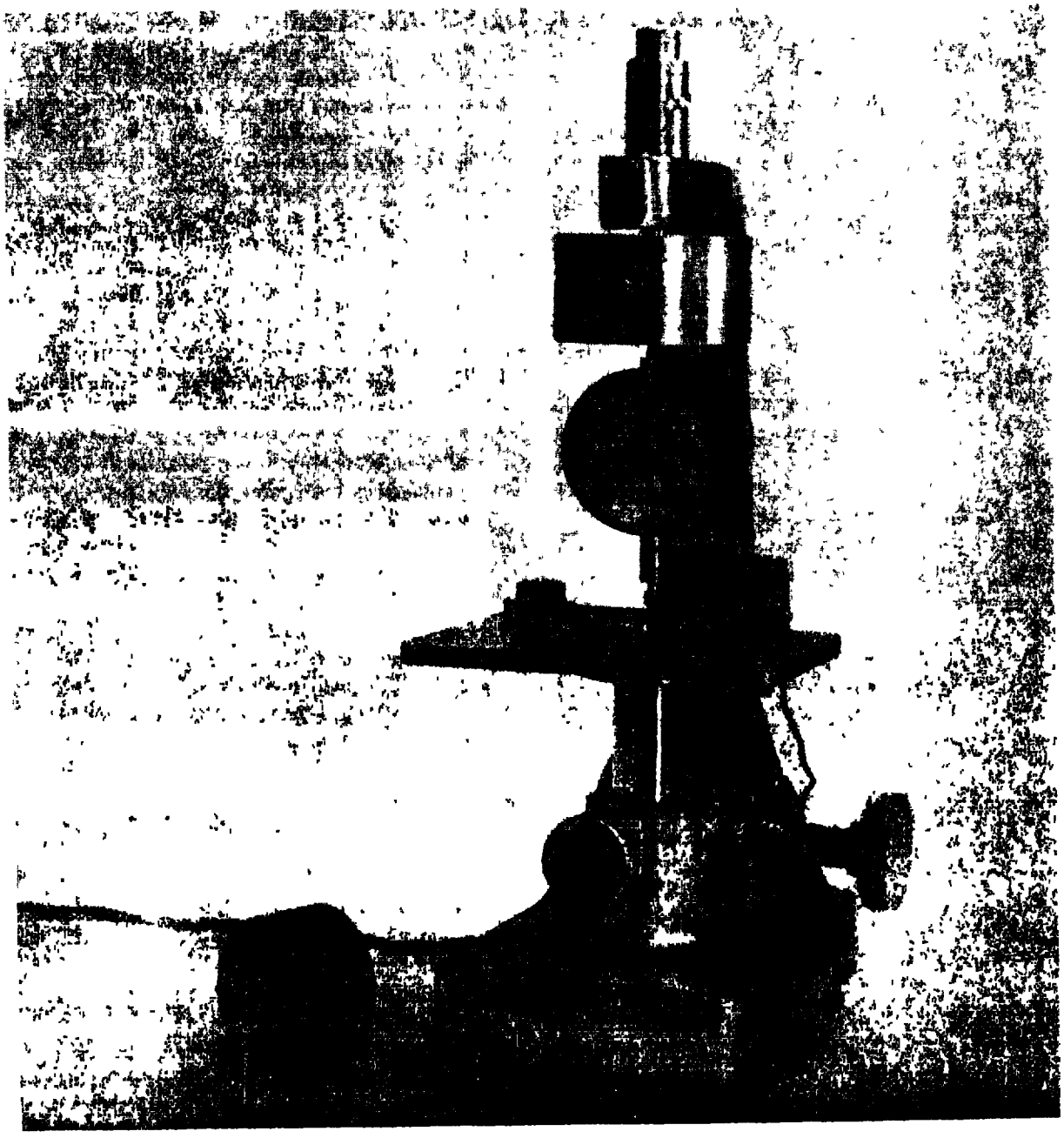

FIGURE 3051A —Indentometer, assembled

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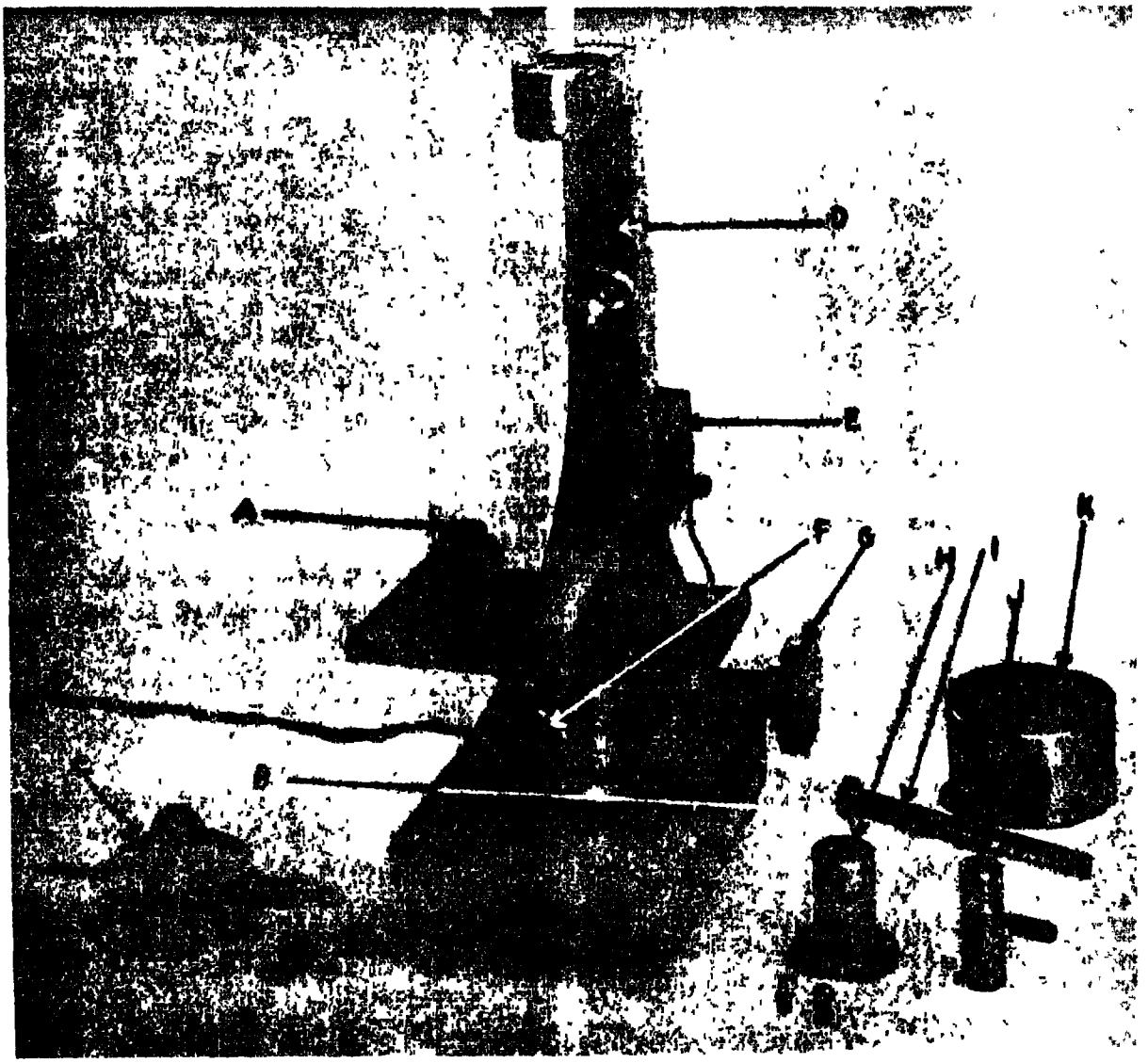

FIGURE 3051B.—Indentometer, parts

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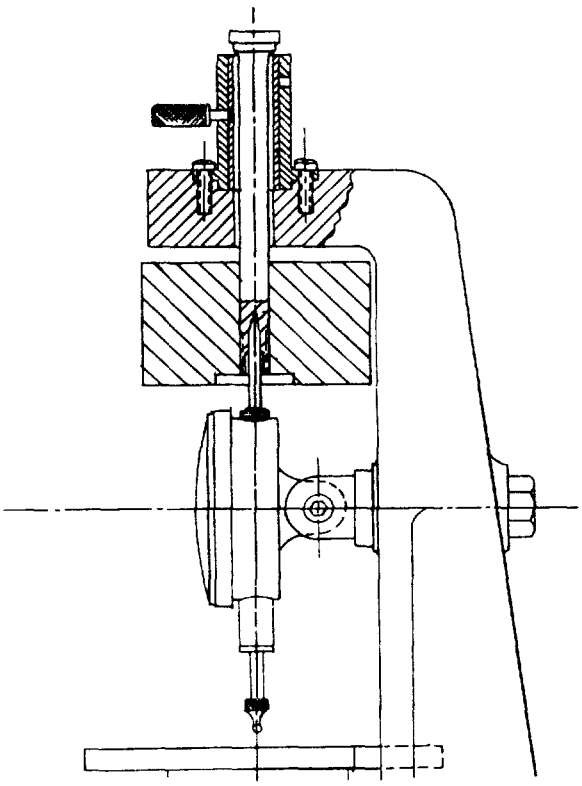

FIGURE 3051C —Cross section of major load application and indentation measurement components

4.3 The specimen shall be free from mechanical damage The apparatus shall be leveled by means of the screws and spirit level The specimen shall be mounted on the platform and the platform raised until the specimen just makes contact with the indenter point. If necessary, the 454-gram weight may be placed on the specimen to keep it flat on the platform. The minor load shall be applied by elevating the platform until the main pointer makes three complete revolutions around the dial, which is equivalent to raising the platform 3 mm The vibrator shall be started and the indicator shall be adjusted so that the zero position on the dial corresponds to the pointer position The major load shall be applied to the indenter shaft by means of the sleeve support. At the end of the required time of application of the major load, 4 1, the indentation shall be read from the dial of the gage and the value recorded to the nearest 0 01 mm The center of the spherical indenter shall be at least ½ inch from any edge of the specimen

5. RESULTS

5.1 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.2 The hardness of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see section 5

5.3 The hardness of the test unit shall be recorded to the nearest 0 01 mm

5.4 The time interval between the application of the major load and reading of the instrument shall be recorded.

5.5 If a plied-up specimen is tested, it shall be recorded

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PLASTIC FLOW

1. SCOPE

1.1 This method is intended for use in determining the deflection of soft rubber compounds under a constant force for a specified period of time

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 2 by 2 inches

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test The specimen shall be of uniform thickness within ±0 005 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An apparatus consisting essentially of a base plate, a rigid vertical shaft, a gage for registering the penetration, weights, and framework for holding the shaft and weights A satisfactory apparatus is shown in figure 3111

3.1.1.1 The base plate is made of flat sheet steel of sufficient strength to withstand the compressive stresses without deformation

3.1.1.2 *Penetrator* —The vertical shaft for penetrating the specimen is made of rigid, cylindrical metal, 1 128 to 1.129 inches in diameter (area of one square inch) The other end of the shaft is equipped to hold the required weights The shaft is supported in the frame in such a manner as to insure that the face of the foot is parallel to the surface of the base plate The edges of the foot shall be beveled smoothly but not rounded

3.1.1.3 *Indicator gage* —The dial gage is attached to the shaft in such a manner that it will register the penetration of the shaft foot into the specimen The gage shall indicate the penetration within 0 001 inch.

3.1.1.4 A set of weights sufficient to furnish the necessary force required for the test, 4 1

3.1.2 A stopwatch or other timing device that will indicate the time in seconds

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the total load, including the shaft, applied to the specimen shall be 100 pounds

4.2 If the specimen is not of uniform thickness and will not lie flat on the base plate, it shall be buffed as described in method 1111 until flat and of uniform thickness.

4.3 The specimen shall be free from mechanical damage The specimen shall be placed on the base plate with one 2- by 2-inch surface in contact with the plate in such a manner that the center of the foot of the penetrating shaft will contact the specimen at the center

4.4 The load shall be lowered gently until the foot of the penetrating shaft is just in contact with the specimen, but not compressing it, the gage set at zero, and the full required load applied in one or two seconds

4.5 Thirty ±1 second after application of the required load, the depth of penetration of the shaft into the specimen shall be read from the dial of the gage and the value recorded to the nearest 0 001 inch as the plastic flow after 30 seconds

4.6 The specimen shall be maintained in position under the required load for a total of 46½±¼ hour At the end of 22½±¼ hour under load, the depth of penetration shall be again read from the dial of the gage and the value recorded to the nearest 0 001 inch as the plastic flow of the specimen at the end of 22½ hours At the end of 46½±¼ hour under load, the depth of penetration shall be read from the dial of the gage and the value recorded to the nearest 0 001 inch as the plastic flow of the specimen after 46½ hours

5. RESULTS

5.1 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.2 The plastic flow of the test unit after 30 seconds, 22½ hours, and 46½ hours, respectively, shall be the average of the values obtained from the specimens tested

5.3 The plastic flow of the test unit after 30 seconds, 22½ hours, and 46½ hours shall be recorded separately to the nearest 0 001 inch

5.4 The total load applied to the specimen shall be recorded

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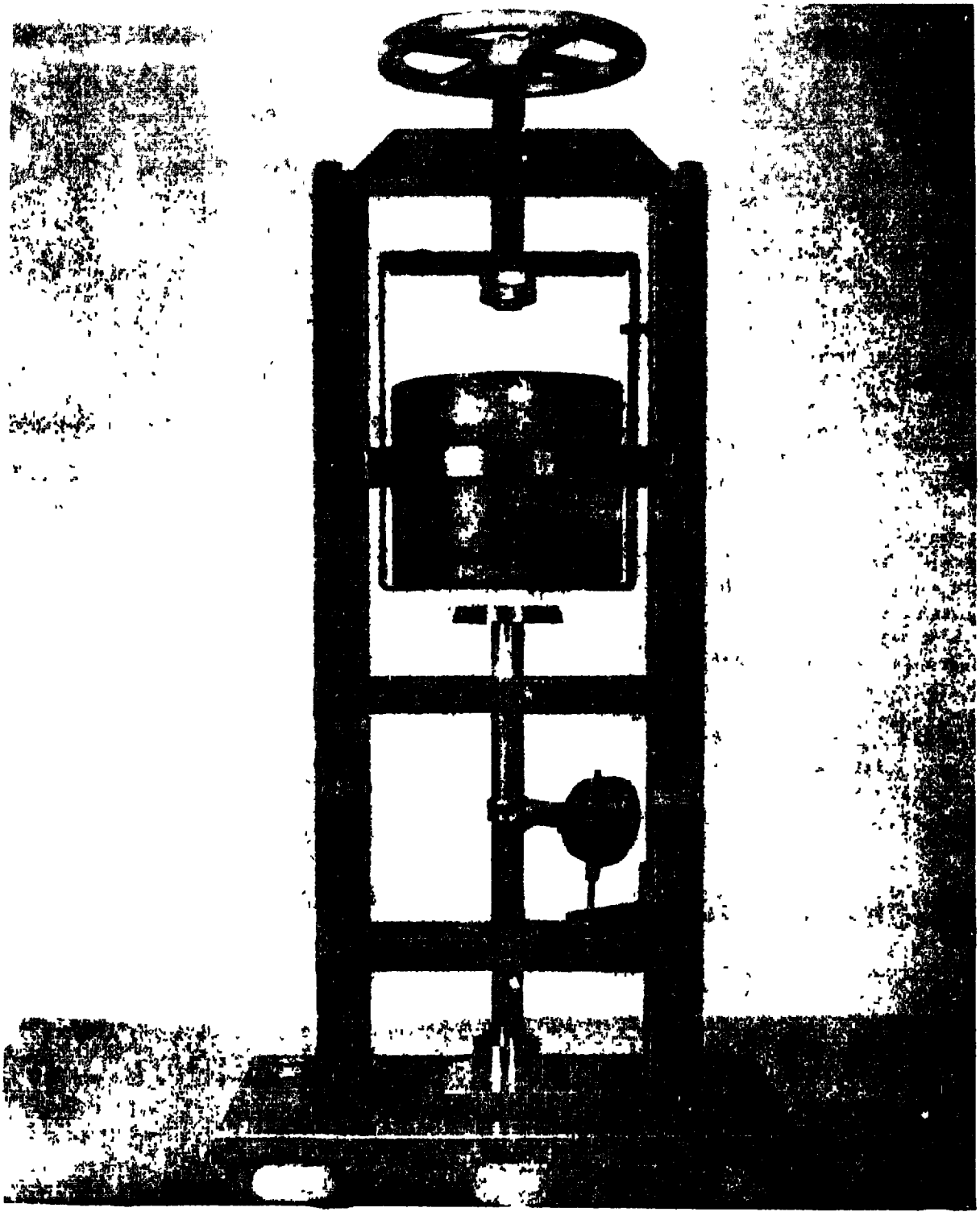

FIGURE 3111 —Plastic flow apparatus

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SEALING PRESSURE

1. SCOPE

1.1 This method is intended for use in determining the sealing pressure of soft vulcanized rubber and rubberlike materials, such as gaskets used for sealing against low air pressure The sealing pressure is determined both before and after aging at an elevated temperature The method is not applicable to materials classed as hard rubber

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1 and 2 1 2

2.1.1 The specimen shall be a circular cylinder cut with the die described in 3 1 9

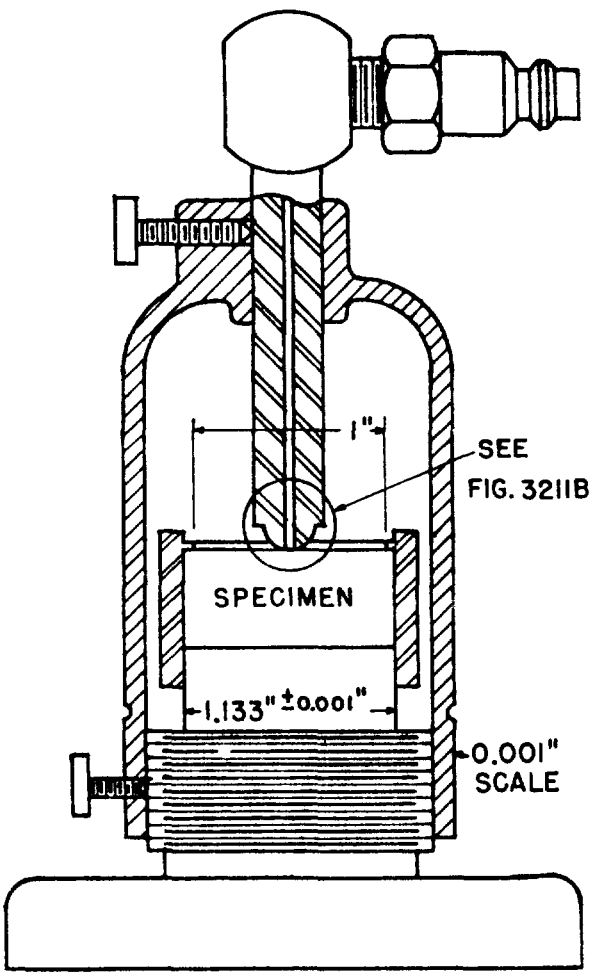

FIGURE 3211A —Indenting apparatus

2.1.2 Unless otherwise specified in the detail specification, the specimen shall be 0 50±0 01 inch in thickness, method 2011

2.2 Unless otherwise specified in the detail specification, the specimen may be composed of two plies neither of which is less than 0 24 inch in thickness

3. APPARATUS

3.1. The apparatus shall be as follows

3.1.1 *Indenting device* —An indenting device of a design capable of indenting the specimen a specified amount within ±0 0005 inch The specimen and indenter may or may not be rotated relative to each other during the indenting operation A suitable apparatus is shown in figure 3211A

3.1.2 *Indenter* —An indenter tip hemispherical in shape 0 250 +0 001, −0 000 inch in diameter having an orifice 0 040 +0 002, −0 000 inch in diameter The indenter is made of corrosion-resistant steel and the tip shall be free from scratches and surface defects The dimensions of the indenter and location of the orifice are shown in figure 3211B

3.1.3 *Specimen holder* —A specimen holder which consists of a holder proper and a base The base may screw or slide into the holder, as shown in figure 3211A The holder shall have an internal diameter of 1 133±0 001 inch and

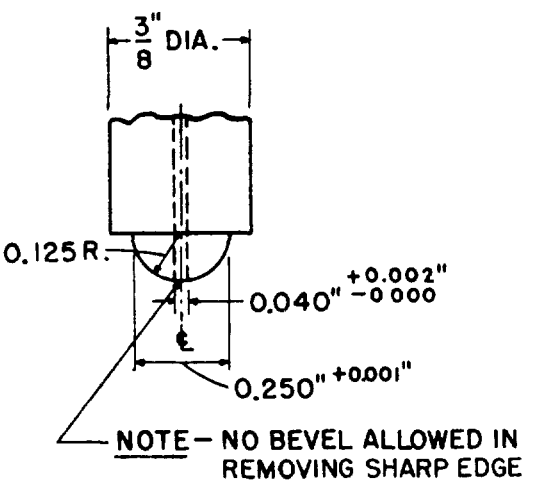

FIGURE 3211B —Indenter point

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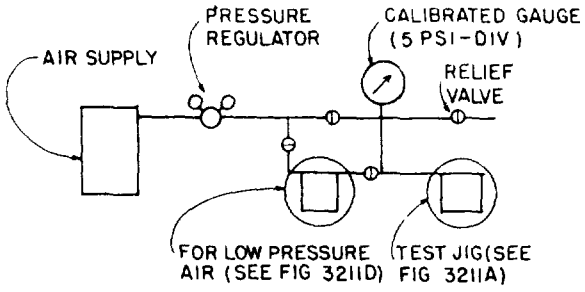

FIGURE 3211C —Air-supply apparatus

an opening 1 000 inch in diameter in the upper end The opening provides a cavity above the specimen that retains sufficient fluid on the surface of the specimen to submerge the indenter tip

3.1.4 *Air-supply apparatus* —An air-supply apparatus which consists of a source of compressed air, compressed air line, pressure-regulator valve, pressure gage, and a vent to release the air after test The pressure gage used to determine the sealing pressure shall be capable of measuring the pressure within ±2 pounds per square inch A suitable apparatus is shown in figure 3211C If a suitable air supply is not available, compressed gas cylinders of nitrogen or carbon dioxide may be used

3.1.5 *Initial leakage point apparatus* —An apparatus for determining the initial leakage point of the specimen prior to indenting which consists of a one-pound-per-square-inch air supply taken from the main air supply A suitable apparatus is shown in figure 3211D

3.1.6 *Air oven* —A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the aging period and of sufficient size to hold the indenting apparatus

3.1.7 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature within 1 0° C (1 8° F)

3 1.8 A medicine dropper or similar equipment for delivering small quantities of liquid

3.1.9 A circular metal die 1 129±0 001 inch in inside diameter for cutting the specimen

3.1.10 A stopwatch or other timing device that will indicate the time in seconds

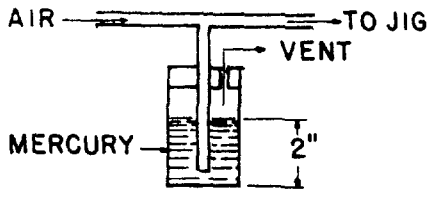

FIGURE 3211D —Initial leaking point apparatus

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be aged at a temperature of 70°±1° C (158°±1 8° F) for 46±¼ hour

4.2 Unless otherwise specified in the detail specification, the specimen shall be indented to a depth of 0 0625 inch

4.3 Preparation of specimen.—The specimen shall be cut by means of the die, 3 1 9, rotated in a drill press or similar device The pressure applied to the die shall be sufficiently small to keep the "cupping" of the cut surfaces to a minimum To facilitate cutting of the specimen with smooth-cut surfaces and square edges, the die may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0 5 percent sodium chromate or with silicone mold release emulsion before each specimen is cut The bases of the specimen shall be parallel to each other and at right angles to the axis of the cylinder

4.3.1 If the material is too thick or has an uneven surface that may interfere with the test, it shall be buffed as described in method 1111 Care shall be taken to keep the bases parallel and at right angles to the axis of the cylinder

4.4 Zero sealing point.—The specimen shall be free from mechanical damage The specimen shall be inserted all the way into the specimen holder and the holder inserted in the indenting device The indenting device shall be adjusted until the specimen just contacts the indenter The surface of the specimen shall be covered with water or alcohol and the indenter connected with the one-pound-per-square-inch air line The indenting device shall be further adjusted until

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leakage occurs at the rate of one bubble in 10 seconds at a pressure of one pound per square inch The initial indentation reading shall be recorded The indenter shall be backed off and the above procedure repeated If the second adjustment falls within 0 0005 inch of the first adjustment it shall be considered satisfactory If not, the operation shall be repeated until two successful adjustments check within 0 0005 inch

4.5 Indentation of specimen.—The specimen shall be indented to the required depth, 4 2, by adjusting the indenting device After indenting the specimen it shall be allowed to rest for 30 minutes before determining the initial sealing pressure

4.6 Initial sealing pressure.—Thirty minutes after indentation of the specimen as described in 4 5, air pressure shall be applied until profuse bubbling occurs and the pressure then decreased to zero The pressure shall be adjusted until the rate of leakage is one bubble per 10 seconds This pressure shall be recorded to the nearest 2 pounds per square inch as the initial sealing pressure

4.7 Aging.—The water or alcohol shall be removed from the specimen and the apparatus with the indented specimen in place shall be placed in the circulating-air oven and maintained for the required time at the required temperature, 4 1

4.8 Aged sealing pressure.—At the end of the aging period, the apparatus with the indented specimen in place shall be removed from the oven and the assembly allowed to cool for $3 \pm \frac{1}{4}$ hour at 23° ±3° C (73 4° ± 5 4° F) At the end of the cooling period, the sealing pressure of the specimen shall be again determined as described in 4 6 The pressure shall be recorded to the nearest 2 pounds per square inch as the aged sealing pressure

5. RESULTS

5.1 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.2 The sealing pressure of the test unit before and after aging, respectively, shall be the average of the values obtained from the specimens tested

5.3 The sealing pressure of the test unit before and after aging shall be recorded separately to the nearest 2 pounds per square inch

5.4 The time and temperature of aging and the depth of indentation shall be recorded

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COMPRESSION SET

1. SCOPE

1.1 This method is intended for use in determining the compression set of soft vulcanized rubber compounds which are to be used in such a manner that the rubber will be subjected to compression stresses or shear The method is particularly applicable to rubber compounds used in machinery mounting and vibration dampers The set is determined under constant deflection

1.2 For the purpose of this method, compression set is the residual decrease in the thickness of a specimen measured 30 minutes after removal from a suitable loading device in which the specimen has been subjected for a definite time to compressive deformation under specified conditions of load application and temperature

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1 and 2 1 2

2.1.1 The specimen shall be a circular cylinder cut with the die described in 3 1 4

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2.1.2 Unless otherwise specified in the detail specification, the specimen shall be 0 500±0 005 inch in thickness The principal faces of the specimen shall be as nearly parallel as practicable

2.2 Unless otherwise specified in the detail specification, materials thinner than ½ inch in thickness shall be built up to the required thickness by plying together thin, smooth, uniform layers of approximately equal thickness, not more than seven plies shall be used

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Compression apparatus*—A compression apparatus consisting of two or more flat steel plates with parallel faces between which the specimen may be compressed, as shown in figure 3311 The plates shall be of sufficient thickness to withstand compression stresses without bending The surface of the plates that comes in contact with the specimen shall be ground to a maximum roughness of 10 microinches, chromium plated, and polished The surfaces of the plates shall be thoroughly cleaned before each use

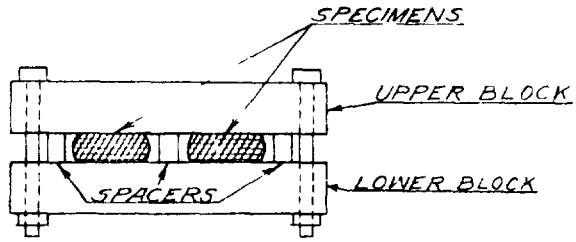

FIGURE 3311 —Compression apparatus

3.1.1.1 The compression apparatus is equipped with steel spacers equal in height to the thickness the specimen is to be compressed as required in 4 2

3.1.2 *Air oven*—A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the exposure period and of sufficient size to hold the compression apparatus

3.1.3 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature within 1° C (1 8° F)

3.1.4 A circular metal die 1 129±0 001 inch in inside diameter for cutting the specimen

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be aged at a temperature of 70°±1° C (158°±1 8° F) for 22±¼ hour

4.2 Unless otherwise specified in the detail specification, the deflection of the specimen shall be in accordance with the following table

Durometer hardness, method 3021	Deflection, percent of original thickness
44 and less	40
45 to 64, inclusive	30
65 to 84, inclusive	25
85 and over	20

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4.3 Preparation of specimen.—The specimen shall be cut by means of the die rotated in a drill press or similar device The pressure applied to the die shall be sufficiently small to keep "cupping" of the cut surfaces to a minimum To facilitate cutting of the specimen with smooth-cut surfaces and square edges, the die may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0 5 percent sodium chromate or with silicone mold release emulsion before each specimen is cut The bases of the specimen shall be parallel to each other and at right angles to the axis of the cylinder

4.3.1 If the material is too thick or has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111 Care shall be taken to keep the bases parallel and at right angles to the axis of the cylinder

4.4 The specimen shall be free from mechanical damage The thickness of the specimen shall be determined as described in method 2021, except that the determination shall be made in such a manner as to indicate the perpendicular distance between the center of the top and bottom faces and the value recorded to the nearest 0 001 inch as t_o

4.5 The specimen shall be placed between the plates of the compression apparatus with the required spacer, 4 2, on each side If 1 specimen is tested, it shall be placed in the center If 2 specimens are tested, 3 spacers shall be used, 1 in the center and 1 on the outside of each specimen, allowing sufficient clearance for bulging of the specimen when compressed The bolts shall be tightened so that the plates are brought together uniformly until the plates are in contact with the spacers Only 1 specimen from a given test unit shall be compressed at one time in the same compression apparatus

4.6 The compression apparatus with specimen under compression shall be placed in the air oven within 2 hours after completion of the assembly and heated for the required time at the required temperature, 4 1

4.7 At the end of the exposure period, the compression apparatus shall be removed from the oven and the specimen removed immediately and allowed to cool for 30 minutes At the end of the cooling period, the thickness of the specimen shall again be determined as described in 4 4 and the value recorded to the nearest 0 001 inch as t_{30}

5. RESULTS

5.1 Calculation.—The compression set of the specimen, as percent of the original deflection, shall be calculated as follows

$$\text{Compression set, percent of original deflection,} = \frac{t_o - t_{30}}{t_o - t_s} \times 100$$

Where

t_o is the thickness of specimen before compression, inch

t_{30} is the thickness of specimen 30 minutes after removal from compression, inch

t_s is the thickness of the spacers used, inch

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The compression set of the test unit shall be the average of the values obtained from the specimens tested

5.4 The compression set of the test unit shall be recorded to the nearest one percent

5.5 If a plied-up specimen is tested, it shall be recorded

5.6 The percent deflection of the specimen shall be recorded

5.7 The temperature and time of exposure shall be recorded

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COMPRESSION AND RECOVERY

1. SCOPE

1.1 This method is intended for use in determining the resistance of soft vulcanized rubber compounds to compression when subjected to a known force for a definite period of time, and the ability of the material to return to its original thickness in a specified period of time, after removal of the compression force It is particularly applicable to testing of gaskets and gasket materials

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1 and 2 1 2

2.1.1 The specimen shall be a circular cylinder cut with the die described in 3 1 2

2.1.2 Unless otherwise specified in the detail specification, the specimen shall be 0 25±0 01 inch in thickness

3. APPARATUS

3.1 The apparatus shall be as follows:

3.1.1 *Compression apparatus*—A compression testing machine capable of producing a force of at least 1,000 pounds per square inch ±1 percent on the specimen One of the anvils shall preferably be self-aligning and arranged so that the specimen is automatically centered and the resultant of the force is through its center The machine shall be equipped with a dial or scale capable of measuring the distance between the anvils to 0.001 inch or less A testing machine of the kind shown in figure 3321 is suitable for the purpose

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3.1.1.1 The apparatus described in figure 3321 consists essentially of a lever arrangement capable of producing a relatively large force on the specimen which is held between two anvils The machine has a lever ratio of 10 to 1 and the diameter of each anvil is greater than that of the specimen

3.1.2 A circular metal die 1 129±0 001 inch in inside diameter for cutting the specimen

3.1.3 A stopwatch or other timing device that will indicate the time in seconds

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, a total load of 1,000±10 pounds shall be applied to the specimen

4.2 Preparation of specimen.—The specimen shall be cut by means of the die rotated in a drill press or similar device The pressure applied to the die shall be sufficiently small to keep "cupping" of the cut surfaces to a minimum To facilitate cutting of the specimen with smooth-cut surfaces and square edges, the die may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0 5 percent sodium chromate or with silicone mold release emulsion before each specimen is cut The bases of the specimen shall be parallel to each other and at right angles to the axis of the cylinder

4.2.1 If the material is too thick or has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111 Care shall be taken to keep the bases parallel and at right angles to the axis of the cylinder

4.3 The specimen shall be free from mechanical damage The specimen shall be mounted between the anvils of the testing machine A load of 25±1 pound shall be applied to the specimen for a period of 15±1 second At the end of the 15-second period, the distance between the anvils shall be read from the dial or scale and the value recorded to the nearest 0 001 inch as the thickness of the specimen, T_1.

4.4 After the thickness of the specimen has been determined, additional loads shall be applied so as to compress the specimen at the rate of 0 5 inch per minute, as measured from the dial or scale, until the required total load, 4 1, has been applied The required load shall be held for 30±1 second, the distance between the anvils read from the dial or scale, and the value recorded to the nearest 0 001 inch as the thickness of the specimen, T_2

4.5 At the end of the 30-second period, the force shall be immediately removed and the specimen allowed to recover for 60±1 second At the end of the one-minute recovery period, a force of 25±1 pound shall be applied to the

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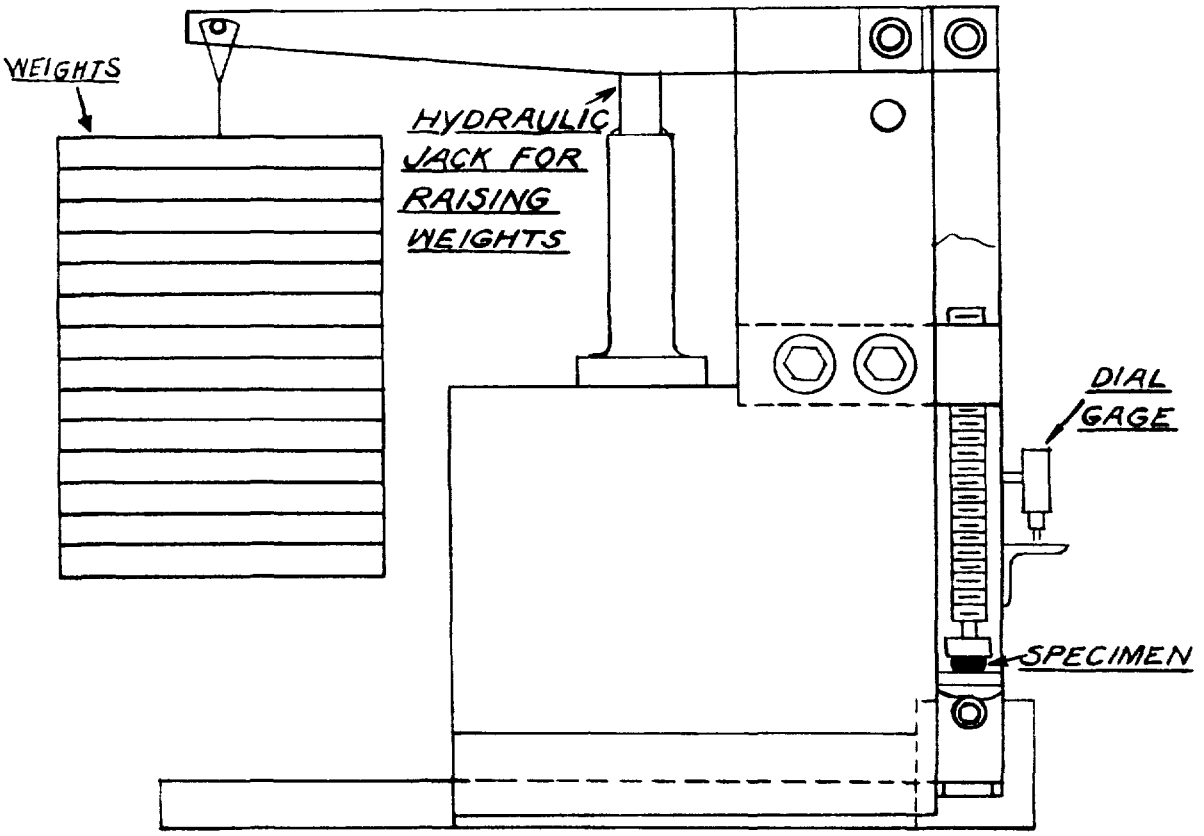

FIGURE 3321 —Compression testing machine

specimen for a period of 15 ± 1 second At the end of the 15–second period, the distance between the anvils shall be read from the dial or scale and the value recorded to the nearest 0 001 inch as thickness of the specimen, T_3

5. RESULTS

5.1 Calculations.—The compression and recovery of the specimen shall be calculated as follows

$$\text{Compression, percent,} = \frac{T_1 - T_2}{T_1} \times 100$$

$$\text{Recovery, percent,} = \frac{T_3}{T_1} \times 100$$

Where
T_1 is the thickness of the specimen under 25-pound load, inch
T_2 is the thickness of the specimen under the required total load, inch
T_3 is the thickness of the specimen under 25-pound load after recovery for one minute, inch

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit unit shall be tested

5.3 The compression and recovery, respectively, of the test unit shall be the average of the values obtained from the specimens tested

5.4 The compression and recovery of the test unit shall be recorded separately to the nearest one percent

5.5 The total load applied to the specimen shall be recorded

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COMPRESSIBILITY AND RECOVERY, GASKET MATERIALS

1. SCOPE

1.1 This method is intended for use in determining the resistance of gaskets and gasket materials to compression when subjected to a known force for a short time at room temperature, and the ability of the material to return to its original thickness in a specified time after removal of the compression force The method is not intended for use in determining the resistance of materials to compression under prolonged stress, generally referred to as "creep," or for determining recovery after prolonged stress, the inverse of which is generally designated as "compression set" This method is not restricted to the testing of rubber compositions, but may be used for cork, paper, fiber, asbestos, and other gasket materials

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit, either circular or square in shape, of the dimensions given in 2.1 1 or 2 1 2

2.1.1 *Cork compositions* —The specimen shall have a minimum area of 4 square inches and a thickness of at least ⅛ inch

2.1.2 *Other materials* —The specimen shall have a minimum area of one square inch and a thickness of at least 1/16 inch, except that rubber compositions containing cork shall have a thickness of at least ⅛ inch

2.1.3 The principal faces of the specimen shall be as nearly parallel as practicable.

2.2 Unless otherwise specified in the detail specification, materials less than the required thickness shall be plied up to obtain the required thickness

3. APPARATUS

3.1 The apparatus shall be as follows.

3.1.1 *Testing machine* —A testing machine consisting essentially of the following parts

3.1.1.1 An anvil with hardened and ground surface having a minimum diameter of 1 250 inch

3.1.1.2 *Penetrator* —A penetrator consisting of a steel cylinder with the end hardened and ground smoothly The diameter of the cylinder shall be as required in table I for the material to be tested, within ±0 001 inch

3.1.1.3 *Indicator gage* —A dial gage so located that it will indicate within ±0 001 inch the thickness of the specimen when compressed by the preload and the preload plus the major load

3.1.1.4 *Loading devices*

3.1.1.4.1 *Preload* —A device for applying the preload required in table I within ±1 percent The preload shall include the weight of the penetrator, 3 1 1 2

3.1.1.4.2 *Major load* —A device for applying the major load required in table I to the penetrator in a period of 10 seconds without removing the preload The major load may be applied by an arrangement of dead weights, hydraulic cylinder, air cylinder, or other device capable of applying the major load at a uniform rate and to within ±1 percent of that required in table I for the material undergoing test

3.1.2 A stopwatch or other timing device that will indicate the time in seconds

3.1.3 *Air oven* —A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the conditioning period for the higher conditioning temperatures

3.1.4 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature within 1 0° C (1 8° F)

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the various types of material shall be preconditioned as required in table I before testing for compressibility and recovery

4.2 Unless otherwise specified in the detail specification, the size of the penetrator used for each type of material, and the preload and major load applied to each material shall be as required in table I

4.3 The specimen shall be free from mechanical damage Immediately after preconditioning as required in 4 1, the specimen shall be centered on the anvil of the testing machine, the preload, table I, applied and maintained for

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TABLE I

Gasket material	Preconditioning of specimen	Penetrator diameter, ±0 001 inch	Preload	Major load
		Inches	*Pounds*	*Pounds*
Rubber-cork compositions	22±¼ hr at 27°±5° C (80°±9° F), 50±4 percent relative humidity	0 504	1	79
Rubber, cork, and fiber compositions	60±5 min at 100°±2° C (212°±3 6° F) Cool in desiccator	252	5	45
Rubber-fiber compositions, other than compressed asbestos sheet	60±5 min at 100°±2° C (212°±3 6° F) Cool in desiccator	252	1	49
Other rubber compositions of 70 durometer hardness or higher	22±¼ hr at 27°±5° C (80°±9° F), 50±4 percent relative humidity	252	1	49
Compressed asbestos sheet	60±5 min at 100°±2° C (212°±3 6° F) Cool in desiccator	252	5	245
Asbestos fiber compositions	4±1/10 hr at 100°±2° C (212°±3 6° F) Cool in desiccator	252	1	49
Cork compositions	46±¼ hr at 27°±5° C (80°±9° F), 50±4 percent relative humidity	1 129	1	99
Treated and untreated papers	22±¼ hr at 27°±5° C (80°±9° F), 50±4 percent relative humidity	252	1	49

15±1 second At the end of the preload time, the thickness of the specimen shall be read from the dial of the gage and the value recorded to the nearest 0 001 inch as T_1

4.4 Immediately after the thickness, T_1, is determined, the major load, table I, shall be applied to the specimen, still under preload, at a slow, uniform rate so that the total load (major load plus preload) is impressed on the specimen within 10±1 second During the time of application of the load, the penetrator shall be guided so that the surface of the end remains parallel to the surface of the anvil The total load shall be maintained on the specimen for 60±1 second At the end of the 60-second period, the thickness of the specimen shall be read from the dial of the gage and the value recorded to the nearest 0 001 inch as T_2

4.5 Immediately after the thickness, T_2, is determined, the major load shall be removed and the specimen allowed to remain under the preload for 60±1 second At the end of the 60-second period under the preload, the thickness of the specimen shall be read from the dial of the gage and the value recorded to the nearest 0 001 inch as T_3

5. RESULTS

5.1 Calculations.— The compressibility and recovery of the specimen shall be calculated as follows

$$\text{Compressibility, percent,} = \frac{T_1 - T_2}{T_1} \times 100$$

$$\text{Recovery, percent,} = \frac{T_3 - T_2}{T_1 - T_2} \times 100$$

Where

T_1 is the thickness of the specimen under preload, 4 3, inch

T_2 is the thickness of the specimen under total load, 4 4, inch

T_3 is the thickness of the specimen under preload, 4 5, inch

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.3 The compressibility and recovery, respectively, of the test unit shall be the average of the values obtained from the specimens tested

5.4 The compressibility and recovery of the test unit shall be recorded separately to the nearest one percent

5.5 The time, temperature, and humidity of conditioning, the preload, the major load, and size of the penetrator shall be recorded

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RESILIENCE, OSCILLOGRAPH

1. SCOPE

1.1 This method is intended for use in determining the resilience of elastomeric vulcanizates in compression No significant test can be made of materials (hard rubber, high-durometer vulcanizates, etc) whose moduli at the temperature of test are such that a force of 280-pound-per-square-inch compression (on standard specimen) will not produce at least 10-percent deformation and/or whose damping characteristics are such that less than 3 full cycles are produced when obtaining the damped sinusoidal curve

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1 and 2 1 2

2.1.1 The specimen shall be a circular cylinder cut with the die described in 3 1 3

2.1.2 Unless otherwise specified in the detail specification, the specimen shall be 0 50 + 0 01 inch in thickness, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An apparatus consisting of a balance beam, pen, and weights A suitable apparatus is shown in figures 3411A and 3411B

3.1.1.1 *Balance beam* —The beam is supported at the center by knife edges It is designed so that a specimen placed beneath the micrometer A of figure 3411B is loaded by placing weights alternately on opposite sides of the pen end of the balance beam A second knife edge and a stabilizing arm are used to maintain parallelism of the loading platens

3.1.1.2 *Pen* —The pen is supported on an arm extending from the end of the beam so as to record automatically the deflections on a chart or oscillogram From figure 3411B, it is apparent that the deflection of the specimen

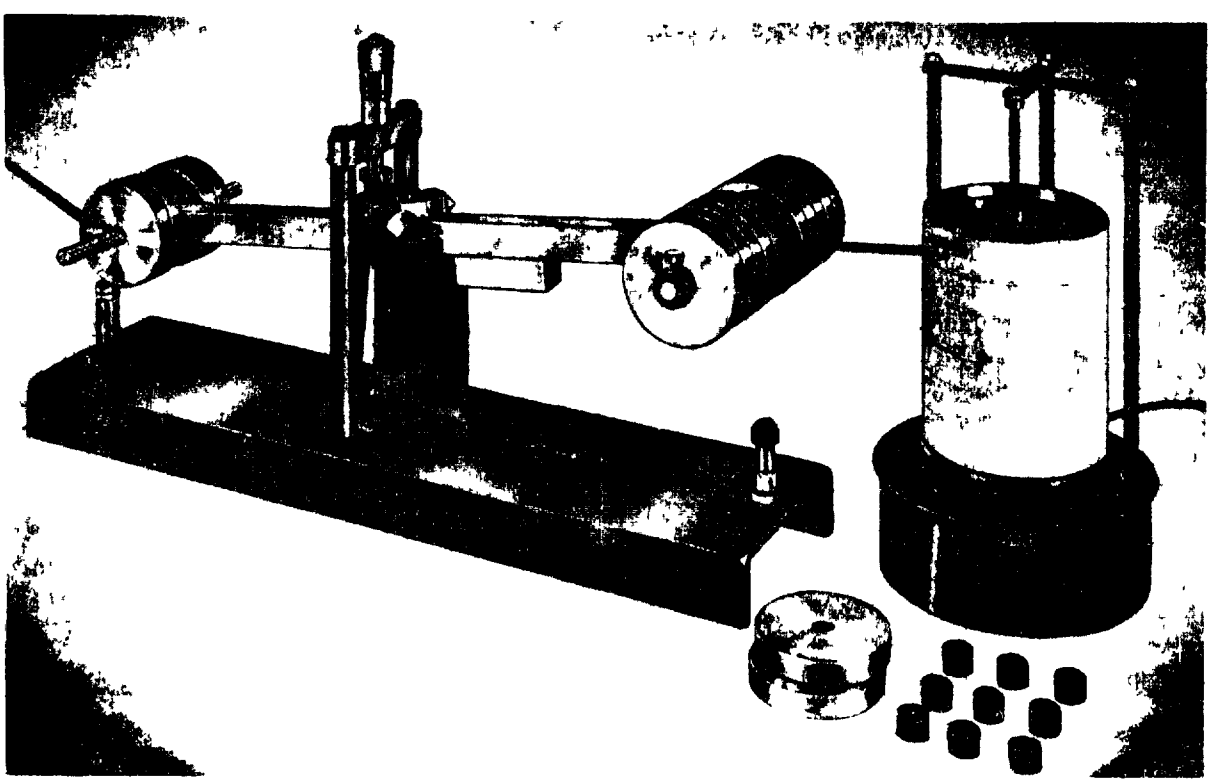

FIGURE 3411A —Oscillograph

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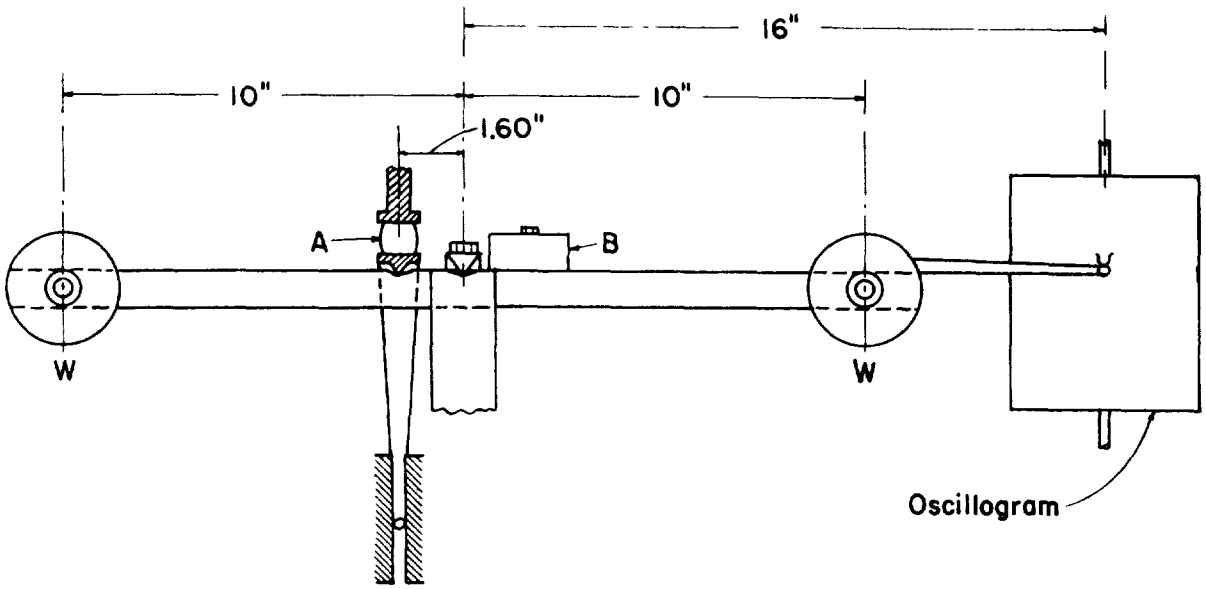

FIGURE 3411B —Diagrammatic sketch of oscillograph

will be magnified in proportion to the lever ratio, which shall be 10 to 1 (16 to 1 6 inches), and a deformation of 0 10 inch will be registered on the oscillogram as a vertical displacement of 1 00 inch

3.1.1.3 *Weights* —The weights, *W*, figure 3411B, consist of several disks each weighing 1 410 pounds and one disk weighing 0 705 pound The lever ratio for the weights is 6 25 to 1 (10 to 1 60 pounds) Each unbalanced 1 410-pound weight on the pen end of the beam therefore will result in a force of 8 82 pounds on the specimen Normally, the oscillograph is operated with weights only on the pen end of the beam, however, the moment of inertia of the system may be increased, with a corresponding decrease in the natural frequency of the system, by adding weights to the hook end of the beam

3.1.1.4 The beam may be checked for friction and balance by removing all weights and the specimen, and disengaging the hook In this condition, the beam should oscillate freely indicating minimum friction at the knife edges and at the stabilizers of the lower platen When the beam comes to rest, the pen end should be below the horizontal This unbalanced condition helps overcome the friction between pen and graph paper The beam shall be kept slightly overbalanced by adjusting the counterweight to apply a positive load to the specimen when all weights are removed The knife edges should be cleaned with cheesecloth and occasionally lubricated with typewriter oil to minimize friction and wear

3.1.2 An engineering grade of graph paper ruled in one-inch squares and then subdivided into 10 or 20 equal squares per inch

3.1.3 A circular metal die 0 75±0 005 inch inside diameter for cutting the specimen

4. PROCEDURE

4.1 Preparation of specimen.—The specimen shall be cut by means of the die rotated in a drill press or similar device The pressure applied to the die shall be sufficiently small to keep "cupping" of the cut surfaces to a minimum To facilitate cutting of specimen with smooth-cut surfaces and square edges, the die may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0 5 percent sodium chromate, or with silicone mold release emulsion before each specimen is cut The bases of the specimen shall be

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parallel to each other and at right angles to the axis of the cylinder

4.1.1 If the material is too thick or has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111 Care shall be taken to keep the bases parallel and at right angles to the axis of the cylinder

4.2 The balance beam of the oscillograph shall be locked in position by means of the spring-loaded hook at the left end of the machine and all weights removed The hook shall be adjusted so that the static equilibrium position of the balance beam will be approximately horizontal when the specimen is under the test deflection desired

4.3 The specimen shall be free from porosity and mechanical damage The specimen shall be placed in the center of the lower platen between the grit side of 2 pieces of 400-grit A sandpaper The micrometer shall be adjusted until the upper platen rests snugly against the sandpaper without deforming the specimen and the micrometer locked in this position by means of the set screw

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4.4 The graph paper shall be placed on the chronograph drum and its position adjusted so that the zero position of the penpoint is on one of the horizontal lines of the paper

4.5 The hook shall be disengaged and sufficient pressure applied by hand on the pen end of the beam to compress the specimen to 30 percent of the required deformation (1 5 inches on the graph for a specimen 0 50 inch in height) and released This operation shall be repeated at least three times to remove any trapped air from the specimen

4.6 With the hook still disengaged, the chart shall be rotated by turning the chronograph drum to the left to displace the chart 4 to 5 small divisions to the left of the penpoint thus marking zero deflection Weights shall then be added to the pen end of the beam sufficient to produce 20 ± 2 percent deformation of the specimen (one inch on the chart)

4.7 The hook shall be reengaged and the drum rotated by hand to the left approximately one small square of the oscillogram and the hook disengaged This operation shall be repeated a sufficient number of times to obtain three successive lines of the same length After the last oscillation, the penpoint should indicate 20 ± 2 percent deformation of the specimen (one inch on the chart)

4.8 After obtaining three successive lines of the same length, the chronograph shall be started with the drum rotating at a speed of four revolutions per minute The hook shall be disengaged and a set of oscillations recorded If the vertical length of the first oscillation is shorter than the length of the last conditioning line, there has been excessive time between successive trials and further conditioning as necessary shall be performed until a satisfactory test is obtained

5. RESULTS

5.1 Calculation.—The resilience of the specimen shall be calculated as follows

$$\text{Resilience, percent,} = \frac{A}{B} \times 100$$

Where

A is the vertical rebound height, first cycle, inch
B is the preceding vertical impact fall, inch

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The resilience of the test unit shall be the average of the values obtained from the specimens tested

5.4 The resilience of the test unit shall be recorded to the nearest one percent

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TENSION TESTS, GENERAL

1. SCOPE

1.1 This group of tests is intended for use in determining the effect of application of a tension load to soft vulcanized rubber compounds and similar rubberlike materials Methods for determining tensile strength, elongation, tensile stress, strain, tear resistance, strength of splice, and tension set are described These methods are not applicable to the testing of rubber compounds ordinarily known as hard rubber

2. DEFINITIONS

2.1 Tensile strength.—Tensile strength is the force per unit of the original cross-sectional area of the unstretched specimen which is applied at the time of rupture of the specimen It is calculated by dividing the breaking force in pounds by the cross section of the unstretched specimen in square inches For example, if a specimen 0 25 by 0 10 inch in cross section broke at a force of 50 pounds, the tensile strength would be 50 divided by (0 25×0 10) which is equal to 2,000 pounds per square inch When a stress is applied to a specimen of soft vulcanized rubber, the rubber stretches to a high degree with an accompanying reduction in cross-sectional area of the specimen Since precise measurements of the cross-sectional area of the specimen cannot be made at the moment the specimen breaks, calculation of the tensile strength at break is based on the cross-sectional area of the specimen before application of any force

2.2 Elongation.—Elongation is the extension between benchmarks produced by a tension force applied to the specimen, and is expressed as a percentage of the original distance between the marks on the unstretched specimen Ultimate elongation is the elongation at the moment of rupture For example, if a one-inch length is marked on an unstretched specimen and the specimen is stretched until the marks are 7 inches apart, elongation is 7—1 = 6 inches, or 600 percent

2.3 Tensile stress.—Tensile stress is the force per unit of the original cross-sectional area of the unstretched specimen required to stretch the specimen to a stated elongation It is expressed in pounds of tension force per square inch at the stated elongation, for example, 1,000 pounds per square inch at 500 percent elongation The tensile stress for a specified elongation is often designated in rubber technology by the term "modulus"

2.4 Strain.—Strain is the elongation produced during a given period of time by a specified force per unit of cross-sectional area of the unstretched specimen

2.5 Tear resistance.—Tear resistance is the ratio of the maximum force applied during tear of the specimen to the thickness of the unstretched specimen For example, if a force of 36 pounds is necessary to tear a specimen 0 082 inch in thickness, the tear resistance is 36 divided by 0 082, or 439 pounds per inch of thickness

2.6 Strength of splice.—Strength of splice is the force per unit of width necessary to separate the union of two pieces of soft vulcanized rubber

2.7 Tension set.—Tension set is the elongation remaining after a specimen has been stretched and held at a specified elongation for a definite period of time, then relieved of the force and allowed to rest for a definite period of time It is expressed as a percentage of the original length or distance between benchmarks on the unstretched specimen For example, a specimen is stretched from 1 to 5 inches for a period of 10 minutes and then released The length after the 10-minute rest is 1 20 inches, so that the set under these conditions is 0 20 inch or 20 percent

2.8 Median.—If numerical values are arranged in an ascending or descending order, the median is obtained as follows

2.8.1 If the number of values is odd, the median is the value in the middle of the series

2.8.2 If the number of values is even, the median is the arithmetic average of the two middle values

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3. SPECIMEN

3.1 Types.

3.1.1 Three types of specimens are used for the determination of tensile strength, elongation, tensile stress, and tension set These are dumbbell, ring, and straight specimens A dumbbell specimen, as described in method 4111, shall be used whenever it is practicable to prepare it from the test unit A ring specimen shall be used when the material is in the form of a ring of convenient size, such as jar rings or rubber bands Ring specimens shall also be used when it is practicable to prepare a ring but not a dumbbell as in the case of finger cots from which a ring may be cut by means of the constricted portion of one of the dies used in cutting the dumbbell specimen Straight specimens in the form of tubes or rods shall be used when it is not practicable to prepare either a dumbbell or a ring as in the case of small tubing or thread

3.1.2 Specimens for use in determining tear resistance are shown in methods 4211 and 4221

3.2 Direction.—Specimens from products made in long lengths, such as dredging sleeves, hose, tubing, or belting, shall be taken with the long dimension of the specimen parallel to the length direction, except that in the case of belts wider than 12 inches and hose or dredging sleeves greater than 4 inches in inside diameter, the specimen shall be taken with the long dimension in the transverse direction Specimens of sheet material, boots, etc , may be taken in any direction

4. PROCEDURE

4.1 When tensile strength, elongation, and tensile stress are specified in the detail specification, these tests shall be made on the same specimen during a single test

4.1.1 The number of specimens required for determining any one of the properties in 4 1 shall be used for determining the other two properties

4.2 Buffing.—If the material is too thick or has a fabric backing or an uneven surface that may interfere with the tension test, the material shall be buffed as described in method 1111 Portions of the test unit from which specimens are to be taken for tension tests shall be buffed in the strip form before cutting with a die

4.3 Cutting of specimen.—To facilitate cutting of the specimen, the edge of the die may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0.5 percent sodium chromate or with silicone mold release emulsion before each specimen is cut

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TENSILE STRENGTH

1. SCOPE

1.1 This method is intended for use in determining the tensile strength of soft vulcanized rubber compounds

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1, 2 1 2, or 2 1 3

2.1.1 *Dumbbell specimen* —Whenever it is practicable to prepare it from the test unit undergoing test, a dumbbell-shaped specimen cut by one of the dies shown in figure 4111 shall be used When the material to be tested permits, the reduced section of the specimen shall be ¼ inch in width and either 1⅝ inches or 2⅝ inches in length Specimens having reduced sections ¼ inch shorter may be used until July 1, 1956 When the material to be tested is narrow, a specimen with a reduced section ⅛ inch in width is permitted When the material to be tested permits, the enlarged ends of the specimen shall be one inch in width When the material is narrow, the enlarged ends of the specimen may be ⅝ inch in width The thickness of the specimen shall not exceed 0 125 inch

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2.1.2 *Ring specimen* —A ring specimen shall be used when the material is in the form of a ring of convenient size, as for example, jar rings, rubber bands, or parachute rings A ring specimen shall also be used when it is possible to prepare a ring but not a dumbbell specimen, as for example, finger cots, in which case the width of the specimen shall be ¼ inch Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test

2.1.3 *Straight specimen* —A straight specimen in the form of a tube or a rod shall be used when it is not practicable to prepare either a dumbbell or a ring specimen from the test unit, as in the case of a narrow rubber strip, small size tubing, and thread Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Testing machine* —A testing machine which meets the following requirements

3.1.1.1 Power driven

3.1.1.2 The applied tension is indicated to within ±1 percent by a dial, scale, or automatic recorder when calibrated according to method 4116

3.1.1.3 The indicator remains at the point of maximum force after rupture of the specimen

3.1.1.4 The rate of travel of the power-actuated grip is 20±1 inch per minute and uniform at all times

3.1.1.5 The machine, when used for a given specimen, is of such capacity that the maximum force required to break the specimen is not greater than 85 percent nor less than 15 percent of the rated capacity

3.1.1.6 If the machine is equipped with a dynamometer head of the compensating type for convenience in eliminating calculations, the head shall have adjustments for variation in thickness of the specimen

3.1.1.7 *Grips*

3.1.1.7.1 *Dumbbell specimen* —For a dumbbell specimen, the grips shall be of a type which tightens automatically as the applied tension increases and which exerts a uniform pressure across the gripping surfaces so as to avoid uneven slipping and to favor failure of the specimen in its reduced section

3.1.1.7.2 *Ring specimen* —For a ring specimen, each grip consists of two rollers, each with a minimum diameter of ¼ inch and a maximum length of ½ inch mounted on ball-bearings The action of these rollers is such that as tension is applied they rotate freely in opposite directions, thus tending to equalize the stress around the periphery of the specimen

3.1.1.7.3 *Straight specimen* —For a straight specimen, either wedge or toggle grips are used The distance between the grips at the start of the test shall be 3±¼ inch

3.1.2 Dies.

3.1.2.1 *Dumbbell specimen* – Metal dies of

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the shape and construction shown in figure 4111 and of the dimensions shown in the accompanying legend, for cutting dumbbell specimens, 2 1 1 It is recommended that the reduced section of the die be equipped throughout its entire length with a spacer to maintain a definite distance between the cutting edges The spacers are held by at least two bolts through the body of the die The difference between the minimum and maximum distance between the cutting edges of the die in the reduced section shall not exceed 0 002 inch The inside faces in the reduced section are polished and perpendicular to the plane formed by the cutting edges for a depth of at least 0 25 inch The angle between the inner and outer faces at the cutting edge shall be not less than 30° and not more than 35° The outer face extends at this angle from the cutting edge for approximately ¼ inch and forms an angle between 18° and 22° with the inner face

3.1.2.2 *Ring specimen* —Metal dies for cutting ring specimens described in 2 1 2 with dimensions which permit the use of the spool grips described in 3 1 1 7 2

3.1.2.3 *Straight specimen* —Metal dies or other suitable apparatus for cutting straight specimens, 2 1 3, with dimensions which permit their installation in the wedge or toggle grips described in 3 1 1 7 3

3.1.2.4 The dies shall be sharp and free from nicks in order to prevent leaving ragged cut edges on the specimen The cutting edges of the die may be kept sharp by daily light honing and touching up with jeweler's hard Arkansas honing stones The condition of the die may be judged by investigating the rupture point of any series of broken specimens When broken specimens are removed from the grips of the testing machine, it is advantageous to pile them and note if there is any tendency to break at or near the same portion of each specimen Rupture points consistently at the same place on the specimen may indicate that the die is dull, nicked, or bent at that particular position

3.1.3 Cutting support.—A smooth, slightly yielding surface for supporting a portion of the test unit from which the specimen is to be cut so that the blade of the die is not injured during cutting of the specimen

3.1.3.1 It is recommended that the cutting support have three parts (1) a solid foundation, such as hardwood or plate glass, (2) a semihard pad, such as plastic, Masonite, etc , and (3) a smooth cutting surface, such as rubber belting, leather belting, light cardboard, etc

3.1.4 A scale or tape graduated to 0 1 cm

4. PROCEDURE

4.1 Preparation of specimen.

4.1.1 *Buffing.*—If the material is too thick or has a fabric backing or an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111 The portion of the test unit from which the specimen is to be taken shall be buffed in the strip form before cutting with the die

4.1.2 *Dumbbell specimen* —The portion of the test unit from which the specimen is to be cut shall be placed on the cutting support and the specimen of the required shape and size cut by means of the required die The cut edges shall be perpendicular to the other surfaces of the specimen, have a minimum of concavity, and be free from ragged edges

4.1.3 *Ring specimen* —In general, the ring specimen will consist of the article as received and no preparation of the specimen is necessary In the case of finger cots and similar articles, rings shall be cut by the procedure described in 4 1 2, except that only the reduced section of the die shall be used and the cut shall be made through both surfaces of the material perpendicular to the length axis Unless otherwise specified in the detail specification, the width of the reduced section of the die used shall be ¼ inch In the case of flat material, the specimen shall be cut with a suitable die, 3 1 2 2, from a portion of the test unit as described in 4 1 2 In the case of heavy-walled tubing greater than 0 062 inch in thickness, the material shall be mounted in a mandrel, rotated in a lathe, and the ring specimen cut with a sharp tool

4 1 4 *Straight specimen* —A straight specimen of proper length for clamping in the grips

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of the testing machine shall be prepared from the piece to be tested with the die or other apparatus, 3 1 2 3, and by the procedure described in 4 1 2

4 1 5 If practicable, all specimens shall be cut with a single stroke of the die or tool so as to obtain smoothly cut surfaces

4.1.6 *Cutting of specimen* —To facilitate cutting of the specimen with smooth cut surfaces, the edge of the die may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0 5 percent sodium chromate or with silicone mold release emulsion before each specimen is cut

4.2 Cross-sectional area of specimen.—The specimen shall be free from mechanical damage

4.2.1 *Dumbbell specimen*

4.2.1.1 *Thickness* —The thickness of the specimen shall be determined as described in method 2011, except that three measurements shall be made, one at the center and one near each end of the reduced section of the specimen The median of the three measurements, 2 8 of method 4001, shall be used as the thickness in calculating the cross-sectional area of the specimen Specimens shall be discarded when the difference between maximum and minimum thickness exceeds 0 003 inch If the specimen is narrower than the diameter of the micrometer foot, measurements shall be made with the center of the micrometer foot coinciding with the longitudinal center line of the specimen so that there will be an equal overlap of the foot on each side

4.2.1.2 *Width* —The width of the specimen shall be the width between the cutting edges of the die in the reduced section

4.2.1.3 The cross-sectional area, C, of the specimen shall be calculated by multiplying the width by the thickness of the specimen

4.2.2 *Ring specimen*

4.2.2.1 *Thickness* —The thickness of the ring specimen shall be determined as described in method 2011 The median of the measurements shall be used as the thickness in calculating the cross-sectional area of the specimen If the specimen is narrower than the diameter of the foot, measurements shall be made with the center of the micrometer foot coinciding with the circumferential center line of the specimen so that there will be an equal overlap of the foot on each side

4.2.2.2 *Width* —When the ring specimen is not cut with a die, the width shall be determined as described in 4 2 of method 2111 The median of the measurements shall be used as the width in calculating the cross-sectional area of the specimen If a die is used to cut the specimen, the width of the specimen shall be the distance between the cutting edges of the die

4.2.2.3 The cross-sectional area, C, of the ring specimen shall be calculated by multiplying the width by the thickness of the specimen

4.2.3 *Straight specimen*

4.2.3.1 The cross-sectional area of the straight specimen shall be calculated as follows

$$\text{Cross-sectional area, } C, \text{ square inch, } = \frac{0.155W}{D \times L}$$

Where

W is the weight of the specimen in air, grams
D is the density of the specimen, grams per cubic centimeter
L is the length of the specimen, centimeters

The weight of the specimen shall be determined and the value recorded as W The length of the specimen shall be measured with the scale or tape and the value recorded to the nearest 0 1 cm as L The density of the specimen shall be determined as described in method 14011 and the value recorded as D

4.3 Determination of tensile strength.

4.3.1 *Dumbbell and straight specimen* —The specimen shall be placed in the grips of the machine and adjusted symmetrically so that the tension will be distributed uniformly over the cross section If the tension is greater on one side of the specimen than on the other, the maximum strength of the specimen will not be developed The machine shall be started and the power-actuated grip shall travel at a uniform speed of 20 ± 1 inch per minute until the specimen ruptures After rupture of the specimen, the breaking force in pounds shall be read from the dial, scale, or automatic recorder and the value recorded as F

4.3.2 *Ring specimen* —The specimen shall be placed around the four spools of the machine

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with a minimum of tension and the portion of the ring that passes over pulleys shall be lubricated with castor oil or other suitable lubricant In other respects the procedure shall be as described in 4 3 1

5. RESULTS

5.1 Calculations.—The tensile strength of the specimen shall be calculated as follows

Dumbbell and straight specimens

Tensile strength, pounds per square inch, $=\frac{F}{C}$

Ring specimens

Tensile strength, pounds per square inch, $=\frac{F}{2C}$

Where

F is the breaking force, pounds

C is the cross-sectional area of the unstretched specimen, square inch

5.2 Unless otherwise specified in the detail specification, 3 specimens from each test unit shall be tested, except that under the following conditions 5 specimens shall be tested·

5.2.1 If the tensile strength of one or more specimens does not meet the specified requirement in the detail specification

5.2.2 If referee tests are being made

5.3 The tensile strength of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see 2 8 of method 4001.

5.4 The tensile strength of the test unit shall be recorded to the nearest 10 pounds per square inch

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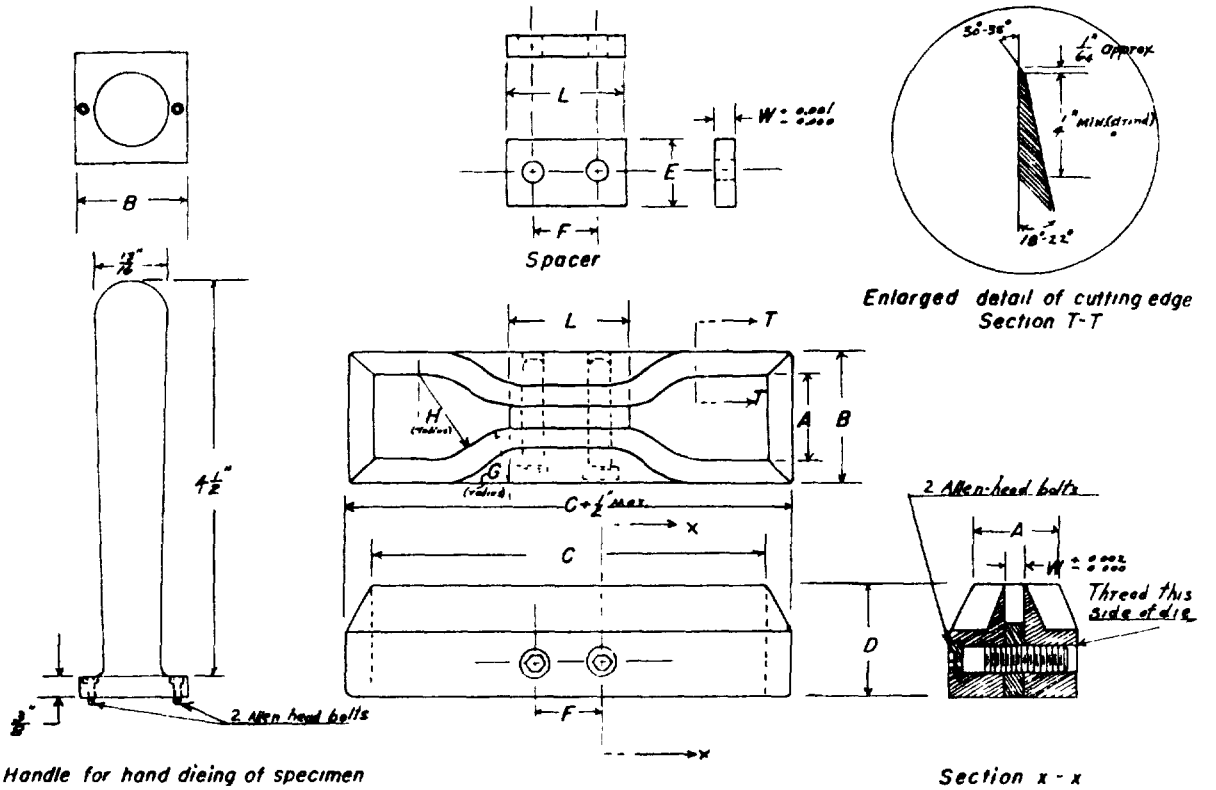

Dimensions of dies, inches

Dimension	Tolerance	Die						
		I	II	III	IV	V	VI	VII
A	±1/32	1	1	1	5/8	5/8	5/8	5/8
B	Maximum	1½	1½	1½	1⅜	1⅜	1⅜	1⅜
C	Minimum	5½	5½	4½	4	5	5	4½
D	±¼ [1]	1¼	1¼	1¼	1¼	1¼	1¼	1¼
D-E	±1/32	½	½	½	½	½	½	½
F	±1/16	1½	1½	¾	¾	1½	1½	¾
G	±1/32	9/16	9/16	9/16	9/16	9/16	9/16	9/16
H	±1/16	1	1	1	5/8	5/8	5/8	5/8
L	±1/16	2 5/16	2 5/16	1 5/16	1 5/16	2 5/16	2 5/16	1 5/16
W	[2]	0.500	0.250	0.250	0.125	0.125	0.250	0.250

[1] For dies used in clicking machines, it is preferable that this tolerance be ±1/64 inch
[2] See figure

FIGURE 4111.—Dies for dumbbell specimen

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CALIBRATION OF TENSION TESTING MACHINE

1. SCOPE

1.1 This method is intended for use in calibrating the testing machine described in method 4111

2. APPARATUS

2.1 The apparatus shall be as follows.

2.1.1 A suitable hook for holding weights which can be attached to the lower grip of the testing machine

2.1.2 Accurately known weight assemblies equal to approximately 10, 20, and 50 percent of the capacity of the machine are recommended

2.1.3 A dumbbell-shaped piece of rubber prepared from a tread compound by means of one of the dies described in 4111.

3. PROCEDURE

3.1 One end of the piece of rubber, 2.1 3, shall be placed in the upper grip of the testing machine The lower grip shall be removed from the machine and attached to the piece of rubber. The hook for holding the weights shall be attached to the lower grip. A weight assembly, 2 1 2, shall be suspended from the hook on the lower grip in such a way as to permit the weight assembly to rest on the grip holder. If the machine has a dynamometer head of the compensating type, it shall be calibrated at two or more settings of the compensator. The motor shall be started as in normal testing and run until the weight assembly is freely suspended by the dumbbell-shaped piece of rubber. If the dial, scale, or automatic recorder, whichever is normally used in testing, does not indicate the weight applied or its equivalent in stress for compensating type of testers within ± 1 percent of the machine capacity, the machine shall be thoroughly checked for excess friction in the bearings and all other moving parts.

3.2 After eliminating as nearly as possible all the excess friction, the machine shall be recalibrated as described above until the machine does indicate correctly The machine shall be calibrated at a minimum of three points, using accurately known weight assemblies of approximately 10, 20, and 50 percent of capacity The weight of the grip and hook shall be included as part of the calibration weight. If pawls or ratchets are used during testing, they shall also be used during calibration The friction in the head may be checked by calibrating with the pawls up Calibration of the machine at least once a month is recommended

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ELONGATION, ULTIMATE

1. SCOPE

1.1 This method is intended for use in determining the elongation of soft vulcanized rubber compounds at break

2. SPECIMEN

2.1 The specimen shall be as described in method 4111

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 The apparatus described in method 4111

3.1.2 *Bench marker*.—A bench marker having two parallel knife edges, ground smooth and true, between 0 002 and 0 003 inch in width at the edge and beveled at an angle of not more than 15° to the perpendicular The distance between the centers of the knife edges shall be $1\ 000 \pm 0\ 003$ inch or $2\ 000 \pm 0\ 003$ inch as required

3.1.3 *Stamp pad*.—A plane unyielding surface, such as hardwood, plate glass, or plastic, covered with a pad containing ink of the desired color and quality for marking the specimen The ink shall have no deteriorating effect on the specimen and shall be of a contrasting color to that of the specimen

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3.1.4 A scale or other device for determining elongation to within 0 1 inch

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the elongation shall be determined on the same specimen during the same test as tensile strength

4.2 Preparation of specimen.—The specimen shall be prepared as described in method 4111 In addition, two parallel benchmarks shall be placed symmetrically on the straight specimen and on the reduced section of the dumbbell specimen perpendicular to the longitudinal axis by means of the bench marker and ink The specimen shall not be injured and the marks shall be not more than 0 010 inch wide The distance between the knife edges of the bench marker, 3 1 2, shall be recorded as D

4.3 Determination of elongation.

4.3.1 *Dumbbell and straight specimens*.—The specimen shall be free from mechanical damage The specimen shall be placed in the grips of the testing machine and the force applied as for tensile strength in method 4111 The distance between the center of the two benchmarks on the specimen shall be noted continuously to the nearest 0 1 inch by means of the scale or other device, which shall be used in such a manner as not to injure or distort the specimen Two operators may be required to conduct this test The distance between the benchmarks when the specimen ruptures shall be read from the scale or other device and the value recorded to the nearest 0 1 inch as D_1

4.3.2 *Ring specimen*.—The specimen shall be free from mechanical damage The inside circumference of the unstretched specimen shall be determined as described in method 2331, 2341, or 2361, as specified, and the value recorded as L The specimen shall be placed around the four spools of the testing machine with a minimum of tension and the portion of the ring which passes over the pulleys lubricated with castor oil The machine shall be started and the force applied as described for tensile strength in method 4111 At the instant of rupture of the specimen, the distance between the spools shall be recorded The inside circumference, L_1, of the ring specimen at any point during the test is equal to two times the vertical distance between centers of the spools plus the circumference of one of the spools

5. RESULTS

5.1 Calculations.—The ultimate elongation of the specimen shall be calculated as follows

Dumbbell and straight specimens

$$\text{Ultimate elongation, percent,} = \frac{D_1 - D}{D} \times 100$$

Where

D_1 is the distance between benchmarks at the moment of rupture of the specimen, inches

D is the distance between the knife edges of the bench marker, inches

Ring specimen

$$\text{Ultimate elongation, percent,} = \frac{L_1 - L}{L} \times 100$$

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Where

L_1 is the inside circumference of the stretched specimen at the moment of rupture, inches

L is the inside circumference of the unstretched specimen, inch

5.2 Unless otherwise specified in the detail specification, 3 specimens from each test unit shall be tested except that under the following conditions 5 specimens shall be tested

5.2.1 If the ultimate elongation of one or more specimens does not meet the specified requirement in the detail specification

5.2.2 If referee tests are being made

5.3 The ultimate elongation of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see 2 8 of method 4001

5.4 The ultimate elongation of the test unit shall be recorded to the nearest five percent

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TENSILE STRESS

1. SCOPE

1.1 This method is intended for use in determining the tensile stress of soft vulcanized rubber compounds Tensile stress is often designated in rubber technology by the term "modulus"

2. SPECIMEN

2.1 The specimen shall be as described in method 4111

3. APPARATUS

3.1 The apparatus shall be as described in methods 4111 and 4121

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the tensile stress shall be determined on the same specimen during the same test as tensile strength and/or elongation

4.2 The elongation in percent at which the tensile stress is to be determined shall be as specified in the detail specification

4.3 Preparation of specimen.—The specimen shall be prepared as described in methods 4111 and 4121

4.4 Cross-sectional area of specimen.—The specimen shall be free from mechanical damage The cross-sectional area, C, of the specimen shall be determined as described in method 4111

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4.5 Determination of tensile stress.—The procedure for determining tensile stress shall be the same as that for determining tensile strength and elongation, methods 4111 and 4121, respectively, except that the force in pounds necessary to stretch the specimen to the specified elongation shall be read from the dial, scale, or automatic recorder and the value recorded as F

5. RESULTS

5.1 Calculations.—The tensile stress of the specimen shall be calculated as follows

$$\text{Tensile stress, pounds per square inch} = \frac{F}{C}$$

Where

F is the force required to stretch the specimen to the required elongation, pounds

C is the cross-sectional area of the unstretched specimen, square inch

5.2 Unless otherwise specified in the detail specification, 3 specimens from each test unit shall be tested except that under the following conditions 5 specimens shall be tested

5.2.1 If the tensile stress of one or more specimens does not meet the specified requirement in the detail specification

5.2.2 If referee tests are being made

5.3 The tensile stress of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see 2 8 of method 4001

5.4 The tensile stress of the test unit shall be recorded to the nearest 10 pounds per square inch

5.5 The percent elongation at which the tensile stress is determined shall be recorded

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STRAIN

1. SCOPE

1.1 This method is intended for use in determining the strain of soft vulcanized rubber compounds under a specified stress The strain is measured after the load has been suspended from the specimen for a specified period of time

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1

2.1.1 The specimen shall be cut with the die described in 3 1 2

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Testing machine* —A testing machine meeting the following requirements

3.1.1.1 The machine extends the specimen at a uniform speed of between 150 and 200 inches per minute until the load is freely suspended

3.1.1.2 The machine is equipped with a timing device which starts automatically at the instant the load is freely suspended by the specimen and signals the operator when the strain is to be measured

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3.1.1.3 The strain is measured to the nearest millimeter by means of a millimeter scale and an indicating device which permits measurements to be made without parallax

3.1.1.4 A weight holder is attached to the lower grip so that weights may be added, preferably automatically, to obtain the desired load

3.1.1.5 *Weights* —The weights correspond to multiples of the load required for 0 01 mm thickness of the specimen and shall be accurate within the load required for 0 001 mm thickness of the specimen The weights are in multiples of 0 01 pound for a load of 100 pounds per square inch

3.1.1.6 The grips which hold the specimen in the tester shall be of a type that tightens automatically as the applied tension increases and exerts a uniform pressure across the gripping surfaces

3.1.1.7 A satisfactory tester is shown in figure 4141A

3.1.2 **Die.**—A metal die for use in cutting the specimen as shown in figure 4141B The cutting blades of the die consist of six strips of razor-blade steel sharpened on one edge Each blade is 6 inches long, ¾ inch deep, and approximately 0 010 inch wide The blades are clamped rigidly between metal spacers, 6 inches long, ⅝ inch deep, and of such thickness that the sharp edges are spaced 0 254 ± 0 001 inch apart The back and sides may be cut from a single piece of metal instead of three pieces as shown in the figure A die so constructed will cut five strips, each 0 254 inch wide Blades of the die shall be replaced when they become dull or nicked

3.1.3 **Bench marker.**—A bench marker having two parallel knife edges ground smooth and true, between 0 002 and 0 003 inch in width at the edge and beveled at an angle of not more than 15° The distance between the centers of the knife edges shall be 10 00 ± 0 01 cm A suitable bench marker is shown in figure 4141C

3.1.4 A stamp pad and ink as described in method 4121

3.1.5 **Thickness gage.**—A thickness gage which will measure the average thickness of the specimen between benchmarks accurately to 0 01 mm under a pressure of approximately 4 pounds per square inch A suitable thickness gage is shown in figure 4141D The instrument is equipped with a dial indicator graduated in 0 01 mm where one revolution of the dial corresponds to one mm and the total range is 2 5 mm The instrument is mounted above an especially constructed base shown in figure 4141E The base is equipped with a pressure bar 10 cm long which is pressed upon the surface of the specimen by means of an adjustable spring The bar is attached to a ball joint which possesses sufficient freedom of movement to permit the dial gage to indicate the average thickness of the specimen

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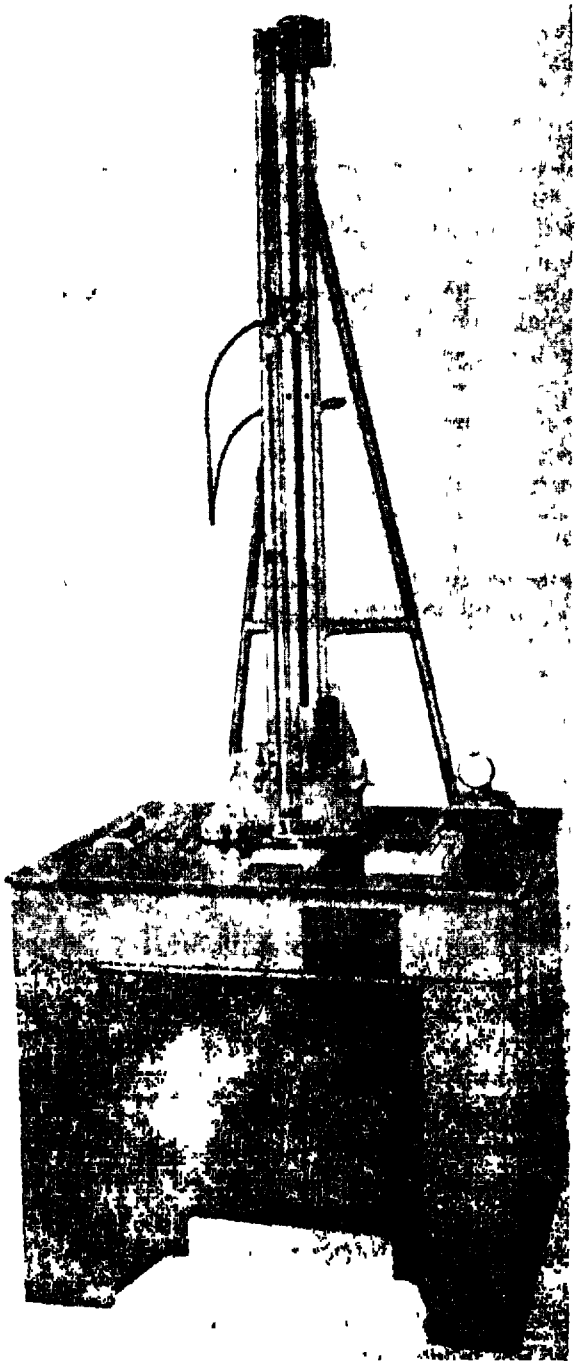

FIGURE 4141A —Strain tester

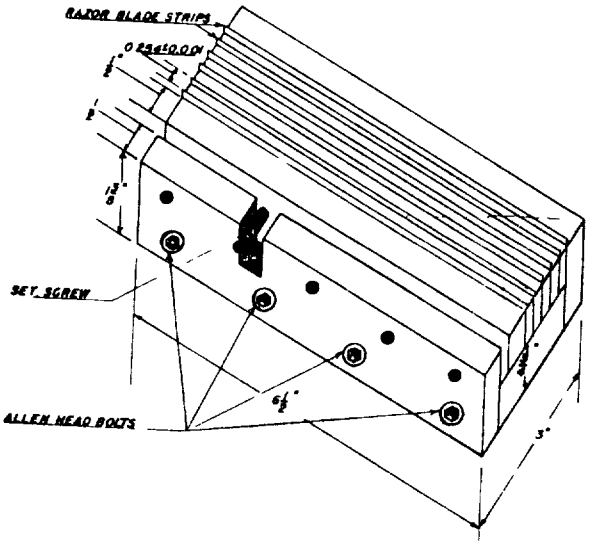

FIGURE 4141B.—Die for specimen.

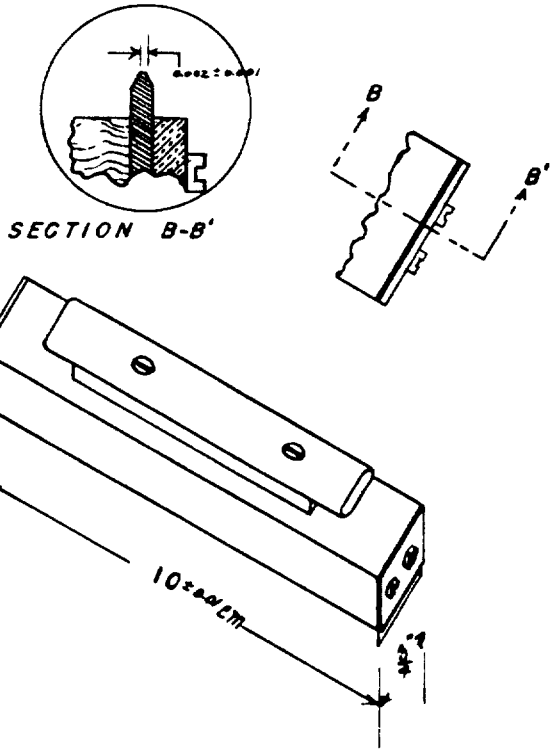

FIGURE 4141C —Bench marker

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FIGURE 4141D —Thickness gage

3.1.6 Cutting support as described in method 4111

4. PROCEDURE

4.1 The stress applied to the specimen shall be as specified in the detail specification

4.2 Preparation of specimen.

4.2.1 *Buffing* —If the material is too thick or has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111 The portion of the test unit from which the specimen is to be taken shall be buffed in the strip form before cutting with the die

4.2.2 The specimen shall be cut with the die described in 3 1 2 If practicable, the specimen shall be cut with a single stroke of the die so as to obtain smooth-cut surfaces To facilitate cutting, the edge of the die may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0 5 percent sodium chromate or with silicone mold release emulsion before each specimen is cut Benchmarks 10 00±0 01 cm apart shall be placed on the straight uniform section of the specimen perpendicular to the longitudinal axis by means of the bench marker and ink

4.3 Determination of load to be applied.

4.3.1 The specimen shall be free from mechanical damage The average thickness of the specimen between bench marks shall be determined to the nearest 0 01 mm by means of the thickness gage described in 3 1 5 The width of the specimen shall be the distance between the cutting edges of the blades of the die The cross-sectional area of the specimen shall be calculated by multiplying the width by the thickness of the specimen

4.3.2 The load to be applied to the specimen shall be determined by multiplying the specified stress, 4 1, by the cross-sectional area, 4.3 1. The force for a stress of 100 pounds per square inch is one pound for each millimeter of thickness

4.4 Measurement of strain.—The specimen shall be placed in the grips of the testing machine with the benchmarks facing the operator The required load, 4 3 2, shall be added to the weight holder on the lower grip The specimen shall be extended at a uniform speed of between 150 and 200 inches per minute until the load on the lower grip is freely suspended The dis-

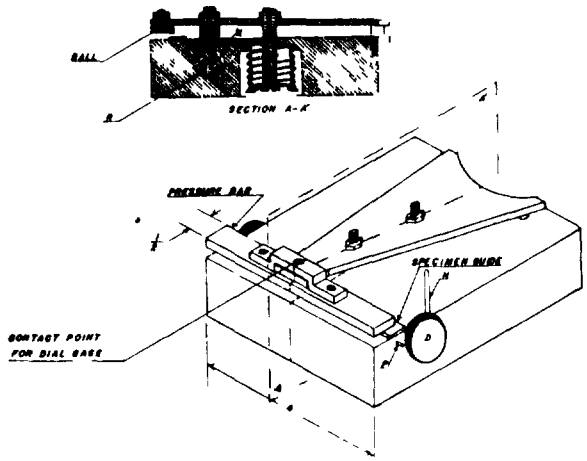

FIGURE 4141E —Averaging mechanism for thickness gage

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tance between bench marks shall be measured 60 ± 3 seconds after the load is freely suspended The increase in the distance between the gage marks due to the application of the load shall be recorded to the nearest one percent

5. RESULTS

5.1 Unless otherwise specified in the detail specification, 3 specimens from each test unit shall be tested except that under the following conditions 5 specimens shall be tested

5.1.1 If the strain of one or more specimens does not meet the specified requirement in the detail specification

5.1.2 If referee tests are being made.

5.2 The strain of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see 2 8 of method 4001

5.3 The strain of the test unit shall be recorded to the nearest one percent

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TEAR RESISTANCE, CRESCENT AND ANGLE

1. SCOPE

1.1 This method is intended for use in determining the resistance to tear of soft vulcanized rubber compounds such as tire tread, carcass, inner tubes, etc It is not applicable to the testing of materials commonly known as hard rubber Since tear resistance is affected to a large extent by mechanical fibering of the rubber under stress as well as by stress distribution, speed of stretching, and size of specimen, results obtained in a tear-resistance test can only be regarded as a measure of the resistance under the conditions of that particular test and not necessarily as having any relation to service value Often grain effects are significant in tearing strength and, if it is desired to evaluate this effect, two sets of specimens should be tested, one at right angles to the other

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1

2.1.1 The specimen shall be cut with die A, die B, or die C, figure 4211, as specified in the detail specification Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test but shall not exceed 0 010 inch.

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Testing machine*—The testing machine and grips as described in method 4111

3.1.2. *Dies*—Metal dies of the shape and dimensions shown in figure 4211 for cutting the specimen The dies shall be kept sharp and free from nicks in order to avoid leaving ragged cut edges on the specimen It is important that the apex of the 90° angle die, C, be sharpened to provide a sharp corner

3.1.3. *Slit cutter*—A suitable device equipped with guides for a knife holder for controlling the depth and position of the slit in the specimen cut with die A or die B The cutting tool is a safety razor blade clamped in the holder The extension of the blade beyond the surface of the holder is regulated by a depth-gage block to produce a slit 0 02 inch in depth

3.1.4 Cutting support as described in method 4111

4. PROCEDURE

4.1 Preparation of specimen.

4.1.1 Unless otherwise specified in the detail specification, the specimen shall be prepared with the grain running parallel to the long way of the specimen

4.1.2. *Buffing*—If the material is too thick or has an uneven surface that may interfere with the test, it shall be buffed as described in method 1111 The portion of the test unit from which specimens are to be taken shall be buffed in the strip form before cutting with the die

4.1.3 The specimen shall be cut by means of die A, die B, or die C, figure 4211, as specified in the detail specification The portion of the test unit from which the specimen is to be cut shall be placed on the cutting support and the specimen cut by means of the required die To facilitate cutting of the specimen with smooth-cut surfaces, the die may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0 5 percent sodium chromate or with silicone mold release before each specimen is cut The cut edges shall be perpendicular to the other surfaces of the specimen, have a minimum of concavity, and be free from ragged edges

4.1.4 Specimens cut with die A or die B shall be slit with extreme care and accuracy across the edge and at the center of the inner concave edge The specimen shall be held in the slit cutter that is equipped with guides for the knife holder to control the depth and centering of the slit The specimen shall be lubricated with the wetting agent, 4 1 3, and the knife holder passed back and forth through the guides of the specimen holder until the slit is cut to a depth of 0 02 inch Specimens cut with die C shall not be slit

4.2 Thickness of specimen.—The specimen shall be free from mechanical damage The

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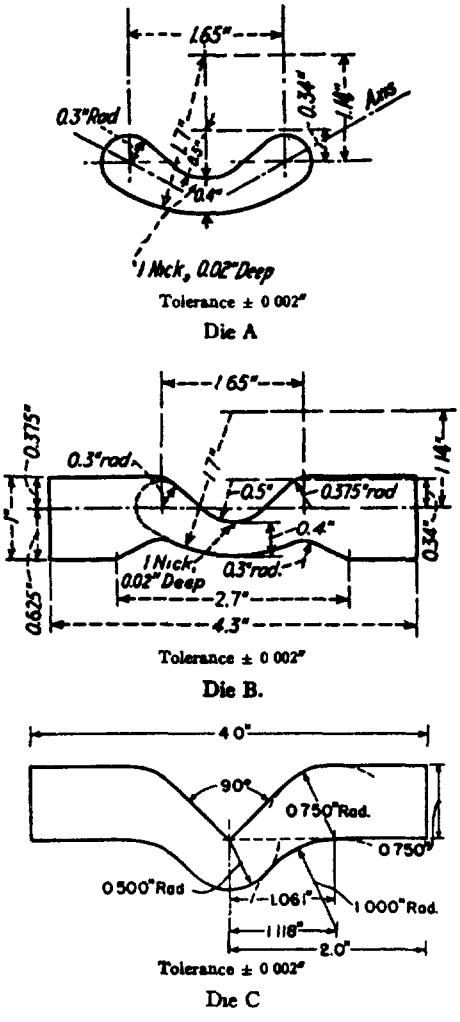

FIGURE 4211.—Dies for tear resistance specimens

thickness shall be determined as described in method 2011, except that one measurement shall be made at the apex of the slit or 90° angle, as applicable, and the value recorded as T

4.3 Determination of tear resistance.—The specimen shall be placed in the grips of the machine and gripped at a distance of approximately 0.5 inch from each end. Specimens cut with die A shall be gripped in the machine so that the axes, as shown in figure 4211, are in line with the direction of application of force. Specimens cut with die B or die C shall be gripped in the machine, so that the "bite" of the grips will be at the center of the tab ends and in line with the direction of the application of force. The machine shall be started and the power-actuated grip shall travel at a uniform speed of 20 ± 1 inch per minute until rupture of the specimen. After rupture of the specimen, the maximum tearing force in pounds shall be read from the dial, scale, or automatic recorder and the value recorded as F

5. RESULTS

5.1 Calculation.—The tear resistance of the specimen shall be calculated as follows.

Tear resistance, pounds per inch of thickness, $= \frac{F}{T}$

Where

F is the maximum tearing force, pounds
T is the thickness of the specimen, inch

5.2 Unless otherwise specified in the detail specification, 3 specimens from each test unit shall be tested except that under the following conditions 5 specimens shall be tested.

5.2.1 If the tear resistance of one or more specimens does not meet the specified requirement in the detail specification.

5.2.2 If referee tests are being made.

5.2.3 If any single value deviates more than 20 percent from the median of the values obtained from the specimens tested. For definition of median, see 2.8 of method 4001.

5.3 The tear resistance of the test unit shall be the median of the values obtained from the specimens tested.

5.4 The tear resistance of the test unit shall be recorded to the nearest one pound per inch of thickness.

5.5 The type of specimen tested shall be recorded.

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TEAR RESISTANCE, STRIP

1. SCOPE

1.1 This method is intended for use in determining the tear resistance of soft vulcanized rubber compounds It is not applicable to the testing of materials commonly known as hard rubber Often grain effects are significant in tearing strength and, if it is desired to evaluate this effect, two sets of specimens should be tested, one at right angles to the other

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1

2.1.1 The specimen shall be of the shape and dimensions shown in figure 4221 and shall be cut with die II in method 4111

3. APPARATUS

3.1 Apparatus described in method 4111

3.2 Sharp razor blade for slitting the specimen

4. PROCEDURE

4.1 Preparation of specimen

4.1.1 Unless otherwise specified in the detail specification, the specimen shall be prepared with the grain running the long way of the specimen

4.1.2 *Buffing*.—If the material is too thick or has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111 The portion of the test unit from which the specimen is to be taken shall be buffed in the strip form before cutting with the die

4.1.3 The portion of the test unit from which the specimen is to be prepared shall be placed on the cutting support and the specimen cut by means of die II described in method 4111 Using a sharp razor blade, a slit shall be cut through the specimen along the center of the reduced section and shall extend to within 0 150 inch of the end of the enlarged part of the specimen The slit shall be cut, as nearly as practicable, 90° to the principal surfaces of the specimen

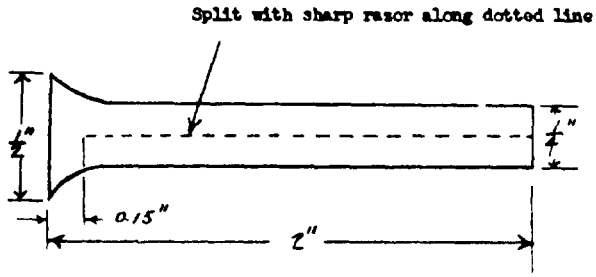

FIGURE 4221.—Strip specimen for tear resistance

4.2 Thickness of specimen.—The specimen shall be free from mechanical damage The thickness shall be determined as described in method 2011, except that one measurement shall be made at the apex of the slit and the value recorded as T.

4.3 Determination of tear resistance.—The specimen shall be centered in the testing machine with one strip of the specimen in each grip of the machine The distance between the grips at the start of the test shall be $3 \pm \frac{1}{8}$ inch The machine shall be started and the power-actuated grip shall travel at a uniform speed of 20 ± 1 inch per minute until rupture of the specimen After rupture of the specimen, the maximum breaking force in pounds shall be read from the dial, scale, or automatic recorder and the value recorded as F

5. RESULTS

5.1 Calculation.—The tear resistance of the specimen shall be calculated as follows

$$\text{Tear resistance, pounds per inch of thickness,} = \frac{F}{T}$$

Where
F is the maximum tearing force, pounds
T is the thickness of the specimen, inch

5.2 Unless otherwise specified in the detail specification, 5 specimens from each test unit shall be tested except that under the following conditions 7 specimens shall be tested

5.2.1 If the tear resistance of one or more specimens does not meet the specified requirements in the detail specification

5.2.2. If referee tests are made

5.3 The tear resistance of the test unit shall be the median of the values obtained from the specimens tested For the definition of median see 2 8 of method 4001

5.4 The tear resistance of the test unit shall be recorded to the nearest one pound per inch of thickness

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ–R–601a)

STRENGTH OF SPLICE

1. SCOPE

1.1 This method is intended for use in determining the strength of the splice or joint of soft vulcanized rubber products such as fire hose or inner tubes

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1

2.1.1 The specimen shall be dumbbell in shape and unless otherwise specified in the detail specification shall be cut with die I described in method 4111 The specimen shall be cut so that the splice or joint will be as near as practicable in the middle of the reduced section

3. APPARATUS

3.1 The apparatus shall be as described in method 4111

4. PROCEDURE

4.1 The specimen shall be prepared as described in method 4111 except that it shall not be buffed The width of the specimen, W, shall be the width of the die The splice or joint shall be as near as practicable to the middle of the reduced section of the specimen Fabric, if present, shall be removed by iso-octane as described in method 1011

4.2 **Determination of strength of splice.**—The specimen shall be placed in the grips of the testing machine as described for tensile strength, method 4111 The machine shall be started and the power-actuated grips shall travel at a uniform speed of 20 ± 1 inch per minute until the specimen ruptures After rupture of the specimen, the breaking force in pounds shall be read from the dial, scale, or automatic recorder and the value recorded as F

5. RESULTS

5.1 **Calculation.**—The strength of splice of the specimen shall be calculated as follows

$$\text{Strength of splice, pounds per inch of width,} = \frac{F}{W}$$

Where
F is the breaking force, pounds
W is the width of the specimen, inch

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.3 The strength of splice of the test unit shall be the smallest value obtained from the specimens tested

5.4 The strength of splice of the test unit shall be recorded to the nearest one pound per inch of width

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TENSION SET

1. SCOPE

1.1 This method is intended for use in determining the elongation remaining in soft vulcanized rubber compounds after a specimen has been held at a specified elongation for a definite period of time, then relieved of the force and allowed to rest for a definite period of time

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1.

2.1.1 Unless otherwise specified in the detail specification, the specimen shall be dumbbell in shape and shall be cut with one of the dies described in method 4111 When tensile strength is required in the detail specification, the same die shall be used for cutting the test specimen that is used for cutting the tensile strength specimen

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Any equipment suitable for stretching the specimen at a uniform speed and holding it at a specified elongation for a definite period of time An apparatus that has been found satisfactory is shown in figure 4411

3.1.2 Grips described in method 4111

3.1.3 A stopwatch or other suitable timing device that will indicate the time in seconds and minutes

3.1.4 A scale or other device graduated to

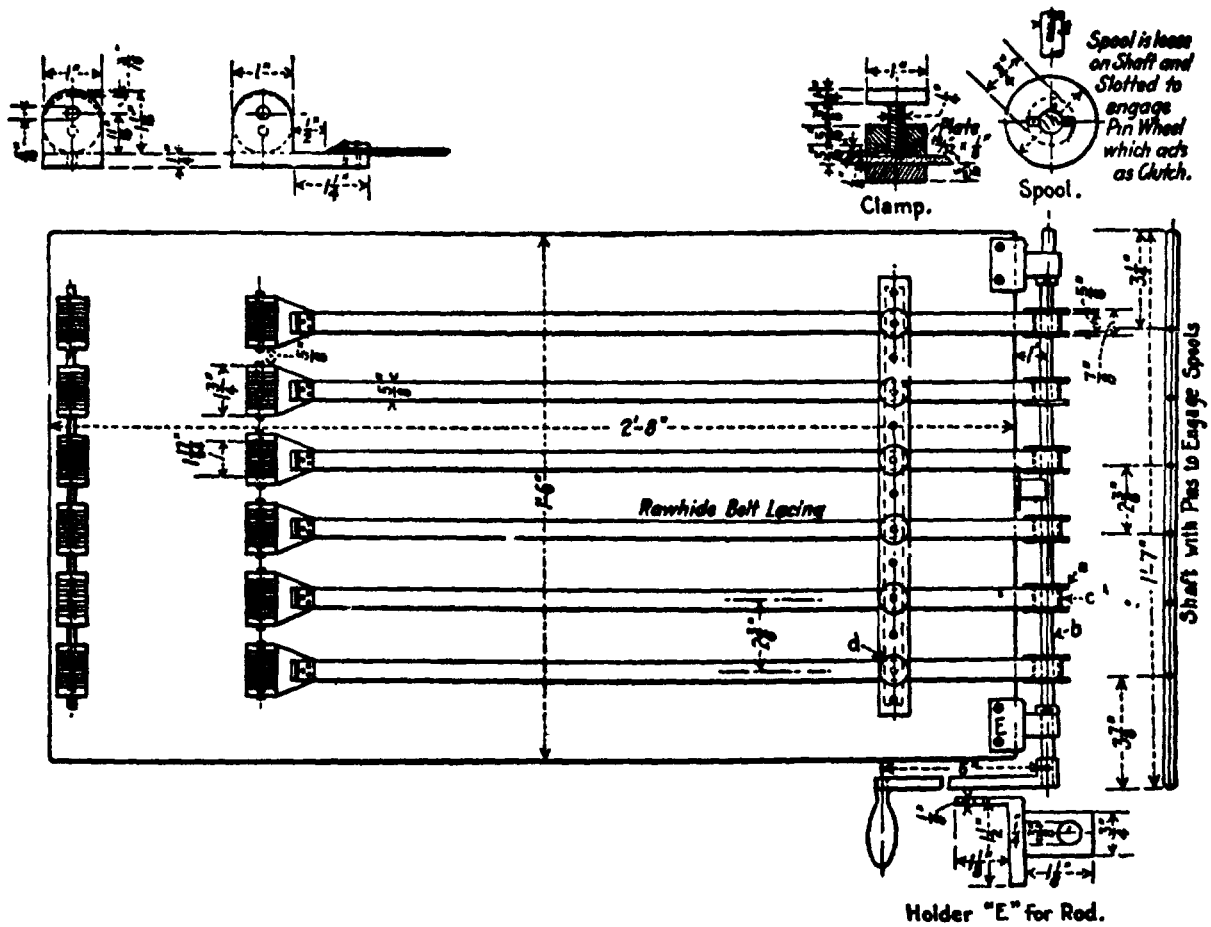

FIGURE 4411 —Tension set apparatus

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0 1 inch for measuring the elongation A straight rod of a length equal to the exact distance required between the bench marks after stretching the specimen to the required elongation has been found suitable for measuring the elongation of dumbbell and straight specimens

3.1.5 A scale or other device graduated in 0 01 inch for measuring the set

3.1.6 Bench marker as described in method 4121

3.1.7 Stamp pad as described in method 4121

4. PROCEDURE

4.1 The elongation or distance between the benchmarks of the stretched specimen shall be as specified in the detail specification If no elongation is specified, the specimen shall be stretched to three-fourths of the ultimate elongation of the compound undergoing test as determined by method 4121

4.2 Unless otherwise specified in the detail specification, the specimen shall be held at the specified elongation for 10 minutes after stretching

4.3 Unless otherwise specified in the detail specification, the specimen shall be allowed to rest for 10 minutes after release before the set is measured

4.4 The specimen shall be prepared as described in method 4111 and marked with the bench marker as described in method 4121

4.5 The specimen shall be free from mechanical damage The distance between the knife edges of the bench marker shall be recorded as L_1

4.6 The specimen shall be placed in the grips of the testing apparatus in the manner described in method 4111 The grips shall be separated at an approximately uniform speed so as to require 15 seconds to reach the required elongation, 4 1

4.7 The specimen shall be held at the required elongation for the required period of time, 4 2, then released immediately without being allowed to snap back, and permitted to rest for the required time, 4 3

4.8 At the end of the rest period, the distance between the benchmarks shall be measured and the value recorded to the nearest 0 01 inch as L_2

4.9 In stretching the specimen to the required elongation, it has been found convenient to use a measured rod of a length equal to the exact distance required between the two benchmarks on the stretched specimen Holding the rod beside the specimen while it is being stretched simplifies the operation and reduces the chance of stretching the specimen more than the required amount.

5. RESULTS

5.1 Calculations.—The tension set of the specimen shall be calculated as follows

$$\text{Tension set, percent, } = \frac{L_2 - L_1}{L_1}$$

Where

L_1 is the distance between the edges of the bench marker, inch

L_2 is the distance between the bench marks of the specimen at the end of the rest period, inch

5.2 Unless otherwise specified in the detail specification, 3 specimens from each test unit shall be tested except that under the following conditions 5 specimens shall be tested

5.2.1 If the tension set of one or more specimens does not meet the specified requirement in the detail specification

5.2.2 If referee tests are being made

5.3 The tension set of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see 2 8 of method 4001

5.4 The tension set of the test unit shall be recorded to the nearest one percent

5.5 The elongation, the time the specimen was held at the specified elongation, and the rest period shall be recorded

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

CONDITIONING OF MATERIALS FOR LOW-TEMPERATURE TESTING, GENERAL

1. SCOPE

1.1 This method describes the conditioning of rubber and rubberlike compounds for low-temperature testing The recommendations herein cover the general theory underlying all low-temperature testing of rubber and rubberlike materials They are particularly applicable to the determination of brittle temperature, stiffness, hardness, and compression set described in this group of tests The purpose is to acquaint individuals making mechanical tests at low temperatures with the characteristics of natural rubber, synthetic rubber, and rubberlike materials, and to outline the conditioning procedures necessary for making these tests within the limit of present knowledge

1.2 These recommendations also cover the various kinds of low-temperature cabinets now in general use for conditioning and testing of rubber and rubberlike materials at low temperatures In addition, a low-temperature cabinet that can be made in the laboratory from inexpensive and readily available materials is described in detail

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2. GENERAL

2.1 Regardless of the test procedure employed, one of the first things necessary in establishing a satisfactory technique is the choice of the time and temperature of exposure of the specimen It is generally known that the following distinct changes take place on lowering the temperature of rubberlike materials (1) simple temperature effects, (2) second-order transitions (vitrification), (3) crystallization (first-order transitions), and (4) effects associated with plasticizers

2.2 **Simple temperature effects.**—Most of the elastic properties of rubberlike materials undergo changes as the temperature is lowered At temperatures above those at which transitions of the second order take place these changes are usually manifested by decreased speed of retraction, loss in resilience, increase in modulus, and increase in hardness As the temperature is decreased, resilience will decrease to a minimum and then increase with further lowering of the temperature Changes due to simple temperature effects are complete when the material has reached thermal equilibrium The magnitude of these changes depends on the composition of the material and the test temperature

2.3 **Second-order transitions (vitrification).**—Transitions of the second order exhibited by all plastics and rubberlike materials do not depend critically on time They take place quickly within a definite temperature range of a few degrees There is considerable change in elastic properties near a second-order transition Measurements of these effects should be limited to exposure times required for thermal equilibrium, otherwise crystallization or plasticizer-time effects may influence the results A convenient differentiation between crystallization and second-order transitions is given in the summary in table I

2.4 **Crystallization (first-order transition).**—Changes in properties resulting from crystallization may take hours, days, weeks, or longer, depending on the material being tested and the temperature employed Crystallization effects begin some time after thermal equilibrium has been established and are manifested by increase in hardness or modulus Only a relatively small number of materials exhibit crystallization, such as natural rubber, Thiokol A, Neoprene GN, Neoprene W, polyethylene, silicone rubbers, and nylon Polyethylene may crystallize in a very short time, in seconds or minutes, at 105° C (221° F) to 112° C (233 6° F) These materials conform to the present concept that only linear polymers made from a single monomer or copolymer containing a very limited amount of a secondary constituent are crystallizable The temperature at which natural rubber crystallizes most rapidly is about −25° C (−13° F) and Neoprene GN is reported to be at about 0° C (32° F) Both above and below

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TABLE I —*Differentiation between crystallization and second-order transition*

Property	Crystallization	Second-order transition
Physical effects	Becomes stiff (hard) but not necessarily brittle	Becomes stiff and brittle
Temperature-volume relation	Significant change in volume	No change in volume, but definite change in coefficient of thermal expansion
Latent heat effect	Heat evolved on crystallization	No heat effect, but definite change in specific heat
Rate	Hours or days required Dependent on temperature In general, as temperature is lowered, rate of crystallization increases to maximum and then decreases Time is important Rate also varies with composition and state of cure	Rapid, takes place within a definite narrow temperature range regardless of thermal history of specimen Time is not important unless plasticizers are present
Temperature of occurrence	Optimum temperature is specific to base polymer	Very wide limits, depending on composition
Effect on molecular structure	Orientation of molecules similar to crystallization of simpler substances	Change in type of motion of segments of molecule

these temperatures, crystallization proceeds at a slower rate Therefore, any attempt to compare materials, particularly those subject to change in properties resulting from crystallization or plasticizer-time effects, on the basis of exposure at a specified temperature for a given time is almost certain to be misleading Such a specified temperature may be near the optimum rate of crystallization of one material and many degrees from the optimum temperature of another material

2.5 Effects associated with plasticizers.—When the material being tested contains considerable amounts of certain plasticizers, time effects not necessarily associated with crystallization may be observed These effects occur over a wide range of time, temperature and composition Some of these effects may be explained on the basis of limited low-temperature solubility of such plasticizers in the compound If the original plasticizer concentration is less than the amount corresponding to saturation at the test temperature, no time effects should be observed The effects consist of delayed stiffening and, in some cases, an elevation of the brittle temperature In the case of elevation of brittle temperature, plasticized compositions may become brittle after an extended exposure to temperatures slightly higher than their brittle temperatures This effect appears to be limited to a very narrow temperature range, in contrast to the relatively wide range in which delayed stiffening may occur

3. CONDITIONING PROCEDURES FOR MECHANICAL TESTS

3.1 Effects of second-order transition (vitrification).—The specimen should be held at each test temperature until it reaches a uniform temperature Longer conditioning times are required for thicker specimens and for certain compositions using thin specimens If allowed to stand too long, crystallization might occur When equilibrium has been reached, the required measurements should be made immediately, while maintaining the specimen at the same temperature

3.2 Effects of crystallization (first-order transition).—These tests should be conducted at the temperature of maximum rate of crystallization, for example, −25° C (−13° F) for natural rubber, 0° C (32° F) for Neoprene GN, etc The temperature of the specimen should be allowed to come to equilibrium as

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described in 3 1 When equilibrium temperature is reached, one set of the required measurements should be made immediately, the specimen aged for at least 72 hours at the specified temperature, and a second set of measurements, comparable to the first, made Increased stiffness is an indication of crystallization or a plasticizer effect

3.3 Effects associated with plasticizers.— The optimum temperature for determining the increased stiffness caused by the time effects of plasticizers has not been determined It is suggested that tests be conducted at temperatures 5° C (9° F) higher than the brittle temperature The procedure shall be as described in 3 2 except that a slight modification is suggested for measuring brittleness that may occur after an extended exposure to a given temperature In this case, the temperature should also be 5° C. (9° F.) higher than the observed brittle point (See 3 1.) Since long exposure in a liquid medium may give erroneous results, a gaseous medium is to be preferred

4. LOW-TEMPERATURE CABINETS

4.1 General requirements.

4.1.1 Low-temperature cabinets may be of the mechanically refrigerated type, dry-ice type, or the "packaged-air" type, but shall conform to the following requirements

4.1.1.1 The heat-transfer medium in the test chamber shall be air or mixture of air and carbon dioxide

4.1.1.2 The temperature variation throughout the effective test chamber should be within ±1° C (1 8° F) of that specified for the test

4.1.1.3 The heat-transfer medium shall be thoroughly circulated in the test chamber by means of mechanical agitation A fan suitably located in the specimen chamber may be used for this purpose

4.1.1.4 Automatic temperature control shall be used A bimetallic thermoregulator is satisfactory

4.1.2 The size of the chamber is optional, but an 8-cubic-foot compartment, 24 by 24 by 24 inches, is sufficient for the low-temperature tests in 1 1

4.1.3 The door may be located in the top or in the side of the cabinet A side-opening door is convenient and is necessary where the equipment involved must be operated from the side However, it has the disadvantage of allowing the cold air to pour out when the door is open In general, a window in the door is desirable for observation and reading of test equipment dials This window should be preferably of 5-ply Thermopane or its equivalent, to prevent frosting

4.1.4 For units that are in continuous use, a 6-inch wall of ground cork, glass fiber, rock wool, or other suitable insulating material will be found satisfactory in reducing thermal losses and in conserving dry ice or refrigeration For operation of equipment involved in the durometer hardness tests, handholes equipped with insulated sleeve may be installed in the walls of the test cabinet

4.2 Mechanically refrigerated units.

4.2.1 In general, the mechanically refrigerated low-temperature cabinets consist of multiple-stage compressor and suitable cooling coils that surround the test chamber Suitable insulation is provided between the test chamber and the outside walls of the cabinet Automatic temperature control is obtained by either a thermostatic control located in the test chamber, which turns the compressor on and off, or by a suitable pressure control, which regulates the refrigerant temperature Air is used as the heat-transfer medium in the specimen compartment

4.2.2 This type of equipment is well adapted to continuous operation at a fixed temperature Except for rather high maintenance and initial costs, this type of equipment is less expensive to operate continuously from a power standpoint than dry-ice units Temperatures as low as −68° C (−90 4° F) can be obtained with commercial units By the installation of electric heaters and suitable automatic controls, temperatures up to room and above can be obtained

4.3 Dry-ice units (direct type).

4.3.1 In this type of cabinet, a suitable fan or blower located in the dry-ice compartment circulates the carbon dioxide vapor from the dry-ice chamber into the specimen chamber

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and back The inlet and outlet openings can be adjusted for maximum efficiency by means of a preset damper between the dry-ice compartment and the test chamber A bimetallic thermoregulator located in the test chamber controls the "on" and "off" operation of the fan thus providing automatic temperature control A fan is located in the test chamber to insure uniformity of temperature

4.3.2 Commercial units are available in 1-cubic-foot and 8-cubic-foot capacities and provide a temperature range of from −73° C (−99 4° F) to 105° C (221° F) These units are particularly adapted for intermittent use where a wide range of testing temperatures is desired Only a short time is required to cool the test chamber to a low temperature

4.4 Dry-ice units (indirect type).

4.4.1 In this type of cabinet, air is used as the heat-transfer medium and no carbon dioxide comes in contact with the specimen The test chamber, which is insulated from the outside of the cold box, is cooled by circulating the carbon dioxide vapor completely around the outside In general this type of cabinet is of more costly construction than the direct type and is not quite as efficient The time required to cool the test chamber to a low temperature is somewhat greater and is more comparable in this respect to the mechanically refrigerated units

4.4.2 Commercial units are available in 1-cubic-foot and 8-cubic-foot test chamber capacities that operate over a temperature range of −73° C. (−99.4° F.) to 105° C (221° F). Two fans are provided, one for circulating the carbon dioxide around the outside of the test chamber and the other for circulating the air heat-transfer medium inside of the test chamber An automatic switch on the door shuts down the fan in the test chamber when the door is opened, thus tending to prevent frosting

4.5 Packaged air units.

4.5.1 It is frequently desirable to enclose a piece of test equipment in a separate test chamber and circulate temperature-regulated cold air or carbon dioxide from a separate unit by means of insulated ducts or pipes This type of unit is somewhat similar to the direct type of dry-ice unit, 4.3, but contains an auxiliary blower to force the cold air through the pipes to the separate test chamber. It has the advantage of being portable and can be connected to different pieces of equipment.

4 5 2 In some cases, the temperature is controlled by a motor-driven damper or its equivalent that by-passes the cold chamber or specimen chamber rather than by the fan motor itself Motor or solenoid-driven dampers that operate in the dry-ice compartment for the control of temperature may cause trouble due to frosting of the mechanism. No method that is entirely satisfactory has been devised to remove moisture from the heat-transfer medium and completely eliminate frosting Desiccants such as calcium chloride and calcium sulfate have been used

4.6 Laboratory-built cabinets.

4.6.1 Figure 5111 shows a schematic diagram of the essential requirements of an inexpensive direct-type, dry-ice, low-temperature cabinet which can be built in any laboratory from materials that are readily available. The cabinet is made of hard structural board and thermal insulating board, with a central dead-air space for insulation. The unit is divided into compartments by the adjustable dampers. The size of the inlet and outlet openings can be adjusted for maximum efficiency by means of these dampers. The fan in the test cabinet insures positive circulation and resultant uniformity of temperature in this chamber. Automatic temperature control to −73° C. (−99.4° F.) is obtained by means of a bimetallic thermoregulator that turns the fan in the ice chamber on and off By the use of electric strip heaters in place of dry ice, the unit will function equally well at elevated temperatures.

4.6.2 This cabinet can be built to any convenient size and may be provided with either top or side opening door, with or without a window The wall thickness is sufficient for good temperature regulation, and dry-ice consumption is not excessive. Where continuous

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operation is desired and dry-ice consumption is a major factor, a 6-inch wall of ground cork, glass fiber, rock wool, or other suitable insulation can be used Where walls are perforated for fixture attachment, bakelite or fiber tubes should be used

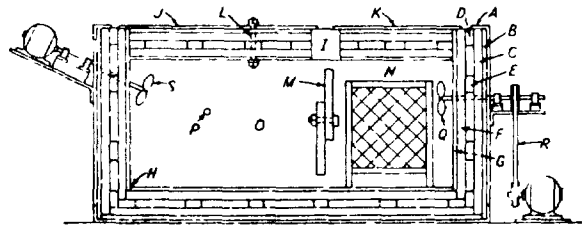

FIGURE 5111 —Schematic diagram of low-temperature cabinet

A—Angle-iron frame of the desired dimensions, constructed from 1½- or 2-in angles with welding at all corners

B—Layer of ¼-in hard structural board

C—Layer of ½-in thermal insulating board

D—½- by 2-in strips of thermal insulating board, glued in place

E—½- by ½-in blocks of thermal insulation, glued in place at intervals of 10 or 12 in to provide an air space

F—Layer of ½-in thermal insulating board

G—Layer of ¼-in hard structural board

H—Gummed paper in the corners and on the top edges (not shown in the sketch) No other bracing of the panels is needed because all but the last layers are held by interlocking at the corners, and the gummed paper serves very well to hold the inner panels and to finish corners

I—Wood support between lids

J and *K*—Lids of the same general construction, except that lid *J* is bolted together

L—Bolts consisting of ½-in wood dowel pins, threaded at both ends and spaced every 10 or 12 in Washers and nuts are of steel

M—Partition arranged so that the top and bottom openings are adjustable

N—Dry-ice compartment with a basket of expanded metal to hold the dry ice

O—Test chamber

P—Thermostat element This may be of the bimetal type except where exposed to vibration which may cause the contacts to chatter A relay is advisable with this type to protect the points Thermocouple control is more expensive but is satisfactory under all conditions

Q—Fan for circulating heat-transfer medium over the dry ice

R—V-belt drive, selected to run the fan *q* because of the ease of changing pulleys to change fan capacity to the correct amount It is necessary to avoid over or under amounts of blowing to control the temperature without leakage at the lids Leakage, of course, uses excess ice and permits the entry of moisture For these reasons it is preferred to use intermittent operations of the fan *q*, rather than automatic control of openings in partition *M* Automatic operation of louvres in partition *M* would be satisfactory if made to close when the drive is shut off and open wide when the drive is running Some control of partition *M* is necessary for relatively high temperatures in chamber *O* Otherwise, convection and conduction through the partition may keep test chamber *O* colder than desired

S—Circulating fan which runs continuously to prevent high- and low-temperature strata in the test chamber

Alternate Construction —Dry-ice compartment *N* and test chamber *O* may be made in separate boxes if desired, connected by two flexible hose lines Blower *q* is then placed in one of the hose lines

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FLEXIBILITY, BENDING BEAM, LOW-TEMPERATURE

1. SCOPE

1.1 This method is intended for use in determining the flexibility of soft vulcanized rubber compounds at low temperatures It is particularly applicable to such rubber items as gaskets and hose

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 10 inches in length. The width shall be one inch and uniform to within 0.01 inch, method 2121

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed 0 085 inch, and shall be uniform to within ±0 005 inch Specimens from the same test unit shall be of the same thickness within ±0 005 inch, method 2011.

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An apparatus consisting essentially of a low-temperature cabinet, specimen rack, spring balance, scale, and temperature measuring equipment. A suitable apparatus is shown in figures 5211A and 5211B

3.1.1.1 *Specimen rack* —The specimen rack with a clamp for holding the specimen is fitted inside the insulated box. A string attached to the free end of the specimen extends through a hole in the cover of the box.

3.1.1.2 *Spring balance* —The spring balance graduated in grams for measuring the load required to deflect the specimen is located outside of the box and is suspended directly over the specimen The string from the free end of the specimen is looped over the hook of the scale The top of the spring balance is attached to a manual or machine-driven pulley in such a manner that the free end of the specimen is raised at a rate of 1 0±0 1 inch per minute

3.1.1.3 *Low-temperature cabinet.*—A cabinet with suitable controls for maintaining the required temperature throughout the cabinet during the exposure period and of sufficient size to hold the specimen rack

3.1.1.4 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within ±1 0° C (1 8° F)

3.1.1.5 A scale graduated to 1/32 inch or finer for measuring the deflection of the specimen

3.1.2 A stopwatch or other timing device that will indicate the time in seconds

4. PROCEDURE

4.1 The specimen shall be exposed at a temperature of −40°±2° C (−40° ±3 6° F) or −55° ± 2° C (−67° ± 3 6° F), as specified in the detail specification

4.2 Unless otherwise specified in the detail specification, the specimen shall be exposed at the low temperature for 16± ¼ hour

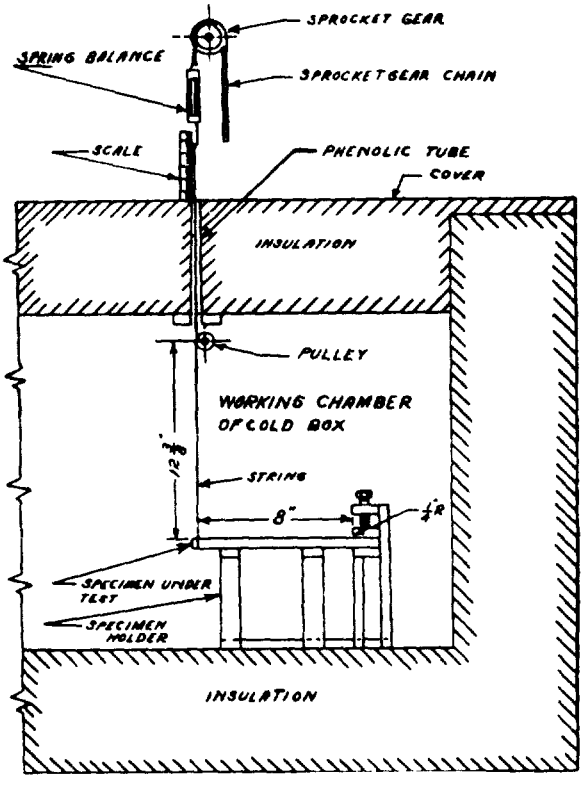

FIGURE 5211A Schematic diagram of low-temperature cabinet

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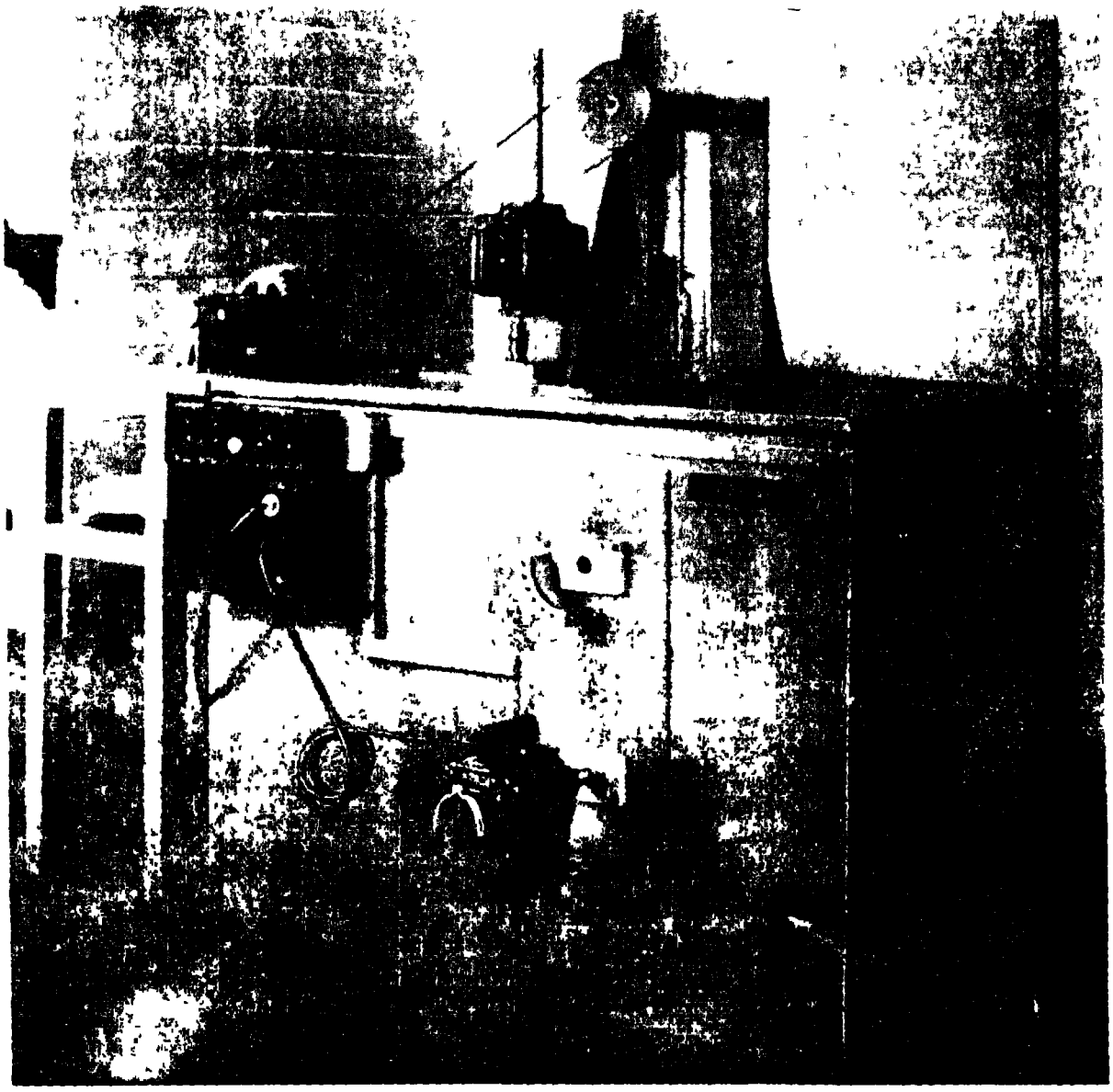

FIGURE 5211B —Cold box showing mechanism for applying load to specimen.

4.3 If the material is too thick or has an uneven surface that may interfere with the test, the material shall be buffed as decribed in method 1111

4.4 The specimen shall be free from mechanical damage The specimen shall be clamped in position in the test rack so that the distance from the edge of the clamp to the point of attachment of the string to the specimen shall be $8 \pm \frac{1}{64}$ inch. The rack with the specimen in position shall be placed in the cabinet and exposed in a medium of air or a mixture of air and carbon dioxide for the required time at the required temperature.

4.5 At the end of the exposure period, with the specimen still at the required temperature, an upward force just sufficient to take up the slack in the string shall be applied to the

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specimen through the spring scale, string, and pulley Sufficient force to deflect the specimen at a rate of 1.0 ± 0.1 inch per minute shall be applied and the force required to deflect the specimen 1 inch and 2 inches shall be read from the scale and the value recorded to the nearest gram

5. RESULTS

5.1 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.2 The flexibility of the test unit shall be the average of the values obtained from the specimens tested

5.3 The flexibility of the test unit shall be recorded to the nearest one gram

5.4 The temperature and time of exposure shall be recorded

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BRITTLENESS, LOW-TEMPERATURE, MOTOR-DRIVEN APPARATUS

1. SCOPE

1.1 This method is intended for use in determining the brittleness of soft vulcanized rubber compounds at low temperature by impact, using a motor-driven apparatus

1.2 For the purpose of this method, failure of the specimen is the occurrence of cracks, fissures, crazing, or rupture

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 1 25±0 25 inch in length The width shall be 0 25 inch and shall be uniform to within ±0 02 inch, method 2121

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed 0 085 inch, and shall be uniform to within ±0 005 inch Specimens from the same test unit shall be of the same thickness within ±0 005 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Clamp and striking arm* —A specimen clamp and striking arm designed to hold the specimen as a cantilever beam The striking arm moves relative to the clamp at a linear speed of 6 5±0 5 feet per second at impact and during at least 0 25 inch of travel after impact The distance between the center line of the striking edge and the edge of the clamp shall be 0 31±0 01 inch at impact The striking edge shall have a radius of 0 062±0 005 inch. The clearance between the striking arm and the specimen clamp shall be 0 25±0 01 inch at and immediately after impact These dimensional requirements are shown in figure 5311B.

3.1.2 A motor for driving the striking arm

3.1.3 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within ±1 0° C (1 8° F)

3.1.4 A suitable apparatus including the above parts and the equipment described in 3 1 5 for use with a liquid heat-transfer medium is shown in figure 5311A

3.1.5 *Liquid heat-transfer medium*

3.1.5.1 A stirrer that insures complete circulation of the liquid medium

3.1.5.2 An insulated tank for holding the liquid medium

3.1.5.3 Equipment with suitable controls for maintaining the required temperature throughout the medium during the exposure period

3.1.5.3.1 Powdered dry ice and an electric heater have been found satisfactory for obtaining the required low temperature

3.1.5.4 *Liquid medium* —The heat-transfer medium consists of any liquid that remains fluid at the temperature of the test and does not materially affect the material under test Liquids that have been found suitable include acetone, ethyl alcohol, methyl alcohol, butyl alcohol, normal hexane, and a mixture consisting of 13 volumes of water and 87 volumes of methyl phosphate For temperatures as low as −120° C (−184° F) methyl cyclohexane and liquid nitrogen have been found suitable

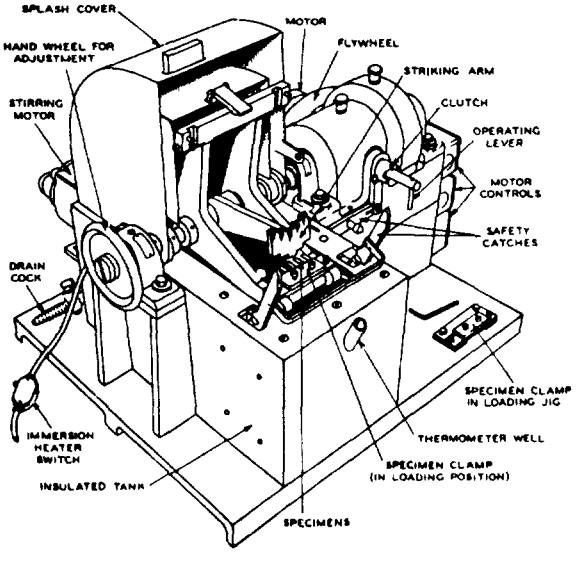

FIGURE 5311A —Motor-driven brittleness tester

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3.1.6 *Air heat-transfer medium*

3.1.6.1 The apparatus described in 3 1 6 2 replaces the equipment described in 3 1 5 when air or a mixture of air and carbon dioxide is used as the heat-transfer medium

3.1.6.2 *Low-temperature cabinet* —A cabinet or testroom with suitable controls for maintaining the required temperature throughout the cabinet or room during the exposure period and of sufficient size to hold the impact apparatus The cabinet is equipped with facilities such as glass windows, handholes with insulated sleeves, etc , that permit operating the brittleness tester at the required temperature

4. PROCEDURE

4.1 If the material is too thick or has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111

4.2 Liquid heat-transfer medium.

4.2.1 The specimen shall be exposed at a temperature of $-40°\pm2°$ C ($-40°\pm3\ 6°$ F) or $-55°\pm2°$ C ($-67\pm3\ 6°$ F), as specified in the detail specification

4.2.2 The liquid medium shall be as specified in the detail specification

4.2.3 The liquid medium shall be adjusted to the required temperature, 4 2 1 This may be accomplished by placing a suitable amount of dry ice in the insulated tank and slowly adding the heat-transfer liquid until the tank is filled to a level of approximately 1 to 2 inches from the top

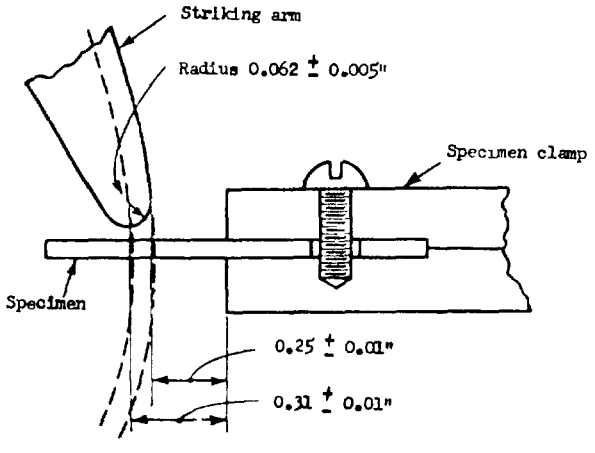

FIGURE 5311B — Dimensional requirements between specimen clamp and striking edge

4.2.4 The specimen shall be free from mechanical damage The specimen shall be mounted in the testing apparatus as shown in figure 5311B and immersed in the liquid medium The specimen shall remain immersed in the liquid medium maintained at the required temperature for $3\pm0\ 5$ minute At the end of the immersion period, the specimen shall be immediately removed from the cooling medium and a single impact blow delivered to the specimen by means of the striking arm Unless otherwise specified in the detail specification, not more than three specimens shall be subjected to a single impact blow of the striking arm

4.2.5 The specimen shall be examined for cracks, fissures, crazing, or rupture

4.3 Air heat-transfer medium.

4.3.1 The specimen shall be exposed at a temperature of $-40°\pm2°$ C ($-40°\pm3\ 6°$ F) or $-55°\pm2°$ C ($-67°\pm3\ 6°$ F), as specified in the detail specification

4.3.2 The specimen shall be free from mechanical damage The specimen shall be mounted in the testing apparatus as shown in figure 5311B, and the apparatus with specimen in position placed in the low-temperature cabinet or testroom The specimen shall be maintained at the required temperature in a medium of air or a mixture of air and carbon dioxide for 60 ± 1 minute At the end of the exposure period, with the specimen still at the required temperature and still in the low-temperature testroom, a single impact blow shall be immediately delivered to the specimen by means of the striking arm of the apparatus Unless otherwise specified in the detail specification, not more than three specimens shall be subjected to a single impact blow of the striking arm

4.3.3 The specimen shall be examined as described in 4 2 5

5. RESULTS

5.1 Unless otherwise specified in the detail specification, five specimens from each test unit shall be tested

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5.2 The number of specimens tested from each test unit, and the number from each test unit that show cracks, fissures, crazing, or rupture shall be recorded

5.3 The heat-transfer medium used shall be recorded

5.4 The temperature of exposure shall be recorded

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

BRITTLENESS, LOW-TEMPERATURE, SOLENOID-ACTUATED APPARATUS

1. SCOPE

1.1 This method is intended for use in determining the brittleness of soft vulcanized rubber compounds at low temperature by impact, using a solenoid-actuated apparatus This method and method 5311 do not necessarily give the same results The definition of failure is given in method 5311

2. SPECIMEN

2.1 The specimen shall be as described in method 5311

3. APPARATUS

3.1 The apparatus shall be as described in method 5311, except that a solenoid-actuated apparatus for driving the striking arm shall be used

4. PROCEDURE

4.1 If the material is too thick or has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111

4.2 Liquid heat-transfer medium.—The procedure shall be as described in 4 2 of method 5311

4.3 Air heat-transfer medium.—The procedure shall be as described in 4 3 of method 5311

5. RESULTS

5.1 The results shall be as described in method 5311

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COMPRESSION SET, LOW-TEMPERATURE

1. SCOPE

1.1 This method is intended for use in determining the ability of soft vulcanized rubber compounds that have been compressed at room temperature and then subjected to a low temperature, to recover from deformation after release from compression while still at low temperature This method is particularly applicable to materials used in such applications as hydraulic seals and brake cups, and hatch gaskets

1.2 For the purpose of this method, compression set is the loss in thickness of the specimen expressed as a percentage of the original deflection This set, unlike compression set at elevated temperature, is temporary since the specimen will regain its original dimension when brought back to room temperature or slightly above

2. SPECIMEN

2.1 The specimen shall be as described in method 3311

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in 3 1 1 and 3 1 4 of method 3311

3.1.2 *Low-temperature cabinet* —A cabinet or testroom with suitable controls for maintaining the required temperature throughout the cabinet or room during the exposure period and of sufficient size to hold the compression apparatus, dial micrometer, and specimen The cabinet is equipped with facilities such as glass windows, handholes with insulated sleeves, etc , that will permit operating the dial micrometer at the required temperature

3.1.3 *Thickness gage* —The dial micrometer described in method 2021 The micrometer may be equipped with an extended dial that can be read outside of the low-temperature cabinet, or it shall be cleaned and lubricated so that the dial will perform satisfactorily in the low-temperature cabinet

3.1.4 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within ±1 0° C (1 8° F)

3.1.5 A stopwatch or other timing device that will indicate the time in seconds

3.1.6 A C-clamp for use in holding the plates of the compression apparatus to facilitate rapid removal of the bolts

4. PROCEDURE

4.1 The specimen shall be exposed at a temperature of −40°±2° C (−40°±3 6° F) or −55°±2° C (−67°±3 6° F), as specified in the detail specification

4.2 The specimen shall be exposed at the low temperature for 22±¼ hour or 94±½ hour, as specified in the detail specification

4.3 The specimen shall be prepared as described in method 3311 and shall be free from mechanical damage The thickness of the specimen shall be determined as described in method 2021, except that the determination shall be made in such a manner as to indicate the perpendicular distance between the center of the top and bottom faces, and the value recorded to the nearest 0 001 inch at T_o

4.4 The specimen shall be compressed to the percent deflection required and in the manner described in 4 2 and 4 5, respectively, of method 3311

4.5 The apparatus with specimen under compression shall be placed in the low-temperature cabinet or room in a medium of air or a mixture of air and carbon dioxide and maintained for the required time at the required temperature At the end of the exposure period, without removing the apparatus or specimen from the cabinet or room, the compression shall be released with the specimen still at the required temperature, the thickness of the specimen shall again be determined as described in 4 3 within 10 seconds after release of the compression, and the value recorded to the nearest 0 001 inch as T_{10} To facilitate rapid removal of the specimen from the compression plates, a C-clamp shall be applied to the plates before removal of the bolts After removal of the bolts, the C-clamp shall be

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quickly removed from the plates, thus permitting the determination of the thickness of the specimen within 10 seconds after release of the compression

4.6 The thickness of the specimen shall again be determined as described in 4.3, 30 minutes after release of the compression and while the specimen is still in the cabinet or room and at the required temperature, and the value recorded to the nearest 0.001 inch as T_{30}

4.7 Thickness measurements in 4.5 and 4.6 shall be made with the presser foot and anvil of the dial micrometer at the same temperature as the specimen

5. RESULTS

5.1 Calculation.—The compression set of the specimen, as percent of the original deflection 10 seconds and 30 minutes after removal from compression, shall be calculated as follows

$$\text{Compression set, percent} = \frac{T_o - T_{10}\ (\text{or } T_{30})}{T_o - T_s} \times 100$$

Where

T_o is the thickness of the specimen before compression, inch

T_{10} is the thickness of the specimen 10 seconds after removal from compression, inch

T_{30} is the thickness of the specimen 30 minutes after removal from compression, inch

T_s is the thickness of the spacer used, inch

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The compression set of the test unit 10 seconds and 30 minutes after compression, respectively, shall be the average of the values obtained from the specimens tested

5.4 The compression set of the test unit 10 seconds and 30 minutes after compression, respectively, shall be recorded to the nearest one percent

5.5 The percent deflection of the specimen shall be recorded

5.6 The temperature and time of exposure shall be recorded

5.7 If a plied-up specimen is tested, it shall be recorded

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HARDNESS, DUROMETER, LOW-TEMPERATURE

1. SCOPE

1.1 This method is intended for use in determining the hardness of soft vulcanized rubber compounds at low temperature by use of the durometer described in method 3021

2. SPECIMEN

2.1 The specimen shall be as described in method 3021

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 The apparatus described in method 3021

3.1.2 *Low-temperature cabinet*—A cabinet or testroom with suitable controls for maintaining the required temperature throughout the cabinet or room during the exposure period and of sufficient size to hold the durometer, specimen, and weight apparatus The cabinet is equipped with facilities such as glass windows, handholes with insulated sleeves, etc , that will permit operating the durometer and load apparatus at the required temperature

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3.1.3 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within ±1 0° C (1 8° F)

3.1.4 Apparatus for applying the durometer to the specimen under a load of 2 pounds, when desired A suitable arrangement is shown in figure 5511

4. PROCEDURE

4.1 The specimen shall be exposed at a temperature of −40°±2° C (−40°±3 6° F.) or −55°±2° C (−67°±3 6° F), as specified in the detail specification

4.2 Unless otherwise specified in the detail specification, the specimen shall be exposed at the low temperature for 22±¼ hour

4.3 If the material has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111

4.4 The specimen shall be free from mechanical damage The specimen shall be placed in the low-temperature cabinet or testroom in a medium of air or a mixture of air and carbon dioxide and maintained for the required time at the requir d temperature. The durometer shall be placed in the cabinet or testroom at least one hour before hardness readings are to be made, in order to permit it to reach temperature equilibrium

4.5 At the end of the exposure period, with the specimen still at the required temperature and without removing the specimen or durometer from the cabinet or testroom, the hardness of the specimen shall be immediately determined as described in method 3021. The durometer may be applied to the specimen by means of the hands, provided they are protected to prevent heating of the instrument, or by means of the load, 3 1 4, sufficient to insure firm contact of the pressure plate to the surface of the specimen The load, if used, shall be 2 pounds

5. RESULTS

5 1 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.2 The hardness of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see section 5

5.3 The hardness of the test unit shall be recorded to the nearest whole scale reading

5.4 The time interval, if any, allowed before reading the instrument shall be recorded

5.5 The temperature and time of exposure shall be recorded

5.6 If a plied-up specimen is tested, it shall be recorded

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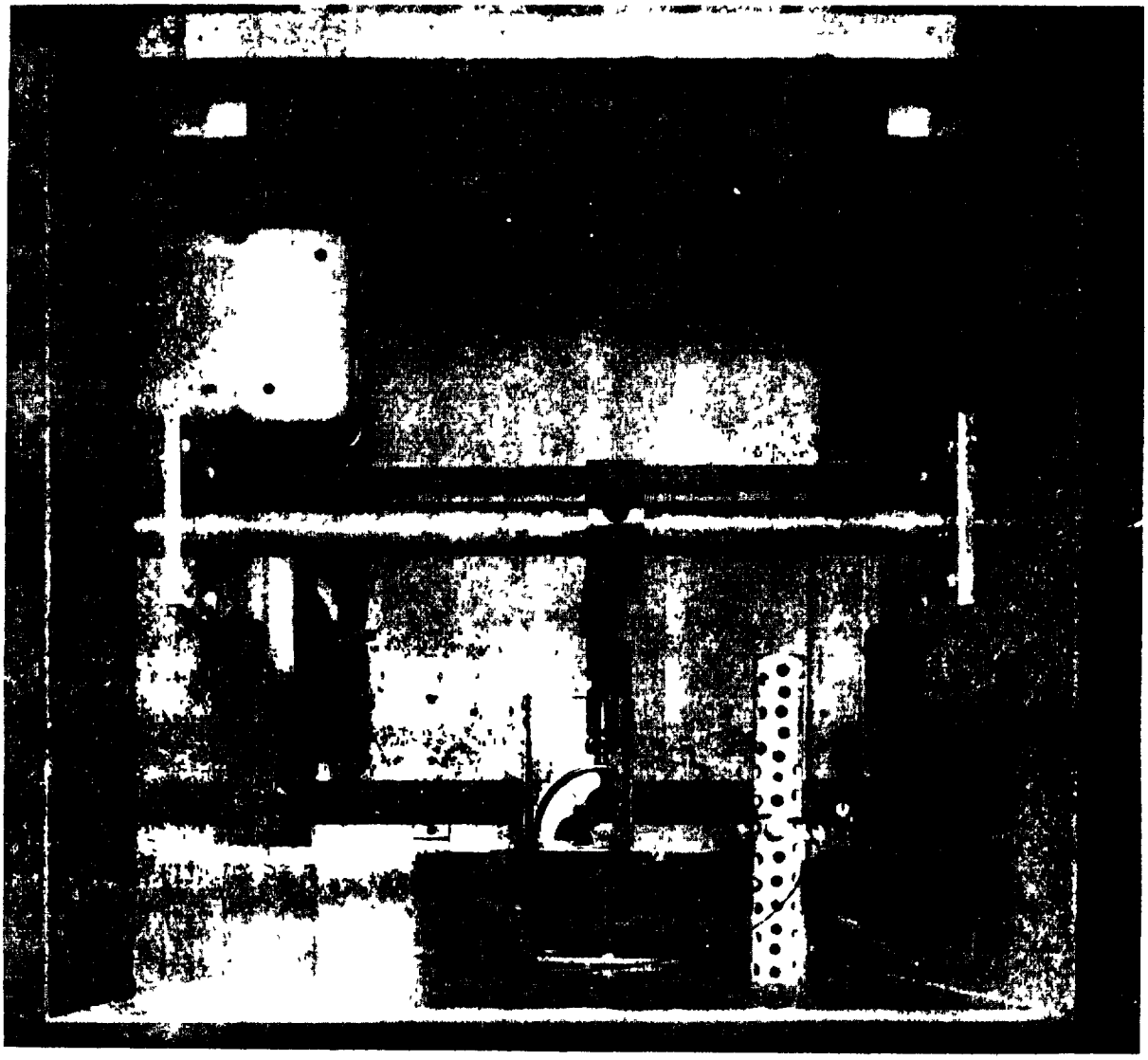

FIGURE 5511.—Apparatus for applying durometer to specimen under a weight

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HARDNESS, INDENTOMETER, LOW-TEMPERATURE

1. SCOPE

1.1 This method is intended for use in determining the hardness of soft vulcanized rubber compounds by use of the indentometer described in method 3051

2. SPECIMEN

2.1 The specimen shall be as described in method 3051

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 The apparatus described in method 3051 The dial indicator of the indentometer shall be cleaned and lubricated so that it will perform satisfactorily at low temperature

3.1.2 *Low-temperature cabinet*—A cabinet or testroom with suitable controls for maintaining the required temperature throughout the cabinet or room during the exposure period and of sufficient size to hold the hardness tester and specimen The cabinet is equipped with facilities such as glass windows, handholes with insulated sleeves, etc , that will permit operating the hardness tester at the required temperature

3.1.3 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within $\pm 1.0°$ C (1 8° F)

4. PROCEDURE

4.1 The specimen shall be exposed at a temperature of $-40° \pm 2°$ C ($-40° \pm 3.6°$ F) or $-55° \pm 2°$ C ($-67° \pm 3.6°$ F), as specified in the detail specification

4.2 Unless otherwise specified in the detail specification, the specimen shall be exposed at the low temperature for $22 \pm \frac{1}{4}$ hour

4.3 If the material has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111

4.4 The specimen shall be free from mechanical damage The specimen shall be placed in the low-temperature cabinet or testroom in a medium of air or a mixture of air and carbon dioxide and maintained for the required time at the required temperature The hardness apparatus shall be cleaned and lubricated so as to operate satisfactorily at the low temperature. The apparatus shall be placed in the cabinet at least one hour before hardness readings are to be made in order to permit it to reach temperature equilibrium At the end of the exposure period, with the specimen still at the required temperature and without removing the apparatus or specimen from the cabinet or testroom, the hardness of the specimen shall be immediately determined as described in method 3051

5. RESULTS

5.1 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.2 The hardness of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see section 5

5.3 The hardness of the test unit shall be recorded to the nearest 0 01 mm

5.4 The time interval between the application of the major load and reading of the instrument shall be recorded

5.5 The temperature and time of exposure shall be recorded

5.6 If a plied-up specimen is tested, it shall be recorded

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HARDNESS, PLASTOMETER, LOW-TEMPERATURE

1. SCOPE

1.1 This method is intended for use in determining the plastometer hardness of soft vulcanized rubber compounds at low temperatures by use of the plastometer described in method 3031 This method is not recommended for temperatures below −10° C (+14° F) when air is used as a cooling medium

2. SPECIMEN

2.1 The specimen shall be as described in method 3031

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 The apparatus described in method 3031

3.1.2 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within ±1 0 C (1 8° F)

3.1.3 *Air heat-transfer medium.*

3.1.3.1 *Low-temperature cabinet* —A cabinet or testroom with suitable controls for maintaining the required temperature throughout the cabinet or room during the exposure period and of sufficient size to hold the plastometer and specimen The cabinet is equipped with facilities such as glass windows, handholes with insulated sleeves, etc , that will permit operating the plastometer at the required temperature

3.1.4 *Liquid heat-transfer medium*

3.1.4.1 *Liquid bath* —A liquid bath with suitable controls for maintaining the required temperature throughout the medium during the exposure period

3.1.4.1.1 *Liquid heat-transfer medium* —The heat-transfer medium consists of any liquid that remains fluid at the temperature of test and does not materially affect the material under test Liquids that have been found suitable include acetone, ethyl alcohol, methyl alcohol, butyl alcohol, normal hexane, and a mixture consisting of 13 volumes of water and 87 volumes of methyl phosphate For temperatures as low as −120° C (−184° F) methyl cyclohexane and liquid nitrogen have been found suitable

3.1.4.2 *Platform* —A platform as shown in figure 5531A The strips on top of the platform for supporting the plastometer are made of any suitable insulating material, such as phenol formaldehyde

3.1.4.3 *Shaft extension* —An extension for lengthening the indenter shaft of the plastometer as shown in figure 5531B, made of the same insulating material used in 3 1 4 2 If an aluminum shaft is used to replace the original shaft of the plastometer, the insulating material may be added without changing the load impressed on the specimen, exclusive of the movable weight

4. PROCEDURE

4.1 If the material has an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111

4.2 Liquid heat-transfer medium.

4.2.1 The specimen shall be exposed at a temperature of −40°±2° C (−40°±3 6° F) or −55°± 2° C (−67°±3 6° F), as specified in the detail specification

4.2.3 Unless otherwise specified in the detail specification, the specimen shall be exposed at the low temperature for 60±5 minutes

4.2.3 The liquid heat-transfer medium shall be as specified in the detail specification

4.2.4 The liquid medium shall be adjusted to the required temperature as described in method 5311 The platform shall be adjusted in the bath so that the surface of the liquid is flush with the top surface of the insulating strips and allowed to reach the temperature of the bath The specimen shall be free from mechanical damage and shall be placed on the platform between the insulating strips, below the surface of the liquid, and allowed to remain for the required time at the required temperature The plastometer with the extension shaft in position shall be placed on the platform as shown in figure 5531C The indenter point shall be immersed in the liquid to permit it to reach temperature equilibrium before making hardness readings At the end of the exposure period, with the specimen still immersed, the

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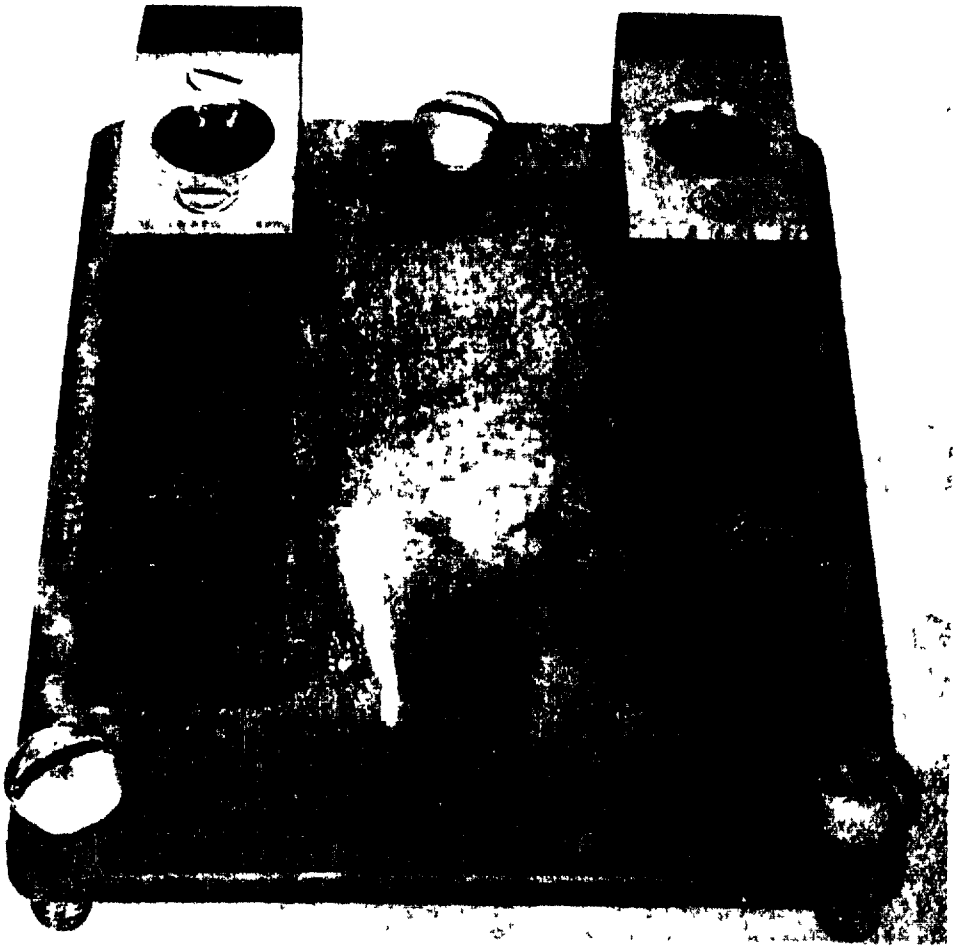

FIGURE 5531A —Platform

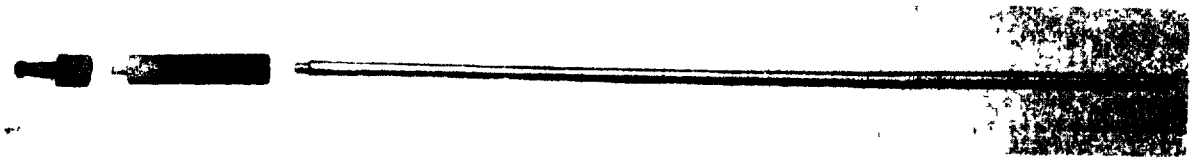

FIGURE 5531B —Indenter shaft extension

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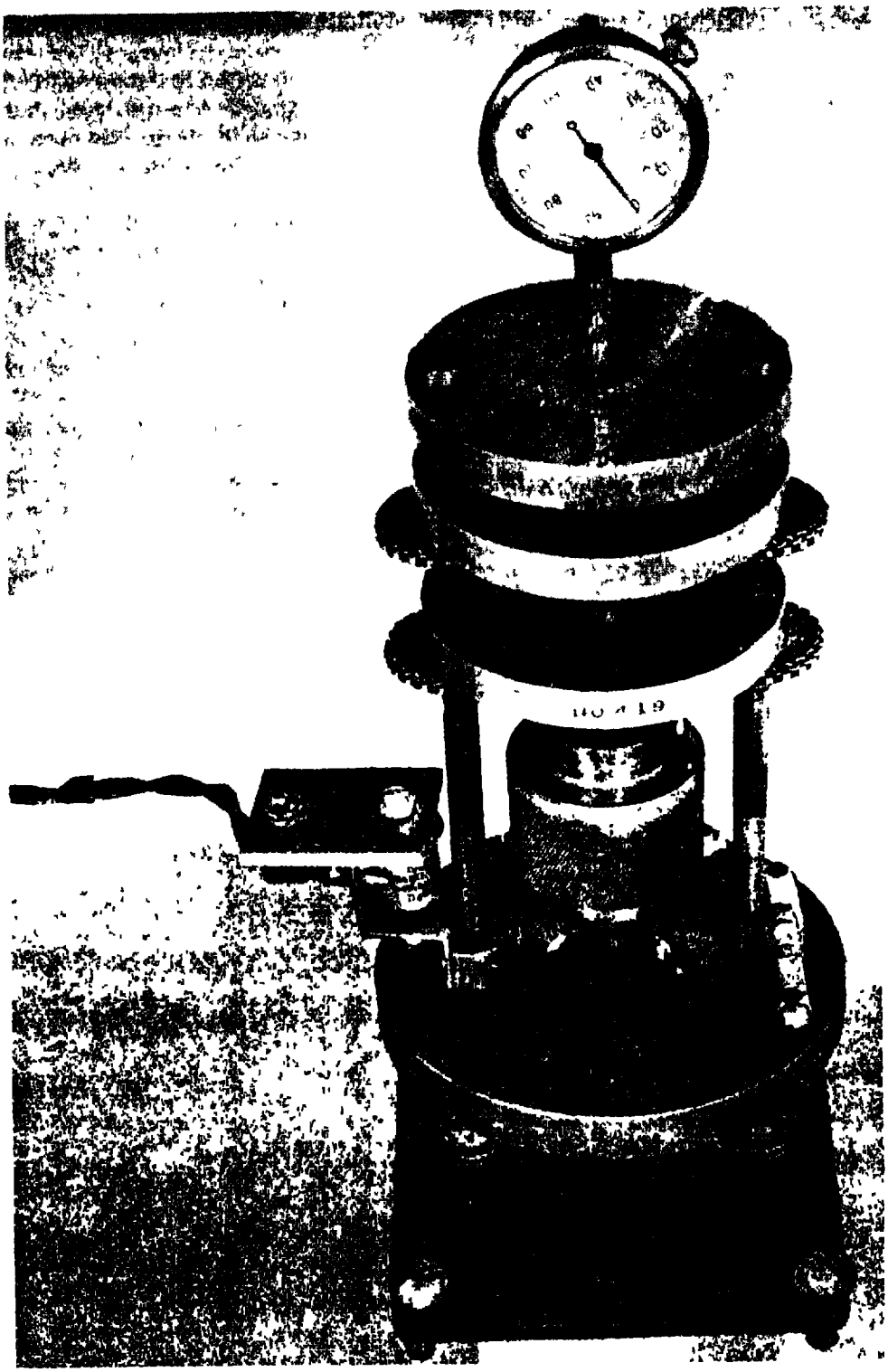

FIGURE 5531 C —Plastometer assembled on platform

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hardness of the specimen shall be immediately determined as described in method 3031, except that the temperature of the indenter point shall be the same as that of the specimen

4.3 Air heat-transfer medium.

4.3.1 The specimen shall be exposed at a temperature of $-40° \pm 2°$ C ($-40° \pm 3.6°$ F) or $-55° \pm 2°$ C ($-67° \pm 3.6°$ F), as specified in the detail specification

4.3.2 Unless otherwise specified in the detail specification, the specimen shall be exposed at the low temperature for $22 \pm \frac{1}{4}$ hour

4.3.3 The specimen shall be free from mechanical damage The specimen shall be placed in the low-temperature cabinet or testroom in a medium of air or a mixture of air and carbon dioxide and maintained for the required time at the required temperature The plastometer shall be cleaned and lubricated so as to operate satisfactorily at low temperature The apparatus shall be placed in the cabinet at least one hour before hardness readings are to be made in order to permit it to reach temperature equilibrium At the end of the exposure period with the specimen still at the required temperature, and without removing the specimen from the cabinet or testroom, the hardness of the specimen shall be immediately determined as described in method 3031

5. RESULTS

5.1 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.2 The hardness of the test unit shall be the median of the values obtained from the specimens tested For definition of median, see section 5

5.3 The hardness of the test unit shall be recorded to the nearest 0.01 mm

5.4 The temperature and time of exposure and the heat-transfer medium used shall be recorded

5.5 The diameter of the indenter point shall be recorded

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STIFFNESS, TORSIONAL, LOW TEMPERATURE

1. SCOPE

1.1 This method is intended for use in determining the relative stiffness of soft vulcanized rubber compounds at low temperature

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit approximately 1 5±0 1 inch in length The width shall be 0 125 inch and shall be uniform to within ±0 005 inch, 4 2 of method 2111

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed 0 085 inch, and shall be uniform to within ±0 005 inch Specimens from the same test unit shall be of the same thickness within ±0 005 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Testing apparatus* —An apparatus consisting essentially of an insulated chamber, cylindrical rack, torsion apparatus, stand, and precooling container A suitable apparatus is shown in figures 5611A and 5611B

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3.1.1.1 *Rack* —The cylindrical rack, *C*, for supporting several specimens consists of top and bottom disks, *D–D*, and grips, *B–B*, made of phenolic plastic composition or other suitable material The hollow centerpost, *E*, supporting the rack and the free turning studs, *F*, on the exterior are made of brass, stainless steel, or other suitable corrosion-resisting material

3.1.1.2 *Insulated chamber* —The insulated chamber, *G*, shall consist of an inner tube, 6 5 inches in length and 3 inches in diameter, made of aluminum, and an outer concentric tube, 4 5 inches in outside diameter, made of chromium-plated steel or other suitable material The space between the tubes is filled with suitable insulating material and held in place by rings of plastic material pressed against the ends of the tubes A plastic disk with a center hole 0 375 inch in diameter is pressed in the bottom of the inner tube The insulating chamber is equipped with an electric fan or other means for circulating the air uniformly throughout the chamber The temperature throughout the chamber shall be uniform to within ±1° C (1 8° F)

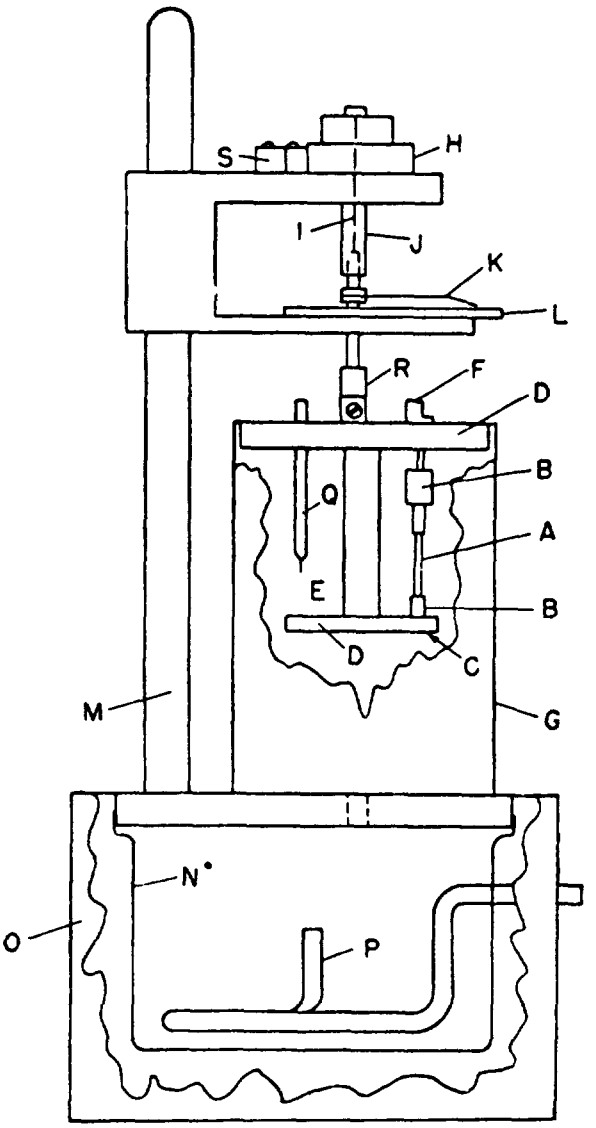

FIGURE 5611A —Schematic diagram of stiffness apparatus

3.1.1.3 *Torsion apparatus* —The apparatus shall consist of a torsion head, *H*, and a torsion wire, *I*, of tempered spring wire 2 5±0 2 inch in length having a torsional constant of 0 500 gm -cm per degree of twist for measuring the

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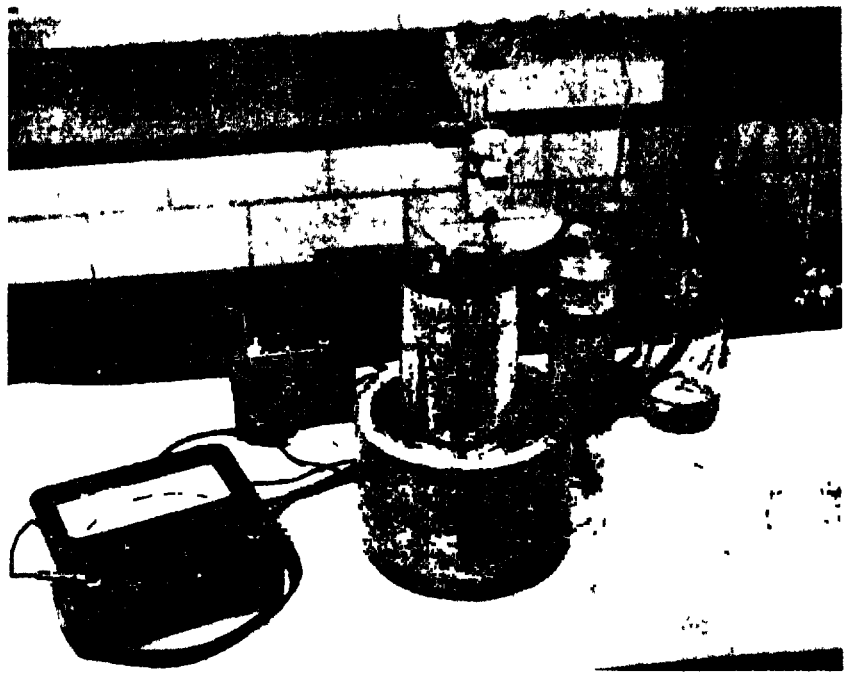

FIGURE 5611 B —Stiffness apparatus assembled

stiffening of the specimen The torsion wire is protected by a loosely fitting sleeve, *J* A pointer, *K*, is attached to the torsion wire A movable protractor, *L*, for convenient adjustment of the pointer to zero, is graduated in degrees for reading the angle of twist of the specimen The torsion wire shall be calibrated as described in 4 1

3.1.1.4 *Stand* —The stand, *M*, for supporting the torsion apparatus and insulated chamber is made of corrosion-resisting steel

3.1.1.5 *Precooling container* —The precooling container, *N*, made of aluminum is equipped with a spiral of copper tubing, *P* A stream of air is passed through the copper tube for precooling before it enters the insulated chamber, *G*, through the 0 375-inch hole in the bottom The container is enclosed in a wooden jacket, *O*

3.1.1.5.1 If available, liquid nitrogen precooling system may be used in place of the usual dry-ice-acetone

3.1.1.6 *Thermocouple and millivoltmeter* — A thermocouple, *Q*, for measuring the temperature in the insulated chamber has one junction connected to a millivoltmeter and the other immersed in ice and water in a thermos bottle Suitable resistance is inserted in the circuit so that the millivoltmeter reads directly in degrees centigrade

3.1.1.7 *Screw connector* —A screw connector, *R*, is used for attaching the specimen to the torsion head

3.1.1.8 *Microswitch* —A microswitch, *S*, mounted on the stand, *M*, is actuated by a cam on the torsion head at the end of 180° rotation

3.1.1.9 *Electric heater* —A small cylindrical 75-watt electric heater that may be inserted in the centerpost, *E*, is used for raising the temperature of the insulated chamber

3.1.1.10 *Electric timer* —An electric timer connected to the microswitch, *S*, gives an audible indication at the end of 10 seconds after being turned on by the microswitch

3.1.2 A thermometer for reading the temperature of the testing room

3.1.3 An air supply under pressure that can be adjusted for rate of flow.

3.1.4 Apparatus for passing the air from the

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air supply through acetone before it enters the cooling chamber

4. PROCEDURE

4.1 Calibration of torsion wire.—One end of the wire shall be inserted in a fixed clamp in a vertical position The lower end of the wire shall be attached at the exact longitudinal center of a rod of known dimensions and weight (For standardization purposes it is suggested that the rod be 8 to 10 inches in length and ¼ inch in diameter) Initially, the rod should not be twisted through more than 90° The rod shall be allowed to oscillate freely in a horizontal plane and the time required for 20 oscillations recorded in seconds An oscillation includes the swing from one extreme to the other and return The mass moment of inertia shall be calculated as follows

$$I=\frac{ML^2}{12}$$

Where

I is the moment of inertia, gm -cm.^{2}
M is the mass of the rod, gm
L is the length of the rod, cm

The torsional constant shall be calculated as follows·

$$\lambda=\frac{4\pi^2 I}{T^2}$$

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Where

T is the period of oscillation, seconds
I is the moment of inertia of the rod, gm -cm ^{2}
λ is the restoring force exerted by the wire, dyne-cm per radian of twist

The dyne-centimeter per radian of twist, λ, shall be converted to gram-centimeter per degree as follows

$$\text{Torsional constant, } K,=\frac{\lambda\times 3\ 1416}{981\times 180}$$

Torsional wires should calibrate within ±3 percent of their specified torsional constants

4.2 If the material is too thick or has a fabric backing or an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111

4.3 The torsion head shall be equipped with a torsion wire The specimen shall be mounted in the rack, the rack placed in the insulated chamber, and the chamber supported on the stand The specimen shall be connected to the torsion head by means of the screw connector The protractor shall be rotated until the pointer is set on zero, the torsion head turned 180°, allowed to remain 10 seconds as indicated by the electric timer, and the pointer reading (twist) read from the scale and the value recorded If the reading at 25° C (77° F) does not fall in the range of 170° to 120°, the standard torsion wire is not suitable for testing the specimen Specimens twisting more than 170° shall l tested with a wire having a torsional constant of 0 125 gm -cm per degree of twist Specimens twisting less than 120° shall be tested with a wire having a torsional constant of 2 000 gm -cm per degree of twist

4.3.1 The torsion head shall be returned to its original position and the specimen disconnected The insulated chamber shall be rotated to bring the next specimen into position for test and the above procedure repeated until all the specimens in the chamber have been tested and the values for twist recorded The room temperature at which tests were made shall be recorded

4.4 The rack with specimen shall be removed from the chamber, dry ice added to fill the chamber to a depth of about 2 inches, and the rack with specimen replaced in the chamber The insulated chamber shall be supported on the stand placed on the precooling container that has been filled with dry ice moistened with acetone Liquid nitrogen or other suitable coolant may be used if available The thermocouple shall be inserted in the chamber and air passed through the copper spiral tubing into the chamber until the lowest desired temperature is reached, as indicated by the thermocouple If the lowest temperature at which the material is to be tested is not known, the test should be started at approximately −70° C (−94° F) When the desired temperature is reached, the airflow shall be regulated to maintain a constant temperature for 5 minutes At the end of the 5-minute period, torsion tests of each specimen shall be made as described in 4 3 and the values recorded

4.5 The temperature of the insulated chamber shall be raised 5° C (9° F) by regulating

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the airflow and maintained for 5 minutes The torsion readings shall be repeated and the values recorded as described in 4 4 This procedure shall be repeated, raising the temperature 5° C (9° F) each time until the pointer reading of the protractor scale is within 5° to 10° C (9° to 18° F) of that of the original readings made at room temperature, 4 3 The electric heater may be used to accelerate the temperature rise

5. RESULTS

5.1 Calculation.

5.1.1 The modulus proportionality factor of the specimen shall be calculated as follows

$$\text{Modulus proportionality factor} = \frac{180° - \text{twist}}{\text{Twist}}$$

5.1.2 The relative modulus at temperature T shall be calculated as follows

$$\text{Relative modulus at temperature } T = \frac{\text{Modulus proportionality factor at temperature T}}{\text{Modulus proportionality factor at 25° C}}$$

5.1.3 The modulus of rigidity or torsional modulus in pounds per square inch shall be calculated as follows

$$\text{Modulus of rigidity, pounds per square inch} = \frac{0\ 795\ K\ (180° - x)}{ab^3\mu(x)}$$

Where

K is the torsional constant of wire, gm -cm per degree
a is the width of specimen, inch
b is the thickness of specimen, inch
μ is the factor based on ratio of a/b, table III
x is the angle of twist of specimen, degrees

5.2 Table I shows the modulus proportionality factor for each degree of twist from 1 to 180

5.3 A twist versus temperature curve similar to that shown in figure 5611C may be prepared for each material under test

5.4 The relative modulus of the material under test at any temperature may be readily obtained from the angle of twist at that temperature and the angle of twist at 25° C (both values taken from the curve prepared in 5 3) and the modulus proportionality factors in table I

5.5 Table II shows the twist angles for designated values of relative modulus The table may be used in conjunction with the twist-temperature curve, figure 5611C, for the material under test to determine the temperature at which the relative modulus is 2, 5, 10, and 100 The first column of the table lists each degree of twist from 120 to 170 at 25° C. Successive columns give the twist angles which correspond to relative modulus values of 2, 5, 10, and 100 The temperatures, T_2, T_5, T_{10}, and T_{100}, respectively, corresponding to these angles may then be read from the twist-temperature curve, 5 3, for the material. Table II can be used during a test to determine when a particular T value has been obtained so that the test may be concluded

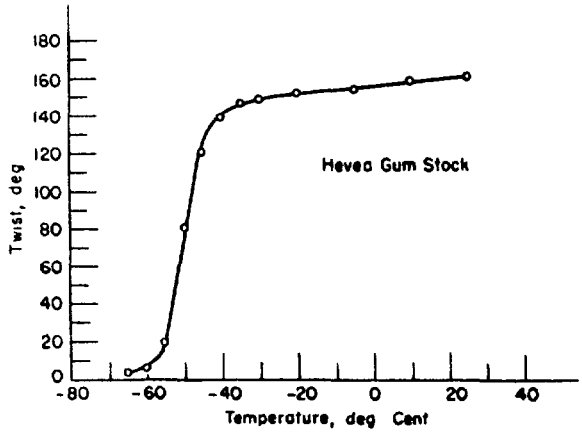

FIGURE 5611C —Illustrative twist versus temperature curve.

5.5.1 *Example of use of table II* —Referring to the twist-temperature curve in figure 5611C, the twist at 25° C is 160° Referring to table II the angles of twist corresponding to relative modulus values of 2, 5, 10, and 100, are 144, 111, 80, and 13, respectively Referring back to figure 5611C, the temperature at which these angles of twist occur are −39° C, −47° C, −50° C, and −56° C, respectively

5.6 Unless otherwise specified in the detailed specification, three specimens from each test unit shall be tested

5.7 The relative modulus of the test unit shall be the average of the values obtained from the specimens tested

5.8 The torsional constant, K, of the wire used in the test shall be recorded

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TABLE I.—*Modulus proportionality factors*

Twist, X deg	$\frac{180-X}{X}$	Twist, X deg	$\frac{180-X}{X}$	Twist, X deg	$\frac{180-X}{X}$	Twist X deg	$\frac{180-X}{X}$	Twist, X deg	$\frac{180-X}{X}$
1	179	41	3 39	81	1 222	121	0 488	161	0 1180
2	89	42	3 29	82	1 195	122	475	162	1111
3	59	43	3 19	83	1 169	123	463	163	1043
4	44	44	3 09	84	1 143	124	452	164	0975
5	35	45	3 00	85	1 118	125	440	165	0909
6	29	46	2 91	86	1 093	126	429	166	0843
7	24 7	47	2 83	87	1 069	127	417	167	0778
8	21 5	48	2 75	88	1 045	128	406	168	0714
9	19 0	49	2 67	89	1 022	129	395	169	0651
10	17 0	50	2 60	90	1 000	130	385	170	0588
11	15 4	51	2 53	91	0 978	131	373	171	0527
12	14 0	52	2 46	92	956	132	364	172	0465
13	12 8	53	2 40	93	935	133	353	173	0405
14	11 86	54	2 33	94	915	134	343	174	0345
15	11 0	55	2 27	95	895	135	333	175	0286
16	10 25	56	2 21	96	875	136	324	176	0227
17	9 59	57	2 16	97	856	137	314	177	0169
18	9 00	58	2 10	98	837	138	304	178	0112
19	8 47	59	2 05	99	818	139	295	179	0056
20	8 00	60	2 00	100	800	140	286	180	----
21	7 57	61	1 95	101	782	141	277		
22	7 18	62	1 90	102	765	142	267		
23	6 83	63	1 86	103	748	143	258		
24	6 50	64	1 81	104	731	144	250		
25	6 20	65	1 77	105	714	145	241		
26	5 92	66	1 73	106	698	146	233		
27	5 67	67	1 69	107	682	147	224		
28	5 43	68	1 65	108	667	148	216		
29	5 21	69	1 61	109	651	149	208		
30	5 00	70	1 571	110	636	150	200		
31	4 81	71	1 535	111	622	151	192		
32	4 62	72	1 500	112	607	152	184		
33	4 45	73	1 466	113	593	153	176		
34	4 29	74	1 432	114	579	154	169		
35	4 14	75	1 400	115	565	155	161		
36	4 00	76	1 368	116	552	156	154		
37	3 86	77	1 337	117	538	157	146		
38	3 74	78	1 308	118	525	158	141		
39	3 62	79	1 278	119	513	159	132		
40	3 50	80	1 250	120	500	160	125		

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TABLE II *Twist angles for designated values of the relative modulus*

Twist at 25° C, deg	Twist for R M = 2, deg	Twist for R M = 5, deg	Twist for R M = 10, deg	Twist for R M = 100, deg
120	90	51	30	3
121	91	52	31	4
122	92	53	31	4
123	93	54	32	4
124	95	55	33	4
125	96	56	33	4
126	97	57	34	4
127	98	58	35	4
128	99	59	36	4
129	101	61	36	5
130	102	62	37	5
131	103	63	38	5
132	104	64	39	5
133	105	65	40	5
134	107	66	41	5
135	108	68	42	5
136	109	69	42	5
137	111	70	43	6
138	112	71	45	6
139	113	72	46	6
140	114	74	47	6
141	116	75	48	6
142	117	77	49	7
143	119	78	50	7
144	120	80	51	7
145	121	82	53	7
146	123	83	54	7
147	124	85	55	7
148	124	85	56	8
149	127	88	58	8
150	129	90	60	9
151	130	92	62	9
152	132	94	62	9
153	133	96	65	10
154	134	97	67	10
155	136	100	69	11
156	138	102	71	11
157	139	104	73	12
158	140	106	75	12

TABLE II —*Twist angles for designated values of the relative modulus*—Continued

Twist at 25° C, deg	Twist for R M = 2, deg	Twist for R M = 5, deg	Twist for R M = 10, deg	Twist for R M = 100 deg
159	142	108	78	13
160	144	111	80	13
161	146	113	82	14
162	147	116	85	15
163	149	118	88	16
164	151	121	91	17
165	152	124	94	18
166	154	126	98	19
167	156	130	101	20
168	158	133	105	22
169	159	136	109	24
170	161	139	113	26

TABLE III —*Values of factor μ for various ratios of a/b*

a/b	μ	a/b	μ
1 00	2 249	2 25	3 842
1 05	2 359	2 50	3 990
1 10	2 464	2 75	4 111
1 15	2. 563	3 00	4 213
1 20	2 658	3 50	4 373
1 25	2 748	4 00	4 493
1 30	2 833	4 50	4 586
1 35	2 914	5 00	4 662
1 40	2 990	6 00	4 773
1 45	3 063	7 00	4 853
1 50	3 152	8 00	4 913
1 60	3 260	9 00	4 960
1 70	3 375	10 00	4 997
1 75	3 428	20 00	5 165
1 80	3 479	50 00	5 226
1 90	3 573	100 00	5 300
2 00	3 659		

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HOSE, FLEXIBILITY, LOW-TEMPERATURE

1. SCOPE

1.1 This method is intended for use in determining the flexibility of hose under hydrostatic pressure at low temperatures The liquid used for immersion and hydrostatic pressure tests should be the same as that used in the hose in service

2. SPECIMEN

2.1 The specimen shall consist of two portions of the test unit, each 18 inches in length and with full circumference

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Low-temperature cabinet* —A cabinet or testroom with suitable controls for maintaining the required temperature throughout the cabinet or room during the exposure period and of sufficient size to permit flexing of the 18-inch length of hose through an angle of 180° The cabinet is equipped with facilities such as glass windows, handholes with insulated sleeves, etc , that will permit flexing the hose at the required temperature

3.1.2 *Air oven* —A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the exposure period and of sufficient size to hold the specimen

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3.1.3 A tank with suitable controls for maintaining the required temperature throughout the immersion medium during the immersion period and of sufficient size to permit immersion of the 18-inch length of hose, 4 3

3.1.4 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature within ±1 0° C (1 8° F)

3.1.5 Hydraulic pressure equipment consisting of a hand pump, gage, release valve, shutoff valve, and accumulator

3.1.6 A satisfactory assembled apparatus is shown in figure 5711

4. PROCEDURE

4.1 The specimen shall be exposed at a temperature of −40° ± 2° C (−40°±3 6° F) or −55°±2° C (−67°±3 6° F), as specified in the detail specification

4.2 Unless otherwise specified in the detail specification, the specimen shall be exposed at the low temperature for 22 ± ¼ hour

4.3 Unless otherwise specified in the detail specification, one length of hose shall be immersed in the liquid medium at a temperature of 70°±2° C (158°±3 6° F) for 166±1 hour

4.4 Unless otherwise specified in the detail specification, the other length shall be aged in the air oven at a temperature of 70°±2° C (158°±3 6° F) for 166±1 hour

4.5 The fluid for immersion and application of the hydrostatic pressure shall be as specified in the detail specification Normally the same fluid is used for both purposes

4.6 The hydraulic pressure applied to the hose at low temperature shall be as specified in the detail specification

4.7 The specimen shall be free from mechanical damage One length of the specimen shall be immersed and allowed to remain in the liquid medium for the required time at the required temperature, 4 3 The other length of the specimen shall be placed in the air oven and allowed to remain for the required time at the required temperature, 4 4

4.8 At the end of the exposure periods, 4 7, both lengths of the specimen shall be filled with the required hydraulic fluid and placed in the low-temperature cabinet or testroom in a medium of air or a mixture of air and carbon dioxide for the required time, 4 2, at the required temperature, 4 1 At the end of the exposure period, with the specimen still in the low temperature cabinet and at the required temperature, hydraulic pressure shall be immediately applied to both lengths and the lengths flexed through 180° to the minimum bend radius of each extreme of travel Each length shall be subjected to 5 cycles at a rate of 1 cycle in 4 seconds The flexed lengths shall be examined for leakage, cracking or other failure

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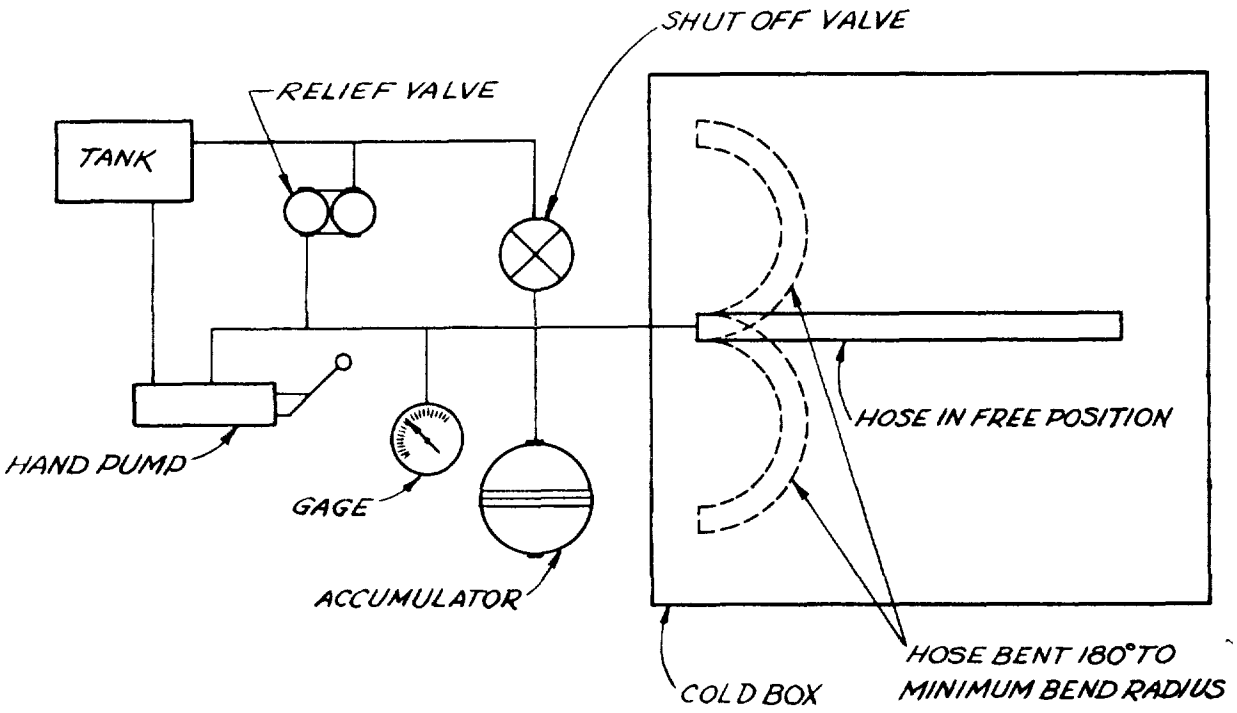

FIGURE 5711 —Apparatus for flexibility of hose

5. RESULTS

5.1 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.2 The number of lengths of hose tested from each test unit and the number from each test unit that showed leakage, cracking, or other failure shall be recorded

5.3 All temperatures and times of exposure and the immersion medium and hydrostatic pressure used shall be recorded

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LIQUID TREATMENT TESTS, GENERAL

1. SCOPE

1.1 This group of methods is intended for use in determining the relative effects of liquids on vulcanized rubber compounds and rubber-like materials In general, these tests consist of exposing a specimen to a liquid under controlled conditions and determining the effects of the exposure by measuring the change in some physical characteristic In some cases, such as change in volume, the effect of exposure to the liquid is determined by comparing the volume of the same specimen before and after exposure In other cases, such as change in tensile strength, comparison is made between the test result for a specified property determined on a portion of the test unit that has not been exposed to liquid and the test result for the same property determined on a portion of the same test unit that has been exposed to the liquid under controlled conditions

2. SPECIMEN

2.1 The specimen shall be as described in the method of test for determining the specified characteristic of the material before and after exposure

3. TEST MEDIA AND APPARATUS

3.1 The liquid to which the material is exposed in service should be used for the immersion tests However, many commercial liquids, particularly petroleum products, vary so much that it is not practicable to use them as test media Hence, it is advisable in such cases to substitute standard test media that are not subject to variation and that cover the range of properties of the particular liquid that may be encountered in service

3.2 The immersion liquid shall be as specified in the detail specification The liquids described in 3 3 to 3 6, inclusive, cover a wide range of properties and it is recommended that wherever practicable the liquid medium be selected from this group

3.3 Petroleum base oils.—Petroleum base oils shall be as follows

Petroleum base oils

	Medium No 1 (low swelling)	*Medium No 2 (medium swelling)*	*Medium No 3 (high swelling)*
Saybolt universal viscosity, second	98 ± 5 at 98 9° C (210° F)	100 ± 5 at 98 9° C (210° F)	155 ± 5 at 37 8° C (100° F)
Aniline point	123 9° ± 1° C (255° ± 1 8° F)	93° ± 3° C (199 4° ± 5 4° F)	69 5° ± 1° C (157 1° ± 1 8° F)
Flash point	243 3°C min (470° F)	246° ± 5 5° C (475° ± 10° F)	166° ± 2 8° C (330° ± 5° F)

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3.4 Fuels.

3.4.1 *Medium No 4 (low swelling)*—Medium No 4 shall be 100 percent iso-octane having a freezing point greater than −107 50° C (−161 50° F)

3.4.2 *Medium No 5 (high swelling)*—Medium No 5 shall consist of a mixture of the following

	Volumes
Iso-octane (3 4 1)	60
Benzene (3 5 1)	5
Toluene (3 5 2)	20
Xylene (3 5 3)	15

3.4.3 *Medium No 6 (high swelling)*—Medium No 6 shall consist of the following

	Volumes
Iso-octane (3 4 1)	70
Toluene (3 5 2)	30

3.5 Aromatics.

3.5.1 *Medium No 7*—Benzene which shall conform to the requirements of Federal Specification VV-B-231, grade 1

3.5.2 *Medium No 8*—Toluene which shall conform to the requirements of Federal Specification TT-T-548

3.5.3 *Medium No 9*—Xylene which shall conform to the requirements of Federal Specification TT-X-916

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3.6 *Medium No 10* — Distilled water

3.7 When a petroleum base oil, 3 3, is used as the medium, the oil specified for use should be the one with aniline point nearest to that of the oil that will come in contact with the rubber in service The aniline point appears to characterize the swelling action of the oil on synthetic rubber In general, the swelling action of the oil becomes more severe as the aniline point becomes lower The aniline points of media numbers 1, 2, and 3 in 3 3 cover a range commonly found in lubricating oils The following suppliers state that their products meet the requirements for these petroleum base oils when ordered as ASTM oils, available only in 5- and 55-gallon quantities

Medium 1 (ASTM Oil 1)	Enjay Co, Inc, 15 W 51st St, New York 19, N Y
Media 2 and 3 (ASTM Oils 2 and 3, respectively)	Sun Oil Co, Industrial Products Dept, Philadelphia 3, Pa

3.8 The media listed in 3 4 represent in general the minimum and maximum swelling effects that are produced by commercial gasolines Medium No 4 has a mild action on rubbers and produces results similar to that of low swelling gasoline of highly paraffinic, straight-run type Media 5 and 6 have a severe action on rubbers and exceed the swelling action of gasolines now in use

3.9 Apparatus such as containers, air condensers, reflux condensers, thermometers, etc, suitable for use in exposing the specimens to liquids at various temperatures

3.9 1 Equipment with suitable controls for maintaining the required temperature throughout the liquid medium during the exposure period, such as a constant temperature bath or air oven with thermostatic control

3.9.2 A cylindrical container having a diameter of not less than 1 5 nor more than 1 8 times the width of the specimen and of sufficient capacity to permit complete immersion of the specimen in a volume of liquid equivalent to 12 to 20 times the volume of the specimen

4. PROCEDURE

4.1 Preparation of specimen.—If the material is too thick or has a fabric backing or an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111 If it is not necessary to buff the material for the above reasons, it shall be lightly buffed on both principal surfaces as described in method 1111 to remove any surface coating before exposing the specimen to the test medium Portions of the test unit from which specimens are to be taken for tension tests shall be buffed in the strip form before cutting with the die

4.2 The time and temperature of exposure shall be as specified in the detail specification and wherever practicable should be selected from the recommendations in 4 2 1 and 4 2 2

4.2.1 *Temperature* —The temperature chosen for the test should depend upon the conditions of service of the compound It should be equal to or higher than the temperature encountered in service In comparative tests, identical temperatures shall be employed The following are recommended

4.2.1.1 23°±2° C (73 4°±3 6° F)

4.2.1.2 70°±2° C (158°±3 6° F)

4.2.1.3 100°±2° C (212°±3 6° F)

4.2.1.4 125°±2° C (257°±3 6° F) and increments of 25° C (45° F)

4.2.2 *Time* —The time of immersion or exposure depends on the compound being tested, the temperature of test, and the liquid used The following are suggested 22±¼ hour, 46±¼ hour, 70±¼ hour, 166±1 hour, and 30 days The period of exposure should be such that the deterioration will not be great enough to prevent determination of the final property.

4.3 Unless otherwise specified in the detail specification, all tests shall be made in the absence of direct light

4.4 In the case of immersion tests, a volume of liquid equivalent to 12 to 20 times the volume of the specimen, but in any case sufficient to completely cover the specimen shall be added to the container, 3 9 2 The specimens shall be

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so immersed that they will not touch each other or the surfaces of the container

4.5 If the specimen is not destroyed in determining the specified characteristic of the material before exposure, the same specimen shall be used in the liquid treatment test If it is necessary to destroy the specimen in determining the specified characteristic, two sets of specimens shall be prepared from the test unit, one set for determining the specified characteristic before exposure, and the other set for determining the specified characteristic after exposure

4.6 Where the specimen is destroyed in determining the specified characteristic of the material, such as tensile strength, the specified characteristic of the unexposed material shall be determined within the period ranging from 24 hours before the start of the exposure period to the time the specified characteristic of the exposed specimen is determined

4.7 The characteristic used for measuring the change due to liquid treatment shall be as specified in the detail specification and shall be determined as described in the applicable method of test

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TENSILE STRENGTH AND ELONGATION IMMEDIATELY AFTER IMMERSION IN LIQUIDS

1. SCOPE

1.1 This method is intended for use in determining the change in tensile strength and elongation of soft vulcanized rubber compounds due to immersion in liquids Tensile strength and elongation after immersion are determined immediately after the specimen is removed from the immersion medium

2. SPECIMEN

2.1 The specimen shall be as described in method 4111

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed $\frac{1}{8}$ inch, and shall be uniform to within ± 0.003 inch Specimens from the same test unit shall be of the same thickness within ± 0.01 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A cylindrical container as described in 3 9 2 of method 6001 A test tube having an outside diameter of approximately 38 mm and a length of 200 to 300 mm has been found suitable

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3.1.2 Air or reflux condenser suitable for attaching to the container for maintaining the volume constant when the temperature of the immersion medium is above the boiling point or when the volatile liquids are used

3.1.3 Equipment with suitable controls for maintaining the required temperature throughout the immersion medium during the immersion period, such as a constant temperature bath or air oven with thermostatic control

3.1.4 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature of the immersion medium within $\pm 1.0°$ C ($1.8°$ F)

3.1.5 Aluminum or corrosion-resistant steel screens or glass framework to prevent the specimens from touching each other or the surface of the container

3.1.6 Filter paper or other absorbent material

3.1.7 Glass beads to prevent bumping of liquid at high temperatures

3.1.8 Acetone

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of $23° \pm 2°$ C ($73.4° \pm 3.6°$ F) for $46 \pm \frac{1}{4}$ hour

4.2 The immersion medium shall be as specified in the detail specification

4.3 Preparation of specimen.—The specimen shall be prepared as described in method 4111 If it is not necessary to buff the material to remove an uneven surface or to reduce the thickness, the material shall be lightly buffed on both principal surfaces as described in method 1111 to remove any surface coating

4.4 The specimen shall be free from mechanical damage The cross-sectional area of the specimen shall be determined before immersion, as described in method 4111

4.5 A volume of the immersion medium sufficient to meet the requirements in 4 4 of method 6001 shall be added to the container After adjusting the immersion medium to the required temperature, the specimen shall be completely immersed so that the liquid will circulate freely around the specimen during the immersion period

4.6 If the immersion medium is appreciably volatile at the temperature required for the immersion, the container shall be fitted with the air or reflux condenser to maintain a constant volume If the condenser is not needed, the container shall be closed with a solid stopper fitted to provide escape for any excess pressure developed and to prevent excessive evaporation The specimen shall remain immersed in the medium for the required time at the required temperature

4.7 At the end of the immersion period, the specimen shall be immediately removed from the liquid If the immersion medium is

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above room temperature, the specimen shall be immediately placed in a fresh supply of the same immersion medium at room temperature, and allowed to remain for 30 ± 5 minutes

4.8 Immediately after removal from the immersion medium or from the cooling liquid, the specimen shall be quickly dipped into acetone, blotted lightly with filter paper or other absorbent material, and benchmarks for determining elongation applied to the specimen as described in method 4121 Tensile strength and elongation of the exposed specimen shall be immediately determined as described in methods 4111 and 4121, respectively The testing of the specimen shall be completed within 3 minutes after it has been removed from the immersion medium at room temperature or from the cooling liquid Tensile strength and elongation, methods 4111 and 4121, respectively, shall be determined of unexposed specimens from the same test unit for the purpose of comparison in determining the amount of deterioration due to immersion

4.8.1 *Tensile strength based on swollen area* — If the tensile strength based on the swollen area of the specimen is specified, it shall be calculated as follows

$$\text{Tensile strength in pounds per square inch based on swollen area} = \frac{T}{\left(1+\frac{V}{100}\right)^{\frac{2}{3}}}$$

Where

T is the tensile strength in pounds per square inch of swollen specimen based on the original unswollen cross-sectional area

V is the percentage increase in volume determined as described in method 6211

5. RESULTS

5.1 Unless otherwise specified in the detail specification, the number of specimens immersed in the test medium from each test unit shall be as required in methods 4111 or 4121.

5.2 Calculation.—The change in tensile strength and elongation of the test unit due to immersion shall be calculated as follows:

$$\text{Change in tensile strength or elongation, percent} = \frac{O-E}{O} \times 100$$

Where

O is the tensile strength or elongation of the test unit before immersion

E is the tensile strength or elongation of the test unit after immersion

5.3 The change in tensile strength and elongation of the test unit shall be recorded separately to the nearest one percent.

5.4 The temperature, time of immersion, and the immersion medium used shall be recorded

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TENSILE STRENGTH AND ELONGATION, LIQUID IMMERSION, AFTER RECOVERY

1. SCOPE

1.1 This method is intended for use in determining the change in tensile strength and elongation of soft vulcanized rubber compounds due to immersion in liquids Tensile strength and elongation after immersion are determined after the specimen has been permitted to recover for a definite period of time after removal from the immersion medium

2. SPECIMEN

2.1 The specimen shall be as described in method 4111

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed $\frac{1}{16}$ inch, and shall be uniform to within ± 0.003 inch Specimens from the same test unit shall be of the same thickness within ± 0.01 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as described in method 6111

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4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of $23° \pm 2°$ C (73 4° ± 3 6° F) for $46 \pm \frac{1}{4}$ hour

4.2 Unless otherwise specified in the detail specification, the specimen shall be permitted to recover $4 \pm \frac{1}{4}$ hour in air after removal from the immersion or cooling medium before determining the tensile strength and elongation after immersion

4.3 The immersion medium shall be as specified in the detail specification

4.4 The remainder of the procedure shall be as described in 4 3 to 4 8, inclusive, of method 6111, except that after immersion the specimen shall be permitted to recover for the required time, 4 2, before determining the tensile strength and elongation

5. RESULTS

5.1 The results shall be as described in 6111, except that the time of recovery shall also be recorded

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CHANGE IN VOLUME, LIQUID IMMERSION

1. SCOPE

1.1 This method is intended for use in determining the change in volume of vulcanized rubber compounds due to immersion in liquids

2. SPECIMENS

2.1 The specimen shall consist of a portion of the test unit 2 by 1 inch

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed $\frac{1}{16}$ inch, and shall be uniform to within ± 0.005 inch Specimens from the same test unit shall be of the same thickness within ± 0.01 inch, method 6011

3. APPARATUS

3.1 The apparatus shall be as follows:

3.1.1 Apparatus described in method 6111

3.1.2 Analytical balance and weights or Jolly balance

3.1.3 Beaker, 250 ml

3.1.4 Distilled water

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of $23° \pm 2°$ C ($73.4° \pm 3.6°$ F) for $46 \pm \frac{1}{4}$ hour

4.2 The immersion medium shall be as specified in the detail specification

4.3 Preparation of specimen.—The specimen shall be prepared as described in 4.1 of method 6001

4.4 The specimen shall be free from mechanical damage The specimen shall be weighed in air and the weight recorded to the nearest milligram as W_1 The specimen shall be dipped in acetone, thoroughly rinsed with distilled water, weighed in distilled water, and the weight recorded to the nearest milligram as W_2

4.5 A volume of the immersion medium sufficient to meet the requirements in 4.4 of method 6001 shall be added to the container After adjusting the immersion medium to the required temperature, the specimen shall be quickly dipped into acetone, blotted with filter paper or other absorbent material, and completely immersed so that the liquid will circulate freely around the specimen during the immersion period

4.6 If the immersion medium is appreciably volatile at the temperature required for the immersion, the container shall be fitted with the air or reflux condenser to maintain a constant volume If the condenser is not needed, the container shall be closed with a solid stopper fitted to provide escape for any excess pressure developed and to prevent excessive evaporation The specimen shall remain immersed in the medium for the required time at the required temperature

4.7 At the end of the immersion period, the specimen shall be immediately removed from the liquid If the immersion medium is above room temperature, the specimen shall be immediately placed in a fresh supply of the same immersion medium at room temperature, and allowed to remain for 30 ± 5 minutes

4.8 Immediately after removal from the immersion medium or from the cooling liquid, the specimen shall be quickly dipped into acetone and blotted lightly with filter paper or other absorbent material The specimen shall be weighed in air and the weight recorded to the nearest milligram as W_3 The specimen shall be dipped into acetone, rinsed with distilled water, weighed in distilled water, and the weight recorded to the nearest milligram as W_4 The final weighing shall be made as quickly as possible from the time the specimen is removed from the immersion medium

4.9 A Jolly balance adequately protected from air currents may be used for weighing

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5. RESULTS

5.1 Calculation.—The change in volume of the specimen due to immersion shall be calculated as follows

$$\text{Change in volume, percent,} = \frac{(W_3 - W_4) - (W_1 - W_2)}{(W_1 - W_2)} \times 100$$

Where

W_1 is the weight of the specimen in air before immersion, grams

W_2 is the weight of the specimen in distilled water before immersion, grams

W_3 is the weight of the specimen in air after immersion, grams

W_4 is the weight of the specimen in distilled water after immersion, grams

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.3 The change in volume of the test unit shall be the average of the values obtained from the specimens tested

5.4 The change in volume of the test unit shall be recorded to the nearest one percent

5.5 The temperature, time of immersion, and the immersion medium used shall be recorded

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CHANGE IN THICKNESS IMMEDIATELY AFTER IMMERSION IN LIQUID

1. SCOPE

1.1 This method is intended for use in determining the change in thickness of vulcanized rubber compounds due to immersion in liquid The thickness after immersion is determined immediately after the specimen is removed from the immersion medium

2. SPECIMEN

2.1 The specimen shall be as described in method 6211

3. APPARATUS

3.1 The apparatus shall be as described in method 6111

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of 23°±2° C (73 4°±3 6° F) for 46±¼ hour

4.2 The immersion medium shall be as specified in the detail specification

4.3 Preparation of specimen.—The specimen shall be prepared as described in 4 1 of method 6001

4.4 The specimen shall be free from mechanical damage The average thickness of the specimen shall be determined as described in method 2011 and the value recorded to the nearest 0 001 inch as the thickness, T_1

4.5 A volume of the immersion medium sufficient to meet the requirements of 4 4 of method 6001 shall be added to the container After adjusting the immersion medium to the required temperature, the specimen shall be completely immersed so that the liquid will circulate freely around the specimen during the immersion period

4.6 If the immersion medium is appreciably volatile at the temperature required for the immersion, the container shall be fitted with the air or reflux condenser to maintain a constant volume If the condenser is not needed, the container shall be closed with a solid stopper fitted to provide escape for any excess pressure developed and to prevent excessive evaporation The specimen shall remain immersed in the medium for the required time at the required temperature

4.7 At the end of the immersion period, the specimen shall be immediately removed from the liquid If the immersion medium is above room temperature, the specimen shall be immediately placed in a fresh supply of the same immersion medium at room temperature, and allowed to remain 30±5 minutes

4.8 Immediately after removal from the immersion medium or from the cooling liquid, the specimen shall be quickly dipped into acetone and blotted lightly with filter paper or other absorbent material The thickness of the specimen shall be immediately determined again as described in 4 4 and the value recorded to the nearest 0 001 inch as T_2 The thickness measurements shall be made as near as practicable at the same points as previously measured in 4 4

5. RESULTS

5.1 Calculation.—The change in thickness of the specimen due to immersion shall be calculated as follows

$$\text{Change in thickness, percent,} = \frac{T_2 - T_1}{T_1} \times 100$$

Where

T_1 is the thickness of the specimen before immersion, inch

T_2 is the thickness of the specimen after immersion, inch

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.3 The change in thickness of the test unit shall be the average of the values obtained from the specimens tested

5.4 The change in thickness of the test unit shall be recorded to the nearest one percent

5.5 The temperature, time of immersion, and the immersion medium used shall be recorded

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

CHANGE IN THICKNESS, LIQUID IMMERSION, AFTER RECOVERY

1. SCOPE

1.1 This method is intended for use in determining the change in thickness of vulcanized rubber compounds due to immersion in liquids The thickness after immersion is determined after the specimen has been permitted to recover for a definite period of time after removal from the immersion medium

2. SPECIMEN

2.1 The specimen shall be as described in method 6211

3. APPARATUS

3.1 The apparatus shall be as described in method 6111

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of 23° ± 2° C (73.4° ± 3 6° F) for 46 ± ¼ hour

4.2 Unless otherwise specified in the detail specification, the specimen shall be permitted to recover for 22 ± ¼ hour in air after removal from the immersion or cooling medium before determining the thickness after immersion.

4.3 The immersion medium shall be as specified in the detail specification

4.4 The remainder of the procedure shall be as described in 4 3 to 4 8, inclusive, of method 6231 except that after blotting the specimen with filter paper or other absorbent material, 4 8 of method 6231, it shall be suspended in air and allowed to recover for the required time, 4 2, before the thickness after immersion is determined

5. RESULTS

5.1 The results shall be as described in 6231 except that the time of recovery shall also be recorded

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CHANGE IN WEIGHT, LIQUID IMMERSION

1. SCOPE

1.1 This method is intended for use in determining the change in weight of vulcanized rubber compounds due to immersion in liquids

2. SPECIMEN

2.1 The specimen shall be as described in method 6211

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in method 6111.

3.1.2 Analytical balance and weights, or Jolly balance

3.1.3 Beaker, 250 ml

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of 23° ± 2° C (73.4° ± 3.6° F.) for 46 ± ¼ hour

4.2 The immersion medium shall be as specified in the detail specification

4.3 Preparation of specimen.—The specimen shall be prepared as described in 4.1 of method 6001

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4.4 The specimen shall be free from mechanical damage. The specimen shall be weighed and the weight recorded to the nearest milligram as W_1. A volume of the immersion medium sufficient to meet the requirements of 4.4 of method 6001 shall be added to the container. After adjusting the immersion medium to the required temperature, the specimen shall be completely immersed so that the liquid will circulate freely around the specimen during the immersion period.

4.5 If the immersion medium is appreciably volatile at the temperature required for the immersion, the container shall be fitted with the air or reflux condenser to maintain a constant volume. If the condenser is not needed, the container shall be closed with a solid stopper fitted to provide escape for any excess pressure developed and to prevent excessive evaporation. The specimen shall remain immersed in the medium for the required time at the required temperature.

4.6 At the end of the immersion period, the specimen shall be immediately removed from the liquid. If the immersion medium is above room temperature, the specimen shall be immediately placed in a fresh supply of the same immersion medium at room temperature, and allowed to remain for 30 ± 5 minutes.

4.7 Immediately after removal from the immersion medium or from the cooling liquid, the specimen shall be quickly dipped into acetone and blotted lightly with filter paper or other absorbent material. The specimen shall be weighed as quickly as possible after it is removed from the immersion medium and the weight recorded to the nearest milligram as W_2.

4.8 A Jolly balance adequately protected from air currents may be used for weighing.

5. RESULTS

5.1 Calculation.—The change in weight of the specimen due to immersion shall be calculated as follows:

$$\text{Increase in weight, percent,} = \frac{W_2 - W_1}{W_1} \times 100$$

$$\text{Decrease in weight, percent,} = \frac{W_1 - W_2}{W_1} \times 100$$

Where

W_1 is the weight of the specimen before immersion, grams

W_2 is the weight of the specimen after immersion, grams

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested.

5.3 The change in weight of the test unit shall be the average of the values obtained from the specimens tested.

5.4 The change in weight of the test unit shall be recorded to the nearest one percent.

5.5 The temperature, time of immersion, and the immersion medium used shall be recorded.

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This method is a part of Fed Test Method Std No 601
(formerly Fed. Spec ZZ-R-601a)

CHANGE IN LAMINATED MATERIALS, LIQUID IMMERSION

1. SCOPE

1.1 This method is intended for use in determining the change in laminated rubber materials due to immersion in liquids

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 4 by 1 inch

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A container of sufficient capacity to permit immersion of the specimen in liquid to a depth of 6 inches

3.1.2 Air or reflux condenser or other device suitable for attaching to the container for maintaining the volume constant when the temperature of the immersion medium is above the boiling point or when volatile liquids are used

3.1.3 Apparatus described in 3 1 3 to 3 1 8, inclusive, of method 6111

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of 23° ± 2° C (73 4° ± 3 6° F) for 22 ± ¼ hour

4.2 Unless otherwise specified in the detail specification, benzene conforming to the requirements of 3 5 1 of method 6001 shall be used as the immersion medium

4.3 The specimen shall be free from mechanical damage A volume of the immersion medium sufficient to cover the specimen to a depth of 6 inches shall be added to the container After adjusting the immersion medium to the required temperature, the specimen shall be immersed to a depth of 6 inches so that the liquid will circulate freely around the specimen during exposure

4.4 If the immersion medium is appreciably volatile at the temperature required for immersion, the container shall be fitted with the air or reflux condenser or other device to maintain a constant volume. If the condenser or other device is not needed, the container shall be closed with a solid stopper or other device fitted to provide escape for any excess pressure developed and to prevent excessive evaporation. The specimen shall remain immersed in the medium for the required time at the required temperature

4.5 At the end of the immersion period, the specimen shall be immediately removed from the liquid If the immersion medium is above room temperature, the specimen shall be immediately placed in a fresh supply of the same immersion medium at room temperature, and allowed to remain for 30 ± 5 minutes.

4.6 Immediately after removal from the immersion medium or from the cooling liquid, the specimen shall be quickly dipped into acetone and blotted lightly with filter paper or other absorbent material. The specimen shall be examined visually for evidence of separation into layers by attempting to separate the layers by hand, or for other damage

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 The number of specimens tested from each test unit and the number from each test unit that showed separation into layers, or other damage shall be recorded

5.3 The temperature, time of immersion, and the immersion medium used shall be recorded

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HOSE, CHANGE IN ADHESION, LIQUID IMMERSION

1. SCOPE

1.1 This method is intended for use in determining the change in adhesion of different parts of rubber hose due to exposure in liquids Two procedures are described, one using a straight specimen and the other using a U-shaped specimen The procedure to be used should be as specified in the detail specification. When the flexibility of the hose will permit, it is recommended that the U-shaped specimen be used

2. SPECIMEN

2.1 **Straight specimen.**—The specimen shall consist of a portion of the test unit approximately 12 inches in length and of full circumference.

2.2 **U-shaped specimen.**—The specimen shall consist of a portion of the test unit of full circumference and of sufficient length so that it can be bent into a U-shape with the following inner diameter of the bend.

Inner diameter of hose	Inner diameter of bend
Inches	*Inches*
¾	6
1	8
1¼	10
1½	16
2	20
2½	30
3	36

3. APPARATUS

3.1 The apparatus shall be as follows:

3.1.1 A container of sufficient capacity to permit complete immersion of the straight specimen, 2 1, in liquid.

3.1.2 A container of sufficient capacity to permit immersion of the base of the U-shaped specimen, 2.2.

3.1.3 Air or reflux condenser or other device suitable for attaching to the container for maintaining the volume constant when the temperature of the immersion medium is above the boiling point or when volatile liquids are used

3.1.4 A hacksaw or other tool for cutting the specimen

3.1.5 A clamp or other equipment for holding U-shaped specimens in position

3.1.6 Apparatus described in 3 1 3 to 3 1 8, inclusive, of method 6111

4. PROCEDURE

4.1 The procedure, 1 1, to be used shall be as specified in the detail specification

4.2 Unless otherwise specified in the detail specification, the specimen shall be exposed at a temperature of 23° ± 2° C (73 4° ± 3.6° F) for 46 ± ¼ hour

4.3 The test or immersion medium shall be as specified in the detail specification

4.4 **Straight specimen.**

4.4.1 The specimen shall be free from mechanical damage A volume of the immersion medium sufficient to permit the complete immersion of the specimen shall be added to the container After adjusting the immersion medium to the required temperature, the specimen shall be completely immersed so that the liquid will circulate freely around the specimen during immersion

4.4.2 If the immersion medium is appreciably volatile at the temperature required for immersion, the container shall be fitted with the air or reflux condenser or other device to maintain a constant volume If the condenser or other device is not needed, the container shall be closed with a cover or other device so as to provide escape for any excess pressure developed and to prevent excessive evaporation. The specimen shall remain immersed in the medium for the required time at the required temperature.

4.4.3 At the end of the immersion period, the specimen shall be immediately removed from the liquid. If the immersion medium is above room temperature, the specimen shall be immediately placed in a fresh supply of the same immersion medium at room temperature, and allowed to remain for 30 ± 5 minutes

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4.4.4 Immediately after removal from the immersion medium or from the cooling liquid, the specimen shall be quickly dipped into acetone, blotted lightly with filter paper or other absorbent material, and allowed to dry for 15 minutes at room temperature before testing

4.4.5 During the 15-minute drying time, friction specimens shall be prepared from the specimen of hose after immersion as described in method 8011 for determining friction between plies, between tube and plies, and between cover and plies

4.4.6 At the end of the 15-minute drying period, friction between plies, between tube and plies, and between cover and plies of the exposed specimen shall be immediately determined as described in method 8011 Friction of the above parts of unexposed specimens from the same test unit shall be determined for the purpose of comparison

4.5 U-shaped specimen.

4.5.1 The specimen shall be free from mechanical damage The specimen shall be placed in the container in the shape of a U The test medium shall be adjusted to the required temperature and poured into the container and specimen until the liquid is at the same level inside and outside of the specimen The ends of the specimen shall be approximately 2 inches above the surfaces of the test medium and the specimen arranged so that the liquid can circulate freely around the base of the U portion during exposure The remainder of the procedure shall be as described in 4 4 2 to 4 4 6, inclusive The specimens for friction tests after immersion shall be taken as close to the base of the U as possible.

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be exposed to the test medium

5.2 Unless otherwise specified in the detail specification, in determining the change in friction, the number of specimens tested from each exposed specimen from each test unit shall be as required in method 8011

5.3 Calculation. —The change in friction of the test unit due to exposure shall be calculated as follows

$$\text{Change in friction, percent,} = \frac{O-E}{O} \times 100$$

Where

O is the friction of the test unit before exposure
E is the friction of the test unit after exposure

5.4 When one specimen from each test unit is exposed to the test medium, the change in friction of the test unit shall be the value obtained from the exposed specimen When more than one specimen from each test unit is exposed to the test medium, the change in friction of the test unit shall be the average of the values obtained from the specimens exposed to the test medium

5.5 The change in friction of the test unit between plies, tube and plies, and cover and plies shall be recorded separately to the nearest one percent

5.6 The temperature, time of exposure, and the test or immersion medium used shall be recorded

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HOSE, CHANGE IN DIAMETER, LIQUID EXPOSURE

1. SCOPE

1.1 This method is intended for use in determining the change in inside diameter of rubber hose due to exposure to liquids

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit approximately 12 inches in length and of full circumference

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Plugs for closing the end of the specimen

3.1.2 Clamp or other equipment for supporting the hose in vertical position

3.1.3 Steel balls having varying known diameters

3.1.4 Apparatus described in 3 1 2 to 3 1 4, inclusive, of method 6111

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be exposed at a temperature of 23° ± 2° C (73 4° ± 3 6° F) for 46 ± ¼ hour

4.2 The test medium shall be as specified in the detail specification

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4.3 The specimen shall be free from mechanical damage The inside diameter of the specimen before exposure to the test medium shall be determined by one of the following methods, as specified in the detail specification, and the value recorded as D_1

(1) The inside diameter of the specimen shall be the diameter of a spherical ball that will just pass through the hose of its own weight when the specimen is held in a vertical position

(2) The average inside diameter of the specimen shall be determined as described in method 2221.

4.4 After the inside diameter of the specimen has been determined, the specimen shall be clamped in an upright position and a plug inserted in the lower end for a distance of 3 inches or less from the bottom so as to close the hose The test medium shall be adjusted to the required temperature and poured into the specimen until the hose is filled to approximately one inch from the top If the test medium is appreciably volatile at the temperature required for the exposure, the hose shall be fitted with the air or reflux condenser to maintain a constant volume If a condenser is not needed, the hose shall be closed with a solid stopper fitted to provide escape for any excess pressure developed and to prevent excessive evaporation The specimen and liquid shall be maintained for the required time at the required temperature

4.5 At the end of the exposure period, the liquid shall be immediately drained from the hose A 3-inch length shall be cut from the bottom end and a 2-inch length from the top end of the specimen and discarded The inside diameter of the remaining portion of the specimen shall again be immediately determined as described in 4 3 and the value recorded as the diameter, D_2 The method used for determining diameter before exposure shall also be used for determining diameter after exposure

5. RESULTS

5.1 Calculation.—The change in diameter of the specimen due to exposure shall be calculated as follows

$$\text{Increase in diameter, percent,} = \frac{D_2 - D_1}{D_1} \times 100$$

$$\text{Decrease in diameter, percent,} = \frac{D_1 - D_2}{D_1} \times 100$$

Where

D_1 is the diameter of the specimen before exposure, inch

D_2 is the diameter of the specimen after exposure, inch

5.2 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.3 When one specimen is tested from each test unit, the change in diameter of the test unit shall be the value obtained from the specimen tested When more than one specimen is

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tested from the test unit, the change in diameter of the test unit shall be the average of the values obtained from the specimens tested

5.4 The change in diameter of the test unit shall be recorded to the nearest one percent.

5.5 The temperature, time of exposure, the test medium used, and the method of test for determining the diameter shall be recorded

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EXTRACTION, ORGANIC SOLVENT

1. SCOPE

1.1 This method is intended for use in determining the amount of material extracted from vulcanized rubber compounds by immersion of the specimen in organic solvents

2. SPECIMEN

2.1 The specimen shall be as described in method 6621

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in method 6621

3.1.2 Apparatus described in 3 1 3 and 3 1 4 of method 6111

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of 23°±2° C (73 4°±3.6° F) for 46±¼ hour

4.2 The solvent shall be as specified in the detail specification

4.3 Preparation of specimen.—The specimen shall be prepared as described in 4 1 of method 6001

4.4 The specimen shall be weighed and the weight recorded to the nearest milligram as W_1 A volume of the immersion medium sufficient to meet the requirements in 4 4 of method 6001 shall be added to the flask After adjusting the immersion medium to the required temperature, the specimen shall be completely immersed so that the solvent will circulate freely around the specimen during the immersion period.

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4.5 If the immersion medium is appreciably volatile at the required temperature for immersion, the flask shall be fitted with the air or reflux condenser to maintain a constant volume If the condenser is not needed, the flask shall be closed with a solid stopper, fitted to provide escape for any excess pressure developed and to prevent excessive evaporation The specimen shall remain immersed in the solvent for the required time at the required temperature

4.6 At the end of the immersion period, the solvent shall be immediately poured from the flask into a weighed container, the flask and specimen rinsed with 25 ml of fresh solvent, and the wash added to the extract The combined solvents shall be evaporated to dryness on a steam bath or hot plate The weighed container and residue shall be dried in the air oven at 105°±3° C (221°±5° F) for one hour, cooled in a desiccator, weighed, and the weight of the residue recorded to the nearest milligram A blank shall be run using the same volume of solvent The weight of the solvent extract after making correction for the blank shall be recorded to the nearest milligram as W_2

5. RESULTS

5.1 Calculation.—The solvent extract of the specimen due to immersion shall be calculated as follows

$$\text{Solvent extract, percent, } = \frac{W_2}{W_1} \times 100$$

Where.

W_1 is the weight of the specimen, grams

W_2 is the weight of the solvent extract, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The solvent extract of the test unit shall be the average of the values obtained from the specimens tested

5.4 The solvent extract from the test unit shall be recorded to the nearest 0 1 percent

5.5 The time and temperature of extraction and the solvent used shall be recorded

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RESISTANCE TO BOILING WATER

1. SCOPE

1.1 This method is intended for use in determining the change in soft vulcanized rubber compounds due to immersion in boiling water It is applicable to compounds which are prepared for use with water at elevated temperatures

2. SPECIMEN

2.1 The specimen shall be as described in method 4111 or 8011, or other method of test specified in the detail specification for determining the amount of deterioration.

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen for tensile tests shall be the thickness of the material undergoing test, but shall not exceed $\frac{1}{16}$ inch, and shall be uniform to within ± 0.003 inch Specimens for tensile tests from the same test unit shall be of the same thickness within ± 0.01 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An apparatus for immersing specimens in boiling water for the required time A suitable apparatus for most materials consists of a 500-ml Erlenmeyer wide-mouth flask equipped with a reflux condenser to maintain a constant volume of water, and supports for suspending the specimen

3.1.2 Hot plate or other source of heat

3.1.3 Distilled water

4. PROCEDURE

4.1 Rubber parts.—Unless otherwise specified in the detail specification, tensile strength and elongation, methods 4111 and 4121, respectively, shall be used to determine the deterioration of the rubber compounds due to boiling in water

4.2 Rubber parts containing plies of other materials.—Unless otherwise specified in the detail specification, the effect of boiling water on the friction between the parts shall be determined as described in method 8011.

4.3 Unless otherwise specified in the detail specification, the specimen shall be immersed in boiling water for 166 ± 1 hour

4.4 Preparation of specimen

4.4.1 When change in tensile strength and elongation of the rubber compound due to boiling water are to be determined the specimen shall be prepared as described in method 4111, except that if it is not necessary to buff the material to remove an uneven surface or to reduce the thick ess, the material shall be lightly buffed on both principal surfaces as described in method 1111 to remove any surface coating When another property is specified in the detail specification for determining the deterioration of the rubber compound, the specimen shall be prepared as described in the specified method of test

4.4.2 When the effect of boiling water on the adhesion between parts is to be determined, the specimen shall be prepared as described in method 8011

4.5 The specimen shall be free from mechanical damage The cross-sectional area of the specimen for tensile strength of rubber parts shall be determined before immersion as described in method 4111

4.6 A volume of distilled water sufficient to completely immerse the specimen shall be added to the flask After adjusting the water to the boiling temperature, the specimen shall be completely immersed so that the water will circulate freely around the specimen during immersion The flask shall be fitted with a reflux condenser to maintain a constant volume The specimen shall remain immersed in the boiling water for the required time Several specimens from the same test unit may be exposed at one time in the same flask provided they are completely immersed in the boiling water so that they will not touch each other or the surfaces of the flask

4.7 At the end of the exposure period, the specimen shall be immediately removed from the boiling water and set aside to rest for not less than 16 hours and not more than 96 hours at room temperature before tests are made

4.8 When elongation is to be determined on the exposed specimen, benchmarks shall be ap-

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plied to the specimen as described in method 4121 after removal from the boiling water

4.9 At the end of the rest period, unless otherwise specified in the detail specification, tensile strength and elongation of the exposed rubber specimen shall be determined as described in methods 4111 and 4121, respectively, and friction on the exposed rubber-fabric or other plied specimen shall be determined as described in method 8011 If another property for determining the deterioration of the rubber compound or adhesion between parts is specified in the detail specification, the exposed specimen shall be tested as described in the specified method The same test shall be conducted on an unexposed specimen from the same test unit for the purpose of comparison in determining the amount of deterioration due to the exposure in boiling water

5. RESULTS

5.1 Unless otherwise specified in the detail specification, the number of specimens from each test unit immersed in boiling water shall be as required in methods 4111 or 4121, and 8011, or other method of test specified for determining the deterioration

5.2 Calculation.—The change in tensile strength and elongation, and friction, or other specified characteristic of the test unit due to immersion in boiling water shall be calculated as follows

$$\text{Change in characteristic, percent, } = \frac{O-E}{O} \times 100$$

Where

O is the characteristic of the test unit before immersion in boiling water

E is the same characteristic of the test unit after immersion in boiling water

5.3 The change in tensile strength and elongation, and friction, or other characteristic of the test unit shall be recorded to the nearest one percent

5.4 The time of immersion and the property used for determining the deterioration shall be recorded

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EXTRACTION, BOILING WATER

1. SCOPE

1.1 This method is intended for use in determining the amount of material extracted from vulcanized rubber compounds by immersion of the specimen in boiling water

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit weighing approximately 10 grams

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed 1/8 inch, and shall be uniform to within ±0 005 inch Specimens from the same test unit shall be of the same thickness within ±0 01 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Aluminum or corrosion-resistant steel screens or glass framework to prevent the specimen from touching the surface of the container

3.1.2 Analytical balance and weights

3.1.3 *Air oven* —A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the drying of the extract in 4 3

3.1.4 A glass container for evaporating the solution and weighing the residue

3.1.5 A glass flask, 250 ml

3.1.6 Desiccator and suitable desiccant

3.1.7 A reflux condenser suitable for attaching to flask by means of stopper

3.1.8 Steam bath and electrical hot plate, or other source of heat

3.1.9 Distilled water

4. PROCEDURE

4.1 **Preparation of specimen.**—The specimen shall be prepared as described in 4 1 of method 6001

4.2 The specimen shall be weighed and the weight recorded to the nearest milligram as W_1 A volume of distilled water sufficient to meet the requirements in 4 4 of method 6001 shall be added to the flask After adjusting the water to the boiling temperature, the specimen shall be completely immersed so that water will circulate freely around the specimen during the extraction The flask shall be fitted with a reflux condenser to maintain a constant volume The specimen shall remain immersed for one hour at the boiling temperature

4.3 At the end of the immersion period, the water extract shall be immediately poured from the flask into a weighed glass container, the flask and specimen rinsed with 25 ml of distilled water, and the wash added to the extract The combined water solutions shall be evaporated to dryness on the steam bath or hot plate The weighed container and residue shall be dried in the air oven at 105° ± 3° C (221° ± 5° F) for one hour, cooled in a desiccator, weighed, and the weight of the residue recorded to the nearest milligram A blank shall be run using the same amount of distilled water The weight of the water extract after making correction for the blank shall be recorded to the nearest milligram as W_2

5. RESULTS

5.1 **Calculation** —The water extract of the specimen due to extraction with boiling water shall be calculated as follows

$$\text{Water extract, percent,} = \frac{W_2}{W_1} \times 100$$

Where

W_1 is the weight of the specimen, grams

W_2 is the weight of the water extract, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The water extract from the test unit shall be the average of the values obtained from the specimens tested

5.4 The water extract from the test unit shall be recorded to the nearest 0 1 percent

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CHANGE IN WEIGHT, WATER IMMERSION

1. SCOPE

1.1 This method is intended for use in determining the change in weight of vulcanized rubber compounds due to immersion in water The change in weight, water soluble material in the compound, and the amount of water absorbed are based on the area of the specimen and expressed in grams per square inch

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit approximately 1 by 2 inches The width and length shall be uniform within ±0.01 inch, method 2121.

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed 1/16 inch, and shall be uniform to within ±0 05 inch Specimens from the same test unit shall be of the same thickness within ±0 01 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Analytical balance and weights

3.1.2 Container with tightly fitting cover for immersion of the specimen

3.1.3 *Air oven* —A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the immersion and drying period.

3.1.4 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature of the water and air oven within ±1 0° C (1 8° F.)

3.1.5 Filter paper or other absorbent material.

3.1.6 Soft brush

3.1.7 Glass rods, ⅜ inch in diameter

3.1.8 Alcohol, ethyl, 95 percent

3.1.9 Distilled water

3.1.10 Soap

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed in distilled water at a temperature of 70° ± 2° C (158° ± 3 6° F.) for 166 ± 1 hour

4.2 Preparation of specimen.—The specimen shall be prepared as described in 4 1 of method 6001

4.3 The average thickness of the specimen shall be determined as described in method 2011 The average width and length of the specimen shall be determined as described in method 2121 The total area of the specimen shall be calculated from the length, width, and thickness and the value recorded to the nearest 0 01 square inch

4.4 The specimen shall be washed with soap and water, using a soft brush to remove loose material, and rinsed with water The washed specimen shall be blotted dry with filter paper or other absorbent material and suspended by means of small glass rods in the oven at a temperature of 70° ± 2° C (158° ± 3 6° F) for 22 ± ¼ hour The specimen shall be removed from the oven, allowed to cool to room temperature in air, weighed, and the weight recorded to the nearest milligram as W_1.

4.5 The specimen shall be dipped into 95 percent ethyl alcohol and suspended in distilled water at least ½ inch below the surface by means of glass rods so that the water will circulate freely around the specimen The container with specimen immersed shall be covered tightly to prevent evaporation of the liquid, placed in the oven, and maintained for the required time at the required temperature, 4 1.

4.6 At the end of the immersion period, the specimen shall be immediately removed from the distilled water, placed in a fresh supply of distilled water at room temperature, and allowed to remain for 30 ± 5 minutes

4.7 The specimen shall be removed from the distilled water at room temperature, the excess water shaken off, dipped quickly into 95 percent ethyl alcohol, blotted dry with filter paper or other absorbent material, weighed immediately, and the weight recorded to the nearest milligram as W_2

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4.8 The weighed specimen shall again be dried in the oven as described in 4 4 At the end of the drying period, the specimen shall be immediately removed from the oven, allowed to cool to room temperature in air, weighed immediately, and the weight recorded to the nearest milligram as W_3

5. RESULTS

5.1 Calculation.

5.1.1 The change in weight of the specimen due to immersion in water shall be calculated as follows

$$\text{Change in weight, grams per square inch,} = \frac{W_2 - W_1}{A}$$

5.1.2 The water-soluble material removed from the specimen by immersion in water shall be calculated as follows

$$\text{Water-soluble material, grams per square inch,} = \frac{W_1 - W_3}{A}$$

5.1.3 The amount of water absorbed by the specimen due to immersion in water shall be calculated as follows·

$$\text{Water absorbed, grams per square inch,} = \frac{W_2 - W_3}{A}$$

Where

W_1 is the weight of the specimen before immersion, grams

W_2 is the weight of the specimen after immersion in water, grams.

W_3 is the weight of the specimen after immersion in water and drying, grams

A is the total area of the specimen, square inches

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested.

5.3 The change in weight of, the water-soluble material in, and the water absorbed by the test unit, respectively, shall be the average of the values obtained from the specimens tested.

5.4 The change in weight of, the water-soluble material in, and the water absorbed by the test unit, shall be recorded separately to the nearest 0 01 gram per square inch

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RESISTANCE TO PHENOL

1. SCOPE

1.1 This method is intended for use in determining the resistance of vulcanized rubber compounds to immersion in a solution of phenol It is particularly applicable to hospital supplies

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 6 by 6 inches by the thickness of the material undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Glass container, with cover, of sufficient capacity to hold the specimen and phenol solution

3.1.2 Filter paper or other absorbent material

3.1.3 Soft brush

3.1.4 Phenol, 5-percent solution

3.1.5 Soap

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed in a 5 percent phenol solution at a temperature of 23° ± 2° C (73 4° ± 3 6° F) for 166 ± 1 hour

4.2 The specimen shall be washed with soap and water, using a soft brush to remove any adhered material, and rinsed with water The washed specimen shall be blotted dry with filter paper or other absorbent material and completely immersed in 5-percent phenol solution in a glass container so that the liquid will circulate freely around the specimen during immersion The container shall be covered tightly to prevent excessive evaporation of the liquid The specimen shall remain immersed in the solution for the required time at the required temperature

4.3 At the end of the immersion period, the specimen shall be removed from the solution, rinsed with water, and dried with filter paper or other absorbent material Unless otherwise specified in the detail specification, the specimen shall be allowed to rest for 2 hours The specimen shall be examined for tackiness, brittleness, or other evidence of deterioration

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 The number of specimens tested from each test unit and the number from each test unit that showed tackiness, brittleness, or other evidence of deterioration shall be recorded

5.3 The temperature and time of immersion shall be recorded

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ACCELERATED AGING TESTS, GENERAL

1. SCOPE

1.1 This group of methods is intended for use in determining the relative resistance of vulcanized rubber compounds to deterioration influenced by heat, light, and oxygen In general, these tests consist of exposing a portion of a test unit to controlled deterioration for a specified period of time and comparing the test result for a specified property obtained on the exposed material, and the test result for the same property determined on a portion of the same test unit that has not been subjected to the controlled deterioration In certain cases, however, the specified property is determined on the same specimen before and after exposure to controlled deterioration Changes in tensile strength and elongation are the properties most commonly used in determining the amount of deterioration and, unless otherwise specified in the detail specification, these properties shall be used.

2. SPECIMEN

2.1 Specimens from the same test unit shall be uniform in thickness within ±0.01 inch, method 2011

3. PROCEDURE

3.1 Preparation of specimen.—If the material is too thick or has a fabric backing or an uneven surface that may interfere with the test, the material shall be buffed as described in method 1111. Portions of the test unit from which specimens are to be taken for tension tests shall be buffed in the strip form before cutting with a die

3.1.1 If buffing is necessary, it shall be done before exposure of the material.

3.2 If the specimen is not destroyed in determining the specified characteristic of the material before aging, the same specimen shall be exposed in the aging test If it is necessary to destroy the specimen in determining the specified characteristic, two sets of specimens shall be prepared from the test unit, one set for determining the specified characteristic before aging and the other set for determining the specified characteristic after aging

3.3 Where the specimen is destroyed in determining the specified characteristic of the material, such as tensile strength, the specified characteristic of the unaged material shall be determined within the period ranging from 24 hours before the start of the exposure period to the time the specified characteristic of the aged specimen is determined

3.4 The characteristic used for measuring the amount of deterioration shall be as specified in the detail specification and shall be determined as described in the applicable method of test

3.5 If change in tensile strength and elongation are used as a measure of the amount of deterioration, these characteristics shall be determined as described in methods 4111 and 4121.

3.6 Where tensile strength or tensile stress is used to determine the amount of deterioration, the cross-sectional area of the specimen shall be determined as described in method 4111 before it is exposed

3.7 Benchmarks for determining the elongation of the aged specimen shall be placed on the specimen as described in method 4121 after it has been exposed

3.8 Aging conditions.

3.8.1 The aging conditions shall be as specified in the detail specification Whenever practicable, the aging conditions should be selected from the recommendations in table I for inclusion in the detail specifications In general, the conditions selected should be such that the deterioration will not be great enough to prevent determination of the final physical characteristic

3.8.2 In the oxygen pressure test, the increase in temperature from 70° to 80° C would be expected to approximately double the rate of oxidation of the rubber or other oxidizable material

3.8.3 For the evaluation of natural rubber compounds intended for service at elevated temperatures, the air heat tests 7221 and 7231 may be used with an operating temperature of

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90° C and for the evaluation of synthetic rubber compounds for all types of service, the above methods may be used with an operating temperature of 100° C By increasing the aging temperature in these tests to 90° C from 70° C , the deterioration of natural rubber compounds may be expected to be approximately quadrupled By increasing the aging temperature to 100° C from 70° C , the rate of deterioration of synthetic rubber compounds may be expected to be greatly increased depending on the type of formulations used

3.9 The contents of the oven, bomb, or other aging chamber shall be restricted to specimens known to be of the same composition

3.10 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device shall be used for measuring the temperature to within ±1 0° C (1 8° F) If the aging chamber is completely surrounded by a liquid heat-transfer medium, the temperature of the specimen may be considered to be the same as that of the liquid medium

3.11 Radiation shields shall be placed between the specimen and any portions of the wall of the aging chamber not maintained within the required temperature limits

3.12 Provisions shall be made for suspending the specimens vertically in the aging chamber without touching each other or the surface of the chamber

3.13 Equipment shall be provided for automatically recording the temperature throughout the aging period

3.14 The aging chamber shall be preheated to the exposure temperature before the specimen is placed in it If the temperature within the chamber changes while the specimen is being placed in it, the chamber and contents shall reach the specified temperature within 15 minutes after the specimen has been placed in the aging chamber

3.15 The aging period shall start at the time the specimen is placed in the aging chamber

3.16 Copper or brass parts shall not be exposed to the atmosphere used in the aging chamber

3.17 If a liquid heating medium that is injurious to the material is used, precautions shall be taken to prevent the liquid or its vapors from coming in contact with the material undergoing test

3.18 Oil or other combustible organic fluids are extremely hazardous in the presence of oxygen and should not be used as a heating medium for the oxygen pressure test

3.19 Oil or other combustible organic fluids may be hazardous at the high temperatures used in the air pressure test If their use is necessary, they shall have a flash point not lower than 200° C.

TABLE I —*Suggested aging conditions*

Test	Medium	Temperature	Pressure	Minimum volume of aging chamber per gram of rubber or other oxidizable materials
		°C ± 1° C	lb./in.²	Milliliters
Oxygen pressure	Oxygen	70	290–310	10
Modified oxygen pressure	do	80	48–52	10
Do	do	80	290–310	10
Air pressure	Air	127	78–82	100
Air oven	Air circulated by convection so as to change at least once per hour	70	Atmospheric	10
Do	do	90	do	10
Do	do	100	do	10
Do	do	121	do	10
Do	do	150	do	10
Test tube	do	150 and increments of 25° C	do	

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OXYGEN PRESSURE TEST

1. SCOPE

1.1 This method is intended for use in determining the effect of oxygen under pressure and at an elevated temperature on vulcanized rubber compounds

2. SPECIMEN

2.1 The specimen shall be as described in method 4111 or other method of test specified in the detail specification for determining the amount of deterioration

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Aging chamber*.—A pressure chamber consisting of a metal vessel designed to retain an internal atmosphere of oxygen under pressure with provisions for placing rubber specimens within it and subjecting the whole to controlled uniform temperature The size of the chamber is optional but shall be such that specimens may be suspended vertically therein without touching each other or the surface of the chamber Provisions are made for rapid opening and closing of the aging chamber for introduction and removal of the specimens Two suitable designs are shown in figures 7111A and 7111B

3.1.2 A pressure gage attached to the aging chamber for registering the oxygen pressure

3.1.3 A safety valve or rupture diaphragm attached to the aging chamber, set to release at approximately 500 pounds per square inch pressure

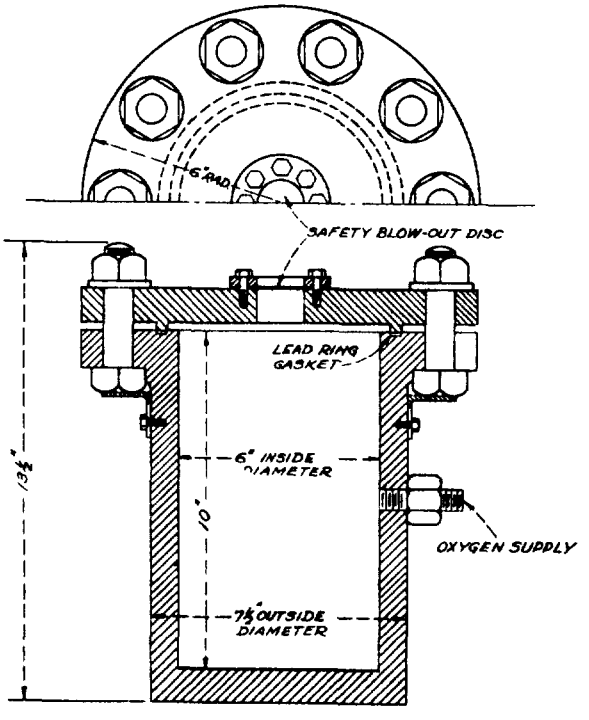

FIGURE 7111A —Pressure aging chamber

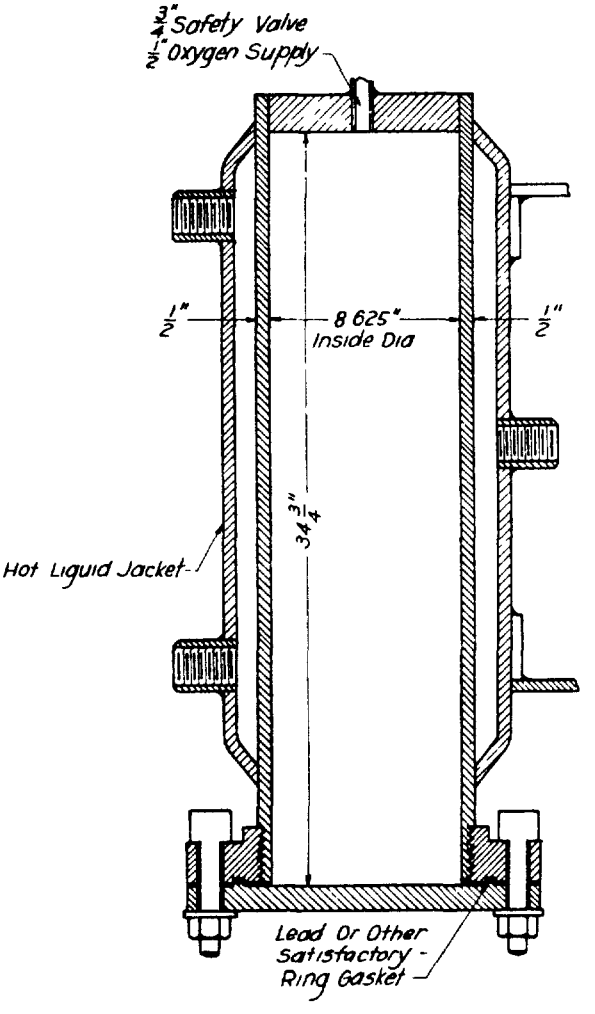

FIGURE 7111B —Pressure aging chamber, hot liquid jacketed

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3.1.4 A source of oxygen under pressure

3.1.5 The source of heat is optional but shall be located outside of the aging chamber

3.1.6 *Heating medium*.—The heating medium is optional but a liquid medium is preferred because of more rapid heat transfer Water, air, or other fluids known to be safe in the presence of oxygen may be used Water has an advantage because of its rapid heat transfer and noncombustible nature Oils or other organic combustible fluids are extremely hazardous in the presence of oxygen and should not be used as heating media for this test When the aging chamber shown in figure 7111A is used, it shall be completely immersed in the heating medium

3.1.6.1 If air is used as the heating medium, the heated air shall be thoroughly circulated around the aging chamber by means of mechanical agitation, baffles shall be used to prevent local overheating, a check of the temperature shall be made in various parts of the heating medium to verify the uniformity of temperature, and the aging chamber shall have a thermometer or thermocouple well, extending into the chamber and filled with water or mercury for measuring the temperature in the aging chamber

3.1.7 Equipment with suitable automatic controls for maintaining the required temperature throughout the heating medium during the aging period

3.1.8 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer or other device for measuring the temperature to within 1 0° C. (1 8° F)

3.1.8.1 The temperature shall be automatically recorded throughout the aging period If a liquid heating medium is used and the aging chamber is completely immersed in the liquid, the temperature within the aging chamber may be taken as that of the heating medium The sensitive element of the temperature measuring device shall be close to the aging chamber but not touching it If the aging chamber is not completely immersed in a liquid heating medium, the sensitive element of the temperature measuring device may be placed in a thermometer well, extending into the aging chamber The thermometer well shall be filled with sufficient liquid medium, such as water or mercury, to cover the element If it is confirmed by actual temperature measurements that the temperature of the oxygen within the aging chamber is the same as that of the heating medium, the temperature may be taken in the heating medium instead of in the thermometer well If air is used as the heating medium, temperature measurements shall be made by means of temperature measuring devices located in various places in the heating medium to verify the uniformity of heating In any case, the recorded temperature shall be verified by checking with a temperature indicating device having its sensitive element directly exposed to the oxygen in the aging chamber.

3.1.9 Equipment for complete and rapid circulation of the heating medium

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be aged at a temperature of 70°±1° C (158°±1.8° F.) and an oxygen pressure of 300±10 pounds per square inch for 94±½ hour

4.2 Unless otherwise specified in the detail specification, tensile strength and elongation, methods 4111 and 4121, respectively, shall be used to determine the amount of deterioration due to aging

4.3 Preparation of specimen.—The specimen shall be prepared as described in method 4111 or other method of test used for determining the amount of deterioration

4.4 Before starting the aging test, the air shall be flushed out of the aging chamber and the apparatus shall be checked to make certain that it does not leak

4.5 The specimen shall be free from mechanical damage When tensile strength or tensile stress is used to determine the amount of deterioration, the cross-sectional area of the specimen shall be determined as described in method 4111 before it is aged After adjusting the aging chamber to the required temperature,

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the specimen shall be suspended vertically in the chamber, the chamber closed immediately, and oxygen pressure applied A volume of at least 10 ml of aging chamber shall be available for each gram of rubber or other oxidizable material The aging period shall start at the time the specimen is placed in the aging chamber and shall continue for the required time at the required temperature and pressure

4.6 At the end of the aging period, the pressure in the aging chamber shall be released at a slow and uniform rate, requiring at least 5 minutes for complete release of the pressure This procedure is necessary to avoid possible formation of porosity in the specimen The specimen shall be removed from the aging chamber immediately after the release of the pressure and set aside to rest for not less than 16 hours and not more than 96 hours at room temperature before determining the required property

4.7 At the end of the rest period, unless otherwise specified in the detail specification, tensile strength and elongation of the aged specimen shall be determined as described in methods 4111 and 4121, respectively If another property for determining the amount of deterioration is specified in the detail specification, the aged specimen shall be tested as described in the specified method The same test shall be conducted on an unaged specimen from the same test unit for the purpose of comparison in determining the amount of deterioration due to aging

4.8 Adequate safety precautions shall be taken when heating oxidizable organic materials in oxygen at an elevated temperature and under pressure since the rate of reaction may become very rapid in some cases and very high pressure may develop, particularly if a large surface area is exposed Oil and grease shall not be permitted to enter the apparatus during the test

5. RESULTS

5.1 Unless otherwise specified in the detail specification, the number of specimens from each test unit exposed in oxygen shall be as required in methods 4111 or 4121, or other method of test specified for determining the amount of deterioration of the material

5.2 Calculation.—The change in tensile strength and elongation, or other specified characteristic of the test unit due to aging shall be calculated as follows

$$\text{Change in characteristic, percent,} = \frac{O-E}{O} \times 100$$

Where

O is the characteristic of the test unit before aging
E is the same characteristic of the test unit after aging

5.3 The change in tensile strength and elongation, or other characteristic of the test unit shall be recorded to the nearest one percent

5.4 The temperature, oxygen pressure, and time of aging and the property used for determining the amount of deterioration shall be recorded

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AIR PRESSURE TEST

1. SCOPE

1.1 This method is intended for use in determining the effect of air under pressure and at an elevated temperature on vulcanized rubber compounds This test is not recommended for rubber compounds that have not been specifically developed to withstand elevated temperatures in service The test is sensitive to small temperature variations

2. SPECIMEN

2.1 The specimen shall be as described in method 4111 or other method of test specified in the detail specification for determining the amount of deterioration

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Aging chamber* —A pressure chamber consisting of a metal vessel designed to retain an internal atmosphere of air under pressure with provisions for placing rubber specimens within it and subjecting the whole to controlled uniform temperature The size of the chamber is optional but shall be such that specimens may be suspended vertically therein without touching each other or the surface of the chamber Provisions are made for rapid opening and closing of the aging chamber for introduction and removal of the specimens Two suitable designs are shown in figures 7111A and 7111B

3.1.2 A pressure gage attached to the chamber for registering the air pressure

3.1.3 A safety valve or rupture diaphragm attached to the aging chamber, set to release at approximately 200 pounds per square inch pressure

3.1.4 A source of compressed air, free from oils, dirt, moisture, etc

3.1.5 The source of heat is optional but if located inside the aging chamber shielding shall be provided so that direct radiation cannot reach the specimen The temperature of the surface of the shield shall be within ±1 0° C. (1 8° F) of the air temperature.

3.1.6 *Heating medium.*—The heating medium is optional Steam, air, or liquid medium is preferred because of more rapid heat transfer Oil or other combustible organic fluids may be hazardous at the elevated temperature required for the test but if their use is necessary they shall have a flash point not lower than 200° C When the aging chamber shown in figure 7111A is used, it shall be completely immersed in the heating medium

3.1.6.1 If air is used as the heating medium, the heated air shall be thoroughly circulated around the aging chamber by means of mechanical agitation, baffles shall be used to prevent local overheating, a check of the temperature shall be made in various parts of the heating medium to verify the uniformity of temperature, and the aging chamber shall have a thermometer or thermocouple well, extending into the chamber and filled with oil or mercury for measuring the temperature in the aging chamber

3.1.7 Equipment with suitable automatic controls for maintaining the required temperature throughout the heating medium during the aging period

3.1.8 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within 1 0° C (1 8° F)

3.1.8.1 The temperature shall be automatically recorded throughout the aging period If a liquid heating medium is used and the aging chamber is completely immersed in the liquid, the temperature within the aging chamber may be taken as that of the heating medium The sensitive element of the temperature measuring device shall be close to the aging chamber but not touching it If the aging chamber is not completely immersed in a liquid heating medium, the sensitive element of the temperature measuring device may be placed in a thermometer well, extending into the aging chamber The thermometer well shall be filled with sufficient liquid medium, such as water or mercury, to cover the element If it is confirmed by actual temperature measurements that the temperature of the air within the aging chamber is the same as that of the heating medium, the temperature may be taken in the

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heating medium instead of in the thermometer well If air is used as the heating medium, temperature measurements shall be made by means of temperature measuring devices placed in various places in the heating medium to verify the uniformity of heating In any case, the recorded temperature shall be verified by checking with a temperature indicating device having its sensitive element directly exposed to the air in the aging chamber

3.1.9 Equipment for complete and rapid circulation of the heating medium

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be aged at a temperature of 127°±1° C (260 6°±1 8° F) and an air pressure of 80±2 pounds per square inch for 20±¼ hour

4.2 Unless otherwise specified in the detail specification, tensile strength and elongation, methods 4111 and 4121, respectively, shall be used to determine the amount of deterioration due to aging

4.3 Preparation of specimen.—The specimen shall be prepared as described in method 4111 or other method of test used for determining the amount of deterioration

4.4 The specimen shall be free from mechanical damage When tensile strength or tensile stress is used to determine the amount of deterioration, the cross-sectional area of the specimen shall be determined as described in method 4111 before it is aged After adjusting the aging chamber to the required temperature, the specimen shall be suspended vertically in the chamber, the chamber closed immediately, and air pressure applied A volume of at least 100 ml of aging chamber shall be available for each gram of rubber or other oxidizable material The aging period shall start at the time the specimen is placed in the aging chamber and shall continue for the required time at the required temperature and pressure

4.5 At the end of the aging period, the pressure in the aging chamber shall be released at a slow uniform rate, requiring at least 5 minutes for complete release of the pressure This procedure is necessary to avoid possible formation of porosity in the specimen The specimen shall be removed from the aging chamber immediately after release of the pressure and set aside to rest for not less than 16 hours and not more than 96 hours at room temperature before determining the required property

4.6 At the end of the rest period, unless otherwise specified in the detail specification, tensile strength and elongation of the aged specimen shall be determined as described in methods 4111 and 4121, respectively If another property for determining the amount of deterioration is specified in the detail specification, the aged specimen shall be tested as described in the specified method The same test shall be conducted on an unaged specimen from the same test unit for the purpose of comparison in determining the amount of deterioration due to aging

4.7 Adequate safety precautions shall be taken when heating oxidizable organic materials in air at an elevated temperature and under pressure since the rate of oxidation may become very rapid in some cases and very high pressures may develop, particularly if a large surface area is exposed Oil and grease shall not be permitted to enter the apparatus during the test

5. RESULTS

5.1 Unless otherwise specified in the detail specification, the number of specimens from each test unit exposed in air shall be as required in methods 4111 or 4121, or other method of test specified for determining the amount of deterioration of the material

5.2 Calculation.—The change in tensile strength and elongation, or other specified characteristic of the test unit due to aging shall be calculated as follows

$$\text{Change in characteristic, percent,} = \frac{O-E}{O} \times 100$$

Where

O is the characteristic of the test unit before aging
E is the same characteristic of the test unit after aging

5.3 The change in tensile strength and elongation or other characteristic of the test unit shall be recorded to the nearest one percent

5.4 The temperature, air pressure, and time of aging and the property used for determining the amount of deterioration shall be recorded

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AIR HEAT TEST, AIR HEATING MEDIUM

1. SCOPE

1.1 This method is intended for use in determining the effect of air at atmospheric pressure and at an elevated temperature on vulcanized rubber compounds

2. SPECIMEN

2.1 The specimen shall be as described in method 4111 or other method of test specified in the detail specification for determining the amount of deterioration

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Aging chamber* —A circulating-air oven having interior dimensions not less than 12 by 12 by 12 inches and not more than 36 by 36 by 48 inches, or equivalent volume The chamber is equipped so that specimens can be suspended vertically therein without touching each other or the surface of the oven

3.1.2 The source of heat is optional but shall be located in the air supply outside of the aging chamber proper

3.1.3 *Heating medium* —The heating medium in the aging chamber shall be air at atmospheric pressure The heated air shall be thoroughly circulated by means of mechanical agitation If a motor-driven fan is used, the air shall not come in contact with the fan motor brush discharge because of ozone formation, baffles shall be used to prevent local overheating, the temperature control shall be located so as to furnish accurate temperature control of the heating medium, the preferred location being adjacent to the sensitive element of the temperature recorder, and checks shall be made by means of maximum reading thermometers or other temperature measuring apparatus located in various places in the heating medium to verify the uniformity of the temperature

3.1.4 Equipment with suitable automatic controls for maintaining the required temperature throughout the heating medium during the aging period

3.1.5 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer or other device for measuring the temperature to within 1 0° C (1 8° F)

3.1.5.1 The temperature shall be automatically recorded throughout the aging period

3.1.6 Equipment for complete and rapid circulation of the air heating medium

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be aged at a temperature of 70°±1° C (158°±1 8° F) for 166±1 hour

4.2 Unless otherwise specified in the detail specification, tensile strength and elongation, methods 4111 and 4121, respectively, shall be used to determine the amount of deterioration due to aging

4.3 Preparation of specimen.—The specimen shall be prepared as described in method 4111 or other method of test used for determining the amount of deterioration

4.4 The specimen shall be free from mechanical damage When tensile strength or tensile stress is used to determine the amount of deterioration, the cross-sectional area of the specimen shall be determined as described in method 4111 before it is aged After adjusting the aging chamber to the required temperature, the specimen shall be suspended vertically in the chamber and the chamber closed immediately A volume of at least 10 ml of aging chamber shall be available for each gram of rubber or other oxidizable material The aging period shall start at the time the specimen is placed in the aging chamber and shall continue for the required time at the required temperature.

4.5 At the end of the aging period, the specimen shall be removed immediately from the aging chamber and set aside to rest for not less than 16 hours and not more than 96 hours at room temperature before determining the required property

4.6 At the end of the rest period, unless otherwise specified in the detail specification,

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tensile strength and elongation of the aged specimen shall be determined as described in methods 4111 and 4121, respectively If another property for determining the deterioration is specified in the detail specification, the aged specimen shall be tested as described in the specified method The same test shall be conducted on an unaged specimen from the same test unit for the purpose of comparison in determining the amount of deterioration due to aging

5. RESULTS

5.1 Unless otherwise specified in the detail specification, the number of specimens from each test unit exposed in air shall be as required in methods 4111 or 4121, or other method of test specified for determining the amount of deterioration of the material

5.2 Calculation.—The change in tensile strength and elongation, or other specified characteristic of the test unit due to aging shall be calculated as follows

$$\text{Change in characteristic, percent,} = \frac{O-E}{O} \times 100$$

Where

O is the characteristic of the test unit before aging
E is the same characteristic of the test unit after aging

5.3 The change in tensile strength and elongation, or other characteristic of the test unit shall be recorded to the nearest one percent

5.4 The temperature and time of aging and the property used for determining the amount of deterioration shall be recorded

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AIR HEAT TEST, LIQUID HEATING MEDIUM

1. SCOPE

1.1 This method is intended for use in determining the effect of air at atmospheric pressure and at an elevated temperature on vulcanized rubber compounds

2. SPECIMEN

2.1 The specimen shall be as described in method 4111 or other method of test specified in the detail specification for determining the amount of deterioration

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Aging chambers* —Circulating-air aging chambers immersed in liquid heating medium so that the space between the surface of the medium and the top of the chambers does not exceed 2 inches The aging chambers are equipped with supports for suspending specimens vertically therein without touching each other or the surface of the vessel Provisions are made for rapid opening and closing of the aging chamber for introduction and removal of the specimen The chambers shall be equipped to permit circulation of air around the specimen during test

3.1.1.1 A suitable aging chamber is shown in figure 7231 It consists of a nickel-plated brass cylinder approximately 4½ inches in diameter and approximately 8⅝ inches in height with a metal cover that rests on a flange inside the cylinder about 2 inches from the top The space above the cover is filled with thermal insulating material Circulation of air through the chamber is by convection by means of tubes ¼ inch in inside diameter, mounted in the metal cover One end of one of the tubes extends downward to within about ½ inch of the bottom of the chamber and the other end extends up through and is flush with the top of the insulating material placed over the cover One end of the other tube is flush with the lower surface of the cover and the other end extends about 6 inches above the top surface of the cover and is insulated with suitable insulating material The cover is equipped with hooks on the under side for attaching specimens

3.1.2 *Heating medium.*—Water is the preferred heating medium If the required aging temperature is such that the use of water is impracticable, other suitable liquids shall be used with precautions to prevent fire hazards and contamination of the aging chamber

3.1.3 A liquid bath and equipment with suitable automatic controls for maintaining the required temperature throughout the heating medium during the aging period

3.1.4 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within ±1 0° C (1 8° F)

3.1.4.1 The temperature of the heating medium shall be automatically recorded throughout the aging period The actual temperature within the aging chamber shall be determined by placing a temperature measuring device in the aging chamber with the sensitive element adjacent to the specimen and the other part of the apparatus extending through the cover of the aging chamber

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be aged at a temperature of 70°±1° C (158°±1 8° F) for 166±1 hour

4.2 Unless otherwise specified in the detail specification, methods 4111 and 4121, respectively, shall be used to determine the amount of deterioration due to aging

4.3 Preparation of specimen.—The specimen shall be prepared as described in method 4111 or other method of test used for determining the amount of deterioration

4.4 The specimen shall be free from mechanical damage When tensile strength or tensile stress is used to determine the amount of deterioration, the cross-sectional area of the specimen shall be determined as described in method 4111 before it is aged After adjusting the aging chamber to the required temperature, the specimen shall be suspended vertically in the chamber and the chamber closed immediately A volume of at least 10 ml of aging

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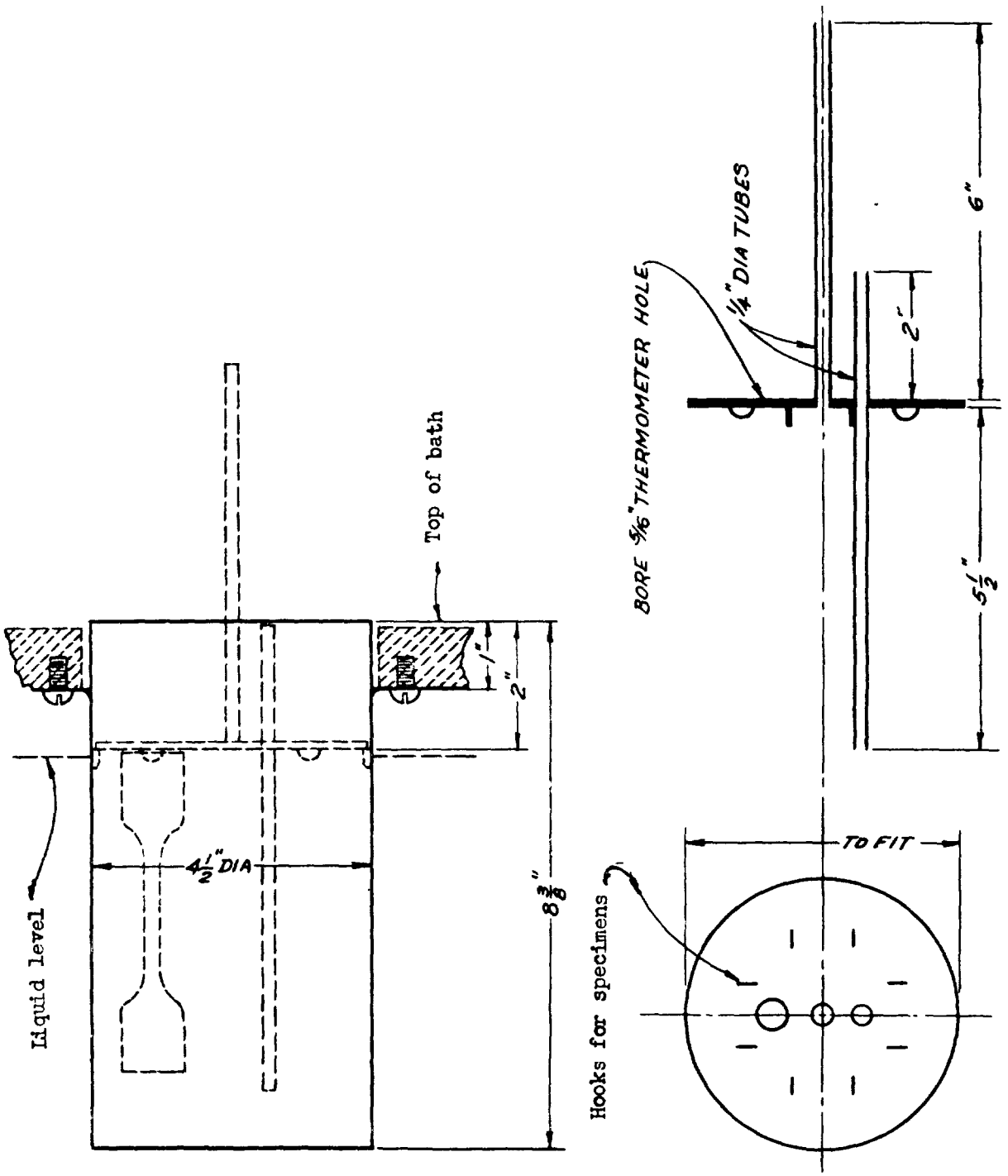

FIGURE 7231.—Aging chamber.

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chamber shall be available for each gram of rubber or other oxidizable material The aging period shall start at the time the specimen is placed in the aging chamber and shall continue for the required time at the required temperature

4.5 At the end of the aging period, the specimen shall be removed immediately from the aging chamber and set aside to rest for not less than 16 hours and not more than 96 hours at room temperature before determining the required property

4.6 At the end of the rest period, unless other specified in the detail specification, tensile strength and elongation of the aged specimen shall be determined as described in methods 4111 and 4121, respectively If another property for determining the amount of deterioration is specified in the detail specification, the aged specimen shall be tested as described in the specified method The same test shall be conducted on an unaged specimen from the same test unit for the purpose of comparison in determining the amount of deterioration due to aging

5. RESULTS

5.1 Unless otherwise specified in the detail specification, the number of specimens from each test unit exposed in air shall be as required in methods 4111 or 4121, or other method of test specified for determining the amount of deterioration of the material

5.2 Calculation —The change in tensile strength and elongation, or other specified characteristic of the test unit due to aging shall be calculated as follows

$$\text{Change in characteristic, percent,} = \frac{O-E}{O} \times 100$$

Where

O is the characteristic of the test unit before aging
E is the same characteristic of the test unit after aging

5.3 The change in tensile strength and elongation, or other characteristic of the test unit shall be recorded to the nearest one percent

5.4 The temperature and time of aging and the property used for determining the amount of deterioration shall be recorded

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TEST-TUBE HEAT-AGING TEST

1. SCOPE

1.1 This method is intended for use in determining the effect of air at atmospheric pressure and at an elevated temperature on vulcanized rubber compounds The temperature and time of aging have not been standardized. See table I, method 7001 Aging periods frequently used are 16, 20, 46, 70, and 166 hours

2. SPECIMEN

2.1 The specimen shall be as described in method 4111 or other method of test specified in the detail specification for determining the amount of deterioration

3. APPARATUS

3.1 The apparatus shall be as follows·

3.1.1 *Aging chamber* —A glass or aluminum test tube approximately 38 mm in diameter and 300 mm in length for holding the specimen The test tube is equipped with a cork stopper fitted with two glass tubes 9 ± 0.6 mm in outside diameter and a wall thickness of approximately one millimeter One of the tubes extends approximately 4½ inches above the top of the stopper and to within about ½ inch of the bottom of the test tube The lower end of the other tube extends approximately 1½ inches below the bottom of the stopper and the upper end approximately 12 inches above the top of the stopper The aging chamber is equipped with supports for suspending specimens vertically as close to the bottom of the tube as possible without touching each other or the surface of the tube

3.1.2 A liquid bath or aluminum block and equipment with suitable automatic controls for maintaining the required temperature throughout the heating medium during the aging period

3.1.2.1 If a liquid heating medium is used, equipment shall be provided for rapid circulation of the liquid medium If an oil medium is used, means shall be provided to remove oil fumes from the oil bath so that the fumes will not enter the inlet tube of the aging chamber

3.1.3 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within $\pm 1.0°$ C ($1.8°$ F)

3.1.3.1 The temperature of the heating medium shall be automatically recorded throughout the aging period The actual temperature within the aging chamber shall be determined by placing a temperature measuring device in the aging chamber with the sensitive element in the location ordinarily occupied by the specimen and the other part of the apparatus extending through a third hole in the cork stopper

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be aged at a temperature of $150° \pm 1°$ C ($300° \pm 1.8°$ F) for $20 \pm ¼$ hour

4.2 Unless otherwise specified in the detail specification, tensile strength and elongation, methods 4111 and 4121, respectively, shall be used to determine the amount of deterioration due to aging

4.3 Preparation of specimen.—The specimen shall be prepared as described in method 4111 or other method of test used for determining the amount of deterioration

4.4 The specimen shall be free from mechanical damage When tensile strength or tensile stress is used to determine the amount of deterioration, the cross-sectional area of the specimen shall be determined as described in method 4111 before it is aged The aging chamber shall be placed in the medium so that the lip of the tube does not extend more than 2 inches above the surface of the medium After adjusting the aging chamber to the required temperature, the specimen shall be suspended vertically in the chamber and the chamber immediately closed The aging period shall start at the time the specimen is placed in the aging chamber and shall continue for the required time at the required temperature

4.5 At the end of the aging period, the specimen shall be removed immediately from the aging chamber and set aside to rest for not less than 16 hours and not more than 96 hours

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at room temperature before determining the required property

4.6 At the end of the rest period, unless otherwise specified in the detail specification, tensile strength and elongation of the aged specimen shall be determined as described in methods 4111 and 4121, respectively If another property for determining the amount of deterioration is specified in the detail specification, the aged specimen shall be tested as described in the specified method. The same test shall be conducted on an unaged specimen from the same test unit for the purpose of comparison in determining the amount of deterioration due to aging.

4.7 The aging chamber and the air circulation tubes shall be carefully cleaned after each test to remove any deposits of volatile material that may have condensed on the sides

5. RESULTS

5.1 Unless otherwise specified in the detail specification, the number of specimens from each test unit exposed to the deteriorating conditions shall be as required in methods 4111 or 4121, or other method of test specified for determining the amount of deterioration of the material

5.2 Calculation.—The change in tensile strength and elongation, or other specified characteristic of the test unit due to aging shall be calculated as follows

$$\text{Change in characteristic, percent,} = \frac{O-E}{O} \times 100$$

Where

O is the characteristic of the test unit before aging
E is the same characteristic of the test unit after aging

5.3 The change in tensile strength and elongation or other characteristic of the test unit shall be recorded to the nearest one percent

5.4 The temperature and time of aging and the property used for determining the amount of deterioration shall be recorded

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RESISTANCE TO LIGHT

1. SCOPE

1.1 This method is intended for use in determining the comparative resistance of soft vulcanized rubber compounds to deterioration when exposed to light having a frequency range approximately that of sunlight but having a greater intensity in the ultraviolet range than sunlight The deterioration resulting from the exposure is determined by observing the nature and degree of cracking and checking and by comparing the tensile strength and elongation of an exposed specimen with that of an unexposed specimen taken from the same test unit The specimen is exposed in the stretched condition The quantity of radiation to which the specimen is exposed is measured by means of the decomposition of uranyl oxalate solution

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 1 by 6 or 2 by 6 inches

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed 0 030 inch, and shall be uniform to within ±0 003 inch Specimens from the same test unit shall be of the same thickness within ±0 01 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Light source* —A light source consisting of a vertical, ventilated, flaming carbon arc designed to accommodate 2 or 3 pairs of carbons, number 22 upper and number 13 lower, the arc to burn between only one pair of carbons at a time The carbons shall be of the cored-type Sunshine carbons of National Carbon Company, or equal, designed to duplicate as closely as possible the spectral distribution of sunlight The arc shall operate on 60 amperes and 50 volts across the arc on alternating current or on 50 amperes and 60 volts across the arc on direct current

3.1.1.1 Unless otherwise specified in the detail specification, the arc shall be enclosed by eight Corex D filters or other enclosure having equivalent absorbing and transmitting properties Each Corex D panel is 1/16 to 3/64 inch in thickness The filters are attached so that they can be removed separately Filters shall be replaced when pronounced discoloration or turbidity develops and in any case after 800 hours of service

3.1.2 *Framework* —A cylindrical rotating framework designed to carry specimen holders in such a way that the surface of the specimen is 18½ ± ½ inch from the center of the arc The framework shall rotate around the arc at a uniform speed of one complete revolution every 2 hours

3.1.3 *Specimen holders* —Specimen holders suitable for mounting specimens vertically while being rotated about the carbon arc to provide uniform distribution of light The holders are designed to accommodate either one strip of rubber 2 inches by 6 inches or 2 strips 1 inch by 6 inches in size and to stretch them to any elongation up to 20 percent A suitable specimen holder is shown in figure 7311A

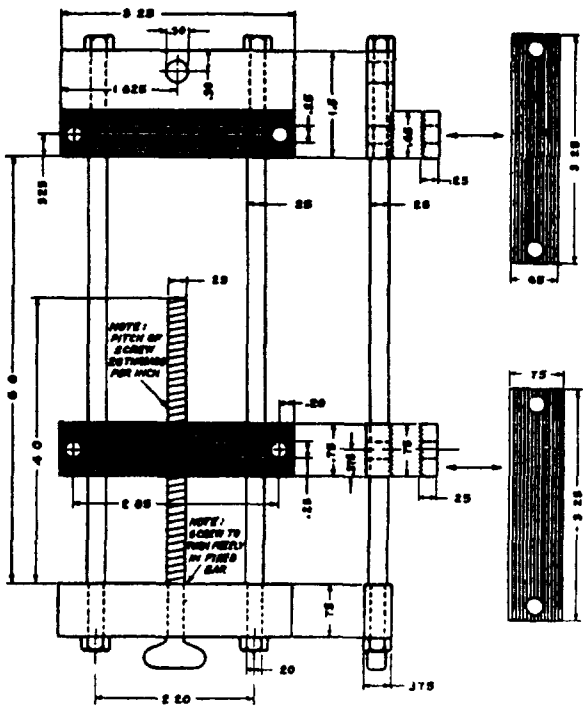

FIGURE 7311A.—Specimen holder for light aging

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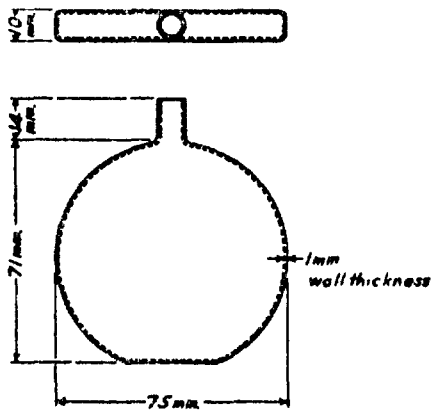

FIGURE 7311B —Transparent fused quartz cell

3.1.4 *Drum* —A cylindrical drum of corrosion-resisting material for enclosing the light source and framework The cylinder is equipped with a protective cover for shielding the operator from radiation from the arc, an overflow for carrying away the water from the spray, and a sliding door to permit access to the specimen

3.1.5 *Spray nozzles* — Fine-spray nozzles mounted inside the cylindrical drum in such a position that each specimen will be exposed to a complete wetting throughout its length one time during each revolution of the framework Each nozzle shall be adjusted so as not to strike the Corex D filters and to deliver between 1 2 and 1 4 gallons of water per hour The pressure of the water entering the spray shall be 25 to 30 pounds per square inch The water shall be maintained at a temperature of 27°±6° C (80 6°±10 8° F) during the test

3.1.6 An exhaust fan for ventilating the arc, with a capacity of 100±20 cubic feet per minute

3.1 7 The temperature of the air in the drum shall be automatically recorded throughout the exposure period The temperature of the air shall be measured to within ±1 0° C (1 8° F) by means of a thermometer with the bulb placed at the position of the specimen The bulb of the thermometer shall be shielded by a

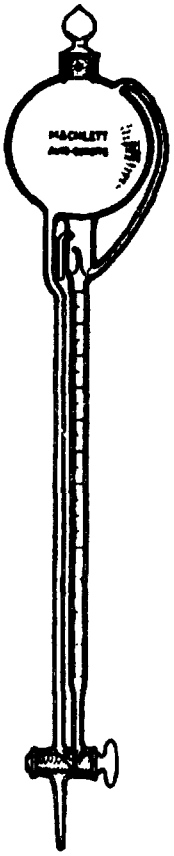

FIGURE 7311C —Dispensing burette

cylinder of bright metal foil 2 inches in diameter and 2 inches in length

3.1.8 A suitable framework, complete with pulleys, cables, and counterweight for raising the lamp for inspection and replacement of the carbons

3.1.9 *Electrical control equipment* —Electrical control equipment for automatically operating and maintaining the current of the arc approximately constant The equipment shall include a constant voltage regulator to maintain the input voltage at 220±1 volt It contains a pushbutton station for starting and stopping the unit, interlocked with the mainline contact, to insure complete electrical isolation of the arc equipment when changing carbons or working on the interior mechanism of the lamp

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3.1.10 *Actinometer cell*—A cell of transparent fused quartz of the design and dimensions shown in figure 7311B, for exposing the actinometer solution to the radiation from the arc

3.1.11 *Cell holder*—A holder for exposing the actinometer cell to the radiation from the arc The holder completely encloses the cell except for the face which is exposed to the light from the arc The exposed cross-sectional area of the cell is measured with an accuracy of at least one percent The inside of the cell holder shall be completely covered with a dull, black enamel paint The holder is designed to take the place of a specimen holder and constructed so that the center of the cell occupies the same relative horizontal position as the center of the rubber specimen

3.1.12 Automatic dispensing burette, figure 7311C

3.1.13 *Radiation actinometer solution*

Substance, reagent grade	Grams per liter	Molar value
Oxalic acid, anhydrous ($H_2C_2O_4$)	4 50	0 050
Uranyl sulfate (UO_2SO_4 $3H_2O$)	4 20	0 0100

3.1.14 *Evaluating solution*

	Grams per liter	Oxidation equivalent
Potassium permanganate ($KMnO_4$)	3 16	0 100 N

4. PROCEDURE

4.1 Preparation, storage and dispensing of solutions.—The solutions described in 3 1 13 and 3 1.14 may be prepared in ordinary diffused light.

4.1.1 *Evaluation solution*—A 0 1 N potassium permanganate solution shall be prepared by dissolving 3.16 grams of reagent grade potassium permanganate in distilled water and diluting the volume to one liter. The solution shall be set aside for one week in a dark place, filtered through a ⅛ inch glass-fiber cloth, and stored in a glass-stoppered bottle protected from the light by a heavy coating of black paint

4.1.2 *Actinometer solution*—The actinometer solution shall be prepared by dissolving 4 50 grams of anhydrous oxalic acid ($H_2C_2O_4$) and 4 20 grams of reagent-grade uranyl sulfate (UO_2SO_4 $3H_2O$) in distilled water and diluting to one liter The solution shall be stored in a glass-stoppered bottle painted with a heavy coat of black paint to protect the solution from light An automatic dispensing burette, shown in figure 7311C, when heavily coated with black paint to exclude light is suitable for storing and dispensing the solutions

4.2 Standardization of solutions.—The evaluating solution, potassium permanganate, shall be standardized against reagent-grade sodium oxalate in the usual manner, with such precautions that its strength will be known within an accuracy of at least ±1 percent The strength of the potassium permanganate shall be expressed in terms of milligrams of anhydrous oxalic acid per milliliter of solution The actinometer solution shall be titrated against the standard potassium permanganate An accurately measured volume of 50 ml of the actinometer solution shall be transferred to a tall-form 200-ml beaker, 20 to 25 ml of distilled water added, and the solution acidified with 5 ml of 1 to 3 sulfuric acid The beaker shall be covered with a porcelain dish and the solution heated to 95° C (203° F) in a lightproof water bath, transferred to an open, glass water bath and maintained at approximately this temperature while resting on a flat, white-glass base The solution shall be clearly seen by light from the glass base illuminated by a "daylight" lamp The hot solution shall be titrated with the standard potassium permanganate evaluating solution from a dispensing burette while stirring constantly until an orange color is obtained which persists for at least 30 seconds. The titration shall be conducted in such a way that the volumes of solution used are precise to within ±0 05 ml

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4.3 Calibration of radiation.—The intensity of radiation of the light shall be measured in terms of the milligrams of oxalic acid decomposed per square decimeter per minute, and the quantity of radiation in any given period of time shall be measured in terms of the milligrams of oxalic acid decomposed per square decimeter An accurately measured volume of the actinometer solution shall be transferred to the quartz cell from a dispensing burette The cell shall be immediately placed in the holder to prevent exposure of the solution to light The cell holder shall be mounted on the cylindrical rotating framework of the apparatus in the same position as that of the specimen holder The solution shall be exposed to the light under normal operating conditions The time of exposure of the actinometer solution shall be sufficient to decompose not less than 10 percent nor more than 30 percent of the oxalic acid in the cell At the end of the exposure period, the actinometer solution shall be immediately transferred from the cell to a 200-ml beaker and titrated with the potassium permanganate evaluating solution as described in 4 2 The quantity of the evaluating solution required for the titration shall be designated as V_1 The same volume of actinometer solution as that used in the light exposure shall be titrated under the same conditions, and the quantity of permanganate solution required for the titration recorded as V_2 The quantity of oxalic acid in milligrams, Q, decomposed by the radiation is given by the equation $Q=a(V_2-V_1)$ where a is the number of milligrams of oxalic acid equivalent to one milliliter of evaluating solution The quantity or dosage of radiation is expressed as a/A and the intensity of radiation as Q/At where A is the exposed area of the cell in square decimeters and t is the time of exposure in minutes

4.4 Unless otherwise specified in the detail specification, the total dosage of radiation shall be 2×10^4 milligrams per square decimeter ±2 percent Approximately 100 hours of exposure should be sufficient for this dosage

4.5 Unless otherwise specified in the detail specification, tensile strength and elongation, methods 4111 and 4121, respectively, and the nature and degree of cracking and checking shall be used to determine the amount of deterioration due to exposure to light

4.6 Unless otherwise specified in the detail specification, the specimen, 4 8 1, mounted in the holder, shall be stretched to an elongation of 10 percent within 3 hours after it has been buffed The specimen shall be held in the stretched position at a temperature of 38°±2° C (100°±3 6° F) for 16 to 24 hours to aid in blooming of waxes

4.7 Unless otherwise specified in the detail specification, 4 of the 8 filters shall be removed, leaving every alternate filter in place in the apparatus during the light exposure

4.8 Preparation of specimen.

4.8.1 *Specimen for light exposure, 4 9 and 4 10* —If the material is too thick or has a fabric backing or an uneven surface that may interfere with the light exposure or physical test for determining the amount of deterioration, the material shall be buffed as described in method 1111 If it is not necessary to buff the material for the above reasons, it shall be lightly buffed on both principal surfaces as described in method 1111 to remove any surface coating The specimen shall be buffed before exposing to light

4.8.2 *Specimen for physical test, 4 11* —The specimen shall be prepared from the exposed specimen, 4 10, as described in method 4111 or other method of test used for determining the amount of deterioration of the material, except that the exposed material shall not be buffed

4.9 The specimen shall be free from mechanical damage Within one hour after the preconditioning period, the stretched specimen in the holder, 4 6, shall be placed in the rotating framework in such a position as to receive full radiation from the arc It shall be exposed to the light until the total exposure is equivalent to that required to decompose the specified quantity of oxalic acid, 4 4 The light shall be calibrated as described in 4 3 to make certain that it is of the proper intensity A measurement of intensity of radiation should be made near the start and near the end of each

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exposure and at such intervals during the exposure period as is necessary to insure that the light is operating at the proper intensity The temperature of the air in the vicinity of the specimen shall be maintained at 45°±5° C (113°±9° F) during the exposure period by controlling the temperature of the room and the ventilation of the space surrounding the specimen The filters or other enclosure shall be cleaned at least once every 24 hours during the period of exposure

4.10 Examination.—At the end of the exposure period, the holder with the exposed specimen shall be immediately removed and the specimen examined immediately by means of a binocular microscope for cracking and checking The specimen shall be removed from the holder and set aside to rest for not less than 16 hours and not more than 96 hours at room temperature before other tests are made

4.11 Tests.—At the end of the rest period, unless otherwise specified in the detail specification, 4 5, tensile strength and elongation of the exposed specimen shall be determined as described in methods 4111 and 4121, respectively If another property for determining the amount of deterioration is specified in the detail specification, the exposed specimen shall be tested as described in the specified method The same test shall be conducted on unexposed material from the same test unit for the purpose of comparison in determining the amount of deterioration due to exposure to light

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5. RESULTS

5.1 Unless otherwise specified in the detail specification, the number of specimens from each test unit exposed to light shall be just sufficient to furnish the number of specimens required in methods 4111 and 4121, or other method of test specified for determining the amount of deterioration of the material.

5.2 Calculation.—The change in tensile strength and elongation, or other specified characteristic of the test unit due to light exposure shall be calculated as follows

$$\text{Change in characteristic, percent,} = \frac{O-E}{O} \times 100$$

Where

O is the characteristic of the test unit before light exposure

E is the same characteristic of the test unit after light exposure

5.3 The change in tensile strength elongation, or other characteristic of the test unit shall be recorded to the nearest one percent

5.4 The number of specimens exposed to light from each test unit and the number from each test unit that showed cracking or checking shall be recorded

5.5 The dosage, percent elongation of the specimen during exposure, the number of filters used, and the characteristic used for determining the amount of deterioration shall be recorded

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STERILIZATION, STEAM

1. SCOPE

1.1 This method is intended for use in determining the effect of steam sterilization on vulcanized rubber compounds such as those used in surgeons' gloves and other hospital equipment.

2. SPECIMEN

2.1 The specimen shall be as described in method 4111 or other method of test specified in the detail specification for determining the amount of deterioration

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A sterilizing chamber of the single shell or jacketed type designed so that the air can be quickly replaced with steam

3.1.2 A source of steam under pressure

3.1.3 Equipment with suitable controls for maintaining the required temperature throughout the sterilization chamber during the sterilization period

3.1.4 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, of other device, fitted into the sterilization chamber, for measuring the temperature of the steam to within ±1 0° C (1 8° F)

3.1.5 Soft brush

3.1.6 Cotton gauze or cheesecloth

3.1.7 Soap

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be exposed to steam at a temperature of 121°±2° C (249 8°±3 6° F) for 20±½ minute

4.2 Unless otherwise specified in the detail specification, tensile strength and elongation, methods 4111 and 4121, respectively, shall be used to determine the amount of deterioration due to sterilization

4.3 Preparation of specimen.—The specimen shall be prepared as described in method 4111 or other method of test used for determining the amount of deterioration

4.4 The specimen shall be free from mechanical damage When tensile strength or tensile stress is used to determine the amount of deterioration, the cross-sectional area of the specimen shall be determined as described in method 4111 before it is exposed to steam The specimen shall be wrapped in 4 or 5 layers of cotton gauze or cheesecloth and then suspended or placed on a perforated shelf in the sterilizing chamber Steam shall be admitted to the chamber at a rate that will raise the temperature to that required for the test, 4 1, within 5 minutes At this temperature, the steam in the chamber shall not contain more than 1 volume of air per 1000 volumes of steam The sterilization period shall start at the time the sterilizing chamber reaches the required temperature and shall continue for the required time at the required temperature, 4 1

4.5 At the end of the sterilization period, the steam shall be blown off and the specimen removed from the sterilizing chamber within 3 minutes If any wax or bloom is present on the specimen after sterilization, the specimen shall be scrubbed with soap, water, and a soft brush to remove it The specimen shall then be dried in air out of direct light

4.6 Unless otherwise specified in the detail specification, the operations in 4 4 and 4 5 shall be repeated until a total of 5 sterilizations has been completed The specimen shall be allowed to rest 20 minutes between sterilizations, from the time it is removed from the steam, before repeating the sterilization operation At the end of the fifth sterilization, the steam shall be blown off and the specimen treated as described in 4 5

4.7 After removal from the steam, 4 6, the specimen shall be set aside, out of direct light, to rest for not less than 16 hours and not more than 96 hours at room temperature before determining the required property

4.8 At the end of the rest period, unless otherwise specified in the detail specification, tensile strength and elongation of the exposed specimen shall be determined as described in methods 4111 and 4121, respectively If another property for determining the amount of deterioration is specified in the detail specifi-

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cation, the exposed specimen shall be tested as described in the specified method The same test shall be conducted on an unexposed specimen from the same test unit for the purpose of comparison in determining the amount of deterioration due to sterilization

5. RESULTS

5.1 Unless otherwise specified in the detail specification, the number of specimens from each test unit exposed to steam shall be as required in methods 4111 or 4121, or other method of test specified for determining the amount of deterioration of the material

5.2 Calculation.—The change in tensile strength and elongation, or other specified characteristic of the test unit due to steam sterilization shall be calculated as follows

$$\text{Change in characteristic, percent,} = \frac{O-E}{O} \times 100$$

Where

O is the characteristic of the test unit before exposure
E is the same characteristic of the test unit after exposure

5.3 The change in tensile strength and elongation, or other characteristic of the test unit shall be recorded to the nearest one percent

5.4 The temperature, time of exposure, and number of sterilizations and the property used for determining the amount of deterioration shall be recorded

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RESISTANCE TO STEAM, DIGESTION METHOD

1. SCOPE

1.1 This method is intended for use in determining the effect of steam on vulcanized rubber compounds The test is more commonly applied to packings and gaskets containing rubber, and the procedure described is primarily for this application

2. SPECIMEN

2.1 **Packing.**—The specimen shall consist of a portion of the test unit at least 6 by 6 inches and the thickness of the material as received

2.2 **Gasket.**—The specimen shall consist of a portion of the test unit approximately 6 inches in length and the full width and thickness of the gasket

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Autoclave* —An autoclave equipped with a steam inlet with a regulating valve and condensate outlet through a steam trap The apparatus is equipped with a petcock air vent for escape of trapped air at the beginning of the test, a pressure gage, and a perforated metal or wire frame for supporting the specimen 6 inches above the condensate outlet

3.1.2 Iron plates of proper size, and bolts

3.1.3 A source of steam under pressure

3.1.4 Equipment with suitable controls for maintaining the required steam pressure throughout the autoclave during the exposure period

3.1.5 A safety valve or rupture diaphragm attached to the autoclave, set to release at approximately 400 pounds per square inch steam pressure

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be subjected to a steam pressure of 150 ± 5 pounds per square inch for $4 \pm \frac{1}{4}$ hour

4.2 Unless otherwise specified in the detail specification, the specimen shall be subjected to a pressure of 200 ± 5 pounds per square inch when bolted between the iron plates

4.3 Unless otherwise specified in the detail specification, visual examination or tensile strength and elongation, methods 4111 and 4121, respectively, shall be used to determine the deterioration due to exposure

4.4 The specimen shall be free from mechanical damage The specimen shall be placed on the supporting frame in the autoclave and steam admitted until the required pressure is obtained The specimen shall remain in the steam for the required time under the required pressure, 4 1 Several specimens may be tested at one time by supporting them on the frame between thin sheets of iron

4.4.1 When testing sheet packing the specimen shall be bolted between iron plates of the proper size and shape before subjecting it to steam

4.5 At the end of the exposure period, the steam pressure shall be released, and the specimen immediately removed from the autoclave and examined as described in 4 6 or 4 7, as specified in the detail specification

4.6 If visual examination of the specimen is specified in the detail specification, the specimen shall be examined for cracking, stiffness, tackiness, disintegration, or any other damage immediately after it has been removed from the autoclave in 4 5

4.7 If other properties are specified in the detail specification, the exposed material shall be set aside, out of direct light, to rest for not less than 16 hours and not more than 96 hours at room temperature before determining the required property

4.7.1 The specimen for test shall be prepared from the exposed material 4 5, as described in method 4111 or other method used for determining the deterioration of the material

4.7.2 At the end of the rest period, unless otherwise specified in the detail specification, 4 3, tensile strength and elongation of the exposed material shall be determined as described in methods 4111 and 4121, respectively If another property for determining deterioration is specified in the detail specification, the specimen shall be tested as described in the specified method The same test shall be conducted

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on an unexposed specimen from the same test unit for the purpose of comparison in determining the amount of deterioration due to exposure in steam

5. RESULTS

5.1 Tests.

5.1.1 Unless otherwise specified in the detail specification, the number of specimens from each test unit exposed to steam shall be just sufficient to furnish the number of specimens required in methods 4111 or 4121, or other method of test specified for determining the amount of deterioration of the material

5.1.2 *Calculation.*—The change in tensile strength, elongation, or other specified characteristic of the test unit due to exposure shall be calculated as follows

$$\text{Change in characteristic, percent} = \frac{O - E}{O} \times 100$$

Where

O is the characteristic of the test unit before exposure

E is the same characteristic of the test unit after exposure

5.1.3 The change in characteristic of the test unit shall be recorded to the nearest one percent

5.2 Visual examination

5.2.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be exposed to steam

5.2.2 The number of specimens tested from each test unit and the number from each test unit that showed cracking, stiffness, tackiness, disintegration, or other damage shall be recorded

5.3 The steam pressure, the time of exposure, and the property used for determining the deterioration shall be recorded

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

RESISTANCE TO STEAM, RACK METHOD

1. SCOPE

1.1 This method is intended for use in determining the effect of steam on hose that has been designed for use with steam Changes in tensile strength, elongation, and friction are commonly used for evaluating the effect of steam

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit of full circumference not less than 16 inches nor more than 24 inches in length The specimen shall be cut to the exact length to fit into the apparatus in 3 1

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An apparatus consisting of two fixed horizontal steam headers with connections for attaching specimens The headers are spaced so that the specimen will just fit between the connections without distortion One header is equipped with a pressure regulating valve and a calibrated pressure gage which will measure the steam pressure within ±2 pounds per square inch The other header is connected to a steam trap Each opening in each header is equipped with a shutoff valve Dry saturated steam at the required pressure shall be supplied through the header equipped with the regulator valve

3.1.2 A source of steam under pressure

3.1.3 Equipment with suitable controls for maintaining the required steam pressure throughout the apparatus during the exposure period

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be subjected to a steam pressure of 72±2 pounds per square inch for a total of 14±¼ hour

4.2 Unless otherwise specified in the detail specification, tensile strength and elongation, methods 4111 and 4121, respectively, shall be used for determining the amount of deterioration of the tube and cover due to exposure to steam

4.3 Friction method 8011 shall be used for determining the change in adhesion between plies of fabric, between cover and fabric, and between tube and fabric due to exposure to steam

4.4 The specimen shall be free from mechanical damage The specimen shall be mounted in the apparatus and steam admitted into the hose until the required internal pressure is obtained The specimen shall be subjected to steam under the required pressure, 4 1, for a period of 7 hours At the end of the exposure period, the steam shall be released and the specimen allowed to stand for 17 hours at room temperature

4.5 The exposure of the specimen to steam as described in 4 4 shall be repeated At the end of the second 7-hour period of steam exposure, the steam shall be released and the specimen removed from the apparatus and set aside to rest for not less than 16 hours and not more than 96 hours at room temperature before determining the required property

4.6 Tube and cover.

4.6.1 Specimens shall be prepared from the tube and cover of the exposed specimen of hose, 4 5, as described in method 4111 or other method of test used for determining the amount of deterioration of the tube and cover

4.6.2 At the end of the rest period, 4 5, unless otherwise specified in the detail specification, 4 2, tensile strength and elongation of the tube and cover of the exposed specimen of hose shall be determined as described in methods 4111 and 4121, respectively If another property is specified in the detail specification for determining the amount of deterioration, the specimen from the exposed hose shall be tested as described in the specified method The same test shall be conducted on unexposed material from the same test unit for the purpose of comparison in determining the amount of deterioration of the tube and cover due to exposure to steam

4.7 Friction between plies.

4.7.1 Specimens for determining friction between plies of fabric, between cover and fabric, and between tube and fabric shall be

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prepared from the exposed specimen of hose, 4 5, as described in method 8011

4.7.2 At the end of the rest period, 4 5, friction between plies, between cover and fabric, and between tube and fabric of the exposed specimen shall be determined as described in method 8011 The same test shall be conducted on an unexposed specimen from the same test unit for the purpose of comparison in determining the amount of change in adhesion due to exposure to steam

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be exposed to steam

5.2 Calculation.—The change in tensile strength and elongation, and friction, or other specified characteristic of the test unit due to exposure to steam shall be calculated as follows

$$\text{Change in characteristic, percent,} = \frac{O-E}{O} \times 100$$

Where

O is the characteristic of the test unit before exposure
E is the same characteristic of the test unit afte exposure

5.3 The change in tensile strength and elongation, and friction, or other characteristic of the test unit shall be recorded to the nearest one percent

5.4 The steam pressure, time of exposure, and the property used for determining the amount of deterioration shall be recorded

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FRICTION, GENERAL

1. SCOPE

1.1 The term "friction" is used to indicate the adhesion between plies of fabric bonded with rubber or the adhesion between rubber and any other material to which it is bonded An adhesion test consists of applying a measured force sufficient to strip a layer of moderate thickness from a specimen at a measured rate, with separation taking place at the adhered surfaces

1.2 Two general methods are described The first employs a tension testing machine and is known as the "machine" method The second involves the application of force by means of weights and is known as the "dead-weight" method

1.3 These methods are applicable only when the adhered surfaces are approximately plane or uniformly circular, as in belting, hose, tire carcasses, or rubber-covered sheet metal If the adhered surfaces contain sharp bends or angles or other large irregularities that cannot be avoided in the specimen, the above methods may not be applicable

1.4 In the dead-weight method, no stripping occurs unless sufficient weight is applied Small increments in weight do not progressively cause proportional increases in rate of separation With increasing weight, the rate increases slowly at first and finally very rapidly

1.5 In the machine method, the rate of separation of the grips of the testing machine is twice the rate of stripping when a strip specimen is tested When a ring specimen is tested, the rate of stripping is the same as the rate of separation of the grips of the testing machine

1.6 The true adhesion value is not determined in case the specimen tears repeatedly The test, however, indicates that the strength of adhesion exceeds the strength of the material and that the adhesion value is not less than the value obtained

1.7 Friction values may differ between different plies of the same product of plied construction and at different points along the same ply The adhesion value of one ply to another is taken as the average result of the test over a sufficient distance of separation to indicate the adhesion value

1.8 In case of dispute between the two methods for measuring friction, the machine method shall be used If equipment is available, the use of the machine method should be specified in the detail specification

1.9 The numerical value of adhesion may be expressed as (1) average tension in pounds required to cause separation at a given rate (1 inch per minute) under stated conditions or (2) average rate of separation in inches per minute caused by a specified tension exerted upon the parts under the stated conditions

2. SPECIMENS

2.1 Two types of specimens, namely, strip and ring, are employed in conducting adhesion tests

2.2 Curved specimens for test in strip form shall have curvature only in the length dimension

3. RESULTS

3.1 Machine method.—In the machine method, the rate of separation is fixed and the adhesion shall be expressed in terms of measured tension, that is, the force required to cause separation

3.2 Dead-weight method.—In the dead-weight method, the average rate of separation produced by a specified load shall be expressed in inches per minute

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FRICTION, MACHINE METHOD

1. SCOPE

1.1 This method is intended for use in determining the friction between rubber and other materials by the use of a tension testing machine It is particularly applicable for determining the friction between plies of fabric bonded with rubber and between rubber and fabrics

2. SPECIMEN

2.1 **Strip.**—The specimen shall consist of a portion of the test unit 1 00 inch in width and uniform within ±0 01 inch, method 2121 If the test unit does not permit a specimen one inch in width, such as square packing or small hose, a narrower specimen may be used

2.1.1 The length shall be such as to permit separation for a sufficient distance to indicate the adhesion value A distance of separation of at least 4 inches is recommended.

2.2 **Ring.**—The specimen shall consist of a closed ring 1 00 inch in width and uniform within ±0 01 inch, method 2121.

2.3 Unless otherwise specified in the detail specification, the specimen may vary in thickness according to the construction of the material undergoing test, but the portion from which a ply or layer is being separated shall not be of less thickness than such a ply or layer and in no case shall the thickness of the latter exceed ¼ inch

2.4 In the case of a strip specimen, when friction between rubber layer and fabric is required, the specimen shall consist of the rubber and 1 ply of fabric When friction between fabric parts is required, the specimen shall consist of 2 plies of fabric. If articles are constructed of lightweight fabric where 1 ply is not strong enough for test, 4 plies shall be taken and 2 plies stripped from 2 plies

3. APPARATUS

3.1 The apparatus shall be as follows·

3.1.1 *Testing machine* —A testing machine which meets the following requirements

3.1.1.1 Power driven

3.1.1.2 The applied tension is indicated to within ±1 percent by an automatic recorder

3.1.1.3 The machine records automatically the value of friction at all points of the specimen, thus producing a chart showing the distance separated on one axis and the applied tension on the other axis.

3.1.1.4 The rate of travel of the power-actuated grip is uniform at all times and shall be 2±⅛ inch per minute for strip specimens and 1±1/16 inch per minute for ring specimens

3.1.1.5 The machine when used for a given specimen is of such capacity that the maximum force required during the test shall not be greater than 85 percent nor less than 15 percent of the rated capacity

3.1.1.6 The machine is operated without any device for maintaining maximum load indication In pendulum-type machines, the weight lever swings as a free pendulum without engagement of pawls

3.1.1.7 *Grips*

3.1.1.7.1 *Strip specimen* —The grips are of the type that will clamp firmly and prevent slipping of the specimen during test

3.1.1.7.2 *Ring specimen* —One grip of the machine shall be as described in 3.1 1 7 1 The grip attached to the recording head of the machine is replaced with a mandrel, mounted in roller or ball bearings, having an outside diameter approximately the same as the inside diameter of the ring specimen undergoing test The mandrel is mounted in such a way that its axis of rotation is in the plane of the ply being separated from the ring and that the applied force is normal to the tangent of the ring circumference at the line of separation

3.1.2 A die or sharp tool for cutting the specimen

3.1.3 A close-fitting, slightly tapered mandrel attached to a lathe for use in cutting ring specimens.

4. PROCEDURE

4.1 **Preparation of specimen.**

4.1.1 Unless otherwise specified in the detail specification, a strip specimen shall be used for testing belting, dredging sleeves, sheet packing,

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and hose less than ½ inch or more than 4 inches in inside diameter

4.1.2 Unless otherwise specified in the detail specification, a ring specimen shall be used for testing hose ½ inch to 4 inches in inside diameter, inclusive

4.1.3 Unless otherwise specified in the detail specification, specimens from belting narrower than 12 inches and hose smaller than ½ inch in inside diameter shall be cut longitudinally, specimens from belting wider than 12 inches and hose more than ½ inch in inside diameter shall be cut transversely For hose over 4 inches in inside diameter, the rings shall be cut through at the point where the outside ply of fabric ends and opened up to form strip specimens Specimens from sheet packing may be cut in any direction In hose constructed with wire-reinforced members, it may be impracticable to prepare a specimen for adhesion tests except for adhesion of the cover

4.1.4 If the material is too thick for test, 2 3, it shall be buffed or sliced as described in method 1111

4.1.5 The specimen shall be cut with a die or sharp tool so as to leave smooth edges If practicable, the specimen shall be cut from the test unit in such a manner that the adhered parts will have the same uniform width of 1 00±0 01 inch

4.1.6 When necessary or when specified in the detail specification, only the layer which is to be separated shall be cut to the 1 00±0 01 inch width In such cases, the portions of the layer remaining outside of the one-inch test width shall be removed from the other layers to avoid edge constriction during separation

4.1.7 A ring specimen is conveniently prepared by mounting the material over a smooth, close-fitting, slightly tapered, wooden mandrel The mandrel shall be rotated in the lathe, and a section or ring 1 00 inch in width and uniform to within ±0 01 inch cut with the point of a wet tool which is forced gradually through the layers of material The pointed tool shall be kept sharp and should be wet before each cutting

4.1.8 *Cutting of specimen* —To facilitate cutting, the edge of the die or tool may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0 5 percent sodium chromate or with silicone mold release emulsion before each specimen is cut

4.2 Determination of friction.

4.2.1 *Strip specimen* —The specimen shall be free from mechanical damage The layer to be tested shall be separated by hand at one end of the specimen a sufficient distance to permit clamping the layer and other part in the grips of the machine The separated end of the specimen with all separate parts except the one under test shall be securely clamped in the grip attached to the recording head of the machine The layer to be stripped from the specimen shall be clamped in the power-actuated grip of the machine The specimen shall be adjusted symmetrically and without twisting in the testing machine so that the tension will be distributed uniformly over the width of the specimen during the test Provision shall be made to maintain the strip during the test approximately in the plane of the clamp The autographic mechanism and chart shall be adjusted to zero and the machine started The power-actuated grip shall travel at a uniform speed of 2 ± ⅛ inch per minute The separating layer shall be stripped from the specimen at approximately an angle of 180° and the separation shall continue for a sufficient distance to indicate the adhesion value If practicable, the separation shall be continued for a distance of at least 4 inches The separation of the grips shall be autographically recorded in inches on one axis of the chart and the force in pounds on the other axis

4.2.1.1 The average force required to separate the layer from the other part shall be determined by drawing a line on the chart parallel to the length of separation axis and as near as practicable midway between the maximum and minimum values for force The line shall be drawn only over the portion of the chart corresponding to actual separation of the parts being tested The average friction of the specimen shall be read from the chart and the value recorded to the nearest ½ pound per inch of width

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4.2.2 *Ring specimen*.—The specimen shall be free from mechanical damage. The layer of the specimen to be tested for adhesion shall be separated by hand for a sufficient distance to permit clamping it in the power-actuated grip of the machine. The specimen shall be placed snugly on the mandrel, the mandrel attached to the recording head of the machine, and the separated layer clamped in the power-actuated grip. The specimen shall be adjusted symmetrically and without twisting in the testing machine so that the tension will be distributed uniformly over the width of the specimen during the test. The autographic mechanism and chart shall be adjusted to zero and the machine started. The power-actuated grip shall travel at a uniform speed of $1 \pm \frac{1}{16}$ inch per minute. The separating layer shall be stripped from the specimen approximately at an angle of 90° to the tangent of the specimen surface and the separation continued for a sufficient distance to indicate the adhesion value. If practicable, the separation shall be continued for a distance of at least 4 inches. During the test, the mandrel shall rotate freely so as to maintain the line of separation at approximately the same position at all times. The separation of the grip and mandrel shall be autographically recorded in inches on one axis of the chart, and the force required recorded in pounds on the other axis. The average force in pounds required to separate the parts of the specimen shall be determined as described in 4.2.1.1.

4.3 Tearing.—If during the test one of the layers begins to tear instead of separating from the other layer of the specimen, the material being torn shall be cut with a knife up to the surface of contact between the two layers and the test started again. If one of the layers repeatedly tears instead of separating from the other layer, the average force at which tear occurs shall be recorded as the friction of the specimen.

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each layer of each test unit for which friction requirements are given in the detail specification shall be tested.

5.2 When one specimen is tested from each layer of the test unit for which friction requirements are specified in the detail specification, the friction of that layer of the test unit shall be the value obtained from the specimen tested. When more than one specimen is tested from each layer of the test unit for which friction requirements are specified in the detail specification, the friction of that layer of the test unit shall be the average of the values obtained from the specimens tested.

5.3 The friction of each layer tested from each test unit shall be recorded separately to the nearest $\frac{1}{2}$ pound per inch of width.

5.4 When tearing repeatedly occurs, 4.3, the number of specimens tested from each layer of each test unit and the number from each layer of each test unit that tear shall be recorded.

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FRICTION, DEAD-WEIGHT METHOD

1. SCOPE

1.1 This method is intended for use in determining the friction between rubber and other materials by the use of dead weights This method is not recommended if it is practicable to use the machine method, 8011

2. SPECIMEN

2.1 The specimen shall be as described in method 8011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Supporting frame*—A supporting frame designed with a grip for holding one end of a strip specimen so that the specimen is suspended vertically and so that the grip may be detached and replaced with a freely rotating mandrel for holding a ring specimen The frame is of sufficient height to permit weights to be suspended from the specimen by means of clamps and to hang freely during the test

3.1.2 *Mandrel*—A mandrel, mounted in roller or ball bearings, having an outside diameter approximately the same as the inside diameter of the ring specimen undergoing test

3.1.3 *Grips*—Two grips for holding the specimen that will clamp firmly and prevent the specimen from slipping One of the grips is equipped for attaching the weight holder

3.1.4 A set of weights and weight holder

3.1.5 A stopwatch or other device that will indicate the time in seconds

3.1.6 A scale graduated to ⅛ inch or finer

3.1.7 A die or sharp tool for cutting the specimen

3.1.8 A closely fitting, slightly tapered mandrel attached to a lathe for cutting ring specimens

4. PROCEDURE

4.1 The weight applied to the specimen shall be as specified in the detail specification

4.2 Preparation of specimen.—The specimen shall be prepared as described in method 8011

4.3 Determination of friction.

4.3.1 *Strip specimen*—The specimen shall be free from mechanical damage The layers to be tested shall be separated by hand at one end of the specimen a sufficient distance to permit clamping the layer and other part in the grips The point at which the separation ends shall be marked on the separated end of the specimen The separated end of the specimen with all separate parts except the one under test shall be securely clamped in the grip attached to the supporting frame and the specimen suspended vertically The layer to be stripped from the specimen shall be clamped in the grip equipped for attaching the weight holder The specimen shall be adjusted symmetrically and without twisting, so that the tension will be distributed uniformly over the width of the specimen during the test The specified weights shall be added to the weight holder gently, without dropping, in such a manner that the layer will be stripped from the specimen at an angle of approximately 180° The weight of the holder and grip shall be included in the total weight, 4 1, required to strip the part from the specimen The time required for the separation of the layer of which the adhesion is to be determined shall be read from the stopwatch or other timing device and the value recorded to the nearest second At the end of the separation, the part attached to the supporting frame shall be removed, the distance between the marked point and the end of the separation measured by means of the scale, and the value recorded to the nearest ⅛ inch as the distance of separation

4.3.2 *Ring specimen*—The specimen shall be free from mechanical damage The layer of the specimen to be tested for adhesion shall be separated from the specimen by hand a sufficient distance to permit clamping it in the grip equipped for attaching the weight holder The point at which the separation ends shall be marked on the part separated The specimen shall be placed snugly on the mandrel attached to the supporting frame so that it will rotate freely The layer to be stripped from the speci-

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men shall be clamped in the grip equipped for attaching the weight holder The specimen shall be adjusted symmetrically and without twisting so that the tension will be distributed uniformly over the width of the specimen during test The specified weight shall be added to the weight holder gently, without dropping, in such a manner that the separating layer will be stripped from the specimen at an angle of approximately 90° The weight of the holder and grip shall be included in the total force, 4 1, required to strip the layer from the specimen The time required for the separation of the layer of which the adhesion is to be determined shall be read from the stopwatch or other timing device and the value recorded to the nearest second At the end of the separation, the layer attached to the weight holder shall be removed, the distance between the marked point and the end of the separation measured by means of the scale, and the value recorded to the nearest ⅛ inch as the distance of separation

4.4 Tearing.—If tearing occurs, the procedure shall be as described in 4 3 of method 8011

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each layer of each test unit for which friction requirements are given in the detail specification shall be tested

5.2 When one specimen is tested from each layer of the test unit for which friction requirements are specified in the detail specification, the friction of that layer of the test unit shall be the value obtained from the specimen tested When more than one specimen is tested from each layer of the test unit for which friction requirements are specified in the detail specification, the friction of that layer of the test unit shall be the average of the values obtained from the specimens tested

5.3 The friction of each layer tested from each test unit shall be recorded separately to the nearest ½ inch of separation per minute

5.4 When tearing repeatedly occurs, 4 4, the number of specimens tested from each layer of each test unit and the number from each layer of each test unit that tear shall be recorded

5.5 The weight applied to the specimen shall be recorded

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ADHESION, RUBBER TO METAL

1. SCOPE

1.1 This method is intended for use in determining the adhesion of rubber to metal by use of a tension testing machine It is applicable to such materials as rubber coatings, rubber-to-metal cements, etc The method measures the force required to separate a rubber strip from metal at a specified speed with the separating force being applied at an angle of 90°

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 6 inches in length and 1⅝ inches in width

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 The testing machine described in method 8011, except that the grips shall be as described in 3 1 1.1

3.1.1.1 *Grips*—A base, *1*, is attached to the loading grip, *2*, of the testing machine as shown in figure 8031 The uprights, *3*, of the base are drilled to form bearings for a square-threaded power shaft, *4* The shaft threads into a rider, *5*, and is keyed to two collars, *6*, and a crank, *7*. Manual rotation of the crank imparts horizontal motion to the rider The rider slides along two guide rods, *8*, which prevent it from rotating An adapter plate, *9*, with two clamps, *10*, secured by means of four studs is bolted to the rider The clamps form a T-slot into which the metal base of the specimen, *11*, is placed A grip, *13*, for holding one end of the rubber strip, *12*, of the specimen is suspended from the load-measuring head of the testing machine by means of a chain, *14*

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3.1.2 A sharp tool for cutting the specimen

4. PROCEDURE

4.1 The specimen shall be free from mechanical damage The rubber coating of the specimen shall be cut through to the metal along 2 parallel lines, leaving a rubber strip 1 00 inch in width and uniform within ±0 01 inch, method 2121, running lengthwise through the center of the specimen The portion of the rubber coating remaining outside of the one-inch strip shall be removed from the metal base One end of the one-inch rubber strip shall be separated from the metal base for a distance sufficient to permit clamping in the grip attached to the recording head of the machine

4.2 The metal base of the specimen shall be placed in the T-slot of the special base described in 3 1 1 1 which replaces the power-actuated grip of the testing machine The separated end of the one-inch rubber strip shall be clamped in the grip attached to the recording head of the machine The specimen shall be assembled in the testing machine so that the rubber strip will be separated from the metal at an angle of 90°, as shown in figure 8031, and so that the tension is distributed uniformly over the width of the rubber strip The machine shall be started and the power-actuated grip shall travel at a uniform speed of 1±1/16 inch per minute As the rubber is separated from the metal, the crank of the special base shall be turned at a rate which will maintain the specimen in a position so that the strip is separated from the metal base at an angle of 90° The rubber shall be separated from the metal for a sufficient distance to indicate the adhesion value If practicable, the separation shall be continued for a distance of at least 4 inches
The separation of the grip and special base shall be autographically recorded in inches on one axis of the chart and the force in pounds on the other axis The average force required to separate the rubber from the metal base shall be determined from the chart as described in 4 2 1 1 of method 8011

4.3 If tearing occurs, the procedure shall be as described in 4 3 of method 8011

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 When one specimen is tested from each test unit, the adhesion of the test unit shall be the value obtained from the specimen tested When more than one specimen is tested from each test unit, the adhesion of the test unit shall

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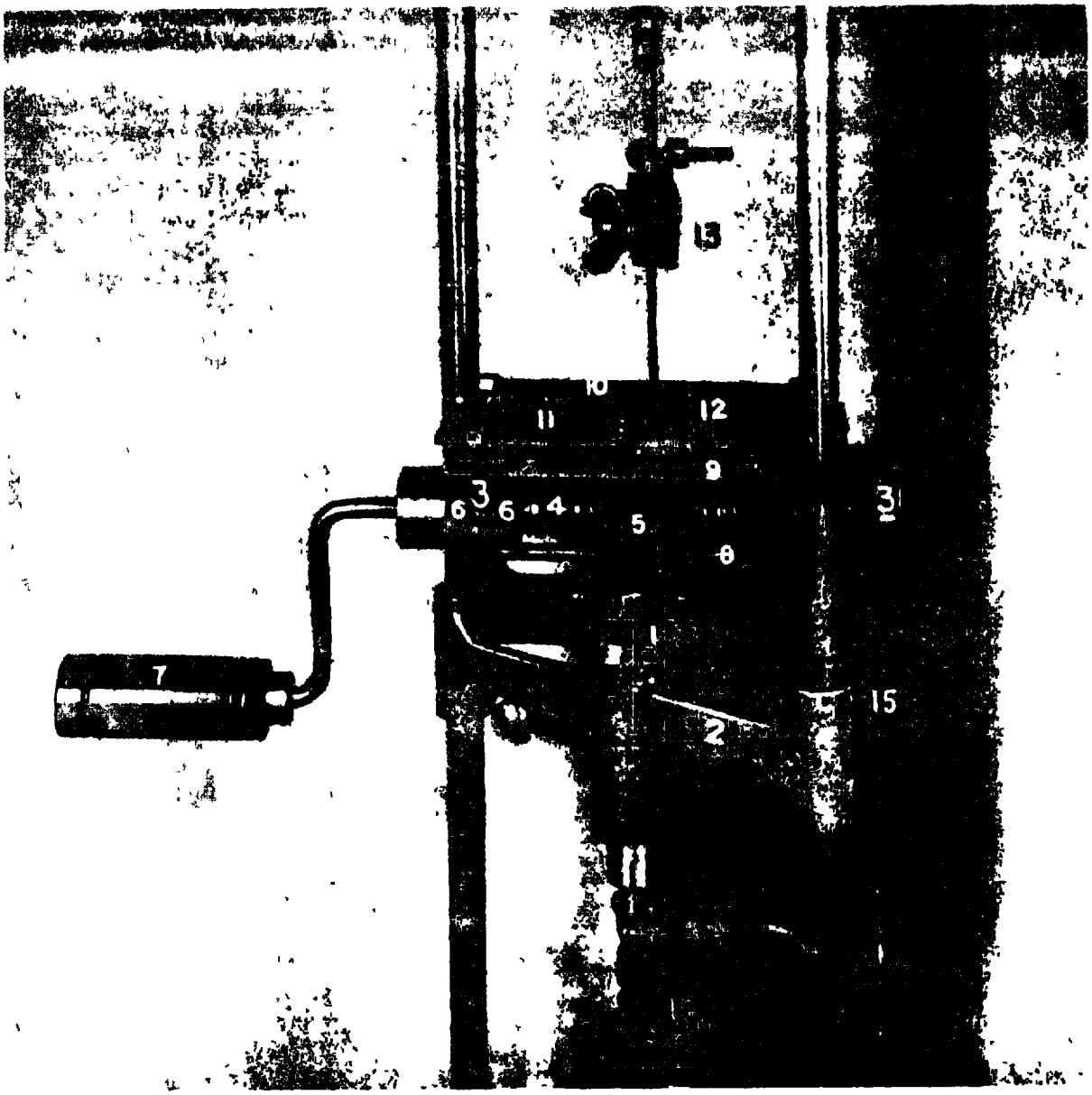

FIGURE 8031 —Special grips for rubber to metal adhesion

be the average of the values obtained from the specimens tested

5.3 The adhesion of the test unit shall be recorded to the nearest ½ pound per inch of width

5.4 When tearing repeatedly occurs, 4 3, the number of specimens tested from each test unit and the number from each test unit that tear shall be recorded

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ADHESION, COATING TO FABRIC

1. SCOPE

1.1 This method is intended for use in determining the force required to strip a coating of rubber compound from a fabric backing The method is applicable to such materials as foul-weather clothing, imitation leather (upholstery materials), etc

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 8 by 2 inches The long dimension shall be parallel to the warp (length) direction of the fabric for determining adhesion in the warp direction and parallel to the filling (width) direction of the fabric for determining adhesion in the filling direction

2.2 Low-strength fabric backing.—In the case of fabric backing too weak for determining the adhesion with a 2-inch strip described in 2 1, the specimen shall be 8 by 3 inches

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 The testing machine described in method 8011, except that the grips shall be as described in 3 1 1 1

3.1.1.1 *Grips* —The grips for holding the specimen have metallic gripping surfaces that will hold the specimen without slipping The width of the grips shall be not less than the width of the portion of the specimen to be gripped The gripping surface of one of the jaws of each grip is 1 by 1 inch and the other gripping surface of each grip is 1 by 1½ inches or more, with the long dimension perpendicular to the application of force All edges that might cause a cutting action shall be rounded to a radius of 1/64 inch

3.1.2 A knife or other sharp tool for cutting the specimen

3.3 Iso-octane conforming to the requirements of 3 4 1 of me hod 6001

4. PROCEDURE

4.1 Preparation of specimen.

4.1.1 *High-strength coating* —When the strength of the coating film exceeds the adhesive strength of the bond to the fabric, the specimen shall be prepared for test by carefully cutting the coating through to the fabric along the two parallel lines *A* and *B*, 1 00±0 01 inch apart, running lengthwise of the specimen, as shown in figure 8211 Cut *A* shall extend

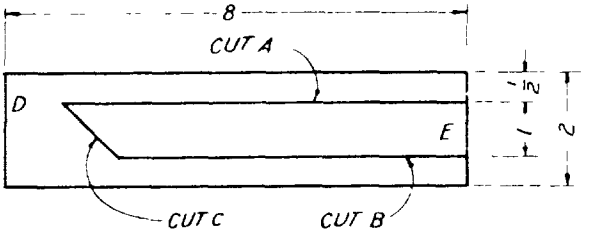

FIGURE 8211 —Specimen for adhesion test

from the end *E* of the specimen to within one inch of the opposite end *D* Cut *B* shall extend from the end *E* to within 2 inches of the opposite end *D* The ends of cuts *A* and *B* shall be joined together with a diagonal cut *C* which shall also be carefully cut through the coating to the fabric The edge of the knife or other sharp tool shall be worked under the point formed by cuts *A* and *C*, and the coating of the one-inch strip separated from the backing for a distance of 2 inches from that point If necessary, a small amount of iso-octane may be used between the coating and the fabric as an aid in separating the coating If iso-octane is used, the specimen shall be permitted to dry in circulating air for 24 hours before testing

4.1.2 *Low-strength coating* — When the strength of the coating is less than the adhesive strength of the bond of the coating to the fabric, two specimens of the coated fabric shall be cemented together, coated-face to coated-face, with an adhesive suitable for adhering the type of coating being tested The cuts shown in figure 8211 shall be made from one side, being careful to cut through only one layer of fabric and to avoid injuring the other layer of fabric The fabric shall then be stripped for a distance of 2 inches, as described in 4 1 1 This will allow the backing fabric to be inserted in one grip of the testing machine and a double thickness of the coating and a layer of fabric in the other grip

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4.1.3 *Low-strength fabric* —When the strength of the fabric is less than the adhesive strength of the bond to the fabric the specimen may be 3 inches in width and the coating strip to be separated may be 2 00±0 01 inch in width The remainder of the preparation of the specimen shall be as described in 4 1 1

4.2 Determination of adhesion.

4.2.1 *High- and low-strength coatings* —The specimen shall be free from mechanical damage The separated portion of the one-inch strip of coating shall be clamped in the power-actuated grip of the testing machine and the end *D* of the specimen clamped in the other grip The specimen shall be adjusted symmetrically in the testing machine in order that tension will be distributed uniformly over the width of the specimen The autographic mechanism and chart shall be adjusted to zero and the machine started The power-actuated grip shall travel at a uniform speed of 12±0 5 inch per minute The separation of the grips shall be autographically recorded in inches on one axis of the chart and the force in pounds on the other axis The coating shall be separated for a distance of not less than 3 inches The adhesion of the specimen shall be the average of the 5 highest peak forces registered on the "force" axis of the chart for separating the coating for a distance of 3 inches and shall be recorded to the nearest 0 1 pound per inch of width

4.2.2 *Low-strength fabric* —The procedure shall be as described in 4 2 1, except that the width of the strip of coating shall be 2 00±0 01 inch and the adhesion of the specimen shall be recorded to the nearest 0 1 pound per 2 inches of width

5. RESULTS

5.1 Unless otherwise specified in the detail specification, 4 specimens from each test unit, 2 with the long dimension parallel to the warp direction of the fabric, and 2 with the long dimension parallel to the filling direction of the fabric, shall be tested

5.2 The adhesion of the warp direction and the filling direction of the test unit shall be the average of the values obtained from the specimens tested in the warp direction and the filling direction, respectively

5.3 The adhesion of the test unit shall be recorded to the nearest 0 1 pound per inch of width, 4 2 1, or per 2 inches of width, 4 2 2, as applicable

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ–R–601a)

ADHESION, SEAMS (SEAM STRENGTH)

1. SCOPE

1.1 This method is intended for use in determining the strength of seams in rubber coated fabrics such as foul-weather clothing The strength of seams is usually compared with the strength of the coated fabric

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 6 by 6 inches with the seam running horizontally across the center

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 The testing machine described in method 4111, except that (1) the rate of travel of the power-actuated grip is 12 0±0 5 inch per minute and is uniform at all times, and (2) the machine is equipped with grips composed of a front and back jaw 3 inches in width, the width being at right angles to the direction of the application of load All edges of the grips that might cause cutting action shall be rounded to a radius of 1/64 inch

4. PROCEDURE

4.1 The specimen shall be free from mechanical damage The specimen shall be placed symmetrically in the grips of the testing machine with the seam running horizontally across the center of the specimen and midway between the grips, and the machine started The distance between the grips at the start of the test shall be 3±1/8 inch The power-actuated grip shall travel at a uniform speed of 12 0±0 5 inch per minute until rupture of the specimen Whether failure of the specimen occurred in the seam or in the fabric shall be recorded

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each seam to be tested in each test unit shall be tested

5.2 The number of specimens tested from each seam from each test unit and the number from each seam of each test unit that failed in the seam shall be recorded

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VOLUME RESISTIVITY

1. SCOPE

1.1 This method is intended for use in determining the volume resistivity of rubber and rubberlike materials having resistivities not to exceed 10 million ohm-centimeters

1.2 For the purpose of this method, volume resistivity is the resistance between two opposite faces of a centimeter cube of the material in absence of surface conduction

2 SPECIMEN

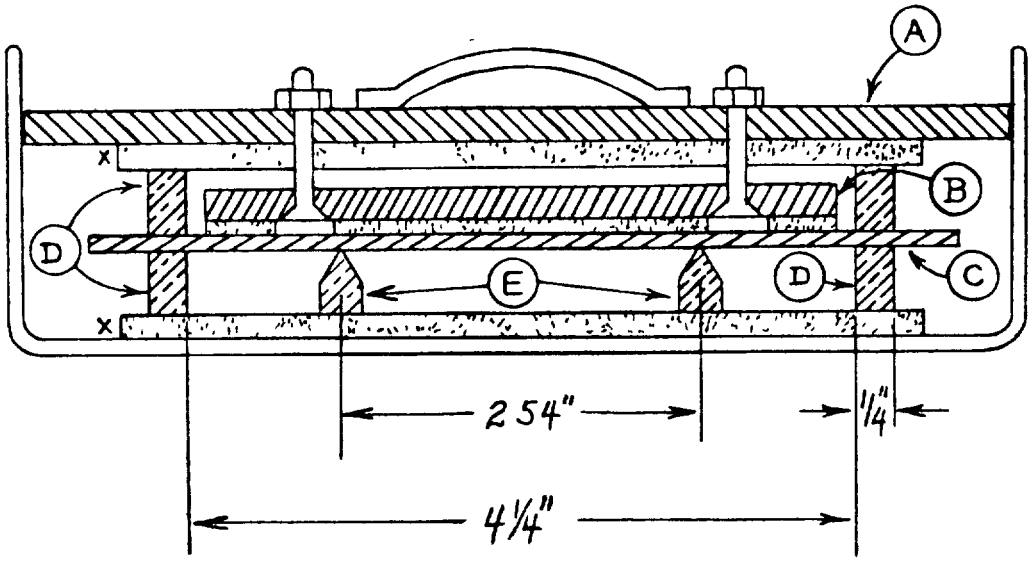
2.1 The specimen shall consist of a portion of the test unit at least 5 inches in length and having a width of at least 1 inch and not greater than 6 inches

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed ⅛ inch, and shall be uniform to within ±0 01 inch

3 APPARATUS

3 1 The apparatus shall be as follows

3.1.1 *Electrode assembly* —The electrode assembly is shown in figure 9111A It consists of current-carrying and potential-measuring electrodes Two brass current-carrying electrodes, 7 by ½ by ¼ inch, are attached to a rigid base of insulating material, such as hard rubber or plastic, so that their inner edges are spaced



X Insulation

A = 10-pound weight

B = 2-pound weight

C = specimen

D = Current electrodes

E = Voltage electrodes

FIGURE 9111A -- Electrode assembly

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4¼ inches apart Two brass potential-measuring electrodes are attached to the same base with knife edges in the plane of the current electrodes The 2 potential electrodes shall be spaced 2 54 inches apart and symmetrically located between the current electrodes A second pair of current-carrying electrodes arranged in a similar manner to the first pair is superimposed on the specimen directly above the first electrodes as shown in figure 9111A A 2-pound weight, *B*, attached to a flat insulating plate rests on the top of the specimen to insure good contact of the specimen with the knife edges of the voltage electrodes, and a 10-pound weight, *A*, is attached to the top part of the assembly to insure good contact of the specimen with the current-carrying electrodes

3.1.2 *Resistance measuring equipment* —This equipment consists of (1) a source of direct current potential to furnish current to the current electrodes and through the specimen and (2) a means for measuring the voltage across the knife edges Since the current through the knife edges must be kept very small, the voltage is measured by means of a null balance arrangement, 3 1 2 1, or by a voltmeter having a resistance of at least 1000 megohms, 3 1 2 2

3.1.2.1 *Equipment using null balance circuit* —The apparatus shall contain the necessary variable resistors and meters required to adjust and evaluate the measuring current and balancing potential A schematic diagram of the circuit is shown in figure 9111B The accompanying table lists parts of equipment suitable for assembling the apparatus

3.1.2.2 *Equipment using high-resistance voltmeter* —A voltmeter having an input resistance of not less than 1000 megohms replaces the null balance arrangement described in 3 1 2 1 without loss of accuracy of the instrument

3 1.3 Cloth

3 1 4 Acetone

4 PROCEDURE

4.1 The specimen shall be conditioned for at least 48 hours at a temperature of 23°±2° C (73 4°±3 6° F) and relative humidity not to exceed 45 percent Conditioning in a desiccator is suitable for this purpose

4.2 Dimensions.—The specimen shall be free from mechanical damage The surface of the specimen shall be free from foreign matter If necessary, the surface of the specimen shall be cleaned with a cloth, wet with acetone or warm water Flexing or distortion of the specimen shall be kept to a minimum The average thickness of the specimen shall be determined as described in method 2011, except that measurements shall be equally spaced over the area between the potential electrodes The width of the specimen shall be determined as described in method 2121, except that the measurements shall be equally spaced over the area between the potential electrodes The width shall be recorded to the nearest ¼₄ inch

4.3 The specimen shall be placed in the electrode assembly so that smooth surfaces are in contact with the current electrodes If the grain direction is known, the specimen shall be arranged in the assembly so that the measuring current flows parallel to the grain The top electrode assembly shall be placed over the specimen so that the current electrodes are directly above the lower pair of electrodes as shown in figure 9111A

4.4 With the specimen in place, switch Sw1 shall be closed and the milliammeter set at zero to 15 milliamperes The adjustment of switch Sw2, together with resistances R1, R2, and R3, shall be such that the current developed will not exceed the following values

Switch contacts	Maximum milli-amperes
1–4	15
5–6	3
7–9	1

4.4.1 *Null balance method, 3 1 2 1* —Switch Sw3 shall be closed and rotary switch Sw4 moved to approximate position, or one or two contacts less than Sw2 Switch Sw7 shall be closed, R7 set for minimum resistance, less sensitive positions for galvanometer, and then

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switch Sw5 closed For null balance, zero reading on galvanometer, R4, R5, and R6, shall be adjusted and the sensitivity of the galvanometer increased by increasing R7, followed by opening switch Sw7 to eliminate R7 altogether Switch Sw6 shall be closed, the voltage read, and the value recorded

4.4.2 *High-resistance voltmeter method, 3 1 2 2* —The circuit shall be closed, the voltage across the voltage electrodes read directly from the voltmeter, and the value recorded

4.5 It is desirable to limit the wattage dissipated in the specimen to 0 1 watt between voltage electrodes This condition can be satisfied by the product of volts by milliamperes being not greater than 100

5. RESULTS

5.1 Calculation.—The volume resistivity of the specimen shall be calculated as follows

$$\text{Volume resistivity, ohm-centimeters,} = \frac{EWT}{I}$$

Where

E is the number of volts across the voltage electrodes
W is the width of the specimen, inches
T is the thickness of the specimen, inch
I is the current passed through the specimen, amperes

5.2 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.3 When one specimen is tested from each test unit, the volume resistivity of the test unit shall be the value obtained from the specimen tested When more than one specimen is tested from each test unit, the volume resistivity of the test unit shall be the average of the values obtained from the specimens tested

5.4 The volume resistivity shall be recorded to three significant figures

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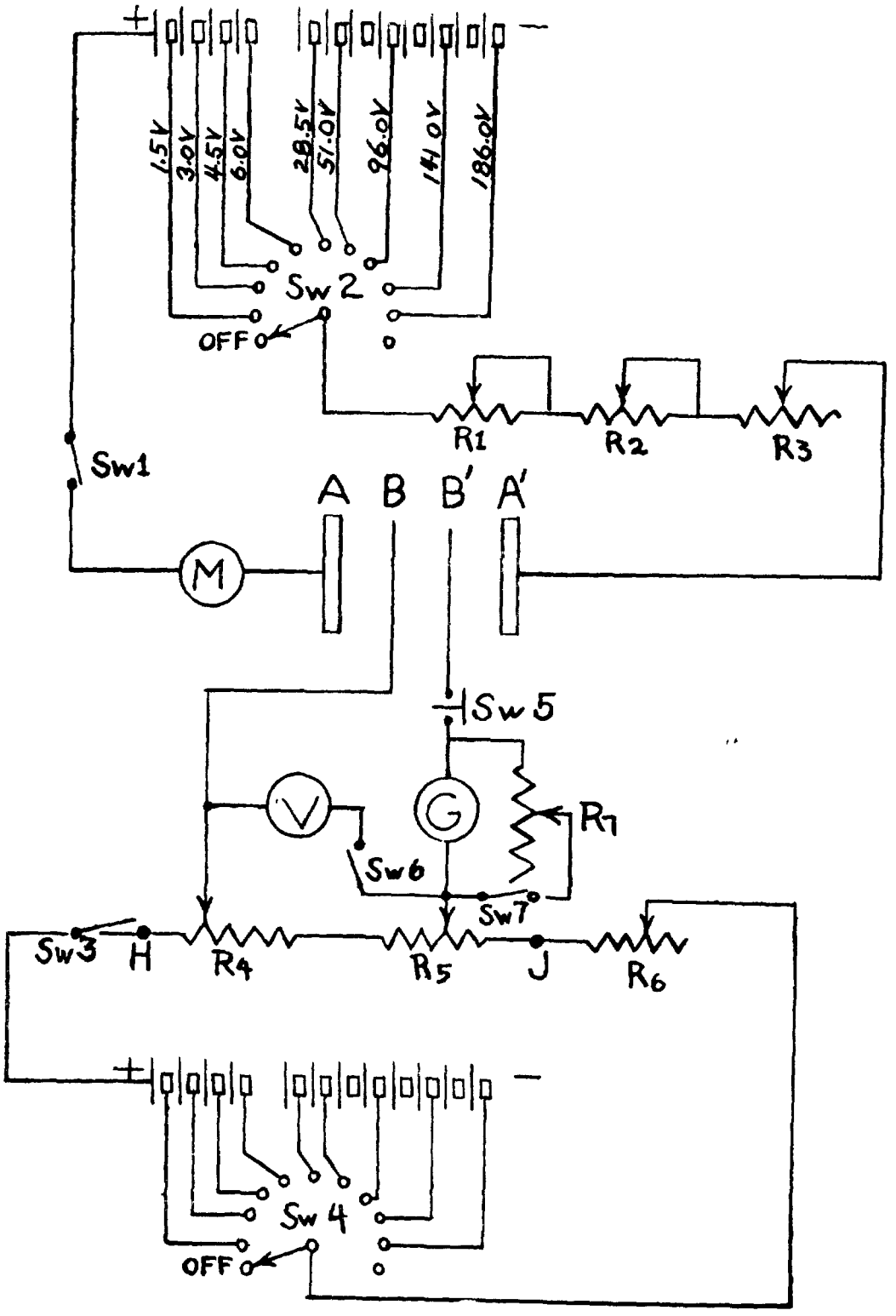

FIGURE 9111B —Resistance measuring apparatus using null balance

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PARTS USED IN FIGURE 9111B

A and A′—Current electrodes
B and B′—Voltage electrodes
Sw1, 3, 6 and 7—On and off toggle switches
Sw2 and 4—Single pole, 11-contact, radio-type rotary selector switches
Sw5—Normally open momentary contact switch
Bank 1 and 3—Four No 6 dry cells each
Bank 2 and 4—Four radio "B" batteries, 45-volt, one connected at 22½ volts
M Milliammeter, Weston DC Model 430, ranges 0–0 15, 0–1 5, and 0–15 milliamperes, scale divisions 150, or equivalent milliammeter
G Galvanometer, Leeds & Northrup Model 2320C, having a sensitivity of one microampere per scale division, or equivalent galvanometer
V Voltmeter—Vacuum Tube Voltohmist Electronic Designs Model 100, Electronic Designs, Inc , New York, N Y , or equivalent vacuum tube voltmeter

If desired a multirange DC voltmeter with a sensitivity of 1,000 ohms per volt or better may be used For protection of this voltmeter it is suggested that a two "gang" 11-contact rotary selector switch be substituted for Sw4 and the resistance multipliers for the voltmeter be connected to the proper points on the second set of the switch contacts In this case, switch Sw6 could be eliminated

R_1, R_2, R_4, and R_5—Two-watt 0–10,000-ohm potentiometers, Mallory wire-wound or equivalent
R_3 and R_6—Two-watt 0–5,000-ohm potentiometers, Mallory wire-wound or equivalent
R_7—Two-watt 0–3,000-ohm potentiometers, Mallory wire-wound or equivalent

NOTE —Where it may be desirable to extend the range of this equipment, more batteries may be added

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Method 9211
April 12, 1955

This method is a part of Fed Test Method Std No 601 (formerly Fed Spec ZZ-R-601•)

ELECTRICAL RESISTANCE OF CASTERS

1. SCOPE

1.1 This method is intended for use in determining the electrical resistance of conductive rubber tires on casters

2. SPECIMEN

2.1 The specimen shall consist of a finished caster from the test unit

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Any resistance-measuring equipment of suitable range which has an open-circuit direct voltage (dc) of about 500 volts For safety of the operator the maximum current delivered by the apparatus shall be less than 10 milliamperes

3.1.2 *Electrodes* —One electrode shall consist of a clean, dry, flat metal plate The metal parts of the caster specimen shall be used as the other electrode

3.1.3 A small brush

3.1.4 Wiping cloth

3.1.5 Soap

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, casters which are a part of other equipment shall be tested while supporting the equipment, 4 4, and casters which are intended for general use shall be tested while supporting 25 percent of their specified rated load

4.2 The specimen shall be conditioned at room temperature and a relative humidity of less than 80 percent for at least 24 hours, and tested under the same conditions

4.3 The surface of the rubber of the specimen shall be free from mechanical damage and shall be cleaned with water, soap, and a soft brush or by other means that will not abrade or otherwise change the surface of the rubber, and dried with a cloth The specimen shall be loaded as required, 4 1, and rolled on the clean, dry, flat, metal-plate electrode The metal plate and the metal part of the specimen acting as electrode shall be attached to the apparatus The resistance shall be measured to the nearest 1,000 ohms Five measurements shall be made, each with a separate area of the rubber part of the specimen in contact with the metal-plate electrode, and the values recorded to the nearest 1,000 ohms

4.4 If the caster is to be tested while supporting equipment, all other casters under the equipment shall be insulated during the test from the surface on which the metal-plate electrode is resting

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 Average resistance.—When one specimen is tested from each test unit, the average resistance of the test unit shall be the average of the values obtained from the specimen tested, 4 3 When more than one specimen is tested from each test unit, the average resistance of the test unit shall be the average of all the values obtained from the specimens tested

5.3 Maximum resistance.—The maximum resistance of the test unit shall be the largest of all the values obtained in determining the average resistance of the test unit

5.4 The resistance of the test unit shall be recorded to the nearest 1,000 ohms

5.5 The load applied shall be recorded

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BURSTING STRENGTH, STRAIGHT SPECIMEN

1. SCOPE

1.1 This method is intended for use in determining the bursting strength of hose by subjecting it to internal water pressure The test is carried out with the specimen in a straight position

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit of full circumference

2.1.1 Unless otherwise specified in the detail specification, the specimen shall be 36 ± 1 inch in length

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Gage* —A calibrated pressure gage with graduations in pounds per square inch and a capacity approximately 35 percent greater than the specified minimum bursting strength of the hose

3.1.2 A hand- or power-driven hydraulic pump or an accumulator system for applying hydrostatic pressure

3.1.3 Two couplings or clamps

3.1.4 A cap or plug equipped with a petcock for closing one coupling

3.1.5 A stopwatch or other timing device that indicates the time in seconds and minutes

4. PROCEDURE

4.1 The specimen shall be free from mechanical damage The ends of the specimen shall be fitted with the couplings or clamps One coupling or clamp shall be connected to the source of hydrostatic pressure and the other closed with the cap or plug equipped with the petcock To insure complete removal of air from the hose, the specimen shall be inclined slightly so that the end with the petcock is at the highest point

4.2 With the petcock open, water shall be admitted slowly into the hose until the air has been completely replaced with water This is important as a safety measure The petcock shall then be closed and the specimen laid out straight on a flat surface and maintained in this position during the test

4.3 For hose with a specified bursting strength of 2,000 pounds per square inch or less, water shall be forced into the specimen in such a quantity as to increase the pressure at a rate of not less than 300 and not more than 1,000 pounds per square inch per minute For hose with a specified bursting strength greater than 2,000 pounds per square inch, the water shall be forced into the specimen in such a quantity as to increase the pressure at a rate of not less than 15,000 and not more than 35,000 pounds per square inch per minute

4.4 During the application of the pressure, the specimen shall be examined for leakage and other defects The pressure shall be applied as described above until rupture of the specimen The pressure necessary to rupture the specimen shall be read from the gage and the value recorded

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 When one specimen is tested from each test unit, the bursting strength of the test unit shall be the value obtained from the specimen tested When more than one specimen is tested from each test unit, the bursting strength of the test unit shall be the average of the values obtained from the specimens tested

5.3 The bursting strength of the test unit shall be recorded to the nearest 10 pounds per square inch for hose bursting under 600 pounds per square inch, 25 pounds per square inch for hose bursting at 600 to 2,000 pounds per square inch, 50 pounds per square inch for hose bursting at 2,000 to 5,000 pounds per square inch, and 100 pounds per square inch for hose bursting at 5,000 pounds per square inch and over

5.4 The number of specimens tested from each test unit and the number from each test unit that showed leakage and other defects during the application of the pressure shall be recorded

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BURSTING STRENGTH, CURVED SPECIMEN

1. SCOPE

1.1 This method is intended for use in determining the bursting strength of hose by subjecting it to internal water pressure The test is carried out with the specimen in a curved position

2. SPECIMEN

2.1 The specimen shall be as described in method 10011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in method 10011

3.1.2 Apparatus for holding a specimen in a circular arc of the radius required, 4 1, during application of the pressure

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the curvature of the specimen during the test shall be a circular arc of 27-inch radius

4.2 The procedure shall be as described in method 10011, except that the specimen shall be supported in the required curved position, 4 1, throughout the test

5. RESULTS

5.1 The results shall be as described in method 10011, except that the curvature of the specimen during test shall also be recorded.

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AIR LEAKAGE

1. SCOPE

1.1 This method is intended for use in determining the leakage of hose by subjecting it to internal air pressure and immersing it in water

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit of full circumference and 10 ± 1 inch in length

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A source of compressed air

3.1.2 A calibrated pressure gage with graduations in pounds per square inch and a capacity approximately 35 percent greater than the specified pressure, 4 1

3.1.3 Two couplings or clamps

3.1.4 A cap or plug for closing one of the couplings

3.1.5 A container suitable for immersing the specimen in water

3.1.6 A stopwatch or other device for indicating the time in seconds and minutes

4. PROCEDURE

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4.1 Unless otherwise specified in the detail specification, the air pressure applied to the specimen shall be 200 ± 5 pounds per square inch.

4.2 Unless otherwise specified in the detail specification, the specimen shall be subjected to the required pressure for 5 minutes ± 5 seconds

4.3 The specimen shall be free from mechanical damage The ends of the specimen shall be fitted with the couplings or clamps One coupling or clamp shall be connected to the source of air pressure and the other closed tightly with the cap or plug The specimen and couplings shall be completely submerged in water in the container so as to permit good visual observation of the specimen The specimen shall be subjected to the required air pressure for the required time

4.4 While under pressure, the specimen shall be examined for leakage of air from the hose and couplings, and other evidence of failure

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 The number of specimens tested from each test unit and the number from each test unit that showed leakage and other evidence of failure shall be recorded

5.3 The air pressure and time of application of the pressure shall be recorded

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PROOF PRESSURE

1. SCOPE

1.1 This method is intended for use in proof pressure testing of hose. The hose is subjected to internal water pressure under specified conditions to determine the effect of hydrostatic pressure in producing changes similar to those occurring in hose and couplings in service

2. SPECIMEN

2.1 The specimen shall be as described in method 10311

3. APPARATUS

3.1 The apparatus shall be as described in 3 1 1, 3 1 2, 3 1 4, and 3 1 5 of method 10011

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the pressure applied to the specimen shall be equal to 40 ± 1 percent of the minimum bursting strength specified in the detail specification for the hose

4.2 Unless otherwise specified in the detail specification, the required pressure, 4 1, shall be maintained on the specimen for a period of 60 ± 1 second

4.3 The specimen shall be free from mechanical damage One coupling or clamp shall be connected to the source of hydrostatic pressure and the other closed with the cap or plug equipped with a petcock To insure complete removal of air from the hose, the specimen shall be inclined slightly so that the end with the petcock is at the highest point

4.4 With the petcock open, water shall be admitted slowly into the hose until the air has been completely replaced with water This is important as a safety measure The petcock shall then be closed and the specimen laid out straight on a flat surface and maintained in this position during the test

4.5 The rate of application of pressure shall be as described in 4 3 of method 10011

4.6 The pressure shall be applied to the specimen until the required pressure, 4 1, is reached The required pressure shall be maintained for 60 seconds or other specified time While under pressure, the hose and couplings shall be examined for leakage and other evidence of failure

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 The number of specimens tested from each test unit and the number from each test unit that showed leakage and other evidence of failure shall be recorded

5.3 The pressure and time of application shall be recorded

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HOLD TEST, STRAIGHT SPECIMEN

1. SCOPE

1.1 This method is intended for use in determining whether hose will withstand internal water pressure under specified conditions without rupture, leakage, and other evidence of failure The test is carried out with the specimen in a straight position

2. SPECIMEN

2.1 The specimen shall be as described in method 10011

3. APPARATUS

3.1 The apparatus shall be as described in method 10011

4. PROCEDURE

4.1 The pressure applied to the specimen shall be as specified in the detail specification

4.2 Unless otherwise specified in the detail specification, the specimen shall be subjected to the specified pressure, 4 1, for 10 minutes ± 5 seconds

4.3 The pressure shall be applied to the specimen as described in 4 1, 4 2, and 4 3 of method 10011

4.4 The pressure shall be applied until the specified pressure, 4 1, is reached The pressure shall be maintained for 10 minutes or other specified time, 4 2 While under pressure, the specimen shall be examined for rupture, leakage, and other evidence of failure If the specimen ruptures, the bursting pressure shall be recorded

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 The number of specimens tested from each test unit and the number from each test unit that showed rupture, leakage, and other evidence of failure shall be recorded

5.3 The pressure applied to the specimen and the time of application shall be recorded

5.4 If the specimen ruptures, the pressure at which rupture occurs shall be recorded as described in 5 3 of method 10011

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HOLD TEST, CURVED SPECIMEN

1. SCOPE

1.1 This method is intended for use in determining whether hose will withstand internal water pressure under specified conditions without rupture, leakage, and other evidence of failure The test is carried out with the specimen in a curved position

2. SPECIMEN

2.1 The specimen shall be as described in method 10011

3. APPARATUS

3.1 The apparatus shall be as described in methods 10011 and 10021

4. PROCEDURE

4.1 The pressure applied to the specimen shall be as specified in the detail specification

4.2 Unless otherwise specified in the detail specification, the specimen shall be subjected to the specified pressure, 4 1, for 10 minutes ±5 seconds

4.3 unless otherwise specified in the detail specification, the curvature of the specimen during the test shall be a circular arc of 27-inch radius

4.4 The pressure shall be applied to the specimen as described in 4 1, 4 2, and 4 3 of method 10011, except that the specimen shall be supported in the required curved position 4 3, throughout the test

4.5 The pressure shall be applied until the specified pressure, 4 1, is reached The pressure shall be maintained for 10 minutes or other specified time, 4 2, with the specimen still in the curved position, 4 3 While the specimen is under pressure, it shall be examined for rupture, leakage, and other evidence of failure. If the specimen ruptures, the bursting pressure shall be recorded

5. RESULT

5.1 The results shall be as described in method 10221, except that the curvature of the specimen during the test shall also be recorded

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ELONGATION OR CONTRACTION

1. SCOPE

1.1 This method is intended for use in determining the change in length of hose due to application of internal water pressure The method is particularly applicable to testing fire hose

2. SPECIMEN

2.1 The specimen shall consist of a full length and circumference of hose from the test unit or the remainder of the length from which the 3-foot specimen is taken for use in method 10011, with couplings attached in either case

2.2 Unless otherwise specified in the detail specification, the same specimen shall be used in methods 10321, 10331, 10341, 10351, and 10311

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Gages*

3.1.1.1 A calibrated pressure gage with graduations in pounds per square inch and a capacity approximately 35 percent greater than the specified final pressure, 4 2

3.1.1.2 A calibrated pressure gage with graduations in pounds per square inch and a capacity of approximately 20 pounds per square inch for use only to indicate the pressure of 10 pounds per square inch, 4 6 The gage shall be equipped with a shutoff valve and a relief cock to protect it against the higher pressure used in the test

3.1.2 A hand- or power-driven hydraulic pump or an accumulator system for applying hydrostatic pressure

3.1.3 A cap or plug equipped with a petcock for closing one coupling

3.1.4 A stopwatch or other timing device that will indicate the time in seconds and minutes

3.1.5 A steel tape graduated to ¼ inch or finer

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, when requirements for elongation or contraction, expansion, twist, warp, and rise are specified, these tests shall be made on the same specimen during a single test

4.2 The final pressure applied to the specimen shall be as specified in the detail specification

4.3 Unless otherwise specified in the detail specification, the final length of the specimen shall be measured 8 minutes after application of the final pressure, 4 2

4.4 Unless otherwise specified in the detail specification, the specimen shall be subjected to the specified final pressure, 4 2, for a total of 10 minutes

4.5 The specimen shall be free from mechanical damage One coupling or clamp shall be connected to the source of hydrostatic pressure and the other closed with the cap or plug equipped with a petcock To insure complete removal of air from the hose, the specimen shall be inclined slightly so that the end with the petcock is at the highest point

4.6 With the petcock open, water shall be admitted slowly into the hose until the air has been completely replaced with water This is important as a safety measure The petcock shall then be closed, the specimen laid out straight on a flat surface, and water forced into the specimen until the 20-pound gage registers 10 pounds per square inch pressure While the hose is held under 10 pounds pressure, the length between the backs of the couplings shall be measured with the steel tape and the value recorded to the nearest ¼ inch as L_1

4.7 When a final pressure of 2,000 pounds per square inch or less is specified in the detail specification, water shall be forced into the specimen in such a quantity as to increase the pressure at a rate of not less than 300 and not more than 1,000 pounds per square inch per minute When a final pressure greater than 2,000 pounds per square inch is specified in the detail specification, the water shall be forced into the specimen in such a quantity as to increase the pressure at a rate of not less than 15,000 and not more than 35,000 pounds per square inch per minute

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4.8 The pressure shall be applied until the specified final pressure, 4 2, is reached The final pressure shall be maintained for a total of 10 minutes or other specified time, 4 4 Eight minutes or other specified time, 4 3, after application of the final pressure, 4 2, the length between the backs of the couplings shall again be measured with the steel tape and the value recorded to the nearest ¼ inch as L_2

4.9 While under the final pressure, the specimen shall be examined for leakage and other evidence of failure

5. RESULTS

5.1 Calculation.—The elongation and contraction of the specimen shall be calculated as follows

$$\text{Elongation, percent,} = \frac{L_2 - L_1}{L_1} \times 100$$

$$\text{Contraction, percent,} = \frac{L_1 - L_2}{100} \times 100$$

Where

L_1 is the length between backs of couplings of the specimen at 10 pounds per square inch pressure, feet and inches

L_2 is the length between backs of couplings of the specimen at the specified final pressure, feet and inches

5.2 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.3 When one specimen is tested from each test unit, the elongation or contraction of the test unit shall be the value obtained from the specimen tested When more than one specimen is tested from each test unit, the elongation or contraction of the test unit shall be the average of the values obtained from the specimens tested

5.4 The elongation or contraction of the test unit shall be recorded to the nearest one percent

5.5 The number of specimens tested from each test unit and the number from each test unit that showed leakage and other evidence of failure shall be recorded

5.6 The final pressure, the time of application of the final pressure, and the time the specimen is under final pressure before measuring the length shall be recorded

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EXPANSION, CIRCUMFERENCE

1. SCOPE

1.1 This method is intended for use in determining the expansion of hose due to application of internal water pressure The method is particularly applicable to testing fire hose

2. SPECIMEN

2.1 The specimen shall be as described in method 10311

2.2 Unless otherwise specified in the detail specification, the same specimen shall be used in methods 10311, 10331, 10341, 10351, and 10321.

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in method 10311, 3 1 1 to 3 1 4, inclusive

3.1.2 A crayon or soft pencil for marking the hose

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, when requirements for expansion, elongation or contraction, twist, warp, and rise are specified, these tests shall be made on the same specimen during a single test

4.2 The final pressure applied to the specimen shall be as specified in the detail specification

4.3 Unless otherwise specified in the detail specification, the final circumference of the specimen shall be measured 8 minutes after application of the final pressure, 4 2

4.4 Unless otherwise specified in the detail specification, the specimen shall be subjected to the specified final pressure, 4 2, for a total of 10 minutes

4.5 The specimen shall be free from mechanical damage One coupling or clamp shall be connected to the source of hydrostatic pressure and the other closed with the cap or plug equipped with the petcock To insure complete removal of air from the hose, the specimen shall be inclined slightly so that the end with the petcock is at the highest point

4.6 With the petcock open, water shall be admitted slowly into the hose until the air has been completely replaced with water This is important as a safety measure The petcock shall then be closed, the specimen laid out straight on a flat surface, and water forced into the specimen until the 20-pound gage registers 10 pounds per square inch pressure Three marks equally spaced along the length of the specimen shall be made by means of a crayon or soft pencil While the hose is held under 10-pound pressure, the circumference shall be measured at the three marked places as described in method 2351 and the average of the values recorded to the nearest $\frac{1}{32}$ inch as C_1

4.7 The pressure shall be applied as described in 4 7 of method 10311

4.8 The pressure shall be applied until the specified final pressure, 4 2, is reached The final pressure shall be maintained for a total of 10 minutes or other specified time, 4 4 Eight minutes or other specified time, 4 3, after application of the final pressure, 4 2, the circumference shall again be measured at the three marked places as described in method 2351 and the average of the values recorded to the nearest $\frac{1}{32}$ inch as C_2

4.9 While under the final pressure, the specimen shall be examined for leakage and other evidence of failure

5. RESULTS

5.1 **Calculation.**--The expansion of the specimen shall be calculated as follows

$$\text{Expansion, percent, } = \frac{C_2 - C_1}{C_1} \times 100$$

Where

C_1 is the circumference of the specimen at 10 pounds per square inch pressure, inches

C_2 is the circumference of the specimen at the specified final pressure, inches

5.2 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.3 When one specimen is tested from each test unit, the expansion of the test unit shall be value obtained from the specimen tested When more than one specimen is tested from each test unit, the expansion of the test unit shall be

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the average of the values obtained from the specimens tested

5.4 The expansion of the test unit shall be recorded to the nearest one percent

5.5 The number of specimens tested from each test unit and the number from each test unit that showed leakage and other evidence of failure shall be recorded.

5.6 The final pressure, time of application of the final pressure, and the time the specimen is under final pressure before measuring the circumference shall be recorded.

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TWIST

1. SCOPE

1.1 This method is intended for use in determining the amount and direction of twist of hose due to the application of internal water pressure The method is particularly applicable to testing fire hose

2. SPECIMEN

2.1 The specimen shall be as described in method 10311

2.2 Unless otherwise specified in the detail specification, the same specimen shall be used in methods 10311, 10321, 10341, 10351, and 10331

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in method 10311

3.1.2 A crayon or soft pencil for marking the hose

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, when requirements for twist, elongation or contraction, expansion, warp, and rise are specified, these tests shall be made on the same specimen during a single test

4.2 The final pressure applied to the specimen shall be as specified in the detail specification

4.3 Unless otherwise specified in the detail specification, the number of turns of the specimen shall be measured 9 minutes after application of the final pressure, 4 2

4.4 Unless otherwise specified in the detail specification, the specimen shall be subjected to the specified final pressure, 4 2, for a total of 10 minutes

4.5 The specimen shall be free from mechanical damage One coupling or clamp shall be connected to the source of hydrostatic pressure and the other closed with the cap or plug equipped with the petcock To insure complete removal of air from the hose, the specimen shall be inclined slightly so that the end with the petcock is at the highest point

4.6 With the petcock open, water shall be admitted slowly into the hose until the air has been completely replaced with water This is important as a safety measure The petcock shall then be closed, the specimen laid out straight on a flat surface, and water forced into the specimen until the 20-pound gage registers 10 pounds per square inch pressure While the hose is held under 10 pounds pressure, the length between the backs of the couplings shall be measured with the steel tape and the value recorded to the nearest ¼ inch as L The specimen shall be marked with a crayon or soft pencil on the top surface of the hose adjacent to the closed coupling

4.7 The pressure shall be applied to the specimen as described in 4 7 of method 10311

4.8 The pressure shall be applied until the specified final pressure, 4 3, is reached The final pressure shall be maintained for a total of 10 minutes or other specified time, 4 4 Nine minutes or other specified time, 4 3, after application of the final pressure, 4 2, the number of turns of the hose as shown by the change in position of the crayon marks shall be determined and the value recorded to the nearest ⅛ turn as N

4.9 While under the final pressure, the specimen shall be examined for the direction of twist and for leakage and other evidence of failure Twist to the right, the direction that would tend to tighten the coupling, is indicated by a clockwise rotation of the coupling looking from the hose toward the coupling Twist to the left is indicated by a counterclockwise rotation

5. RESULTS

5.1 Calculation.--The amount of twist of the specimen shall be calculated as follows

$$\text{Twist, degrees per foot, } = \frac{360 \times N}{L}$$

Where

N is the number of turns of the specimen

L is the length between backs of couplings of the specimen at 10 pounds per square inch pressure, feet and inches

5.2 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.3 When one specimen is tested from each test unit, the twist of the test unit shall be

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the value obtained from the specimen tested. When more than one specimen is tested from each test unit, the twist of the test unit shall be the average of the values obtained from the specimens tested

5.4 The twist of the test unit to the nearest 5 degrees per foot and the direction of the twist as right or left shall be recorded.

5.5 The number of specimens tested from each test unit and the number from each test unit that showed leakage, and other evidence of failure shall be recorded.

5.6 The final pressure, time of application of the final pressure, and the time the specimen is under final pressure before determining the number of turns shall be recorded.

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WARP

1. SCOPE

1.1 This method is intended for use in determining the warp or deviation from a straight line of hose due to the application of internal water pressure The method is particularly applicable to testing fire hose

2. SPECIMEN

2.1 The specimen shall be as described in method 10311

2.2 Unless otherwise specified in the detail specification, the same specimen shall be used in methods 10311, 10321, 10331, 10351, and 10341

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in method 10311

3.1.2 A cord for a reference point in measuring the warp of the specimen

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, when requirements for warp, elongation or contraction, expansion, twist, and rise are specified, these tests shall be made on the same specimen during a single test

4.2 The final pressure applied to the specimen shall be as specified in the detail specification

4.3 Unless otherwise specified in the detail specification, the warp of the specimen shall be measured 9 minutes after application of the final pressure, 4 2

4.4 Unless otherwise specified in the detail specification, the specimen shall be subjected to the specified final pressure, 4 2, for a total of 10 minutes

4.5 The specimen shall be free from mechanical damage One coupling or clamp shall be connected to the source of hydrostatic pressure and the other closed with the cap or plug equipped with the petcock To insure complete removal of air from the hose, the specimen shall be inclined slightly so that the end with the petcock is at the highest point

4.6 With the petcock open, water shall be admitted slowly into the hose until the air has been completely replaced with water This is important as a safety measure The petcock shall be closed and the specimen laid out straight on a flat surface

4.7 The pressure shall be applied to the specimen as described in 4 7 of method 10311

4.8 The pressure shall be applied until the final pressure is reached The final pressure, 4 2, shall be maintained for a total of 10 minutes or other specified time, 4 4 Nine minutes or other specified time, 4 3, after application of the final pressure, 4 2, the amount of warp shall be determined by stretching a cord from center to center of the backs of the two couplings and measuring the distance from the hose to the cord by means of the tape, at the point where the hose is the greatest distance from the cord, and the value recorded to the nearest ¼ inch

4.9 While under the final pressure, the specimen shall be examined for leakage and other evidence of failure.

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 When one specimen is tested from each test unit, the warp of the test unit shall be the value obtained from the specimen tested When more than one specimen is tested from each test unit, the warp of the test unit shall be the average of the values obtained from the specimens tested

5.3 The warp of the test unit shall be recorded to the nearest ¼ inch

5.4 The number of specimens tested from each test unit and the number from each test unit that showed leakage and other evidence of failure shall be recorded.

5.5 The final pressure, the time of application of the final pressure, and the time the specimen is under final pressure before measuring the warp shall be recorded

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RISE

1. SCOPE

1.1 This method is intended for use in determining the rise of hose due to the application of internal water pressure The method is particularly applicable to testing fire hose

2. SPECIMEN

2.1 The specimen shall be as described in method 10311

2.2 Unless otherwise specified in the detail specification, the same specimen shall be used in methods 10311, 10321, 10331. 10341, and 10351

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 The apparatus described in method 10311, 3 1 1 to 3 1 4, inclusive

3.1.2 A steel scale graduated to ⅛ inch or finer

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, when requirements for rise, elongation or contraction, expansion, twist, and warp are specified, these tests shall be made on the same specimen during a single test

4.2 The final pressure applied to the specimen shall be as specified in the detail specification

4.3 Unless otherwise specified in the detail specification, the rise of the specimen shall be measured 9 minutes after application of the final pressure 4 2

4.4 Unless otherwise specified in the detail specification, the specimen shall be subjected to the specified final pressure, 4 2, for a total of 10 minutes

4.5 The specimen shall be free from mechanical damage One coupling or clamp shall be connected to the source of hydrostatic pressure and the other closed with the cap or plug equipped with the petcock To insure complete removal of air from the hose, the specimen shall be inclined slightly so that the end with the petcock is at the highest point

4.6 With the petcock open, water shall be admitted slowly into the hose until the air has been completely replaced with water. This is important as a safety measure The petcock shall then be closed and the specimen laid out straight on a flat surface

4.7 The pressure shall be applied to the specimen as described in 4.7 of method 10311

4.8 The pressure shall be applied until the specified final pressure, 4 2, is reached The final pressure shall be maintained for a total of 10 minutes or other specified time, 4 4 Nine minutes or other specified time, 4 3, after application of the final pressure, 4 2, the greatest distance any part of the hose rises from the flat surface on which it rests shall be measured with the steel scale and the value recorded to the nearest ⅛ inch

4.9 While under the final pressure, the specimen shall be examined for leakage and other evidence of failure

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested.

5.2 When one specimen is tested from each test unit, the rise of the test unit shall be the value obtained from the specimen tested When more than one specimen is tested from each test unit, the rise of the test unit shall be the average of the values obtained from the specimens tested.

5.3 The rise of the test unit shall be recorded to the nearest ⅛ inch

5.4 The number of specimens tested from each test unit and the number from each test unit that showed leakage, and other evidence of failure shall be recorded.

5.5 The final pressure, time of application of the final pressure, and the time the specimen is under final pressure before measuring the rise shall be recorded

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KINK

1. SCOPE

1.1 This method is intended for use in determining the effect of applying internal water pressure to hose when it is kinked In general, the pressure is somewhat less than that which is used for determining bursting strength

2. SPECIMEN

2.1 The specimen shall be as described in method 10311

2.2 Unless otherwise specified in the detail specification, the specimen shall be that which has previously been used in methods 10311, 10321, 10331, 10341, and 10351

3. APPARATUS

3.1 The apparatus shall be as described in method 10311, 3 1 1 to 3 1 4, inclusive

4. PROCEDURE

4.1 The final pressure applied to the specimen shall be as specified in the detail specification

4.2 Unless otherwise specified in the detail specification, the specimen shall be subjected to the final pressure, 4 1, for 5 seconds

4.3 The specimen shall be free from mechanical damage One coupling or clamp shall be connected to the source of hydrostatic pressure and the other closed with the cap or plug equipped with the petcock To insure complete removal of air from the hose, the specimen shall be inclined slightly so that the end with the petcock is at the highest point

4.4 With the petcock open, water shall be admitted slowly into the hose until the air has been completely replaced with water This is important as a safety measure The petcock shall be closed, the specimen laid out straight on a flat surface, and water forced into the specimen until the 20-pound pressure gage registers approximately, but nor more than, 10 pounds per square inch pressure While this pressure is maintained, the coupling with the petcock attached shall be tied back against the hose so that there will be a sharp kink in the hose about 18 inches from the coupling

4.5 The pressure shall be applied to the specimen as described in 4 7 of method 10311

4.6 The pressure shall be applied until the specified final pressure, 4 1, is reached The final pressure, shall be maintained for a total of 5 seconds or other specified time, 4 2 While under the final pressure, 4 1, the specimen shall be examined for rupture, leakage, and other evidence of failure

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 The number of specimens tested from each test unit and the number from each test unit that showed rupture, leakage, and other evidence of failure shall be recorded

5.3 The final pressure and the time of application of the final pressure shall be recorded

5.4 If the specimen ruptures, the pressure at which rupture occurs shall be recorded as described in 5 3 of method 10011

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COATED FABRICS, HYDROSTATIC RESISTANCE

1. SCOPE

1.1 This method is intended for use in determining the resistance of coated fabrics to hydrostatic pressure The method is also applicable to seams in coated fabrics Two procedures are described for applying the pressure The procedure to be used shall be as specified in the detail specification

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit, the smallest dimension of which is at least 0 5 inch greater than the outside diameter of the ring clamp mechanism of the bursting strength tester, 3 1 1

2.2 In case the seam is to be tested, the specimen shall be cut so that the edge of the seam will be as near as practicable on the diameter of the aperture of the bursting strength tester

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Bursting strength tester* —A hand- or power-driven bursting strength tester equipped as follows

3.1.1.1 The machine holds the specimen firmly, without slippage, between two annular, plane, unpolished (matte) surfaces that may have fine, concentric tool marks not over 0 002 inch in depth The upper clamping surface shall have a circular opening 1 200 ± 0 001 inch in diameter The edges of the openings that come in contact with the specimen are rounded to a radius of not over 1/64 inch to prevent any cutting action The lower clamping surface is 0 125 inch thick and has an opening of 1 25 ± 0 05 inch in diameter The clamping ring is connected to the clamping mechanism through a swivel joint to insure an even clamping pressure During the test, the circular edges of the openings in the two clamping plates are substantially concentric with no overlapping of any point

3.1.1.2 The apparatus is equipped with a Bourdon tube, maximum-reading-type pressure gage, graduated in pounds per square inch, which will indicate the pressure throughout the entire range of its scale to within a value equal to one percent of its maximum capacity The capacity of the gage is such that the individual readings will not be less than 25 percent nor more than 75 percent of the capacity of the gage

3.1.1.3 The power-driven machine is equipped with a motor and the hand-driven machine with a handwheel for applying controlled hydraulic pressure to the underside of the specimen This pressure is generated by a piston forcing a liquid into the pressure chamber of the apparatus at a rate of 50 ± 5 ml per minute

3.1.2 Thermometer graduated in degrees centigrade

3.1.3 Water as a pressure medium

3.1.4 Equipment for testing materials with bursting strengths less than 30 pounds per square inch and for materials in which the stretch of the fabric exceeds the elastic limit of the coating shall be as follows A 16-mesh wire screen and an interchangeable upper clamp having a glass plate not less than 0 125 inch in thickness fitted in its aperture, for use with the screen

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the coated side of the fabric shall be placed next to the water

4.2 If the seam is to be tested, the edge that is exposed to the water in service shall be placed in contact with the water

4.3 The water temperature shall be the same as that of the atmosphere in the testing room The water level shall be brought up flush with the surface of the circular opening in the testing machine so that no air pockets exist between the surface of water and the surface of the coated fabric The specimen shall be free from mechanical damage The specimen shall be securely clamped in the testing machine so that the material will not be damaged If the seam is to be tested, the specimen shall be clamped so that the edge of the seam not in contact with water in service is held in such a manner that the water may penetrate the seam

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4.4 The pressure shall be applied as described in procedure A or B as specified in the detail specification, except that procedure B shall be used if the screen and glass plate, 3 1 4, are used

4.4.1 *Procedure A, increased pressure* —The pressure shall be applied to the specimen by forcing water into the pressure chamber at a uniform rate of 50±5 ml per minute until water penetrates through the fabric or seam At the moment of the first appearance of water, the machine shall be stopped, the applied pressure read from the gage, and the value recorded to the nearest one pound per square inch as the hydrostatic pressure resistance of the specimen

4.4.2 *Procedure B, constant pressure*

4.4.2.1 The pressure applied to the specimen shall be as specified in the detail specification

4.4.2.2 The pressure shall be applied to the specimen as described in 4 4.1, until the specified pressure, 4 4 2 1, is reached. The specimen shall remain at the required pressure within ±2 pounds per square inch for 5 minutes ±5 seconds At the end of the 5-minute period, the specimen shall be examined for leakage through the coated fabric or seam.

5. RESULTS

5.1 Unless otherwise specified in the detail specification, five specimens from each test unit shall be tested

5.2 Procedure A.

5.2.1 The hydrostatic pressure resistance of the test unit shall be the average of the values obtained from the specimens tested

5.2.2 The hydrostatic pressure resistance of the test unit shall be recorded to the nearest one pound per square inch

5.3 Procedure B.

5.3.1 The number of specimens tested from each test unit and the number from each test unit that showed leakage of the coated fabric or seams shall be recorded

5.3.2 The pressure applied to the specimen shall be recorded.

5.4 The side of the specimen, coated or uncoated, placed in contact with the water shall be recorded

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HARD RUBBER, GENERAL

1. SCOPE

1.1 This group of methods is intended for use in determining the various characteristics of hard rubber compounds and products

1.2 For the purpose of these methods, a hard rubber compound is a vulcanized rubber compound having a ratio of combined sulfur to rubber hydrocarbon in excess of 15 percent

2. SPECIMEN

2.1 The specimen shall be as described in the method of test used for determining the specified characteristic of the material Unless otherwise specified in the detail specification, the specimen shall be cut from the finished material to be tested

3. CONDITIONING

3.1 The conditioning of the specimen shall be as described in section 7, except that the time of conditioning before testing shall be not less than 16 hours

4. PROCEDURE

4.1 When tensile strength and elongation are specified in the detail specification, these tests shall be made on the same specimen during a single test

4.2 When flexural strength and deflection are specified in the detail specification, these tests shall be made on the same specimen during a single test

4.3 The number of specimens required for determining one of the properties in either 4 1 or 4 2 shall be used for determining the other property

4.4 Preparation of specimen.

4.4.1 The specimen shall be cut, sawed, machined, filed, or ground to the required dimension and shape The specimen shall have smooth surfaces free from nicks, scratches, grooves, or ragged edges If the material has been calendered in processing, the specimen shall be cut, whenever practicable, so that its longitudinal axis is in the direction of the calender grain

4.4.2 If the material is too thick or has an uneven surface that may interfere with the test, the material shall be machined, filed, or ground to the desired thickness The final machined, filed, or ground surface shall be finished uniformly and smoothly

4.4.3 Care shall be taken that the material is not overheated during the preparation The specimen shall be permitted to rest for not less than 16 hours between preparation and testing

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TENSILE STRENGTH, HARD RUBBER

1. SCOPE

1.1 This method is intended for use in determining the tensile strength of hard rubber compounds The definition of tensile strength is given in method 4001 and section 5

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit of the shape and dimensions shown in figure 11011 The width of the reduced section shall be uniform to within ±0 005 inch, 4 2 of method 2111

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed 0 130 inch, and shall be uniform to within ±0 005 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Testing machine* —A testing machine which meets the following requirements

3.1.1.1 Power driven

3.1.1.2 The applied tension is indicated to within ±1 percent by a dial, scale, or automatic recorder when calibrated according to 4 1

3.1.1.3 The indicator remains at the point of maximum force after rupture of the specimen

3.1.1.4 The rate of travel of the power-actuated grip is such that the average rate of force application is 3 0±0 1 pound per second when measured with a nonextensible steel bar locked in the grips

3.1.1.5 The machine, when used for a given specimen, is of such capacity that the maximum force required to break the specimen is not greater than 85 percent nor less than 15 percent of the rated capacity

3.1.1.6 The grips for holding the specimen in the testing machine are of a type which will tighten automatically as the applied tension increases, such as wedge grips which will exert a uniform pressure across the gripping surfaces so as to avoid slipping and to favor failure of the specimen in its reduced section, or of a type which may be tightened by screw pressure The gripping surfaces are knurled to prevent slipping of the specimen

3.1.1.6.1 The grips are of the self-aligning type, that is, they are attached to the head and moving member, respectively, so that they will move freely into alignment as soon as the force is applied and the long axis of the specimen will coincide with the direction of the applied pull through the center line of the grip assembly

4. PROCEDURE

4.1 Calibration of testing machine.—The machine shall be calibrated by means of dead weights applied in an ascending order while the machine is otherwise arranged in the manner normally used in the testing of hard rubber In each case when an additional weight is added, the indicator shall be moved to the zero position, and the indicator allowed to slowly move to the point where it will go no farther The dial, scale, or automatic recorder should then correctly indicate the amount of weight applied

4.2 Preparation of specimen.—The specimen shall be prepared as described in 4 4 of method 11001

4.3 Cross-sectional area of specimen.

4.3.1 *Thickness* —The specimen shall be free from mechanical damage The minimum thickness of the specimen in the reduced section shall be determined as described in method 2011

4.3.2 *Width* —The minimum width of the specimen in the reduced section shall be determined as described in 4 2 of method 2111.

4 3 3 The cross-sectional area, C, of the specimen shall be calculated by multiplying the minimum thickness by the minimum width of the specimen

4.4 Determination of tensile strength.—The specimen shall be placed in the grips of the machine and adjusted symmetrically so that the tension will be distributed uniformly over the cross section If the tension is greater on one side of the specimen than on the other, the maximum strength of the specimen will not be developed The machine shall be started and

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the power-actuated grip shall travel at a speed so that the average rate of application of the force shall be as required in 3 1 1 4 until the specimen ruptures After rupture of the specimen, the breaking force in pounds shall be read from the dial, scale, or automatic recorder and the value recorded as F

5. RESULTS

5.1 Calculation.—The tensile strength of the specimen shall be calculated as follows:

$$\text{Tensile strength, pounds per square inch,} = \frac{F}{C}$$

Where

F is the breaking force, pounds

C is the cross-sectional area of the unstretched specimen, square inch

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.3 The tensile strength of the test unit shall be the average of the values obtained from the specimens tested

5.4 The tensile strength of the test unit shall be recorded to the nearest 20 pounds per square inch

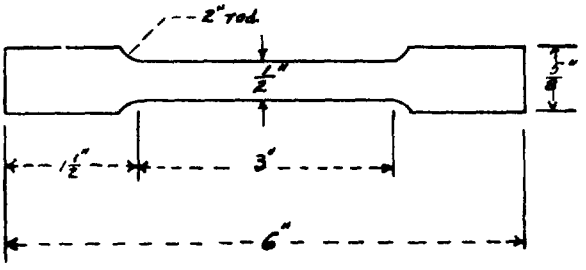

FIGURE 11011 —Specimen for tension test

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ELONGATION, HARD RUBBER

1. SCOPE

1.1 This method is intended for use in determining the elongation of hard rubber compounds The definition of elongation is given in method 4001 and section 5

2. SPECIMEN

2.1 The specimen shall be as described in method 11011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in method 11011

3.1.2 A bench marker as described in method 4121, except that the distance between the knife edges shall be 3 000±0 003 inch, or other equipment such as sharp-pointed dividers for marking the specimen as required in 4 2

3.1.3 The stamp pad described in method 4121 for use with the bench marker

3.1.4 A pair of dividers in which the distance between the points can be accurately adjusted to within 0 01 inch

3.1.5 A scale or other device graduated to 0 01 inch or finer

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the elongation shall be determined on the same specimen during the same test as tensile strength

4.2 Preparation of specimen.—The specimen shall be prepared as described in 4 4 of method 11001 and shall be free from mechanical damage In addition, two parallel marks 3 000±0 003 inch apart shall be placed symmetrically on the reduced section of the specimen perpendicular to the longitudinal axis by means of the bench marker and ink, or a sharp instrument such as a pair of dividers The specimen shall not be injured and the marks shall be as fine as practicable, consistent with legibility

4.3 If the benchmarker is used to make the marks on the specimen, the distance between the knife edges of the marker shall be taken as the distance between the marks on the unstretched specimen If the marks are made on the specimen by other means, the distance between the marks on the specimen shall be recorded to the nearest 0 01 inch as D.

4.4 The specimen shall be placed in the grips of the testing machine, the machine started, and the force applied as described for tensile strength As the power-actuated grip moves, the distance between the two marks shall be followed continuously by means of the dividers or other device which shall be used in such a manner as to avoid injuring the specimen The distance between the marks when the specimen ruptures shall be recorded to the nearest 0 01 inch as D_1 Dividers, if used, shall be placed on the scale and the distance between the pointers read from the scale

5. RESULTS

5.1 Calculation.—The elongation of the specimen shall be calculated as follows

$$\text{Elongation, percent} = \frac{D_1 - D}{D} \times 100$$

Where

D_1 is the distance between the marks on the specimen at the moment of rupture, inches

D is the distance between the knife edges of the bench marker or between marks on the unstretched specimen, inches

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.3 The elongation of the test unit shall be the average of the values obtained from the specimens tested

5.4 The elongation of the test unit shall be recorded to the nearest 0 1 percent

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FLEXURAL STRENGTH, HARD RUBBER

1. SCOPE

1.1 This method is intended for use in determining the flexural strength of hard rubber compounds

1.2 For the purpose of this method, flexural strength is the transverse force expressed as fiber stress required to rupture a simple beam of specified dimensions when subjected to a transverse force

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 0 875 inch in width and uniform to within ±0 005 inch, 4 2 of method 2111

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed ½ inch, and shall be uniform to within ±0 005 inch, method 2011

2.1.2 The length of the specimen shall be at least 16 times the thickness of the specimen plus 2 inches

3. APPARATUS

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3.1 The apparatus shall be as follows

3.1.1 Testing machine described in method 11011, except that the grips shall be replaced with the apparatus described in 3 1 2

3.1.2 An apparatus consisting of two hard metal supports for supporting the specimen as a simple beam and a hard metal loading bar through which force is applied to the specimen midway between the supports The supports and bar are longer than the width of the specimen and have bearing surfaces rounded to a radius of not less than 0 125 and not more than 0.5 inch The supports are adjustable so that the distance between the lines of contact with the specimen can be adjusted to 16 times the thickness of the specimen within ±1/64 inch The supports replace one grip of the machine described in 3 1 1 and the loading bar replaces the other The supports and bar are mounted parallel to each other in the machine and in such a manner that a force can be applied to the specimen midway between the supports when the specimen rests on the supports The total elastic deformation of the system should not exceed one percent of the total deflection of the specimen

3.1.3 A scale or tape graduated to 1/64 inch

4. PROCEDURE

4.1 Preparation of specimen.—The specimen shall be prepared as described in 4 4 of method 11001

4.2 The specimen shall be free from mechanical damage The minimum width of the part of the specimen between the supports shall be determined as described in 4 2 of method 2111 and the value recorded as b The minimum thickness of the part of the specimen between the supports shall be determined as described in method 2011 and the value recorded as d

4.3 By means of the scale or tape, the distance between the lines of support of the specimen shall be adjusted to 16 times the thickness of the specimen and the value recorded to the nearest 1/64 inch as L The specimen shall be centered in the testing machine with one of the principal faces on the supports The loading bar shall be placed in position so that the force will be applied to the specimen midway between the lines of support of the specimen. The machine shall be started and the force applied through the loading bar perpendicular to the specimen and midway between the lines of support so that the average rate of application of the force shall be as required in 3 1 1 4 of method 11011 until the specimen ruptures After rupture of the specimen, the force in pounds shall be read from the dial, scale, or automatic recorder and the value recorded as P

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5. RESULTS

5.1 Calculation.—The flexural strength of the specimen shall be calculated as follows

Flexural strength, pounds per square inch, $= \frac{3PL}{2bd^2}$

Where

P is the force required to rupture the specimen, pounds
L is the distance between lines of support of the specimen, inches
b is the minimum width of the specimen, inch
d is the minimum thickness of the specimen, inch

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested.

5.3 The flexural strength of the test unit shall be the average of the values obtained from the specimens tested

5.4 The flexural strength of the test unit shall be recorded to the nearest 20 pounds per square inch

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DEFLECTION, HARD RUBBER

1. SCOPE

1.1 This method is intended for use in determining the deflection of hard rubber compounds

2. SPECIMEN

2.1 The specimen shall be as described in method 11041

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 The apparatus described in method 11041

3.1.2 Extensometer or dial scale, graduated to read to 0 01 inch, for determining the deflection.

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the deflection shall be determined on the same specimen during the same test as flexural strength

4.2 Preparation of specimen.—The specimen shall be prepared as described in 4.4 of method 11001

4.3 The specimen shall be free from mechanical damage The minimum width of the part of the specimen between the supports shall be determined as described in 4 2 of method 2111 and the value recorded as b The minimum thickness of the part of the specimen between the supports shall be determined as described in method 2011 and the value recorded as d

4.4 By means of the scale or tape, the distance between the lines of support of the specimen shall be adjusted to 16 times the thickness of the specimen and the value recorded to the nearest $\frac{1}{64}$ inch as L The specimen shall be centered in the testing machine with one of the principal faces of the specimen on the supports The loading bar shall be placed in position so that the force will be applied to the specimen midway between the lines of support of the specimen The machine shall be started and the force applied through the loading bar perpendicular to the specimen and midway between the lines of support so that the average rate of application of the force shall be as required in 3 1 1 4 of method 11011 until the specimen ruptures The amount of deflection at the center of the span shall be followed continuously by means of an extensometer or dial scale At the moment of rupture of the specimen, the deflection shall be read from the extensometer or dial scale and the value recorded to the nearest 0 01 inch

5. RESULTS

5.1 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.2 The deflection of the test unit shall be the average of the values obtained from the specimens tested

5.3 The deflection of the test unit shall be recorded to the nearest 0 01 inch

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Method 11121
April 12, 1955

This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

COLD FLOW, HARD RUBBER

1. SCOPE

1.1 This method is intended for use in determining the cold flow of hard rubber compounds, that is, the change in thickness caused by the application of a known force at a specified temperature for a specified time The principle is that of the parallel plate plastometer in which a specimen is placed between the parallel plates of a constant force device and the thickness observed after a specified time at a specified temperature

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 0 500 by 0 500 by 0 500 inch and all dimensions shall be uniform within ±0 005 inch, method 2011

2.2 Unless otherwise specified in the detail specification, materials thinner than ½ inch shall be plied up with thin pieces of uniform thickness, accurately aligned, to obtain the required thickness, the surfaces of the pieces shall be in contact throughout

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 The compression apparatus described in 3 1 1 of method 3321, except that the machine shall be capable of producing on the specimen the force required in 4 1

3.1.2 A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the exposure period and of sufficient size to hold the compression apparatus

3.1.3 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within ±1 0° C (1 8° F)

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be subjected to a force of 1,000±10 pounds at a temperature of 50°±2° C (122°±3 6° F) for 22±¼ hour

4.2 Preparation of specimen.—The specimen shall be prepared as described in 4.4 of method 11001

4.3 The specimen shall be free from mechanical damage The specimen shall be mounted between the anvils of the testing machine The required force, 4 1, shall be applied to the specimen, the distance between the anvils read immediately from the dial or scale, and the value recorded to the nearest 0 001 inch as the thickness of the specimen, T_1

4.4 The testing machine with the specimen still under the applied force shall be placed in the air oven and maintained for the required time at the required temperature At the end of the heating period, with the specimen still under the required force, the distance between the anvils of the testing machine, shall be read immediately from the dial or scale and the value recorded to the nearest 0 001 inch as the thickness of the specimen, T_2

5. RESULTS

5.1 Calculation.—The cold flow of the specimen shall be calculated as follows

$$\text{Cold flow, percent} = \frac{T_1 - T_2}{T_1} \times 100$$

Where
T_1 is the thickness of the specimen immediately after application of the force, inch
T_2 is the thickness of the specimen at the end of the heating period and while still under the force, inch

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The cold flow of the test unit shall be the average of the values obtained from the specimens tested

5.4 The cold flow of the test unit shall be recorded to the nearest one percent

5.5 The time and temperature of exposure and the force applied shall be recorded

5.6 If a plied-up specimen is tested, it shall be recorded

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IMPACT RESISTANCE, HARD RUBBER, GENERAL

1. SCOPE

1.1 These methods are intended for use in determining the relative susceptibility of hard rubber compounds to fracture by shock as indicated by the energy expended by an impact machine in breaking a specimen in a single blow

2. METHODS

2.1 Two types of equipment are used for determining impact resistance, namely, pendulum-type machines and equipment for dropping a ball Pendulum machines are of two kinds, the cantilever beam and the simple beam machines

2.2 **Cantilever beam or Izod method.**—In this method the specimen is held as a cantilever beam in a vertical position and is broken by a single blow delivered at a fixed distance from the edge of the specimen holder A notched specimen is used so as to produce a standard degree of stress concentration

2.3 **Simple beam or Charpy method.**—In this method the specimen is supported at both ends in a horizontal position as a simple beam The specimen is broken by a single blow delivered midway between the two supports

2.4 **Ball methods.**—These methods measure the resistance of hard rubber compounds to the impact caused by a falling ball A spherical steel ball is dropped by an electromagnet or by other means The specimen may be the same as that used in the pendulum-type machines or of some other suitable shape and size depending upon the material to be tested The specimen is held in a horizontal position and may be supported either at one or both ends

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IMPACT RESISTANCE, HARD RUBBER, CANTILEVER

1. SCOPE

1.1 This method is intended for use in determining the impact resistance of hard rubber compounds The specimen, which is always notched, is supported in a vertical position as a cantilever beam and broken by a blow delivered from a fixed distance This method is also known as the Izod impact test.

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit of the shape and dimensions shown in figure 11221B The dimensions of the specimen shall be uniform to within the tolerances shown in the figure, method 2111

2.2 Unless otherwise specified in the detail specification, materials thinner than ½ inch shall be plied up using pieces of uniform thickness, accurately aligned, to obtain the required thickness, the surfaces of the pieces shall be in contact throughout

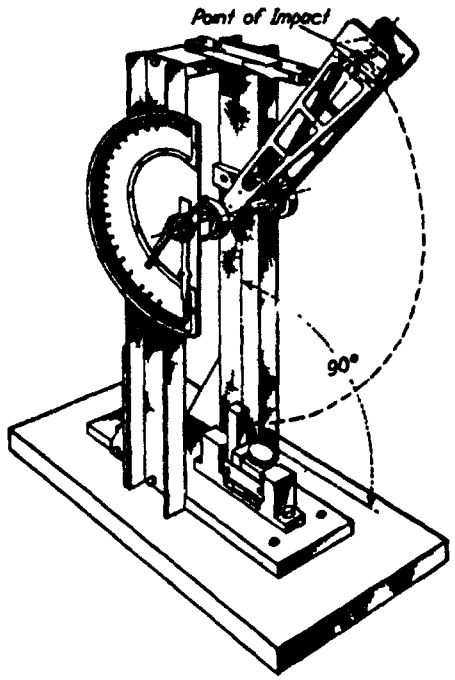

FIGURE 11221A —Cantilever beam impact machine

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A cantilever-beam impact machine as shown in figure 11221A

3.1.1.1 The machine is of the pendulum type It is of rigid construction and indicates the impact resistance to within ±0 01 foot-pounds for readings of less than 1 foot-pound and to within ±1 percent for higher readings A correction shall be made for friction and windage losses

3.1.1.2 *Dimensions* —The dimensions of the machine are such that the center of percussion of the striker is at the point of impact, that is, the center of the striking edge The distance from the axis of support to the center of percussion, L, is determined experimentally from the period of oscillation of the pendulum through a small angle by means of the expression $L = 0\ 81P^2$ where P is the time in seconds of one complete swing, to and fro

3.1.1.3 *Pendulum.*—The pendulum is released from such a position that the linear velocity of the center of the striking edge at the instant of impact is approximately 11 feet per second which corresponds to an initial elevation of 2 feet at this point

3.1.1.4 *Striking edge* —The striking edge of the pendulum is a circular cylindrical surface of 1/32-inch radius with its axis horizontal When the pendulum is hanging free, the cylindrical surface is tangent to the specimen in a line 0 866 inch above the top surface of the specimen clamp The pendulum above the

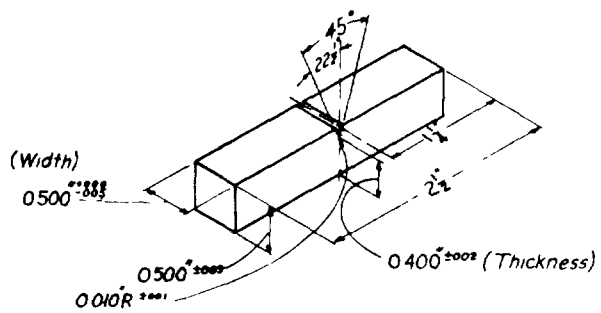

FIGURE 11221B —Cantilever beam specimen

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cylindrical portion of the striking edge is recessed or inclined at a suitable angle so that there is no chance of its coming in contact with the specimen during the break

3.1.1.5 *Specimen clamp*—The machine is equipped with a clamp for holding the specimen rigidly in a vertical position with the edges of the supporting surfaces at 90° angles

3.1.1.6 The machine is equipped with means for indicating the impact value of the specimen, that is, the energy expended by the machine in breaking the specimen After suitable corrections have been made, this value is equal to the difference between the energy in the pendulum blow and the energy remaining in the pendulum after breaking the specimen

3.1.2 *Notching equipment*—A milling cutter shall be provided for notching the specimen The cutting edges shall be sharp and free from nicks and burrs The included cutting angle of the tool is 45° ± 1° with a radius of 0 010 ± 0 001 inch at the intersection of the arms of this angle Tools with no rake and with a work-relief angle of 15° to 20° have been found satisfactory The tool is mounted so that the plane which bisects the included angle is perpendicular to the axis of rotation within ±2° The linear speed of the cutting tool shall not exceed 600 feet per minute The feed shall be slow enough to result in a smooth cut Speeds in the range of 0 5 to 5 inches per minute have been found satisfactory for some materials

3.1.3 A dial micrometer as described in method 2011 with the presser foot replaced by a foot having a V-shaped tip, the angle of which is less than 45°, so that it will penetrate to the apex of the notch in the specimen

4. PROCEDURE

4.1 Preparation of specimen.

4.1.1 The specimen shall be prepared as described in 4 4 of method 11001 with the exception of the direction of cutting which shall be as described in 4 1 2

4.1.2 Unless otherwise specified in the detail specification, specimens from sheet material shall be cut in both the lengthwise and crosswise directions If the specimen is cut from a thick sheet, the portion of the sheet from which the specimen is cut, that is, top, bottom, or center, shall be noted When the sheet from which the specimen is to be cut has the same length and width, one direction shall be arbitrarily designated as lengthwise and the other as crosswise Plied-up specimens shall be prepared by placing thin pieces together, with edges aligned smoothly, and shall be tested edgewise Before testing, the specimen shall be permitted to rest for not less than 16 hours after the material has been machined, filed, or ground

4.1.3 *Notching of specimen*

4.1.3.1 To insure correct contour of the specified notch, all specimens shall be notched with the special milling cutter The bottom of the notch shall be smooth, straight, and free from scratches

4.1.3.2 If the material is compression molded, the notch shall be cut on the side parallel to the direction of application of the molding pressure Specimens that have been prepared from materials ½ inch thick or thicker than ½ inch shall be notched either edgewise or flatwise as specified in the detail specification If the specimen is machined only on one side and if it is to be tested flatwise, the notch shall be made on the machined surface Specimens of plied-up materials shall be notched edgewise

4.2 The specimen shall be free from mechanical damage The length of the notch, that is, the width of the specimen at the notch, shall be determined as described in 4 2 of method 2111, except that one measurement shall be made at the notch, and the value recorded as L The thickness of the specimen below the notch shall be determined by means of the dial micrometer described in 3 1 3 The specimen shall be placed on the anvil of the micrometer, the presser foot lowered until the tip of the V-shaped foot rests on the apex of the 45° notch of the specimen, and the thickness read from the dial One measurement shall be made and the value recorded to the nearest 0.001 inch

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4.3 The specimen shall be clamped firmly in the testing machine with the center line of the notch on a level with the top of the clamping surfaces The specimen shall be placed so that the notched side will be struck by the pendulum A jig or template may be used to locate the specimen in the jaws of the clamp If a plied-up specimen is used, the individual pieces shall be held closely in contact, accurately aligned with each other, and firmly clamped in position so that the specimen will be struck edgewise The pendulum shall be released from the position described in 3 1 1 3 After rupture of the specimen, the force in foot-pounds shall be read from the indicating device and the value recorded as F

5. RESULTS

5.1. Calculation.—The impact resistance of the specimen shall be calculated as follows

$$\text{Impact resistance, foot-pounds per inch of notch length,} = \frac{F}{L}$$

where

F is the force required to rupture the specimen, foot-pounds

L is the length of notch in the specimen, inch

5.2 Unless otherwise specified in the detail specification, five specimens from each test unit shall be tested

5.3 The impact resistance of the test unit shall be the average of the values obtained from the specimens tested

5.4 The impact resistance of the test unit shall be recorded to the nearest 0 1 foot-pound per inch of notch length

5.5 The direction, edgewise or flatwise, in which the specimen was tested shall be recorded

5.6 If a plied-up specimen is tested, it shall be recorded

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IMPACT RESISTANCE, HARD RUBBER, SIMPLE BEAM

1. SCOPE

1.1 This method is intended for use in determining the impact resistance of hard-rubber compounds The specimen, which is always notched, is supported in a horizontal position as a simple beam and is broken by a blow delivered from a fixed distance midway between the supports This method is also known as the Charpy impact test

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit of the shape and dimensions shown in figure 11231B The dimensions of the specimen shall be uniform to within the tolerances shown in the figure, method 2111

2.2 Unless otherwise specified in the detail specification, materials thinner than ½ inch shall be plied up with thin pieces of uniform thickness, accurately aligned, to obtain the required thickness, the surfaces of the pieces shall be in contact throughout

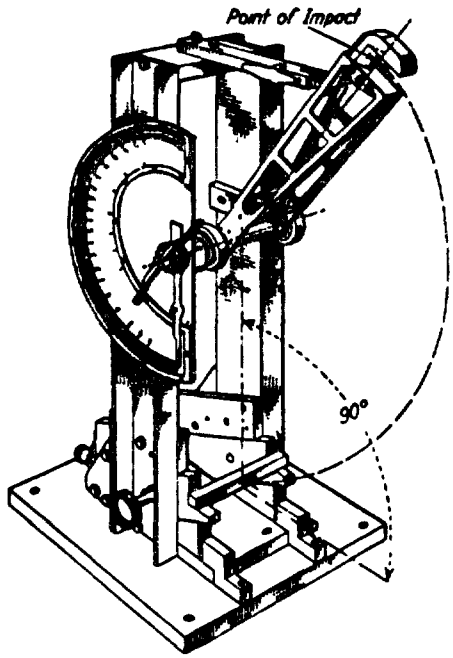

FIGURE 11231A —Simple beam impact machine

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A simple beam impact machine as shown in figure 11231A

3.1.1.1 The machine is of the pendulum type It is of rigid construction and indicates the impact resistance to within ±0 01 foot-pound for readings of less than 1 foot-pound and to within ±1 percent for higher readings Corrections shall be made for friction and windage losses

3.1.1.2 *Dimensions* —The dimensions of the machine are such that the center of percussion of the striker is at the point of impact, that is, the center of the striking edge The distance from the axis of support to the center of percussion shall be determined as described in 3 1 1 2 of method 11221

3.1.1.3 *Pendulum* —The pendulum is released from such a position that the linear velocity of the center of the striking edge at the instant of impact is approximately 11 feet per second which corresponds to an initial elevation of 2 feet at this point

3.1.1.4 *Striking edge* —The striking edge of the pendulum is tapered to have an included angle of 45 degrees and is rounded to a radius of 0 125±0 001 inch and the points of support shall be 4±0 05 inch apart

3.1.1.5 *Specimen support* —The machine is equipped with two steel blocks for supporting the specimen in such a position that the center of gravity of the specimen lies on a tangent to the arc of travel of the center of percussion of the pendulum drawn at the position of impact The edges of the blocks are rounded to a radius of 0 125±0 001 inch and the points of support shall be 4±0 05 inch apart

3.1.1.6 The machine is equipped with means for indicating the impact value of the specimen, that is, the energy expended by the machine in breaking the specimen After suitable corrections have been made for friction and windage, this value is equal to the difference between the energy in the pendulum blow and the energy

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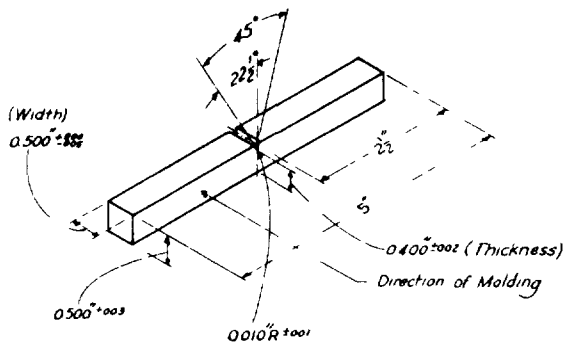

FIGURE 11231B —Simple beam specimen.

remaining in the pendulum after breaking the specimen

4. PROCEDURE

4.1 Preparation of specimen.

4.1.1 The specimen shall be prepared as described in 4.4 of method 11001 with the exception of the direction of cutting which shall be as described in 4 1 2

4.1.2 Unless otherwise specified in the detail specification, specimens from sheet material shall be cut in both the lengthwise and crosswise directions If the specimen is cut from a thick sheet, the portion of the sheet from which the specimen is cut, that is, top, bottom, or center, shall be noted When the sheet from which the specimen is to be cut has the same length and width, one direction shall be arbitrarily designated as lengthwise and the other as crosswise Plied-up specimens shall be prepared by placing thin pieces together with the edges aligned smoothly and shall be tested edgewise Before testing, the specimen shall be permitted to rest for not less than 16 hours after the material has been machined, filed, or ground

4.1.3 The specimen shall be notched as described in 4 1 3 of method 11221

4.2 The specimen shall be free from mechanical damage The length of the notch, that is, the width of the specimen at the notch, shall be determined as described in 4 2 of method 2111, except that one measurement shall be made at the notch and the value recorded as L The thickness of the specimen below the notch shall be determined as described in 4 2 of method 11221

4.3 The specimen shall be fastened against the two blocks on the testing machine so that the pendulum will strike the center of the specimen The specimen shall be placed so that the notch will be on the opposite side from the blow of the pendulum If a plied-up specimen is used, the individual pieces shall be held closely in contact, accurately aligned with each other, and tightly clamped in position so that the specimen will be struck edgewise The pendulum shall be released from the position described in 3 1 1 3 After rupture of the specimen, the force in foot-pounds shall be read from the indicating device and the value recorded as F

5. RESULTS

5.1 Calculation.—The impact resistance of the specimen shall be calculated as follows

$$\text{Impact resistance, foot-pounds per inch of notch length, } = \frac{F}{L}$$

Where

F is the force required to rupture the specimen foot-pounds

L is the length of the notch in the specimen, inch

5.2 Unless otherwise specified in the detail specification, five specimens from each test unit shall be tested

5.3 The impact resistance of the test unit shall be the average of the values obtained from the specimens tested

5.4 The impact resistance of the test unit shall be recorded to the nearest 0 1 foot-pound per inch of notch length

5.5 The direction, edgewise or flatwise, in which the specimen was tested shall be recorded

5.6 If a plied-up specimen is tested, it shall be recorded

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IMPACT RESISTANCE, HARD RUBBER, VARIABLE SIZE BALL

1. SCOPE

1.1 This method is intended for use in determining the impact resistance of hard rubber compounds This method employs the specimen used in method 11221 The ball is released by demagnetizing an electromagnet Varying size balls are dropped from a specified height

2. SPECIMEN

2.1 The specimen shall be as described in method 11221

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A drop-ball testing apparatus consisting of an electromagnet and specimen holder

3.1.1.1 *Electromagnet* —An electromagnet which when demagnetized will release a steel ball instantly The ball hits the specimen at a distance of 0 866 ± 0 001 inch from the line of support The electromagnet is provided with a means for centering the ball on the magnet and is mounted so that it can be moved up or down and can be adjusted to keep the height of drop constant for any size ball A suitable apparatus is shown in figure 11241A

3.1.1.2 *Specimen holder* —A specimen holder as shown in figure 11241B that can be raised or lowered so that the specified height of drop can be varied between 12 and 36 inches in 1-inch increments The specimen is clamped so that the point at which it is struck by the falling ball is 0.866 ± 0 001 inch from the edge of the clamp

3.1.2 *Steel balls* —A set of steel balls of the following diameters and weights The diameters shall be within ±0 0001 inch

Diameter	Weight	Diameter	Weight
Inches	*Pounds*	*Inches*	*Pounds*
1	0 148	1 15/16	1 07
1 1/16	183	2	1 17
1 1/8	214	2 1/16	1 30
1 3/16	250	2 1/8	1 41
1 1/4	287	2 3/16	1 55
1 5/16	332	2 1/4	1 68
1 3/8	382	2 5/16	1 82
1 7/16	436	2 3/8	1 97
1 1/2	497	2 7/16	2 14
1 9/16	554	2 1/2	2 30
1 5/8	630	2 3/4	3 05
1 11/16	707	3	3 95
1 3/4	787	3 1/4	5 06
1 13/16	875	3 1/2	6 34
1 7/8	990	3 3/4	7 81

3.1.3 A means for guiding the falling ball The ball may be confined within but should not contact the guide.

4. PROCEDURE

4.1 The size of the ball used for the initial drop and the height of the drop shall be as specified in the detail specification

4.2 Preparation of specimen.

4.2.1 The specimen shall be prepared as described in 4 4 of method 11001 with the exception of the direction of cutting which shall be as described in 4 2 2

4.2.2 Unless otherwise specified in the detail specification, specimens from sheet material shall be cut in both the lengthwise and crosswise directions If the specimen is cut from a thick sheet, the portion of the sheet from which the specimen is cut, that is top, bottom, or center, shall be noted When the sheet from which the specimen is to be cut has the same length and width, one direction shall be arbitrarily

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FIGURE 11241A —Drop-ball impact machine

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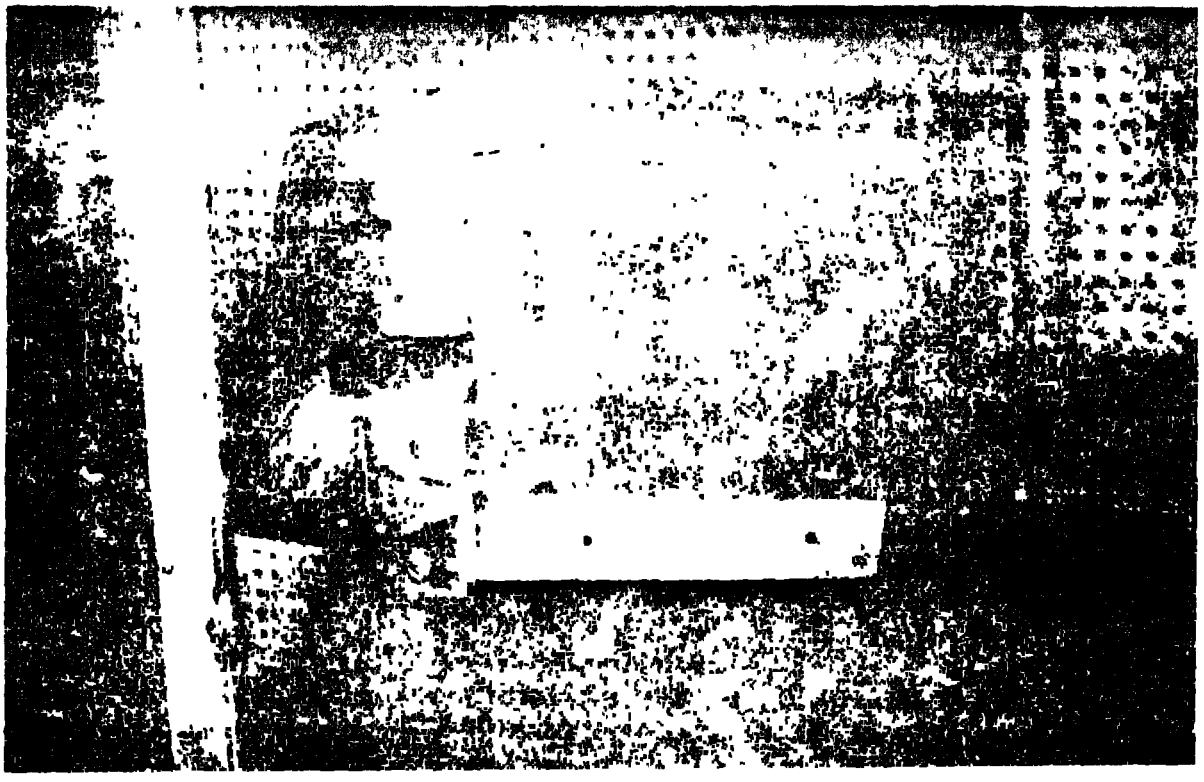

FIGURE 112-1B —Specimen holder for ball-drop machine

designated as lengthwise and the other as crosswise Plied-up specimens shall be prepared by placing thin pieces together with the edges aligned smoothly and shall be tested edgewise Before testing, the specimen shall be permitted to rest for not less than 16 hours after the material has been machined, filed, or ground

4.2.3 The specimen shall be notched as described in 4 1 3 of method 11221

4.3 The specimen shall be free from mechanical damage The length of the notch, that is, the width of the specimen at the notch, shall be determined as described in 4 2 of method 2111, except that one measurement shall be made at the notch, and the value recorded as L The thickness of the specimen below the notch shall be determined as described in 4 2 of method 11221

4.4 The specimen shall be clamped in the apparatus with the center line of the notch on a level with the edges of the clamping surfaces so that the notched side will be struck by the falling ball at a distance of 0.866 ± 0.001 inch from the edge of the clamp A ball of the required size shall be dropped once from the required height, 4 1, by breaking the circuit of the electromagnet The height of drop shall remain constant and the size of the ball shall be increased by increments shown in 3 1 2 until the specimen is broken, dropping each size ball only once The weight of the ball used to rupture the specimen shall be recorded

5. RESULTS

5.1 Unless otherwise specified in the detail specification, five specimens from each test unit shall be tested

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5.2 The impact resistance of the test unit shall be the average of the values obtained from the specimens tested

5.3 The impact resistance of the test unit shall be recorded to the nearest 0 01 pound

5.4 If a plied-up specimen is tested, it shall be recorded

5.5 The direction, edgewise or flatwise, in which the specimen was tested shall be recorded

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IMPACT RESISTANCE, HARD RUBBER, FIXED SIZE BALL

1. SCOPE

1.1 This method is intended for use in determining the impact resistance of hard rubber compounds This method is similar to method 11241 except that the specimen is larger and is supported on both ends, the size of the ball is fixed, and the height of drop of the ball is variable

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 4 inches in length The width shall be 2 00 inches and shall be uniform to within ±0 01 inch, method 2121

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, and shall be uniform to within ±0 005 inch Specimens from the same test unit shall be of the same thickness within ±0 005 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Electromagnet* —An electromagnet which when demagnetized will release a steel ball instantly The magnet is mounted in a frame so that it can be moved vertically in increments of one inch

3.1.2 A support for the specimen as shown in figure 11251

3.1.3 A means for guiding the falling ball and adjusting the magnet The ball may be confined within but should not contact the guide

3.1.4 A spherical steel ball weighing 2 30 ±0 02 pound, approximately 2 5 inches in diameter

3.1.5 A scale or tape graduated to 1/16 inch or finer, or other equipment for measuring the height of drop of the ball

4. PROCEDURE

4.1 The specimen shall be prepared as described in 4 4 of method 11001

4.2 The specimen shall be free from mechanical damage The specimen shall be placed symmetrically on the supports in a horizontal position so that the falling ball will strike the specimen midway between the lines of support The ball shall be adjusted to a distance of 1 ± 1/16 inch above the center of the specimen and then released by breaking the electric circuit of the electromagnet This procedure shall be repeated increasing the height of the ball by 1 ± 1/16 inch each time until the specimen ruptures The ball shall be prevented from striking the specimen more than once on each fall by catching it on the rebound

4.3 The above procedure shall be repeated on each specimen tested except that if the first specimen breaks at 15 inches or higher, the other specimens shall be started at a height equal to 1/2 the height required to rupture the first specimen The height of drop of the ball required to rupture the specimen shall be measured with the scale or tape and the value recorded to the nearest 1/16 inch as the impact resistance of the specimen

5. RESULTS

5.1 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.2 The impact resistance of the test unit shall be the average of the values obtained from the specimens tested

5.3 The impact resistance of the test unit shall be recorded to the nearest one inch

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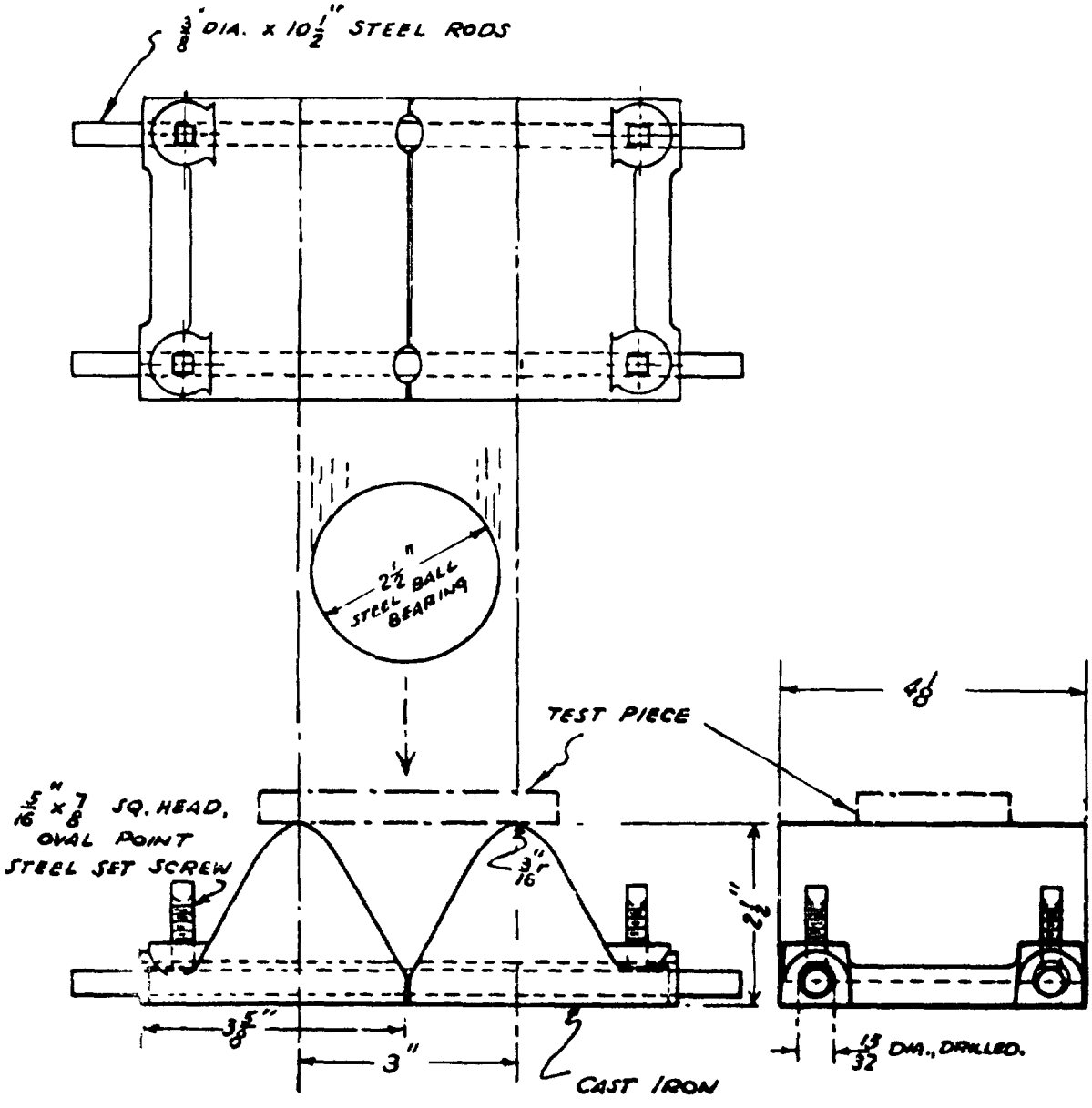

FIGURE 11251 —Specimen support

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IMPACT RESISTANCE, BATTERY CONTAINERS

1. SCOPE

1.1 This method is intended for use in determining the impact resistance of battery jar walls The method is similar to method 11241, except that the specimen is larger and is supported on both ends, the size of the ball is fixed, and the height of drop of the ball may be varied.

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 12 by 12 inches

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A specimen support consisting of a hardwood frame mounted on a ½-inch steel plate as shown in figure 11261

3.1.2 A steel ball weighing 3 00 ± 0.02 pound.

3.1.3 A scale or tape graduated to 1/16 inch or finer

3.1.4 A means for holding the ball above the center of the specimen and releasing it from the required height

3.1.5 A means for guiding the falling ball The ball may be confined within but should not contact the guide

4. PROCEDURE

4.1 **Preparation of specimen.**—The specimen shall be cut, sawed, machined, filed, or ground to the required dimension and shape Care shall be taken that the material is not overheated during the preparation The specimen shall be permitted to rest for not less than 16 hours between preparation and testing

4.2 **Specified height.**

4.2.1 The height of drop of the ball shall be as specified in the detail specification

4.2.2 The specimen shall be free from mechanical damage The specimen shall be placed symmetrically on the support in a horizontal position The steel ball shall be dropped one time from the specified height so that it will strike the specimen approximately at the center of one of the principal faces The specimen shall be examined on both sides for cracks or breaks

4.3 **Height required to break specimen.**

4.3.1 The specimen shall be free from mechanical damage The specimen shall be placed symmetrically on the support in a horizontal position The steel ball shall be dropped from a height of 24 ± 1/16 inch so that it will strike the specimen approximately at the center of one of the principal faces The specimen shall be examined on both sides for cracks and breaks If the specimen does not crack or break, this procedure shall be repeated, increasing the height of drop by 6 ± 1/16 inch each time until the specimen breaks or cracks on either side The ball shall be prevented from striking the specimen more than once on the same drop by catching it on the rebound The height of drop at which the specimen broke or cracked shall be measured with the scale or tape and the value recorded to the nearest 1/16 inch as impact resistance of the specimen

5. RESULTS

5.1 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.2 **Specified height.**

5.2.1 The number of specimens tested from each test unit and the number from each test unit that cracked or broke shall be recorded

5.3 **Height required to break specimen.**

5.3.1 The impact resistance of the test unit shall be the average of the values obtained from the specimens tested

5.3.2 The impact resistance of the test unit shall be recorded to the nearest one inch

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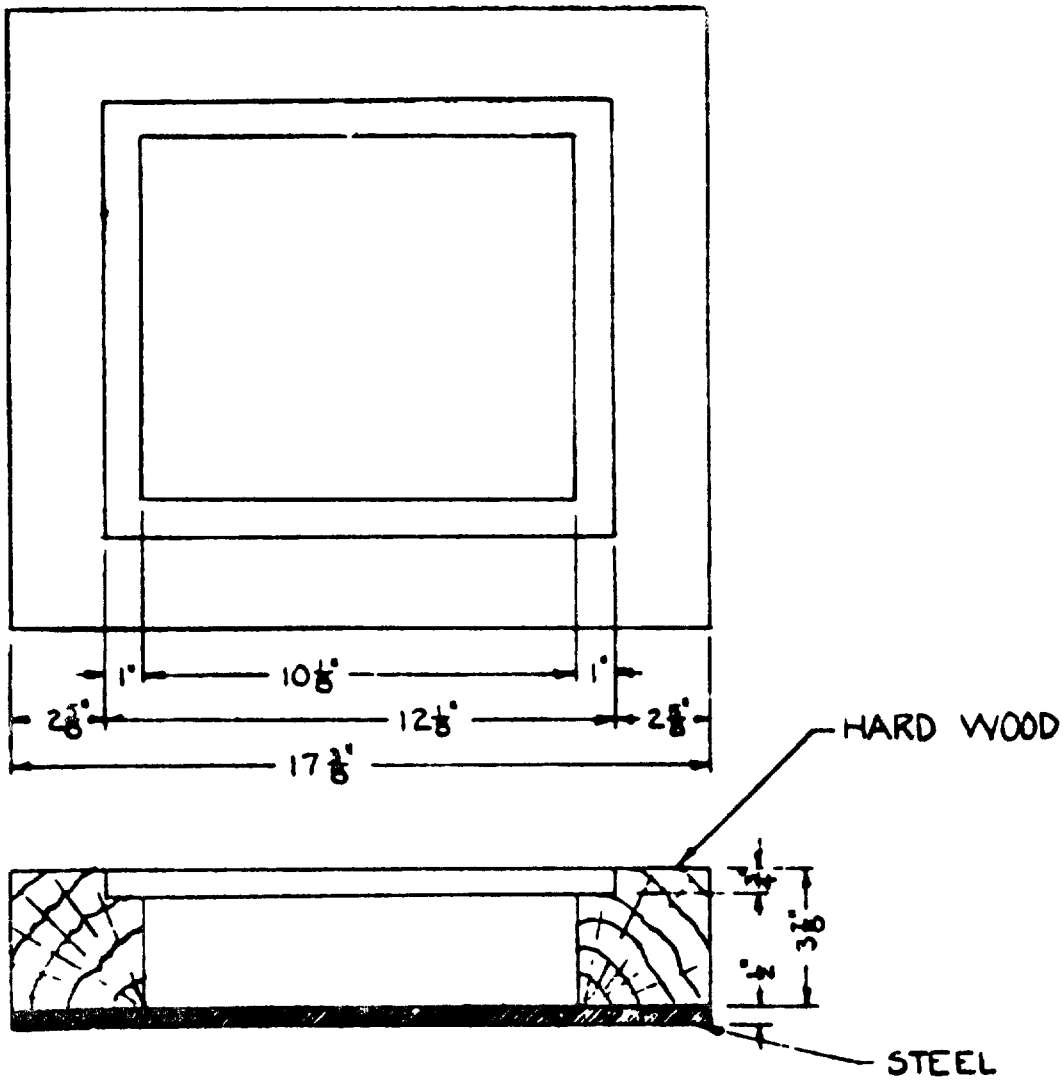

FIGURE 11261 —Specimen support

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SOFTENING POINT, HARD RUBBER

1. SCOPE

1.1 This method is intended for use in determining the softening point of hard rubber compounds The temperature at which deformation occurs is determined when the material is subjected to specified testing conditions The method is applicable to the testing of hard, heat-resisting rubber compounds and is not considered suitable for semihard materials

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 5 inches in length The width shall be 0 500 inch and shall be uniform to within ±0 005 inch, 4 2 of method 2111

2.1.1 The thickness of the specimen shall be 0 187 inch and shall be uniform to within ±0 005 inch, method 2011

2.2 Unless otherwise specified in the detail specification, materials thinner than 0 187 inch shall be plied up using pieces of uniform width and thickness, accurately aligned, to obtain the required thickness

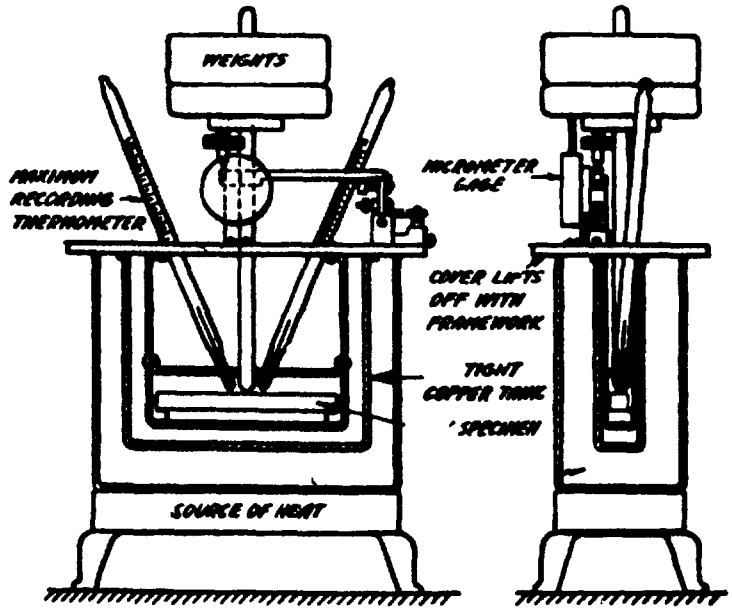
3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A softening point apparatus consisting essentially of the following parts A suitable apparatus is shown in figure 11311

3.1.1.1 An insulated metal tank for holding the liquid heat-transfer medium

3.1.1.2 *Specimen holder* —The specimen holder consists of a metal frame on which are mounted specimen supports 4 00 ± 0 05 inch apart, a loading rod through which force is applied on top of the specimen vertically and mid-



NOTES

1 Friction of the weighted shaft has been greatly decreased by guiding the shaft between two pairs of roller bearings. The bearings operate in two diametrically opposite grooves machined in the shaft
2. Thermal expansion errors have been minimized by the use of Invar steel for construction of the weighted shaft and all supporting members
3 The shaft and structural members were made larger to provide the required rigidity
4. The width of the contact shoulder on the shaft has been increased to ½ inch so as to bear on the full width of specimen
5 Additional clearance has been provided between the ends of the specimens and the vertical supports to facilitate manipulation of the specimens.

FIGURE 11311 —Apparatus for softening point

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way between the lines of support by means of weights, and a dial gage or other device for measuring the deflection of the specimen to within 0 0005 inch The contact surfaces of the specimen supports and the loading rod are rounded to a radius of ⅛ inch The framework is attached to the metal cover plate which fits over the insulated tank, 3 1 1 1 Provisions shall be made for alignment of the loading rod Means shall be provided for lifting the rod and weights out of contact with the specimen and holding them in a raised position A hole in the top of the specimen holder permits inserting a thermometer so that the bulb is located about ⅛ inch from the specimen near the midpoint

3.1.1.2.1 The vertical members of the frame that connect the specimen supports with the cover plate and the loading rod are of a material having the same coefficient of thermal expansion These parts must have the same coefficient of expansion so as to avoid a differential change in length with change in temperature that would introduce an error in the reading of the apparent deformation of the specimen The coefficient of expansion of the materials in the holder shall be determined over the temperature range to be used, by substituting for the specimen a test bar having a low coefficient of expansion, such as Invar or glass If the deflection is 0 0005 inch or greater at any temperature, a correction factor shall be applied to each test by adding the value algebraically to the reading of the apparent deflection of the specimen at [illegible] temperature

3.1.1.3 *[illegible]ghts* —A set of weights sufficient to bring the total force imposed on the specimen to 5 5±0 1 pounds when placed on the loading [illegible] The total force of 5 5 pounds includes the weight of the loading rod and any force exerted by the spring on the dial gage or other device In some designs of this apparatus, the force of the dial gage spring is directed upward as it must be subtracted from the load, while in others the force of the spring is downward and must be added The force exerted by the spring in certain dial gages varies over the stroke, therefore, this force should be measured in that part of the stroke which is to be used

3.1.1.4 Equipment with suitable controls for increasing the temperature of the liquid heat-transfer medium at a uniform rate of 2°±0 2° C (3 6°±0 4° F)

3.1.1.5 A thermometer with a suitable temperature range for determining the temperature of the heat-transfer medium

3.1.2 A stirrer that will insure complete circulation of the heat-transfer medium

3.1.3 *Heat-transfer medium* —A liquid heat-transfer medium that will not affect the rigidity of the material to be tested The insulated tank shall be filled with the liquid medium to a depth that will cover the thermometer stem to the point specified in the calibration

4. PROCEDURE

4.1 The liquid heat-transfer medium shall be as specified in the detail specification

4.2 Preparation of specimen.—The specimen shall be prepared as described in 4 4 of method 11001

4.3 The medium in the apparatus shall be at room temperature The specimen shall be placed symmetrically on the supports so that it is completely immersed in the heat-transfer medium with one of the wide faces on the supports A total load of 5 5±0 1 pound shall be applied to the specimen midway between the lines of support After the dial gage or other device for measuring the deflection has come to rest, it shall be set at zero A thermometer shall be placed in the liquid and extended to within ⅛ inch of the specimen but shall not be allowed to touch it

4.4 Heat shall be applied so that the temperature of the heat-transfer medium is increased at a uniform rate of 2°±0 2° C (3 6°±0 4° F) per minute The temperature shall be recorded for each 0 015-inch deflection of the specimen as shown on the dial gage or other device The heating shall be continued until the deflection of the specimen increases so rapidly that it is difficult to read the corresponding temperature

4.5 The deflection readings shall be plotted against the corresponding temperature and a curve drawn through the points A tangent

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shall be drawn to the linear portion of the curve The temperature at the point where the tangent intersects the temperature axis shall be recorded to the nearest degree centigrade as the softening point of the specimen

5. RESULTS

5.1 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.2 The softening point of the test unit shall be the average of the values obtained from the specimens tested

5.3 The softening point of the test unit shall be recorded to the nearest one degree centigrade

5.4 If a plied-up specimen is used, it shall be recorded

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

BULGE TEST, BATTERY CONTAINERS

1. SCOPE

1.1 This method is intended for use in determining the tendency of battery jars to soften or bulge due to heat generated when in service

2. SPECIMEN

2.1 The specimen shall consist of a battery jar from the test unit

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Calipers and scale or other device for measuring the width and length of the specimen to at least 1/64 inch

3.1.2 Equipment with suitable controls for maintaining the required temperature throughout the liquid medium during the exposure period

3.1.3 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within ±1 0° C (1 8°F)

3.1.4 A heat-transfer medium consisting of a liquid in which the battery container material is not soluble

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be exposed at a temperature of 70°±2° C (158°±3 6° F) for 3±¼ hour

4.2 The specimen shall be free from mechanical damage The width and length of the specimen shall be measured at room temperature by means of calipers and scale or other device and the value recorded to the nearest 1/64 inch as W_1 and L_1, respectively One measurement each for length and width shall be made from the middle point of one panel to the middle point of the opposite panel

4.3 The liquid heat-transfer medium shall be heated to the required temperature, 4 1 Each cell of the battery container shall be filled with the liquid medium to within 5/8±1/8 inch of the top The liquid medium shall be maintained at the required temperature for the required time, 4 1 Any method of maintaining the liquid at the required uniform temperature throughout the cells is satisfactory provided heat is not applied to the outside of the container

4.4 At the end of the heating period, the container and its liquid content shall be set aside overnight to cool to a temperature of 23°±2° C (73 5°±3 6° F) The width and length of the jar shall again be measured as described in 4 2 and at the same points initially measured, and the values recorded to the nearest 1/64 inch as W_2 and L_2 respectively The specimen shall be examined for cracks, blisters, or other visible defects

5. RESULTS

5.1 **Calculation.**—The bulge of the specimen shall be calculated as follows

$$\text{Bulge in length, inch,} = L_2 - L_1$$
$$\text{Bulge in width, inch,} = W_2 - W_1$$

Where

L_1 is the initial length of the specimen, inches
L_2 is the length of the specimen after heating, inches
W_1 is the initial width of the specimen, inches
W_2 is the width of the specimen after heating, inches

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The bulge of the test unit in length and width, respectively, shall be the average of the values obtained from the specimens tested

5.4 The bulge of the test unit in length and in width shall be recorded separately to the nearest 1/64 inch

5.5 The number of specimens tested from each test unit and the number from each test unit that showed blisters, cracks, or other visible defects shall be recorded

5.6 The time and temperature of exposure shall be recorded

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WEIGHT CHANGE, HARD RUBBER IN BATTERY ACID

1. SCOPE

1.1 This method is intended for use in determining the change in weight of hard rubber compounds used in battery jars and separators due to immersion in sulfuric acid

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 2 by 2 inches

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed ¼ inch Specimens from the same test unit shall be of the same thickness within ±0 01 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3 1.1 Analytical balance and weights

3.1.2 Desiccator with desiccant

3.1.3 *Air oven* —A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the immersion period and of sufficient size to hold the container in 3 1 7

3.1.4 A drying oven with suitable controls for maintaining a temperature of 103°±2° C (217 4°±3 6° F) for one hour

3.1.5 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature of the ovens to within ±1 0° C (1 8° F)

3.1.6 Glass rods, ⅛ inch in diameter, or other device for immersing the specimens in the liquid so that they will not touch each other or the surface of the container

3.1.7 Glass container with a tightly fitted cover for immersion of specimen in sulfuric acid

3.1.8 Soft brush

3.1.9 Immersion liquid, sulfuric acid, specific gravity 1 30 to 1 40

3.1.10 Soap

3.1.11 Litmus paper

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of 70°±2° C (158°±3 6° F) for 166±1 hour

4.2 Preparation of specimen.—The specimen shall be prepared as described in 4 4 of method 11001

4.3 The specimen shall be free from mechanical damage and shall be washed with soap and water, using a soft brush to remove all loose particles, and rinsed with water The washed specimen shall be suspended in the drying oven and dried for one hour at 103°± 2° C (217 4°±3 6° F), cooled to room temperature in the dessicator, weighed, and the weight recorded to the nearest milligram as W_1

4.4 The weighed specimen shall be completely immersed in the sulfuric acid in the glass container by means of glass rods or other device so that the acid will circulate freely around the specimen during immersion The container with the specimen immersed shall be tightly covered to prevent evaporation and placed in the circulating air oven for the required time at the required temperature, 4 1 When practicable, the immersion medium shall be stirred every 24 hours At the end of the immersion period, the specimen shall be immediately removed from the acid, rinsed quickly with water, dried by means of filter paper or soft cotton cloth, immediately weighed, and the weight recorded to the nearest milligram as W_2

4.5 The specimen from 4 4 shall be washed with running water until the washings are neutral to litmus paper The washed specimen shall again be suspended in the drying oven and dried for one hour at 103°±2° C (217 4°±3 6° F), cooled to room temperature in the dessicator, weighed, and the weight recorded to the nearest milligram as W_3

4.6 The specimen shall be examined for blisters, cracks, and other visible damage

5. RESULTS

5.1 Calculation.

5.1.1 The change in weight of the specimen

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due to immersion in sulfuric acid shall be calculated as follows

$$\text{Change in weight, percent,} = \frac{W_2 - W_1}{W_1} \times 100$$

5.1.2 The amount of acid-soluble material removed from the specimen by immersion in sulfuric acid shall be calculated as follows

$$\text{Acid-soluble material, percent} = \frac{W_1 - W_3}{W_1} \times 100$$

5.1.3 The amount of acid absorbed by the specimen due to immersion in sulfuric acid shall be calculated as follows

$$\text{Acid absorbed, percent,} = \frac{W_3 - W_2}{W_1} \times 100$$

Where

W_1 is the original weight of the dried specimen, grams

W_2 is the weight of the specimen after immersion in acid, grams

W_3 is the weight of the specimen after immersion in acid, washing free of acid, and drying, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The change in weight, the sulfuric acid soluble material, and the acid absorbed, respectively, of the test unit shall be the average of the values obtained from the specimens tested

5.4 The change in weight, acid-soluble material, and acid absorbed shall be recorded separately to the nearest 0 1 percent

5.5 The number of specimens tested from each test unit and the number from each test unit that showed blisters, cracks, or other visible damage shall be recorded

5.6 The time and temperature of immersion shall be recorded

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DIMENSIONAL CHANGES, HARD RUBBER IN BATTERY ACID

1. SCOPE

1.1 This method is intended for use in determining the dimensional changes in hard-rubber compounds used in battery jars and separators due to immersion in sulfuric acid

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 2 by 2 inches and these dimensions shall be uniform within ±0 005 inch

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test but shall not exceed ¼ inch The thickness of the specimen shall be uniform within ±0 005 inch Specimens from the same test unit shall be of the same thickness within ±0 01 inch, method 2011

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus as described in 3 1 3 to 3 1 11, inclusive, of method 11411

3.1.2 Vernier caliper graduated to 1/10 inch or less with a vernier to read to 0 001 inch

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of 70°±2° C (158°±3 6° F) for 166±1 hour

4.2 Preparation of specimen.—The specimen shall be prepared as described in 4 4 of method 11001

4.3 The specimen shall be washed with soap and water, using a soft brush to remove all loose particles, and rinsed with water The washed specimen shall be suspended in the drying oven, dried for one hour at 103°±2° C (217 4°±3 6° F), and cooled to room temperature

4.4 The average thickness of the specimen shall be determined as described in method 2011 The average width and length of the specimen shall be determined by means of the vernier caliper described in 3 1 2 The calipers shall be applied directly to the specimen so that the jaws of the calipers are just brought into contact with the specimen in such a manner that the material is not distorted The criterion of contact is the initial development of frictional resistance to movement of the specimen between the jaws of the calipers The width or length of the specimen shall be read from the caliper scale Three measurements each, for width and length equally spaced along the specimen shall be made and the average of the values recorded separately to the nearest 0 001 inch

4.5 The specimen shall be completely immersed in the sulfuric acid in the glass container by means of glass rods or other device, so that the acid will circulate freely around the specimen during immersion The container with the specimen immersed shall be tightly covered to prevent evaporation and placed in the circulating air oven for the required time at the required temperature, 4 1 When practicable, the acid shall be stirred every 24 hours

4.6 At the end of the immersion period, the specimen shall be immediately removed from the acid and washed in running tap water for approximately one minute to remove the acid from the surface of the specimen The surface shall be dried by blotting with a soft cotton cloth or filter paper and the dimensions again determined as described in 4 4, measurements being made as near as practicable at the same places

4.7 The specimen shall be examined for blisters, cracks, or other visible damage

5. RESULTS

5.1 Calculation.—The change in each dimension of the specimen shall be calculated as follows

$$\text{Change in dimension, percent} = \frac{I-O}{O} \times 100$$

Where

O is the dimension of the specimen before immersion, inches

I is the dimension of the specimen after immersion in acid, inches

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5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The change in length, width, and thickness, respectively, of the test unit shall be the average of the values obtained from the specimens tested

5.4 The change in length, width, and thickness of the test unit shall be recorded separately to the nearest 0 1 percent.

5.5 The number of specimens tested from each test unit and the number from each test unit that showed blisters, cracks, or other visible damage shall be recorded

5.6 The temperature and time of immersion shall be recorded

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PENETRATION, ACID, HARD RUBBER

1. SCOPE

1.1 This method is intended for use in determining the penetration of sulfuric acid into hard rubber compounds used in battery jars and separators due to immersion in sulfuric acid

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 2 by 2 inches

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Glass container with a tightly fitting cover for immersion of the specimen in sulfuric acid

3.1.2 *Air oven*.—Circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the immersion period and of sufficient size to hold the container in 3 1 1

3.1.3 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature of the immersion medium to within ±1 0 C (1 8° F)

3.1.4 Glass rods, ⅛ inch in diameter, or other device for immersing the specimens in a liquid so that they will not touch each other or the surface of the container

3.1.5 Soft brush

3.1.6 Scale, graduated to 1/64 inch or finer

3.1.7 Immersion liquid, sulfuric acid, specific gravity 1 30 to 1 40

3.1.8 Benzene or other similar solvent

3.1.9 Acidproof adhesive for sealing the edges of the specimen

3.1.10 Soap

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed in sulfuric acid at a temperature of 70°±2° C (158°±3 6° F) for 28 days

4.2 **Preparation of specimen.**—The specimen shall be prepared as described in 4 4 of method 11001

4.3 The specimen shall be washed with soap and water, using a soft brush to remove all loose particles, rinsed with water, and dried with a soft cotton cloth or blotting paper The cut edges of the specimen shall be sealed with a high melting, acid-resistant adhesive to prevent penetration of acid

4.4 The specimen shall be completely immersed in the sulfuric acid in the glass container by means of glass rods or other device so that the acid will circulate freely around the specimen during immersion The container with the specimen immersed shall be tightly covered to prevent evaporation and placed in the circulating-air oven for the required time at the required temperature, 4 1 When practicable, the acid shall be stirred every 24 hours

4.5 At the end of the immersion period, the specimen shall be immediately removed from the acid, rinsed quickly with water, and dried with a soft cotton cloth or blotting paper The specimen shall be cut through the center into two pieces, each about 2 inches long and 1 inch wide The freshly cut edge of one piece of the specimen shall be polished with benzene or other suitable solvent

4.6 The depth of penetration of the acid shall be measured with the scale to the nearest 1/64 inch Three measurements shall be made in the area within 0 5 inch of the middle point and the average of the three values recorded to the nearest 1/64 inch

4.7 The specimen shall be examined for cracks, blisters, or other visible damage

5. RESULTS

5.1 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

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5.2 The acid penetration of the test unit shall be the average of the values obtained from the specimens tested

5.3 The acid penetration of the test unit shall be recorded to the nearest 1/4 inch

5.4 The number of specimens tested from each test unit and the number from each test unit that showed blisters, cracks, or other visible damage shall be recorded

5.5 The temperature and time of exposure shall be recorded

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VOLTAGE WITHSTAND, BATTERY CONTAINERS

1. SCOPE

1 1 This method is intended for use in determining the voltage withstand of battery jars

2. SPECIMEN

2.1 Wet test.—The specimen shall consist of a finished battery jar from the test unit

2.2 Dry test.—The specimen shall consist of any part of the jar or container or cell partitions from a test unit

3. APPARATUS

3 1 The apparatus shall be as follows

3 1 1 *Dry*

3 1 1 1 *Electrodes* —For testing sheets and plates, the electrodes consist of metal cylinders 2 inches in diameter and 1 inch in length with the edges rounded to a radius of ¼ inch For testing thick materials, the electrodes consist of metal cylinders 1 inch in diameter and 1 inch in length with edges rounded to a radius of ⅛ inch The faces of the electrodes are kept smooth, polished, and free from pitting

3 1 1 2 *Transformer* —The desired test voltage may be most readily obtained by a step-up transformer energized from a variable low-voltage source The transformer and its controlling equipment are of such size and design that with the specimen in the circuit the crest factor of the test voltage does not differ by more than ±5 percent from that of a sinusoidal wave over the upper half of the range of the test voltage For specimens of small capacitance, a transformer as small as 500-volt-ampere rating or a well-designed potential transformer may be adequate For specimens of high capacitance, a transformer of ample kilovolt-ampere rating must be used Where the wave-form cannot be determined conveniently, a transformer having a rating of 2 kilovolt-amperes or more is used for voltages not exceeding 50,000 volts, and one with a rating of not less than 5 kilovolt-amperes for voltages exceeding 50,000 When a transformer is used at voltages lower than its full rating, the current drawn from the high voltage winding should not exceed the full-load, full-voltage current rating Tests are made at commercial power frequencies

3 1 1 3 *Circuit breaker* —An automatic circuit breaker for protecting the transformer circuit is designed to open instantly on the current produced by breakdown of the specimen

3 1 1 4 *Voltage control* —Voltage control may be accomplished by one of the following ways (1) variable-ratio autotransformer, (2) resistance-potential divider, (3) generator-field regulation, or (4) induction regulator The rate of voltage rise shall not vary by more than ±25 percent of the specified rate Preferences should be given to equipment having an approximately straight-line, voltage-time curve over the desired operating range

3 1 1 5 *Voltmeter* —The voltage is determined in root-mean-square values by one of the following arrangement of voltmeters (1) a voltmeter connected to the secondary of a separate potential transformer, (2) an electrostatic voltmeter in the high-tension circuit, or (3) a voltmeter connected to a well-designed tertiary coil in the transformer

3.1.1.6 *Rate of rise of voltage* —The rate of voltage rise may be calculated by measuring the time required to raise voltage between two prescribed values When motor-driven regulating equipment is used, the speed-control rheostat may be calibrated in terms of voltage rise for any particular test transformer

3.1.2 *Wet test*

3.1.2.1 Apparatus described in 3 1 1 2 through 3 1 1 6

3.1.2.2 Container for holding the specimen and water

3.1.2.3 The electrodes for the wet test shall consist of the water placed in the container and in the battery jar

4. PROCEDURE

4.1 Wet test.

4.1.1 Unless otherwise specified in the detail specification, the voltage shall be increased at

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a rate of 1 kilovolt per second until a voltage of 10,000±100 volts is reached

4.1.2 Unless otherwise specified in the detail specification, the voltage of 10,000±100 volts or other specified voltage shall be applied for a period of 5 seconds

4.1.3 The specimen shall be free from mechanical damage and shall be placed in the container Water shall be added to the container and battery jar so as to bring the water level to within 1½ inches of the top on the inside and outside of the battery jar Any water on the top edges of the jar shall be wiped from the jar before the test One lead of the transformer shall be placed in the water in the jar and the other lead placed in the water in the container (outside the jar) Voltage shall be applied uniformly at the required rate and held for the required time Whether the specimen failed or withstood the applied voltage shall be recorded

4.2 Dry test.

4.2.1 Unless otherwise specified in the detail specification, the voltage shall be increased at a rate of 1 kilovolt per second until a voltage of 20,000±200 volts is reached.

4.1.2 Unless otherwise specified in the detail specification, the voltage of 20,000±200 volts or other specified voltage shall be applied for a period of 5 seconds.

4.2.3 The electrodes shall be placed on the surface of each side of the specimen at least 1½ inches from the edge of the specimen. The required voltage shall be applied uniformly at the required rate and held for the required period of time Whether the specimen failed or withstood the applied voltage shall be recorded

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested.

5.2 The number of specimens tested from each test unit and the number from each test unit that withstood the specified voltage without failure shall be recorded

5.3 The test voltage, rate of increase of voltage, and time of application of the test voltage shall be recorded.

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CELLULAR RUBBER, GENERAL

1. SCOPE

1.1 These methods are intended for use in determining the various characteristics of cellular rubber materials except hard rubbers These methods are applicable for testing cellular rubber compounds made from natural rubber, synthetic rubber, rubberlike materials, or a combination of these materials

2. SPECIMEN

2.1 Unless otherwise specified in the detail specification, the specimen shall be taken from the finished unit Specimens shall be cut with a sharp die or other instrument free from nicks If cylindrical specimens are required, they may be cut with a die suitably rotated in a drill press or similar device The pressure applied to the die shall be sufficiently small to keep cupping of the cut surface to a minimum To facilitate cutting, the edge of the die may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0 5 percent sodium chromate or with silicone mold release emulsion before each specimen is cut In some materials, it may be necessary to freeze the material in order to cut a specimen having parallel cut surfaces When practicable, the finished unit shall be used as the specimen

3. DEFINITIONS

3.1 **Cellular rubber.**—Cellular rubber is a material containing cells or small hollow receptacles The cells may be either open and interconnecting or closed and not interconnecting

3.2 **Skin or rind.**—The skin or rind is the smooth outer surface of cellular rubber material It may be formed by contact with the surface of the mold or cover plates The skin or rind may also be formed by applying a solid rubber coating to the uncured rubber and vulcanizing them together The skin or rind formed by contact with the surface mold or cover plate is termed the natural skin

3.3 **Sponge rubber.**—Sponge rubber is an open-cell material made by incorporating into the rubber compound an inflating agent such as sodium bicarbonate that gives off a gas which expands the rubber during vulcanization

3.4 **Expanded rubber.**—Expanded rubber is a closed-cell material usually made by subjecting a rubber compound to a gas, such as nitrogen, under high pressure It may also be made by incorporating gas-forming ingredients in the compound

3.5 **Latex foam rubber.**—Latex foam rubber is a material with open or interconnecting cells either cored or solid made from rubber latices or liquid rubber

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GEOMETRICAL MEASUREMENTS, CELLULAR RUBBER, GENERAL

1. SCOPE

1.1 These methods are intended for use in determining the geometrical properties of cellular rubber, cellular rubber products, or specimens of cellular rubber for physical tests. Geometrical measurements are made for one of the following purposes

1.1.1 To determine conformance with specification requirements regarding geometrical properties of whole units such as the size of a cellular rubber cushion In such a case, determination of the geometrical property is considered as an inspection, and measurements are made on one or more locations on each unit in the sample for inspection

1.1.2 To determine conformance with specification requirements regarding particular functional parts of units, such as the thickness of cellular rubber parts of vibration absorbing units. In such a case, determination of the geometrical property is considered as a test since it necessitates destruction of the unit, and measurements are made on one or more specimens taken from each test unit

1.1.3 To determine geometrical values that are necessary in the calculation of test results for a physical property for which requirements are specified, such as the thickness of a specimen for compression set In such a case, measurements are made on the specimens on which the physical property in question is measured and the geometrical values found are only an adjunct to the determination of the physical property

1.2 Methods are described for measuring length, width, thickness, and diameter One method each is described for length, width, and diameter Two methods are described for thickness, one for specimens over 1 inch and the other for specimens 1 inch and less Geometrical measurements shall not be made at places where any damage is apparent that may interfere with the geometrical property.

2. SPECIMEN

2.1 **Units or test units.**—The specimen shall consist of a unit or test unit, or a portion of a unit or test unit, 1.1 1 and 1 1 2

2.1.1 It is not practicable to specify the size and shape of specimens for geometrical properties in all cases In general, the specimen shall consist of the largest single piece of material in the unit or in the specified part of the test unit which it is practicable to measure.

2.2 **Physical test.**—The specimen shall be as described in the applicable test method, 1 1 3

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LENGTH, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the length of cellular rubber units, test units, or specimens for physical tests

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 12005

3. APPARATUS

3.1 The apparatus shall consist of a steel scale or tape graudated to 1/32 inch or finer

4. PROCEDURE

4.1 **Length measurement.**—The specimen shall be placed on a smooth surface so as to expose the full length without distortion of the material The scale or tape shall be applied over the specimen in such a manner that the specimen is not distorted The length shall be read from the scale or tape to the nearest 1/32 of an inch If more than one measurement is made on the specimen, the measurements shall be equally spaced over the specimen

4.2 **Average length.**

4.2.1 *Unit or test unit*—Unless otherwise specified in the detail specification, a total of nine measurements shall be made on the unit or test unit The measurements shall be made as described in 4 1 and if made on more than one specimen they shall be equally distributed as nearly as practicable among all the specimens from the unit or test unit

4.2.2 *Specimens for physical test*—Unless otherwise specified in the applicable test method, three measurements shall be made on the specimen The measurements shall be made as described in 4 1

4.3 **Minimum length.**

4.3.1 *Unit or test unit*—The unit or test unit shall be examined for the shortest place If the shortest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 1 If no shortest place is found, measurements shall be made as described in 4 2 1 and the smallest value obtained shall be the minimum length of the unit or test unit

4.3.2 *Specimen for physical test*—The specimen shall be examined for the shortest place If the shortest place is found, a measurement shall be made at that place as described in 4 1 If no shortest place is found, measurements shall be made as described in 4 2 2 and the smallest value obtained shall be the minimum length of the specimen

4.4 **Maximum length.**

4.4.1 *Unit or test unit*—The unit or test unit shall be examined for the longest place If the longest place is found, the specimen shall be taken so as to include it and a measurement shall be made at that place as described in 4 1 If no longest place is found, measurements shall be made as described in 4 2 1 and the largest value obtained shall be the maximum length of the unit or test unit

4.4.2 *Specimen for physical test*—The specimen shall be examined for the longest place If the longest place is found, a measurement shall be made at that place as described in 4 1 If no longest place is found, measurements shall be made as described in 4 2 2 and the largest value obtained shall be the maximum length of the specimen

5. RESULTS

5.1 **Unit or test unit.**

5.1.1 *Average length*—The average length of the unit or test unit shall be the average of all the values obtained in 4 2 1

5.1.2 *Minimum length*

5.1.2.1 If a shortest place is found in the unit or test unit, the minimum length of the unit or test unit shall be the length obtained at that place

5.1.2.2 If no shortest place is found in the unit or test unit, the minimum length of the unit or test unit shall be the smallest of all the values obtained when measurements are made as described in 4 2 1

5.1.3 *Maximum length*

5.1.3.1 If a longest place is found in the

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unit or test unit, the maximum length of the unit or test unit shall be the length obtained at that place

5.1.3.2 If no longest place is found in the unit or test unit, the maximum length of the unit or test unit shall be the largest of all the values obtained when measurements are made as described in 4 2 1

5.1.4 The average, minimum, and maximum length of the unit or test unit shall be recorded to the nearest $\frac{1}{32}$ inch

5.2 Specimen for physical test.—The number of specimens to be tested and the treatment of data (average, etc.) shall be as required in the applicable test method

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THICKNESS, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the thickness of cellular rubber units, test units, or specimens for physical tests Two procedures are described one for specimens over 1 inch and one for specimens 1 inch and less

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 12005

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A dial micrometer having a flat anvil not less than 1 25 inch in diameter and a flat presser foot 1 25 ± 0 05 inch in diameter which exerts a total force of 25 ± 1 gram on the specimen, the force being applied by means of a weight The surfaces of the anvil and presser foot shall be parallel to within 0 0001 inch The dial shall be graduated to read to 0 001 inch

3.1.2 A vernier caliper graduated to 0 025 inch with a sliding vernier to read to 0 005 inch or finer

4. PROCEDURE

4.1 Thickness measurements

4.1.1 *Over one inch* —The vernier caliper gage shall be set at the estimated thickness of the specimen The specimen shall be placed between the jaws of the previously set gage This operation shall be repeated until the jaws just contact the surfaces of the specimen without distortion of the material The thickness shall be read from the vernier caliper to the nearest 0 005 inch If more than one measurement is made on a specimen, the measurements shall be equally spaced over the specimen

4.1.2 *One inch and less* —The specimen shall be placed on the anvil of the dial micrometer and the presser foot lowered gently until it just contacts the surface of the specimen The thickness shall be read from the dial to the nearest 0 001 inch If more than one measurement is made on a specimen, the measurements shall be equally spaced over the specimen

4.2 Average thickness.

4.2.1 *Unit or test unit* —Unless otherwise specified in the detail specification, a total of nine measurements shall be made on the unit or test unit The measurements shall be made as described in 4 1 1 or 4 1 2, as applicable, and if made on more than one specimen they shall be equally distributed as nearly as practicable among all the specimens from the unit or test unit

4.2.2 *Specimen for physical test* —Unless otherwise specified in the applicable test method, three measurements shall be made on the specimen The measurements shall be made as described in 4 1 1 or 4 1 2, as applicable

4.3 Minimum thickness.

4.3.1 *Unit or test unit* —The unit or test unit shall be examined for the thinnest place If the thinnest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 1 1 or 4 1 2, as applicable If no thinnest place is found, measurements shall be made as described in 4 2 1 and the smallest value obtained shall be the minimum thickness of the unit or test unit

4.3.2 *Specimen for physical test* —The specimen shall be examined for the thinnest place If the thinnest place is found, a measurement shall be made at that place as described in 4 1 1 or 4 1 2, as applicable If no thinnest place is found, measurements shall be made as described in 4 2 2 and the smallest value obtained shall be the minimum thickness of the specimen

4.4 Maximum thickness.

4.4.1 *Unit or test unit* —The unit or test unit shall be examined for the thickest place If the thickest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 1 1 or 4 1 2, as applicable If no thickest place is found, measurements shall be made as described in 4 2 1 and the largest value obtained shall be the maximum thickness of the unit or test unit

4.4.2 *Specimen for physical test* —The specimen shall be examined for the thickest place If the thickest place is found, a measurement

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shall be made at that place as described in 4 1 1 or 4 1 2, as applicable If no thickest place is found, measurements shall be made as described in 4 2.2 and the largest value obtained shall be the maximum thickness of the specimen

5. RESULTS

5.1 Unit or test unit.

5.1.1 *Average thickness* —The average thickness of the unit or test unit shall be the average of the values obtained in 4 2 1

5.1.2 *Minimum thickness*

5.1.2.1 If a thinnest place is found in the unit or test unit, the minimum thickness of the unit or test unit shall be the thickness obtained at that place

5.1.2.2 If no thinnest place is found in the unit or test unit, the minimum thickness of the unit or test unit shall be the smallest of all the values obtained when measurements are made as described in 4 2 1

5.1.3 *Maximum thickness*

5.1.3.1 If a thickest place is found in the unit or test unit, the maximum thickness of the unit or test unit shall be the thickness obtained at that place

5.1.3.2 If no thickest place is found in the unit or test unit, the maximum thickness of the unit or test unit shall be the largest of all the values obtained when measurements are made as described in 4 2 1

5.1.4 The average, minimum, and maximum thickness of units and test units shall be recorded to the nearest 0 001 inch for units and test units one inch and less in thickness and to the nearest 0 005 inch for units and test units over one inch in thickness

5.2 Specimen for physical test.—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method

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Method 12021
April 12, 1955

WIDTH, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the width of cellular rubber units, test units, or specimens for physical tests

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 12005

3. APPARATUS

3.1 The apparatus shall consist of a steel scale or tape graduated to 1/32 inch or finer

4. PROCEDURE

4.1 Width measurement.—The specimen shall be placed on a smooth surface so as to expose the full width without distortion of the material The graduated scale or tape shall be applied over the specimen in such a manner that the specimen is not distorted The width shall be read from the scale or tape to the nearest 1/32 inch If more than one measurement is made on the specimen, the measurements shall be equally spaced over the specimen

4.2 Average width.

4.2.1 *Unit or test unit* —Unless otherwise specified in the detail specification, a total of nine measurements shall be made on the unit or test unit The measurements shall be made as described in 4 1 and if made on more than one specimen they shall be equally distributed as nearly as practicable among all the specimens from the unit or test unit

4.2.2 *Specimen for physical test* —Unless otherwise specified in the applicable test method three measurements shall be made on the specimen The measurements shall be made as described in 4 1

4.3 Minimum width.

4.3.1 *Unit or test unit* —The unit or test unit shall be examined for the narrowest place If the narrowest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 1 If no narrowest place is found, measurements shall be made as described in 4 2 1 and the smallest value obtained shall be taken as the minimum width of the unit or test unit

4.3.2 *Specimen for physical test* —The specimen shall be examined for the narrowest place If the narrowest place is found, a measurement shall be made at that place as described in 4 1 If no narrowest place is found, measurements shall be made as described in 4 2 2 and the smallest value obtained shall be the minimum width of the specimen

4.4 Maximum width.

4.4.1 *Unit or test unit* —The unit or test unit shall be examined for the widest place If the widest place is found, the specimen shall be taken so as to include it, and a measurement shall be made at that place as described in 4 1 If no widest place is found, measurements shall be made as described in 4 2 1 and the largest value obtained shall be the maximum width of the unit or test unit

4.4.2 *Specimen for physical test* —The specimen shall be examined for the widest place If the widest place is found, a measurement shall be made at that place as described in 4 1 If no widest place is found, measurements shall be made as described in 4 2 2 and the largest value obtained shall be the maximum width of the specimen

5. RESULTS

5.1 Unit or test unit.

5.1.1 *Average width* —The average width of the unit or test unit shall be the average of the values obtained in 4 2 1

5.1.2 *Minimum width*

5.1.2.1 If a narrowest place is found in the unit or test unit, the minimum width of the unit or test unit shall be the width obtained at that place

5.1.2.2 If no narrowest place is found in the unit or test unit, the minimum width of the unit or test unit shall be the smallest of all the values obtained when measurements are made as described in 4 2 1

5.1.3 *Maximum width*

5.1.3.1 If a widest place is found in the unit or test unit, the maximum width of the unit or

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test unit shall be the width obtained at that place

5.1.3.2 If no widest place is found in the unit or test unit, the maximum width of the unit or test unit shall be the largest of all the values obtained when measurements are made as described in 4 2 1

5.1.4 The average, minimum, and maximum width of the unit or test unit shall be recorded to the nearest 1/32 inch.

5.2 Specimen for physical test. The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method.

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DIAMETER, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the diameter of cellular rubber units, test units, or specimens for physical tests

2. SPECIMEN

2.1 The specimen shall be as described in 2 of method 12005

3. APPARATUS

3.1 The apparatus shall consist of the vernier caliper described in method 12031

4. PROCEDURE

4.1 The vernier caliper gage shall be set at the estimated diameter of the specimen at the place to be measured The specimen shall be placed between the jaws of the previously set gage This operation shall be repeated until the jaws of the gage just contact the surfaces of the specimen without distortion of the material The diameter of the specimen shall be read from the vernier caliper to the nearest 0 005 inch One set of measurements shall be made at each cross section measured A set of measurements is two measurements made on the same cross section at right angles to each other If more than one measurement (set) is made on the specimen, the measurements shall be equally spaced over the specimen

4.2 Unit or test unit.—Unless otherwise specified in the detail specification, a total of 12 measurements (6 sets) shall be made on the unit or test unit The measurements shall be made as described in 4 1 and if made on more than one specimen they shall be equally distributed as nearly as practicable among all the specimens from the unit or test unit

4.3 Specimen for physical test.—Unless otherwise specified in the applicable test method, 6 measurements (3 sets) shall be made on the specimen The measurements shall be made as described in 4 1

5. RESULTS

5.1 Units or test units.

5.1.1 The diameter of the unit or test unit shall be the average of all the values obtained in 4 2

5.1.2 The diameter of the unit or test unit shall be recorded to the nearest 0 005 inch

5.2 Specimen for physical test.—The number of specimens to be tested and the treatment of data (average, etc) shall be as required in the applicable test method

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FLEXING ENDURANCE, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the effect of repeated compression on cellular rubber compounds It is particularly applicable to foam latex rubber compounds At the end of the flexing period, the specimen is examined for physical breakdown, and the set of the material is determined after the specimen has been allowed to rest for a specified time

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit approximately 12 by 12 inches

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Compression machine*—A machine equipped with two flat parallel plates, one for supporting the specimen in a horizontal position and the other for compressing it One of the plates is perforated with ¼-inch holes placed on ¾-inch centers The supporting plate is adjustable and held in a fixed position in the machine while the other plate is mounted in the machine so that it travels with reciprocating motion at the rate of 60±5 cycles per minute alternately compressing the specimen and releasing the force The length of stroke of the movable plate is adjustable to accommodate specimens of different thicknesses and to provide for a length of travel from the point where the plate just contacts the surface of the specimen to the point where the percent of compression, 4 1, is obtained Both plates are at least ¼ inch larger on each side than the specimen undergoing test The apparatus is equipped with a counting device that records the number of flexing cycles to which the specimen is subjected

3.1.2 A vernier caliper graduated to 0 025 inch, with a sliding vernier to read to at least 0 01 inch

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, for materials having an indentation of 67 pounds per 50 square inches or less, method 12121, the amplitude of compression and decompression shall be 50 percent of their original thickness, and for materials having an indentation greater than 67 pounds per 50 square inches, the amplitude of compression and decompression shall be 25 percent of their original thickness

4.2 Unless otherwise specified in the detail specification, the specimen shall be subjected to 250,000 cycles of flexing

4.3 The specimen shall be allowed to rest undeflected and undistorted for at least 12 hours before testing for flexing endurance

4.4 The specimen shall be free from mechanical damage The specimen shall be placed centrally between the plates of the testing machine with the cored or honeycomb side, if present, in contact with the perforated plate The plate positions shall be adjusted so that the loading plate is brought just into contact with any part of the specimen The distance between the two plates shall be determined by means of the calipers and the value recorded to the nearest 0 01 inch as the original thickness, T_o

4.5 The length of stroke of the movable plate shall be adjusted to give the required amplitude of compression and decompression of the specimen, 4 1 The machine shall then be operated at the required amplitude of compression and decompression and at the rate of 60±5 cycles per minute for the required number of flexing cycles or until breakdown of the specimen

4.6 At the completion of the required number of flexing cycles, the specimen shall be examined for physical breakdown of the cellular structure by visual examination and comparison with the unflexed specimen The specimen shall be allowed to rest out of contact with the compression plate for 30 minutes

4.7 At the end of the 30-minute rest period,

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the specimen shall be placed between the plates, the thickness of the specimen again determined as described in 4 4, and the value recorded as T_{30}

5. RESULTS

5.1 Calculation.—The set of the specimen shall be calculated as follows

$$\text{Set, percent, } = \frac{T_o - T_{30}}{T_o - T_t} \times 100$$

Where

T_o is the original thickness of the specimen before flexing, inches

T_{30} is the thickness of the specimen 30 minutes after flexing, inches

T_t is the thickness to which the specimen is compressed, 4 1

5.2 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.3 When one specimen is tested from each test unit, the set of the test unit after flexing shall be the value obtained from the specimen tested When more than one specimen is tested from each test unit, the set of the test unit after flexing shall be the average of the values obtained from the specimens tested

5.4 The set of the test unit shall be recorded to the nearest one percent.

5.5 The number of specimens tested from each test unit and the number from each test unit that showed physical breakdown of the cellular structure shall be recorded

5.6 The number of flexing cycles and the amplitude of compression and decompression used shall be recorded

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INDENTATION, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the indentation of cellular rubber compounds The method consists of measuring the force required to produce a specified indentation of the material It is particularly applicable to testing latex foam rubber compounds

2. SPECIMEN

2.1 The specimen shall consist of the test unit or a portion of the test unit at least 12 by 12 inches

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Indentation apparatus* —The apparatus is equipped with 2 flat, parallel plates, 1 for supporting and 1 for indenting the specimen The indenter plate is circular and has an area of 50 0±0 5 square inch The supporting plate is arranged to support the specimen in a level horizontal position One of the plates is perforated with ¼-inch-diameter holes on ¾ inch centers to permit rapid escape of air during the test The indenter plate is connected to the load measuring device and is mounted in such a manner that the specimen can be deflected at a specified rate

3.2 Vernier caliper described in method 12111

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be compressed 25 percent of the original thickness

4.2 Unless otherwise specified in the detail specification, the specimen shall be compressed at the rate of 25±10 inches per minute

4.3 The specimen shall be allowed to rest undeflected and undistorted for at least 12 hours before testing for indentation

4.4 The specimen shall be free from mechanical damage The specimen shall be placed between the plates of the indenting apparatus with the cored or honeycomb side, if present, in contact with the perforated plate The position of the specimen on the supporting plate shall be such that the indentation will be made at the center of the specimen The indenter plate shall be brought just into contact with any part of the specimen A force of one pound shall be applied to the indenter plate, the distance between the plates measured with the calipers, and the value recorded to the nearest 0 01 inch as the thickness of the specimen Force shall then be applied to the specimen at the required rate, 4 2, until the specimen is compressed the required amount, 4 1, and the value recorded immediately to the nearest 0 5 pound

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 When one specimen is tested from each test unit, the indentation of the test unit shall be the value obtained from the specimen tested When more than one specimen is tested from each test unit, the indentation of the test unit shall be the average of the values obtained from the specimens tested

5.3 The indentation of the test unit shall be recorded to the nearest 0 5 pound per 50 square inches

5.4 The amount the specimen was compressed and the rate at which the specimen was compressed shall be recorded

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COMPRESSION SET, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the compression set of cellular rubber compounds under constant deflection The method is used to measure the ability of cellular rubbers to retain elastic properties during prolonged action of compressive stresses

2. SPECIMEN

2.1 Cored material.—The specimen shall consist of a portion of the test unit, either round or rectangular in shape, as described in 2 1.1, 2 1 2, and 2.3

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test

2.1.2 The minimum dimension of the top and bottom surfaces shall be greater than the thickness of the specimen, and the top and bottom surfaces each shall have a minimum area of 16 square inches

2.2 Uncored or slab material.—The specimen shall consist of a portion of the test unit as described in 2 2 1, 2 2 2, and 2 3

2.2.1 The specimen shall be a circular cylinder cut with the die described in method 12141

2.2.2 Unless otherwise specified in the detail specification the specimen shall be not less than ¼ inch and not more than 2 inches in thickness

2.2.3 Unless otherwise specified in the detail specification, materials thinner than ¼ inch shall be plied up to obtain the required thickness

2.3 The specimen shall have parallel top and bottom surfaces

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Compression apparatus* —A compression apparatus consisting of two or more flat plates with parallel faces, between which specimens may be compressed, as shown in figure 12131 The plates may be of steel, aluminum, or any rigid smooth metal of sufficient thickness to withstand the required compression stresses without bending The plates shall be thoroughly cleaned before using

3.1.1.1 *Spacers* —The compression apparatus shall be equipped with steel spacers equal in thickness to the thickness the specimen is to be compressed as required in 4 2

3.1.2 *Air oven* —A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the exposure period and of sufficient size to hold the compression apparatus

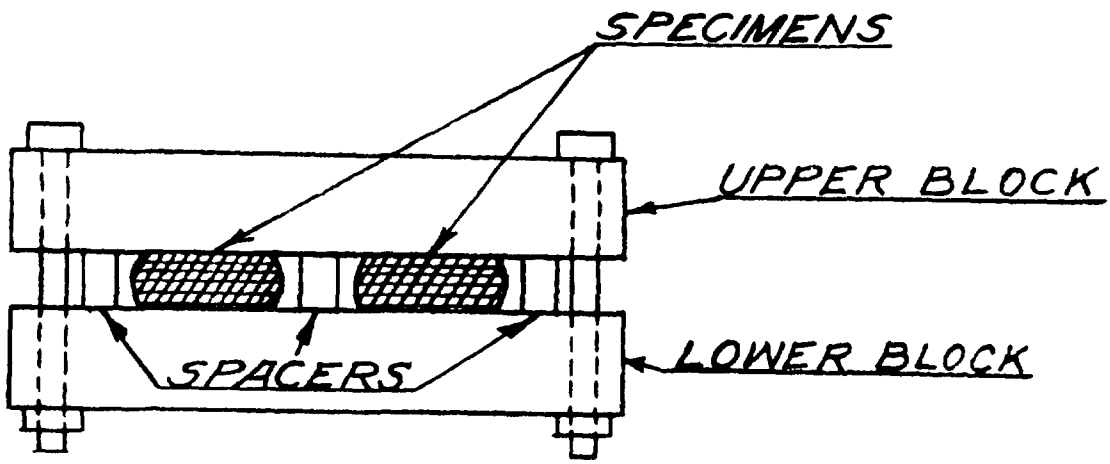

FIGURE 12131 —Compression apparatus.

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3.1.3 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer or other device for measuring the temperature of the oven to within ±1.0° C. (1.8° F.)

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be exposed at a temperature of 70°±1° C (158°±1.8° F.) for 46±¼ hour.

4.2 Unless otherwise specified in the detail specifications, the specimen shall be compressed to 50 percent of its original thickness.

4.3 The specimen from uncored or slab material shall be prepared as described in method 12141.

4.4 Thickness.—The specimen shall be free from mechanical damage.

4.4.1 *Cored material.*—The average thickness of the specimen shall be determined as described in method 12031 and the value recorded to the nearest 0.001 inch or 0.005 inch, as applicable, as T_o.

4.4.2 *Uncored or slab material.*—The thickness of the specimen shall be determined as described in method 12031 except that measurements shall be made in such a manner as to indicate the perpendicular distance between the center portions of the top and bottom faces and the value recorded to the nearest 0.001 inch or 0.005 inch, as applicable, as T_o.

4.5 The specimen shall be placed between the plates of the compression apparatus with the required spacer on each side necessary to provide the compression required in 4.2. If one specimen is tested in the apparatus, the specimen shall be placed in the center. If 2 specimens are tested, 3 spacers shall be used, 1 in the center and 1 on the outside of each specimen, allowing sufficient clearance for bulging of the specimen when compressed. The bolts shall be tightened so that the plates are brought together uniformly until the plates are in contact with the spacers. Only one specimen from a given test unit shall be compressed at one time in the same compression apparatus.

4.6 The compression apparatus with specimens under compression shall be placed in the air oven and exposed for the required time at the required temperature, 4.1.

4.7 At the end of the exposure period, the compression device shall be removed from the oven and the specimen removed immediately and allowed to cool for 30 minutes. At the end of the cooling period, the thickness of the specimen shall again be determined as described in 4.4 and the value recorded as T_{30}. Thickness measurements shall be made as near as practicable at the same location on the specimen as previously measured in 4.4.

5. RESULTS

5.1 Calculation.—The compression set of the specimen, as percent of the original deflection, shall be calculated as follows.

$$\text{Compression set, percent of original deflection,} = \frac{T_o - T_{30}}{T_o - T_s} \times 100$$

Where

T_o is the thickness of the specimen before compression, inches

T_{30} is the thickness of the specimen 30 minutes after removal from compression, inches

T_s is the thickness of the spacers used, inches

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested.

5.3 The compression set of the test unit shall be the average of the values obtained from the specimens tested.

5.4 The compression set of the test unit shall be recorded to the nearest one percent.

5.5 If a plied-up specimen is used, it shall be recorded.

5.6 The percent deflection of the specimen shall be recorded.

5.7 The temperature and time of exposure shall be recorded.

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COMPRESSION RESISTANCE, CELLULAR RUBBER OSCILLOGRAPH

1. SCOPE

1.1 This method is intended for use in determining the resistance of cellular rubber compounds to compression by means of the oscillograph

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1 and 2 1 2.

2.1.1 The specimen shall be a circular cylinder cut with the die described in 3 1 3.

2.1.2 The specimen shall be not less than ¼ inch and not more than 1⅛ inches in thickness

2.1.3 Unless otherwise specified in the detail specification, materials thinner than ¼ inch shall be plied up to obtain the required thickness

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An apparatus consisting of a balance beam, pen, and weights A suitable apparatus is shown in figures 3411A and 3411B

3.1.1.1 *Balance Beam* —The beam is supported at the center by knife edges It is designed so that a specimen placed beneath the micrometer *A* of figure 3411B is loaded by placing weights alternately on opposite sides of the pen end of the balance beam A second knife edge and a stabilizing arm are used to maintain parallelism of the loading platens A perforated plate 2½ inches square and a circular depressor plate 1⅛ inches in diameter fits into the micrometer for compressing the specimen

3.1.1.2 *Pen* —The pen is supported on an arm extending from the end of the beam so as to record automatically the deflection on a chart or oscillogram From figure 3411B, it is apparent that the deflection of the specimen will be magnified in proportion to the lever ratio, which is 10 to 1 (16 inches to 1 6 inches) and a deformation of 0 10 inch will be registered on the oscillogram as a vertical displacement of 1.00 inch

3.1.1.3 *Weights* —The weights, *W*, figure 3411B, consist of several disks that are added to compress the specimen The weights are selected so as to produce a known compressive stress, in pounds per square inch, on the specimen The lever ratio for the weights is 6 25 to 1 (10 pounds to 1 6 pounds) Each one-pound weight added shall produce a compressive stress of 3 014 pounds per square inch on the specimen Normally the oscillograph is operated with weights only on the pen end of the beam, however, the moment of inertia of the system may be increased, with a corresponding decrease in the natural frequency of the system, by adding weights to the hook end of the beam

3.1.1.4 The beam may be checked for friction and balance by removing all weights and the specimen and disengaging the hook In this condition the beam should oscillate freely indicating minimum friction at the knife edges and the stabilizers of the lower platen When the beam comes to rest, the pen end should be below the horizontal This unbalanced condition helps overcome the friction between the pen and graph paper The beam shall be kept slightly overbalanced by adjusting the counterweight to apply a positive load to the specimen when all weights are removed The knife edges should be cleaned with cheesecloth and occasionally lubricated with typewriter oil to minimize friction and wear

3.1.2 An engineering grade of graph paper ruled in one-inch squares and then subdivided into 20 equal squares per inch

3.1.3 A circular metal die 1.625 ± 0.005 inch in inside diameter for cutting the specimen

3.1.4 A drill press or similar device for rotating the die

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the compression resistance of the specimen shall be determined at a compression of 25 percent of its original thickness

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4.2 Preparation of specimen.—If the material is too thick, it shall be sliced to the required thickness as described in method 1111 The specimen shall be cut by means of the die rotated in a drill press or similar device The pressure applied to the die shall be sufficiently small to keep "cupping" of the cut surfaces to a minimum In some cases, it may be necessary to freeze the cellular rubber before cutting the specimen in order to obtain parallel cut surfaces To facilitate cutting of the specimen with smooth-cut surfaces and square edges, the die may be lubricated with water containing a wetting agent and a corrosion inhibitor such as 0 5 percent sodium chromate or with silicone mold release emulsion before each specimen is cut If a lubricant is used, the specimen shall be permitted to dry before testing The circular bases of the specimen shall be parallel to each other and at right angles to the axis of the cylinder

4.3 The specimen shall be allowed to rest undeflected and undistorted for at least 12 hours before testing for compression resistance

4.4 The specimen shall be free from mechanical damage The thickness of the specimen shall be determined as described in method 12031, except that measurements shall be made in such a manner as to indicate the perpendicular distance between the center portion of the top and bottom faces and the value recorded to the nearest 0 001 or 0 005 inch, as applicable, as T

4.5 The balance beam of the oscillograph shall be locked in position by means of the hook at the left end of the machine and all weights removed The hook shall be adjusted so that the static equilibrium position of the balance beam will be approximately horizontal when the specimen is under the test deflection desired

4.6 The specimen shall be placed between the perforated plate and the depressor plate, the micrometer adjusted until it rests on the depressor plate without distorting the specimen, and the micrometer locked in this position by means of the set screw

4.7 The graph paper shall be placed on the chronograph drum and the position adjusted so that the zero position of the penpoint is on one of the horizontal lines of the paper

4.8 The hook shall be disengaged and sufficient pressure applied by hand on the pen end of the beam to compress the specimen about 30 percent of the required deformation, 4 1, and released This operation shall be repeated three times to remove any trapped air from the specimen

4.9 With the hook still disengaged, the drum chart shall be rotated by turning the chronograph drum to the left to displace the chart 4 to 5 small divisions to the left of the penpoint, thus making zero deflection

4.10 At least four deflection readings shall be obtained by applying approximately equal weights to the beam at intervals of one minute and recording the corresponding deflections The weights applied shall be selected to give deflection readings to include values on both sides of the required deflection, 4 1 One minute after the weight is applied, the oscillogram shall be rotated to the left by two small divisions and the deflection in divisions recorded as D The total weight in pounds that produced the deflection D shall be recorded as W

5. RESULTS

5.1 Calculation.

5.1.1 The percent deflection of the specimen for each weight shall be calculated as follows

$$\text{Deflection, percent, } = \frac{D \times 0\,5}{T}$$

Where

D is the deflection recorded on the oscillogram for each weight, W, divisions

T is the thickness of the original specimen, inch

5.1.2 The compressive stress of the specimen for each weight shall be calculated as follows

$$\text{Compressive stress, pounds per square inch, } = W \times 3\,014$$

Where

W is the total weight in pounds for each deflection, D

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.3 The average deflection in percent of the specimens tested for each weight shall be plotted against the average compressive

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strength in pounds per square inch of the specimens tested for each weight and a curve drawn through the points

5.4 The compression resistance of the test unit shall be the compressive strength required to produce 25-percent deflection as read from the curve

5.5 The compression resistance of the test unit shall be recorded to the nearest 0 1 pound per square inch

5.6 The percent the specimen was compressed shall be recorded

5.7 If a plied-up specimen is tested, it shall be recorded

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Method 12151
April 12, 1955

This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ–R–601a)

COMPRESSION DEFLECTION, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the compression deflection of cellular rubber compounds It is particularly applicable to sponge and expanded rubber

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1 and 2 1 2

2.1.1 The specimen shall be a circular cylinder cut with the die described in 3 1 3 of method 12141

2.1.2 Unless otherwise specified in the detail specification, the specimen shall be not less than ¼ inch and not more than 2 inches in thickness

2.2 Unless otherwise specified in the detail specification, materials thinner than ¼ inch shall be plied up to form a specimen as near ½ inch in thickness as practicable

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Testing machine* —A compression testing machine capable of compressing the specimen gently without impact at a uniform rate of 0 5 to 2 0 inches per minute The apparatus is equipped with two horizontal flat metal plates at least 2 0 inches in diameter The machine is equipped with a dial gage, or equivalent, which will indicate the deflection of the specimen to within ±0 001 inch The applied force in pounds is indicated to within ±1 percent by a dial or scale or other means The machine may be hand or motor driven

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be compressed 25 percent of its original thickness

4.2 The specimen shall be prepared as described in method 12141

4.3 The specimen shall be allowed to rest undeflected and undistorted for at least 12 hours before testing for compression deflection

4.4 The specimen shall be free from mechanical damage The thickness of the specimen shall be determined as described in method 12031 except that measurements shall be made in such a manner as to indicate the perpendicular distance between the center portion of the top and bottom faces and the value recorded to the nearest 0 001 or 0 005 inch, as applicable

4.5 The specimen shall be centered between the plates of the testing machine with the specimen just in contact with the compression plates The dial gage shall be set at zero and force applied to compress the specimen at the rate of 0 5 to 2 0 inches per minute until the specimen has been compressed the required amount, 4 1 The force required to compress the specimen the required amount shall be read from the dial or scale and the value recorded as F

5. RESULTS

5.1 **Calculation.**—The compression deflection of the specimen shall be calculated as follows

$$\text{Compression deflection, pounds per square inch,} = \frac{F}{2\ 074}$$

Where

F is the force required to deflect the specimen the required amount, pounds

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The compression deflection of the test unit shall be the average of the values obtained from the specimens tested

5.4 The compression deflection of the test unit shall be recorded to the nearest 0 1 pound per square inch

5.5 The percent the specimen was compressed shall be recorded

5.6 If a plied-up specimen is used, it shall be recorded

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AIR HEAT TEST, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the effect of air at atmospheric pressure and at an elevated temperature on cellular rubber The test consists of determining the amount of deterioration by comparing the values obtained for some specified characteristic on aged and unaged specimens from the same test unit Compression deflection, compression resistance, identation, and flexing endurance are the methods of test most commonly used In the first three methods the same specimen is tested both before and after aging When flexural endurance is used to determine the amount of deterioration it is necessary to use two specimens from the test unit One specimen is tested in the unaged condition and the other is exposed to specified conditions of deterioration and then tested for flexing endurance

2. SPECIMEN

2.1 The specimen shall be as described in method 12111, 12121, 12141, or 12151 as specified in the detail specification for determining the amount of deterioration

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus for exposing the specimen to air at an elevated temperature described in method 7221 or 7231

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be exposed in air at a temperature of 100°±2° C (212°±3 6° F) for 70°±¼ hour

4.2 Flexing endurance, method 12111, indentation, method 12121; compression resistance, method 12141, or compression deflection, method 12151, as specified in the detail specification shall be used for determining the amount of deterioration

4.3 The specimen shall be prepared as described in the method of test specified in the detail specification for determining the amount of deterioration

4.4 The specified characteristic of the specimen, 4 2, shall be determined as described in the applicable method of test and the value recorded as O

4.5 When indentation, compression resistance, or compression deflection is specified in the detail specification for determining the amount of deteri ation, the specimen that was tested as required in 4 4 shall be exposed in air as described in method 7221 or 7231, except that the time and temperature of exposure shall be as required in 4 1

4.6 When flexing endurance, method 12111, is specified in the detail specification for determining the amount of deterioration, a second specimen shall be taken from the test unit undergoing the test This specimen shall be exposed in air as described in method 7221 or 7231, except that the time and temperature of exposure shall be as required in 4 1

4.7 At the end of the aging period, the specimen shall be immediately removed from the aging chamber and set aside to rest for not less than 16 hours and not more than 96 hours at room temperature before determining the required property

4.8 At the end of the rest period, the specified characteristic, 4 2, of the aged specimen shall be determined as described in the applicable method of test and the value recorded as E

5. RESULTS

5.1 Unless otherwise specified in the detail specification, the number of specimens subjected to the aging test from each test unit shall be as required in the method of test specified for determining the deterioration of the material

5.2 Calculation.—The change in indentation, compression resistance, compression deflection, or flexing endurance of the test unit due to aging shall be calculated as follows

$$\text{Change in characteristic, percent} = \frac{O-E}{O} \times 100$$

Where

O is the characteristic of the test unit before aging
E is the same characteristic of the test unit after aging

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5.3 The change in indentation, compression resistance, compression deflection, or flexing endurance of the test unit shall be recorded to the nearest one percent

5.4 The temperature and time of aging and the property used for determining the deterioration shall be recorded.

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DEFLECTION AT LOW TEMPERATURE, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the effect of low temperature exposure on cellular rubber compounds The test consists of determining the load necessary to compress the specimen a specified amount, exposing the specimen to a low temperature, and determining the amount the specimen is deflected under the same load while the specimen is still at low temperature

2. SPECIMEN

2.1 The specimen shall be as described in method 12141 or 12151, as specified in the detail specification for determining the effect of low temperature exposure

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Low-temperature cabinet* —A cabinet or testroom with suitable controls for maintaining the required temperature throughout the cabinet or room during the exposure period and of sufficient size to hold the compression apparatus described in method 12151 or 12141 The cabinet is equipped with facilities such as glass windows, handholes with insulated sleeves, etc , that permit operating the testing apparatus at the required temperature If the apparatus described in method 12141 is used for determining the effects of low-temperature exposure, means should be provided for extending the pen arm of the oscillograph through the cabinet so that the deflection is recorded on the chart located outside of the cabinet

3.1.2 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature within ±1 0° C (1 8° F)

3.1.3 A desiccator and calcium chloride or other suitable desiccant

4. PROCEDURE

4.1 The specimen shall be exposed at a temperature of −40°±2° C (−40°±3 6° F) or −55°±2° C (−67°±3 6° F), as specified in the detail specification

4.2 Unless otherwise specified in the detail specification, the specimen shall be exposed at the low temperature for 5±¼ hour

4.3 Compression resistance, method 12141, or compression deflection, method 12151, as specified in the detail specification, shall be used for determining the effects of exposure to low temperature

4.4 Unless otherwise specified, in the detail specification, the specimen shall be compressed 25 percent of its original thickness

4.5 The specimen shall be prepared as described in method 12141

4.6 Compression resistance or compression deflection of the specimen, as specified in the detail specification, shall be determined as described in method 12141 or 12151, respectively, and the value recorded The specimen shall be compressed as required in 4 4

4.7 The specimen that was tested as required in 4 6 shall be dried in a desiccator over calcium chloride or other suitable desiccant for at least 16 hours

4.8 The dried specimen shall be placed in the low-temperature cabinet or testroom in a medium of air or a mixture of air and carbon dioxide and maintained for the required time at the required temperature The compression testing apparatus shall be placed in the cabinet or room at least one hour before compression resistance or compression deflection is to be determined so as to permit it to reach temperature equilibrium

4.9 At the end of the exposure period, with the specimen and testing apparatus at the specified low temperature, the force previously determined in 4 6 to compress the specimen the required amount shall be immediately applied to the specimen as required in method 12141 or 12151, as applicable, and the deflection or the amount the specimen is compressed by the load recorded as E

5. RESULTS

5.1 **Calculation.**—The change in deflection or the change in the amount the specimen is

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compressed due to exposure to low temperature shall be calculated as follows

Change in deflection or amount of compression,

$$\text{percent, } = \frac{O-E}{O} \times 100$$

Where

O is the deflection or the amount the specimen is compressed before exposure, percent, 4 4

E is the deflection or the amount the specimen is compressed at low temperature, percent, 4 9

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.3 The change in deflection or the amount of compression of the test unit shall be the average of the values obtained from the specimens tested

5.4 The change in the deflection or the amount of compression of the test unit shall be recorded to the nearest one percent

5.5 The temperature and time of exposure of the specimen at low temperature shall be recorded

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AIR PRESSURE TEST, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the effect of air under pressure and at an elevated temperature on cellular rubber compounds The method is particularly applicable to sponge rubber

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit of any convenient size to permit visual examination

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus for exposing the specimen to air under pressure and at elevated temperature described in method 7211

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be exposed at a temperature of 127°±2° C (260 6°±3 6° F) and an air pressure of 60±2 pounds per square inch for 2 0±0 1 hour

4.2 The specimen shall be free from mechanical damage The specimen shall be aged as described in method 7211, except that the air pressure and time and temperature of aging shall be as required in 4 1

4.3 At the end of the exposure period, the aged specimen shall be set aside to rest for at least one hour at room temperature before visual examination is made

4.4 At the end of the rest period, the specimen shall be examined for hardening, brittleness, tackiness, or any other evidence of deterioration

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 The number of specimens tested from each test unit and the number from each test unit that showed hardening, brittleness, tackiness, or any other evidence of deterioration shall be recorded

5.3 The temperature, air pressure, and time of aging shall be recorded

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OIL IMMERSION TEST, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the change in volume of cellular rubber compounds due to immersion in oil

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1 and 2 1 2

2.1.1 The specimen shall be a circular cylinder cut with the die described in 3 1 3 of method 12141

2.1.2 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test but shall not exceed ½ inch

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in method 6111

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed at a temperature of 100°±2° C (212°±3 6° F) for 70±¼ hour

4.2 The test medium shall be as specified in the detail specification, method 6001

4.3 The specimen shall be prepared as described in method 12141

4.4 The specimen shall be free from mechanical damage The average diameter of the specimen shall be determined as described in method 12041, except that 4 measurements (2 sets) shall be made The thickness of the specimen shall be determined as described in method 12031, except that one measurement of the specimen shall be made The volume of the specimen shall be calculated from the thickness and the average diameter and the value recorded as V_1

4.5 A volume of immersion medium equal to five times the volume of the specimen shall be added to the container After adjusting the liquid to the required temperature, the specimen shall be completely immersed in the liquid so that the liquid will circulate freely around the specimen during the immersion period

4.6 If the immersion medium is appreciably volatile at the temperature required for immersion, the container shall be fitted with the air or reflux condenser to maintain a constant volume If the condenser is not needed, the container shall be closed with a solid stopper, fitted to provide escape for any excess pressure developed and to prevent excessive evaporation The specimen shall remain immersed in the medium for the required time at the required temperature

4.7 At the end of the immersion period, the specimen shall be immediately removed from the liquid If the immersion medium is above room temperature, the specimen shall be placed immediately in a fresh supply of the same immersion medium at room temperature, and allowed to remain for 30±5 minutes

4.8 Immediately after removal from the immersion medium or from the cooling liquid, the specimen shall be quickly dipped into acetone and blotted lightly with filter paper or other absorbent material The diameter and thickness of the specimen shall again be determined, the volume calculated as described in 4 4, and the value recorded as V_2 The measurements for diameter and thickness, respectively, shall be made as near as practicable at the same places as measured in 4 4

5. RESULTS

5.1 Calculation.—The change in volume of the specimen due to immersion shall be calculated as follows

$$\text{Volume change, percent,} = \frac{V_2 - V_1}{V_1} \times 100$$

Where

V_1 is the volume of the specimen before immersion
V_2 is the volume of the specimen after immersion

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.3 The change in volume of the test unit shall be the average of the values obtained from the specimens tested

5.4 The change in volume of the test unit shall be recorded to the nearest one percent

5.5 The temperature and time of immersion and the immersion medium used shall be recorded

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WATER ABSORPTION, CELLULAR RUBBER

1. SCOPE

1.1 This method is intended for use in determining the change in weight of expanded rubber compounds due to immersion in water The method is not recommended for use in testing open-cell rubber compounds unless they are completely encased in an added skin

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1 through 2 1 3

2.1.1 The specimen shall be a circular cylinder cut with the die described in 3 1 3 of method 12141

2.1.2 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test but shall not exceed ½ inch

2.1.3 Unless otherwise specified in the detail specification, the specimen shall have the natural skin on the top and bottom surfaces

3 APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Analytical balance and weights

3.1.2 Glass container at least 8 inches deep and of sufficient width to accommodate the specimen

3.1.3 Filter paper or other absorbent material

3.1.4 Distilled water

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be immersed in distilled water at a temperature of 23° ± 2° C (73 4° ± 3 6° F) for 22 ± ¼ hour

4.2 The specimen shall be prepared as described in method 12141

4.3 The specimen shall be free from mechanical damage The specimen shall be weighed and the weight recorded to the nearest milligram as W_1

4.4 Distilled water shall be placed in the container and the specimen immersed to a depth of 6 inches below the surface and maintained for the required time at the required temperature At the end of the immersion period, the specimen shall be immediately removed from the water, blotted with filter paper or other absorbent material, dried off with air blow, weighed, and the weight recorded to the nearest milligram as W_2

5. RESULTS

5.1 Calculation.—The water absorption of the specimen due to immersion shall be calculated as follows

$$\text{Water absorption, percent,} = \frac{W_2 - W_1}{W_1} \times 100$$

Where

W_1 is the weight of the specimen before immersion, grams

W_2 is the weight of the specimen after immersion in water, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3 The water absorption of the test unit shall be the average of the values obtained from the specimens tested

5.4 The water absorption of the test unit shall be recorded to the nearest one percent

5.5 The time and temperature of immersion shall be recorded

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TAPE, GENERAL

1. SCOPE

1.1 This group of methods is intended for use in determining the various properties of friction and rubber insulating tapes Methods are described for determining the breaking strength, adhesion, tackiness, pinholes, dielectric strength, and sulfur content of friction tape; and for tensile strength, elongation, fusion, tackiness, and dielectric strength of rubber insulating tape

1.2 Friction tape is used for protecting and binding in place insulation applied to joints of electrical wires and cables and for other electrical and mechanical purposes Friction tape consists of a cotton sheeting impregnated with a tacky adhesive insulating compound and cut into narrow widths The impregnating compound contains practically no free sulfur or other substance which would have a deteriorating effect on copper or other metals or on the fabric

1.3 Rubber insulating tape is used for the insulation of joints in electrical wires and cables and for other insulating purposes Rubber insulating tape consists of an unvulcanized or partially vulcanized rubber compound with a low free sulfur content

2. SPECIMEN

2.1 At least 2 feet of tape shall be discarded from the outside of each roll selected for test before taking the test specimen

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BREAKING STRENGTH, FRICTION TAPE

1. SCOPE

1.1 This method is intended for use in determining the breaking strength of friction tape

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit at least 16 inches in length

2.1.1 The thickness and width of the specimen shall be the thickness and width of the tape undergoing test

3. APPARATUS

3.1 The apparatus shall consist of the testing machine described in method 4111, equipped with grips that will hold the specimen without slipping or injury during the application of the force

4. PROCEDURE

4.1 The specimen shall be free from mechanical damage The minimum width of the portion of the specimen between the grips of the testing machine shall be determined as described in method 2121 and the value recorded to the nearest $\frac{1}{64}$ inch as W

4.2 The distance between the grips of the testing machine shall be set at $12 \pm \frac{1}{4}$ inch at the start of the test The specimen shall be placed in the grips of the testing machine and adjusted symmetrically so that the tension will be distributed uniformly over the cross section If the tension is greater on one side than on the other, the maximum strength of the specimen will not be developed The machine shall be started and the power-actuated grip shall travel at a uniform speed of 20 ± 1 inch per minute until the specimen ruptures After rupture of the specimen, the breaking force in pounds shall be read from the dial, scale, or automatic recorder and the value recorded as F

5. RESULTS

5.1 Calculation.—The breaking strength of the specimen shall be calculated as follows

$$\text{Breaking strength, pounds per inch of width,} = \frac{F}{W}$$

Where

F is the breaking force of the specimen, pounds
W is the minimum width of the specimen, inches

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested

5.3 The breaking strength of the test unit shall be the average of the values obtained from the specimens tested

5.4 The breaking strength of the test unit shall be recorded to the nearest one pound per inch of width

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TENSILE STRENGTH, INSULATING TAPE

1. SCOPE

1.1 This method is intended for use in determining the tensile strength of rubber insulating tape The definition of tensile strength is given in method 4001

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit as described in 2 1 1 and 2 1 2

2.1.1 The specimen shall be dumbbell in shape and cut with die I described in method 4111

2.1.2 The thickness of the specimen shall be the thickness of the tape undergoing test

2.2 If the tape to be tested is less than 1 inch in width, the ends of the specimen need not be the full 1-inch width of the die

3. APPARATUS

3.1 The apparatus shall be as described in method 4111

4. PROCEDURE

4.1 Preparation of specimen.—The specimen shall be prepared as described in method 4111 for a dumbbell specimen, except that it shall be cut from the tape before the separator is removed and shall not be buffed

4.2 The specimen shall be free from mechanical damage The cross-sectional area and tensile strength of the specimen shall be determined as described in method 4111 for a dumbbell specimen

5. RESULTS

5.1 The results shall be as described in method 4111 for a dumbbell specimen

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ELONGATION, INSULATING TAPE

1. SCOPE

1.1 This method is intended for use in determining the elongation of rubber insulating tape The definition of elongation is given in method 4001

2. SPECIMEN

2.1 The specimen shall be as described in method 13021

3. APPARATUS

3.1 The apparatus shall be as described in methods 4111 and 4121

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the elongation shall be determined on the same specimen during the same test as tensile strength.

4.2 Preparation of specimen.—The specimen shall be prepared as described in method 4111 for a dumbbell specimen, except that it shall be cut from the tape before the separator is removed and shall not be buffed

4.3 The specimen shall be free from mechanical damage Bench marks 2 inches apart shall be placed on the specimen as described in method 4121 for a dumbbell specimen. The specimen shall be placed in the grips of the testing machine, the machine started, and the force applied The remainder of the procedure shall be as described in 4 3 1 of method 4121

5. RESULTS

5.1 The results shall be as described in method 4121 for a dumbbell specimen

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ADHESION, FRICTION TAPE

1. SCOPE

1.1 This method is intended for use in determining the adhesion between adjacent layers of friction tape as taken from the roll

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 23 inches in length

2.1.1 The thickness and width of the specimen shall be the thickness and width of the tape undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Adhesion tester* The tester consists of a mandrel ¼ inch in diameter with a slot approximately 1/16 inch in width and long enough to accommodate the full width of the specimen The mandrel is mounted in a level position in ball bearings so that it turns freely under a weight of ¼ ounce suspended from a cotton thread wound in a single layer around the center of the mandrel A suitable apparatus is shown in figure 13111

3.1.2 Weights that will provide forces of 4 and 10 pounds per inch of width of the tape undergoing test

3.1.3 A grip for attaching the weights to the tape.

3.1.4 Stopwatch or other timing device that will indicate the time in minutes and seconds

3.1.5 A scale or tape graduated to 1/32 inch

4. PROCEDURE

4.1 The specimen shall be tested immediately after removing it from the roll During the preparation and testing of the specimen, the surface to be tested shall not come in contact with the hands or foreign matter

4.2 The specimen shall be free from mechanical damage The average width of the specimen shall be determined as described in method 2121 and the value recorded to the nearest 1/64 inch

4.3 One end of the specimen shall be inserted in the slot of the mandrel and 2 inches of the tape wound on the mandrel in such a manner that the layers of tape are superimposed on each other A load of 10 0±0 1 pound per inch of width shall then be attached to the other end of the specimen by means of the grip and 19 inches more of the specimen shall be wound around the mandrel at a uniform rate of 12±1 inch per minute in such a manner that the layers are superimposed on each other The specimen shall be allowed to remain wound around the mandrel, with the 10-pound weight attached, for 3 minutes±5 seconds

4.4 At the end of the 3-minute period, the 10-pound weight shall be replaced immediately with a weight of 4 00±0 05 pound per inch of width of the specimen and the tape allowed to unwind After 2 inches have unwound, the length of tape that unwinds in the next 60±1-

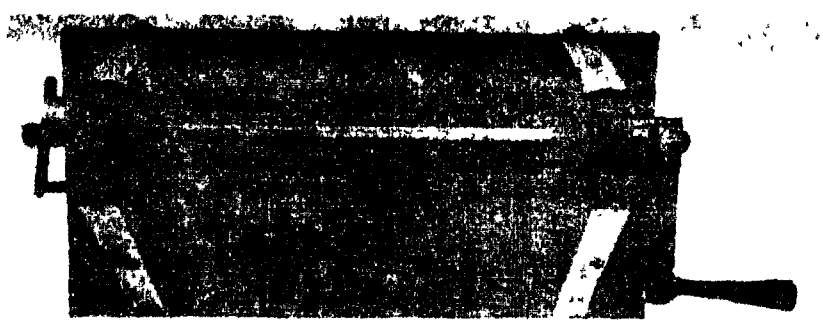

FIGURE 13111 –Apparatus for adhesion test

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second period shall be measured with the tape or scale and the value recorded to the nearest ½₂ inch The adhesion of the specimen shall be the length of tape that unwound in the 60-second period.

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested.

5.2 When one specimen is tested from each test unit, the adhesion of the test unit shall be the value obtained from the specimen tested. When more than one specimen is tested from each test unit, the adhesion of the test unit shall be the average of the values obtained from the specimens tested

5.3 The adhesion of the test unit shall be recorded to the nearest ½₂ inch

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ADHESION, FRICTION TAPE, AGED

1. SCOPE

1.1 This method is intended for use in determining the effect of air at atmospheric pressure and at an elevated temperature on the adhesion of friction tape

2. SPECIMEN

2.1 The specimen shall be as described in method 13111

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 Apparatus described in method 13111, except that a 3-pound weight replaces the 4-pound weight

3.1.2 Apparatus described in method 7221 for exposing the specimen to air at an elevated temperature

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specimen shall be aged in air at atmospheric pressure and at a temperature of $100° \pm 2°$ C ($212° \pm 3.6°$ F.) for $16 \pm \frac{1}{4}$ hour

4.2 The specimen shall be free from mechanical damage The average width of the specimen shall be determined as described in method 2121 and the value recorded to the nearest $\frac{1}{64}$ inch

4.3 After adjusting the aging chamber to the required temperature, the specimen shall be supported in the aging chamber by means of clips or other devices attached to the ends so that the adhesive surface of 19 consecutive inches of the specimen will hang free and not come in contact with the hands, the aging chamber, other tape specimens, or foreign material The specimen shall remain in the aging chamber for the required time at the required temperature 4 1 At the end of the aging period, the specimen shall be removed from the aging chamber and allowed to rest at room temperature for not less than 4 hours nor more than 8 hours During the rest period, the specimen shall not come in contact with the hands or foreign material

4.4 At the end of the rest period, the adhesion of the specimen shall be determined as described in method 13111, except that the weight applied to unwind the tape shall be 3.00 ± 0.05 pound per inch of width in place of 4.00 pounds per inch of width The adhesion of the specimen shall be the length of the tape that unwound in the 60-second period after the first 2 inches have unwound under a load of 3 pounds per inch of width

5. RESULTS

5.1 The results shall be as described in method 13111

5.2 The time and temperature of aging shall be recorded

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FUSION, INSULATING TAPE

1. SCOPE

1.1 This method is intended for use in determining the adhesion between adjacent layers of rubber insulating tape

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 7 inches in length

2.1.1 The thickness and width of the specimen shall be the thickness and width of the tape undergoing test

3. APPARATUS

3.1 The apparatus shall be as described in method 13111

4. PROCEDURE

4.1 The specimen shall be tested immediately after removing it from the roll During the preparation and testing of the specimen, the surfaces to be tested shall not come in contact with the hands or foreign matter

4.2 The specimen shall be free from mechanical damage The average width of the specimen shall be determined as described in method 2121 and the value recorded to the nearest ¼ inch

4.3 One end of the specimen shall be inserted in the slot of the mandrel and 2 inches of the tape wound on the mandrel in such a manner that the layers of tape are superimposed on each other A load of 4 00 ± 0 05 pound per inch of width of the tape shall be attached to the other end of the specimen by means of the grip If this weight is insufficient to elongate the tape 200 percent or more, additional downward force shall be applied by hand until an elongation of 200 percent has been produced The specimen shall then be wound around the mandrel at a uniform rate of 12 ± 1 inch per minute in such a manner that the layers are superimposed on each other. The specimen shall be allowed to remain wound around the mandrel for 3 minutes ± 5 seconds with the 4-pound weight attached.

4.4 At the end of the 3-minute period, with the 4-pound weight still attached, the tape shall be allowed to unwind slowly and the number of turns the mandrel makes during the first 120 ± 1 second recorded to the nearest ¼ turn During the 120 seconds of unwinding, the tape shall be observed to determine whether it sustains the load or breaks If the specimen breaks, the location of the break, that is, at the point of unwinding or in the free length shall be recorded

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 When one specimen is tested from each test unit and does not break, the adhesion of the test unit shall be the value obtained from the specimen tested When more than one specimen is tested from each test unit without breaking, the adhesion of the test unit shall be the average of the values obtained from the specimens tested

5.3 The fusion of the test unit if the specimen does not break shall be recorded to the nearest ¼ turn

5.4 If any of the specimens break, the number of specimens tested from each test unit, the number that break, and the location of the break shall be recorded

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TACKINESS, FRICTION TAPE

1. SCOPE

1.1 This method is intended for use in determining the tackiness of friction tape, that is, the ability of tape to adhere to itself

2. SPECIMEN

2.1 The specimen shall consist of 2 portions of the test unit, each 8 inches in length

2.1.1 The thickness and width of the specimen shall be the thickness and width of the tape undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Roller* —The roller, not less than ½ inch and not more than 1 inch in diameter, is made of glass or chromium-plated metal The length is at least ¼ inch greater than the width of the tape to be tested The roller weighs 9 0±0 5 gram for testing tape ¾ inch in width and weighs proportionately more or less for wider or narrower tapes, respectively

3.1.1.1 A suitable apparatus may be made from a squeegee handle and a roller of proper weight and dimensions The shaft at the end of the squeegee handle may be used to propel the roller by coming in contact with the inside surface of the roller along the horizontal diameter Small projections or feet at the ends of the shaft and perpendicular to the plane of the handle rest on the smooth surface on which the tests are made These prevent any downward pressure on the tape except the weight of the roller

3.1.2 Two pairs of tweezers for handling the specimen.

3.1.3 A piece of plate glass or other clean, smooth horizontal surface for supporting the specimen

3.1.4 A stopwatch or other timing device that will indicate the time in seconds

3.1.5 A scale or tape graduated to ⅛ inch.

4. PROCEDURE

4.1 The specimen shall be tested immediately after removing it from the roll During preparation and testing, the specimen shall be handled with tweezers so that the surface to be tested shall not come in contact with the hands or foreign matter

4.2 One of the 8-inch pieces of the specimen shall be placed lightly on a clean, smooth, horizontal surface, such as plate glass The second length shall be placed evenly on the first piece, allowing a ½-inch overlap lengthwise at one end Light pressure shall be applied by rolling the roller, 3 1 1, over the specimen one time at a uniform rate of about 1 inch per second The roller shall be applied so as to avoid the application of any force to the specimen other than the weight of the roller

4.3 The specimen shall be picked up by the end last in contact with the roller, using the two pairs of tweezers, one for each piece of the specimen, and the two pieces separated immediately by pulling them apart In pulling the pieces apart, one hand shall rest on a rigid support, such as a table edge, and the other hand moved slowly along the edge of the table to bring about separation of the pieces The rate of separation shall be uniform and shall not exceed one inch per second The last inch of the specimen shall be discarded The relative position of the two pieces of the specimen and the tweezers shall be observed Tackiness is indicated in combinations (1) and (2) of 4 4 if the point of separation remains in the same horizontal position as the tweezers Tackiness is indicated in combination (3) of 4 4 if the point of separation travels along the pieces when pulled apart

4.4 Unless otherwise specified in the detail specification, the tape shall be tested in each of the following combinations (1) front-to-front, (2) back-to-back, and (3) back-to-front

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5. RESULTS

5.1 Unless otherwise specified in the detail specification, three specimens for each combination in 4.4 from each test unit shall be tested

5.2 The number of specimens tested for each combination from each test unit and the number for each combination in 4 4 from each test unit that failed to show tackiness shall be recorded

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TACKINESS, INSULATING TAPE

1. SCOPE

1.1 This method is intended for use in determining the tackiness of rubber insulating tape, that is, the ability of the tape to adhere to itself

2. SPECIMEN

2.1 The specimen shall be described in method 13131

3. APPARATUS

3.1 The apparatus shall be as described in method 13111.

4. PROCEDURE

4.1 The procedure shall be as described in method 13131, except that instead of leaving the 4-pound weight suspended for 3 minutes it shall be removed immediately and the specimen observed to determine whether it shows any tendency to unwind from the mandrel

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 The number of specimens tested from each test unit and the number from each test unit that showed any tendency to unwind from the mandrel shall be recorded

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PINHOLES, FRICTION TAPE

1. SCOPE

1.1 This method is intended for use in determining the uniformity of coating on friction tape by determining the number of small holes in a given length of tape

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit 36 inches in length

2.1.1 The thickness and width of the specimen shall be the thickness and width of the tape undergoing test

3. APPARATUS

3.1 The apparatus shall consist of a box approximately 8 inches in height by 8 inches in width by 18 inches in length, inside dimensions The box shall contain a slot in the top, 2 inches in width and 12 inches in length The box shall be painted white inside and illuminated with a 25-watt lamp The slot shall be covered with a clear glass set flush with the top of the box The apparatus shall be equipped with an adjustable cover to limit the light through the slot to the width of the tape being tested

4. PROCEDURE

4.1 The specimen shall be passed slowly over the slot in the top of the illuminated box The pinholes or openings permitting light to pass through the specimen shall be counted and the value recorded Any ruptures in the specimen due to cuts or tears shall not be counted

5. RESULTS

5.1 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested.

5.2 The maximum number of pinholes in any single specimen from the test unit shall be recorded

5.3 The number of specimens tested from each test unit and the total number of pinholes in all of the specimens from each test unit shall be recorded

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DIELECTRIC STRENGTH, FRICTION TAPE

1. SCOPE

1.1 This method is intended for use in determining the voltage required to produce electrical breakdown of friction tape

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit approximately 6 inches in length

2.1.1 The thickness and width of the specimen shall be the thickness and width of the tape undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows:

3.1.1 *Electrodes*.—Two flat electrodes ¼ inch in width and 4¼ inches in length The edges of the electrodes shall be square and the ends rounded to ⅛-inch-radius cylinders whose axes are parallel to the contact faces of the electrodes, thus giving an area of contact with the surfaces of the specimen of exactly ¼ inch in width by 4 inches in length

3.1.2 *Electrical equipment*.—Electrical equipment having a capacity of at least 2 kilovolt-amperes which will furnish a 60-cycle alternating current voltage of practically sine wave form Means shall be provided for increasing the voltage at a rate of 100 volts per second during the test

4. PROCEDURE

4.1 The specimen shall be free from mechanical damage and shall be placed between the electrodes under an electrode pressure of one pound per square inch A 60-cycle alternating-current voltage of practically sine wave form shall be applied to the specimen at a value not to exceed 100 volts The voltage shall be raised at the rate of 100 volts per second until puncture of the specimen occurs and the breakdown voltage recorded In order to prevent flashover, the width of the specimen may be increased by attaching to each side of the specimen an added piece of tape, making a ⅛-inch lap seam carefully rolled down

5. RESULTS

5.1 Unless otherwise specified in the detail specification, five specimens from each test unit shall be tested

5.2 The dielectric strength of the test unit shall be the average of the values obtained from the specimens tested.

5.3 The breakdown voltage of the test unit shall be recorded to the nearest 100 volts

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DIELECTRIC STRENGTH, INSULATING TAPE

1. SCOPE

1.1 This method is intended for use in determining the voltage required to produce electrical breakdown of rubber insulating tape

2. SPECIMEN

2.1 The specimen shall be as described in method 13311

3. APPARATUS

3.1 The apparatus shall be as described in method 13311, except that means shall be provided for increasing the voltage at the rate of 1,000 volts per second during the test

4. PROCEDURE

4.1 The specimen shall be free from mechanical damage and shall be placed between the electrodes under an electrode pressure of one pound per square inch A 60-cycle alternating-current voltage of practically sine wave form shall be applied to the specimen at a value not to exceed 1,000 volts. The voltage shall be increased at a rate of 1,000 volts per second until puncture of the specimen occurs, and the breakdown voltage recorded to the nearest 50 volts as V. In order to prevent flashover, the width of the specimen may be increased by attaching to each side of the specimen an added piece of tape, making a ⅛-inch lap seam carefully rolled down

4.2 The thickness of the specimen shall be determined as described in method 2011 except that one measurement shall be made at the point of puncture of the specimen The thickness shall be recorded to the nearest mil (0 001 inch) as T.

5. RESULTS

5.1 **Calculation.**—The dielectric strength of the specimen shall be calculated as follows.

$$\text{Dielectric strength, volts per mil} = \frac{V}{T}$$

Where

V is the voltage required to rupture the specimen
T is the thickness of the specimen at the point of puncture, mils

5.2 Unless otherwise specified in the detail specification, five specimens from each test unit shall be tested

5.3 The dielectric strength of the test unit shall be the average of the values obtained from the specimens tested

5.4 The dielectric strength of the test unit shall be recorded to the nearest 50 volts per mil of thickness.

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SULFUR, FRICTION TAPE

1. SCOPE

1.1 This method is intended for use in determining the presence of sulfur in friction tape

2. SPECIMEN

2.1 The specimen shall consist of a portion of the test unit of sufficient length to permit five superimposed layers to be wound around a copper rod ¼ inch in diameter

2.1.1 The thickness and width of the specimen shall be the thickness and width of the tape undergoing test

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 A copper rod approximately ¼ inch in diameter and 3 inches in length

3.1.2 A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven during the exposure period

3.1.3 A test tube 4 inches in length and approximately ¾ inch in diameter

3.1.4 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature of the oven to within ±1 0° C (1 8° F)

4. PROCEDURE

4.1 The specimen shall be exposed in the oven at a temperature of 100° ± 2° C (212° ± 3 6° F) for 16 ± ¼ hour

4.2 The copper rod shall be cleaned to a smooth bright surface Five superimposed layers of the specimen shall be wound around the copper rod, and the copper rod and specimen sealed in the test tube

4.3 The test tube containing the specimen shall be placed in the oven and maintained for the required time at the required temperature At the end of the exposure period, the copper rod shall be removed from the test tube, the tape removed, and the copper surface examined for the characteristic blue or black coloration, indicating the presence of copper sulfide

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 The number of specimens tested from each test unit and the number from each test unit that showed the presence of sulfur shall be recorded

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SPECIFIC GRAVITY, HYDROSTATIC

1. SCOPE

1.1 This method is intended for use in determining the specific gravity of rubber compounds through the loss in weight of the specimen when suspended in water. It is applicable to materials that are in the form of one piece This method is more rapid than method 14021 In the case of dispute between this method and method 14021, method 14011 shall be used for determining the specific gravity for use in determining the rubber content of rubber compounds

2. SPECIMEN

2.1 Rubber content.—The specimen shall consist of approximately one gram of material cut from a portion of the subsample prepared for specific gravity tests as described in 4 2 1

2.2 Other cases.—The specimen shall consist of a portion of the test unit weighing approximately one gram

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An analytical balance and weights

3.1.2 A fine wire approximately 0 004 inch in diameter or other equivalent material for supporting the specimen when weighed in distilled water

3.1.3 Ethyl alcohol, 95 percent

3.1.4 Distilled water

3.1.5 Filter paper or other absorbent material

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specific gravity shall be determined at a temperature of 25.0° ± 0 5° C. (77 0° ± 0 9° F), unless the coefficient of expansion of the rubber compound is known, in which case the determination may be made at any convenient temperature and the value corrected to 25 0° C (77 0° F)

4.2 Preparation of specimen.

4.2.1 *Specific gravity for rubber content* —A subsample for specific gravity tests shall be prepared from the portions designated for the composite sample in method 1211 The number of portions to make up the subsample shall be selected at random from all the portions designated for the composite sample, as follows

Number of portions designated for composite sample	Number of portions in subsample for specific gravity
2 to 10	2
11 to 20	3
21 to 30	4
31 to 40	5
41 to 50	6
51 to 60	7
61 to 70	8
71 to 80	9

One specimen of the required size, 2 1, for specific gravity test shall be taken from each of the portions in the subsample (second column)

4.2.2 *Specific gravity for other purposes* —A specimen of the required size, 2 2, shall be taken from the test unit

4.3 The specimen shall be weighed in air on the analytical balance, and the weight recorded to the nearest 0 1 mg as W_1 The wire for suspending the specimen shall be attached to one arm of the balance, weighed in distilled water, and the weight recorded to the nearest 0 1 mg as W_2, noting the depth to which the wire was immersed The specimen shall be dipped in 95 percent alcohol and blotted with filter paper or other absorbent material to remove the alcohol

4.4 The specimen shall be suspended from the arm of the balance by means of the wire, weighed in distilled water, and the weight recorded to the nearest 0.1 mg as W_3 The latter weighing shall be made with the wire immersed in the water to the same depth as when the wire was weighed without the specimen attached

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5. RESULTS

5.1 Calculation.—The specific gravity of the specimen at 25/4° C (77/39 2° F) shall be calculated as follows

$$\text{Specific gravity (25/4° C)} = \frac{W_1}{W_1-(W_3-W_2)} \times 0\ 9971$$

Where

W_1 is the weight of the specimen in air, grams
W_2 is the weight of the supporting wire in distilled water, grams
W_3 is the weight of the specimen and supporting wire in distilled water, grams

5.2 Rubber content.

5.2.1 One specimen from each portion of the subsample, 4 2 1, shall be tested

5.2.2 The specific gravity of the sample for test shall be the average of the values obtained from the specimens tested

5.3 Other cases.

5.3.1 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3.2 The specific gravity of the test unit shall be the average of the values obtained from the specimens tested

5.4 The specific gravity of the test unit or sample for test shall be recorded to the nearest 0 001 unit

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SPECIFIC GRAVITY, PYCNOMETER

1. SCOPE

1.1 This method is intended for use in determining the specific gravity of rubber compounds by use of the pycnometer It is applicable to materials that are not in the form of a continuous piece This method is not as rapid as method 14011

2. SPECIMEN

2.1 Rubber content.—The specimen shall consist of approximately one gram of material cut from a portion of the subsample prepared for specific gravity tests as described in 4 2 1

2.2 Other cases.—The specimen shall consist of a portion of the test unit weighing approximately one gram

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 An analytical balance and weights

3.1.2 A pycnometer

3.1.3 Ethyl alcohol, 95 percent

4. PROCEDURE

4.1 Unless otherwise specified in the detail specification, the specific gravity determination shall be determined at a temperature of 25 0° ± 0 5° C (77 0° ± 0 9° F), unless the coefficient of expansion of the rubber compound is known, in which case the determination may be made at any convenient temperature and the value corrected to 25 0° C (77 0° F)

4.2 Preparation of specimen.

4.2.1 *Specific gravity for rubber content* —A subsample for specific gravity tests shall be prepared from the portions designated for the composite sample in method 1211 The number of portions to make up the subsample shall be taken at random from all the portions designated for the composite sample, as follows

Number of portions designated for composite sample	Number of portions in subsample for specific gravity
2 to 10	2
11 to 20	3
21 to 30	4
31 to 40	5
41 to 50	6
51 to 60	7
61 to 70	8
71 to 80	9

One specimen of the required size, 2 1, for specific gravity test shall be taken from each of the portions in the subsample (second column)

4.2.2 *Specific gravity for other purposes* —A specimen of the required size, 2 2, shall be taken from the test unit

4.3 The specimen shall be weighed in air on the analytical balance and the weight recorded to the nearest 0.1 mg as W_1 The pycnometer shall be filled with 95-percent ethyl alcohol, weighed, and the weight recorded to the nearest 0 1 mg as W_2 The specimen shall be placed in the empty pycnometer, the pycnometer filled with alcohol, the whole weighed, and the weight recorded to the nearest 0 1 mg as W_3

5. RESULTS

5.1 Calculation.—The specific gravity of the specimen at 25°/4° C (77°/39 2° F) shall be calculated as follows

$$\text{Specific gravity } (25°/4° \text{ C}) = 0\,9971 \times \frac{W_1}{W_1 - (W_3 - W_2)} \times 8$$

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Where

W_1 is the weight of the specimen in air, grams.
W_2 is the weight of the pycnometer filled with alcohol, grams
W_3 is the weight of the pycnometer filled with specimen and alcohol, grams.
S is the specific gravity of alcohol at 25°/25° C

5.2 Rubber content.

5.2.1 One specimen from each portion of the subsample, 4 2 1, shall be tested

5.2.2 The specific gravity of the sample for test shall be the average of the values obtained from the specimens tested

5.3 Other cases.

5.3.1 Unless otherwise specified in the detail specification, two specimens from each test unit shall be tested

5.3.2 The specific gravity of the test unit shall be the average of the values obtained from the specimens tested.

5.4 The specific gravity of the test unit or sample for test shall be recorded to the nearest 0.0001 unit

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ABRASION

1. SCOPE

1.1 This method is intended for use in determining the resistance to abrasion of soft vulcanized rubber compounds; it is not recommended for materials thinner than 0 1 inch.

2. SPECIMEN

2.1 Unless otherwise specified in the detail specification, the specimen shall consist of a portion of the test unit 1 by 1 inch by approximately 0.25 inch in thickness.

2.1.1 Unless otherwise specified in the detail specification, materials thinner than 0.25 inch shall be plied up using thin pieces, accurately aligned to obtain the desired thickness; the surfaces of the pieces shall be in contact throughout.

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Abrasion machine.*—An abrasion machine as shown in figure 14111 consisting of the following parts·

3.1.1.1 A rubber-coated metal drum, 6 inches in diameter, which is rotated at a speed of 45 ± 5 revolutions per minute by means of an electric motor with a reducing mechanism The number of revolutions of the drum is indicated by a counter attached to one end of the shaft.

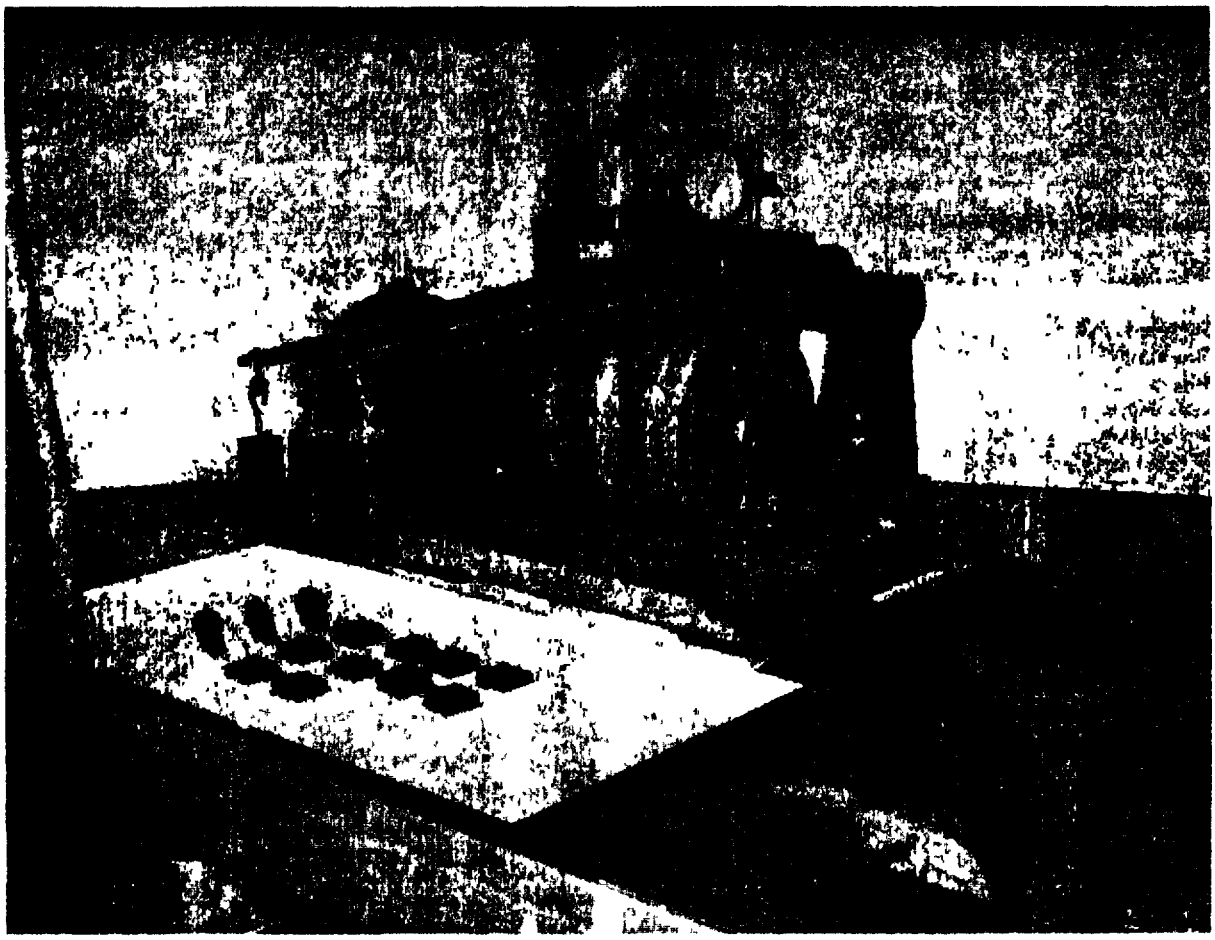

FIGURE 14111 —Abrasion test machine.

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3.1.1.2 Three arms, each pivoted at one end and each having a weight suspended from the other end The weight is attached to the arm so that a downward force of 5 pounds is exerted directly on the specimen which is in contact with the abrasive.

3.1.1.3 Three dial thickness gages, graduated to read to 0 001 inch, are attached to a bridge so that one gage contacts each arm at a point directly over the specimen The bridge is hinged at one end to allow the arms to swing back for mounting the specimen

3.1.1.4 Moisture- and grease-free air from a compressed air line, for cleaning the surface of the abrasive, is delivered to a manifold or nozzles The pressure of air in the manifold or nozzles shall be 40 ± 5 pounds per square inch

3.1.1.5 An arm stop, for each arm

3.1.1.6 An abrasive consisting of No. 2½ garnet paper or cloth 6 inches wide is held around the rubber-covered drum by 4 rubber bands or metal clamps The ends of the abrasive sheets are cut at an angle of about 80° and when in place permit a clearance of about 1/16 inch

3.1.1.7 Small fiber disks which can be attached to the arms directly above the abrasive and to which the specimens are cemented.

3.1.2 *Standard reference compounds.*

3.1.2.1 *Composition and curing conditions.*— The composition of the standard reference compounds and the time and temperature of cure shall be as given in table I

TABLE I —*Composition and curing conditions*

Ingredients	Parts by weight, grams		
	Standard No 1	Standard No. 2	Standard No. 3
GR-S	400		
Rubber No 1 smoked sheet		400	
Rubber No 1 smoked sheet			400.
EPC Black	160	200	
MPC Black			120
Zinc oxide	20	20	80.
Stearic acid		12	8.
Sulfur	8	12	14.
Phenyl-beta-naphthylamine		4	4
Benzothiazyl disulfide	7	4	
Di-ortho-tolylguanidine			5
Cure	40 minutes at 150° C (302° F)	40 minutes at 140° C. (284° F.)	60 minutes at 142° C (287° F.).

3.1.2.1.1 *Compounding ingredients* —Certain of the ingredients required in table I are available as standard compounding ingredients at the National Bureau of Standards In the preparation of standard abrasion compounds, it is recommended that the ingredients available at the National Bureau of Standards be used If materials from other sources are used, they shall have properties identical with those supplied by the National Bureau of Standards. However, in the case of dispute as to the properties of the ingredients from other sources, compounding ingredients available at the National Bureau of Standards shall be used.

3.1.2.1.2 *Weights.*—The standard reference compounds shall be prepared by weighing the ingredients to within ± 1 percent of the weight required The weight of the mixed batch and the sum of the weights of the ingredients shall not differ by an amount exceeding 0.6 percent.

3.1.2.2 *Mixing of compounds.*

3.1.2.2.1 *Mills* —The laboratory mill used for mixing the compounds shall have rolls 6 ± 1/4

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inch in outside diameter with a working distance between the guides of 10 5±0 25 inch The mill shall have a slow roll speed of 24 revolutions per minute and a gear ratio of 1 4 The mill rolls shall be equipped with facilities for controlling the roll temperature and means for indicating the temperature to within the tolerances required in 3 1 2 2 2 and 3 1 2 2 3

3.1.2.2.2 *Standard reference compound No 1* —The compound shall be mixed with the mill rolls at a temperature of 50°±5° C (122°±9° F) The mixing procedure shall be as follows:

Operation	*Time, minutes*
3.1.2.2.2.1 The rubber shall be passed through the mill rolls twice, without banding, at a mill roll opening of 0 008 inch	1
3.1.2.2.2.2 The rubber shall be banded with the mill opening set at 0 055 inch and broken down	10
3.1.2.2.2.3 The carbon black shall be added evenly across the rolls at a uniform rate, and the mill shall be opened at intervals so as to maintain an approximately constant bank When about one-half of the carbon black has been incorporated and all the dry pigment in the bank has disappeared, one ¾-cut shall be made from each side The remainder of the carbon black shall be added	11
3.1.2.2.2.4 All other ingredients, in one mixture, shall be added evenly across the mill at a uniform rate	4
3.1.2.2.2.5 The batch shall be cut three-fourths the way across the mill three times from each side	2
3.1.2.2.2.6 The batch shall be cut from the mill The mill shall be set with an opening of 0 030 inch and the rolled stock passed edgewise through the mill six times	2
3.1.2.2.2.7 The batch shall be weighed to ascertain that materials have not been lost, 3 1 2 1 2, and sheeted to a thickness of approximately 0 030 inch but in no case less than 5 percent more than the mold cavity depth, 3 1 2 3 2	1

3.1.2.2.3 *Standard reference compounds Nos 2 and 3* —The compounds shall be mixed, with the mill rolls at a temperature of 70°±5° C (158°±9° F) The mixing procedure shall be as follows

Operation	*Time, minutes*
3.1.2.2.3.1 The rubber shall be passed through the mill rolls twice, without banding, at a mill roll opening of 0 008 inch	1
3.1.2.2.3.2 The rubber shall be banded with the mill opening set at 0 055 inch and broken down, opening the mill to 0 075 inch during breakdown as the band becomes smooth	4
3.1.2.2.3.3 The carbon black shall be added evenly across the rolls at a uniform rate, and the mill shall be opened at intervals so as to maintain an approximately constant bank When about one-half of the carbon black has been incorporated and all the dry pigment in the bank has disappeared, one ¾-cut shall be made from each side The remainder of the carbon black shall be added	11
3.1.2.2.3.4 The stearic acid shall be added	2
3.1.2.2.3.5 All other ingredients, in one mixture, shall be added evenly across the mill at a uniform rate	4
3.1.2.2.3.6 The batch shall be cut three-fourths the way across the roll three times from each side	2
3.1.2.2.3.7 The batch shall be cut from the mill The mill shall be set with an opening of 0 030 inch and the rolled stock passed edgewise through the mill six times	2
3.1.2.2.3.8 The batch shall be weighed to ascertain that materials have not been lost, 3 1 2 1 2, and sheeted to a thickness of approximately 0 30 inch but in no case less than 5 percent more than the mold cavity depth, 3 1 2 3 2	1

3.1.2.2.4 *Preparation of stock* —At the end of the milling operations 3 1 2 2 2 and 3 1 2 2 3, the batch shall be cut from the rolls and placed on a clean metal surface to cool for not less than

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30 minutes nor more than 60 minutes before curing, 3 1 2 3

3.1.2.3 *Curing*

3.1.2.3.1 *Press* —The press shall be capable of exerting a pressure of not less than 500 pounds per square inch on the molding surface during the entire period of vulcanization The press shall have heated plates of sufficient size so that no portion of the rubber will be nearer than 3 inches to the edge of the platens during vulcanization The platens shall preferably be made of rolled steel, bored for steam heating A small vent shall be placed in the exit steam line to allow steam to flow continuously through the platens If chamber-type platens are used, the steam outlet shall be placed slightly below the steam chamber to insure good drainage The conduction of heat from the hot platens to the press crosshead shall be reduced as much as practicable by a steel grid between them or by other means The pressing surfaces of the platen shall be plane parallel to within 0 003 inch per foot when the plates are at 149° C (300° F) and closed under full pressure with a grid of soft solder or lead wire between them

3.1.2.3.2 *Mold* —The cavities to within ¼ inch of the edges shall be uniform in depth to within ±0 001 inch and shall be between 0 25 and 0 28 inch in depth The corners of the cavities may be rounded with a radius not greater than ¼ inch The molding surface shall be clean, highly polished, and hard chromium plated The cover of the mold shall be a flat plate between 0 5 and 1 0 inch in thickness and preferably hinged to the cavity section to minimize scratching of mold surfaces Instead of a separate mold and cover, the cavities may be cut directly into the hot platen of the press A mold lubricant shall not be used on the mold surfaces

3.1.2.3.3 *Procedure* —The stock, 3 1 2 2 4, shall be cut into strips at least ⅛ inch less in width and length than the corresponding dimensions of the mold. The mold shall be brought to the specified curing temperature within ±0 5° C (1 0° F), 3 1 2 1, in the closed press and maintained for at least 20 minutes before the uncured sheets are inserted The temperature of the mold shall be verified by means of thermocouples or other suitable temperature measuring device inserted in one of the overflow grooves and in intimate contact with the mold The press shall be opened and the uncured sheets inserted in the mold and the press closed in the minimum possible time

3.1.2.3.3.1 The stock shall remain in the mold for the required time under the required conditions, 3 1.2 1 The time of cure shall be considered to be the period between the instant the pressure is applied fully and the instant the pressure is released The time between the instant the sheets are placed in the hot mold and the instant the full pressure is applied to the mold shall be kept to a minimum A minimum pressure of 500 pounds per square inch shall be maintained during the curing period At the end of the curing period, the sheets shall be removed immediately from the mold and cooled for 10 to 15 minutes in water at room temperature or lower The cooled sheets shall be removed from the water, wiped dry, avoiding stretching or flexing through inadvertent handling, and stored out of direct light

3.1.2.4 *Maintenance of standard reference compounds* —The use of standard reference compounds more than 6 months old is not recommended Subsequent batches of reference compounds shall be tested for abrasion resistance, and shall be within 5 percent of the standard compound being used at the time The standard reference compound shall be stored in an airtight container out of direct light at room temperature or lower

4. PROCEDURE

4.1 The standard reference compound used for comparison purposes shall be as specified in the detail specification

4.2 Preparation of specimen.—If the material is too thick or has a fabric backing, surface coating, or an uneven surface that may interfere with the abrasion test, it shall be buffed as described in method 1111

4.3 The specimen shall be free from mechanical damage and shall be cemented to one of the fiber disks. One specimen from the test unit

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shall be mounted on each arm of the testing machine The machine shall be set in motion at 45 ± 5 revolutions per minute, the air pressure turned on, and the machine allowed to run until the surface of the specimen is worn to the shape of the drum At that point, the machine shall be stopped, the gage bridge locked in place, and the gages and the revolution counter set at zero The machine shall again be started and run until approximately 0 1 inch thickness has been abraded from the specimens as recorded on the gages The machine shall be stopped, the number of revolutions recorded from the counter, and the gage readings recorded for each specimen From the data, the number of revolutions required to abrade 0.1 inch thickness from each specimen shall be calculated and the value recorded as R_1 The thickness readings shall be taken with the drum as near as practicable in the same position as it was when the gages were set at zero

4.4 The specimens shall be removed from the machine and one specimen from the standard reference compound, 4 1, mounted on each arm of the machine with the direction of the grain parallel to the direction of abrasion. The standard specimens shall be tested and the readings recorded as previously described in 4 3, except that the number of revolutions required to abrade 0 1 inch thickness from each standard specimen shall be recorded as R_2

5. RESULTS

5.1 Calculation.—The abrasive resistance of the specimen shall be expressed numerically by an abrasive index which shall be calculated as follows

$$\text{Abrasive index} = \frac{R_1}{R_2} \times 100$$

Where

R_1 is the number of revolutions required to abrade 0 1 inch thickness from the specimen being tested

R_2 is the number of revolutions required to abrade 0 1 inch thickness of the specimen from the reference compound

5.2 Unless otherwise specified in the detail specification, three specimens from each test unit shall be tested, one on each arm

5.3 The abrasive index of the test unit shall be the average of the values obtained from the specimens tested

5.4 The abrasive index of the test unit shall be recorded to the nearest one unit

5.5 If a plied-up specimen is used, it shall be recorded

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CORROSION OF METAL BY RUBBER

1. SCOPE

1.1 This method is intended for use in determining whether metals are corroded by rubber when in contact under pressure The method is applicable to rubber material with thicknesses up to ¼ inch The metals used are copper and mild steel, however, this method is applicable to other metals

2. SPECIMEN

2.1 The specimen shall consist of two portions of the test unit, each one inch square

2.1.1 Unless otherwise specified in the detail specification, the thickness of the specimen shall be the thickness of the material undergoing test, but shall not exceed ¼ inch

3. APPARATUS

3.1 The apparatus shall be as follows

3.1.1 *Compression apparatus* —The apparatus consists of a steel baseplate, 2 steel studs, 2 compression springs, and 3 test plates The studs are threaded into the base plate and pass through the test plates, 3 1 2, and the compression springs Compression springs are placed over the studs and means provided for depressing them so as to exert a pressure of not less than 25 pounds per square inch on the specimen when held between the test plates The springs shall be calibrated at room temperature

3.1.2 *Test plates* —Two steel test plates made of mild carbon steel and one test plate made of copper of 99 5 percent purity The plates are approximately 3 inches long, 1 inch wide, and ¼ inch thick, with holes at each end for inserting the studs of the compression apparatus The finish of the plates shall be equal to or better than root-mean-square of 16 microinches

3.1.3 *Air oven* —A circulating-air oven with suitable controls for maintaining the required temperature throughout the oven and of sufficient size to hold the compression apparatus

3.1.4 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within 1 0° C (1 8° F)

3.1.5 Wiping cloth

3.1.6 Benzene, technical grade, or other suitable solvent

4. PROCEDURE

4.1 Preparation of specimen.—If the material is too thick or has a fabric backing, surface coating, or an uneven surface that may interfere with the test, it shall be buffed as described in method 1111

4.2 Unless otherwise specified in the detail specification, the specimen while in contact with the metal shall be exposed at a temperature of 70° ± 2° C (158° ± 3 6° F) for 166 ± 1 hour

4.3 The test plates shall be cleaned with benzene and wiping cloth The copper test plate shall be placed between the two pieces of the specimen, one steel test plate placed outside of each rubber piece and the assembly mounted in the compression apparatus The two springs shall be compressed a distance sufficient to produce a pressure of not less than 25 pounds per square inch on the specimen The principal surfaces of the specimen shall be in contact with the plates at all places The apparatus with specimen compressed shall be placed in the oven and allowed to remain for the required time at the required temperature, 4 2

4.4 At the end of the exposure period, the specimen shall be removed from the test plates and the surface of the plates that were in contact with the rubber examined visually for corrosion or pitting of the metal test plates A slight discoloration of the metallic surfaces shall be ignored

5. RESULTS

5.1 Unless otherwise specified in the detail specification, one specimen from each test unit shall be tested

5.2 The number of specimens tested from each test unit and the number from each test unit that showed any corrosion or pitting of the metal test plates shall be recorded

5.3 The time and temperature of exposure shall be recorded

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

Method 15001
April 12, 1955

CHEMICAL ANALYSIS OF SYNTHETIC RUBBER COMPOUNDS, GENERAL

1. SCOPE

1.1 These methods are intended for use in the identification and quantitative determination of synthetic rubber in rubber compounds The methods of identification include tests for natural rubber for the purpose of a more positive identification The quantitative methods described are direct methods that may be used either alone or in conjunction with methods described in the 16000 group Certain applications and limitations of these methods are given in the scope of the individual methods

2. DEFINITIONS

2.1 The definitions of terms given in group 16000 shall apply to the same terms appearing in the methods in the 15000 group

3. COMPOSITE SAMPLE AND SPECIMENS

3.1 Unless otherwise specified in the detail specification, the specimen shall be taken from the composite sample prepared as described in method 1211

3.2 In methods where the specimen is subjected to extraction previous to analysis, the weight of the specimen used in calculating the results shall be the weight of the specimen prior to the extraction

4. APPARATUS AND REAGENTS

4.1 The apparatus and reagents shall be as follows

4.1.1 Unless otherwise indicated, volumetric glassware used in the analysis shall be in accordance with National Bureau of Standards Circular 434

4.1.2 Unless otherwise indicated, reagents used in the analysis shall conform to the specifications of the Committee on Analytical Reagents of the American Chemical Society when such specifications are available

4.1.3 When solutions are required to be standardized, the normality shall be determined to the fourth decimal place

4.1.4 When reagents such as acids and ammonium hydroxide are required by name only in the individual methods, concentrated solutions of the specific gravities or percentage compositions given in 3 1 6 of method 16001 shall be used

4.1.4.1 When the concentration of reagents such as acids and ammonium hydroxide is required in the individual methods as a ratio, the ratio shall be the number of volumes of concentrated reagent, 4 1 4, that is diluted with a given number of volumes of distilled water For example, 1 to 7 hydrochloric acid means 1 volume of hydrochloric acid that is diluted with 7 volumes of distilled water

4.1.4.2 When the concentration of reagents is required as percent, the value represents the number of grams of the reagent per 100 ml of solution For example, 10 percent barium chloride solution means 10 grams of barium chloride per 100 ml of solution

4.1.5 General laboratory apparatus and reagents shall be as described in method 16001 Special apparatus and reagents are described in the individual methods

5. PROCEDURE

5.1 Unless otherwise indicated, all weighings shall be made to 1 part in 1000

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IDENTIFICATION OF SYNTHETIC RUBBER

1. SCOPE

1.1 This method is intended for use in identifying natural and synthetic rubbers, and some polyvinyl resins when each is present alone in a compound It may also be used to identify the components in some mixtures containing more than one type of rubber In those cases where ambiguous colors are obtained, it will be necessary to use a specimen that has been extracted with acetone

2. SPECIMEN

2.1 The specimen shall consist of a portion of the rubber compound from the composite sample, prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 *Distillation apparatus* —This apparatus shall consist of a test tube, 10 by 75 mm , equipped with a side arm and glass condenser tube about 4 mm in outside diameter attached to the side arm by means of a cork stopper The condenser tube shall be bent at least 90° and shall extend about 100 mm beyond the bend

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3.1.2 Reflux condenser

3.1.3 Test tubes, 10 by 75 mm , and 16 by 150 mm

3.1.4 Copper wire

3.1.5 Ice bath

3.1.6 *Solution No 1.*—One gram of p-dimethylaminobenzaldehyde and 0 01 gram of hydroquinone shall be dissolved in 100 ml of absolute methyl alcohol. Five milliliters of hydrochloric acid and 10 ml of ethylene glycol shall be added and the specific gravity adjusted to 0 851 at 25°/4° C (77/39 2° F) by the addition of methyl alcohol or ethylene glycol The solution remains stable for several months when stored in a brown bottle

3.1.7 *Solution No 2* —Two grams of sodium citrate, $2Na_3C_6H_5O_7\ 11H_2O$, 0 2 gram of citric acid, 0 03 gram of bromocresol green, and 0 03 gram of Metanil yellow shall be dissolved in 500 ml of distilled water

3.1.8 *Solution No 3, chloroprene-nitrile wetting solution* —Two and five-tenths grams of benzidine dihydrochloride shall be dissolved in a mixture of 500 ml of methyl alcohol and 500 ml of distilled water, and 10 ml of 0 1 percent aqueous solution of hydroquinone added The solution remains stable for several months when stored in a brown bottle A precipitate which usually forms on standing does not affect the efficiency of the solution

3 1 9 *Spot test paper No 1, chloroprene-nitrile* —Two grams of cupric acetate and 0 25 gram of Metanil yellow shall be dissolved in 500 ml of methyl alcohol Filter paper shall be impregnated with the solution, dried, and then cut into strips to form test paper To conduct the test for the presence of chloroprene and nitrile rubber, this paper shall be moistened, just prior to the test, with wetting solution, 3 1 8

3.1.10 *Solution No 4, polyisobutylene wetting solution* —Five grams of yellow mercuric oxide shall be added to a mixture of 15 ml of sulfuric acid and 80 ml of distilled water, and the mixture boiled until the mercuric oxide has dissolved The solution shall be cooled and diluted to 100 ml with distilled water Blank filter paper strips shall be moistened in this solution To conduct the test for polyisobutylene, the paper shall be wet with the solution just prior to the test

3.1.11 *Solution No 5 styrene, wetting solution.*—Thirty grams of trichloracetic acid shall be dissolved in isopropyl alcohol and the solution diluted to 100 ml with the same solvent This solution should not be allowed to come in direct contact with the skin

3.1.12 *Spot test paper No 2, natural rubber-styrene* —Filter paper squares shall be impregnated with a solution prepared by dissolving 3 grams of p-dimethylaminobenzaldehyde and 0 05 gram of hydroquinone in 100 ml of ethyl ether The papers shall be dried, cut into strips, and stored in a brown glass bottle Papers stored under such conditions are stable for several weeks, but lose their efficiency if stored in light To conduct the test for natural

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rubber and styrene polymer these papers shall be moistened, just prior to the test, with the wetting solution described in 3 1 11

3.1.13 Iodine solution, 0 2 gram of iodine in 1 liter of carbon tetrachloride

3.1.14 Sodium hydroxide solutions, 5-percent solution, 20-percent solution

3.1.15 Zinc, granulated

3.1.16 Sodium nitrite, 0 5 N solution

3.1.17 Phenol

3.1.18 β–naphthol in 5-percent sodium hydroxide solution

3.1.19 Mercuric acetate in 5-percent methyl alcohol solution

3.1.20 Petroleum ether, boiling range 50° to 60° C (122° to 140° F)

3.1.21 Sulfuric acid, 1 to 1 4 solution

4. PROCEDURE

4.1 Pyrolysis tests.—The specimen containing about 0 5 gram of the rubber compound, shall be transferred to the test tube of the distillation apparatus, 3 1 1 The test tube shall be fastened in position by means of a clamp and then heated with a very low flame until the specimen begins to decompose When vapor appears at the mouth of the side arm, the end of the condenser tube shall be immersed in 1 5 ml. of solution No 2, contained in a receiving tube, and the distillation continued until it has been determined whether a color change will develop in the solution The condenser tube shall then be removed from solution No 2, transferred to a second receiving tube, and immersed in 1 5 ml of solution No 1, and the distillation continued The distillate in the two receiving tubes shall be permitted to cool and then shaken The second receiving tube shall be examined to determine whether the drops of distillate sink or float in solution No 1 Both solution No 1 and solution No. 2 shall be observed for any color change Solution No 1 shall be transferred to a 16- by 150-mm test tube and 5 ml of absolute methyl alcohol added The mixture shall be heated on a water bath at about 100° C (212° F) for 3 minutes, and any color that develops recorded. All observations made on both solutions shall be recorded and the material classified by means of comparison with table I

TABLE I

Material	Solution 1		Solution 2
	Initial color	Color after heating	Color
Blank	Pale yellow	Pale yellow	Green
Polyvinyl chloride	Yellow	Yellow	Red
Chloroprene	do	Pale yellow green	Do
Nitrile	Orange red	Red	Green
Chloroprene rubber	do	do	Yellow to red
Styrene	Yellow green	Green	Green
Natural rubber	Brown	Violet blue	Do
50 styrene 50 natural rubber	Olive green	Green blue	Do
Polyisobutylene	Yellow droplet floats	Pale blue green	Do
Polyvinyl acetate	Yellow	Pale yellow green	Yellow

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4.2 Spot tests.

4.2.1 The material shall be tested as described in 4 2 2, 4 2 3, and 4 2 4, and classified by reference to the color listed in table II The chloroprene-nitrile rubbers test shall be carried out first If both of these rubbers are absent the other tests will normally be successful

TABLE II

Material	Tests		
	Chloroprene-nitrile	Polyisobutylene	Rubber-styrene
Chloroprene	Red	Blank [1]	Green
Nitrile	Green	Pale brown	Yellow green
Chloroprene-nitrile mixture	Red/green	do	Green
Polyisobutylene	Blank [1]	Yellow	Pale lavender
Natural rubber	do [1]	Brown	Blue
Styrene	do [1]	do	Blue green

[1] Blank color tests may be pale brown rather than colorless

4.2.2 *Chloroprene-nitrile spot test* —A metal surface shall be heated, either electrically or by means of a gas flame, to a temperature sufficient to produce dense fumes of pyrolysis of the specimen but not high enough to ignite the rubber The specimen shall be pressed against the heated surface until fumes are liberated A strip of the spot test paper No 1 shall be moistened with solution No 3 and then held in a parallel position about 5 mm above the hot metal surface The test shall be continued until a good color is produced on the side of the paper facing the fumes without scorching the paper or the impregnating materials

4.2.3 *Polyisobutylene spot test* —This test shall be conducted as described for the chloroprene-nitrile test, 4 2 2, except that blank filter paper moistened with solution No 4 just before the test shall be used

4.2.4 *Rubber-styrene spot test* —The rubber-styrene spot test shall be conducted as described in 4 2 2, except that spot test paper No 2, moistened with solution No. 5 shall be used.

4.2.5 *Mixtures of rubbers* —Chloroprene-nitrile mixtures containing more than 30 percent of chloroprene-type rubber give a green color when tested as described in 4 2 2 and a red color when spot test paper No 1, without wetting, is used Chloroprene in quantities less than 30 percent cannot usually be detected in this mixture Chloroprene will not normally mask natural or polyisobutylene-type rubber, but may mask a styrene type Nitrile rubber will mask a styrene-type rubber and may also cause some difficulty in detecting natural rubber Polyisobutylene rubber normally can be detected in the presence of moderate amounts of the other four rubbers The chloroprene-nitrile test is not masked by the presence of the other three rubbers Styrene rubber is quite difficult to detect unless alone or with only small amounts of other rubbers Styrene and natural rubber together can only be identified in the range of 50–50 mixtures These spot tests are not always sufficient for detecting all rubbers However, a combination of these spot tests and the confirmatory tests, 4 3, are usually adequate for identifying most all mixtures of rubbers.

4.3 Confirmatory tests.—When necessary, these tests shall be used to confirm the tests described in 4 1 and 4 2

4.3.1 *Chloroprene rubber*

4.3.1.1 *Iodine test* —Chloroprene rubber is distinguished from saturated vinyl plastic by the following iodine test A specimen of

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approximately 0 2 gram of the rubber compound shall be shaken with 2 ml of iodine solution, 3 1 13 The formation of a violet color which fades noticeably in 2 or 3 minutes indicates the presence of a chloroprene

4.3.1.2 *Flame test* —A specimen of approximately 0 1 gram shall be burned in contact with a clean copper wire A persistent green flame indicates the presence of chlorine-containing materials This test is particularly applicable for testing for chlorine-containing rubber in the presence of much nitrile rubber

4.3.2 *Polyvinyl acetate* —A specimen of approximately 0 2 gram of the compound shall be transferred to a test tube containing 2 ml of 1 to 1 4 sulfuric acid and the mixture warmed gently If decomposition occurs, polyvinyl acetate is indicated

4.3.3 *Natural rubber and styrene rubber*

4.3.3.1 *Pyrolysis test* —Asphaltic extenders, if present, may interfere with distinguishing between natural rubber and styrene rubber by pyrolysis

4.3.3.1.1 *Qualitative test for asphaltic extenders* —A specimen of approximately 0 2 gram of the compound shall be shaken with 2 ml of chloroform The presence of asphaltic extenders in the specimen is indicated by an appreciable darkening of the chloroform

4.3.3.1.2 The asphaltic extenders, if present, shall be removed from a fresh specimen by extraction with chloroform as described in method 16331 and the specimen dried in a vacuum for one hour at 70°±2° C (158°±3 6° F) The extracted specimen shall be subjected to the pyrolysis test in 4 1 and the material classified by reference to table I

4.3.3.2 *Color test for natural rubber* —This test may be conducted as further confirmation of the presence of natural rubber in a mixture with styrene rubber A specimen of approximately one gram of the compound shall be extracted with acetone as described in method 16331 and dried for one hour at 70°±2° C (158°± 3 6° F) A few milligrams of the dried specimen shall be placed in an evaporating dish, covered with carbon tetrachloride, and allowed to swell A few drops of bromine shall be added and the specimen allowed to stand for about 2 minutes One gram of phenol shall be added and the mixture warmed on a steam bath to remove the carbon tetrachloride A blue to red-violet color indicates the presence of natural rubber Oil-type reclaimed rubber gives the same color reaction but alkali-type reclaim does not Polyisoprene also gives a positive test

4.3.3.3 *Color test for styrene rubber* —The presence of styrene rubber in mixtures with natural rubber may be further confirmed by this color test A 1-or 2-gram specimen of the compound shall be extracted with acetone as described in method 16321 and dried for one hour at 70°±2° C (158°±3 6° F) The dried specimen shall be placed in a small flask, 20 ml of nitric acid added and the mixture refluxed for one hour The refluxed mixture shall be diluted by pouring into 100 ml of distilled water, and extracted with 50-, 25- and 25-ml portions of ether The ether extracts shall be combined, washed twice with 15 ml of distilled water, and the washings rejected The ether solution shall be extracted with three 15-ml portions of 5-percent sodium hydroxide solution, followed by extraction with 20 ml of distilled water, and the ether discarded The sodium hydroxide extracts and washing shall be combined, made just acid with hydrochloric acid, and 20 ml added in excess The solution shall be heated on the steam bath and the nitrobenzoic acid reduced with 5 grams of granulated zinc The solution shall be made alkaline with 20-percent sodium hydroxide solution, and sufficient excess added to just dissolve the zinc hydroxide precipitate that forms The solution shall be extracted twice with ether and the ether discarded The aqueous solution shall be made acid with hydrochloric acid, cooled to room temperature, and 2 ml of 0 5 *N* sodium nitrite added The diazotized solution shall be poured into an excess of a solution of β-naphthol in 5-percent sodium hydroxide solution A vivid scarlet color indicates the presence of styrene in the rubber

4.3.4 *Polyisobutylene rubber* —The destructive distillation of polyisobutylene types yields a white difficultly condensable vapor A light

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yellow mobile oil is obtained The presence of polyisobutylene rubber may be confirmed by the following A bent delivery tube shall be attached to a test tube by means of a stopper The delivery tube shall pass through a stopper almost to the bottom of a second test tube having a side arm This second test tube shall be placed in an ice bath A second delivery tube shall be attached to the side arm of the second test tube and extended into a third open test tube containing 0 5 gram of mercuric acetate in 10 ml of methyl alcohol A one-gram specimen of the rubber compound that has been extracted with acetone as described in method 16321, and dried for one hour at 70° ± 2° C , (158° ± 3 6° F) shall be placed in the first test tube and heated strongly until decomposed, and the compound distilled off into the second and third test tubes The liquid that collects in the second test tube shall be rejected The third test tube shall be heated to evaporate the methyl alcohol, avoiding excessive heating near the end of the evaporation Twenty-five milliliters of petroleum ether, 3 1 20, shall be added to the test tube, and the mixture boiled and filtered The filtrate shall be evaporated to a small volume, chilled in ice and the sides of the vessel scratched to crystallize the mercury derivative The crystals shall be dried at 30° to 40° C (86° to 104° F) and the melting point determined The derivative is believed to be methoxy-isobutyl-mercuriacetate with a melting point of about 55° C (131° F) The test shall be further confirmed by a mixed melting point determination with the mercury derivative of known polyisobutylene or isobutylene

4.3.5 *Polysulfide rubbers* —The polysulfide rubbers are easily identified by means of odor and high sulfur content Additional qualitative tests include the pronounced swelling action of acetone, unlike other rubbers except unvulcanized nitriles, and the extremely rapid reaction at 40° C (104° F) with a mixture of equal volumes of nitric and sulfuric acids Polysulfide rubber will start to decompose in from 5 to 35 seconds while other rubbers usually take longer

5. RESULTS

5.1 All of the rubbers found shall be recorded

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POLYCHLOROPRENE RUBBER

1. SCOPE

1.1 This method is intended for use in determining the amount of polychloroprene rubber in rubber compounds The procedure consists of determining the chlorine in the specimen and from this calculating the amount of polychloroprene rubber This method assumes that the polychloroprene contains 9 1 percent of nonrubber components and has a chlorine content of 36 5 percent, therefore, the type of chlorine compound must be determined from additional information obtained from other analyses

2. SPECIMEN

2.1 The specimen shall consist of approximately 0 2 gram of the rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Monel metal crucibles 10 ml and 25 ml

3.1.2 Silver nitrate, 0 113 *N* solution, standardized

3.1.3 Sodium thiocyanate, 0 056 *N* solution, standardized

3.1.4 Ferric ammonium sulfate, saturated solution

4. PROCEDURE

4.1 The finely cut specimen shall be weighed to the nearest milligram, placed in the 10-ml monel metal crucible and sodium carbonate packed around it to fill the crucible The 10-ml crucible and contents shall be inverted into the 25-ml monel metal crucible The crucible shall be packed to the top with anhydrous sodium carbonate, then placed in a muffle furnace at about 725° C (1,337° F) and the temperature increased to 800° C (1,472° F) in about 5 minutes This temperature shall be maintained for 15 to 20 minutes, and the crucible removed from the furnace and cooled to room temperature The contents shall be transferred to a 300-ml beaker with distilled water The solution shall be neutralized with nitric acid, 5 ml added in excess, and diluted to 150 ml with distilled water Twenty milliliters of silver nitrate solution, 3 1 2, shall be added and the solution boiled on the hot plate to coagulate the silver chloride The solution shall be cooled to room temperature and then titrated to a permanent pink end point with the sodium thiocyanate solution, 3 1 3, using 2 ml of the saturated ferric ammonium sulfate solution as indicator

5. RESULTS

5.1 Calculations

5.1.1 The chlorine in the specimen shall be calculated as follows

$$\text{Chlorine, percent,} = \frac{(A \times N_1) - (B \times N_2) \times 0.0355 \times 100}{W}$$

Where

A is the number of milliliters of silver nitrate solution added

B is the number of milliliters of sodium thiocyanate solution required to titrate the excess silver nitrate

N_1 is the normality of the silver nitrate solution

N_2 is the normality of the sodium thiocyanate solution

W is the weight of the specimen, gram

5.1.2 The polychloroprene rubber in the specimen shall be calculated as follows

$$\text{Polychloroprene rubber, percent,} = \frac{\text{Percent chlorine in the specimen}}{0\ 365}$$

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The polychloroprene rubber in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The polychloroprene rubber in the sample for test shall be recorded to the nearest one percent

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ACRYLONITRILE RUBBER

1. SCOPE

1.1 This method is intended for use in determining the amount of acrylonitrile rubber in rubber compounds The procedure consists of determining the nitrogen in the specimen and from this calculating the amount of acrylonitrile rubber present The method assumes that the amount of nitrogen present in acrylonitrile rubber is 26 4 percent, therefore, the type of nitrogen compound must be determined from additional information obtained from other analyses

2. SPECIMEN

2.1 The specimen shall consist of approximately one gram of the rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Distillation unit

3.1.2 Kjeldahl digestion unit

3.1.3 Kjeldahl flask, 750 ml

3.1.4 Potassium sulfate

3.1.5 Copper sulfate

3.1.6 Powdered selenium

3.1.7 Zinc powder, 20 mesh

3.1.8 Paraffin

3.1.9 Sodium hydroxide, 40-percent solution

3.1.10 Boric acid indicator solution containing 43 grams of H_3BO_3 (free from borax), 6 ml of methyl red indicator, 3 1 11, and 4 ml of methylene blue indicator, 3 1 12, per liter of freshly distilled water The solution shall be preserved in a pyrex glass container

3.1.11 Methyl red indicator containing one gram per liter of 95 percent ethyl alcohol

3.1.12 Methylene blue indicator, containing one gram per liter of 95 percent ethyl alcohol

3.1.13 Sulfuric acid, 0 5 N solution, standardized

3.1.14 Sucrose

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram, extracted without acetone as described in method 16321, except that the time of extraction shall be 8 hours, and the residue dried in a circulating-air bath for one hour at 70°±2° C (158°±3 6° F)

4.2 The extracted specimen shall be transferred to a 750-ml Kjeldahl flask and 10 grams of potassium sulfate, 0 3 gram of copper sulfate, 0 1 gram of powdered selenium, and 25 ml of sulfuric acid added The flask shall be placed in the digestion unit and heated gently until any frothing subsides The mixture shall then be boiled until the liquid is a clear, light-greenish-blue color and then boiled for 30 minutes longer The flask and contents shall be allowed to cool to room temperature, then diluted by cautiously adding about 300 ml of distilled water, and again cooled to room temperature

4.3 The cooled flask shall be connected to the distillation unit, one gram of zinc powder and a small piece of paraffin added The Kjeldahl flask shall be tilted and 100 ml of the 40-percent sodium hydroxide added by cautiously pouring it down the neck of the flask with a minimum of mixing of the solutions The flask shall be stoppered quickly with the stopper connected to the distillation spray trap and the contents mixed thoroughly by swirling The solution in the flask shall be heated to boiling and boiled at such a rate as to distill over about 250 ml in 1½ hours, catching the evolved ammonia in a 500-ml Erlenmeyer flask containing 100 ml of the boric acid indicator solution, 3 1 10 During the distillation process the delivery tube attached to the end of the condenser shall be kept below the surface of the boric acid solution

4.4 After about 250 ml has been distilled over, the Kjeldahl flask shall be disconnected from the spray trap, the delivery tube removed from the condenser and rinsed with distilled water, adding the wash to the Erlenmeyer flask The solution shall be titrated with the 0 5 N sulfuric acid solution to a pink end point During the titration, the color changes from

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green to gray to pink, the intensity of the pink increasing to red with further addition of acid

4.5 A blank shall be run by substituting one gram of sucrose for the specimen and following the procedure in detail The titration value obtained shall be deducted from the acid titration obtained for the specimen

5. RESULTS.

5.1 Calculation.

5.1.1 The nitrogen in the specimen shall be calculated as follows

$$\text{Nitrogen, percent} = \frac{A \times N \times 0.014}{W} \times 100$$

Where

A is the number of milliliters of standard sulfuric acid required to titrate the specimen

N is the normality of the standard sulfuric acid

W is the weight of the specimen before extraction, grams

5.1.2 The acrylonitrile rubber in the specimen shall be calculated as follows

Acrylonitrile rubber, percent = percent nitrogen × 3 79

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The acrylonitrile rubber in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The acrylonitrile rubber in the sample for test shall be recorded to the nearest one percent

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STYRENE RUBBER

1. SCOPE

1.1 This method is intended for use in determining the amount of styrene rubber in rubber compounds The procedure is based on the oxidation of the styrene rubber by nitric acid and by alkaline potassium permanganate, followed by the isolation and titration of the ether-soluble acids The acid mixture produced is primarily *p*-nitrobenzoic acid It has been found experimentally that on oxidation by this method each styrene molecule forms one equivalent of ether-soluble acid The results obtained by this method are at best an estimate, ± 2 percent, and in heavily compounded rubbers or mixtures of rubbers the error is frequently greater If natural rubber is present, an interference to the extent of about 3 percent of the rubber present should be subtracted from the styrene content

2. SPECIMEN

2.1 The specimen shall consist of approximately 1 gram of the rubber, or 2 grams of rubber compound, if the material is compounded with fillers, from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Flask, 250-ml, with condenser attached by means of ground glass joint

3.1.2 Sodium hydroxide, 20-percent solution

3.1.3 Sodium hydroxide, 4-percent solution

3.1.4 Sodium hydroxide, 0 1 N solution, standardized

3.1.5 Potassium permanganate, 4-percent solution

3.1.6 Sulfuric acid, 25-percent solution

3.1.7 Sodium acid sulfite, 4-percent solution

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram, extracted with acetone as described in method 16321, and dried in a circulating-air oven at 100°±2°C (212°±3 6°F)

4.2 The extracted specimen shall be transferred to the 250-ml flask fitted with a condenser by means of a ground glass joint, and 20 ml of nitric acid added to the flask The condenser shall be attached and the flask heated until the reaction starts The heat shall be removed until the reaction subsides, using external cooling if necessary The mixture shall then be boiled gently for about 6 hours, 20 ml of distilled water added to the condenser, and the flask cooled Twenty-percent sodium hydroxide shall be slowly added to the flask until the solution is neutral to litmus and 1 but not more than 2 ml in excess

4.3 The mixture in the flask shall be transferred to a 500-ml Erlenmeyer flask, the flask rinsed, and the washings added to the mixture Fifty milliliters of the 4-percent potassium permanganate solution shall be added and the flask swirled and placed on a steam bath for 5 hours, with frequent swirling If the pink color fades, 10 ml of the potassium permanganate solution in addition shall be added and this continued for the entire 5 hours of heating on steam bath If more than two additions of the permanganate solution are necessary, the solution may have been too alkaline and the results may be low

4.4 The hot solution shall be filtered through a Buchner funnel, and precipitate and flask washed several times with distilled water The filtrate and washings shall be transferred to a 500-ml beaker, cooled, and 25 percent sulfuric acid solution added until just acid to Congo-red paper Four-percent solution of sodium acid sulfite shall be added dropwise, while stirring, until the pink color just disappears The solution shall be diluted to 250 to 300 ml if necessary

4.5 The solution shall be transferred to a 500-ml separatory funnel and extracted with three 100-ml portions of ether The three ether extracts shall be combined in the separatory funnel and extracted with 10 ml of distilled water, discarding the aqueous layer The ether solution shall be extracted with 25-ml, 12 5-ml, and 12 5-ml portions, respectively, of

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4-percent sodium hydroxide solution, and finally 15 ml of distilled water, and the ether layer discarded

4.6 The sodium hydroxide extractions shall be combined, transferred to a separatory funnel, and acidified with 25-percent sulfuric acid solution The solution shall be extracted with three 50-ml portions of ether, and the ether extracts combined and extracted with two 10-ml portions of distilled water to remove the mineral acid

4.7 The ether extract shall be transferred to a 600-ml beaker and evaporated, the residue treated with acetone, the acetone evaporated, and the residue dried at 60°±2° C (140°±3 6° F) until the weight does not change more than 2 mg in 30 minutes' heating Styrene polymers give a crystalline residue The residue shall be dissolved in 25 to 75 ml of neutral 95-percent ethyl alcohol and titrated with 0 1 *N* sodium hydroxide solution using phenolphthalein as indicator

4.8 If it is necessary to confirm the presence of *p*-nitrobenzoic acid, the titrated solution shall be made acid and alcohol added The *p*-nitrobenzoic acid shall be recrystallized and a mixed melting point made on the crystals A second method for identifying the crystals is by reducing, diazotizing, and coupling as described in 4 3 3 3 of method 15011

5. RESULTS

5.1 The amount of styrene rubber in the specimen shall be calculated as follows

$$\text{Styrene rubber, percent} = \frac{A \times N \times 0\ 104}{W} \times 100$$

Where

A is the number of milliliters of standard sodium hydroxide required to titrate the specimen
N is the normality of the standard sodium hydroxide
W is the weight of the specimen before extraction, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The styrene rubber in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The styrene rubber in the sample for test shall be recorded to the nearest one percent

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POLYSULFIDE RUBBER (THIOPLASTS)

1. SCOPE

1.1 This method is intended for use in determining the amount of polysulfide rubber in rubber compounds The procedure consists of determining the sulfur content of the specimen and from this calculating the amount of polysulfide rubber in the compound The type of sulfur compound must be determined from additional information obtained from other analysis in order to obtain the theoretical value for the sulfur content of the polysulfide rubber being analyzed Two procedures for determining sulfur are described, one is a wet oxidation procedure and the other a combustion procedure

2. SPECIMEN

2.1 The specimen shall consist of approximately 0 5 gram of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows:

3.1.1 Zinc - nitric acid solution as described in method 16221

3.1.2 Fuming nitric acid

3.1.3 Bromine, saturated solution

3.1.4 Picric acid, saturated solution

4. PROCEDURE

4.1 Wet oxidation procedure.

4.1.1 The specimen shall be weighed to the nearest milligram and transferred to a 500-ml chemically resistant Erlenmeyer destruction flask Ten milliliters of zinc - nitric acid solution shall be added, the specimen moistened thoroughly, and the flask set aside overnight. By allowing the mixture to stand overnight, the specimen becomes partially decomposed and thus permits the addition of fuming nitric acid without danger of ignition Fifteen milliliters of fuming nitric acid shall be added and the flask swirled rapidly to keep the specimen immersed to avoid ignition by too rapid oxidation With some specimens it may be necessary to cool the flask under tap water

4.1.2 After solution of the rubber is complete, 5 ml of the saturated water solution of bromine shall be added and the mixture slowly evaporated to a foamy syrup If particles of organic matter remain at this point, a few milliliters of fuming nitric acid shall be added and the evaporation repeated

4.1.3 The mixture shall be cooled Twenty milliliters of nitric acid and 10 ml of perchloric acid shall be added, and the mixture then evaporated to dense white fumes of perchloric acid The evaporation shall be continued until the volume is reduced to about 3 to 5 ml , using a hotplate behind a safety screen The flask shall be cooled, 100 ml of distilled water added, and then heated until solution is as complete as possible If the compound contains barium salts, the barium will be precipitated as barium sulfate at this point

4.1.4 The solution shall be filtered, diluted to 300 ml , made just acid to Congo-red paper with hydrochloric acid, and 2 ml added in excess Ten milliliters of saturated picric acid solution shall be added and the solution heated to 90° to 100° C (194° to 212° F) The sulfate shall be precipitated as barium sulfate by the dropwise addition of a slight excess of 10-percent barium chloride solution while stirring vigorously and set aside for 3 hours or overnight on a steam bath The barium sulfate shall be filtered and the filtrate tested for complete precipitation of barium sulfate by the addition of a few milliliters of barium chloride The precipitate shall be washed free from acid with distilled water, transferred to a crucible, ignited at 650° to 700° C (1,202° to 1,292° F) in a muffle furnace, cooled to room temperature in a desiccator, and weighed

4.2 Combustion method.—Compounds containing polyisobutylene cannot be analyzed by the wet-oxidation method, therefore, the sulfur in such materials must be determined by a combustion method The sulfur shall be determined by a combustion method with oxygen, or in a Parr peroxide bomb In the case of the Parr peroxide bomb method, the

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specimen shall be finely divided in order to obtain complete oxidation. After oxidation of the sulfur to sulfate by either of these methods, the sulfate shall be determined as barium sulfate as described above

5. RESULTS

5.1 Calculation.

5.1.1 The sulfur in the specimen shall be calculated as follows

$$\text{Sulfur, percent} = \frac{A \times 0.1373}{W} \times 100$$

Where

A is the weight of the barium sulfate, gram
W is the weight of the specimen, gram

5.1.2 The polysulfide rubber in the specimen shall be calculated as follows

Polysulfide rubber,

$$\text{percent,} = \frac{\text{Percent sulfur in the specimen}}{F}$$

Where

F is the percent sulfur in the polysulfide rubber

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The polysulfide rubber in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The polysulfide rubber in the sample for test shall be recorded to the nearest one percent

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POLYISOBUTYLENE RUBBER

1. SCOPE

1.1 This method is intended for use in determining the amount of polyisobutylene rubber in rubber compounds The method is fairly successful for determining polyisobutylene rubber in mixtures If the mixture is compounded or vulcanized, the upper limit is about 25 percent polyisobutylene

2. SPECIMEN

2.1 The specimen shall consist of approximately one gram of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Reflux condenser

3.1.2 Ammonium hydroxide, 1 to 2 solution

4. PROCEDURE

4.1 The finely cut specimen shall be weighed to the nearest milligram, transferred to a 250-ml Erlenmeyer flask, 20 ml of nitric acid added, and a reflux condenser attached to the flask The specimen shall be refluxed for 30 minutes on a hot plate The liquid shall be decanted while hot and the residue washed by decantation several times with cold nitric acid and then washed with distilled water Fifty milliliters of 1 to 2 ammonium hydroxide shall be added to the flask and the flask warmed for 30 minutes on a water bath The liquid shall be decanted and the residue washed several times with hot distilled water The insoluble material shall be dried at 100°±2° C (212°±3 6° F) to a constant weight and weighed as polyisobutylene

5. RESULTS

5.1 The polyisobutylene rubber in the specimen shall be calculated as follows·

$$\text{Polyisobutylene rubber, percent,} = \frac{A}{W} \times 100$$

Where

A is the weight of the polyisobutylene, grams
W is the weight of the specimen, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The polyisobutylene rubber in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The polyisobutylene rubber in the sample for test shall be recorded to the nearest one percent

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CARBON BLACK

1. SCOPE

1.1 This method is intended for use in determining the amount of carbon black in most rubber compounds It is not applicable to compounds containing polyisobutylene rubber

2. SPECIMEN

2.1 The specimen shall consist of approximately ½ gram of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Combustion apparatus for carbon dioxide determination

3.1.2 Carbon dioxide absorption bulbs

3.1.3 Oxygen supply

3.1.4 Nitrobenzene

3.1.5 Nitric acid, 1 to 3 solution

3.1.6 Xylene

3.1.7 Soda-lime carbon dioxide absorption material

3.1.8 Calcium chloride, anhydrous

4. PROCEDURE

4 1 The specimen shall be weighed to the nearest milligram and the weight recorded as W. The specimen shall be extracted with acetone-chloroform mixture as described in method 16311 and dried in an air oven at 100°±2° C (212°±3 6° F)

4.2 Absence of fillers containing combined water.—The extracted specimen shall be transferred to a 400-ml beaker, and 20 ml of nitrobenzene added The beaker shall be heated on a hot plate for a few minutes to allow the specimen to swell Twenty milliliters of 1 to 3 nitric acid shall be added and the beaker placed on the hot plate until the specimen disintegrates and dissolves The beaker shall be heated on the steam bath for one hour, 100 ml of xylene added, and the mixture, while hot, filtered through a Gooch crucible

4.3 The carbon residue shall be washed with hot xylene until free from decomposed material and then with 95-percent ethyl alcohol The Gooch crucible containing the carbon shall be dried to a constant weight at 240°±10° C (464°±18° F), cooled in a desiccator, and weighed, and the weight recorded as W_1 The weighed crucible and contents shall be placed in a muffle furnace, and the carbon burned off at 650° to 700° C (1,202° to 1,292° F) The crucible shall be cooled and weighed to the nearest milligram, and the weight recorded as W_2

4.4 Fillers containing combined water.—The procedure shall be as described in 4 1, 4 2, and 4 3 through drying the crucible and contents to a constant weight The contents of the dried crucible shall be transferred to a combustion boat and the boat placed in the tube of a combustion apparatus Oxygen freed from carbon dioxide by passing it through soda lime shall be passed through the combustion tube until all carbon dioxide has been removed from the tube The combustion tube shall then be slowly heated to approximately 600° C (1,202° F) while oxygen is passed through it at the rate of about 200 ml per minute The issuing gases shall be passed through drying tubes containing anhydrous calcium chloride to remove the water and then into a weighed absorption bulb, W_3, containing soda-lime carbon dioxide absorption material The heating shall be continued until all the carbon has burned off and then for 10 minutes longer The carbon dioxide absorption bulb shall be disconnected, cooled to room temperature, weighed, and the weight recorded as W_4

5. RESULTS

5.1 Calculation.—The carbon black in the specimen shall be calculated as follows

5.1.1 Carbon black, absence of fillers containing combined water

$$\text{Carbon black, percent} = \frac{W_1 - W_2}{W} \times 100$$

Where

W is the weight of the specimen before extraction, gram
W_1 is the weight of the crucible plus carbon dried at 240° C (464° F), grams
W_2 is the weight of the crucible after ignition in furnace, grams

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5.1.2 Carbon black, fillers containing combined water

$$\text{Carbon black, percent,} = \frac{W_4 - W_3}{W} \times 100$$

Where

W_3 is the weight of the soda-lime absorption bulb, grams

W_4 is the weight of the soda-lime absorption bulb plus carbon dioxide, grams

W is the weight of the specimen before extraction, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The carbon black in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The carbon black in the sample for test shall be recorded to the nearest one percent.

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PHOSPHATE PLASTICIZER, FUSION METHOD

1. SCOPE

1.1 This method is intended for use in determining the amount of the phosphate plasticizer in synthetic rubber compounds The specimen is brought into solution by fusion with sodium carbonate, the phosphorous pentoxide determined, and the plasticizer calculated as tricresylphosphate

2. SPECIMEN

2.1 The specimen shall consist of approximately 0 3 gram of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Nickel crucibles, 25 by 25 mm and 35 by 37 mm

3.1.2 Nitric acid, 10-percent solution

3.1.3 Potassium nitrate, 2-percent solution

3.1.4 Sodium hydroxide 0 1874 *N* solution, standardized

3.1.5 Nitric acid, 0 1874 *N* solution, standardized

3.1.6 Ammonium molybdate solution prepared by adding 100 grams of ammonium molybdate and 100 grams of ammonium nitrate to 80 ml of ammonium hydroxide and making the volume to 4000 ml with distilled water

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram, placed in the 25-mm nickel crucible containing a layer of anhydrous sodium carbonate 5 mm deep, and the crucible filled with sodium carbonate The crucible and contents shall be inverted into the 37-mm nickel crucible and sodium carbonate added to the top The crucible shall be placed in a muffle furnace at about 100° C (212° F) and the temperature slowly increased to about 800° C (1,472° F) Avoid fusing the sodium carbonate The crucible and contents shall be cooled to about room temperature, the contents emptied into a 500-ml beaker, and the remainder of the material washed out by means of distilled water and 1 to 2 nitric acid The solution shall be made slightly acid with nitric acid, then boiled for 10 minutes, and filtered if large quantities of carbon are present

4.2 The solution shall be diluted to about 300 ml , 40 ml of ammonium molybdate solution added, and the solution neutralized with ammonium hydroxide and then acidified with nitric acid The temperature at this point should be about 50° C (122° F) The solution shall be stirred until precipitation is complete, and then set aside for at least 6 hours The solution shall be filtered through a filter paper and the precipitate washed free from acid with the 2 percent potassium nitrate solution The filter paper and precipitate shall be returned to the beaker, 250 ml of distilled water and a few drops of phenolphthalein indicator added, an excess of the standard sodium hydroxide solution added, the mixture stirred until the yellow precipitate has dissolved, and the excess standard sodium hydroxide titrated with standard nitric acid solution, until the phenolphthalein color has just disappeared

5. RESULTS

5.1 Calculation.—The tricresylphosphate in the specimen shall be calculated as follows

$$\text{Tricresylphosphate, percent,} = \frac{(B \times N_1) - (A \times N_2) \times 0.016}{W} \times 100$$

Where

B is the number of milliliters of standard sodium hydroxide solution added
A is the number of milliliters of standard nitric acid to back titrate the excess sodium hydroxide
N_1 is the normality of the sodium hydroxide solution
N_2 is the normality of the nitric acid solution
W is the weight of the specimen, gram

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The tricresylphosphate in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The tricresylphosphate in the sample for test shall be recorded to the nearest one percent

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PHOSPHATE PLASTICIZER, SODIUM METHOD

1. SCOPE

1.1 This method is intended for use in determining the phosphate plasticizer in rubber compounds The specimen is treated with metallic sodium, the phosphorous pentoxide determined in the solution, and the plasticizer calculated as tricresylphosphate.

2. SPECIMEN

2.1 The specimen shall be as described in method 15821

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Test tubes about 3 inches in length and ⅜-inch outside diameter

3.1.2 Metallic sodium

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and placed in a test tube containing a small piece of metallic sodium The contents of the tube shall be carefully heated until the specimen is completely decomposed and sodium vapor deposited an inch up the sides of the tube The lower end of the hot tube shall be dipped into 10 ml of distilled water contained in a small beaker so as to crack the tube and permit water to react with the excess sodium (CARE) The bottom of the tube containing the specimen shall be crushed in the beaker, the solution boiled for a few minutes and filtered The filtrate shall be made slightly acid with nitric acid, boiled for 10 minutes, and filtered The remainder of the procedure shall be as described in 4 2 of method 15821

5. RESULTS

5.1 The results shall be as described in method 15821

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CHEMICAL ANALYSIS OF RUBBER COMPOUNDS, GENERAL

1. SCOPE

1.1 These methods are intended for use in the analysis of natural rubber compounds and some types of synthetic rubber compounds to determine whether they conform to the composition requirements of Federal specifications Tests are described for the identification of certain constituents of rubber compounds and quantitative methods are included for the determination of rubber content, various forms of sulfur, extractable materials, fillers, fibrous materials, free carbon, glue, and packing materials General methods are included as an introduction to groups of methods of similar tests such as fillers, sulfur, extractable materials, etc , which give definitions, certain limitations and applications of the methods, and a general scheme relating the methods in the group These general methods shall be considered a part of the individual methods wherever applicable

1.2 **Rubber content.**—Two methods are described for the determination of rubber content, namely, the indirect and direct methods

1.2.1 *Indirect method* —The indirect method consists of determining the nonrubber constituents and calculating the rubber content by difference This method involves certain arbitrary features, hence the results obtained may not in all cases be in precise agreement with the amount of rubber employed in the manufacturing process Also, it may not give accurate results when applied to compounds containing a high percentage of factice, hard asphalt, or mineral rubber Reclaimed rubber in the compound shows up in the results of the analysis simply as "rubber "

1.2.2 *Direct method* —The direct method consists of oxidizing the natural rubber by digestion in chromic acid solution, distilling off the acetic acid formed, and titrating the acetic acid with standard sodium hydroxide The direct method is applicable only to natural rubber

1.2.3 *Hard rubber* —No accurate method is given for the determination of the amount of rubber in hard rubber compounds if the compound contains fillers that decompose at 550° C (1,022° F)

1.3 In general, the procedures described in this group are written for the primary objective of determining the individual constituents normally found in rubber compounds However, the scheme of analysis given in some of the groups of tests relates the methods for use in determining the nonrubber constituents for use in calculating the rubber content by the indirect method

1.4 Certain applications and limitations of these methods to the analysis of compounds containing synthetic rubber and other materials are given in the general methods for related tests and in the scope of the individual methods

1.5 No method is described for determining the amount of reclaimed rubber in a rubber compound

1.6 A list of common reagents and equipment which should be available in the laboratory for conducting an analysis of rubber compounds is given in section 3 of this method Reagents and apparatus common only to an individual method are listed in the individual method

1.7 Section 4 of this method provides general requirements for weighings

2. SPECIMEN

2.1 Unless otherwise indicated, the specimen shall be taken from the composite sample prepared as described in method 1211

2.2 In methods where an extracted specimen is used for further analysis, the weight of the specimen prior to extraction shall be used for calculating the results

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Unless otherwise indicated, volumetric glassware used in the analysis shall be in

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accordance with National Bureau of Standards Circular 434

3.1.2 Unless otherwise indicated, reagents used in the analysis shall conform to the specifications for Analytical Reagents of the American Chemical Society where such specifications are available

3.1.3 When solutions are required to be standardized, the normality shall be determined to the fourth decimal place

3.1.4 When reagents such as acids and ammonium hydroxide are required in the individual methods by name only, concentrated solutions of the specific gravities or percentage composition given in 3 1 6 shall be used.

3.1.4.1 When the concentration of reagents such as acids and ammonium hydroxide is required in the individual methods as a ratio, it shall be the number of volumes of concentrated reagent, 3 1 4, that is diluted with a given number of volumes of distilled water For example, 1 to 7 hydrochloric acid means 1 volume of concentrated hydrochloric acid, specific gravity 1 18 to 1 19, that is diluted with 7 volumes of distilled water

3.1.4.2 When the concentration of reagents is required as percent, the value represents the number of grams of the reagent per 100 ml of solution For example, 10-percent barium chloride solution means 10 grams of barium chloride per 100 ml of solution

3.1.5 *Apparatus* —The following general laboratory apparatus shall be made available Apparatus applicable only to an individual method is described in the individual methods

3.1.5.1 Analytical balance and weights
3.1.5.2 Baths, water, steam, and air
3.1.5.3 Beakers
3.1.5.4 Burettes
3.1.5.5 Burners, gas
3.1.5.6 Crucibles, ignition, Gooch, extraction, etc
3.1.5.7 Crucible, porcelain, of varying capacities
3.1.5.8 Crucible tongs
3.1.5.9 Cylinders, graduated, etc
3.1.5.10 Desiccators and desiccants
3.1.5.11 Dishes, evaporating, etc
3.1.5.12 Extraction apparatus shown in figure 16001
3.1.5.13 Filter paper
3.1.5.14 Flasks, chemical-resistant, Erlenmeyer, suction, volumetric, etc
3.1.5.15 Funnels, filter, Buchner, separatory, etc
3.1.5.16 Furnaces, muffle
3.1.5.17 Gauze
3.1.5.18 *Gooch crucible with asbestos pad* — The pad for the crucible shall be prepared as follows Amphibole asbestos shall be cut into pieces about ⅛ inch long, then digested with 10-percent sodium hydroxide solution, washed with water, and digested a few hours with hydrochloric acid After it has been washed free from acid by decantation, the asbestos shall be shaken with distilled water in order to break up the fibers The finer fibers shall be separated from the coarser material by decantation The pad shall be prepared in the crucible with gentle suction, using the coarse fibers as the base Ridges that may appear on the pad shall be leveled by tamping with a stirring rod having a flattened end The pad shall be completed by adding several layers of finer fibers In some cases it is advisable to use two such pads, one built on top of the other, in order to retain some of the reinforcing blacks used in the rubber compounds After the Gooch has been ignited to constant weight, at a temperature of 650° to 900° C (1,202° to 1,652° F), it is ready for use
3.1.5.19 Hot plates, gas, and electric
3.1.5.20 Hydrogen sulfide generating apparatus
3.1.5.21 Oven, circulating-air, with suitable controls for maintaining the required temperature throughout the oven
3.1.5.22 Pipettes
3.1.5.23 Spatulas
3.1.5.24 Stirring rods
3.1.5.25 Test tubes
3.1.5.26 Apparatus such as thermometers, copper-constantan thermocouple and potentiometer, or other device for measuring the temperature to within ±1 0° C (1 8° F)

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3.1.5.27 Watch glasses
3.1.5.28 Weighing bottles
3.1.5.29 Wire triangles

3.1.6 *Reagents* —The following general laboratory reagents shall be made available Reagents applicable only to an individual method are described in the individual methods

3.1.6.1 Acetic acid, glacial

3.1.6.2 Acetone meeting the requirements of Federal Specification O–A–51, or U S P grade acetone distilled over anhydrous potassium carbonate not more than 10 days before use The fraction that distills at 56° to 57° C shall be used

3.1.6.3 Alcohol, ethyl, methyl

3.1.6.4 Ammonium hydroxide, specific gravity, 0 90 to 0 91

3.1.6.5 Barium chloride

3.1.6.6 *Barium chloride, 10-percent solution* —The solution shall be prepared by dissolving 100 grams of barium chloride in distilled water, diluting the solution to one liter with distilled water, and adding 2 or 3 drops of hydrochloric acid If there is any insoluble matter or cloudiness, the solution shall be heated overnight on a steam bath and filtered

3.1.6.7 Benzene

3.1.6.8 Bromine

3.1.6.9 Carbon tetrachloride

3.1.6.10 Chloroform, U S P grade

3.1.6.11 Congo-red indicator solution and paper.

3.1.6.12 Distilled water.

3.1.6.13 Ether, ethyl

3.1.6.14 Hydrochloric acid, specific gravity 1 18 to 1 19

3.1.6.15 Hydrogen sulfide

3.1.6.16 Iodine

3.1.6.17 Litmus paper

3.1.6.18 Methyl orange indicator

3.1.6.19 Methyl red indicator

3.1.6.20 Nitric acid, specific gravity 1 42 to 1 43

3.1.6.21 Perchloric acid, 70 percent

3.1.6.22 Petroleum ether

3.1.6.23 Phenolphthalein indicator solution

3.1.6.24 Potassium chlorate

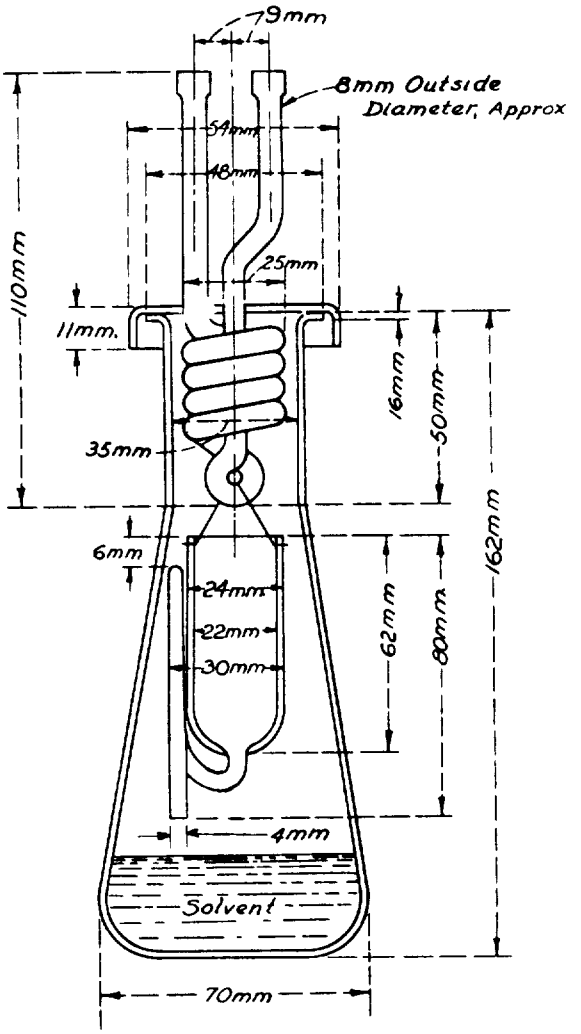

FIGURE 16001 —Extraction apparatus

3.1.6.25 *Mineral oil solvent for rubber* —The rubber solvent shall be prepared by mixing "300° mineral seal" oils from two or more sources to obtain a suitable solvent having the following characteristics

Saybolt Universal viscosity at 68° F	56 seconds
Saybolt Universal viscosity at 100° F	45 seconds
Flashpoint	132 2° C (270° F)
Fire point	176 7° C (350° F)
Specific gravity	0 853
Color	Colorless

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An oil having an aniline point of 84° ± 1° C (183 2° ± 1 8° F) has been found satisfactory

3.1.6.26 Sodium carbonate

3.1.6.27 Sodium hydroxide

3.1.6.28 Starch

3.1.6.29 Sulfuric acid, specific gravity 1 83 to 1 84

4. PROCEDURE

4.1 Unless otherwise indicated, all weighings shall be made to 1 part in 1,000

360

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PRELIMINARY EXAMINATION OF SAMPLE FOR TEST

1. SCOPE

1.1 This method is intended to aid the analyst in the selection of the proper procedures to be used in the analysis of rubber compounds Methods of analyses are described for identifying certain constituents, and quantitative methods are referenced which will enable the analysis to be conducted in the presence or absence of these constituents Certain applications and limitations of the methods are given in the general method preceding a group of similar methods and in the scope of the individual methods

2. SPECIMEN

2.1 The specimen for each method of analysis shall consist of a portion of the rubber compound from the composite sample prepared as described in method 1211

3. PROCEDURE

3.1 The procedure for the preliminary examination shall be as follows

3.1.1 *Carbonates* —A specimen shall be dropped into a test tube containing hydrochloric acid saturated with bromine If a stream of bubbles is given off, fillers shall be determined by method 16411 or 16421

3.1.2 *Antimony and lead* —A 0 2- to 0.3-gram specimen shall be placed in a porcelain crucible and the rubber burned off as described in method 16431 The ash shall be treated with 10 ml of hydrochloric acid and the solution evaporated to dryness The residue shall be dissolved in 1 ml of hydrochloric acid and the solution diluted with distilled water to a volume of 100 ml The solution shall be filtered and the filtrate saturated with hydrogen sulfide If a precipitate of antimony sulfide or lead sulfide forms, the organic plus the inorganic sulfur shall be determined by the fusion method 16241 If a precipitate of antimony sulfide is formed, the inorganic sulfur in fillers shall be determined as described in method 16251 If antimony sulfide is present, fillers shall be determined by method 16411 or 16421

3.1.3 *Carbon black* —A specimen shall be heated with nitric acid until there is no more frothing If black particles are suspended in the liquid, the free carbon shall be determined by method 16511

3.1.4 *Barium salts* —A qualitative test for acid-soluble barium salts shall be made if the sample contains carbonates, 3 1 1 A specimen shall be ashed in a porcelain crucible The ash shall be digested with 1-to-1 hydrochloric acid and the mixture filtered If a white precipitate which is insoluble in an excess of hydrochloric acid is obtained when a few drops of 1-to-5 sulfuric acid is added to the filtrate, the presence of barium is indicated and the organic sulfur shall be determined by the fusion method 16241

3.1.5 *Glue* —The specimen shall be extracted with a mixture of 32 percent acetone and 68 percent chloroform by volume for at least 8 hours, as described in method 16311 The specimen shall be dried at a temperature of 70°±2° C (158°±3 6° F) and digested for one hour with hot distilled water The mixture shall be filtered and the filtrate cooled A few drops of freshly prepared 2-percent tannic acid solution shall be added to the filtered solution and the solution allowed to stand for a few minutes If the solution becomes turbid and the turbidity persists on the addition of an excess of tannic acid, glue shall be determined as described in method 16521

3.1.6 *Factice* —The specimen remaining from the test for glue shall be digested with hot 15-percent sodium hydroxide solution The liquid shall be decanted and acidified with hydrochloric acid Any cloudiness or precipitate indicates the presence of factice and the alcoholic potassium hydroxide extract shall be determined as described in method 16341

3.1.7 *Waxy hydrocarbon* —The specimen shall be extracted with acetone for 8 hours as described in method 16321 If waxy hydrocarbons are present, they will solidify in the cold acetone extract as a white flocculent precipitate clinging to the sides of the flask Waxy hydrocarbon shall be determined as described in method 16361

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RUBBER CONTENT, GENERAL

1. SCOPE

1.1 These methods are intended for use in determining the amount of rubber in rubber compounds They are applicable to the determination of the rubber content of all soft natural rubber compounds and to some compounds containing synthetic rubber Two methods are described for determining the rubber content—the indirect and direct An outline of the steps required in determining rubber content by the indirect method is given in method 16111 and the detailed procedure for the direct method is given in method 16121 Certain limitations on the use and applicability of these methods are given in the following paragraph and in the individual methods.

1.2 Rubber content, indirect method.—In this method the nonrubber constituents are determined individually or in groups and the rubber content is determined by difference If, in using this method, fillers are determined by the ash method 16431, satisfactory results on most rubber compounds will be obtained except where decomposable compounding ingredients such as carbonates that decompose at 550° C (1,022° F), clay, asbestos, asbestine, talc, and antimony sulfide are found to be present. In the presence of carbonates decomposing at or below 550° C (1,022° F) or antimony sulfide, the mineral-oil solvent method 16411 or the paranitrotoluene-orthodichlorobenzene solvent method 16421 will give satisfactory results in most cases No method for filler determination is described which will give accurate results in the presence of clay, asbestos, asbestine, or talc unless a correction is made for loss of water of hydration on ashing.

1.3 Rubber content, direct method.—In this method, the rubber content is determined directly The method is applicable to the determination of natural rubber content of crude and vulcanized natural rubber compounds and mixtures of natural rubber with certain rubberlike materials It is also applicable to reclaimed natural rubber, but has been found to give consistently low results when compared to previously accepted estimates of the rubber hydrocarbon content of reclaimed rubber If the material is unvulcanized or reclaimed rubber suspected of containing mineral rubber or factice, the material shall be vulcanized by a simple known recipe before analysis The procedure consists essentially of quantitatively oxidizing the rubber hydrocarbon to acetic acid by means of digestion in chromic acid solution, separating the acetic acid by distillation, and titrating the acetic acid with standard sodium hydroxide solution after carbon dioxide has been removed by aeration The degree of interference of certain materials in the application of this method is given in table I

TABLE I —*Interfering materials and degree of interference*

Material	Degree of interference
Combined sulfur	None in normal soft cures.
Carbon black	None as tested in tread stocks
Cellulose	Negligible, 2 percent or less of its weight reacts as if it were rubber hydrocarbon
Aliphatic hydrocarbon (mineral rubber)	Removed by acetone and chloroform extraction If not extracted, approximately 45 percent of its weight reacts as rubber hydrocarbon
Factice, brown	Negligible, after acetone and chloroform extraction
Balata	Approximately equivalent to rubber
Hard rubber	Approximately 50 percent of its weight reacts as rubber hydrocarbon
Nitrile rubber	Approximately 1 5 to 2 percent of its weight reacts as rubber hydrocarbon
Styrene rubber	Approximately 3 percent of its weight reacts as rubber hydrocarbon

1.4 Hard rubber.—The mineral oil and paranitrotoluene-orthodichlorobenzene solvent methods are not applicable to the analysis of hard rubber compounds The ash method 16431 is applicable to the determination of

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fillers in hard rubber compounds if fillers do not decompose at a temperature of 550° C (1,022° F.) No accurate method is given for the determination of rubber in hard rubber compounds if fillers decomposing at 550° C. (1,022° F) are present.

1.5 Unless otherwise specified in the detail specification, the indirect method 16111 shall be used for determining the rubber content of rubber compounds analyzed for conformance with Federal Specifications

2. DEFINITIONS

2.1 Natural rubber hydrocarbon.—Natural rubber hydrocarbon is the characteristic and major component of crude natural rubber and is considered to have the chemical formula $(C_5H_8)_x$ It is approximately 94/97 of the rubber content obtained by the indirect method of analysis

2.2 Natural rubber as compounded.—Natural rubber as compounded is approximately equivalent to the crude rubber used in the manufacture of a rubber compound. Its computation is based on the arbitrary assumption that crude natural rubber consists of 94 percent rubber hydrocarbon, and 6 percent resins, proteins, and other substances

2.3 Natural rubber by volume.—Natural rubber by volume is the percentage of the volume of a natural rubber compound occupied by the natural rubber as compounded, assuming the latter to have a specific gravity of 0 911 at 25° C (77° F.)

2.4 Natural rubber hydrocarbon by volume.—Natural rubber hydrocarbon by volume is the percentage of the volume of natural rubber compound occupied by the natural rubber hydrocarbon, assuming the latter to have a specific gravity of 0 911 at 25° C (77° F)

2.5 Synthetic rubber as compounded.—Synthetic rubber as compounded is considered to be the rubber content as determined by the indirect method of analysis 16111 This differs from the true synthetic rubber content by the amount of the nonrubber constituents present in the crude synthetic rubber Since this quantity is variable for different rubbers, the results are approximate

2.6 Synthetic rubber by volume.—Synthetic rubber by volume is the percentage of a rubber compound occupied by the synthetic rubber as compounded.

2.7 In methods where an extracted specimen is used for further analysis, the weight of the specimen prior to extraction shall be used for calculating the results

3. RESULTS

3.1 Calculation.

3.1.1 *Natural rubber.*—The natural rubber content of the specimen shall be calculated as follows

3.1.1.1 Natural rubber hydrocarbon by weight, percent, $A = \frac{94}{97}(100 - C)$

Where

C is the sum of the following constituents Total extract, alcoholic potassium hydroxide extract, organic sulfur, free carbon, glue, cellulose, and inorganic fillers, method 16111

The factor 94/97 serves to correct for the natural proteins

3.1.1.2 Natural rubber as compounded by weight, percent, $B, = \frac{100}{94} \times A$

3.1.1.3 Natural rubber by volume,

$$\text{percent}, = \frac{B \times \text{specific gravity of compound}}{0\ 911}$$

3.1.1.4 Natural rubber hydrocarbon by volume, percent, $= \frac{A \times \text{specific gravity of compound}}{0\ 911}$

3.1.2 *Synthetic rubber.*—The synthetic rubber content of the specimen shall be calculated as follows

3.1.2.1 Synthetic rubber as compounded by weight, percent, $D, = 100 - C$

Where

C is the sum of the following constituents total extract, alcoholic potassium hydroxide extract, organic sulfur, free carbon, glue, cellulose, and organic fillers, method 16111

3.1.2.2 Synthetic rubber by volume,

$$\text{percent}, = \frac{D \times \text{specific gravity of compound}}{X}$$

Where

X is the specific gravity of synthetic rubber, 0 933 for GR-S, 0 917 for GR-I, and 1 27 for chloroprene rubber

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3.2 The natural rubber content shall be recorded as rubber hydrocarbon, rubber as compounded, rubber by volume, or rubber hydrocarbon by volume, as specified in the detail specification

3.3 The synthetic rubber content shall be recorded as rubber as compounded or rubber by volume as specified in the detail specification

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RUBBER CONTENT, INDIRECT METHOD

1. SCOPE

1.1 This method is intended for use in determining the amount of natural or synthetic rubber in rubber compounds by determining the amount of nonrubber constituents and calculating the amount of rubber The constituents required for determining the rubber content by difference are listed The procedures for determining the required constituents are referenced and where more than one method is referenced, the application of each method is discussed A general scheme, figure 16111, is presented for the convenience of the analyst which shows the relationship and general separation of the various components that are to be determined Certain applications and limitations of the methods used for determining the nonrubber constituents are given in 4 and also in the scope of the individual methods This method is applicable to the determination of the rubber content of all types of soft natural rubber compounds and to some compounds containing synthetic rubber Any reclaimed rubber used in the rubber compound shows up in the results simply as "rubber" This method may not give accurate results when applied to compounds containing high percentages of factice, hard asphalt, or "mineral rubber" It is applicable to the determination of rubber content of synthetic rubber compounds that can be successfully extracted with acetone or chloroform. Nitrile-type rubber compounds shall not be extracted with acetone or chloroform Analysis of this type of compound is therefore not practicable by this method except to estimate the rubber content Compounds containing isobutylene rubber will not extract well with acetone but can be more successfully extracted with methyl ethyl ketone

2. SPECIMEN

2.1 The specimen shall consist of a portion of the rubber compound from the composite sample prepared as described in method 1211, as required in the applicable method listed in 4 1

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as required in the methods referenced in 4 1

4. PROCEDURE

4.1 The procedure shall consist of determining the following nonrubber constituents and calculating the rubber content by difference The specific gravity shall also be determined for use in calculating the amount of rubber by volume

4.1.1 Specific gravity, method 14011

4.1.2 Total extract, method 16311

4.1.3 Inorganic fillers, methods 16411, 16421, or 16431, see 4 3

4.1.4 Organic sulfur, see 4 4

4.1.5 Alcoholic potassium hydroxide extract, method 16341

4.1.6 Free carbon, method 16511

4.1.7 Cellulose (fibrous materials), methods 16531 or 16535, see 4 5

4.1.8 Glue, method 16521.

4.2 **Scheme of analysis.**—The scheme of analysis for determining rubber content by the indirect method is given in figure 16111 for the convenience of the analyst in planning his procedure for the determination of the necessary nonrubber constituents

4.3 **Inorganic fillers.**—The inorganic fillers in rubber compounds are those inorganic constituents other than free or elemental sulfur and free carbon that have been added to the compounds Fillers are discussed in method 16401 and in the following paragraphs

4.3.1 The ash method 16431 shall be used for the determination of fillers if carbonates, antimony sulfide, or other inorganic materials that decompose at the required ashing temperature are absent This method is not applicable to chloroprene rubber when zinc or other metal compounds that form volatile substances are present

4.3.2 The oil solvent method 16411 shall be used for determining fillers if antimony sulfide, carbonates, or other decomposable inorganic materials are present in the rubber compound and if the rubber is soluble in the mineral oil

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solvent The oil-solvent method is not applicable to compounds containing nitrile rubber or to any synthetic rubber that will not dissolve in the oil solvent, or to hard rubber compounds

4.3.3 The paranitrotoluene-orthodichlorobenzene solvent, method 16421, shall be used for the determination of fillers if antimony sulfide, carbonates, or other decomposable inorganic materials are present in the rubber compounds and if the rubber is soluble in the paranitrotoluene-orthodichlorobenzene solvent but is not soluble in the mineral oil solvent The paranitrotoluene-orthodichlorobenzene solvent method is not applicable to compounds containing nitrile rubber or hard rubber compounds

4.3.4 In case of dispute or disagreement between the ash and the solvent methods, method 16411 shall be used for rubber compounds where the rubber is soluble in the mineral oil solvent, and method 16421 shall be used for rubber compounds where the rubber is soluble in the paranitrotoluene-orthodichlorobenzene mixture and not soluble in the mineral oil solvent

4 4 Organic sulfur.—The definition of organic sulfur is given in method 16201

4.4.1 *Organic plus inorganic sulfur* —The organic sulfur is obtained by determining the organic sulfur plus the inorganic sulfur in an extracted specimen of the rubber compound, then determining the inorganic sulfur in the fillers in another specimen from the same compound, and subtracting the inorganic sulfur in the fillers from the organic plus inorganic sulfur in the compound If the rubber compound contains acetone soluble barium salts or antimony or lead compounds, method 16241 shall be used for determining the organic plus inorganic sulfur If the rubber compound does not contain acetone soluble barium salts or antimony or lead compounds, method 16231, which is less time consuming than method 16241, may be used for determining the organic plus the inorganic sulfur in the compound

4.4.2 *Inorganic sulfur* —The inorganic sulfur in fillers shall be determined as described in method 16251 if the rubber compound contains acetone-soluble barium salts or antimony or lead compounds and if the rubber is soluble in the mineral oil solvent or paranitrotoluene-orthodichlorobenzene mixture If the rubber compound does not contain antimony, the inorganic sulfur in fillers may be determined as described in method 16261, which is less time consuming than method 16251.

4.4.3 Whichever method, 16231 or 16241, is used for determining organic sulfur plus inorganic sulfur in the compound, shall also be used for determining inorganic sulfur in fillers, methods 16251 or 16261

4.4.4 These methods are not applicable to the determination of organic sulfur in polysulfide rubber or in compounds which contain nitrile or polyisobutylene rubber

4.5 Fibrous materials.—For the determination of fibrous materials, method 16531 shall be used for compounds containing natural rubber and other rubbers that are soluble in the mineral oil solvent, and method 16535 shall be used if the rubber is soluble in paranitrotoluene-orthodichlorobenzene solvent but is not soluble in the mineral oil solvent

5. RESULTS

5.1 Calculation.

5.1.1 *Natural rubber* —The natural rubber content of the specimen shall be calculated as follows

5.1.1.1 Natural rubber hydrocarbon by weight, percent, $A, = \frac{94}{97}(100 - C)$

Where

C is the sum of the nonrubber constituents listed in 4 1 2 to 4 1 8, inclusive, percent

5.1.1.2 Natural rubber as compounded by weight, percent, $B, = \frac{100}{94} \times A$

5.1.1.3 Natural rubber by volume, percent, $= \frac{B \times \text{specific gravity of compound}}{0\ 911}$

5.1.1.4 Natural rubber hydrocarbon by volume, percent, $= \frac{A \times \text{specific gravity of compound}}{0\ 911}$

5.1.2 *Synthetic rubber* —The synthetic rubber content of the specimen shall be calculated as follows

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5.1.2.1 Synthetic rubber as compounded by weight, percent, $B, = 100 - C$

Where

C is the sum of the nonrubber constituents listed in 4 1 2 to 4 1 8, inclusive, percent

5.1.2.2 Synthetic rubber by volume, percent, $= \frac{B \times \text{specific gravity of compound}}{X}$

Where

X is the specific gravity of the synthetic rubber and is 0 933 for GR-S, 0 917 for GR–I, and 1 27 for chloroprene

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The rubber content of the sample for test shall be the average of the values obtained from the specimens tested

5.4 The rubber hydrocarbon, rubber as compounded, rubber by volume, or rubber hydrocarbon by volume of the sample for test shall be recorded to the nearest one percent

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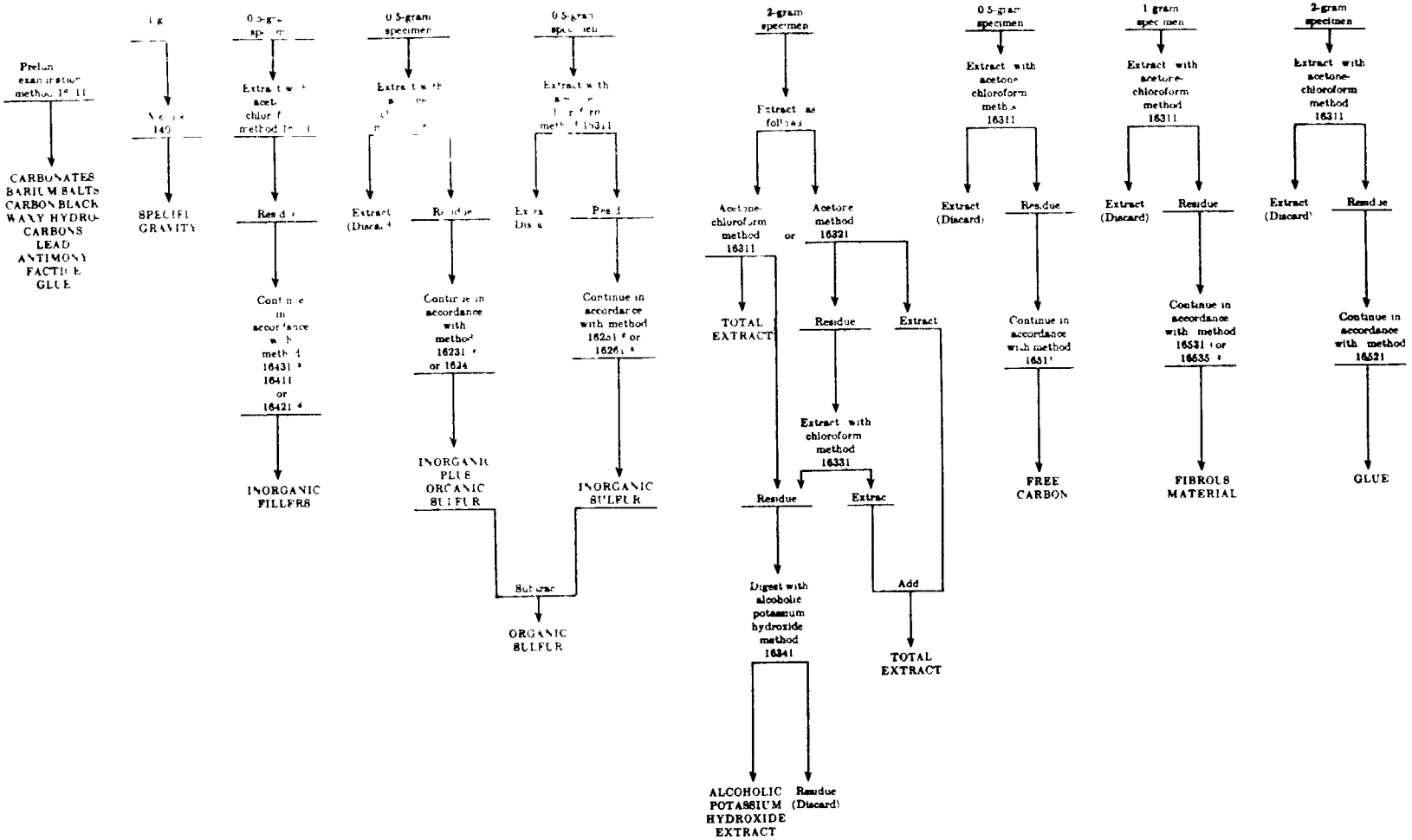
SCHEME OF ANALYSIS FOR DETERMINING RUBBER CONTENT BY THE INDIRECT METHOD



[a] Whenever method 16431 is used for determining inorganic fillers a 2-gram specimen shall be tested.
[b] No decomposable materials at ashing temperature present.
[c] Decomposable materials present, rubber soluble in mineral oil solvent.
[d] Decomposable materials present, rubber not soluble in mineral oil solvent but soluble in paranitrotoluene-orthodichlorobenzene solvent.
[e] Acid-soluble barium salts, antimony compounds or lead compounds absent.
[f] Acid-soluble barium salts, antimony compounds or lead compounds present.
[g] Antimony compounds present, rubber soluble in mineral oil solvent or paranitrotoluene-orthodichlorobenzene solvent.
[h] Antimony compounds absent.
[i] Rubber soluble in mineral oil solvent.
[k] Rubber not soluble in mineral oil solvent but soluble in paranitrotoluene-orthodichlorobenzene solvent.

FIGURE 16111.

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RUBBER CONTENT, DIRECT METHOD

1. SCOPE

1.1 This method is intended for use in determining the natural rubber content of crude and vulcanized natural rubber compounds, and mixtures of natural rubber with certain rubber-like materials In the case of reclaimed rubber, the results are consistently lower than accepted estimates of rubber hydrocarbon content of reclaimed rubber The results will also be low in the case of hard rubber since only approximately 50 percent of its weight reacts as if it were rubber hydrocarbon The degree of interference of certain materials is given in table I of method 16101 Definitions relating to rubber content and methods of calculation are given in method 16101

2. SPECIMEN

2.1 The specimen shall consist of a portion of the rubber compound from the composite sample prepared as described in method 1211 containing approximately 0.3 gram of rubber hydrocarbon The specimen shall be sheeted to a thickness of not more than 0 5 mm

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents described in method 16001 as required

3.1.2 *Digestion, distillation, and aeration apparatus* —The digestion, distillation, and aeration apparatus are shown in figure 16121 Rubber connections shall not be used where they may come in contact with the digestion mixture Spring clamps are used to secure joints wherever possible Pinch clamps are provided for positions 1 and 2 A glass ball and socket joint is desirable between *I* and *J*

3.1.2.1 The steam generator consists of a 3-liter flask *B* provided with an electric immersion heater *C* The heating element is prepared by winding a length of No 25 A W. G Chromel wire having a resistance of about 25 ohms into a coil ¼ inch in diameter This coil is mounted in the bottom of the flask as shown in figure 16121 and is provided with leads of No 10 A W G copper wire which is passed through the stopper of the flask The heater is connected in series with an 11-ohm slide-wire resistance A to a 110-volt outlet The rate of generation of steam can be controlled precisely by changing the setting of the resistance It is recommended that 110 volts a c be used in the heater coils of the steam generator.

3.1.2.2 The aeration assembly *M* contains a capillary tube which when connected to a vacuum line maintains an airflow through the receiving flask *L* of approximately 2 liters per minute If the vacuum is less than 30 ml of mercury, a capillary tube approximately 10 cm in length with an 0 75-mm bore will maintain the required airflow Since it is essential that the aeration be maintained at a rate of within 10 to 20 percent of 2 liters per minute, each capillary tube shall be tested before use The following method may be used for testing the tube A graduate shall be inverted over a beaker filled with water and the air evacuated through the capillary by means of a tube extending up into the graduate The rate of airflow will be the same as the rate at which the water fills the graduate After the initial calibration, it is convenient to place a locking device on the vacuum-line valve so that the valve may be opened to a position which will maintain an airflow at the same rate for subsequent determinations

3.1.3 *Chromic acid digestion mixture* —The digestion mixture shall be prepared by dissolving 200 grams of chromic acid in 500 ml of distilled water, adding 150 ml of sulfuric acid, and mixing well.

3.1.4 Sodium hydroxide solution 0 1 *N* standardized

4. PROCEDURE

4 1 The specimen shall be weighed to the nearest milligram and wrapped loosely in filter or lens paper The specimen from vulcanized-

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rubber compounds shall be extracted with acetone, method 16321, followed by extraction with chloroform, method 16331, or the specimen shall be extracted with acetone-chloroform mixture, method 16311 Unvulcanized rubber or reclaimed rubber containing mineral rubber or factice shall be vulcanized before extracting with acetone, followed by chloroform, or before extracting with acetone-chloroform mixture The specimen from unvulcanized rubber or reclaimed rubber which does not contain these ingredients shall be extracted with acetone only

4 2 After extraction, the specimen shall be removed from the filter or lens paper If the lens or filter paper adheres to the specimen, as much of the paper as practicable shall be removed The extracted specimen shall be dried for one hour at a temperature of 100° ±2° C (212°±3 6° F) From 1,200 to 1,500 ml of distilled water shall be added to the steam generating flask *B*, figure 16121, and sufficient distilled water to cover the end of the adapter shall be added to the flask *L* The steam generator shall be started at least ½ hour before the digestion period is completed so that steam is available at the end of the digestion period During the digestion period, the steam tube shall be closed at position 2 with a pinch clamp The digestion flask *F* shall be marked at a point indicating the liquid level when it contains 75 ml The steam tube shall be lifted from the digestion flask and 50±1 ml of the chromic acid digestion mixture added to the digestion flask

4.3 The extracted and dried specimen shall be inserted in the digestion flask and the steam tube replaced The water bath *G* surrounding the digestion flask shall be heated to boiling and the boiling continued for at least one hour from the time the digestion mixture in the flask starts to boil Care should be taken that no particles of rubber or filter or lens paper adhere to the sides of the digestion flask during the digestion period If particles do adhere, the supporting clamp for *H* should be loosened and the digestion flask carefully shaken so that the particles are swirled into the digestion solution Again care should be taken not to loosen the spring clamps at the junctions of *F* and *H*, *E* and *H*, and *H* and *I*, to prevent the loss of acetic acid

4.4 At the end of the digestion period, the beaker of boiling water shall be removed and the digestion flask cooled for 5 minutes The volume of the digestion mixture 50 ml. shall be increased to 75 ml by the direct addition of 25 ml of distilled water The steam tube shall be closed at position 1 with a pinch clamp, the steam tube shall be opened at position 2 by releasing the pinch clamp, and 25 ml of distilled water shall be added simultaneously through the funnel *D* taking care that the water is directed down into the digestion flask *F* This operation should be completed as rapidly as possible to avoid the loss of acetic acid The steam tube shall be closed at position 3 with the pinch clamp, and the pinch clamp opened at position 1, thus allowing steam to flow from the steam generator into the digestion flask A Bunsen burner with a small flame shall be placed under the digestion flask, keeping the flame at a point where it will maintain the liquid in the flask at a volume of 75 ml throughout the distillation The distillation shall be continued until 500 ml has been collected in the receiving flask *L* At the completion of the distillation, the burner shall be removed, followed by removal of the receiving flask The condenser *J* and adapter *K* shall be rinsed with distilled water added at the top of the condenser. The washings shall be collected and added to the distillate

4.5 The temperature of the liquid in the receiving flask shall be adjusted to 25°±5° C and the aeration assembly attached to the receiving flask which is connected to the vacuum line A stream of air shall be drawn through the liquid for 30 minutes at a rate of 2 liters per minute ***Caution:*** Variations in the temperature, type of apparatus, and rate of airflow may lead to erroneous results.

4.6 After 30 minutes' aeration, the aeration assembly shall be disconnected from the receiving flask and the glass tubing rinsed with water from the wash bottle, catching the rinsings in the receiving flask Phenolphthalein indicator solution shall be added to the receiving flask

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and the solution titrated with 0 1 *N* standardized sodium hydroxide solution

4.7 In order to check for complete digestion of the rubber after the analysis has been completed, the digestion solution shall be poured into a beaker of water If particles of rubber hydrocarbon float to the surface, digestion is incomplete and it will then be necessary to repeat the procedure using a longer digestion period

4.8 A blank determination shall be made using the procedure described in 4.2 through 4 6. If filter or lens paper remains on the specimen after extraction, a blank should be determined using a similar amount of filter or lens paper The volume of the solution resulting from the determination of the blank shall be the same as that involved in the actual determination

5. RESULTS.

5.1 Calculation.

5.1.1 The natural rubber hydrocarbon content of the specimen shall be calculated as follows

5.1.1.1 Natural rubber hydrocarbon by weight, percent, $A=\frac{9\,08\times(A-B)\times N}{C}$

The above calculation is based on the observation that 75-percent yield is always obtained in the reaction $(C_5H_8)_n=n\ CH_3COOH$, using purified rubber

Where

A is the number of milliliters of the standard sodium hydroxide solution required to titrate the specimen

B is the number of milliliters of the standard sodium hydroxide solution required to titrate the blank

C is the weight of the specimen before extraction, grams

N is the normality of the sodium hydroxide solution

5.1.1.2 Natural rubber as compounded by weight, percent. $E,=\frac{100}{94}\times A$

5.1.1.3 Natural rubber by volume, percent, $=\frac{E\times\text{specific gravity of compound}}{0\,911}$

5.1.1.4 Natural rubber hydrocarbon by volume, percent, $=\frac{A\times\text{specific gravity of compound}}{0\,911}$

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The natural rubber content of the sample for test shall be the average of the values obtained from the specimens tested

5.4 The rubber hydrocarbon, rubber as compounded, rubber by volume, or rubber hydrocarbon by volume of the sample for test shall be recorded to the nearest one percent

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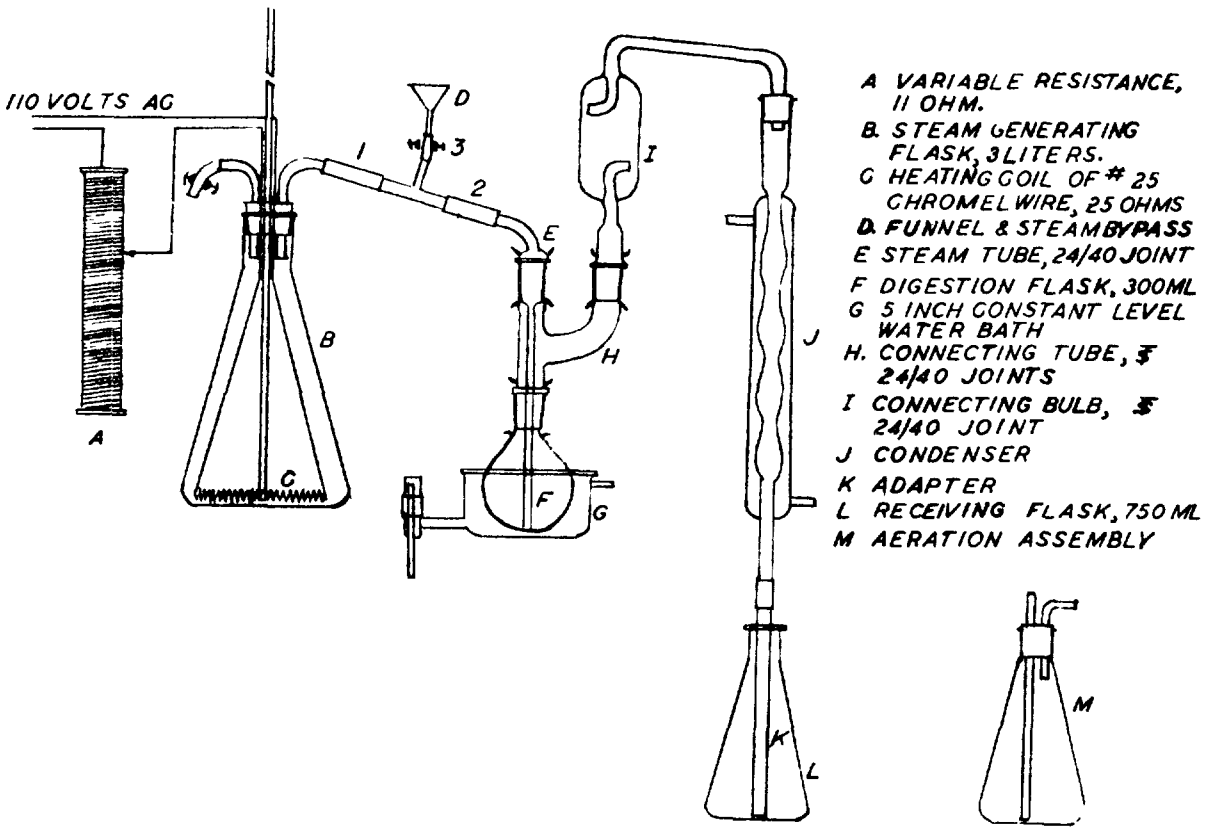

FIGURE 16121 —Apparatus for the direct determination of rubber

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SULFUR, GENERAL

1. SCOPE

1.1 This method gives definitions for the various forms of sulfur found in rubber compounds and a general scheme relating the determination of the various forms of sulfur in order to arrive at the organic sulfur for use in calculating the rubber content in the indirect method 16111 Methods of analysis for determining the various forms of sulfur are referenced in the definitions and scheme of analysis Certain applications and limitations of the procedure are given in the definitions Other applications and limitations are given in the scopes of the individual methods These methods are not applicable to thioplasts and polyisobutylene rubber

2. DEFINITIONS

2.1 Free sulfur.—The free or elemental sulfur in a rubber compound is that sulfur which is removed by extraction with a 5-percent solution of sodium sulfite It includes only the elemental sulfur and does not include any sulfur in organic compounds except in rare instances where a slight interference from such compounds is encountered Free sulfur shall be determined by method 16211

2.2 Sulfur in extract.—Sulfur in extract is that sulfur which is removed from a rubber compound by extraction with acetone, method 16321, or acetone-chloroform mixture, method 16311, and includes not only elemental sulfur but sulfur present in soluble organic compounds It is necessary to determine sulfur in extract when acetone extract corrected, organic extract, or organic acetone extract is required Sulfur in extract shall be determined by method 16221

2.3 Organic sulfur.—The organic sulfur is that sulfur in a rubber compound which is not removed during the extraction with acetone, method 16321, or with the acetone-chloroform mixture, method 16311, and which is present in the rubber compound exclusive of that contained in inorganic sulfates or sulfides which may be present The organic sulfur is not intended to represent the sulfur of vulcanization because the methods used to obtain it are not adequate to give an exact value for this sulfur The organic sulfur as determined includes (1) sulfur combined with rubber, (2) sulfur present in accelerators which is not removed by extraction with acetone or acetone-chloroform mixture, and (3) part of the sulfur present in factice and mineral rubber which is not removed by extraction Two methods intended for use in determining the amount of organic sulfur in rubber are described in this group of tests In the absence of acid-soluble barium salts or antimony or lead compounds, method 16231 may be used for determining organic plus inorganic sulfur, but if any of these compounds are present, method 16241 shall be used Organic sulfur shall be calculated by subtracting the inorganic sulfur from the organic plus inorganic sulfur

2.4 Inorganic sulfur.—The inorganic sulfur in a rubber compound includes (1) a portion of the sulfur originally added to the rubber compound but which has since become combined with the fillers and (2) sulfur that is determined as inorganic sulfides or sulfates The inorganic sulfur in the fillers is determined for use in calculating the organic sulfur Method 16251 shall be used for determining inorganic sulfur in the fillers when antimony is present and method 16261 may be used when antimony is absent

2.5 Total sulfur.—The total sulfur is all the sulfur present in the rubber compound exclusive of that contained in the inorganic sulfides and sulfates which may be present in the compound Total sulfur is obtained by subtracting the inorganic sulfur in fillers from the total plus inorganic sulfur Total plus inorganic sulfur shall be determined by method 16231 or 16241, using an unextracted specimen The inorganic sulfur in fillers is determined by method 16251 or 16261

3. PROCEDURE

3.1 The general scheme relating the determination of the various forms of sulfur in rubber compounds is given in figure 16201

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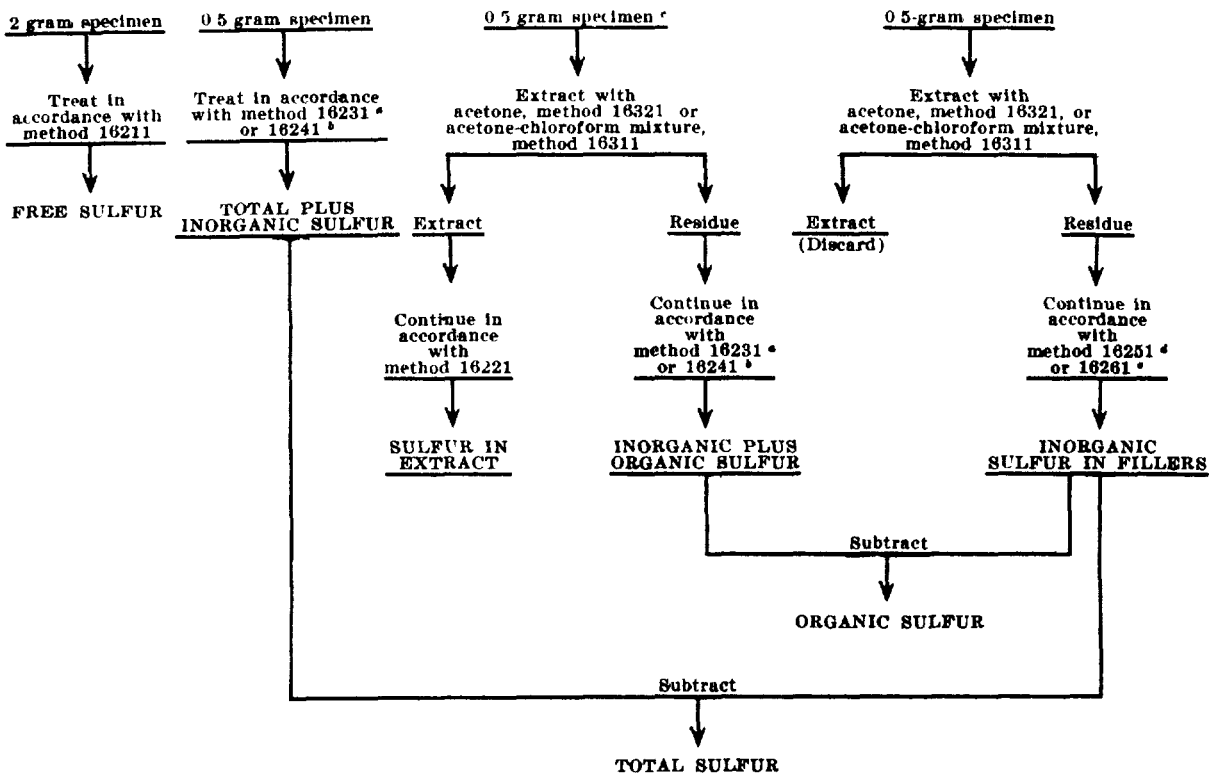
GENERAL SCHEME RELATING THE VARIOUS FORMS OF SULFUR IN RUBBER COMPOUNDS



[a] Acid soluble barium salts, antimony compounds or lead compounds absent
[b] Acid-soluble barium salts, antimony compounds, or lead compounds present
[c] Whenever sulfur in extract is required, a 2 gram specimen shall be tested
[d] Antimony compounds present, rubber soluble in mineral oil solvent or paranitrotoluene-orthodichlorobenzene solvent
[e] Antimony compounds absent

FIGURE 16201

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SULFUR, FREE

1. SCOPE

1.1 This method is intended for use in determining the amount of free or elemental sulfur in rubber compounds The sulfur determined does not include any sulfur in organic compounds except in rare instances where a slight interference from such compounds is encountered Free sulfur content does not enter into the calculation of rubber content by the indirect method This method is applicable to control work or in predicting the possibility of corrosion of metals which may come in contact with the rubber compound This method is not applicable to the determination of free or elemental sulfur in rubber compounds containing polyisobutylene rubber because of incomplete extraction of the sulfur in the rubber See method 16201 for definitions of sulfur

2. SPECIMEN

2.1 The specimen shall consist of approximately 2 grams of rubber compound taken from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Sodium sulfite, 5-percent solution

3.1.3 Sodium stearate, 0 1-percent suspension

3.1.4 Paraffin

3.1.5 Strontium chloride, 0 5-percent solution

3.1.6 Cadmium acetate, 3-percent solution

3.1.7 Starch, 1-percent solution

3.1.8 Formaldehyde, 40-percent solution

3.1.9 Iodine, 0 05 to 0 1 *N* solution, standardized

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and placed in a 400-ml , thin-walled, chemically resistant flask One hundred milliliters of 5-percent sodium sulfite solution, 3 ml of the 0 1-percent sodium stearate suspension, and approximately one gram of paraffin shall be added The flask shall be covered with a watch glass and the mixture boiled gently for at least 4 hours, or digested just below the boiling point for 16 hours The flask shall be removed from the hot plate and 100 ml of 0 5-percent strontium chloride and 10 ml of 3-percent cadmium acetate shall be added

4.2 The rubber and precipitates shall be separated by filtering through a Buchner funnel with suction The Buchner funnel shall be prepared by forming a thin asbestos pad or a layer of diatomaceous earth filter over a single layer of filter paper The residue in the funnel shall be washed twice with 75 to 100 ml of a wash solution containing 40 ml of 3-percent cadmium acetate per liter of wash solution, and the wash added to the filtrate Ten ml of 40-percent formaldehyde, 10 ml of glacial acetic acid, and 5 ml of one-percent starch solution shall be added to the filtrate while stirring The temperature of the filtrate shall be adjusted to below 15° C (59° F) by addition of crushed ice The filtrate shall then be titrated with the 0 05 to 0 1 *N* standardized iodine solution to a blue end point

4.3 A blank determination shall be made following the above procedure and using the same amount of all the reagents The amount of iodine solution required for the titration, usually 0 2 to 0 3 ml , shall be subtracted from the amount used for the titration of the specimen

5. RESULTS

5.1 Calculation.—The free sulfur in the specimen shall be calculated as follows

$$\text{Free sulfur, percent} = \frac{(A - B) \times C}{W} \times 100$$

Where

A is the number of milliliters of the standard iodine solution required to titrate the specimen
B is the number of milliliters of the standard iodine solution required to titrate the blank
C is the weight of sulfur equivalent to one milliliter of the standard iodine solution, gram
W is the weight of the specimen, grams

5.2 Unless otherwise specified in the detail specification, two specimens from the composite sample shall be tested

5.3 The free sulfur in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The free sulfur in the sample for test shall be recorded to the nearest 0 1 percent

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SULFUR IN EXTRACT

1. SCOPE

1.1 This method is intended for use in determining the amount of sulfur that is removed from rubber compounds during extraction with acetone or with an acetone-chloroform mixture The sulfur determined by this method includes the free or elemental sulfur and that present in soluble organic compounds which have been added to the rubber compound It is also applicable to the determination of sulfur in the total extract whenever organic extract is required and to the determination of sulfur in acetone extract whenever acetone extract corrected or organic acetone extract is required It is not applicable to compounds containing polyisobutylene rubber because of incomplete extraction of sulfur in the rubber or to compounds containing nitrile rubber because of swelling of the rubber by acetone See method 16201 for definitions of sulfur

2. SPECIMEN

2.1 The specimen shall consist of approximately 2 grams of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Picric acid, saturated solution.

3.1.3 Zinc - nitric acid solution prepared by dissolving 200 grams of zinc oxide in 1000 ml of nitric acid

3.1.4 Fuming nitric acid

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone-chloroform mixture as described in method 16311 or with acetone as described in method 16321

4.2 The flask containing the extract shall be placed on a steam bath and the solvent distilled off, passing a gentle current of filtered air through the flask to prevent boiling of the liquid The flask shall be removed from the steam bath just before the last trace of solvent has disappeared and the current of air continued for about 10 minutes Ten milliliters of the zinc - nitric acid solution and 2 to 3 ml of bromine shall be added to the flask containing the extract and the flask covered with a watch glass The covered flask shall be placed on the side of the steam bath for 30 minutes and then heated cautiously over the direct steam until the solution is a foamy sirup If particles of organic matter or carbon remain at the end of the evaporation, a few milliliters of fuming nitric acid shall be added and the evaporation repeated until all the carbon is gone and the solution is clear, colorless, or light yellow

4.3 The mixture shall be cooled and a few crystals of potassium chlorate added with caution to assist in the oxidation of sulfur and decomposition of any nitrates The mixture shall be evaporated to dryness, cooled, and 10 ml of hydrochloric acid added The mixture shall be evaporated to dryness avoiding spattering The procedure shall be repeated until all oxides of nitrogen are decomposed

4.4 The flask shall be cooled and 50 ml of 1 to 6 hydrochloric acid added, and the mixture heated until solution is as complete as possible The solution shall be filtered while hot, the filter shall be washed, and the filtrate diluted to about 300 ml Ten milliliters of the saturated picric acid solution shall be added and the solution heated to 90° to 100° C (194° to 212° F) The sulfate shall be precipitated as barium sulfate by the dropwise addition of a slight excess of 10-percent barium chloride solution while stirring vigorously The flask shall be covered with a watch glass and then set aside for 3 hours or preferably overnight on a steam bath The barium sulfate shall be filtered and the filtrate tested for complete precipitation of the sulfate by adding a few milliliters of barium chloride The precipitate shall be washed free of acid with distilled water, transferred to a crucible, ignited

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at a temperature of 650° to 700° C (1,202° to 1,292° F.) in a muffle furnace, cooled in a desiccator, and weighed

5. RESULTS

5.1 Calculation.—The sulfur in extract of the specimen shall be calculated as follows

$$\text{Sulfur, percent,} = \frac{A \times 0.1373}{W} \times 100$$

Where

A is the weight of the barium sulfate, gram
W is the weight of the specimen before extraction, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The sulfur in extract of the sample for test shall be the average of the values obtained from the specimens tested.

5.4 The sulfur in extract of the sample for test shall be recorded to the nearest 0.1 percent

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SULFUR, ZINC - NITRIC ACID METHOD

1. SCOPE

1.1 This method is intended for use in determining all the sulfur in rubber compounds except that contained in barium sulfate If acid-soluble barium salts, antimony sulfide, or lead compounds are present, this method will give erroneous results, in which case the fusion method 16241 should be used This method may be used for the determination of organic plus inorganic sulfur in an extracted specimen, total plus inorganic sulfur in an unextracted specimen When this method is used for the determination of total plus inorganic sulfur or organic sulfur plus inorganic sulfur, it must also be used for the determination of inorganic sulfur in fillers, method 16261 This method is not applicable to compounds containing polyisobutylene rubber or to the determination of organic sulfur in compounds containing nitrile rubbers. See method 16201 for definition of sulfur.

2. SPECIMEN

2.1 The specimen shall consist of approximately 0 5 gram of soft rubber compound or 0.2 gram of hard rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Zinc-nitric acid solution as described in method 16221

3.1.3 Fuming nitric acid

3.1.4 Bromine water, saturated solution.

3.1.5 Picric acid, saturated solution

4. PROCEDURE

4.1 If organic plus inorganic sulfur is to be determined, the specimen shall be weighed to the nearest milligram, extracted with acetone-chloroform mixture as described in method 16311 or with acetone as described in method 16321, and dried for one hour at a temperature of 70° ± 2° C (158° ± 3 6° F)

4.2 If total plus inorganic sulfur is to be determined in the compound, an unextracted specimen shall be used The specimen shall be weighed to the nearest milligram

4.3 Organic plus inorganic, extracted specimen, and total plus inorganic sulfur, unextracted specimen.

4.3.1 The specimen shall be placed in a 500-ml Erlenmeyer chemically resistant destruction flask and 10 ml of the zinc - nitric acid solution shall be added The specimen shall be thoroughly moistened and the mixture set aside overnight By allowing the mixture to stand overnight, the specimen becomes partly decomposed and thus permits the addition of fuming nitric acid without the danger of ignition Fifteen milliliters of fuming nitric acid shall be added to the flask and the flask swirled rapidly to keep the specimen immersed to avoid ignition by too rapid oxidation With some specimens it may be found necessary to cool the flask in running water

4.3.2 After the solution of the rubber appears to be complete, 5 ml of the saturated bromine solution shall be added and the mixture slowly evaporated to a foamy sirup For the determination of sulfur in unvulcanized mixtures, 3 ml. of bromine is used in place of the bromine water If particles of organic matter or carbon remain at the end of the evaporation, a few milliliters of fuming nitric acid shall be added and the evaporation repeated until all the carbon is gone and the solution is clear, colorless, or light yellow

4.3.3 The mixture shall be cooled and a few crystals of potassium chlorate added with caution to assist in the oxidation of sulfur and decomposition of any nitrates The flask shall be placed on an asbestos gauze The mixture shall be evaporated to dryness, cooled, and 10 ml of hydrochloric acid added The mixture shall be evaporated to dryness avoiding spattering The addition of acid and evaporation

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to dryness shall be repeated until all the oxides of nitrogen have been dispelled

4.3.4 The flask shall be cooled Fifty milliliters of 1 to 6 hydrochloric acid shall be added and the mixture heated until solution is as complete as possible If the specimen contains barium sulfate, a white residue will remain undissolved The solution shall be filtered while hot to prevent the loss of lead sulfate The filter shall be washed and the filtrate diluted to about 300 ml Ten milliliters of the saturated picric acid solution shall be added and the solution heated to a temperature of 90° to 100° C (194° to 212° F) The sulfate shall be precipitated as barium sulfate by the dropwise addition of a slight excess of 10-percent barium chloride solution while stirring vigorously The flask shall be covered with a watch glass and then set aside for 3 hours or overnight on a steam bath The barium sulfate shall be filtered and the filtrate tested for complete precipitation of the sulfate by the addition of a few milliliters of barium-chloride solution The precipitate shall be washed free from acid with distilled water, transferred to a crucible, ignited at a temperature of 650° to 700° C. (1,202° to 1,292° F) in a muffle furnace, cooled in a desiccator, and weighed.

5. RESULTS

5.1 Calculation.—The organic plus inorganic or total plus inorganic sulfur, as applicable, of the specimen shall be calculated as follows

$$\text{Sulfur, percent,} = \frac{A \times 0.1373}{W} \times 100$$

Where

A is the weight of the barium sulfate, gram
W is the weight of the specimen before extraction, gram

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The organic plus inorganic or total plus inorganic sulfur of the sample for test shall be the average of the values obtained from the specimens tested

5.4 The organic plus inorganic or total plus inorganic sulfur of the sample for test shall be recorded to the nearest 0 1 percent

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SULFUR, FUSION METHOD

1. SCOPE

1.1 This method is intended for use in determining all of the sulfur in a rubber compound including that present in barium sulfate It is applicable to the determination of organic plus inorganic sulfur in an extracted specimen or total plus inorganic sulfur in an unextracted specimen This method is longer and more time consuming than the zinc-nitric acid method 16231 but must be used if the rubber compound contains acid-soluble barium salts, antimony compounds, or lead compounds When this method is used for the determination of organic plus inorganic sulfur or total plus inorganic sulfur, it must also be used for the determination of inorganic sulfur in fillers, method 16251 or 16261 This method is not applicable to the determination of sulfur in compounds containing polyisobutylene rubber It can be used for the determination of total plus inorganic sulfur in compounds containing nitrile rubber where extraction with acetone or acetone-chloroform mixture is not required See method 16201 for definitions of sulfur.

2. SPECIMEN

2.1 The specimen shall consist of approximately 0 5 gram of soft rubber compound or 0 2 gram of hard rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Nitric acid-bromine mixture prepared by adding an excess of bromine to nitric acid so that a layer of bromine is always present in the bottom of the bottle The solution shall be shaken and allowed to stand for 24 hours before using

3.1.3 Picric acid, saturated solution

3.1.4 *Sulfur-free gas for fusion* —A suitable sulfur-free gas may be obtained by bubbling air through petroleum ether contained in a metal cylinder specially fitted with metal gas cocks

3.1.5 Five-percent solution of sodium carbonate

3.1.6 Bromine water, saturated solution

4. PROCEDURE

4.1 If organic plus inorganic sulfur in the compound is to be determined, the specimen shall be weighed to the nearest milligram, extracted with acetone-chloroform mixture as described in method 16311, or with acetone as described in method 16321, and dried for one hour at a temperature of 70° ± 2° C (158° ± 3 6° F)

4.2 If total plus inorganic sulfur is to be determined in the rubber compound, an unextracted specimen shall be used The specimen shall be weighed to the nearest milligram

4.3 Organic plus inorganic, extracted specimen, and total plus inorganic sulfur, unextracted specimen.

4.3.1 The specimen shall be placed in a porcelain crucible and 15 ml of the nitric acid-bromine mixture added The crucible shall be covered with a watch glass and set aside in the cold for one hour The mixture shall be heated for one hour on a steam bath The cover shall be rinsed into the crucible with a small amount of distilled water and the mixture evaporated to dryness Three milliliters of nitric acid shall be added and the watch glass replaced The mixture shall be warmed for a short time on the steam bath and then set aside to cool

4.3.2 Approximately 5 grams of anhydrous sodium carbonate shall be carefully added to the mixture in small portions by means of a glass spatula The watch glass shall be raised slightly to permit the carbonate to slide down the side of the crucible The carbonate shall not be dropped directly into the acid The watch glass shall be rinsed with 2 to 3 ml of hot distilled water and the mixture thoroughly stirred with a glass rod The mixture shall be digested for a few minutes, then spread halfway up the side of the crucible to facilitate drying, and dried on a steam bath

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4.3.3 The mixture shall be fused by heating over a sulfur-free flame The crucible shall be placed in an inclined position on a wire triangle over a low flame The tendency of organic matter to burn too rapidly shall be controlled by the use of a glass stirring rod with which the burning portion is scraped away from the rest of the material When part of the mass has burned white, a fresh portion shall be worked into it until all the organic matter is destroyed During this operation, the crucible shall be held in position by means of a pair of tongs Toward the last half of the operation, the flame should be increased It is not necessary to heat the crucible to redness.

4.3.4 After the fusion is complete, the crucible and contents shall be permitted to cool The crucible and contents shall be placed in a 400-ml. beaker, covered with distilled water, and with occasional stirring digested on a steam bath for at least 2 hours The solution shall be filtered into a 400-ml beaker containing 10 ml of hydrochloric acid The residue shall be washed thoroughly with a hot 5-percent solution of sodium carbonate and the wash added to the filtrate A qualitative test for barium may be made on the residue, but no analysis for barium or correction because of its presence is necessary unless a detailed ash analysis is desired

4.3.5 The total volume of the solution shall be made to about 300 ml , and 10 ml of saturated picric acid solution added The beaker shall be uncovered and heated on a steam bath at a temperature of 90° to 100° C (194° to 212° F.) The sulfate shall be precipitated as barium sulfate by the dropwise addition of a slight excess of 10-percent barium chloride solution while stirring vigorously The beaker shall be covered with a watch glass and then set aside for 3 hours or overnight on a steam bath The barium sulfate shall be filtered and the filtrate tested for complete precipitation of the sulfate by the addition of a few milliliters of barium chloride solution. The precipitate shall be washed free from acid with distilled water, transferred to a crucible, ignited at a temperature of 650° to 700° C (1,202° to 1,292° F.) in a muffle furnace, cooled to room temperature in a desiccator, and weighed.

5. RESULTS

5.1 Calculation.—The organic plus inorganic or total plus inorganic sulfur of the specimen shall be calculated as follows:

$$\text{Sulfur, percent,} = \frac{A \times 0.1373}{W} \times 100$$

Where

A is the weight of the barium sulfate, gram
W is the weight of the specimen before extraction, gram

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested.

5.3 The organic plus inorganic or total plus inorganic sulfur of the sample for test shall be the average of the values obtained from the specimens tested

5.4 The organic plus inorganic or total plus inorganic sulfur of the sample for test shall be recorded to the nearest 0 1 percent.

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SULFUR, INORGANIC, ANTIMONY PRESENT

1. SCOPE

1.1 This method is intended for use in determining the amount of inorganic sulfur in the fillers from rubber compounds where antimony compounds are present All sulfur in the fillers, including that present in barium sulfate, is determined This method is applicable to all rubber compounds in which the rubber is soluble in the mineral oil solvent, method 16411, or the paranitrotoluene-orthodichlorobenzene mixture, method 16421 It is not applicable to hard rubber compounds, compounds containing nitrile or polyisobutylene rubber, or other compounds in which the rubber is not soluble in the mineral oil solvent or paranitrotoluene-orthodichlorobenzene mixture In the application of the method, the inorganic fillers are separated from the remainder of the compound by the mineral oil solvent or mixture of paranitrotoluene and orthodichlorobenzene, and the inorganic sulfur in the fillers determined by the fusion method 16241. See method 16201 for definitions of sulfurs.

2. SPECIMEN

2.1 The specimen shall consist of approximately 0 5 gram of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Absorbent cotton

3.1.3 Bromine water, saturated solution

3.1.4 Nitric acid-bromine mixture as described in method 16241

3.1.5 Paranitrotoluene

3.1.6 Orthodichlorobenzene

3.1.7 Assay flask, 125-ml, or 250-ml Erlenmeyer flask with standard taper joint.

3.1.8 Air condenser with standard taper joint

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone-chloroform mixture as described in method 16311 or with acetone as described in method 16321, and dried in air

4.2 Mineral oil solvent.

4.2.1 The dried specimen shall be placed in a 125-ml assay flask and 25 ml of rubber solvent conforming to 3 1 6 25 of method 16001 shall be added The flask shall be covered with a watch glass and the mixture digested in an air bath at a temperature of 150° to 165° C (302° to 329° F) until the rubber colloid has broken down and the oil appears quite clear

Caution: Keep away from the flame

4.2.2 The flask shall be removed from the air bath, cooled to about 90° C (194° F), and 10 to 15 ml of benzene added in small stream while mixing thoroughly The flask and contents shall be allowed to cool to room temperature and petroleum ether added to fill the flask to within 2 centimeters of the top The contents of the flask shall be thoroughly mixed, and the mixture allowed to stand until the filler particles have settled sufficiently to permit rapid decantation

4.2.3 The mixture shall be filtered by decantation through a Gooch crucible, 3 1 5 18 of method 16001, using suction The residue in the flask shall be washed well with petroleum ether and as much as possible of the residue transferred to the crucible The suction shall be turned off and warm acetone added to both the crucible and the flask The warm acetone shall be allowed to soak through the Gooch pad until the filtrate is colorless Gentle suction shall be applied and the flask and crucible washed well with acetone followed by a warm mixture of equal parts of acetone and chloroform if the filtrate is dark Finally, the flask and crucible shall be washed with hot alcohol During this part of the procedure, the material in the Gooch should not be allowed to become dry and crack

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4.2.4 The air-dried pad shall be transferred to the flask, and the last traces of fillers shall be removed from the crucible by means of absorbent cotton moistened with water and transferred to the flask. Ten milliliters of bromine water shall be added and the flask swirled to moisten the contents. Twenty milliliters of the nitric acid-bromine mixture shall be added, the flask allowed to stand for 15 minutes, and then heated on the steam bath for one hour. The contents of the flask shall be transferred to a 75-ml crucible and evaporated to dryness Three milliliters of nitric acid shall be added and the crucible covered with a watch glass. The mixture shall be warmed for a short time on the steam bath and set aside to cool The remainder of the procedure shall be as described in 4 3 2 to 4.3.5, inclusive, of method 16241

4.3 Paranitrotoluene - orthodichlorobenzene solvent.

4.3.1 If the rubber in the specimen is not soluble in the mineral oil solvent, 4 2, but is soluble in paranitrotoluene-orthodichlorobenzene mixture, the following procedure shall be used The dried specimen from 4 1 shall be placed in the 250-ml Erlenmeyer flask The flask shall then be fitted with the air condenser Ten grams of paranitrotoluene and 25 ml. of orthodichlorobenzene shall be added to the flask and the mixture heated to a temperature of 180° to 190° C (356° to 374° F.) on a hot plate under a hood with occasional stirring until the rubber dissolves. From 4 to 10 hours' digestion at this temperature is usually sufficient to complete the solution of the rubber The remainder of the procedure shall be as described in 4.2.2 to 4.2.4, inclusive.

5. RESULTS

5.1 Calculation.—The inorganic sulfur in fillers of the specimen shall be calculated as follows

$$\text{Sulfur, percent,} = \frac{A \times 0.1373}{W} \times 100$$

Where

A is the weight of the barium sulfate, gram
W is the weight of the specimen before extraction, gram

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested.

5.3 The inorganic sulfur in the fillers of the sample for test shall be the average of the values obtained from the specimens tested

5.4 The inorganic sulfur in the fillers of the sample for test shall be recorded to the nearest 0 1 percent

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SULFUR, INORGANIC, ANTIMONY ABSENT

1. SCOPE

1.1 This method is intended for use in determining the amount of inorganic sulfur in the fillers from rubber compounds when there is no antimony in the compound The method is not applicable to compounds containing nitrile or polyisobutylene rubber In the application of the method, the inorganic fillers are separated from the remainder of the compound by burning the rubber The inorganic sulfur in the fillers is then determined by method 16241 if acid-soluble barium salts or lead are present Method 16231 may be used for determining the inorganic sulfur in the fillers if acid-soluble barium salts or lead are absent The same method, that is 16231 or 16241, must be used for determining the inorganic sulfur in the fillers that was used for determining the organic plus inorganic sulfur or total plus inorganic sulfur in the compound See method 16201 for definitions of sulfur

2. SPECIMEN

2.1 The specimen shall consist of approximately 0 5 gram of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Nitric acid-bromine mixture as described in method 16241

3.1.3 Fuming nitric acid

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram, extracted with acetone-chloroform mixture as described in method 16311 or acetone as described in method 16321, and dried in air

4.2 The extracted specimen shall be placed in a porcelain crucible of about 75-ml capacity and the rubber hydrocarbon distilled off in a muffle furnace at a temperature of not more than 450° C (842° F) It is not necessary to burn off all the carbon during this ignition The crucible shall be cooled

4.3 Acid-soluble barium salts or lead compounds present.

4.3.1 If acid-soluble barium salts or lead compounds are present, 15 ml of bromine-nitric acid mixture shall be added to the crucible The crucible shall be covered with a watch glass and allowed to stand for one hour in the cold The crucible and contents shall be heated for one hour on the steam bath, the cover removed and rinsed with a small amount of distilled water, and the contents of the crucible evaporated to dryness Three milliliters of nitric acid shall be added to the crucible and the crucible covered with a watch glass The crucible and contents shall be warmed for a short time on the steam bath and allowed to cool The remainder of the procedure shall be as described in 4 3 2 to 4 3 5, inclusive, of method 16241

4.4 Acid-soluble barium salts and lead compounds absent.

4.4.1 If acid-soluble barium salts and lead compounds are absent, either the procedure in 4 3 1 or 4 4 2 shall be used for determining the sulfur in the fillers The procedure described in 4 4 2 is less time consuming than the procedure described in 4 3 1

4.4.2 Three milliliters of nitric acid-bromine mixture shall be added to the ash, 4 2, the crucible covered with a watch glass, and the crucible and contents heated for one hour The contents of the crucible shall be transferred with washing into a 500-ml Erlenmeyer destruction flask of chemically resistant material and evaporated to a small volume If particles of organic matter or carbon remain at the end of the evaporation, a few milliliters of fuming nitric acid shall be added and the evaporation repeated until all the carbon is gone and the solution is clear, colorless, or light yellow The remainder of the procedure shall be described in 4 3 3 and 4 3 4 of method 16231

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5. RESULTS

5.1 Calculation.—The inorganic sulfur in the fillers of the specimen shall be calculated as follows:

$$\text{Sulfur, percent,} = \frac{A \times 0.1373}{W} \times 100$$

Where

A is the weight of the barium sulfate, gram

W is the weight of the specimen before extraction, gram

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The inorganic sulfur in the fillers of the sample for test shall be the average of the values obtained from the specimens tested.

5.4 The inorganic sulfur in the fillers of the sample for test shall be recorded to the nearest 0.1 percent.

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

EXTRACTABLE MATERIALS, GENERAL

1. SCOPE

1.1 This method gives definitions for the various types of extracts that are usually made on rubber compounds by the use of organic solvents A general scheme relating the various extracts and references to the procedures for determining them are also given Insofar as practicable, the definitions list the materials removed by extractions The tests are applicable in their entirety to natural rubber compounds and to synthetic rubber compounds containing styrene rubbers Compounds containing nitrile rubber cannot be analyzed by this group of tests because of swelling and partial solubility of the rubber Compounds containing polyisobutylene rubber do not extract well with acetone

2. DEFINITIONS

2.1 Total extract.—The total extract is the material removed from a vulcanized rubber compound by extracting with a mixture consisting of 32 percent acetone and 68 percent chloroform by volume as described in method 16311 It is also approximately equal to the sum of the acetone and chloroform extracts

2.2 Acetone extract.—The acetone extract is the material removed from the rubber compound by extraction with acetone as described in method 16321 The acetone removes resins, free sulfur, softeners, cellulose ethers and esters, naphthalene, part of any bituminous materials, mineral oil, and waxes, factice, and part or all of organic dyes, accelerators, and antioxidants or their decomposition products.

2.3 Chloroform extract.—The chloroform extract is the material removed from a vulcanized rubber compound by extraction with chloroform, method 16331, after the specimen has been extracted with acetone as described in method 16321 The chloroform removes a portion of the bituminous materials and may include a small amount of rubber for which no correction is made

2.4 Alcoholic potassium hydroxide extract.—The alcoholic potassium hydroxide extract is the material removed from the rubber compound by extraction with alcoholic potassium hydroxide as described in method 16341 after the specimen has been extracted with acetone and chloroform The extract consists of a portion of the factice and any saponifiable material not removed by previous extractions

2.5 Unsaponifiable acetone extract.—The unsaponifiable acetone extract is the portion of the acetone extract that is not saponified by a 1 *N* alcoholic potassium hydroxide solution

2.6 Waxy hydrocarbons.—The waxy hydrocarbons are contained in the portion of the unsaponifiable acetone extract that is removed by extraction with absolute alcohol and that separates from the alcoholic solution on cooling to −5° C (23° F)

2.7 Mineral oil extract.—Mineral oil extract is the portion of unsaponifiable acetone extract which is soluble in absolute alcohol at −5° C (23° F) and which is soluble in carbon tetrachloride and is not decomposed by sulfuric acid

2.8 Organic extract.—Organic extract is calculated by subtracting from the total extract the amount of sulfur found in the total extract by the use of method 16221

2.9 Organic acetone extract.—Organic acetone extract is calculated by subtracting from the acetone extract the amount of sulfur found in the acetone extract by the use of method 16221

2.10 Acetone extract corrected.—Acetone extract corrected is calculated by subtracting the sum of the sulfur in the acetone extract, waxy hydrocarbon, and mineral oil from the acetone extract

3. PROCEDURE

3.1 The general scheme relating the separation of organic solvent-soluble materials is given in figure 16301

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GENERAL SCHEME RELATING EXTRACTABLE MATERIALS.

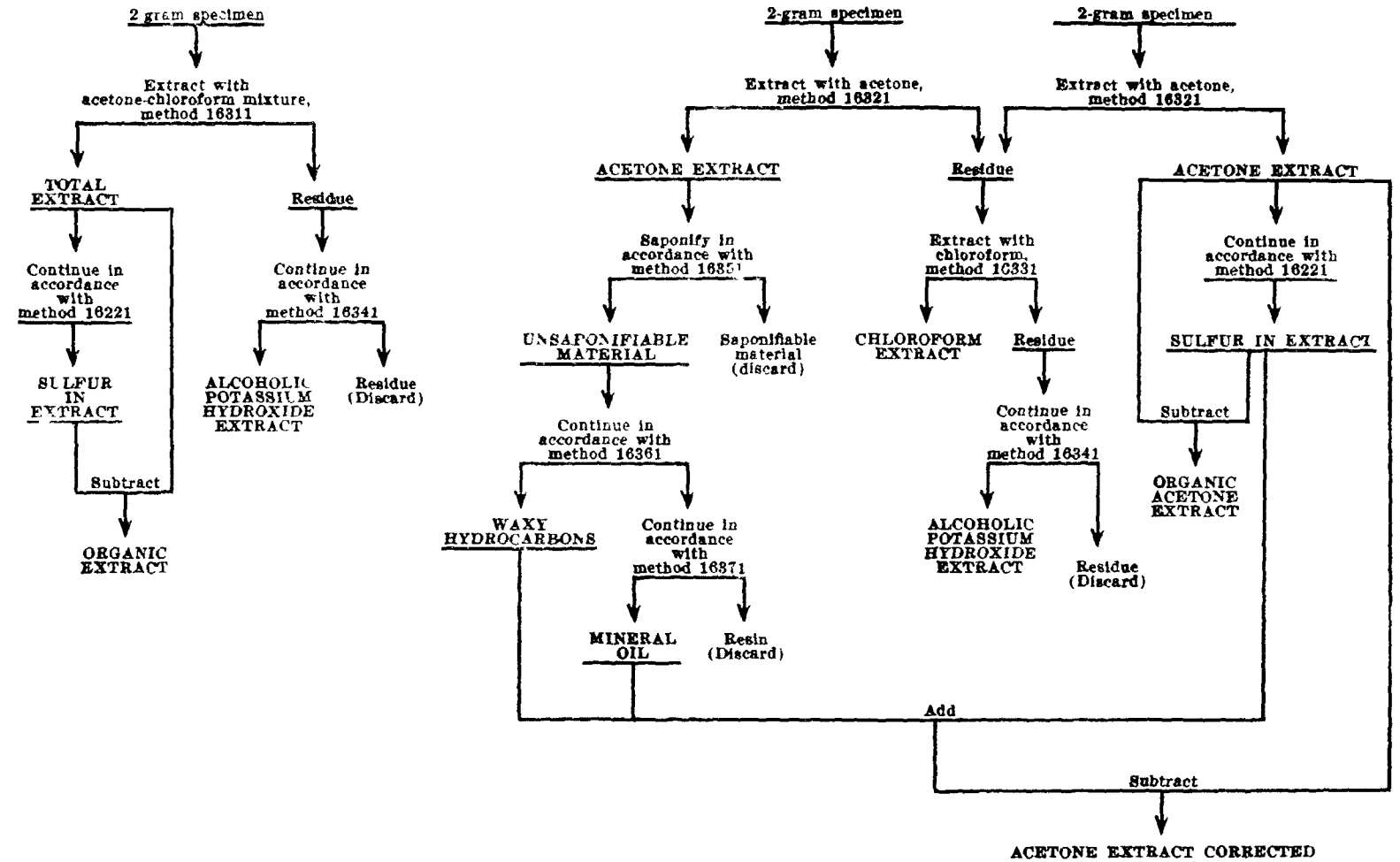

FIGURE 16301.

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EXTRACT, TOTAL

1. SCOPE

1.1 This method is intended for use in determining the amount of material removed from a rubber compound by extraction with a mixture of acetone and chloroform It is not applicable to crude rubber, reclaimed rubber, or unvulcanized rubber compounds The total extract may be used to replace the sum of the acetone extract, method 16321, and chloroform extract, method 16331, where analysis is to be made of an extracted specimen or where rubber content is being determined The method is applicable when it is necessary to obtain the total extract for use in calculating organic extract It is not applicable to compounds containing nitrile or polyisobutylene rubber but is applicable to compounds containing styrene rubber See method 16301 for definitions of extracts

2. SPECIMEN

2.1 The specimen shall consist of approximately 2 grams of rubber compound from the composite sample prepared as described in method 1211

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3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Extraction apparatus shown in figure 16001

3.1.3 Source of compressed air

3.1.4 Extraction mixture consisting of 32-percent acetone and 68-percent chloroform by volume

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and, if in the form of a sheet, it shall be cut into strips 3 to 5 mm in width, and placed in a thimble made by folding a filter paper so that it will fit into the extraction cup If the specimen may become tacky during the extraction, care shall be taken that adjacent portions are separated by the paper The extraction flask shall be dried for 2 hours at a temperature of 70°±2°C (158°±3 6°F), cooled in a desiccator, and weighed Sixty to seventy milliliters of the extraction mixture shall be added to the flask, the extraction cup containing the specimen suspended in position, and the apparatus assembled as shown in figure 16001

4.2 The specimen shall be extracted for 16 hours at such a rate that from 2 5 to 3 5 minutes are required to fill and empty the extraction cup Hard rubber, and any soft rubber compound having a ratio of total sulfur to rubber hydrocarbon in excess of 10 percent, shall be extracted for 72 hours The characteristics of the extract shall be noted both when hot and cold If the color of the extract is black, the specimen shall be extracted with chloroform as described in method 16331, and the weight of the chloroform extract added to that of the total extract The extracted specimen should be reserved for further analysis

4.3 The solvent shall be distilled off on a steam bath, passing a gentle current of filtered air through the flask to prevent boiling of the liquid The flask shall be removed from the steam bath just before the last trace of solvent has disappeared and the current of air continued for about 10 minutes The flask and contents shall be dried in circulating air for 2 hours at a temperature of 70°±2° C (158°±3 6° F); cooled in a desiccator, and weighed

5. RESULTS

5.1 Calculation.—The total extract of the specimen shall be calculated as follows

$$\text{Total extract, percent,} = \frac{A}{W} \times 100$$

Where

A is the weight of the extract, gram

W is the weight of the specimen before extraction, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The total extract of the sample for test shall be the average of the values obtained from the specimens tested

5.4 The total extract of the sample for test shall be recorded to the nearest 0 1 percent

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EXTRACT, ACETONE

1. SCOPE

1.1 This method is intended for use in determining the amount of acetone-soluble material in rubber compounds. The method is also used when any of the following are to be determined. Unsaponifiable acetone extract, waxy hydrocarbons, or mineral oil. The method is applicable when it is necessary to obtain an analysis of the acetone extract in order to calculate the organic acetone extract or acetone extract corrected It is also applicable to the determination of the acetone-soluble material in crude, reclaimed, unvulcanized, or vulcanized natural rubber compounds and compounds containing styrene synthetic rubber This method is not applicable to compounds containing nitrile or polyisobutylene rubber See method 16301 for definitions of extracts.

2. SPECIMEN

2.1 The specimen shall consist of approximately 2 grams of rubber compound from the composite sample prepared as described in method 1211.

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Extraction apparatus shown in figure 16001

3.1.3 A source of compressed air.

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and, if in the form of a sheet, it shall be cut into strips 3 to 5 mm in width and placed in a thimble made by folding a filter paper so that it will fit into the extraction cup If the specimen may become tacky during the extraction, care shall be taken that adjacent portions are separated by the paper The extraction flask shall be dried for 2 hours at a temperature of 70°±2° C (158°±3 6° F.), cooled in a desiccator, and weighed Sixty to seventy-five milliliters of acetone shall be added to the flask, the extraction cup containing the specimen suspended in position, and the apparatus assembled as shown in figure 16001

4.2 The specimen shall be extracted for 16 hours at such a rate that from 2.5 to 3 5 minutes are required to fill and empty the extraction cup Hard rubber, and any soft rubber compound having a ratio of total sulfur to rubber hydrocarbon in excess of 10 percent, shall be extracted for 72 hours The characteristics of the extract shall be noted both when hot and cold. The extracted specimen should be reserved for further analysis

4.3 The acetone shall be distilled off on a steam bath, passing a gentle current of filtered air through the flask to prevent boiling of the liquid The flask shall be removed from the steam bath just before the last trace of acetone has disappeared and the current of air continued for about 10 minutes The flask and contents shall be dried in circulating air for 2 hours at a temperature of 70°±2° C. (158°± 3 6° F), cooled in a desiccator, and weighed The extract should be reserved for further analysis.

5. RESULTS

5.1 **Calculation.**—The acetone extract of the specimen shall be calculated as follows.

$$\text{Acetone extract, percent,} = \frac{A}{W} \times 100$$

Where

A is the weight of the extract, gram

W is the weight of the specimen before extraction, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested.

5.3 The acetone extract of the sample for test shall be the average of the values obtained from the specimens tested.

5.4 The acetone extract of the sample for test shall be recorded to the nearest 0.1 percent.

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EXTRACT, CHLOROFORM

1. SCOPE

1.1 This method is intended for use in determining the amount of material in a rubber compound that is removed by extraction with chloroform after the specimen has been extracted with acetone This method is not applicable to natural, crude, unvulcanized, or reclaimed rubber or compounds containing nitrile or polyisobutylene rubber It is applicable to compounds containing styrene rubber See method 16301 for definitions of extracts

2. SPECIMEN

2.1 The specimen shall consist of approximately 2 grams of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required.

3.1.2 Extraction apparatus shown in figure 16001

3.1.3 A source of compressed air

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone as described in method 16321 The extraction flask containing the acetone extract shall be removed from the extraction apparatus and replaced with another flask that has been dried for 2 hours at a temperature of 70°±2° C (158°±3 6° F), cooled in a desiccator, and weighed Sixty to seventy-five milliliters of chloroform shall be added to the flask

4.2 The specimen shall be extracted for 4 hours, at such a rate that from 2 5 to 3 5 minutes are required to fill and empty the extraction cup Hard rubber, and any soft rubber compound having a ratio of total sulfur to rubber hydrocarbon in excess of 10 percent, shall be extracted for 24 hours The characteristics of the extract shall be noted both when hot and cold The extracted specimen should be reserved for further analysis in method 16341

4.3 The chloroform shall be distilled off on a steam bath, passing a gentle current of filtered air through the flask to prevent boiling of the liquid The flask shall be removed from the steam bath just before the last traces of chloroform have disappeared and the current of air continued for about 10 minutes The flask and contents shall then be dried in circulating air for 2 hours at a temperature of 70°±2° C (158°±3 6° F), cooled in a desiccator, and weighed

5. RESULTS

5.1 Calculation.—The chloroform extract of the specimen shall be calculated as follows

$$\text{Chloroform extract, percent,} = \frac{A}{W} \times 100$$

Where

A is the weight of the extract, gram

W is the weight of the specimen before extraction with acetone, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The chloroform extract of the sample for test shall be the average of the values obtained from the specimens tested

5.4 The chloroform extract of the sample for test shall be recorded to the nearest 0 1 percent

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EXTRACT, ALCOHOLIC POTASSIUM HYDROXIDE

1. SCOPE

1.1 This method is intended for use in determining the amount of material in a rubber compound that is removed by extraction with alcoholic potassium hydroxide after the compound has been extracted with an acetone-chloroform mixture or with acetone followed by chloroform This method is applicable to compounds containing styrene rubber but is not applicable to compounds containing nitrile or polyisobutylene rubber. See method 16301 for definitions of extracts

2. SPECIMEN

2.1 The specimen shall consist of approximately 2 grams of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows:

3.1.1 Apparatus and reagents in method 16001 as required.

3.1.2 Reflux condenser.

3.1.3 Absorbent cotton.

3.1.4 Hydrochloric acid, 10-percent solution

3.1.5 Alcoholic potassium hydroxide —A 1 *N* alcoholic potassium hydroxide solution prepared by dissolving potassium hydroxide in alcohol that has been purified as follows One and one-half grams of silver nitrate shall be dissolved in 3 ml of distilled water and the solution added to one liter of ethyl alcohol Three grams of potassium hydroxide shall be dissolved in the smallest amount of hot water and the solution cooled. The potassium hydroxide solution shall be added to the alcohol-silver nitrate solution, the two solutions thoroughly shaken, and the mixture set aside for 24 hours, the mixture shall then be filtered and distilled Ethyl alcohol denatured with 10 percent by volume of methyl alcohol may be used in place of pure ethyl alcohol.

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone-chloroform mixture as described in method 16311 or with acetone as described in method 16321 followed by extraction with chloroform as described in method 16331 and dried at a temperature of 70°±2° C (158°±3 6° F) to remove the solvent.

4.2 The extracted specimen shall be placed in a 250-ml Erlenmeyer flask and 50 ml of the alcoholic potassium hydroxide solution shall be added to the flask A reflux condenser shall be attached to the flask and the mixture boiled for 4 hours In the case of hard rubber, the specimen shall be digested for at least 16 hours The mixture shall be filtered into a 250-ml beaker, the residue washed with two 25-ml portions of boiling ethyl alcohol and then with three 25-ml portions of boiling distilled water The filtrate shall be evaporated just to dryness

4.3 The residue from the evaporation shall be transferred to a separatory funnel using about 75 ml of distilled water and the solution made acid to Congo red paper with a 1 to 2 hydrochloric acid solution The solution shall be extracted in the separatory funnel with four 25-ml portions of ethyl ether If the fourth extraction is colored, the extraction shall be repeated until the extract is colorless The ether extracts shall be combined in the separatory funnel and washed free from acid with distilled water The ether solution shall be filtered through a previously washed plug of absorbent cotton into a flask which has been dried at a temperature of 100°±2°C (212°±3 6° F.), cooled in a desiccator, and weighed The separatory funnel and plug shall be washed with ether and the wash added to the flask. The ether shall be distilled off on a steam bath, passing a gentle current of filtered air through the flask to prevent boiling of the liquid The flask shall be removed from the steam bath just before the last trace of the ether has disappeared, and the current of air continued for about 10 minutes. The flask and contents shall be dried to a constant weight in circulating air at a temperature of 100°±2° C (212°±3.6° F), cooled in a desiccator, and weighed A blank determination shall be made

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5. RESULTS

5.1 Calculation.—The alcoholic potassium hydroxide extract of the specimen shall be calculated as follows.

$$\text{Alcoholic potassium hydroxide extract, percent} = \frac{A-B}{W} \times 100$$

Where

A is the weight of the extract, gram
B is the weight of the blank, gram
W is the weight of the specimen before extraction with acetone and chloroform, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested.

5.3 The alcoholic potassium hydroxide extract of the sample for test shall be the average of the values obtained from the specimens tested.

5.4 The alcoholic potassium hydroxide extract of the sample for test shall be recorded to the nearest 0 1 percent

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EXTRACT, ACETONE, UNSAPONIFIABLE

1. SCOPE

1.1 This method is intended for use in determining the amount of unsaponifiable material contained in the acetone-soluble portion of a rubber compound. The unsaponifiable acetone extract will include waxes and mineral oils The method is applicable to compounds containing styrene rubber but is not applicable to compounds containing nitrile or polyisobutylene rubber See method 16301 for definitions of extracts.

2. SPECIMEN

2.1 The specimen shall consist of approximately 2 grams of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows:

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Reflux condenser.

3.1.3 Alcoholic potassium hydroxide, 1 *N* solution, method 16341.

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone as described in method 16321

4.2 Fifty milliliters of the 1 *N* alcoholic potassium hydroxide solution shall be added to the flask containing the dry acetone extract The flask shall be attached to a reflux condenser and the mixture boiled for at least 2 hours The condenser shall be removed and the mixture evaporated to dryness on a steam bath The contents of the flask shall be transferred to a separatory funnel using about 100 ml of distilled water Twenty-five milliliters of ether shall be added to the funnel and the funnel shaken well. The ether and water layers shall be allowed to separate thoroughly and the water layer drawn off The extraction of the water layer shall be repeated with fresh portions of ether until no more unsaponifiable matter is removed Four extractions are usually sufficient

4.3 The ether extracts shall be combined in the separatory funnel and then washed with distilled water until the wash water is free from alkali as shown by phenolphthalein The ether extract shall be filtered through a plug of cotton into a flask which has been dried at a temperature of 70° ± 2° C (158° ± 3 6° F), cooled in a desiccator, and weighed The plug shall be washed with ether and the wash added to the flask The ether shall be distilled off on a steam bath, passing a gentle current of filtered air through the flask to prevent boiling of the liquid The flask shall be removed from the steam bath just before the last trace of the ether has disappeared, and the current of air continued for about 10 minutes The flask and contents shall be dried to a constant weight in circulating air at a temperature of 70° ± 2° C (158° ± 3 6° F) cooled in a desiccator, and weighed The contents of the flask should be reserved for use in method 16361

5. RESULTS

5.1 Calculation.—The unsaponifiable acetone extract of the specimen shall be calculated as follows

$$\text{Unsaponifiable acetone extract, percent,} = \frac{A}{W} \times 100$$

Where

A is the weight of the unsaponifiable extract, gram
W is the weight of the specimen before extraction, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested.

5.3 The unsaponifiable acetone extract of the sample for test shall be the average of the values obtained from the specimens tested.

5.4 The unsaponifiable acetone extract of the sample for test shall be recorded to the nearest 0 1 percent

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EXTRACT, WAXY HYDROCARBONS

1. SCOPE

1.1 This method is intended for use in determining the amount of waxy hydrocarbons in the unsaponifiable acetone extract from a rubber compound that is soluble in alcohol and that separates from alcoholic solution on cooling to −5° C (23° F) The method is applicable to compounds containing styrene rubber, but is not applicable to compounds containing nitrile or polyisobutylene rubber See method 16301 for definitions of extracts

2. SPECIMEN

2.1 The specimen shall consist of approximately 2 grams of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Ice-salt bath or cooling bath

3.1.3 Absolute ethyl alcohol

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone as described in method 16321

4.2 The dry acetone extract shall be treated as described in method 16351 to obtain the unsaponifiable acetone extract

4.3 Fifty milliliters of absolute ethyl alcohol shall be added to the flask containing the unsaponifiable acetone extract The mixture shall be digested on a steam bath for at least ½ hour and cooled The flask shall be placed in an ice-salt mixture and kept at a temperature of −5° C (23° F) or lower for at least one hour The separated waxy hydrocarbon shall be filtered from the mixture through a filter paper by applying gentle suction while keeping the funnel surrounded by an ice-salt mixture at a temperature of −5° C (23° F) or lower. The flask and precipitate of waxy hydrocarbon shall be washed with 95-percent alcohol that has been cooled to a temperature of −5° C (23° F) or lower The filtrate and washings should be reserved for use in method 16371

4.4 The precipitate shall be dissolved from the filter paper with hot chloroform, catching the filtrate in a weighed 150-ml container which has been dried at a temperature of 70°±2° C (158°±3.6° F.), cooled in a desiccator, and weighed The flask shall be washed with hot chloroform which is added to the 150-ml container The chloroform shall be distilled off on a steam bath, passing a gentle current of filtered air through the container to prevent boiling of the liquid The container shall be removed from the steam bath just before the last trace of chloroform has disappeared and the current of air continued for about 10 minutes The container and contents shall be dried to a constant weight in circulating air at a temperature of 70°±2° C (158°±3.6° F), cooled in a desiccator, and weighed.

5. RESULTS

5.1 Calculation.—The waxy hydrocarbons of the specimen shall be calculated as follows

$$\text{Waxy hydrocarbons, percent, } = \frac{A}{W} \times 100$$

Where

A is the weight of the extract, gram

W is the weight of the specimen before extraction with acetone, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The waxy hydrocarbons of the sample for test shall be the average of the values obtained from the specimens tested

5.4 The waxy hydrocarbons of the sample for the test shall be recorded to the nearest 0 1 percent

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EXTRACT, MINERAL OIL

1. SCOPE

1.1 This method is intended for use in determining the amount of mineral oil that is extracted from a rubber compound with acetone The mineral oils are saturated hydrocarbons which are soluble in alcohol at −5° C (23° F), are soluble in carbon tetrachloride, and are not attacked by sulfuric acid. This method is applicable to compounds containing styrene rubber but is not applicable to compounds containing nitrile and polyisobutylene rubber See method 16301 for definitions of extracts

2. SPECIMEN

2.1 The specimen shall consist of approximately 2 grams of rubber compound from the composite sample prepared as described in method 1211.

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as described in method 16001 as required

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone as described in method 16321

4.2 The dry acetone extract shall be treated as described in method 16351 to obtain the unsaponifiable acetone extract

4.3 The waxy hydrocarbons shall be removed from the unsaponifiable acetone extract as decribed in 4 3 of method 16361.

4.4 The container with filtrate and washings, 4 3, shall be placed on a steam bath and the alcohol distilled off, passing a gentle current of filtered air through the container to prevent boiling of the liquid The container shall be removed from the steam bath just before the last traces of alcohol have disappeared and the current of air continued for about 10 minutes The residue shall be taken up with 15 to 25 ml of carbon tetrachloride and the solution transferred to a separatory funnel, using small portions of carbon tetrachloride to wash the last traces of solution from the container Fifteen milliliters of sulfuric acid shall be added to the separatory funnel, the mixture shaken well, and the colored acid layer drawn off This operation shall be repeated until the acid layer is no longer colored The last extraction shall be drawn off The acid removes the unsaponified resin present

4.5 About 75 ml of distilled water and sufficient ether to form an ether-carbon tetrachloride layer above the water shall be added to the contents of the separatory funnel The mixture shall be shaken well and the water layer drawn off This washing operation shall be repeated with distilled water until the water layer no longer shows the presence of acid upon the addition of 2 or 3 drops of methyl red indicator The ether - carbon tetrachloride solution shall be transferred to a flask which has been dried at a temperature of 100°±2° C (212°±3 6° F), cooled in a desiccator, and weighed The solvent shall be distilled off on a steam bath, passing a gentle current of filtered air through the flask to prevent boiling of the liquid The flask shall be removed from the steam bath just before the last trace of solvent has disappeared and the current of air continued for about 10 minutes The flask and contents shall be dried to a constant weight in circulating air at a temperature of 100°±2° C (212°±3 6° F), cooled in a desiccator, and weighed

5. RESULTS

5.1 Calculation.—The mineral oil in the specimen shall be calculated as follows

$$\text{Mineral oil, percent,} = \frac{A}{W} \times 100$$

Where

A is the weight of the extract, gram
W is the weight of the specimen before extraction with acetone, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The mineral oil in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The mineral oil in the sample for test shall be recorded to the nearest 0 1 percent

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FILLERS, GENERAL

1. SCOPE

1.1 The fillers in rubber compounds are those inorganic materials other than free or elemental sulfur and free carbon that have been added to a rubber compound This method lists three procedures, which are later described in detail, and some of their applications and limitations In general, the methods are applicable to both natural and synthetic rubber, but only the ash method is applicable to hard rubber In the determination of fillers for use in determining rubber content by the indirect method, the methods are not applicable to compounds containing nitrile rubber Other applications and limitations are given in the scope of the individual methods

1.2 Unless otherwise specified in the detail specification or preliminary examination of the sample indicates the presence of decomposable materials, the ash method 16431 shall be used for determining the fillers for use in calculating the rubber content of soft-rubber compounds by the indirect method to determine their conformance with Federal Specifications

1.2.1 The ash method 16431 shall not be used if carbonates, antimony sulfide, and other inorganic materials that decompose at the ashing temperature are present The ash method is also not applicable to compounds containing chloroprene rubber if zinc compounds or other metals which form volatile compounds are present

1.2.2 Method 16411 shall be used if antimony sulfide, carbonates, or other decomposable inorganic materials are present in the rubber compound, providing the rubber is soluble in the mineral oil solvent This method is not applicable to hard rubber compounds or to compounds containing nitrile rubber.

1.2.3 Method 16421 shall be used if antimony sulfide, carbonates, or other decomposable inorganic materials are present in the rubber compound, and if the rubber is soluble in a paranitrotoluene-orthodichlorobenzene mixture and not soluble in mineral oil This method is not applicable to hard rubber compounds or compounds containing nitrile rubber

1.2.4 In the case of dispute or disagreement between the ash and the solvent methods, method 16411 shall be used for rubber compounds where the rubber is soluble in the mineral oil solvent, and method 16421 shall be used for rubber compounds where the rubber is soluble in the paranitrotoluene-orthodichlorobenzene mixture and not soluble in the mineral-oil solvent

1.3 Fillers in hard rubber shall be determined as described in method 16431 if materials decomposing at 550° C (1,022° F) are absent. None of these methods is applicable to hard-rubber compounds that contain inorganic materials decomposing at 550° C. (1,022° F)

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FILLERS, MINERAL OIL

1. SCOPE

1.1 This method is intended for use in determining the amount of inorganic materials other than free or elemental sulfur and free carbon that is present in rubber compounds This method consists of dissolving the rubber in a mixture of mineral oils, separating the inorganic residue by filtration and weighing the residue. A correction is applied for the organic materials that are not completely removed by the solvent. The correction is made by removing the decomposable ingredients from the inorganic residue with hydrochloric acid, determining the organic material remaining by ignition, and subtracting this value from the residue obtained This method is not applicable to compounds containing nitrile rubber or to any compound in which the rubber will not dissolve in the mineral oil solvent, nor to hard-rubber compounds

2. SPECIMEN

2.1 The specimen shall consist of approximately 0.5 gram of rubber compound from the composite sample prepared as described in method 1211.

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Assay flask, 125 ml.

3.1.3 Rubber solvent described in 3 1 6.25 of method 16001

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram, extracted for at least 8 hours with acetone-chloroform mixture as described in method 16311, and dried in air

4.2 The specimen shall be placed in the 125-ml. assay flask containing 25 ml of the mineral oil solvent. The flask shall be covered with a watch glass and the mixture digested in an air bath at a temperature of 150° to 165° C. (302° to 329° F) until the rubber colloid has broken down and the oil seems quite clear ***Caution:*** Keep away from flame.

4.3 The flask shall be removed from the air bath, cooled to about 90° C (194° F), and 10 to 15 ml of benzene shall be added in a stream while mixing thoroughly The mixture shall be allowed to cool and petroleum ether added to fill the flask to within 2 centimeters of the top The contents of the flask shall be thoroughly mixed, the flask covered with a watch glass to prevent evaporation, and the mixture allowed to stand until the filler particles have settled sufficiently to permit rapid filtration

4.4 The Gooch crucible, 3 1 5 18 of method 16001, with asbestos pad shall be ignited at a temperature of 650° to 900° C (1,202° to 1,652° F), cooled in a desiccator, and weighed and the weight recorded as *A*. The mixture shall be filtered by decantation through the Gooch crucible using suction The residue in the flask shall be washed well with petroleum ether, transferring as much of the residue as possible to the crucible The suction shall be turned off and warm acetone added to both the crucible and the flask The warm acetone shall be allowed to soak through the Gooch pad until the filtrate is colorless Gentle suction shall be applied and the Gooch crucible and flask washed well with acetone, followed by a warm mixture of equal volumes of acetone and chloroform if the filtrate is dark. Finally the flask and crucible shall be washed with hot alcohol A portion of the fillers will remain in the flask. During this part of the procedure the mass on the Gooch crucible should not be permitted to become dry and crack

4.5 The Gooch crucible shall be cleaned on the outside with solvent and cloth, dried for one hour at a temperature of 105° to 110° C (221° to 230° F), cooled in a desiccator, weighed, and the weight recorded as *B* The flask shall be cleaned on the outside with solvent and cloth, dried for one hour at a temperature of 105° to 110° C (221° to 230° F), cooled in a desiccator, weighed, and the weight recorded as *C* At this point of the procedure, practically all of the rubber and organic compounds have been separated from the inorganic filler and free

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carbon The subsequent operation described in 4 6 is necessary to determine the small amount of organic material, including free carbon, remaining

4.6 The contents of the flask and Gooch crucible shall be moistened with a few milliliters of alcohol, allowed to soak for a few minutes, and then washed three times with boiling distilled water The inside of the flask shall be washed down with 10 to 15 ml of 1 to 3 hydrochloric acid, and the flask swirled to bring the acid into contact with the fillers A few drops of the 1 to 3 hydrochloric acid shall be poured from the flask to the crucible and sucked through the pad using gentle suction to prevent excessive frothing. After the violent action has ceased, the suction shall be turned off and the remainder of the 1 to 3 hydrochloric acid in the flask added to the crucible and allowed to remain until no more bubbles rise through the liquid When no more gas is evolved, acid shall be drawn through the pad with gentle suction and the contents washed thoroughly with fresh acid If antimony is present, this washing procedure should be repeated using hydrochloric acid instead of 1 to 3 hydrochloric acid The residue shall then be washed well with hot distilled water and all the solid material transferred from the flask to the crucible The crucible shall be dried at a temperature of 105° to 110° C (221° to 230° F) for one hour, cooled in a desiccator, weighed, and the weight recorded as *D* The flask shall be cleaned on the outside with solvent, dried for one hour at a temperature of 105° to 110° C (221° to 230° F), cooled in a desiccator, and weighed and the weight recorded as *E* The Gooch crucible and contents shall be ignited in a muffle furnace at a temperature of 650° to 900° C (1,202° to 1,652° F) to remove the organic residue, cooled in a desiccator, and weighed, and the weight recorded as *F*.

5. RESULTS

5.1 Calculation.—The fillers in the specimen shall be calculated as follows

$$\text{Fillers, percent,} = \frac{(B+C+F)-(A+E+D)}{W} \times 100$$

Where

A is the weight of Gooch crucible, grams (4 4)
B is the weight of Gooch crucible and contents, grams (4 5)
C is the weight of flask and contents, grams (4 5)
D is the weight of Gooch crucible and contents, grams (4 6)
E is the weight of flask, grams (4 6)
F is the weight of Gooch crucible after ignition at 650° to 900° C (1,202° to 1,652° F), grams (4 6)
W is the weight of the specimen before extraction, gram

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested.

5.3 The fillers in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The fillers in the sample for test shall be recorded to the nearest 0 1 percent.

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FILLERS, PARANITROTOLUENE-ORTHODICHLOROBENZENE

1. SCOPE

1.1 This method is intended for use in determining the amount of inorganic materials other than free or elemental sulfur and free carbon that is present in rubber compounds This method consists of dissolving the rubber in a mixture of paranitrotoluene and orthodichlorobenzene, separating the inorganic residue by filtration, and weighing the residue. A correction is applied for the organic materials that are not completely removed by the solvent. The correction is made by removing the decomposable ingredients from the inorganic residue with hydrochloric acid, determining the organic material remaining by ignition, and subtracting this value from the residue obtained This method is not applicable to compounds containing nitrile rubber or to any compounds in which the rubber will not dissolve in paranitrotoluene-orthodichlorobenzene mixture or to hard rubber compounds.

2. SPECIMEN

2.1 The specimen shall consist of 0 5 gram of rubber compound from the composite sample prepared as described in method 1211.

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3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Erlenmeyer flask, 250 ml , with standard taper joint

3.1.3 Air condenser with standard taper joint.

3.1.4 Orthodichlorobenzene.

3.1.5 Paranitrotoluene.

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram, extracted for at least 8 hours with acetone-chloroform mixture as described in method 16311, and dried in air

4.2 The specimen shall be placed in a 250-ml. Erlenmeyer flask. The flask shall then be fitted with the air condenser Ten grams of paranitrotoluene and 25 ml of orthodichlorobenzene shall be added to the flask and the mixture heated to 180° to 190° C. (356° to 374° F) on a hot plate under a hood with occasional stirring until the rubber dissolves The remainder of the procedure shall be as described in 4 3 to 4 6, inclusive, of method 16411

5. RESULTS

5.1 The results shall be as described in method 16411

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This method is a part of Fed Test Method Std No 601
(formerly Fed Spec ZZ-R-601a)

FILLERS, ASH METHOD

1. SCOPE

1.1 This method is intended for use in determining the amount of inorganic material other than free or elemental sulfur and free carbon that is present in rubber compounds when carbonates, antimony sulfide, and other inorganic materials that decompose at ashing temperatures are absent This method is not applicable to chloroprene synthetic rubber when zinc compounds or other metal compounds that form volatile chlorides are present This method is applicable to hard rubber compounds that do not contain inorganic materials that decompose at the ashing temperature

2. SPECIMEN

2.1 The specimen shall consist of approximately 2 grams of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 A muffle furnace equipped with a copper-constantan thermocouple and potentiometer, or other device for measuring the temperature

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted for 8 hours with acetone-chloroform mixture as described in method 16011 The specimen shall be transferred to a weighed porcelain crucible of approximately 50-ml capacity For hard rubber, extraction is not necessary

4.2 The specimen shall be heated in a muffle furnace, taking care that the rubber does not catch fire This can be done by gradually heating the crucible until the rubber begins to distill off and then adjusting the temperature until the rubber just continues to distill off. When smoking ceases, the temperature shall be raised rapidly until the specimen is burned clean The following temperature-time scale may be used.

Time, minutes	5	10	15	70	75	80	85	145
Temperature, °C	100	200	300	300	400	500	550	550

4.2.1 The crucible shall be removed from the furnace, cooled in a desiccator, and weighed

5. RESULTS

5.1 Calculation.—The fillers from the specimen shall be calculated as follows

$$\text{Fillers, percent,} = \frac{A}{W} \times 100$$

Where

A is the weight of the fillers, gram
W is the weight of the specimen before extraction, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The fillers in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The fillers in the sample for test shall be recorded to the nearest 0 1 percent

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FREE CARBON

1. SCOPE

1.1 This method is intended for use in determining the amount of free carbon in a rubber compound This method is not applicable to compounds containing polyisobutylene rubber

2. SPECIMEN

2.1 The specimen shall consist of 0 5 gram of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Assay flask, 125 ml

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone-chloroform mixture as described in method 16311

4.2 The extracted specimen shall be transferred to the 125-ml assay flask and heated on a steam bath until the solvent has been removed The flask and specimen shall be cooled, a few milliliters of nitric acid added, and the flask set aside until the violent action has subsided Sixty milliliters more of the nitric acid shall be added, taking care to wash down the sides of the flask, and the mixture heated on a steam bath until no more bubbles or foam appear on the surface of the mixture At least one hour of digestion on the steam bath is required

4.3 The hot liquid from the flask shall be poured into a Gooch crucible, 3 1 5 18 of method 16001, taking care to keep as much of the insoluble material as practicable in the flask The hot mixture shall be filtered slowly, with gentle suction, catching the filtrate in a filter flask containing 2 inches of cold water The insoluble material shall be washed well by decantation with hot nitric acid

4.4 ***Caution:*** The filter flask should be emptied and rinsed, and 2 inches of water put into the flask before proceeding with the determination The insoluble material shall be washed by decantation with acetone, followed by a 1 1 acetone-chloroform mixture until the filtrate is colorless The insoluble material which has been carefully retained in the flask shall be digested on a steam bath with 10 to 15 ml of 30-percent sodium hydroxide solution for 30 minutes, diluted with 60 ml of hot distilled water, and f tered The flask shall be rinsed well with hot 15-percent sodium hydroxide solution and any remaining residue transferred quantitatively to the Gooch crucible The residue in the Gooch crucible shall then be washed four times with hot hydrochloric acid solution and then with warm 1 to 7 hydrochloric acid until the filtrate is colorless, followed by a wash with a few milliliters of denatured alcohol to facilitate drying

4.5 The Gooch crucible shall be removed from the funnel and the outside cleaned thoroughly with acetone The crucible and contents shall be dried to a constant weight in an air bath at a temperature of 240° ± 10° C (464° ± 18° F), cooled in a desiccator, and weighed and the weight recorded as A The Gooch crucible and contents shall be ignited in a muffle furnace at a temperature of 650° to 900° C (1,202° to 1,652° F), cooled in a desiccator, and weighed, and the weight recorded as *B*

5. RESULTS

5.1 Calculation.—The free carbon in the specimen shall be calculated as follows

$$\text{Free carbon, percent,} = \frac{A-B}{W} \times 100$$

Where

A is the weight of the crucible plus carbon, grams
B is the weight of the crucible after ignition, grams
W is the weight of the specimen before extraction, gram

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The free carbon in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The free carbon in the sample for test shall be recorded to the nearest 0 1 percent

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GLUE

1. SCOPE

1.1 This method is intended for use in determining the amount of glue in rubber compounds which are decomposed by sulfuric acid In calculating the amount of glue from the amount of nitrogen found in a natural rubber compound, a correction is applied for the nitrogen content of the natural rubber, assuming that the rubber as compounded contains 0 4-percent protein nitrogen This value is used to calculate a more exact figure for rubber hydrocarbon. In the absence of any other nitrogenous material, including nitrile rubber, the method can be applied to synthetic rubber In the case of synthetic rubber, the calculation of total nitrogen is sufficient and no correction is necessary.

2. SPECIMEN

2.1 The specimen shall consist of 2 grams of rubber compound from the composite sample prepared as described in method 1211.

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Kjeldahl flask, 750 ml.

3.1.3 Distillation unit

3.1.4 Digestion unit for Kjeldahl flask.

3.1.5 Sulfuric acid, 0.1 *N* solution, standardized

3.1.6 Sodium hydroxide, 50-percent solution

3.1.7 Sodium hydroxide, 0.1 *N* solution, standardized

3.1.8 Sodium sulfate.

3.1.9 Copper sulfate

3.1.10 Granulated zinc

3.1.11 Paraffin

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted for 8 hours with acetone-chloroform mixture as described in method 16311.

4.2 The specimen shall be removed from the filter paper, placed in a 750-ml Kjeldahl flask, and heated on a steam bath until the solvent has been removed Twenty-five milliliters of sulfuric acid, 10 to 12 grams of sodium sulfate, and about one gram of copper sulfate shall be added to the flask The flask and contents shall be heated gently until the frothing ceases, then the temperature increased gradually until the liquid boils, and the boiling continued until the solution is clear The flask shall be removed from the flame and allowed to cool to room temperature One hundred and fifty milliliters of distilled water shall be carefully added and the flask again allowed to cool One hundred milliliters of 50 percent sodium hydroxide shall be added by slowly pouring the solution down the side of the flask so that it does not mix immediately with the sulfuric acid solution One gram of granulated zinc to prevent bumping and a piece of paraffin the size of a pea to prevent excessive frothing shall be added

4.3 The Kjeldahl flask shall be connected quickly with a condenser, the delivery tube of which dips into a 500-ml Erlenmeyer flask containing 50 ml of 0 1 *N* sulfuric acid solution and 50 ml of distilled water The Kjeldahl flask shall be swirled carefully to mix the contents and heated gently The heat shall be increased as the danger of foaming over diminishes and the mixture finally boiled briskly until about one-half of the liquid in the Kjeldahl flask has passed over into the Erlenmeyer receiving flask A few drops of methyl red indicator solution shall be added to the Erlenmeyer flask and the excess sulfuric acid titrated with 0 1 *N* sodium hydroxide solution A blank determination shall be made

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5. RESULTS

5.1 Calculations

5.1.1 The glue content from total nitrogen determination in the specimen shall be calculated as follows:

Glue from total nitrogen,

$$\text{percent, } A, = \frac{(B-C)N_1 - DN_2 \times 0.014 \times 6.5}{W} \times 100$$

Where

B is the number of milliliters of sulfuric acid to titrate specimen
C is the number of milliliters of sulfuric acid to titrate blank
D is the number of milliliters of sodium hydroxide to back titrate
N_1 is the normality of sulfuric acid
N_2 is the normality of sodium hydroxide
W is the weight of the specimen before extraction, grams

5.1.2 The natural rubber hydrocarbon of the specimen shall be calculated as follows

Natural rubber hydrocarbon by weight,

$$\text{percent, } E, = \frac{94}{97}(100 - F)$$

Where

F is the sum of the percentages of total extract, fillers, organic sulfur, alcoholic potassium hydroxide extract, free carbon, and cellulose from 4.1 of method 16111, and the glue, A, determined in 5.1.1

5.1.3 The natural rubber as compounded of the specimen shall be calculated as follows

$$\text{Natural rubber as compounded, percent, } G, = \frac{100}{94} \times E$$

5.1.4 The glue corrected in the specimen shall be calculated as follows

$$\text{Glue corrected, percent, } = A - (0.026 \times G)$$

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested.

5.3 The glue in the sample for test shall be the average of the values obtained from the specimens tested.

5.4 The glue in the sample for test shall be recorded to the nearest 0.1 percent.

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FIBROUS MATERIALS, MINERAL OIL

1. SCOPE

1.1 This method is intended for use in determining the total amount of fibrous material in rubber compounds The fibrous material is that material which is separated from the rubber compound by removing the rubber with mineral oil solvent and which will not pass through a No 100 sieve This method is not applicable to compounds containing nitrile rubber or to any rubber compound that is not soluble in the mineral oil solvent.

2. SPECIMEN

2.1 The specimen shall consist of one gram of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Assay flask, 125 ml

3.1.3 No 100 sieve, U S Standard Sieve Series, Federal Specification RR-S-366

3.1.4 Rubber solvent described in 3 1 6 25 of method 16001.

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted for at least 8 hours with acetone-chloroform mixture as described in method 16311

4.2 The specimen shall be placed in the assay flask, heated on a steam bath until the solvent has been removed, and 25 ml of the mineral oil solvent added. The flask shall be covered with a watch glass and the mixture digested in an air bath at a temperature of 150° to 165° C (302° to 329° F) until the rubber compound has formed a colloidal solution with the fillers still in suspension This requires from 30 to 90 minutes ***Caution:*** Keep away from the flame.

4.3 The mixture shall be decanted through the No 100 sieve and the residue washed well with chloroform The cycle of heating the specimen with rubber solvent, decanting through the sieve, nd washing with chloroform shall be repeated until the rubber solvent remains clear This indicates that all the rubber compound has been removed

4.4 The filtrate and washings shall then be poured through a portion of the sieve not occupied by fibers in order to recover any fibrous material that may have slipped through The residue shall then be extracted with chloroform as described in method 16331 for 4 hours to remove all the mineral oil solvent from the last treatment The residue shall be dried to a constant weight at a temperature of 105° to 110° C (221° to 230° F) for one hour, cooled in a desiccator, and weighed

5. RESULTS

5.1 Calculation. The fibrous material in the specimen shall be calculated as follows

$$\text{Fibrous material, percent,} = \frac{A}{W} \times 100$$

Where

A is the weight of the fibrous material, gram

W is the weight of the specimen before extraction, gram

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The fibrous material in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The fibrous material in the sample for test shall be recorded to the nearest 0 1 percent.

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(formerly Fed Spec ZZ–R–601a)

FIBROUS MATERIALS, PARANITROTOLUENE-ORTHODICHLOROBENZENE

1. SCOPE

1.1 This method is intended for use in determining the total amount of fibrous material in rubber compounds The fibrous material is that material which is separated from the rubber compound by removing the rubber with paranitrotoluene-orthodichlorobenzene solvent and which will not pass through a No 100 sieve

2. SPECIMEN

2.1 The specimen shall consist of one gram of rubber compound from the composite sample prepared as described in method 1211

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Erlenmeyer flask, 250 ml , with standard taper joint

3.1.3 Air condenser with standard taper joint

3.1.4 Sieve No 100, U S Standard Sieve Series, Federal Specification RR–S–366

3.1.5 Orthodichlorobenzene

3.1.6 Paranitrotoluene

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted for at least 8 hours with acetone-chloroform mixture as described in method 16311

4.2 The specimen shall be placed in the 250-ml Erlenmeyer flask and heated on a steam bath until the solvent has been removed. The flask shall then be fitted with the air condenser Ten grams of paranitrotoluene and 25 ml of orthodichlorobenzene shall be added to the flask and the mixture heated to 180° to 190° C (356° to 374° F) on a hot plate under a hood with occasional stirring until the rubber dissolves The flask and contents shall be cooled and 10 ml of chloroform added The remainder of the procedure shall be as described in 4.3 and 4 4 of method 16531

5. RESULTS

5.1 The results shall be as described in method 16531.

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CHEMICAL ANALYSIS OF PACKINGS, GENERAL

1. SCOPE

1.1 These methods are intended for use in the analysis of the usual types of mechanical packing containing rubber and asbestos for conformance with Federal specifications The methods are based on the use of the mineral oil solvent and paranitrotoluene-orthodichlorobenzene solvents for removing the rubber compound from the packing, determining the amount of fibrous materials directly, and calculating the amount of rubber compound by difference The packings may contain lubricants, asbestos, and metallic members in addition to the rubber compound. Since Federal specifications usually require that asbestos packings contain a minimum amount of asbestos in the fibrous materials, a method is also given for determining the amount of asbestos In these methods, only the fibrous material retained on a No 100 sieve conforming to the requirements of Federal Specification RR–S–366 is considered fiber A general scheme relating the different methods is given in figure 16601 for the convenience of the analyst in running a complete analysis of the packing Methods are described for determining lubricants, rubber compound and rubber hydrocarbon, fillers, fibrous material and metal, asbestos content of fibrous materials; and chemically combined water in the asbestos

2. COMPOSITE SAMPLE

2.1 Rubber packing.—A composite sample shall be prepared as described in method 1211

2.2 Other packing.—A composite sample shall be prepared by taking equal portions from each test unit in the sample for test so as to include a representative cross section of each test unit Each portion from each test unit shall be cleaned of any foreign material, weighed on a balance, and adjusted to within 0 5 gram of the same weight All portions taken shall be mixed thoroughly

2.2.1 If the packing contains a rubber core, rubber spring back, or metal, the portions shall be stored in a suitable airtight container

2.2.2 If a rubber core, rubber spring back, or metal is not present, the portions shall be cut into small strips or small cross sections and thoroughly mixed and stored in a suitable airtight container

3. SPECIMEN

3.1 Unless otherwise indicated, the specimen shall be taken from the composite sample prepared as described in 2

4. PROCEDURE

4.1 If a complete analysis of the packing is required, the detail specification should be reviewed to ascertain the type of packing undergoing test, that is, asbestos with graphite lubricant, asbestos and rubber, etc , and compared with the general scheme of analysis to determine the most expeditious procedure to follow

4.2 If the packing contains a rubber core or spring back and graphite is absent, the specimen shall be weighed, the rubber core or spring back removed, and the remainder of the packing cut into small strips or pieces before continuing the analysis If the packing contains a rubber core or spring back and graphite is present, the rubber core or spring back shall be removed after removal of the graphite Care shall be taken that none of the fibers are lost in removing the rubber core or spring back or in cutting the specimen Unless otherwise indicated, the weight of the specimen before removal of the rubber core or spring back shall be used in calculating the results of the analysis

4.3 If organic solvent-soluble lubricant only is present, method 16621 shall be used If the packing contains graphite lubricant, and total lubricant is required, method 16625 shall be used If the packing contains graphite, and both graphite and organic solvent-soluble lubricants are required, the two shall be determined as described in method 16627 If the packing contains rubber and asbestos, and lubricant is not required, the total extract of the specimen shall be determined as required in method 16621 before determining the asbestos or rubber content

4.4 If the amount of rubber compound or rubber hydrocarbon in the packing, or asbestos

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in the fibrous portion of the packing is required, the metal and fibrous materials shall be separated from the rubber and fillers as described in method 16631 or 16635 and the fillers in the rubber compound determined as described in method 16641 or 16645 The amount of rubber compound in the packing shall be calculated as described in method 16611 by subtracting the sum of the total lubricant or total extract and metal and fibrous materials from 100 The amount of rubber hydrocarbon in the packing shall be calculated as described in method 16611 by subtracting the fillers from the rubber compound

4.5 If the amount of asbestos in the fibrous material is required, it shall be determined as described in method 16651 If the chemically combined water of the asbestos and the asbestos fiber are both required, the chemically combined water shall be determined as described in method 16651 If the asbestos in the fiber is not required, the chemically combined water shall be determined as described in method 16661.

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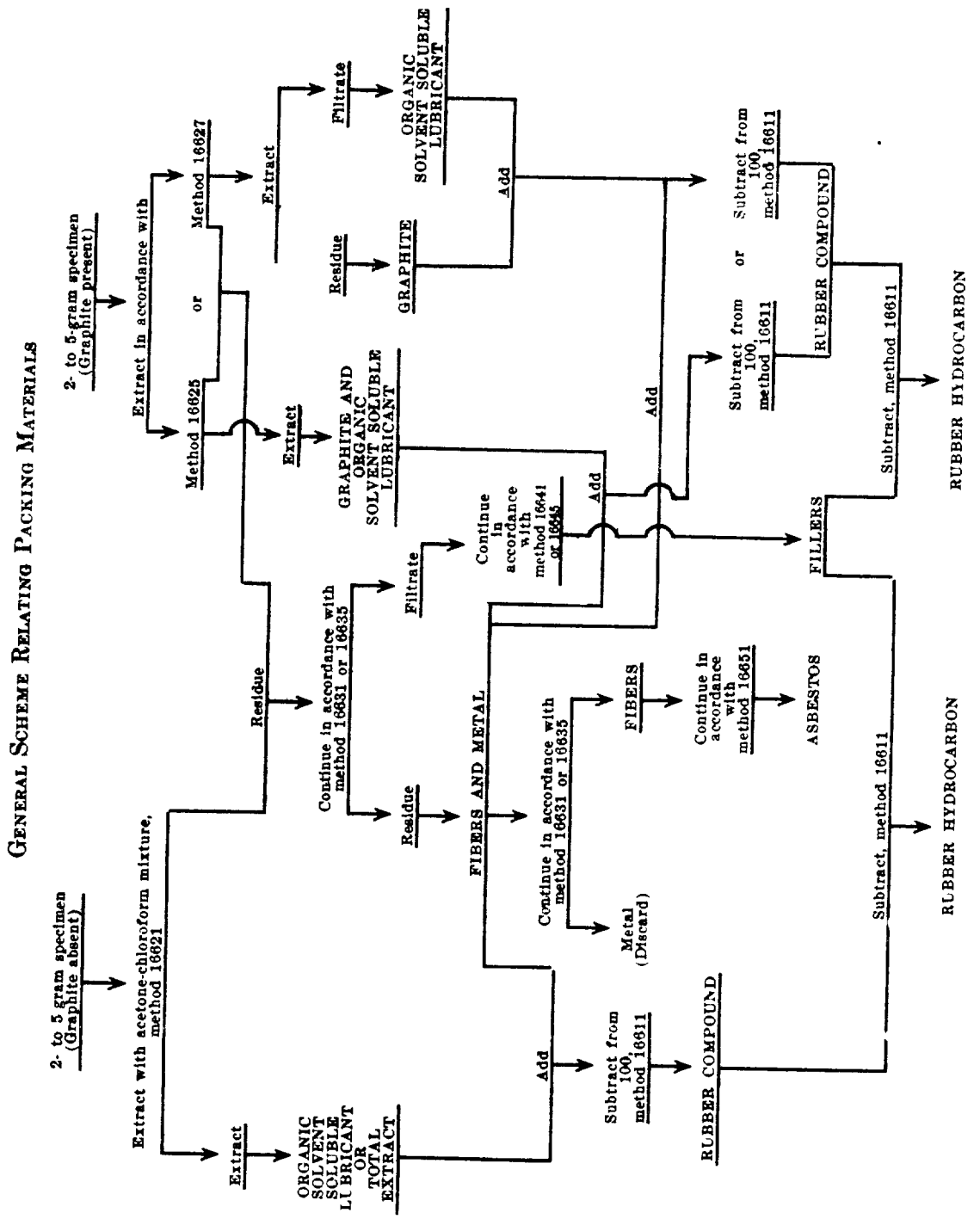

GENERAL SCHEME RELATING PACKING MATERIALS

FIGURE 16601.

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PACKINGS, RUBBER COMPOUND AND RUBBER HYDROCARBON

1. SCOPE

1.1 This method is intended for use in determining the amount of rubber compound and rubber hydrocarbon in packings The nonrubber constituents are determined and the amount of rubber and rubber hydrocarbon calculated by difference

2. SPECIMEN

2.1 The specimen shall consist of a portion of the packing from the composite sample prepared as described in method 16601, as required in the applicable method referenced in 4

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as required in the methods referenced in 4

4. PROCEDURE

4.1 The procedure shall consist of determining the following nonrubber constituents and calculating the rubber content by difference

4.1.1 Total lubricant or total extract, method 16621, 16625, or 16627

4.1.2 Fibers and metal, method 16631 or 16635

4.1.3 Fillers, method 16641 or 16645

5. RESULTS

5.1 Calculations.

5.1.1 The rubber compound of the specimen shall be calculated as follows

Graphite present
Rubber compound by weight, percent, $E = 100 - (A + C)$
Graphite absent
Rubber compound by weight, percent, $F = 100 - (B + C)$

Where
A is the percent of total lubricant, graphite plus organic solvent-soluble lubricant, in the specimen, method 16625 or 16627
B is the percent of total extract or organic solvent-soluble lubricant in the specimen, method 16621
C is the percent of fiber and metal in the specimen, method 16631 or 16635

5.1.2 The rubber hydrocarbon in the specimen shall be calculated as follows

Graphite present
Rubber hydrocarbon by weight, percent, $G = E - D$
Graphite absent
Rubber hydrocarbon by weight, percent, $G = F - D$

Where
D is the percent fillers in the specimen, method 16641 or 16645

5.1.2.1 The rubber hydrocarbon by volume shall be calculated as follows

$$\text{Rubber hydrocarbon by volume, percent,} = \frac{G \times \text{specific gravity of compound}}{X}$$

Where
X is the specific gravity of the rubber, and is 0 911 for natural rubber, 0 933 for GR-S, 0 917 for GR-I, and 1 27 for chloroprene

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The rubber compound of the sample for test or the rubber hydrocarbon in the rubber compound shall be the average of the values obtained from the specimens tested

5.4 The rubber compound of the sample for test or the rubber hydrocarbon in the rubber compound shall be recorded to the nearest one percent

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PACKINGS, LUBRICANT, GRAPHITE ABSENT

1. SCOPE

1.1 This method is intended for use in determining the amount of lubricating materials removed from packings by extracting them with a mixture of acetone and chloroform. If rubber is present in the packing, the acetone-chloroform-soluble materials in the rubber are neglected because of the uncertainty involved in the separation of the lubricating materials from the soluble resins, antioxidants, sulfur, and accelerators that may be present in the rubber compound. This method is not applicable to packings containing nitrile or polyisobutylene rubber or to packings containing graphite as a lubricant.

2. SPECIMEN

2.1 Packings containing a rubber core, rubber spring back, or metal.—The specimen shall consist of 3 to 5 grams of packing from the composite sample prepared as described in method 16601.

2.2 Other packings.—The specimen shall consist of approximately 2 grams of packing from the composite sample prepared as described in method 16601.

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows:

3.1.1 Apparatus and reagents in method 16001 as required.

3.1.2 Source of compressed air.

3.1.3 Extraction apparatus shown in figure 16001.

3.1.4 Extraction mixture consisting of 32-percent acetone and 68-percent chloroform by volume.

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram. If the specimen contains a rubber core or rubber spring back, the rubber core or rubber spring back shall be removed. The remainder of the specimen shall be cut into small strips or pieces. Care shall be taken that none of the fibers are lost in removing the rubber core, rubber spring back, or in cutting the specimen. The specimen shall be placed in the siphon cup.

4.2 An extraction flask shall be dried for 2 hours at a temperature of 100° to 105° C (212° to 221° F), cooled in a desiccator, and weighed. Sixty to seventy-five milliliters of the extraction mixture shall be added to the flask, the extraction cup containing the specimen suspended in position, and the apparatus assembled as shown in figure 16001. The specimen shall be extracted for 2 hours at such a rate that from 2.5 to 3.5 minutes are required to fill and empty the extraction cup. The extracted specimen should be reserved for use in method 16631 or 16635.

4.3 The acetone and chloroform shall be distilled off on a steam bath passing a gentle current of filtered air through the flask to prevent boiling of the liquid. The flask shall be removed from the steam bath just before the last trace of solvent has disappeared and the current of air continued for about 10 minutes. The flask and contents shall be dried in circulating air for one hour at a temperature of 105°±2° C (221°±3.6° F), cooled in a desiccator, and weighed.

5. RESULTS

5.1 Calculation.—The lubricant content of the specimen shall be calculated as follows:

$$\text{Organic solvent-soluble lubricant or total extract, percent,} = \frac{L}{W}$$

Where

L is the weight of the extract, gram

W is the weight of the specimen before removal of rubber core or rubber spring back and before extraction, grams, 4.1

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested.

5.3 The organic solvent-soluble lubricant or total extract of the sample for test shall be the average of the values obtained from the specimens tested.

5.4 The organic solvent-soluble lubricant or total extract of the sample for test shall be recorded to the nearest 0.1 percent.

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PACKINGS, LUBRICANT, GRAPHITE PRESENT

1. SCOPE

1.1 This method is intended for use in determining the total amount of lubricant in packing when packing contains graphite as a lubricant If rubber is present in the packing, the acetone-chloroform-soluble materials in the rubber are neglected because of the uncertainty involved in the separation of the lubricating materials from the soluble resins, antioxidants, sulfur, and accelerators that may be present in the rubber compound This method is not applicable to packings containing nitrile or polyisobutylene rubber

2 SPECIMEN

2.1 Packings containing a rubber core, rubber spring back, or metal.—The specimen shall consist of 3 to 5 grams of packing from the composite sample prepared as described in method 16601

2.2 Other packings.—The specimen shall consist of approximately 2 grams of packing from the composite sample prepared as described in method 16601

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Source of compressed air

3.1.3 Extraction apparatus shown in figure 16001

3.1.4 Extraction mixture consisting of 32-percent acetone and 68-percent chloroform by volume

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and placed in the siphon cup An extraction flask shall be dried at a temperature of 100° to 105° C (212° to 221° F), cooled in a desiccator, and weighed A few milliliters of acetone-chloroform mixture shall be added to the cup and the cup agitated gently to dislodge the bulk of the graphite which shall be removed by decanting into the extraction flask This procedure is necessary to prevent the siphon cup from becoming clogged Usually 2 or 3 such treatments are sufficient to remove most of the graphite which shall be collected in the extraction flask along with the acetone-chloroform-soluble materials

4.2 Sufficient acetone-chloroform mixture shall be added to the flask to bring the volume up to about 60 to 75 ml The extraction cup containing the specimen and the flask shall be assembled in the extraction apparatus as shown in figure 16001 The specimen shall be extracted for 2 hours with the acetone-chloroform mixture at such a rate that from 2.5 to 3.5 minutes are required to fill and empty the extraction cup At the end of the extraction period, the extracted specimen shall be transferred to a watch glass and air dried. The dry fibers shall be carefully separated, and the remaining graphite brushed off and collected on the watch glass and then added to the extract in the flask.

4.3 If the remaining portion of the extracted specimen is to be used for further analysis in method 16631 or 16635, care should be taken not to lose any of the fibers in the separation of the graphite If a rubber core or rubber spring back is present in the specimen, the core or spring back shall be removed from the fibrous materials at the same time as the graphite

4.4 The remainder of the procedure shall be as described in 4 3 of method 16621

5. RESULTS

5.1 Calculation.—The lubricant content of the specimen shall be calculated as follows

$$\text{Total lubricant or organic solvent-soluble lubricant plus graphite, percent,} = \frac{L}{W}$$

Where

L is the weight of graphite and acetone-chloroform-soluble material, grams, 4 4

W is the weight of the specimen before extraction, grams, 4 1

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The total lubricant content of the sample for test shall be the average of the values obtained from the specimen tested

5.4 The total lubricant content of the sample for test shall be recorded to the nearest 0 5 percent

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PACKINGS, LUBRICANT, ORGANIC SOLVENT-SOLUBLE, AND GRAPHITE

1. SCOPE

1.1 This method is intended for use in determining the amount of organic solvent-soluble lubricant and the amount of graphite lubricant used in packings If rubber is present in the packing, the acetone-chloroform-soluble materials in the rubber are neglected because of the uncertainty involved in separation of the organic solvent-soluble lubricants from the soluble resins, antioxidants, sulfur, and accelerators present in the rubber compound This method is not applicable to packings containing nitrile or polyisobutylene rubber

2. SPECIMEN

2.1 Packings containing a rubber core, rubber spring back, or metal.—The specimen shall consist of 3 to 5 grams of packing from the composite sample prepared as described in method 16601

2.2 Other packings.—The specimen shall consist of approximately 2 grams of packing from the composite sample prepared as described in method 16601

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone-chloroform mixture as described in 4 1 to 4 3, inclusive, of method 16625 to separate the graphite and organic solvent-soluble material from the packing The extracted residue should be reserved for use in method 16631 or 16635

4.2 The mixture of graphite and organic solvent-soluble material in the flask shall be diluted to a volume of approximately 100 ml with the acetone-chloroform mixture The graphite shall be separated from the mixture by filtering with suction, dried to a constant weight at a temperature of 100° to 105° C (212° to 221° F), cooled in a desiccator, and weighed

4.3 The filtrate from 4 2 shall be returned to the weighed extraction flask and treated as described in 4 3 of method 16621

5. RESULTS

5.1 Calculation.

5.1.1 *Graphite*—The graphite content of the specimen shall be calculated as follows

$$\text{Graphite, percent,} = \frac{G}{W}$$

Where

G is the weight of the graphite, gram, 4 2

W is the weight of the specimen before extraction, grams, 4 1

5.1.2 *Organic solvent-soluble lubricant* — The organic solvent-soluble lubricant content of the specimen shall be calculated as follows

$$\text{Organic solvent-soluble lubricant, percent,} = \frac{L}{W}$$

Where

L is the weight of the acetone-chloroform-soluble material, gram, 4 3

W is the weight of the specimen before extraction, grams, 4 1

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The graphite content and the organic solvent-soluble lubricant content, respectively, of the sample for test shall be the average of the values obtained from the specimen tested

5.4 The graphite content and the organic solvent-soluble lubricant content, respectively, of the sample for test shall be recorded to the nearest 0 5 percent

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PACKINGS, METAL AND FIBROUS MATERIALS, MINERAL OIL

1. SCOPE

1.1 This method is intended for use in determining the amount of metal and fibrous materials in packings containing rubber The rubber and fillers are removed from the metal and fibrous material by dissolving the rubber in mineral oil Only the fibrous material that will not pass through a No 100 sieve is considered as fiber This method is not applicable to packings containing nitrile or polyisobutylene rubber or rubber which is not soluble in the mineral oil solvent

2. SPECIMEN

2.1 Packings containing a rubber core, rubber spring back, or metal.—The specimen shall consist of 3 to 5 grams of packing from the composite sample prepared as described in method 16601

2.2 Other packings.—The specimen shall consist of approximately 2 grams of packing from the composite sample prepared as described in method 16601

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Assay flask, 125-ml capacity

3.1.3 No 100 sieve, U S Standard Sieve Series, Federal Specification RR-S-366

3.1.4 Mineral oil solvent as described in 3 1 6 25 of method 16001

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone-chloroform mixture as described in method 16621 or 16625

4.2 The extracted specimen shall be placed in the 125-ml assay flask containing 10 to 15 ml of the mineral-oil solvent The flask shall be covered with a watch glass and the mixture digested in an air bath at a temperature of 150° to 165° C (302° to 329° F) until the rubber has formed a colloidal solution with the fillers still in suspension Solution is considered complete when the rubber compound has broken down and the oil appears quite clear, which requires approximately 1½ hours of heating ***Caution:*** Keep away from the flame The mixture shall be poured onto the No 100 sieve which retains the fiber and metal, and the residue washed well with chloroform The washed residue shall be transferred from the sieve to the assay flask and the mineral oil treatment, filtering, and chloroform washing repeated until the fillers are completely removed

4.3 The combined filtrates and washings from 4 2 shall be poured through an area of the sieve not covered by fibers to collect any fibers that may have passed through the sieve during the filtering and washing operations The residue shall be extracted with chloroform as described in method 16331 except that the period of extraction shall be one hour and the chloroform extract shall be added to the filtrate and washings The filtrate and washings containing the rubber compound should be reserved for use in method 16641

4 4 The residue shall be transferred to a weighed crucible, dried to a constant weight at a temperature of 105° to 110° C (221° to 230° F), cooled in a desiccator, and weighed The weight of the dry residue shall be recorded as *A* If the residue does not contain metal, it should be reserved for use in method 16651

4.5 If the residue contains metal, the dried residue from 4 4 shall be placed on a sheet of glazed paper and the pieces of wire or other metal removed with the aid of a lens and a pair of tweezers, care being taken not to lose any of the fibrous material The fibrous material shall be placed in a weighed crucible, dried to a constant weight at a temperature of 105° to 110° C (221° to 230° F), cooled in a desiccator, and weighed The weight of the dry fibers shall be recorded as *F* The dry fibers should be reserved for use in method 16651

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5. RESULTS

5.1 Calculation.

5.1.1 The metallic and fibrous materials in the specimen shall be calculated as follows

$$\text{Fibrous materials and metal, percent, } C = \frac{A}{W}$$

Where

A is the weight of the fibers and metal, grams

W is the weight of the specimen before extraction, grams, 4 1

5.1.2 The fibrous materials in the specimen shall be calculated as follows:

$$\text{Fibrous materials, percent, } D = \frac{F}{W}$$

Where

F is the weight of the fibrous materials, grams, 4 5

W is the weight of the specimen before extraction, grams, 4 1

5.1.3 The amount of metallic material in the specimen shall be calculated as follows

$$\text{Metal, percent,} = C - D,$$

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The metal plus fibrous materials, metal, or fibrous materials in the sample for test shall be the average of the values obtained from the specimens tested

5.4 The metal plus fibrous materials, metal, or fibrous materials in the sample for test shall be recorded to the nearest 0 1 percent

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PACKINGS, METAL AND FIBROUS MATERIALS, PARANITROTOLUENE-ORTHODICHLOROBENZENE

1. SCOPE

1.1 This method is intended for use in determining the amount of metal and fibrous materials in packing containing rubber The rubber and fillers are removed from the metal and fibrous materials by dissolving the rubber in a mixture of paranitrotoluene and orthodichlorobenzene Only the fibrous material that will not pass through a No 100 sieve is considered as fiber This method is not applicable to packings containing nitrile rubber

2. SPECIMEN

2.1 Packings containing a rubber core, rubber spring back, or metal.—The specimen shall consist of 3 to 5 grams of packing from the composite sample prepared as described in method 16601

2.2 Other packings.—The specimen shall consist of approximately 2 grams of packing from the composite sample prepared as described in method 16601

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Erlenmeyer flask, 250 ml , with standard taper joint

3.1.3 Air condenser with standard taper joint

3.1.4 No 100 sieve, U S. Standard Sieve Series, Federal Specification RR-S-566

3.1.5 Paranitrotoluene

3.1.6 Orthodichlorobenzene.

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone-chloroform mixture as described in method 16621 or 16625

4.2 The extracted specimen shall be placed in the 250-ml Erlenmeyer flask The flask shall then be fitted with the air condenser Ten grams of paranitrotoluene and 25 ml of orthodichlorobenzene shall be added to the flask and the mixture heated to a temperature of 180° to 190° (336° to 374° F) on a hot plate under a hood with occasional stirring until the rubber dissolves From 4 to 10 hours is usually sufficient to effect solution The flask and contents shall be cooled, 10 ml of chloroform added, and the mixture poured through the No 100 sieve which retains the fibers and metal The residue shall be washed with chloroform until the insoluble fillers are removed as indicated by a clear filtrate If undissolved rubber remains, the fibers and metal shall be returned to the digestion flask and the paranitrotoluene-orthodichlorobenzene treatment, filtering, and chloroform washing repeated until the rubber has dissolved

4.3 The combined filtrates and washings from 4 2 shall be poured through an area of the sieve not covered by fibers to collect any fibers that may have passed through the sieve during the filtering and washing operations The residue shall be extracted with chloroform as prescribed in method 16331, except that the period of extraction shall be one hour and the chloroform extract shall be added to the filtrate and washings The filtrate and washings containing the rubber compound should be reserved for use in method 16645

4 4 The residue shall be transferred to a weighed crucible, dried to a constant weight at a temperature of 105° to 110° C (221° to 230° F), cooled in a desiccator, and weighed The weight of the dry residue shall be recorded as *A* If the residue does not contain metal, it should be reserved for use in method 16651

4.5 If the residue contains metal, the dried residue from 4 4 shall be placed on a sheet of glazed paper and the pieces of wire or other metal removed with the aid of a lens and a pair of tweezers, care being taken not to lose any of the fibrous materials The fibrous material shall be placed in a weighed crucible, dried to a constant weight at a temperature of 105° to 110° C (221° to 230° F), cooled in a desiccator, and weighed The weight of the

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dry fibers shall be recorded as *F* The dry fibers should be reserved for use in method 16651

5. RESULTS

5.1 The results shall be as described in method 16631

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PACKINGS, FILLERS IN RUBBER COMPOUND, MINERAL OIL

1. SCOPE

1.1 This method is intended for use in determining the amount of fillers which have been added to the rubber compound used in packing The fillers consist of free carbon and inorganic materials other than free or elemental sulfur which is removed by extraction This method consists of dissolving the rubber in a mixture of mineral oil, separating the fillers by filtration, and weighing the residue. This method is not applicable to compounds containing nitrile rubber.

2. SPECIMEN

2.1 Packings containing a rubber core, rubber spring back, or metal.—The specimen shall consist of 3 to 5 grams of packing from the composite sample prepared as described in method 16601.

2.2 Other packings.—The specimen shall consist of approximately 2 grams of packing from the composite sample prepared as described in method 16601

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as described in method 16001 as required

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone-chloroform mixture as described in method 16621 or 16625.

4.2 The extracted specimens shall be treated as described in 4 2 and 4 3 of method 16631 to remove the fibers and metal

4.3 The flask containing the combined filtrate and washings from the removal of the fibers and metal shall be heated gently to remove the chloroform and digested again at a temperature of 150° to 165° C. (302° to 329° F) as described in 4 2 of method 16631, in order to insure complete solution of the rubber.

4.4 The flask shall be removed from the air bath, cooled to about 90° C. (194° F.), and 10 to 15 ml. of benzene added in a stream while mixing thoroughly The mixture shall be allowed to cool and petroleum ether added to fill the flask to within 2 cm of the top The contents of the flask shall be thoroughly mixed, the flask covered with a watch glass to prevent evaporation, and the mixture allowed to stand until the filler particles have settled sufficiently to permit rapid filtration

4.5 The Gooch crucible, 3.1 5.18 of method 16001, with asbestos pad shall be dried for one hour at a temperature of 105° to 110° C (221° to 230° F), cooled in a desiccator, and weighed, and the weight recorded as *A*. The mixture shall be filtered by decantation through the Gooch crucible using suction The residue in the flask shall be washed well with petroleum ether, transferring as much of the residue as possible to the crucible The suction shall be turned off and warm acetone added to both the flask and crucible The remainder of the material in the flask shall be transferred to the crucible with hot alcohol Finally the crucible shall be washed with hot alcohol

4.6 The Gooch crucible shall be cleaned on the outside with solvent and cloth, and the crucible and contents dried for one hour at a temperature of 105° to 110° C. (221° to 230° F), cooled in a desiccator, and weighed, and the weight recorded as *B*.

5. RESULTS

5.1 Calculation.—The fillers in the specimen shall be calculated as follows

$$\text{Fillers, percent,} = \frac{B-A}{W} \times 100$$

Where

A is the weight of the Gooch crucible, grams, 4 5

B is the weight of the Gooch crucible and contents, grams, 4 6

W is the weight of the specimen before extraction, grams, 4 1

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested.

5.4 The fillers in the sample for test shall be recorded to the nearest 0.1 percent.

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(formerly Fed Spec ZZ-R-601a)

PACKINGS, FILLERS IN RUBBER COMPOUND, PARANITROTOLUENE-ORTHODICHLOROBENZENE

1. SCOPE

1.1 This method is intended for use in determining the amount of fillers which have been added to the rubber compound used in packing The fillers consist of free carbon and inorganic materials other than free or elemental sulfur which is removed by extraction This method consists of dissolving the rubber in a mixture of paranitrotoluene-orthodichlorobenzene, separating the fillers by filtration, and weighing the residue The method is not applicable to compounds containing rubber that is not soluble in paranitrotoluene-orthodichlorobenzene solvent or to compounds containing nitrile rubber

2. SPECIMEN

2.1 **Packings containing a rubber core, rubber spring back, or metal.**—The specimen shall consist of 3 to 5 grams of packing from the composite sample prepared as described in method 16601

2.2 **Other packings.**—The specimen shall consist of approximately 2 grams of packing from the composite sample prepared as described in method 16601

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as described in method 16001 as required

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone-chloroform mixture as described in method 16621 or 16625

4.2 The extracted specimen shall be treated as described in 4.2 and 4,3 of method 16635 to remove the fibers and metal.

4.3 The flask containing the combined filtrate and washings from the removal of the fiber and metal shall be heated gently to remove the chloroform and digested again at a temperature of 180° to 190° C. (336° to 374° F.) as described in 4.2 of method 16635, in order to insure complete solution of the rubber. The mixture shall be cooled and 10 ml. of chloroform and approximately 60 ml. of acetone added. The contents of the flask shall be thoroughly mixed and the flask covered with a watch glass to prevent evaporation. The covered flask and contents shall be set aside and the mixture allowed to stand until the fillers have settled sufficiently to permit rapid filtration. The remainder of the procedure shall be as described in 4 5 and 4.6 of method 16641

5. RESULTS

5.1 The results shall be as described in method 16641.

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PACKINGS, ASBESTOS

1. SCOPE

1.1 This method is intended for use in determining the amount of asbestos in the fibrous portions of packings The method consists of determining directly the amount of cotton in the fibrous materials and calculating the asbestos content by difference The chemically combined water of the asbestos fiber is also calculated. If graphite, carbon black, or other material insoluble in the mineral oil, method 16631, or the paranitrotoluene-orthodichlorobenzene mixture, method 16635, remains in the fibrous materials, this method will not give reliable results and in such cases the results should be considered as approximations.

2. SPECIMEN

2.1 Packings containing a rubber core, rubber spring back, or metal.—The specimen shall consist of 3 to 5 grams of packing from the composite sample prepared as described in method 16601

2.2 Other packings.—The specimen shall consist of approximately 2 grams of packing from the composite sample prepared as described in method 16601

2.3 In any case, the specimen shall be large enough to provide one gram of fibrous material

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents in method 16001 as required

3.1.2 Organic combustion furnace

3.1.3 Vanier or similar carbon-dioxide absorption bulb.

3.1.4 Carbon-dioxide-free oxygen supply.

3.1.5 Combustion boat, porcelain or platinum

3.1.6 U-tubes or other drying tubes

3.1.7 Calcium chloride, magnesium perchlorate, or calcium sulfate for absorbing moisture

3.1.8 Sodium hydroxide impregnated base material (the absorbent having the trade name "Ascarite" is of this type) or caustic potash for absorbing carbon dioxide

4. PROCEDURE

4.1 The specimen shall be weighed to the nearest milligram and extracted with acetone-chloroform mixture as described in method 16621 or 16625.

4.2 The extracted specimen shall be treated as described in 4 2 to 4 5, inclusive, of method 16631 or 16635 to obtain the weight of the dry fibrous material, *F*

4.3 The combustion boat shall be heated to constant weight at a temperature of 900° ± 50° C. (1,652 ± 122° F), cooled in a desiccator, and weighed Approximately one gram of the fibrous material, 4 2, shall be placed in the combustion boat, dried for one hour at a temperature of 105° to 110° C (221° to 230° F), cooled in a desiccator, and weighed The weight of the dry specimen shall be recorded to the nearest milligram as *E*

4.4 The dried specimen of fiber in the boat shall be inserted into the combustion tube of the organic combustion furnace. A current of carbon-dioxide-free oxygen shall be passed through the combustion tube until all carbon dioxide has been removed from the tube The tube shall be connected to drying tubes and carbon-dioxide absorption apparatus and heated to a temperature of 900° ± 50° C. (1,652° ± 122° F) for approximately 30 minutes or until the combustion of the cotton is complete The carbon-dioxide-free oxygen shall be passed through the combustion tube during the combustion period at the rate of approximately 200 ml per minute

4.5 The combustion gases shall be passed through either two U-tubes containing calcium chloride or through a drying tube containing anhydrous magnesium perchlorate or calcium sulfate to remove the moisture, and finally the gases shall be passed into either a weighed Vanier tube or similar absorption bulb containing a solution of caustic potash or into a weighed carbon-dioxide absorption bulb con-

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taining a sodium hydroxide impregnated base to absorb the carbon dioxide

4.6 At the end of the combustion period, the Vanier or other carbon-dioxide absorption bulb shall be disconnected, cooled to room temperature, and weighed and the weight of the carbon dioxide recorded as C The combustion boat containing the ignited residue shall be removed from the combustion tube and cooled in a desiccator and the weight of the residue after ignition recorded as R

4.7 Three-elevenths of the weight of the carbon dioxide formed, 4 6, shall represent the weight of the carbon in the fibrous material This shall be considered 44 40 percent of any cotton These factors may be combined to give a constant of 0.614

5. RESULTS

5.1 Calculation.

5.1.1 The cotton content of the specimen of fiber shall be calculated as follows

$$\text{Cotton, percent, } A = \frac{61\,4 \times C}{E}$$

Where

C is the weight of the carbon dioxide, gram, 4 6
E is the weight of the specimen of fibers, grams, 4 3

5.1.2 The asbestos content of the fiber in the specimen shall be calculated as follows.

$$\text{Asbestos in fiber, percent, } D = 100 - A$$

5.1.3 The ignited asbestos in the specimen of fiber shall be calculated as follows.

$$\text{Ignited asbestos, percent, } B = \frac{R}{E} \times 100$$

Where

R is the weight of the ignited residue, grams, 4 6
E is the weight of the specimen of fibers, grams, 4 3

5.1.4 The chemically combined water of the asbestos in the specimen shall be calculated as follows

$$\text{Chemically combined water of asbestos, percent,} = \frac{D - B}{D} \times 100$$

5.1.5 The asbestos in the specimen of packing shall be calculated as follows

$$\text{Asbestos content of specimen of packing, percent,} = \frac{F \times D}{S}$$

Where.

F is the weight of the fibrous material in the specimen of packing, grams, 4.2
D is the asbestos in the fiber, percent, 5 1 2
S is the weight of the specimen of packing before extraction, grams, 4 1

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested.

5.3 The asbestos content and chemically combined water of the sample for test shall be the average of the values obtained from the specimens tested

5.4 The asbestos content of the sample for test shall be recorded to the nearest one percent

5.5 The chemically combined water of the sample for test shall be recorded to the nearest 0.1 percent

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PACKINGS, CHEMICALLY COMBINED WATER, ASBESTOS

1. SCOPE

1.1 This method is intended for use in determining the chemically combined water of the asbestos used in packing containing asbestos It is applicable for determining the chemically combined water in asbestos that does not contain cotton or other organic material When cotton is present, the chemically combined water is determined at the same time as the asbestos content in method 16651

2. SPECIMEN

2.1 Packings containing a rubber core, rubber spring back, or metal.—The specimen shall consist of 3 to 5 grams of packing from the composite sample prepared as described in method 16601

2.2 Other packings.—The specimen shall consist of approximately 2 grams of packing from the composite sample prepared as described in method 16601

3. APPARATUS AND REAGENTS

3.1 The apparatus and reagents shall be as follows

3.1.1 Apparatus and reagents from method 16001 as required

3.1.2 Platinum crucible

4. PROCEDURE

4.1 Organic material present.

4.1.1 The specimen shall be extracted with acetone-chloroform mixture as described in method 16621 or 16625

4.1.2 The extracted specimen shall be treated as described in 4 2 to 4 5, inclusive, of method 16631 or 16635 to obtain the fibrous material

4.1.3 Approximately 1 gram of the fibrous material from 4 1 2 shall be transferred to a weighed platinum crucible The crucible and contents shall be dried for one hour at a temperature of 105° to 110° C (221° to 230° F) cooled in a desiccator, and weighed, and the weight of the dry fibrous material recorded as W

4.1.4 The crucible and contents shall be ignited in a muffle furnace at a temperature of 800° to 825° C (1,472° to 1,517° F) to a constant weight, cooled in a desiccator, and weighed, and the weight of the ignited specimen recorded as I

4.2 Organic material absent.—If the packing does not contain organic material other than lubricant, the procedure shall be as described in 4 1 1, 4 1 3, and 4 1 4

5. RESULTS

5.1 Calculation.—The chemically combined water of the specimen shall be calculated as follows

$$\text{Chemically combined water, percent,} = \frac{W-I}{W}$$

Where

W is the weight of the dry specimen of fiber before ignition, grams

I is the weight of the specimen of fiber after ignition, grams

5.2 Unless otherwise specified in the detail specification, two specimens from each composite sample shall be tested

5.3 The chemically combined water of the sample for test shall be the average of the values obtained from the specimens tested

5.4 The chemically combined water shall be recorded to the nearest 0 1 percent

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